CHAPTER

78

EXHAUST

(LEAP-1B ENGINES)

BOEING

737-7/8/8200/9/10
AIRCRAFT MAINTENANCE MANUAL

CHAPTER 78
EXHAUST

Subject/Page	Date	COC
78-EFFECTIVE PAGES		
1 thru 9	Sep 15/2026	
10	BLANK	
78-CONTENTS		
1	May 15/2024	
O 2	Sep 15/2026	
3	Sep 15/2025	
4	Jan 15/2025	
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6	May 15/2024	
7	Jan 15/2026	
8	May 15/2024	
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O 13	Sep 15/2026	
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O 409	Sep 15/2026	
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407	May 15/2024	
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409	Jan 15/2026	
410	Jan 15/2026	
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406	Jan 15/2025	
407	May 15/2024	
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78-31-00		
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R 214	Sep 15/2026	
R 215	Sep 15/2026	
R 216	Sep 15/2026	
R 217	Sep 15/2026	
R 218	Sep 15/2026	
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221	Sep 15/2025	
222	Sep 15/2025	
223	Jan 15/2026	
224	Jan 15/2026	
225	Jan 15/2026	
R 226	Sep 15/2026	
227	Jan 15/2026	
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R 229	Sep 15/2026	
230	Jan 15/2026	
231	Sep 15/2025	

A = Added, R = Revised, D = Deleted, O = Overflow, C = Customer Originated Change

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78-31-00 (cont)		
232	Sep 15/2025	
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234	Sep 15/2025	
235	Sep 15/2025	
236	Sep 15/2025	
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R 238	Sep 15/2026	
O 239	Sep 15/2026	
R 240	Sep 15/2026	
O 241	Sep 15/2026	
O 242	Sep 15/2026	
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O 247	Sep 15/2026	
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R 505	Sep 15/2026	
O 506	Sep 15/2026	
O 507	Sep 15/2026	
O 508	Sep 15/2026	
509	May 15/2024	
R 510	Sep 15/2026	
O 511	Sep 15/2026	
O 512	Sep 15/2026	

Subject/Page	Date	COC
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Subject/Page	Date	COC
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R 433	Sep 15/2026	
O 434	Sep 15/2026	
O 435	Sep 15/2026	
O 436	Sep 15/2026	
O 437	Sep 15/2026	
O 438	Sep 15/2026	
78-31-01		
501	Jan 15/2025	
502	Jan 15/2025	

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636	May 15/2024	
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813	May 15/2024	
814	May 15/2024	
815	May 15/2024	
816	May 15/2024	
78-31-02		
401	May 15/2026	
402	May 15/2024	

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410	May 15/2024	
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619	May 15/2025	

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623	May 15/2024	
624	May 15/2024	
78-31-03		
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427	Sep 15/2025	
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430	May 15/2024	

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78-31-03 (cont)		
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452	Sep 15/2025	
453	Sep 15/2025	
454	Sep 15/2025	
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456	Sep 15/2025	
78-31-03		
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78-31-03		
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802	May 15/2024	
803	May 15/2024	

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805	May 15/2024	
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807	May 15/2024	
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810	May 15/2024	
78-31-04		
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78-31-04		
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402	May 15/2024	
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404	May 15/2024	
405	Sep 15/2024	

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208	May 15/2024	

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403	Jan 15/2026	
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405	May 15/2024	
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R 407	Sep 15/2026	
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R 409	Sep 15/2026	
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78-31-08		
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78-31-08		
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78-31-09		
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EXHAUST

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403	May 15/2024	
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405	May 15/2024	
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414	May 15/2024	
415	May 15/2024	
R 416	Sep 15/2026	
R 417	Sep 15/2026	
418	May 15/2024	
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78-31-11		
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78-31-12		
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402	May 15/2024	

Subject/Page	Date	COC
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R 432	Sep 15/2026	
433	Jan 15/2026	
R 434	Sep 15/2026	
R 435	Sep 15/2026	
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437	May 15/2024	
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78-31-13		
401	Jan 15/2026	

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78-31-13 (cont)		
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818	May 15/2024	
78-31-14		
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403	Jan 15/2025	
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78-31-15		
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405	May 15/2025	
406	May 15/2025	
407	Jan 15/2025	
408	BLANK	
78-31-16		
401	May 15/2024	
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404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	May 15/2024	
78-31-23		
401	May 15/2024	
R 402	Sep 15/2026	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	BLANK	
78-31-23		
601	May 15/2024	
602	Jan 15/2026	
603	Jan 15/2026	
604	Jan 15/2026	

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78-31-23 (cont)		
605	Jan 15/2026	
606	BLANK	
78-31-24		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
78-34-01		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2026	
407	May 15/2026	
408	May 15/2026	
409	May 15/2024	
410	May 15/2024	
78-34-03		
401	May 15/2024	
402	May 15/2026	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2026	
409	May 15/2024	
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412	May 15/2026	
413	Sep 15/2025	
414	Sep 15/2025	
415	Sep 15/2025	
416	BLANK	

Subject/Page	Date	COC
78-34-03		
501	May 15/2026	
502	May 15/2024	
503	May 15/2024	
504	May 15/2024	
505	May 15/2024	
506	May 15/2024	
507	May 15/2024	
508	May 15/2024	
78-34-04		
R 401	Sep 15/2026	
O 402	Sep 15/2026	
R 403	Sep 15/2026	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	Sep 15/2025	
408	May 15/2024	
409	May 15/2024	
410	May 15/2024	
411	May 15/2024	
412	BLANK	
78-34-04		
R 501	Sep 15/2026	
502	Sep 15/2025	
503	May 15/2024	
R 504	Sep 15/2026	
505	Sep 15/2025	
506	Sep 15/2025	
507	May 15/2024	
508	May 15/2024	
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510	May 15/2024	
78-34-05		
401	May 15/2024	
402	BLANK	

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78-34-07		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
78-34-08		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
78-34-09		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	BLANK	
78-34-09		
601	May 15/2024	
602	May 15/2024	
78-34-10		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	Jan 15/2026	
407	May 15/2025	
408	Jan 15/2026	
78-34-10		
501	May 15/2025	
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504	May 15/2025	

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505	May 15/2024	
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78-34-10		
R 601	Sep 15/2026	
602	Sep 15/2025	
603	May 15/2024	
604	May 15/2024	
605	May 15/2024	
R 606	Sep 15/2026	
A 607	Sep 15/2026	
A 608	Sep 15/2026	
78-34-11		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
R 406	Sep 15/2026	
407	May 15/2024	
408	May 15/2024	
78-36-02		
401	Sep 15/2025	
402	Jan 15/2026	
403	Jan 15/2026	
404	May 15/2024	
405	Jan 15/2026	
406	Jan 15/2026	
407	Jan 15/2026	
408	Jan 15/2026	
78-36-02		
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502	May 15/2024	
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78-36-02 (cont)		
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737-7/8/8200/9/10
AIRCRAFT MAINTENANCE MANUAL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
EXHAUST - DDG MAINTENANCE PROCEDURES	78-00-00		901	PTF ALL
DDG 78-31-01 Thrust Reversers Inoperative - Preparation TASK 78-00-00-040-801-G00			901	PTF ALL
DDG 78-31-01 Thrust Reversers Inoperative - Restoration TASK 78-00-00-440-801-G00			908	PTF ALL
TURBINE EXHAUST SYSTEM - INSPECTION/CHECK	78-11-00		601	PTF ALL
Labyrinth Seals Inspection TASK 78-11-00-210-801-G00			601	PTF ALL
Primary Nozzle Assembly and Primary Plug Assembly Inspection TASK 78-11-00-210-802-G00			604	PTF ALL
PRIMARY NOZZLE ASSEMBLY - REMOVAL/INSTALLATION	78-11-01	2	401	PTF ALL
Primary Nozzle Assembly Removal TASK 78-11-01-000-801-G00		2	401	PTF ALL
Primary Nozzle Assembly Installation TASK 78-11-01-400-801-G00		2	408	PTF ALL
PRIMARY PLUG ASSEMBLY - REMOVAL/INSTALLATION	78-11-02	2	401	PTF ALL
Primary Plug Assembly Removal TASK 78-11-02-000-801-G00		2	401	PTF ALL
Primary Plug Assembly Installation TASK 78-11-02-400-801-G00		2	408	PTF ALL
CENTER VENT TUBE EXTENSION - REMOVAL/INSTALLATION	78-11-03		401	PTF ALL
Center Vent Tube Extension Removal TASK 78-11-03-000-801-G00			401	PTF ALL
Center Vent Tube Extension Installation TASK 78-11-03-400-801-G00			405	PTF ALL
CENTER VENT TUBE - INSPECTION/CHECK	78-11-03		601	PTF ALL
Center Vent Tube Extension (CVTE) Inspection TASK 78-11-03-210-801-G00			601	PTF ALL

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THRUST REVERSER SYSTEM - MAINTENANCE PRACTICES	78-31-00		201	PTF ALL
Open the Thrust Reverser (Selection) TASK 78-31-00-010-801-G00			201	PTF ALL
Close the Thrust Reverser (Selection) TASK 78-31-00-010-802-G00			201	PTF ALL
Open the Thrust Reverser (Hydraulic Pump Procedure) TASK 78-31-00-000-801-G00			202	PTF ALL
Close the Thrust Reverser (Hydraulic Pump Procedure) TASK 78-31-00-410-801-G00			214	PTF ALL
Open the Thrust Reverser (Sling Procedure) TASK 78-31-00-010-803-G00			219	PTF ALL
Close the Thrust Reverser (Sling Procedure) TASK 78-31-00-410-802-G00			222	PTF ALL
Thrust Reverser Operation - Extend (Selection) TASK 78-31-00-980-801-G00			226	PTF ALL
Thrust Reverser Operation - Retract (Selection) TASK 78-31-00-980-802-G00			226	PTF ALL
Thrust Reverser Operation - Extend (Manual Procedure) TASK 78-31-00-980-803-G00			227	PTF ALL
Thrust Reverser Operation - Retract (Manual Procedure) TASK 78-31-00-980-804-G00			233	PTF ALL
Thrust Reverser ,Operation - Extend (Power Procedure) TASK 78-31-00-980-805-G00			238	PTF ALL
Thrust Reverser Operation - Retract (Power Procedure) TASK 78-31-00-980-806-G00			243	PTF ALL
Thrust Reverser Deactivation For Ground Maintenance TASK 78-31-00-040-801-G00			248	PTF ALL

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Thrust Reverser Activation After Ground Maintenance TASK 78-31-00-440-801-G00			253	PTF ALL
THRUST REVERSER SYSTEM - ADJUSTMENT/TEST	78-31-00		501	PTF ALL
Thrust Reverser Operational Test (Selection) TASK 78-31-00-700-801-G00			501	PTF ALL
Thrust Reverser System Normal Operational Test TASK 78-31-00-700-802-G00			501	PTF ALL
Thrust Reverser Operational Test (Standby Hydraulic System) TASK 78-31-00-700-803-G00			510	PTF ALL
Thrust Reverser Locks Operational Test TASK 78-31-00-700-804-G00			520	PTF ALL
Thrust Reverser Proximity Sensors Operational Test TASK 78-31-00-700-805-G00			532	PTF ALL
Thrust Reverser Annunciator Lights Operational Test TASK 78-31-00-700-806-G00			538	PTF ALL
Scheduled Maintenance Task for EEC Faults - Reverser Sensor TASK 78-31-00-700-807-G00			540	PTF ALL
THRUST REVERSER - REMOVAL/INSTALLATION	78-31-01		401	PTF ALL
Thrust Reverser Removal TASK 78-31-01-000-801-G00			401	PTF ALL
Thrust Reverser Installation TASK 78-31-01-400-801-G00			429	PTF ALL
THRUST REVERSER - ADJUSTMENT/TEST	78-31-01		501	PTF ALL
Thrust Reverser Adjustment TASK 78-31-01-820-801-G00			501	PTF ALL
THRUST REVERSER - INSPECTION/CHECK	78-31-01		601	PTF ALL
Thrust Reverser Fan Duct Wall Inspection TASK 78-31-01-200-801-G00			601	PTF ALL

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Thrust Reverser Cowl Hinge Inspection TASK 78-31-01-211-801-G00			605	PTF ALL
Precooler Kiss Seal Inspection TASK 78-31-01-211-802-G00			609	PTF ALL
Thrust Reverser Inner and Outer V-Blade Inspection TASK 78-31-01-211-803-G00			614	PTF ALL
Thrust Reverser Outer V-Blade Seal Inspection TASK 78-31-01-211-804-G00			620	PTF ALL
Thrust Reverser Outer V-Blade Inspection TASK 78-31-01-211-805-G00			626	PTF ALL
Thrust Reverser Fire Seal Inspection TASK 78-31-01-211-806-G00			629	PTF ALL
Thrust Reverser Rigging Check TASK 78-31-01-280-801-G00			632	PTF ALL
Composite Inner Wall Inspection TASK 78-31-01-211-807-G00			637	PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL
Composite Inner Wall Inlet and Exhaust Ports TASK 78-31-01-210-801-G00			639	PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL
THRUST REVERSER- CLEANING/PAINTING	78-31-01		701	PTF ALL
Thrust Reverser Witness Mark Installation TASK 78-31-01-370-801-G00			701	PTF ALL
Thrust Reverser Acoustic Surfaces Cleaning TASK 78-31-01-160-801-G00			705	PTF ALL
THRUST REVERSER - REPAIRS	78-31-01		801	PTF ALL
Thrust Reverser Hinge Beam Fire Wall Replacement TASK 78-31-01-960-801-G00			801	PTF ALL
Blocker Door Support Assembly Replacement TASK 78-31-01-960-802-G00			806	PTF ALL
Through Stud Replacement TASK 78-31-01-960-803-G00			809	PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL
TRANSLATING SLEEVE - REMOVAL/INSTALLATION	78-31-02		401	PTF ALL
Translating Sleeve Removal TASK 78-31-02-000-801-G00			401	PTF ALL

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Translating Sleeve Installation TASK 78-31-02-400-801-G00			412	PTF ALL
TRANSLATING SLEEVE - INSPECTION/CHECK	78-31-02		601	PTF ALL
Translating Sleeve Inspection TASK 78-31-02-200-801-G00			601	PTF ALL
Main Slider and Auxiliary Slider Inspection TASK 78-31-02-200-802-G00			606	PTF ALL
Engine Thrust Reverser Hinge Beam and Latch Beam Primary Track and Translating Sleeve Primary Slider General Visual Inspection for Wear with the Thrust Reverser Deployed TASK 78-31-02-210-801-G00			608	PTF ALL
Thrust Reverser Actuator Attach Fitting Visual Inspection TASK 78-31-02-200-803-G00			611	PTF ALL
Thrust Reverser Actuator Attach Fitting Detailed Inspection TASK 78-31-02-200-804-G00			615	PTF ALL
Engine Thrust Reverser Hinge Beam and Latch Beam Secondary Track and Translating Sleeve Secondary Slider General Visual Inspection for Wear with the Thrust Reverser Deployed TASK 78-31-02-210-802-G00			622	PTF ALL
THRUST REVERSER HYDRAULIC ACTUATORS - REMOVAL/INSTALLATION	78-31-03		401	PTF ALL
Thrust Reverser Hydraulic Actuator Removal (Selection) TASK 78-31-03-000-801-G00			401	PTF ALL
Thrust Reverser Hydraulic Actuator Installation (Selection) TASK 78-31-03-400-801-G00			401	PTF ALL
Upper Locking Hydraulic Actuator Removal TASK 78-31-03-000-802-G00			401	PTF ALL
Upper Locking Hydraulic Actuator Installation TASK 78-31-03-400-802-G00			415	PTF ALL
Middle Locking Hydraulic Actuator Removal TASK 78-31-03-000-803-G00			421	PTF ALL

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Middle Locking Hydraulic Actuator Installation TASK 78-31-03-400-803-G00			434	PTF ALL
Lower Hydraulic Actuator Removal TASK 78-31-03-000-804-G00			439	PTF ALL
Lower Hydraulic Actuator Installation TASK 78-31-03-400-804-G00			451	PTF ALL
THRUST REVERSER HYDRAULIC ACTUATORS - INSPECTION/CHECK	78-31-03		601	PTF ALL
Thrust Reverser Hydraulic Actuator Rod End Inspection TASK 78-31-03-610-801-G00			601	PTF ALL
THRUST REVERSER HYDRAULIC ACTUATORS - REPAIRS	78-31-03		801	PTF ALL
Manual Unlock Handle Assembly Replacement TASK 78-31-03-300-801-G00			801	PTF ALL
SYNC SHAFT - REMOVAL/INSTALLATION	78-31-04		401	PTF ALL
Sync Shaft Removal TASK 78-31-04-000-801-G00			401	PTF ALL
Sync Shaft Installation TASK 78-31-04-400-801-G00			411	PTF ALL
SYNC SHAFT - INSPECTION/CHECK	78-31-04		601	PTF ALL
Sync Shaft Inspection TASK 78-31-04-200-801-G00			601	PTF ALL
CASCADES - REMOVAL/INSTALLATION	78-31-05		401	PTF ALL
Cascade Removal TASK 78-31-05-000-801-G00			401	PTF ALL
Cascade Installation TASK 78-31-05-400-801-G00			405	PTF ALL
CASCADE - INSPECTION/CHECK	78-31-05		601	PTF ALL
Thrust Reverser Cascade Segments Inspection TASK 78-31-05-200-801-G00			601	PTF ALL
BLOCKER DOORS - REMOVAL/INSTALLATION	78-31-06		401	PTF ALL
Blocker Door Removal TASK 78-31-06-000-801-G00			401	PTF ALL

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Blocker Door Installation TASK 78-31-06-400-801-G00			406	PTF ALL
BLOCKER DOORS - INSPECTION/CHECK	78-31-06		601	PTF ALL
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Blocker Door Hinge Attachments and Bearings Inspection TASK 78-31-06-211-801-G00			608	PTF ALL
BLOCKER DOOR DRAG LINKS - MAINTENANCE PRACTICES	78-31-07		201	PTF ALL
Drag Link Spherical Bearing Inspection TASK 78-31-07-900-801-G00			201	PTF ALL
BLOCKER DOOR DRAG LINKS - REMOVAL/INSTALLATION	78-31-07		401	PTF ALL
Blocker Door Drag Link Removal TASK 78-31-07-000-801-G00			401	PTF ALL
Blocker Door Drag Link Installation TASK 78-31-07-400-801-G00			407	PTF ALL
THRUST REVERSER OPENING ACTUATOR - MAINTENANCE PRACTICES	78-31-08		201	PTF ALL
Fill and Bleed Procedure TASK 78-31-08-870-801-G00			201	PTF ALL
THRUST REVERSER OPENING ACTUATOR - REMOVAL/INSTALLATION	78-31-08		401	PTF ALL
Thrust Reverser Opening Actuator Removal TASK 78-31-08-000-801-G00			401	PTF ALL
Thrust Reverser Opening Actuator Installation TASK 78-31-08-400-801-G00			405	PTF ALL
KRUEGER FLAP DEFLECTOR AND PLUG - REMOVAL/INSTALLATION	78-31-09		401	PTF ALL
Krueger Flap Deflector Removal TASK 78-31-09-010-801-G00			401	PTF ALL
Krueger Flap Deflector Installation TASK 78-31-09-420-801-G00			405	PTF ALL

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KRUEGER FLAP DEFLECTOR - INSPECTION/CHECK	78-31-09		601	PTF ALL
Krueger Flap Deflector Inspection TASK 78-31-09-200-801-G00			601	PTF ALL
RUBSTRIP - REMOVAL/INSTALLATION	78-31-10		401	PTF ALL
Rubstrip Removal TASK 78-31-10-000-801-G00			401	PTF ALL
Rubstrip Installation TASK 78-31-10-400-801-G00			412	PTF ALL
TENSION LATCHES - REMOVAL/INSTALLATION	78-31-11		401	PTF ALL
Tension Latch Removal TASK 78-31-11-000-801-G00			401	PTF ALL
Tension Latch Installation TASK 78-31-11-400-801-G00			407	PTF ALL
TENSION LATCHES - ADJUSTMENT/TEST	78-31-11		501	PTF ALL
Latches Adjustment TASK 78-31-11-820-801-G00			501	PTF ALL
FIRE SEAL - REMOVAL/INSTALLATION	78-31-12		401	PTF ALL
Fire Seal Removal TASK 78-31-12-000-801-G00			401	PTF ALL
Fire Seal Installation TASK 78-31-12-400-801-G00			420	PTF ALL
FIRE SEAL - INSPECTION/CHECK	78-31-12		601	PTF ALL
Fire Seal Inspection (Visual Check) TASK 78-31-12-200-801-G00			601	PTF ALL
Fire Seal Inspection (Detailed) TASK 78-31-12-200-802-G00			603	PTF ALL
Fire Seal Compression Check TASK 78-31-12-800-801-G00			609	PTF ALL
FIRE SEAL - REPAIR	78-31-12		801	PTF ALL
Fire Seal Repair TASK 78-31-12-300-801-G00			801	PTF ALL

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INSULATION BLANKET - REMOVAL/INSTALLATION	78-31-13		401	PTF ALL
Insulation Blankets - Removal TASK 78-31-13-000-801-G00			401	PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL
Insulation Blankets - Installation TASK 78-31-13-400-801-G00			411	PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL
Insulation Blankets - Removal TASK 78-31-13-000-802-G00			420	PTF ALL; AIRPLANES WITH TITANIUM INNER WALL
Insulation Blankets - Installation TASK 78-31-13-400-802-G00			428	PTF ALL; AIRPLANES WITH TITANIUM INNER WALL
INSULATION BLANKET - INSPECTION/CHECK	78-31-13		601	PTF ALL
Insulation Blankets Inspection TASK 78-31-13-200-802-G00			601	PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL
Insulation Blankets Inspection TASK 78-31-13-200-803-G00			607	PTF ALL; AIRPLANES WITH TITANIUM INNER WALL
Insulation Blanket Seal Inspection TASK 78-31-13-211-801-G00			613	PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL
INSULATION BLANKET - REPAIRS	78-31-13		801	PTF ALL
Insulation Blanket Repair TASK 78-31-13-300-801-G00			801	PTF ALL
PRECOOLER INLET DUCT GUIDE VANE - REMOVAL/INSTALLATION	78-31-14		401	PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL
Precooler Inlet Duct Guide Vane Removal TASK 78-31-14-000-801-G00			401	PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL
Precooler Inlet Duct Guide Vane Installation TASK 78-31-14-400-801-G00			406	PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL
PRECOOLER INLET DUCT SEAL	78-31-15		401	PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL
Precooler Inlet Duct Seal - Removal TASK 78-31-15-000-801-G00			401	PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL
Precooler Inlet Duct Seal - Installation TASK 78-31-15-400-801-G00			405	PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL

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PRESSURE RELIEF DOOR - REMOVAL/INSTALLATION	78-31-16		401	PTF ALL
Pressure Relief Door Removal TASK 78-31-16-000-801-G00			401	PTF ALL
Pressure Relief Door Installation TASK 78-31-16-400-801-G00			407	PTF ALL
BULLNOSE SEAL - REMOVAL/INSTALLATION	78-31-23		401	PTF ALL
Bullnose Seal Removal TASK 78-31-23-000-801-G00			401	PTF ALL
Bullnose Seal Installation TASK 78-31-23-400-801-G00			404	PTF ALL
BULLNOSE SEAL - INSPECTION/CHECK	78-31-23		601	PTF ALL
Bullnose Seal Inspection TASK 78-31-23-200-801-G00			601	PTF ALL
BLOCKER DOOR AERO SEAL - REMOVAL/INSTALLATION	78-31-24		401	PTF ALL
Blocker Door Aero Seal Removal TASK 78-31-24-000-801-G00			401	PTF ALL
Blocker Door Aero Seal Installation TASK 78-31-24-400-801-G00			404	PTF ALL
THRUST REVERSER CONTROL VALVE MODULE - REMOVAL/INSTALLATION	78-34-01		401	PTF ALL
Control Valve Module Removal TASK 78-34-01-000-801-G00			401	PTF ALL
Control Valve Module Installation TASK 78-34-01-400-801-G00			406	PTF ALL
THRUST REVERSER SLEEVE LOCK PROXIMITY SENSOR - REMOVAL/INSTALLATION	78-34-03		401	PTF ALL
Thrust Reverser Sleeve Lock Proximity Sensor Removal TASK 78-34-03-000-801-G00			401	PTF ALL
Thrust Reverser Sleeve Lock Proximity Sensor Installation TASK 78-34-03-400-801-G00			412	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
THRUST REVERSER SLEEVE LOCK PROXIMITY SENSOR - ADJUSTMENT/TEST	78-34-03		501	PTF ALL
Sleeve Lock Proximity Sensor Adjustment and Test TASK 78-34-03-800-801-G00			501	PTF ALL
THRUST REVERSER CONTROL SWITCH - REMOVAL/INSTALLATION	78-34-04		401	PTF ALL
Control Switch Removal TASK 78-34-04-000-801-G00			401	PTF ALL
Control Switch Installation TASK 78-34-04-400-801-G00			407	PTF ALL
THRUST REVERSER CONTROL SWITCH - ADJUSTMENT/TEST	78-34-04		501	PTF ALL
Control Switch Adjustment and Test TASK 78-34-04-700-801-G00			501	PTF ALL
THRUST REVERSER ARM, STOW AND SYNC LOCK SWITCHES - REMOVAL/INSTALLATION	78-34-05		401	PTF ALL
Thrust Reverser Arm, Stow and Sync Lock Switches Removal TASK 78-34-05-000-801-G00			401	PTF ALL
VOLUMETRIC HYDRAULIC FUSES - REMOVAL/INSTALLATION	78-34-07		401	PTF ALL
Standby System Volumetric Hydraulic Fuse Removal TASK 78-34-07-000-802-G00			401	PTF ALL
Standby System Volumetric Hydraulic Fuse Installation TASK 78-34-07-400-802-G00			405	PTF ALL
THRUST REVERSER SHUTTLE VALVE - REMOVAL/INSTALLATION	78-34-08		401	PTF ALL
Thrust Reverser Shuttle Valve Removal TASK 78-34-08-000-801-G00			401	PTF ALL
Thrust Reverser Shuttle Valve Installation TASK 78-34-08-400-801-G00			407	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
THRUST REVERSER CONTROL SWITCH ACTUATION SPRING - REMOVAL/INSTALLATION	78-34-09		401	PTF ALL
Actuation Spring Removal TASK 78-34-09-000-801-G00			401	PTF ALL
Actuation Spring Installation TASK 78-34-09-400-801-G00			403	PTF ALL
THRUST REVERSER CONTROL SWITCH ACTUATION SPRING - INSPECTION/CHECK	78-34-09		601	PTF ALL
Actuation Spring Free Angle Check TASK 78-34-09-200-801-G00			601	PTF ALL
THRUST REVERSER SYNC LOCK - REMOVAL/INSTALLATION	78-34-10		401	PTF ALL
Thrust Reverser Sync Lock Removal TASK 78-34-10-000-801-G00			401	PTF ALL
Thrust Reverser Sync Lock Installation TASK 78-34-10-400-801-G00			406	PTF ALL
THRUST REVERSER SYNC LOCK - ADJUSTMENT/TEST	78-34-10		501	PTF ALL
Sync Lock Manual Integrity Test When Hydraulic Power Is Not Available TASK 78-34-10-700-801-G00			501	PTF ALL
Sync Lock Manual Integrity Test When Hydraulic Power Is Available TASK 78-34-10-700-802-G00			507	PTF ALL
THRUST REVERSER SYNC LOCK - INSPECTION/CHECK	78-34-10		601	PTF ALL
Sync Lock Control Relay Hot Short Protection Check TASK 78-34-10-710-801-G00			601	PTF ALL
Thrust Reverser Sequence Relay Electrical Resistance Check TASK 78-34-10-710-802-G00			606	PTF ALL
STOW HYDRAULIC PRESSURE SWITCH - REMOVAL/INSTALLATION	78-34-11		401	PTF ALL
Stow Hydraulic Pressure Switch Removal TASK 78-34-11-000-801-G00			401	PTF ALL

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CHAPTER 78
EXHAUST

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Stow Hydraulic Pressure Switch Installation TASK 78-34-11-400-801-G00			406	PTF ALL
LINEAR VARIABLE DIFFERENTIAL TRANSFORMER (LVDT) - REMOVAL/INSTALLATION	78-36-02		401	PTF ALL
LVDT Removal TASK 78-36-02-000-801-G00			401	PTF ALL
LVDT Installation TASK 78-36-02-400-801-G00			405	PTF ALL
LINEAR VARIABLE DIFFERENTIAL TRANSFORMER (LVDT) - ADJUSTMENT/TEST	78-36-02		501	PTF ALL
Linear Variable Differential Transformer (LVDT) - Adjustment TASK 78-36-02-820-801-G00			501	PTF ALL

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EXHAUST - DDG MAINTENANCE PROCEDURES

1. General

A. This procedure has maintenance tasks for the Master Minimum Equipment List (MMEL) maintenance requirements as shown in the Dispatch Deviations Guide (DDG). These tasks are used to prepare the airplane for flight with certain systems/components inoperative.

B. This procedure also has the tasks that put the airplane back to its usual condition.

C. This procedure has these tasks:

(1) DDG 78-31-01 Thrust Reversers Inoperative - Preparation

(2) DDG 78-31-01 Thrust Reversers Inoperative - Restoration.

TASK 78-00-00-040-801-G00

2. DDG 78-31-01 Thrust Reversers Inoperative - Preparation

(Figure 901)

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with one thrust reverser inoperative.

(2) There are two methods you can use to prepare the airplane for flight with the thrust reverser inoperative:

(a) Use Method 1 when you are able to deploy and stow the thrust reverser with hydraulic power.

(b) Use Method 2 when you are not able to deploy and stow the thrust reverser with hydraulic power.

(3) These are the conditions for this task:

NOTE: These conditions are applicable to Method 1 and Method 2.

(a) One thrust reverser may be inoperative provided the inoperative thrust reverser is secured in the forward thrust position.

(b) Thrust reverser deactivation can result in the illumination of the MASTER CAUTION, ENG annunciation, and REVERSER LIMITED indication light associated with the deactivated thrust reverser during engine start or when performing a MASTER CAUTION recall.

(c) On the P5 aft overhead panel:

1) The REVERSER LIMITED indication light for the inoperative thrust reverser can be illuminated.

2) The REVERSER LIMITED indication light for the operational thrust reverser must not be illuminated.

(d) Make sure that the two sync locks on the inoperative thrust reverser are in the locked position.

(e) Install the deactivation pins in the left and right thrust reverser sleeves of the inoperative thrust reverser.

(f) Make sure that the applicable thrust reverser sleeve position indications on the EPCS maintenance data page are correct.

(g) Lockwire the reverse thrust lever for the inoperative thrust reverser to the applicable forward thrust lever with MS20995C32 lockwire, G01048.

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(h) Attach a REVERSER INOP tag to the applicable reverse thrust lever.

(i) Attach an INOP placard on the applicable REVERSER LIMITED indication light.

(j) Open the applicable circuit breakers and attach lock collars.

B. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
31-65-00-750-801	Fault Interrogation for a Maintenance Message (P/B 201)
46-13-02-860-802	Special Functions (P/B 201)
71-00-00-700-816-G00	Test No. 38 - Thrust Reverser Cycle Special Functions (P/B 501)
71-00-00-970-801-G00	Maintenance Data Pages from the Engine Propulsion Control System (EPCS) (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-21-00-800-801-G00	EEC Maintenance Power Selection (P/B 201)
78-34-10-700-801-G00	Sync Lock Manual Integrity Test When Hydraulic Power Is Not Available (P/B 501)

C. Consumable Materials

Reference	Description	Specification
D00006	Compound - Antiseize, Pure Nickel Special - Never-Seez NSBT-8N/-16N	
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

E. Prepare the Airplane for Dispatch with One Thrust Reverser Inoperative - Method 1

SUBTASK 78-00-00-040-003-G00

(1) If you are able to deploy and stow the thrust reverser with hydraulic power, do the steps that follow to prepare the airplane for dispatch with one thrust reverser inoperative:

(a) Do this task: Test No. 38 - Thrust Reverser Cycle Special Functions, TASK 71-00-00-700-816-G00.

NOTE: This task will permit ground operation of the thrust reverser.

WARNING MAKE SURE THAT ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE AREA AFT OF THE APPLICABLE THRUST REVERSER. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(b) Move the applicable reverse thrust lever up and aft to fully deploy the thrust reverser.

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(c) For the left engine, open this circuit breaker and install a safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

(d) For the right engine, open this circuit breaker and install a safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK

(e) Move the applicable reverse thrust lever down and forward to the stow position.

1) Make sure that the thrust reverser sleeves do not move to the stowed position.

NOTE: This is the indication that the sync locks are in the locked position.

CAUTION

MAKE SURE THAT YOU PUT THE THRUST REVERSER LEVER TO THE DEPLOY POSITION BEFORE YOU CLOSE THE CIRCUIT BREAKERS. IF YOU DO NOT PUT THE THRUST REVERSER LEVER TO THE DEPLOY POSITION, YOU CAN CAUSE DAMAGE THE SYNC LOCK UNIT.

(f) Move the applicable reverse thrust lever up and aft to the deploy position.

(g) For the left engine, remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

(h) For the right engine, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK

(i) Move the applicable reverse thrust lever down and forward to move the thrust reverser to the fully stowed position.

(j) After you stow the thrust reverser, select the STOP FUNCTION to complete the special function (TASK 46-13-02-860-802).

NOTE: The EEC MAINT POWER function on the MISC SYSTEM CTRLS page on the Multi Function Display (MFD) will remain in TEST. Electronic Engine Control (EEC) power is necessary for the steps that follow.

(k) Erase the thrust reverser related MAX Display System (MDS) status messages that are currently displayed.

NOTE: It is acceptable for the T/R related faults to appear again.

(l) Make sure that the thrust reverser sleeve indications on the EPCS maintenance data page are correct as follows:

1) Get access to the EPCS maintenance data page (TASK 71-00-00-970-801-G00).

2) On EPCS page 1, make sure that the T/R SLEEVE L and T/R SLEEVE R indications on the applicable engine show 0.0 ± 4.0.

(m) Lockwire the reverse thrust lever for the inoperative thrust reverser to the applicable forward thrust lever with MS20995C32 lockwire, G01048.

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(n) Install a REVERSER INOP tag on the reverse thrust lever for the inoperative thrust reverser.

(o) Install an INOP placard on the applicable REVERSER LIMITED indication light on the P5 aft overhead panel.

(p) For the left engine, open these circuit breakers and install safety locks:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	5	C00276	ENGINE 1 THRUST REVERSER CONT
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

(q) For the right engine, open these circuit breakers and install safety locks:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK
C	7	C00277	ENGINE 2 THRUST REVERSER CONT

(r) Install the 315A6258 deactivation pins [1] in the left and right thrust reverser sleeves on the applicable engine:

NOTE: The two deactivation pins are painted red and are part of the 012A8102 fly-away kit. The fly-away kit bag is marked 737MAX GROUND SUPPORT EQUIPMENT. You must install one deactivation pin in each of the thrust reverser sleeves.

1) Get the two deactivation pins [1] that are in the fly-away kit.

2) Put Pure Nickel Special compound, D00006, on the threads of the deactivation pins [1].

3) Install one deactivation pin [1] in the right thrust reverser sleeve and one deactivation pin [1] in the left thrust reverser sleeve.

a) Make sure that the deactivation pins [1] go through the deactivation pin holes and into the deactivation fitting on the thrust reverser torque box.

b) Tighten the deactivation pins [1] to 160 ±5 in-lb (18 ±1 N·m).

NOTE: When installed correctly, the gap between the head of the deactivation pin and outer cowl is approximately 0.4 in. (10.2 mm).

(s) Do a check of the MAINT light, on the P5 aft overhead panel, and ENGINE CONTROL light, on the engine control panel.

1) If the MAINT light or the ENGINE CONTROL light is on, do a check on the Onboard Maintenance Function (OMF) for recent EEC faults.

2) Go to the IFIM and do the applicable procedure(s).

SUBTASK 78-00-00-860-006-G00

(2) Set the applicable EEC MAINT POWER function on the MISC SYSTEM CTRLS page, on the MFD, to NORM (TASK 73-21-00-800-801-G00).

F. Prepare the Airplane for Dispatch with One Thrust Reverser Inoperative - Method 2

SUBTASK 78-00-00-040-001-G00

(1) Prepare the airplane for dispatch with one thrust reverser inoperative as follows:

NOTE: You will do the steps that follow if you are not able to operate the thrust reverser.

(a) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

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(b) Make sure that the sync locks on the left and right thrust reverser halves are in the locked position (TASK 78-34-10-700-801-G00).

(c) Do the steps that follow to make sure that the thrust reverser sleeve indications on the EPCS maintenance data page are correct:

1) Get access to the EPCS maintenance data page.

2) On EPCS page 1, make sure that the T/R SLEEVE L and T/R SLEEVE R indications on the applicable engine show 0.0 ± 4.0.

(d) Lockwire the reverse thrust lever for the inoperative thrust reverser to the applicable forward thrust lever with MS20995C32 lockwire, G01048.

(e) Install a REVERSER INOP tag on the reverse thrust lever for the inoperative thrust reverser.

(f) Install an INOP placard on the applicable REVERSER LIMITED indication light on the P5 aft overhead panel.

(g) For the left engine, open these circuit breakers and install safety locks:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	5	C00276	ENGINE 1 THRUST REVERSER CONT
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

(h) For the right engine, open these circuit breakers and install safety locks:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK
C	7	C00277	ENGINE 2 THRUST REVERSER CONT

(i) Do these steps to install the deactivation pins [1] on the thrust reverser sleeves:

1) Install the deactivation pin [1] in the left and right thrust reverser sleeves on the applicable engine as follows:

NOTE: The two deactivation pins are painted red and are part of the 012A8102 fly-away kit. The fly-away kit bag is marked 737MAX GROUND SUPPORT EQUIPMENT. You must install one deactivation pin in each of the thrust reverser sleeves.

a) Get the two deactivation pins [1] that are in the fly-away kit.

b) Put Pure Nickel Special compound, D00006, on the threads of the two deactivation pins [1].

c) Install one deactivation pin [1] in the right thrust reverser sleeve and one deactivation pin [1] in the left thrust reverser sleeve.

<1> Make sure that the deactivation pins [1] go through the deactivation pin holes and into the deactivation fitting on the thrust reverser torque box.

<2> Tighten the deactivation pins [1] to 160 ±5 in-lb (18 ±1 N·m).

NOTE: When installed correctly, the gap between the head of the deactivation pin and outer cowl is approximately 0.4 in. (10.2 mm).

(j) Pressurize the applicable hydraulic system (TASK 29-11-00-860-801).

(k) Make sure that the actuator lock handles on either the upper or middle locking actuator of both thrust reverser halves is in the locked position.

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(l) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

(m) Do a check of the MAINT light, on the P5 aft overhead panel, and ENGINE CONTROL light, on the engine control panel.

1) If the MAINT light or the ENGINE CONTROL light is on, do a check on the OMF for recent EEC faults.

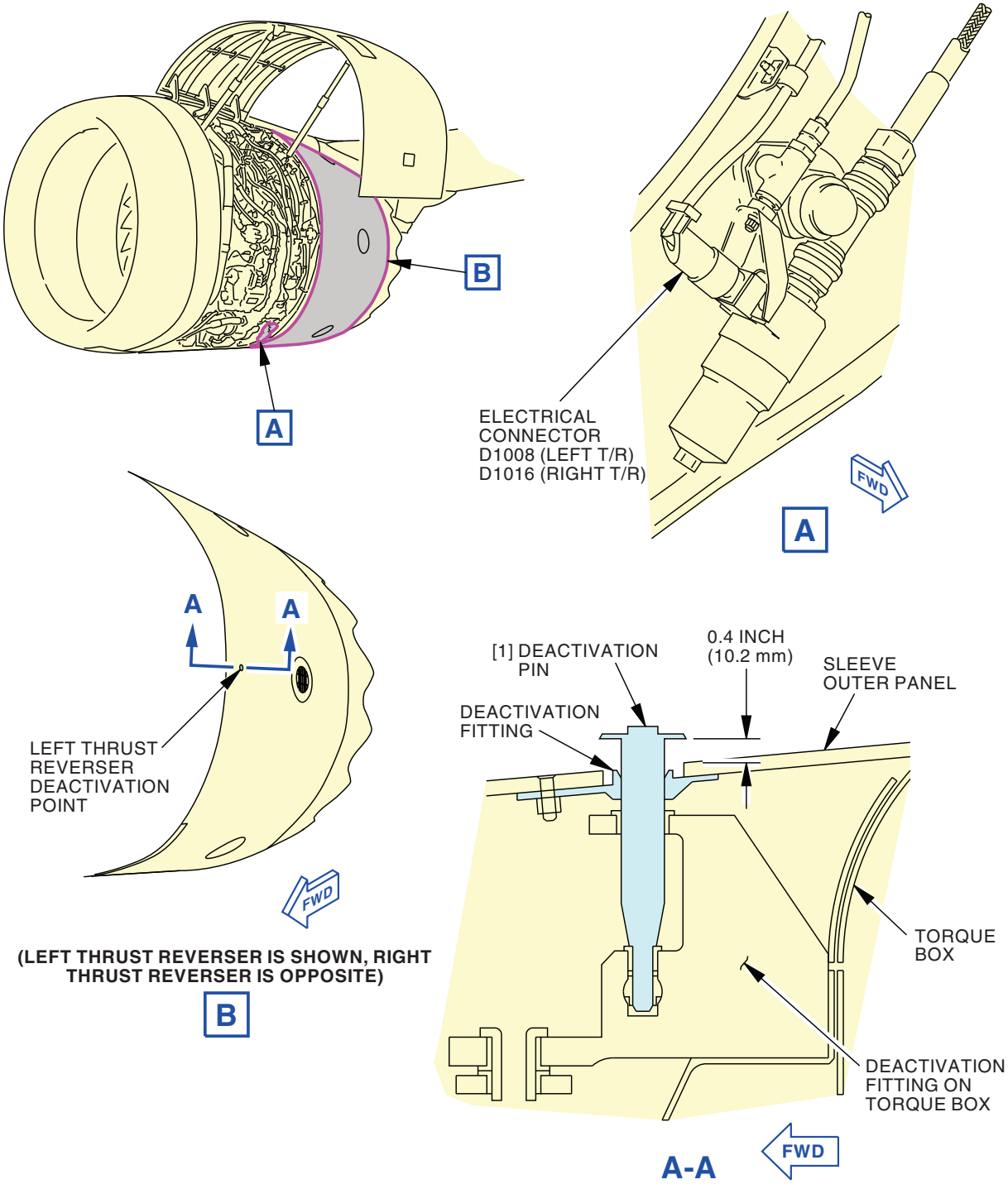
2) Go to the IFIM and do the applicable procedure(s).

SUBTASK 78-00-00-810-002-G00

(2) If it is necessary, extinguish the MAINT light on the P5-1 panel, do this task: Fault Interrogation for a Maintenance Message, TASK 31-65-00-750-801.

——— END OF TASK ———

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Thrust Reverser Sleeve Deactivation
Figure 901/78-00-00-990-801-G00

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TASK 78-00-00-440-801-G00

3. DDG 78-31-01 Thrust Reversers Inoperative - Restoration

(Figure 901)

A. General

(1) This task puts the airplane back to its usual condition after operation with an inoperative thrust reverser.

B. References

Reference	Title
31-65-00-750-801	Fault Interrogation for a Maintenance Message (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

D. Procedure

SUBTASK 78-00-00-440-001-G00

(1) Remove the 315A6258 deactivation pins [1] from the left and right thrust reverser sleeves on the applicable engine:

NOTE: You must remove the deactivation pins from each of the thrust reverser sleeves.

(a) Put the deactivation pins [1] in the 012A8102 fly-away kit and stow on the airplane.

NOTE: The fly-away kit bag is marked 737MAX GROUND SUPPORT EQUIPMENT.

SUBTASK 78-00-00-860-003-G00

(2) For the left engine, remove the safety locks and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	5	C00276	ENGINE 1 THRUST REVERSER CONT
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

For the right engine, remove the safety locks and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK
C	7	C00277	ENGINE 2 THRUST REVERSER CONT

SUBTASK 78-00-00-860-004-G00

(3) Remove the lockwire and REVERSER INOP tag from the applicable reverse thrust lever.

SUBTASK 78-00-00-860-005-G00

(4) Remove the INOP placard from the REVERSER LIMITED light, on the P5 aft overhead panel.

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E. MAINT Light Status Message

SUBTASK 78-00-00-750-001-G00

(1) Uncheck the applicable MAINT light Status Message(s) on the MAINT LIGHT menu (TASK 31-65-00-750-801).

——— END OF TASK ———

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TURBINE EXHAUST SYSTEM - INSPECTION/CHECK

1. General

A. This procedure has these tasks:

(1) A visual check of the labyrinth seals on the primary nozzle that are on the aft cowl panels on the left and right thrust reversers.

(2) A visual check of the primary nozzle and plug for damage.

TASK 78-11-00-210-801-G00

2. Labyrinth Seals Inspection

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions for a visual check of the labyrinth seals that are on the primary nozzle.

B. References

Reference	Title
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
SRM 54-40-02	PRIMARY EXHAUST SYSTEM

C. Prepare for the Inspection

SUBTASK 78-11-00-010-001-G00

(1) Do this task: Open the Thrust Reverser (Selection), TASK 78-31-00-010-801-G00.

D. Labyrinth Seals Inspection

SUBTASK 78-11-00-210-002-G00

(1) Do a visual check for missing labyrinth seals:

(a) If you find missing labyrinth seals, replace the labyrinth seals.

SUBTASK 78-11-00-900-001-G00

(2) Do a visual check for missing bolts, nuts, and washers that attach the labyrinth seal to the primary nozzle assembly.

(a) Missing bolts, nuts, and washers are not permitted.

(b) Replace the missing components.

SUBTASK 78-11-00-210-001-G00

(3) Do a visual check of the primary nozzle labyrinth seals for damage:

(a) Cracks, nicks, gouges, scratches, dents, holes or punctures.

1) If you find damage, refer to the SRM 54-40-02 for the permitted limits.

2) If you find damage that is more than the limits, replace the labyrinth seals.

NOTE: Report to Boeing via Service Request when the seals are replaced.

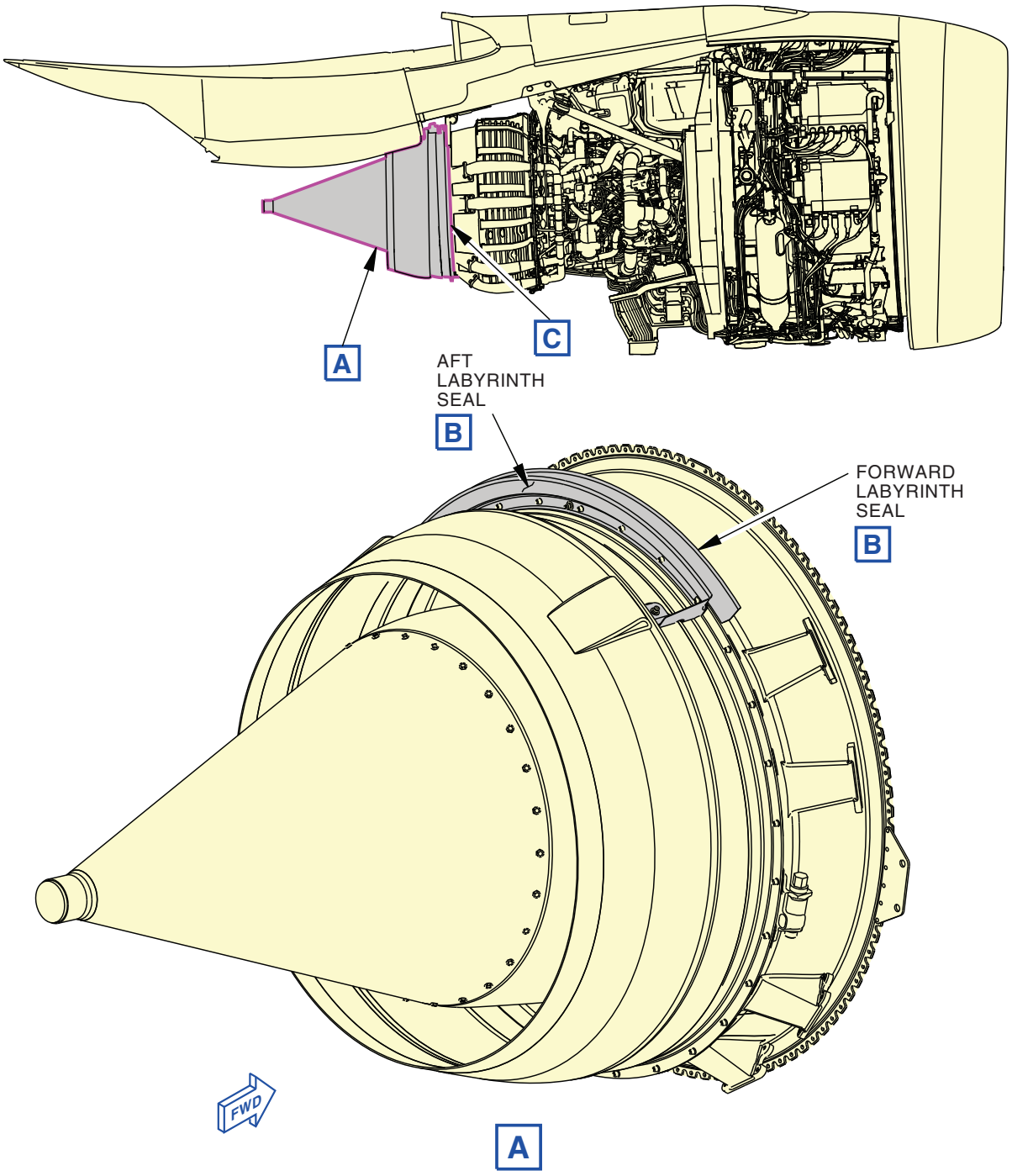
E. Put the Airplane Back to Its Usual Condition

SUBTASK 78-11-00-410-001-G00

(1) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

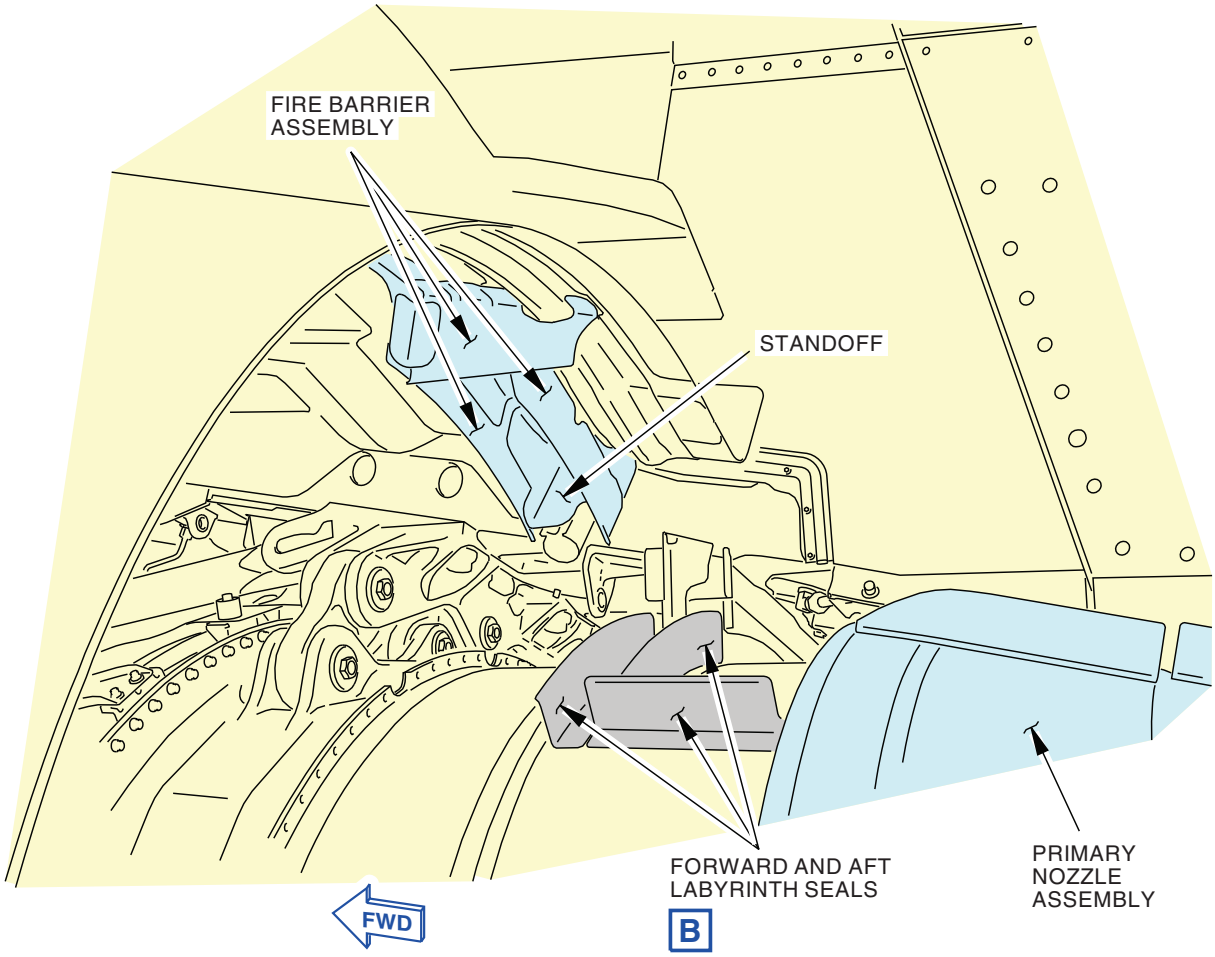
END OF TASK

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Labyrinth Seals Inspection
Figure 601/78-11-00-990-804-G00 (Sheet 1 of 2)

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Labyrinth Seals Inspection
Figure 601/78-11-00-990-804-G00 (Sheet 2 of 2)

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TASK 78-11-00-210-802-G00

3. Primary Nozzle Assembly and Primary Plug Assembly Inspection

(Figure 602)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions on how to examine the primary nozzle and plug assemblies.

B. References

Reference	Title
78-11-01-000-801-G00	Primary Nozzle Assembly Removal (P/B 401)
78-11-01-400-801-G00	Primary Nozzle Assembly Installation (P/B 401)
78-11-02-000-801-G00	Primary Plug Assembly Removal (P/B 401)
78-11-02-400-801-G00	Primary Plug Assembly Installation (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Nozzle side rubstrip	78-11-01-02-110	PTF ALL

E. Prepare for the Inspection

SUBTASK 78-11-00-860-004-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 78-11-00-010-005-G00

(2) Do this task: Open the Thrust Reverser (Selection), TASK 78-31-00-010-801-G00.

F. Primary Nozzle Assembly and Primary Plug Assembly Inspection

SUBTASK 78-11-00-210-005-G00

(1) Do a visual check of the primary nozzle assembly and primary plug assembly for damage as follows:

NOTE: The gap between the plug assembly and the Center Vent Tube Extension (CVTE) averages 0.0200 in. (0.5080 mm). The CVTE may contact the plug assembly. This is permitted.

(a) Nicks, gouges, scratches, cracks, dents, holes, or punctures.

1) If you find nicks, gouges, scratches, cracks, or dents, refer to THE BOEING COMPANY 737-MAX SRM 54-40-02 for the permitted limits and repair procedures.

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2) For the primary nozzle assembly, if you find holes or punctures, replace the primary nozzle (TASK 78-11-01-000-801-G00 and TASK 78-11-01-400-801-G00).

PTF ALL POST SB 737-78-1101

a) Remove and replace the nozzle side rubstrip [1] when 30% or less of the original material thickness remains.

NOTE: Wearing of the fastener heads is unacceptable.

PTF ALL

3) For the primary plug assembly, if you find holes or punctures, replace the primary plug (TASK 78-11-02-000-801-G00 and TASK 78-11-02-400-801-G00).

SUBTASK 78-11-00-210-006-G00

(2) Do a visual check for loose or missing bolts that hold the primary nozzle assembly:

(a) Loose or missing bolts are not permitted.

(b) If necessary, replace and tighten the bolts (TASK 78-11-01-400-801-G00).

SUBTASK 78-11-00-210-008-G00

(3) Do a visual check for loose or missing bolts that hold the forward plug:

(a) Loose or missing bolts are not permitted.

(b) Remove the aft plug (TASK 78-11-02-000-801-G00).

(c) If necessary, replace and tighten the bolts that hold the forward plug (TASK 78-11-02-400-801-G00).

(d) Install the aft plug (TASK 78-11-02-400-801-G00).

G. Put the Airplane Back to its Usual Condition

SUBTASK 78-11-00-010-006-G00

(1) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

SUBTASK 78-11-00-860-005-G00

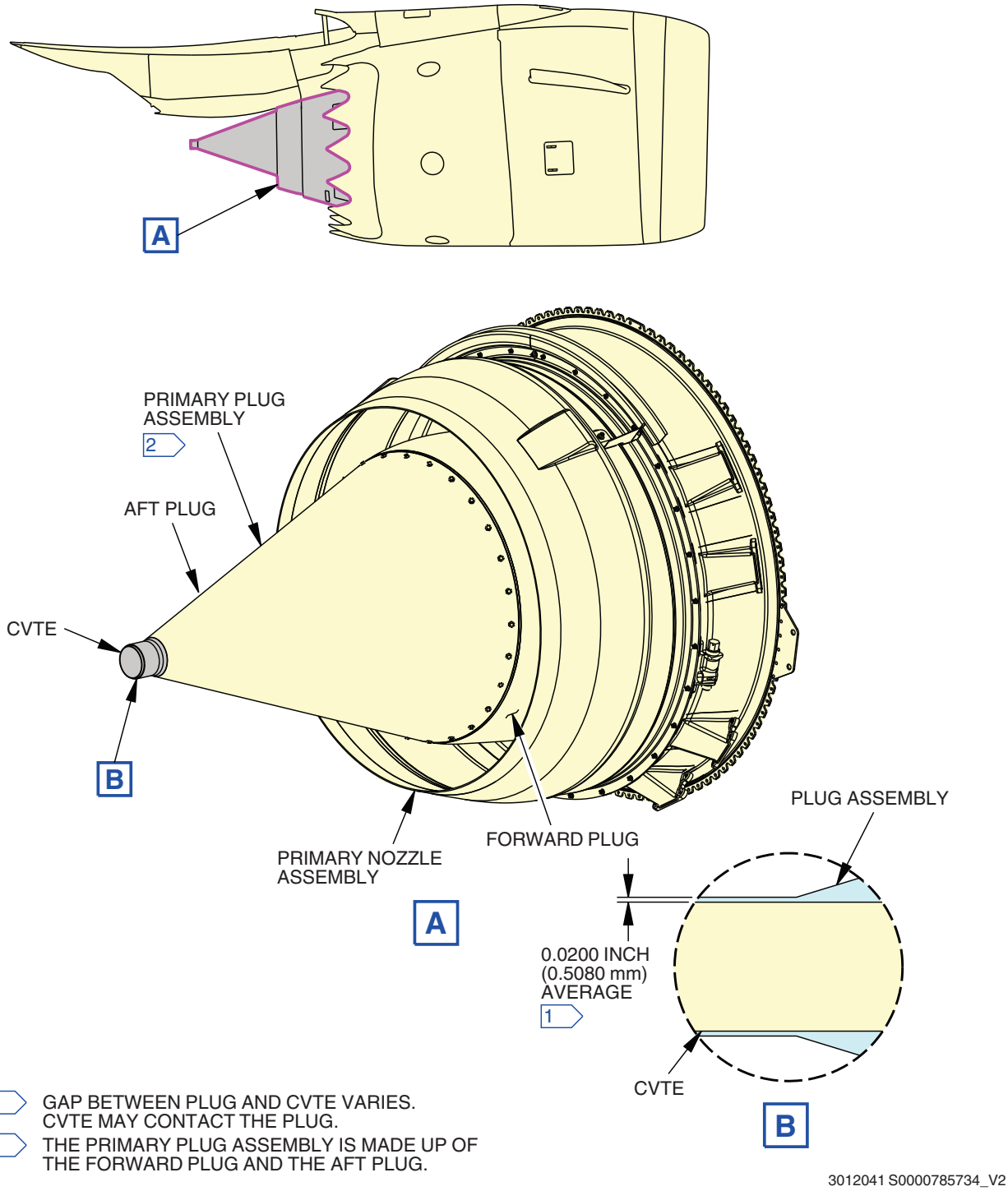
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

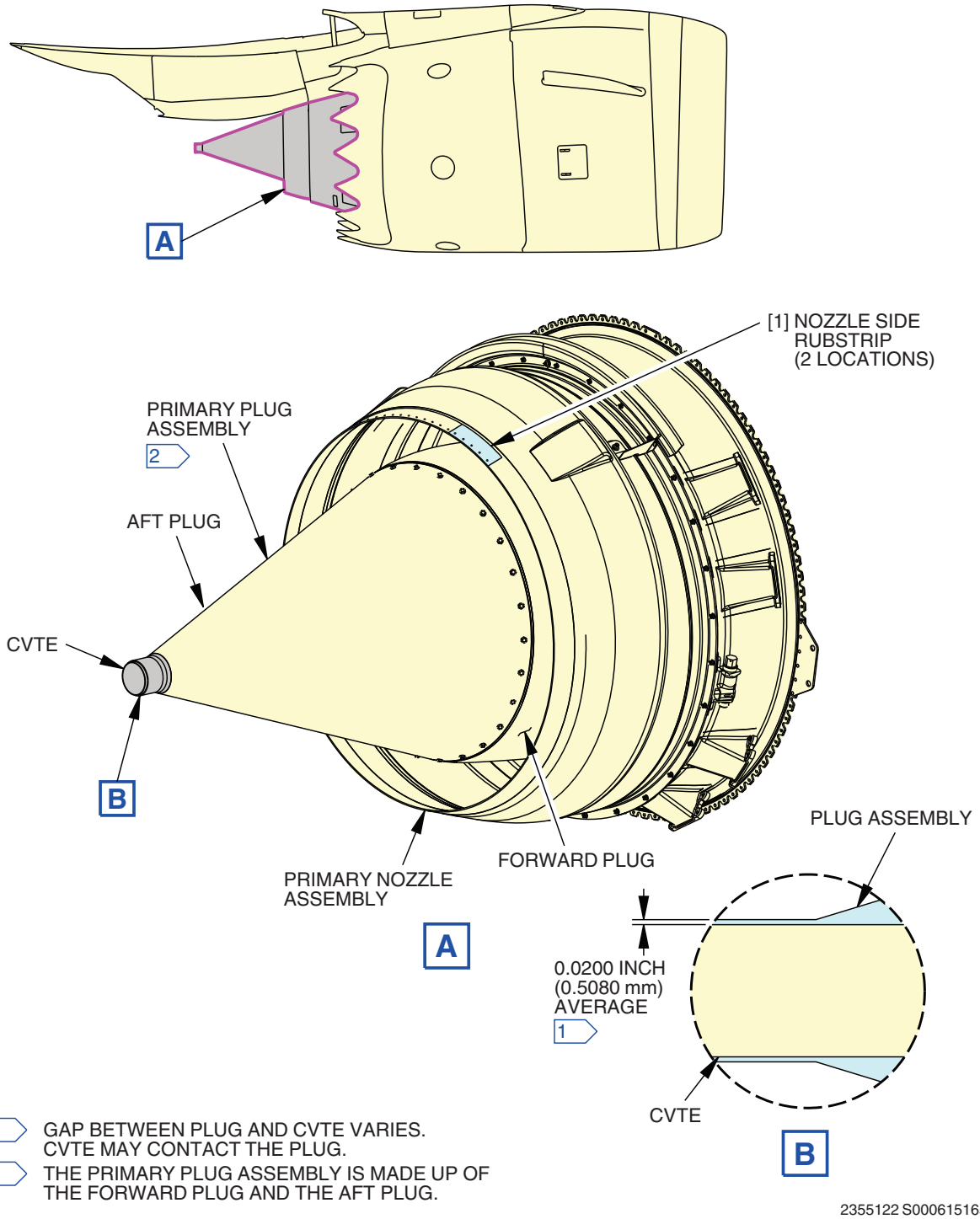
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Primary Nozzle Assembly and Primary Plug Assembly Inspection
Figure 602/78-11-00-990-802-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 GAP BETWEEN PLUG AND CVTE VARIES. CVTE MAY CONTACT THE PLUG.

2 THE PRIMARY PLUG ASSEMBLY IS MADE UP OF THE FORWARD PLUG AND THE AFT PLUG.

2355122 S00061516587_V5

Primary Nozzle Assembly and Primary Plug Assembly Inspection
Figure 602/78-11-00-990-802-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PRIMARY NOZZLE ASSEMBLY - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the primary nozzle assembly

(2) An installation of the primary nozzle assembly.

TASK 78-11-01-000-801-G00

2. Primary Nozzle Assembly Removal

(Figure 401 and Figure 402)

A. General

(1) This task gives the instructions on how to remove the primary nozzle assembly.

(2) The primary nozzle is on the aft end of the power plant and attached to the turbine rear frame outer flange.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1568	Jack - Hydraulic, General Low Profile Part #: HW93718 Supplier: 28047 Opt Part #: B67563 Supplier: 36251 Opt Part #: W93718 Supplier: 36251
SPL-2419	Equipment - Handling, Primary Exhaust Sleeve and Plug Part #: C78009-88 Supplier: 81205
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
417	Engine 1 - Primary Exhaust Nozzle and Plug
427	Engine 2 - Primary Exhaust Nozzle and Plug

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 78-11-01-040-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 78-11-01-010-001-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Primary Nozzle Assembly Removal

SUBTASK 78-11-01-020-001-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING GET SUFFICIENT AID FROM OTHER PERSONNEL AND EQUIPMENT TO HOLD THE COMPONENT DURING THE REMOVAL, AND INSTALLATION. THE COMPONENT IS HEAVY. THIS WILL PREVENT INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(1) Remove the primary nozzle [12], forward labyrinth seal [17], and duct vent valve [18] as follows:

(a) Attach the jack adapter from the equipment, SPL-2419, to the low profile hydraulic jack, COM-1568.

(b) Attach the jack adapter to the primary nozzle [12] with the straps.

CAUTION MAKE SURE TO ATTACH THE STRAPS ALONG THE CONTINUOUS SURFACE OF THE PRIMARY NOZZLE ONLY. DO NOT ATTACH THE STRAPS ACROSS THE COMPONENTS ON TOP OF THE PRIMARY NOZZLE. IF YOU DO NOT OBEY, DAMAGE TO THE PRIMARY NOZZLE CAN OCCUR.

(c) Make sure that the jack supports the weight of the primary primary nozzle [12].

(d) Disconnect the pneumatic tubes from the duct vent valve [18].

(e) Remove the two bolts [13], two countersunk washers [14], two flat washers [15], two nuts [16], and duct vent valve [18].

(f) Remove the two bolts [1], two countersunk washers [2], two flat washers [3], and two nuts [4] from two holes, one at each end of the forward labyrinth seal [17].

(g) Remove the six bolts [5], six countersunk washers [6], and six nuts [7] from the top 6 holes of the primary nozzle [12].

CAUTION BE CAREFUL WHEN YOU REMOVE/TOUCH THE FORWARD LABYRINTH SEAL. IF YOU DO NOT OBEY, DAMAGE TO THE FORWARD LABYRINTH SEAL CAN OCCUR.

(h) Remove the forward labyrinth seal [17].

(i) Remove the remaining 20 bolts [8], 20 countersunk washers [9], 20 flat washers [10], and 20 nuts [11] from the lower 20 holes of the primary nozzle [12].

(j) Move the primary nozzle [12] slightly aft to disengage the alignment pin from the turbine rear frame.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 78-11-01-020-002-G00

CAUTION

LOWER THE PRIMARY NOZZLE UNTIL THE AFT LABYRINTH SEALS ARE DISENGAGED. IF YOU DO NOT LOWER THE NOZZLE SUFFICIENTLY, DAMAGE TO THE AFT LABYRINTH SEALS CAN OCCUR. IF YOU LOWER THE PRIMARY NOZZLE TOO MUCH, DAMAGE TO THE PRIMARY PLUG ASSEMBLY CAN OCCUR.

(2) Lower the primary nozzle [12] until the two labyrinth seals are disengaged.

SUBTASK 78-11-01-020-003-G00

(3) Move the primary nozzle [12] aft until it is clear of the primary plug assembly.

SUBTASK 78-11-01-020-004-G00

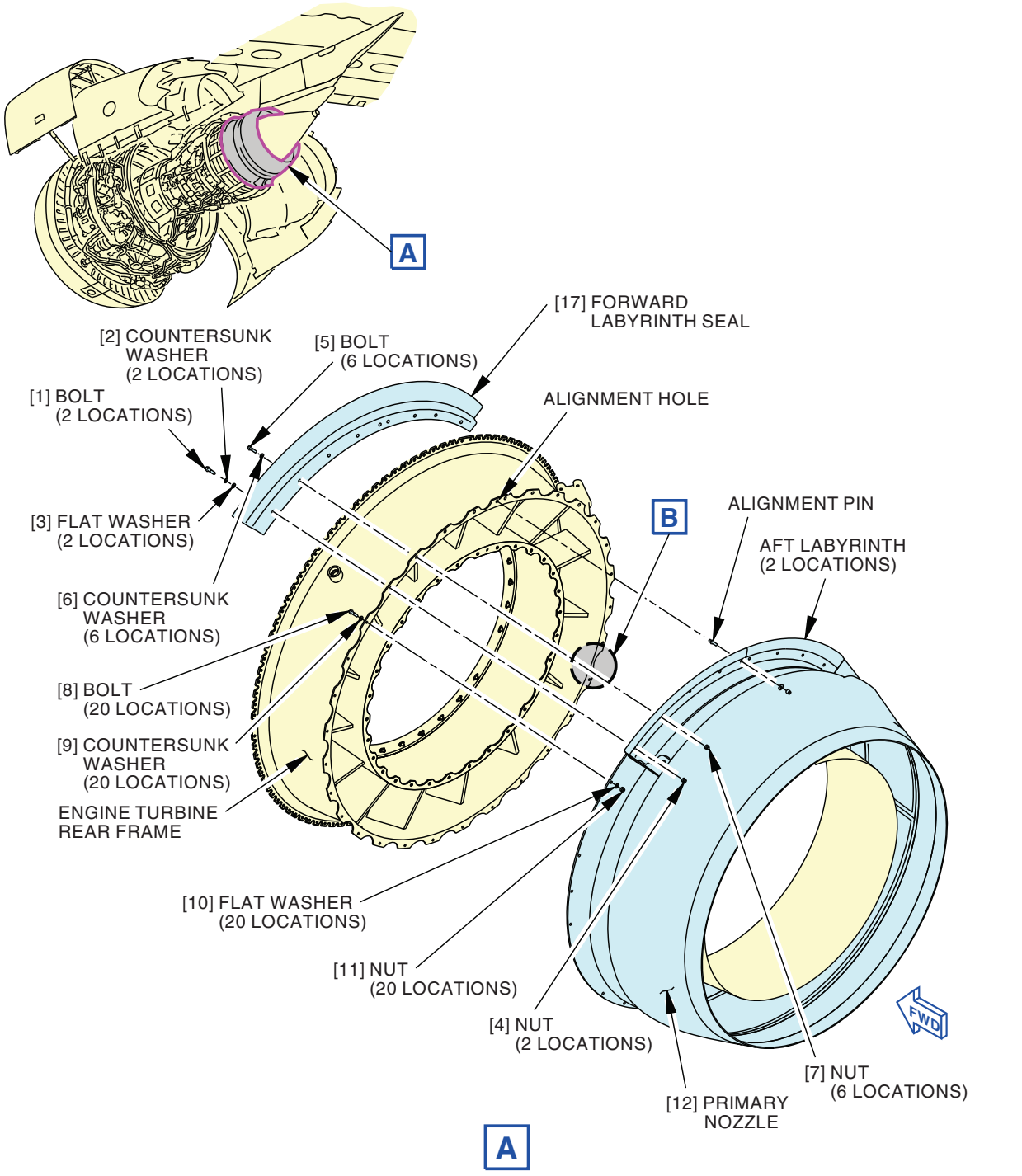
WARNING

GET SUFFICIENT AID FROM OTHER PERSONNEL AND EQUIPMENT TO HOLD THE COMPONENT DURING THE REMOVAL, AND INSTALLATION. THE COMPONENT IS HEAVY. THIS WILL PREVENT INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(4) Remove the primary nozzle [12] from the jack adapter.

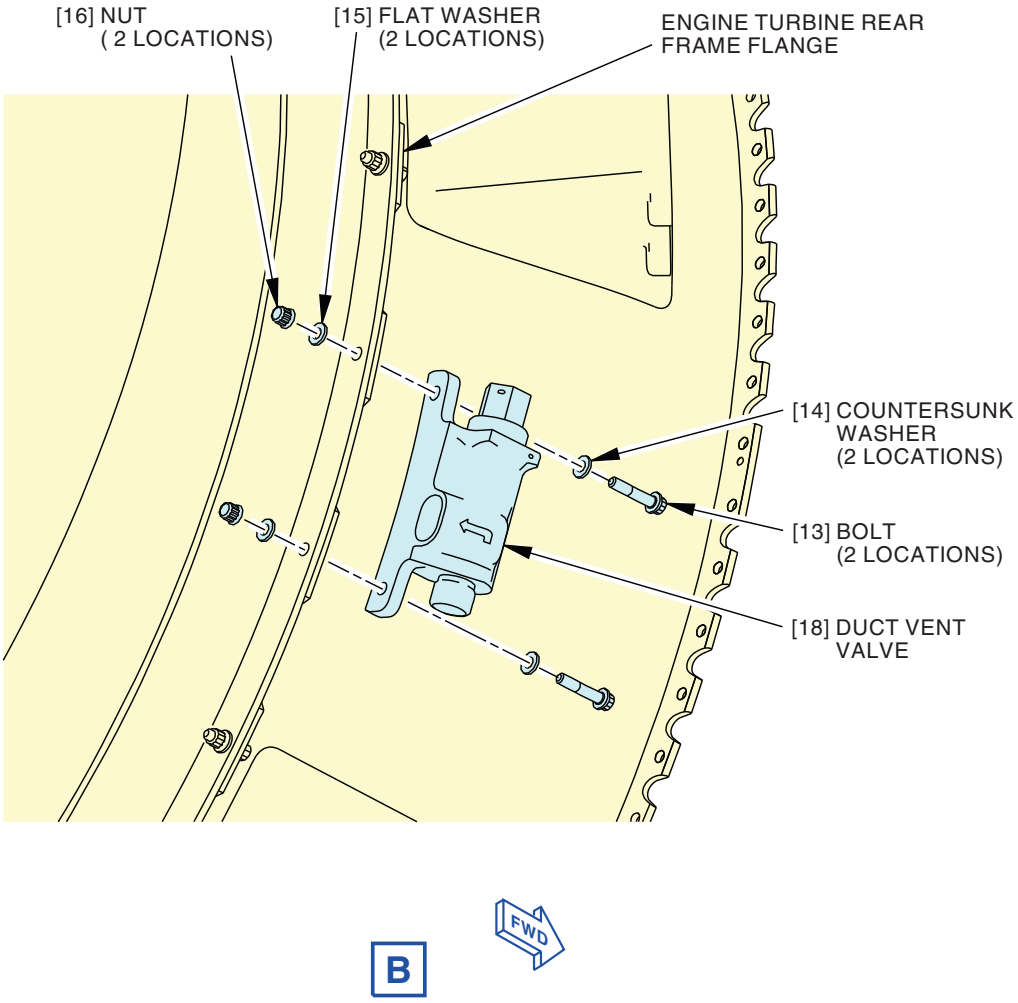
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



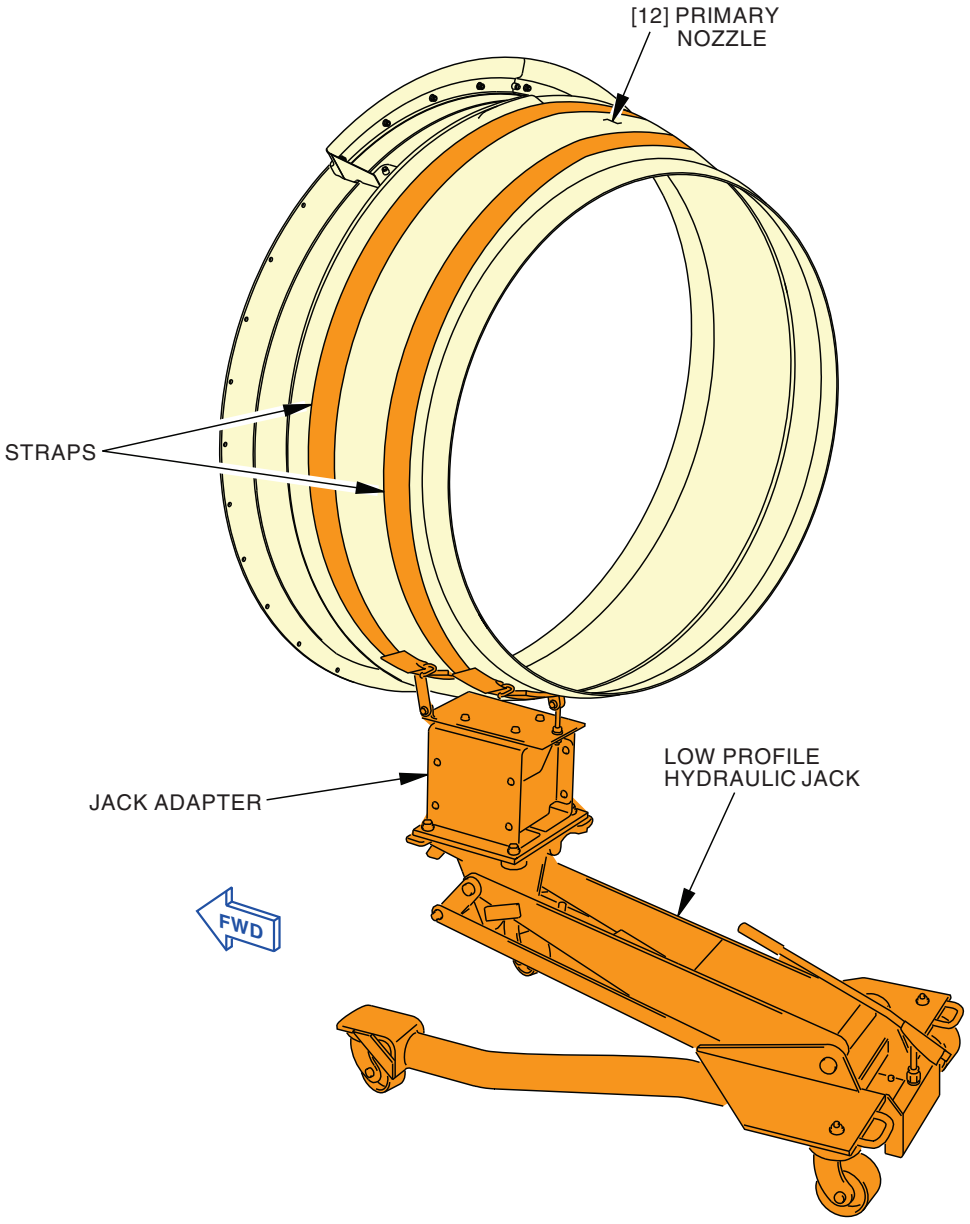
Primary Nozzle Assembly Installation
Figure 401/78-11-01-990-804-G02 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Primary Nozzle Assembly Installation
Figure 401/78-11-01-990-804-G02 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Primary Nozzle Assembly Handling Equipment
Figure 402/78-11-01-990-805-G02

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-11-01-400-801-G00

3. Primary Nozzle Assembly Installation

(Figure 401 and Figure 402)

A. General

(1) This gives the instructions on how to install the primary nozzle assembly.

NOTE: The total acoustic panel blockage for the engine installation must be recalculated when components are changed per SRM SUBJECT 54-00-01.

B. References

Reference	Title
54-52-07-200-801	Leaf Spring Seal - Inspection/Check (P/B 601)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1568	Jack - Hydraulic, General Low Profile Part #: HW93718 Supplier: 28047 Opt Part #: B67563 Supplier: 36251 Opt Part #: W93718 Supplier: 36251
SPL-2419	Equipment - Handling, Primary Exhaust Sleeve and Plug Part #: C78009-88 Supplier: 81205
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00006	Compound - Antiseize, Pure Nickel Special - Never-Seez NSBT-8N/-16N	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
12	Primary nozzle	78-11-01-02-055	PTF ALL

F. Location Zones

Zone	Area
417	Engine 1 - Primary Exhaust Nozzle and Plug
427	Engine 2 - Primary Exhaust Nozzle and Plug

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Primary Nozzle Assembly Installation

SUBTASK 78-11-01-200-001-G02

(1) Make sure that the leaf spring seals are serviceable, do this task: Leaf Spring Seal - Inspection/Check, TASK 54-52-07-200-801.

SUBTASK 78-11-01-480-002-G02

(2) Attach the jack adapter from the equipment, SPL-2419, to the low profile hydraulic jack, COM-1568.

SUBTASK 78-11-01-480-003-G02

WARNING GET SUFFICIENT AID FROM OTHER PERSONNEL AND EQUIPMENT TO HOLD THE COMPONENT DURING THE REMOVAL, AND INSTALLATION. THE COMPONENT IS HEAVY. THIS WILL PREVENT INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(3) Put the primary nozzle [12] on the jack adapter as follows:

(a) Make sure to orient the primary nozzle [12] with the alignment pin near the 12 o'clock position so that it can align with the alignment hole on the engine turbine rear frame flange.

NOTE: The alignment hole is the larger of the two holes in the widest flange near the 12:00 o'clock position.

CAUTION MAKE SURE TO ATTACH THE STRAPS ALONG THE CONTINUOUS SURFACE OF THE PRIMARY NOZZLE ONLY. DO NOT ATTACH THE STRAPS ACROSS THE COMPONENTS ON TOP OF THE PRIMARY NOZZLE. IF YOU DO NOT OBEY, DAMAGE TO THE PRIMARY NOZZLE CAN OCCUR.

(b) Attach the primary nozzle [12] to the jack adapter with the straps.

SUBTASK 78-11-01-980-001-G02

CAUTION MAKE SURE THAT YOU LIFT THE PRIMARY NOZZLE UNTIL THE PRIMARY NOZZLE CLEARS THE PRIMARY PLUG ASSEMBLY. IF YOU LIFT THE PRIMARY NOZZLE TOO MUCH, DAMAGE TO THE AFT LABYRINTH SEALS CAN OCCUR. IF YOU DO NOT LIFT THE PRIMARY NOZZLE SUFFICIENTLY, DAMAGE TO THE PRIMARY PLUG ASSEMBLY CAN OCCUR.

(4) Lift and move forward the primary nozzle [12].

SUBTASK 78-11-01-820-001-G00

(5) Align the primary nozzle [12] to the engine as follows:

(a) Make sure that the primary nozzle [12] is oriented with the alignment pin at the 12 o'clock position.

(b) Make sure that the alignment pin on the primary nozzle [12] engages with the alignment hole on the engine turbine rear frame flange.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 78-11-01-420-002-G02

(6) Attach the primary nozzle [12], forward labyrinth seal [17], and duct vent valve [18] to the turbine rear frame flange as follows:

(a) Make sure that the countersunk washers under the bolt heads have countersinks oriented towards the head of the bolts.

(b) Loosely install the two bolts [1], two countersunk washers [2], two flat washers [3], and two nuts [4] into two holes, one at each end of the forward labyrinth seal.

NOTE: Countersunk washers and flat washers are located under bolt.

1) Locate the flat washers [3] on flat surface.

2) Make sure that the edges of the flat washers [3] do not interfere with labyrinth bend radius.

NOTE: Fasteners are required to protrude a minimum of 1.5 threads beyond the top of the nut.

NOTE: It is acceptable to apply Pure Nickel Special compound, D00006, to threads.

(c) Loosely install the six bolts [5], six countersunk washers [6], and six nuts [7] at the top of the primary nozzle [12].

NOTE: The top 6 hole locations do not have a flat washer under the nut.

NOTE: Fasteners are required to protrude a minimum of 1.5 threads beyond the top of the nut.

NOTE: It is acceptable to apply Pure Nickel Special compound, D00006, to threads.

(d) Loosely install the 20 bolts [8], 20 countersunk washers [9], 20 flat washers [10], and 20 nuts [11].

NOTE: The 20 hole locations use countersunk washers under the bolt and flat washers under the nut.

NOTE: It is acceptable to apply Pure Nickel Special compound, D00006, to threads.

(e) Loosely install the duct vent valve [18], two bolts [13], two countersunk washers [14], two flat washers [15], and two nuts [16].

NOTE: The 2 hole locations use countersunk washers under the bolt and flat washers under the nut.

NOTE: It is acceptable to apply Pure Nickel Special compound, D00006, to threads.

(f) Tighten the nuts [11] at the 3 o'clock, 9 o'clock, and 6 o'clock locations, and nut [7] at the 12 o'clock location in that sequence until they are snug and there is no gap between the engine turbine rear frame flange and primary nozzle [12].

NOTE: A minimum of one fastener must be snug fit at each position. Starting point clocking is optional.

(g) Tighten the remaining 17 nuts [11], two nuts [16], two nuts [4], and five nuts [7] until they are snug.

(h) Sequentially tighten nuts at the 3 o'clock, 9 o'clock, 6 o'clock, and 12 o'clock locations in that sequence to final torque.

(i) Sequentially tighten all remaining nuts to final torque.

(j) Make sure that torque of the nut [11] at the 3 o'clock position required torque range.

(k) If the torque is within the required torque range, no further action is required.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(l) If the torque is not within the required torque range, re-tighten the nuts [4], nuts [7], nuts [11], and nuts [16] as required until all fasteners reach the required torque range.

(m) Connect the applicable pneumatic tubes to the duct vent valve [18].

SUBTASK 78-11-01-080-001-G02

(7) Remove the primary exhaust sleeve and plug equipment, SPL-2419, from the primary nozzle [12] as follows:

(a) Remove the straps that attach the jack adapter to the primary nozzle [12].

(b) Lower and remove the low profile hydraulic jack, COM-1568, from the primary nozzle [12].

(c) Remove the jack adapter from the low profile hydraulic jack, COM-1568.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 78-11-01-410-001-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 78-11-01-440-001-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PRIMARY PLUG ASSEMBLY - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Removal of the primary plug assembly.

(2) Installation of the primary plug assembly.

TASK 78-11-02-000-801-G00

2. Primary Plug Assembly Removal

(Figure 401, Figure 402)

A. General

(1) This task is the removal of the primary plug assembly.

(2) The primary plug assembly has two separate pieces: the forward plug and aft plug.

(3) The primary plug assembly is referred to as the forward plug and aft plug in this task.

(4) The primary plug assembly is installed on the aft end of the power plant, attached to the turbine rear frame inner flange.

(5) To get access to the forward plug, you must remove the exhaust nozzle assembly.

B. References

Reference	Title
78-11-01-000-801-G00	Primary Nozzle Assembly Removal (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1568	Jack - Hydraulic, General Low Profile Part #: HW93718 Supplier: 28047 Opt Part #: B67563 Supplier: 36251 Opt Part #: W93718 Supplier: 36251
SPL-2419	Equipment - Handling, Primary Exhaust Sleeve and Plug Part #: C78009-88 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
D50186 [CP2691]	Fluid - Penetrating - Aerokroil	

E. Location Zones

Zone	Area
417	Engine 1 - Primary Exhaust Nozzle and Plug
427	Engine 2 - Primary Exhaust Nozzle and Plug

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Prepare for the Removal

SUBTASK 78-11-02-860-001-G00

(1) For engine 1, open this circuit breaker and install a safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 78-11-02-860-002-G00

(2) For engine 2, open this circuit breaker and install a safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 78-11-02-010-001-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Open the thrust reverser (TASK 78-31-00-010-801-G00).

SUBTASK 78-11-02-010-002-G00

(4) If it is necessary to get access to the forward plug [1], remove the primary nozzle assembly (TASK 78-11-01-000-801-G00).

G. Primary Plug Assembly Removal

SUBTASK 78-11-02-020-001-G00

(1) Do these steps to remove the aft plug [3]:

(a) Remove the bolts [2].

1) Make sure that the aft plug [3] is fully supported before you remove the last bolt [2].

NOTE: The aft plug weighs approximately 10.0 lb (4.5 kg).

(b) Remove the aft plug [3].

SUBTASK 78-11-02-480-001-G00

(2) Use the exhaust sleeve and plug removal equipment, SPL-2419, to remove the forward plug [1].

(a) Attach the jack adapter to the low profile hydraulic jack, COM-1568 (or equivalent).

(b) Attach the jack adapter to the forward plug [1] with the straps.

PTF ALL; AIRPLANES WITH TURBINE REAR VANE (TRV) SPACER

SUBTASK 78-11-02-020-003-G02

WARNING GET SUFFICIENT AID FROM OTHER PERSONNEL AND EQUIPMENT TO HOLD THE COMPONENT DURING THE REMOVAL, AND INSTALLATION. THE COMPONENT IS HEAVY. THIS WILL PREVENT INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(3) Remove the forward plug [1]:

(a) Apply penetrating fluid, D50186 [CP2691], to the nuts [9], bolts [8], and washers [5].

(b) Remove the nuts [9], bolts [8], and washers [5].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH TURBINE REAR VANE (TRV) SPACER (Continued)

1) Make sure that the forward plug [1] is fully supported before you remove the last bolt [8].

NOTE: The forward plug weighs approximately 61 lb (28 kg).

(c) Move the forward plug [1] aft until the alignment pin at the top of the forward plug [1] is clear of the spacer flange, and lower the forward plug [1].

PTF ALL; AIRPLANES WITHOUT TURBINE REAR VANE (TRV) SPACER

SUBTASK 78-11-02-020-002-G00

WARNING GET SUFFICIENT AID FROM OTHER PERSONNEL AND EQUIPMENT TO HOLD THE COMPONENT DURING THE REMOVAL, AND INSTALLATION. THE COMPONENT IS HEAVY. THIS WILL PREVENT INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(4) Remove the forward plug [1]:

(a) Apply penetrating fluid, D50186 [CP2691], to the nuts [6], bolts [4], washers [5], and washers [7].

(b) Remove the nuts [6], washers [7], bolts [4], and washers [5].

1) Make sure that the forward plug [1] is fully supported before you remove the last bolt [4].

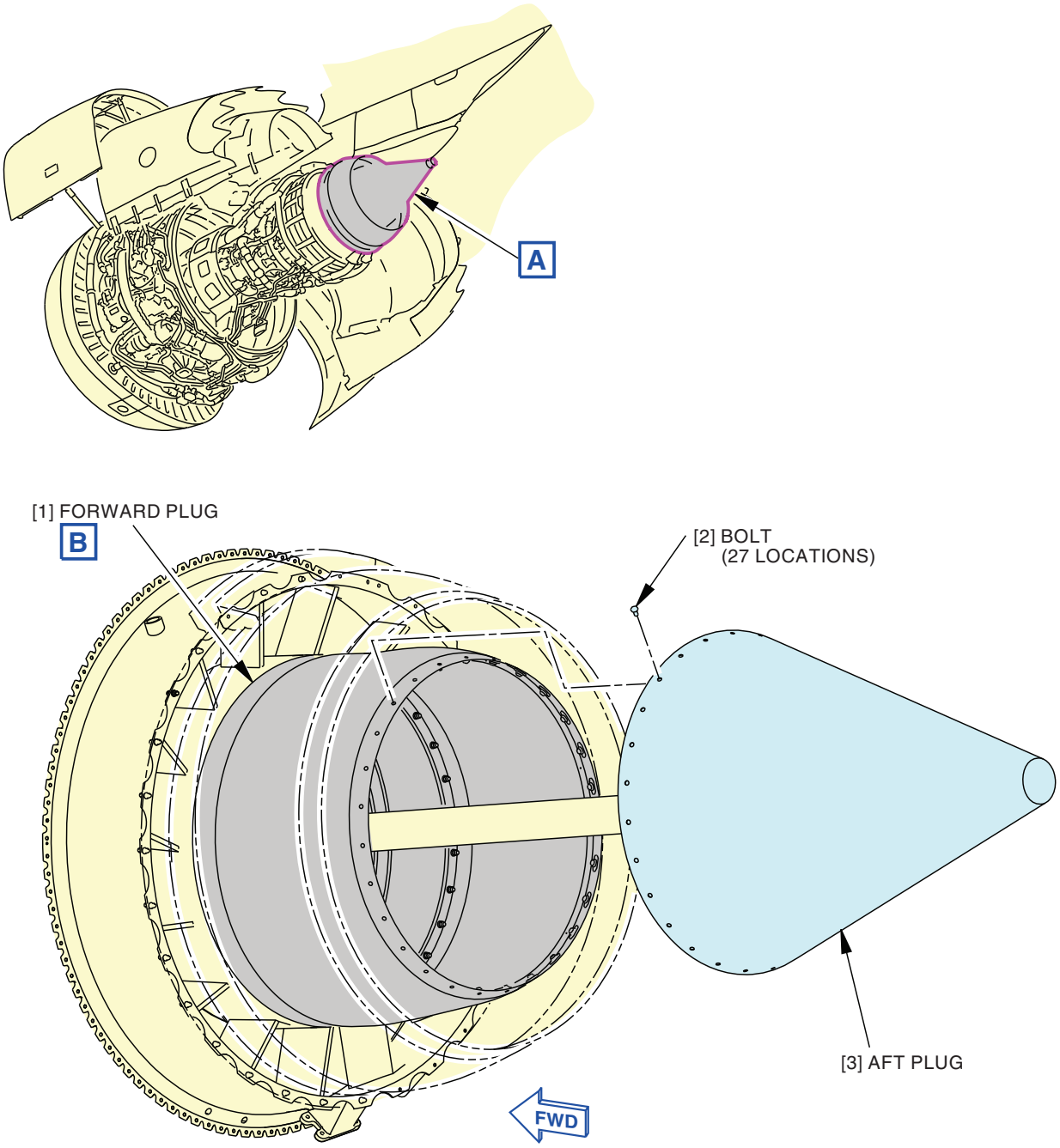
NOTE: The forward plug weighs approximately 61 lb (28 kg).

(c) Move the forward plug [1] aft until the alignment pin at the top of the forward plug [1] is clear of the turbine rear frame flange, and lower the forward plug [1].

PTF ALL

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

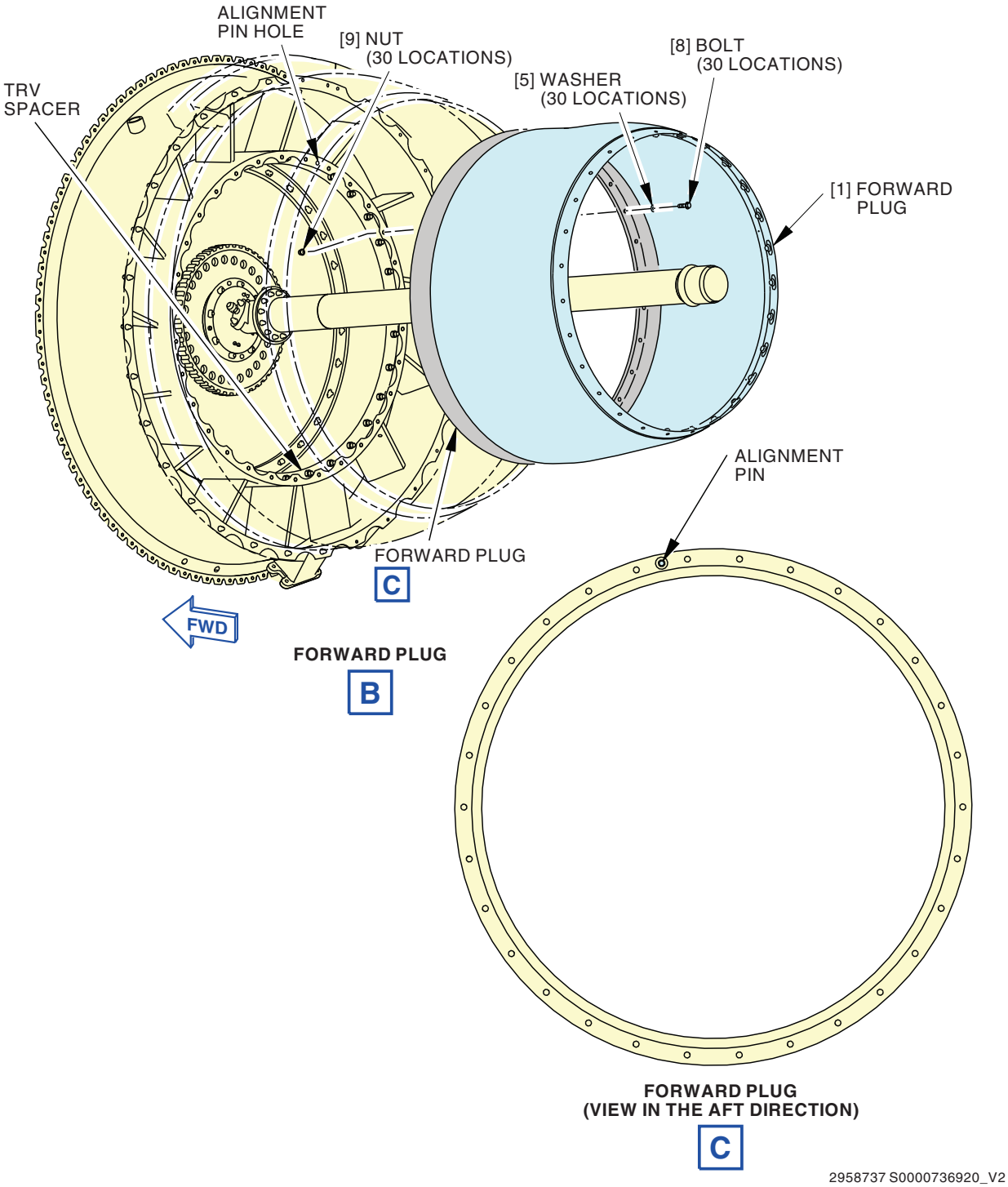


NOTE: SHOWN WITH PRIMARY NOZZLE ASSEMBLY REMOVED.

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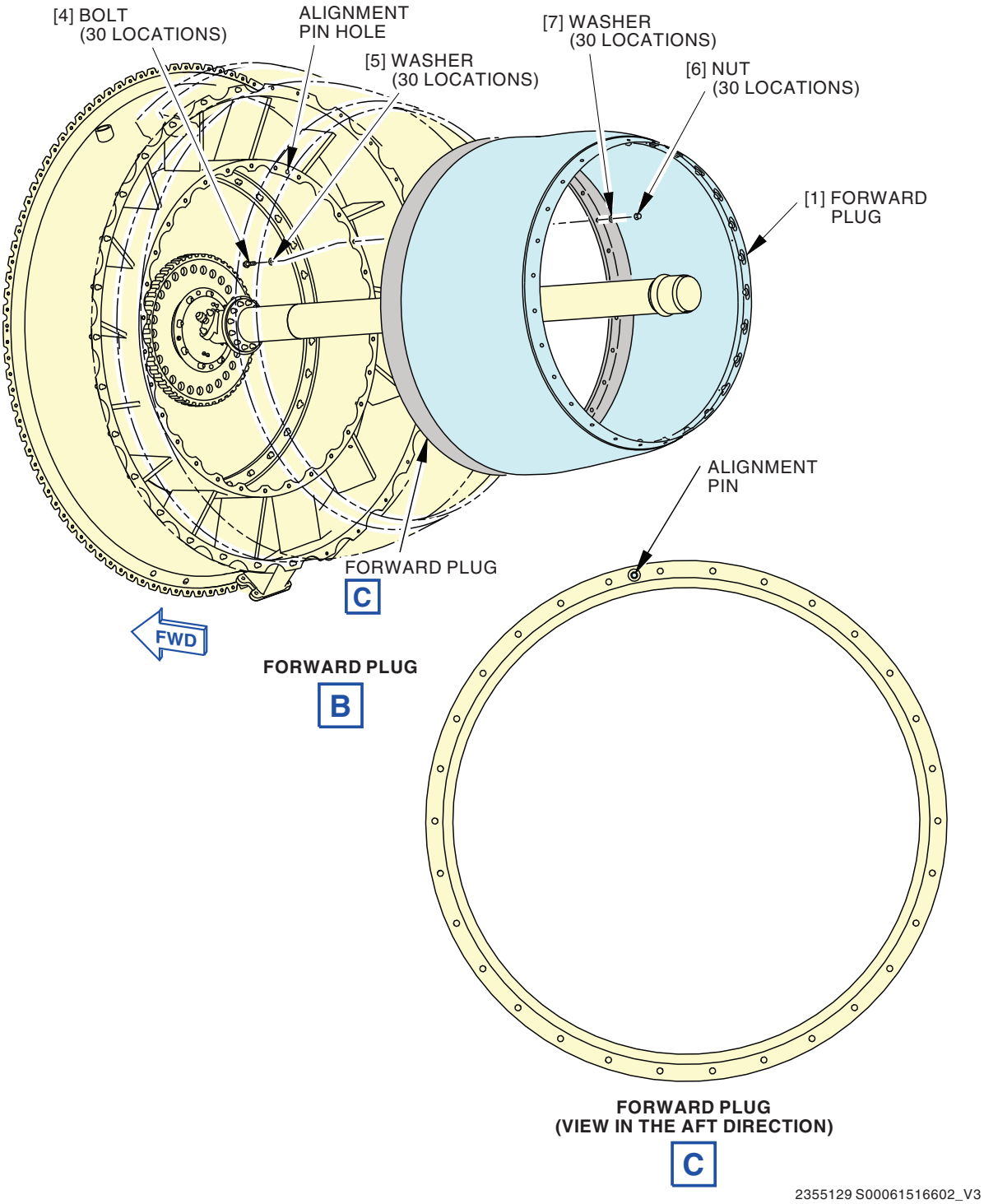
Primary Plug Assembly Installation
Figure 401/78-11-02-990-806-G02 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



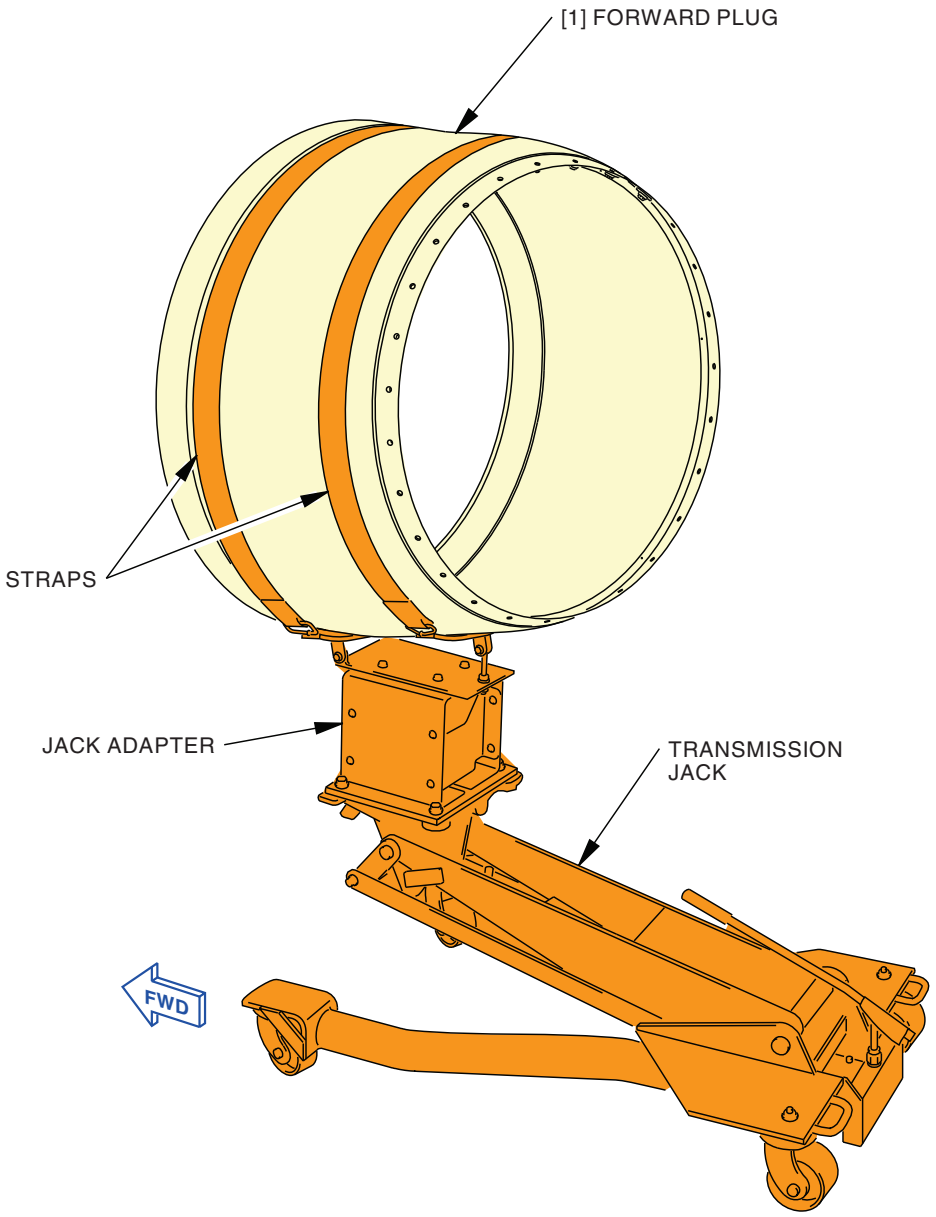
Primary Plug Assembly Installation
Figure 401/78-11-02-990-806-G02 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Primary Plug Assembly Installation
Figure 401/78-11-02-990-806-G02 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2355130 S00061516603_V2

Primary Plug Assembly Handling Equipment
Figure 402/78-11-02-990-807-G02

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-11-02-400-801-G00

3. Primary Plug Assembly Installation

(Figure 401, Figure 402)

A. General

(1) This task is the installation of the primary plug assembly.

NOTE: The total acoustic panel blockage for the engine installation must be recalculated when components are changed per SRM SUBJECT 54-00-01.

(a) The primary plug assembly is attached to the inner flange of the engine turbine rear frame, at the aft end of the power plant.

(b) The primary plug assembly has two separate pieces: the forward plug and aft plug.

B. References

Reference	Title
78-11-01-400-801-G00	Primary Nozzle Assembly Installation (P/B 401)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1568	Jack - Hydraulic, General Low Profile Part #: HW93718 Supplier: 28047 Opt Part #: B67563 Supplier: 36251 Opt Part #: W93718 Supplier: 36251
SPL-2419	Equipment - Handling, Primary Exhaust Sleeve and Plug Part #: C78009-88 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)
D00006	Compound - Antiseize, Pure Nickel Special - Never-Seez NSBT-8N/-16N	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Forward plug	78-11-02-01-043	PTF ALL
3	Aft plug	78-11-02-01-100	PTF ALL

F. Location Zones

Zone	Area
417	Engine 1 - Primary Exhaust Nozzle and Plug
427	Engine 2 - Primary Exhaust Nozzle and Plug

G. Primary Plug Assembly Installation

SUBTASK 78-11-02-210-001-G00

(1) Make sure that the forward plug [1] and aft plug [3] flange surfaces are clean and free of contamination.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) If it is necessary, clean the surfaces with solvent, B00062.

SUBTASK 78-11-02-480-002-G00

WARNING GET SUFFICIENT AID FROM OTHER PERSONNEL AND EQUIPMENT TO HOLD THE COMPONENT DURING THE REMOVAL, AND INSTALLATION. THE COMPONENT IS HEAVY. THIS WILL PREVENT INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(2) Use the primary exhaust sleeve and plug equipment, SPL-2419, to install the forward plug [1].

NOTE: The forward plug weighs approximately 61.0 lb (27.7 kg).

(a) Attach the jack adapter to the low profile hydraulic jack, COM-1568, or equivalent.

(b) Put the forward plug [1] on the jack adapter.

1) Make sure to orient the forward plug [1] so that the alignment pin will align with the alignment hole on the inner flange of the turbine rear frame.

(c) Attach the forward plug [1] to the jack adapter with the straps.

PTF ALL; AIRPLANES WITH TURBINE REAR VANE (TRV) SPACER

SUBTASK 78-11-02-420-001-G00

(3) Install the forward plug [1].

(a) Lift the forward plug [1] and move it into its position on the turbine rear frame.

1) Make sure that the forward plug [1] is securely supported before you lift it.

2) Align the alignment pin with the alignment hole in the inner flange of the turbine rear frame or spacer.

(b) If it is necessary, apply Pure Nickel Special compound, D00006, to the threads of the bolts [8].

(c) Loosely install all the bolts [8], washers [5], and nuts [9].

(d) Starting at the 3 o'clock position, tighten the bolts [8] at the 3, 9, 6, and 12 o'clock positions in that sequence until they are snug.

(e) Tighten the remaining bolts [8] until they are snug.

(f) Starting at the 3 o'clock position, tighten the bolts [8] at the 3, 9, 6, and 12 o'clock positions to 170 ±5 in-lb (19 ±1 N·m).

(g) Tighten the remaining bolts [8] to 170 ±5 in-lb (19 ±1 N·m).

(h) Make sure that the bolt [8] at the 3 o'clock position is still tightened to 170 ±5 in-lb (19 ±1 N·m).

1) If the bolt [8] at the 3 o'clock position is tightened to less than 170 ±5 in-lb (19 ±1 N·m), do this step:

a) Retighten the bolts [8] to 170 ±5 in-lb (19 ±1 N·m), in the same sequence described above, until you find a bolt [8] that has held its torque.

PTF ALL; AIRPLANES WITHOUT TURBINE REAR VANE (TRV) SPACER

SUBTASK 78-11-02-420-003-G02

(4) Install the forward plug [1].

(a) Lift the forward plug [1] and move it into its position on the turbine rear frame.

1) Make sure that the forward plug [1] is securely supported before you lift it.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITHOUT TURBINE REAR VANE (TRV) SPACER (Continued)

2) Align the alignment pin with the alignment hole in the inner flange of the turbine rear frame or spacer.

(b) If it is necessary, apply Pure Nickel Special compound, D00006, to the threads of the bolts [4].

(c) Loosely install all the bolts [4], washers [5], washers [7], and nuts [6].

(d) Starting at the 3 o'clock position, tighten the nuts [6] at the 3, 9, 6, and 12 o'clock positions in that sequence until they are snug.

(e) Tighten the remaining nuts [6] until they are snug.

(f) Starting at the 3 o'clock position, tighten the nuts [6] at the 3, 9, 6, and 12 o'clock positions to 170 ±5 in-lb (19 ±1 N·m).

(g) Tighten the remaining nuts [6] to 170 ±5 in-lb (19 ±1 N·m).

(h) Make sure that the nut [6] at the 3 o'clock position is still tightened to 170 ±5 in-lb (19 ±1 N·m).

1) If the nut [6] at the 3 o'clock position is tightened to less than 170 ±5 in-lb (19 ±1 N·m), do this step:

a) Retighten the nuts [6] to 170 ±5 in-lb (19 ±1 N·m), in the same sequence described above, until you find a nut [6] that has held its torque.

PTF ALL

SUBTASK 78-11-02-000-001-G02

(5) Remove the primary exhaust sleeve and plug equipment, SPL-2419, from the forward plug [1].

(a) Remove the straps that attach the jack adapter to the forward plug [1].

(b) Lower and remove low profile hydraulic jack, COM-1568, from the forward plug [1].

(c) Remove the jack adapter from the low profile hydraulic jack, COM-1568.

SUBTASK 78-11-02-420-002-G00

(6) Install the aft plug [3]:

NOTE: When the aft plug is installed, there may be contact between the center vent tube extension and the inner radius of the aft plug. This is an acceptable condition.

(a) Apply Pure Nickel Special compound, D00006, to the threads of the 27 bolts [2].

(b) Move the aft plug [3] forward and over the attach flange of the forward plug [1].

(c) Install the 27 bolts [2] to attach the aft plug [3]:

1) Loosely install all the bolts [2].

2) Starting at the 3 o'clock position, tighten the bolts [2] at the 3, 9, 6, and 12 o'clock positions in that sequence until they are snug,

3) Tighten the remaining bolts [2] until they are snug.

4) Starting at the 3 o'clock position, tighten the bolts [2] at the 3, 9, 6, and 12 o'clock positions to 180 ±5 in-lb (20 ±1 N·m).

5) Tighten the remaining bolts [2] to 180 ±5 in-lb (20 ±1 N·m).

6) Make sure that the bolt [2] at the 3 o'clock position is still tightened to 180 ±5 in-lb (20 ±1 N·m).

a) If the bolt [2] at the 3 o'clock position is tightened to less than 180 ±5 in-lb (20 ±1 N·m), do this step:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

<1> Retighten the bolts [2] to 180 ±5 in-lb (20 ±1 N·m), in the same sequence described above, until you find a bolt [2] that has held its torque.

SUBTASK 78-11-02-210-002-G00

(7) Do a visual check of the gap between the center vent tube extension and the inner radius of the aft plug [3].

(a) It is acceptable for the gap between the center vent tube and the aft plug [3] to vary around the radius of the center vent tube.

(b) It is acceptable for the center vent tube to make contact with the inner radius of the aft plug [3].

H. Put the Airplane Back to Its Usual Condition

SUBTASK 78-11-02-410-001-G00

(1) If the primary nozzle assembly was removed, install the nozzle (TASK 78-11-01-400-801-G00).

SUBTASK 78-11-02-940-001-G02

(2) Make sure that the work area is clean of all tools and other items.

SUBTASK 78-11-02-410-002-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Close the thrust reverser (TASK 78-31-00-010-802-G00).

SUBTASK 78-11-02-440-001-G00

(4) For engine 1, remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 78-11-02-440-002-G00

(5) For engine 2, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CENTER VENT TUBE EXTENSION - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Center vent tube extension removal

(2) Center vent tube extension installation.

TASK 78-11-03-000-801-G00

2. Center Vent Tube Extension Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the center vent tube extension.

(2) The center vent tube extension is referred to as the vent tube in this task.

(3) The vent tube is two separate pieces, the mixer and the outer tube.

(a) The mixer is installed inside the outer tube.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-11-02-000-801-G00	Primary Plug Assembly Removal (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
417	Engine 1 - Primary Exhaust Nozzle and Plug
427	Engine 2 - Primary Exhaust Nozzle and Plug

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 78-11-03-040-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 78-11-03-010-001-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

SUBTASK 78-11-03-000-001-G00

(3) Do this task: Primary Plug Assembly Removal, TASK 78-11-02-000-801-G00.

G. Center Vent Tube Extension Removal

SUBTASK 78-11-03-000-002-G00

(1) Remove the vent tube as follows:

(a) Remove the 12 bolts [1], 12 washers [2], 12 washers [8] and 12 nuts [7] that attach the mixer [5] and outer tube [6] to the turbine rear frame flange [3].

1) Make sure that you support the vent tube before you remove the last bolt [1].

NOTE: The vent tube weighs approximately 10.0 lb (4.5 kg).

(b) Remove the mixer [5] and outer tube [6].

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(c) Remove the gasket [4].

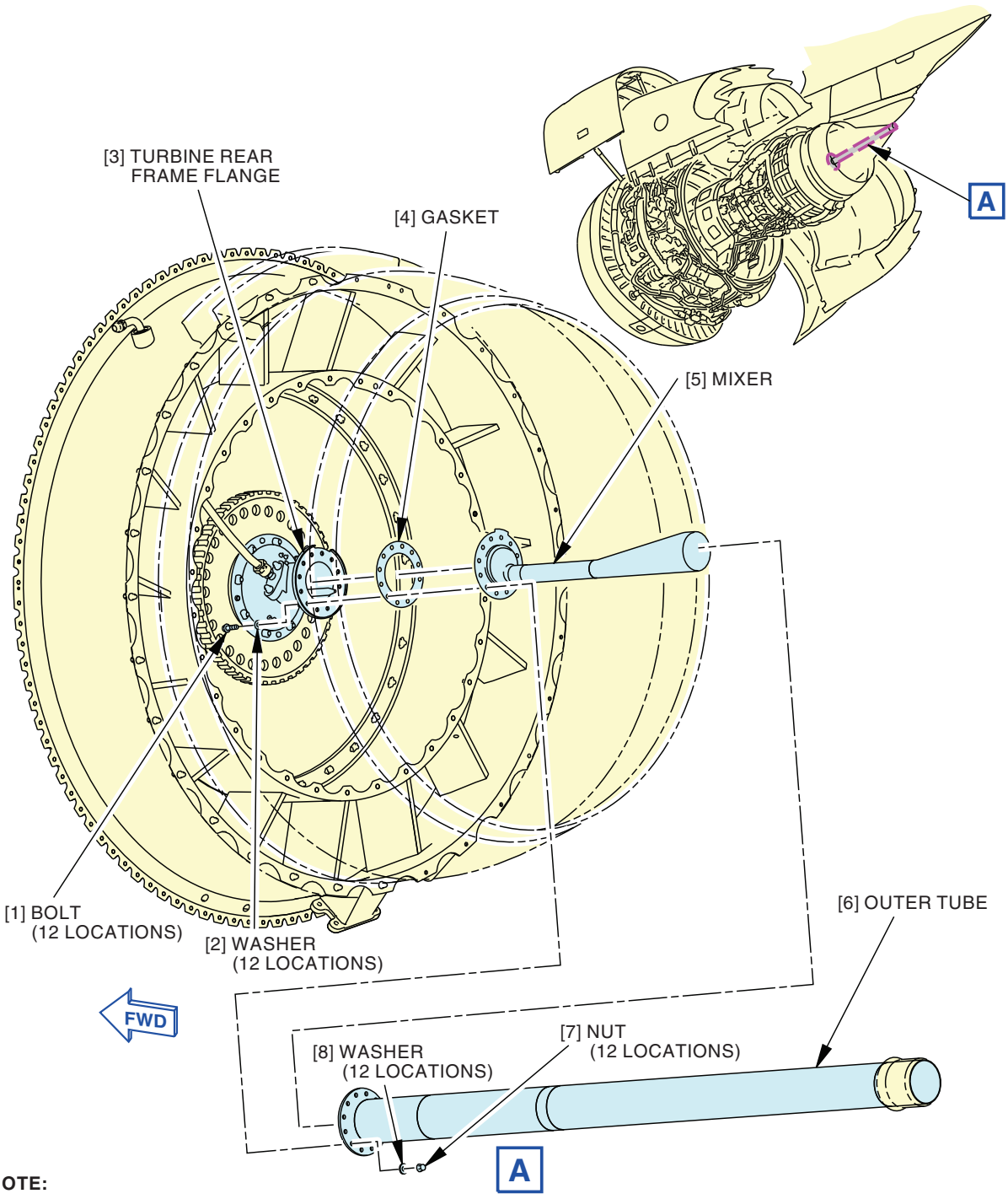
(d) Examine the gasket [4].

1) If there is damage to the gasket [4], discard it.

2) If there is no damage to the gasket [4], use it for the installation.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

SHOWN WITH PRIMARY NOZZLE ASSEMBLY REMOVED.

2459340 S0000570283_V3

Center Vent Tube Extension
Figure 401/78-11-03-990-801-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-11-03-400-801-G00

3. Center Vent Tube Extension Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the center vent tube extension.

(2) The center vent tube extension is referred to as the vent tube in this task.

(3) The vent tube is two separate pieces, the mixer and the outer tube.

(a) The mixer is installed inside the outer tube.

B. References

Reference	Title
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-11-02-400-801-G00	Primary Plug Assembly Installation (P/B 401)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)
D50237	Compound - Antiseize, Pure Nickel Special - Never Seez NSBT-8N/-16N	BAC5008

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
4	Gasket	78-11-03-01-040	PTF ALL
5	Mixer	78-11-03-01-035	PTF ALL
6	Outer tube	78-11-03-01-030	PTF ALL

F. Location Zones

Zone	Area
417	Engine 1 - Primary Exhaust Nozzle and Plug
427	Engine 2 - Primary Exhaust Nozzle and Plug

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Center Vent Tube Extension Installation

SUBTASK 78-11-03-160-001-G00

(1) Make sure that the turbine rear frame flange [3] and the flange surfaces of the mixer [5] and outer tube [6] are clean and free of contamination.

(a) If it is necessary, clean the surfaces with solvent, B00062.

SUBTASK 78-11-03-400-001-G00

(2) If it is necessary, install the mixer [5] inside the outer tube [6].

(a) Install the outer tube [6] in a position so that the part marking is oriented on the side for visibility.

NOTE: The outer tube [6] does not have a clocking feature.

SUBTASK 78-11-03-400-002-G00

(3) Install the vent tube as follows:

(a) Install the gasket [4] onto the turbine rear frame flange [3].

(b) Lift the vent tube into its position on the turbine rear frame flange [3].

(c) Make sure that the alignment tab on the vent tube flange is at the top.

(d) Lubricate the 12 bolts [1] with Pure Nickel Special compound, D50237.

NOTE: It is optional to lubricate the 12 bolts [1] with Pure Nickel Special compound, D50237.

(e) Loosely install the 12 bolts [1], washers [2], 12 washers [8] and nuts [7].

(f) Tighten the nuts [7] at the 3, 6, 9, and 12:00 o'clock locations in that sequence until they are snug and there is no gap between the turbine rear frame flange [3] and the vent tube flange.

(g) Tighten the nuts [7] at the remaining 8 locations until they are snug.

(h) If the 12 bolts [1] was lubricated, do these steps:

1) Tighten the nuts [7] at the 3, 6, 9, and 12 o'clock positions to 65 +2 / -0 in-lb (7 +1 / -0 N·m).

2) Tighten the nuts [7] at the remaining 8 locations to 65 +2 / -0 in-lb (7 +1 / -0 N·m).

3) Make sure that the nut [7] at the 3 o'clock position is still tighten to 65 +2 / -0 in-lb (7 +1 / -0 N·m).

a) If the nut [7] at the 3:00 o'clock position is not tight at 65 +2 / -0 in-lb (7 +1 / -0 N·m), do these steps:

<1> Tighten the nut [7] to 65 +2 / -0 in-lb (7 +1 / -0 N·m).

<2> Continue to tighten the remaining nuts [7] in the sequence used above until you find a nut [7] that is in the specified torque range.

(i) If the 12 bolts [1] was not lubricated, do these steps:

1) Tighten the nuts [7] at the 3, 6, 9, and 12 o'clock positions to 85 +2 / -0 in-lb (10 +1 / -0 N·m).

2) Tighten the nuts [7] at the remaining 8 locations to 85 +2 / -0 in-lb (10 +1 / -0 N·m).

3) Make sure that the nut [7] at the 3 o'clock position is still tighten to 85 +2 / -0 in-lb (10 +1 / -0 N·m).

a) If the nut [7] at the 3:00 o'clock position is not tight at 85 +2 / -0 in-lb (10 +1 / -0 N·m), do these steps:

<1> Tighten the nut [7] to 85 +2 / -0 in-lb (10 +1 / -0 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

<2> Continue to tighten the remaining nuts [7] in the sequence used above until you find a nut [7] that is in the specified torque range.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 78-11-03-400-003-G00

(1) Do this task: Primary Plug Assembly Installation, TASK 78-11-02-400-801-G00.

SUBTASK 78-11-03-410-001-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 78-11-03-440-001-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CENTER VENT TUBE - INSPECTION/CHECK

1. General

A. This procedure has this task:

(1) Center Vent Tube Extension (CVTE) Inspection

TASK 78-11-03-210-801-G00

2. Center Vent Tube Extension (CVTE) Inspection

(Figure 601)

A. General

(1) The Center Vent Tube Extension (CVTE) and mixer assemblies are used to vent the engine A-sump. The vent flow from the engine A-sump includes a mixture of air and engine oil.

(2) It is normal to see oil puddles and coked oil in the CVTE and mixer assemblies. Coked oil that decreases the cross-sectional areas up to 2.500 in. (63.500 mm) of the CVTE and mixer up to 1.000 in. (25.400 mm) will have a negligible impact on the vent performance for the engine A-sump.

(3) This task gives instructions to examine the CVTE and mixer for the coking condition.

B. References

Reference	Title
78-11-03-000-801-G00	Center Vent Tube Extension Removal (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-25834	Coking Checking Inspection Equipment - Vent Tube Part #: 956A8462G01 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
417	Engine 1 - Primary Exhaust Nozzle and Plug
427	Engine 2 - Primary Exhaust Nozzle and Plug

E. Prepare for the Inspection

SUBTASK 78-11-03-480-001-G00

(1) Do these steps to make sure that the ENGINE START switch and START LEVER switch are not operated:

(a) On the P5 overhead panel, make sure that the applicable ENGINE START switch is in the OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) On the P10 control stand, make sure that the applicable START LEVER switch is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Center Vent Tube Extension (CVTE) Inspection

SUBTASK 78-11-03-210-001-G00

(1) Do a visual check of the CVTE and mixer by looking in the aft end of the CVTE.

(a) Use the coking checking equipment, SPL-25834, to inspect the rear vent tube (View C and View D, Figure 601).

(b) Oil puddles in the CVTE are normal and no further maintenance is required if the engine oil consumption remains within Aircraft Maintenance Manual (AMM) limits.

(c) Coking in the CVTE and mixer assemblies are normal if the remaining unobstructed diameter in the area of the coking is in the following limits:

1) The mixer has a minimum inner diameter of 1.0 in. (25.4 mm) that remains unobstructed in the area of the coking.

NOTE: The smallest cross-sectional area of the mixer has a nominal inner diameter of 1.181 in. (30.0 mm) when there is no coking present.

2) The CVTE has a minimum inner diameter (ID) of 2.5 in. (63.5 mm) that remains unobstructed in the area of the coking.

NOTE: The cross-sectional area at the aft end of the CVTE has a nominal inner diameter of 3.0 in. (76.2 mm) when there is no coking present.

(d) If coking in the CVTE or mixer is out of these limits, remove the CVTE and mixer (TASK 78-11-03-000-801-G00).

NOTE: The CVTE and mixer removal is necessary so they can be cleaned in the shop.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 78-11-03-080-001-G00

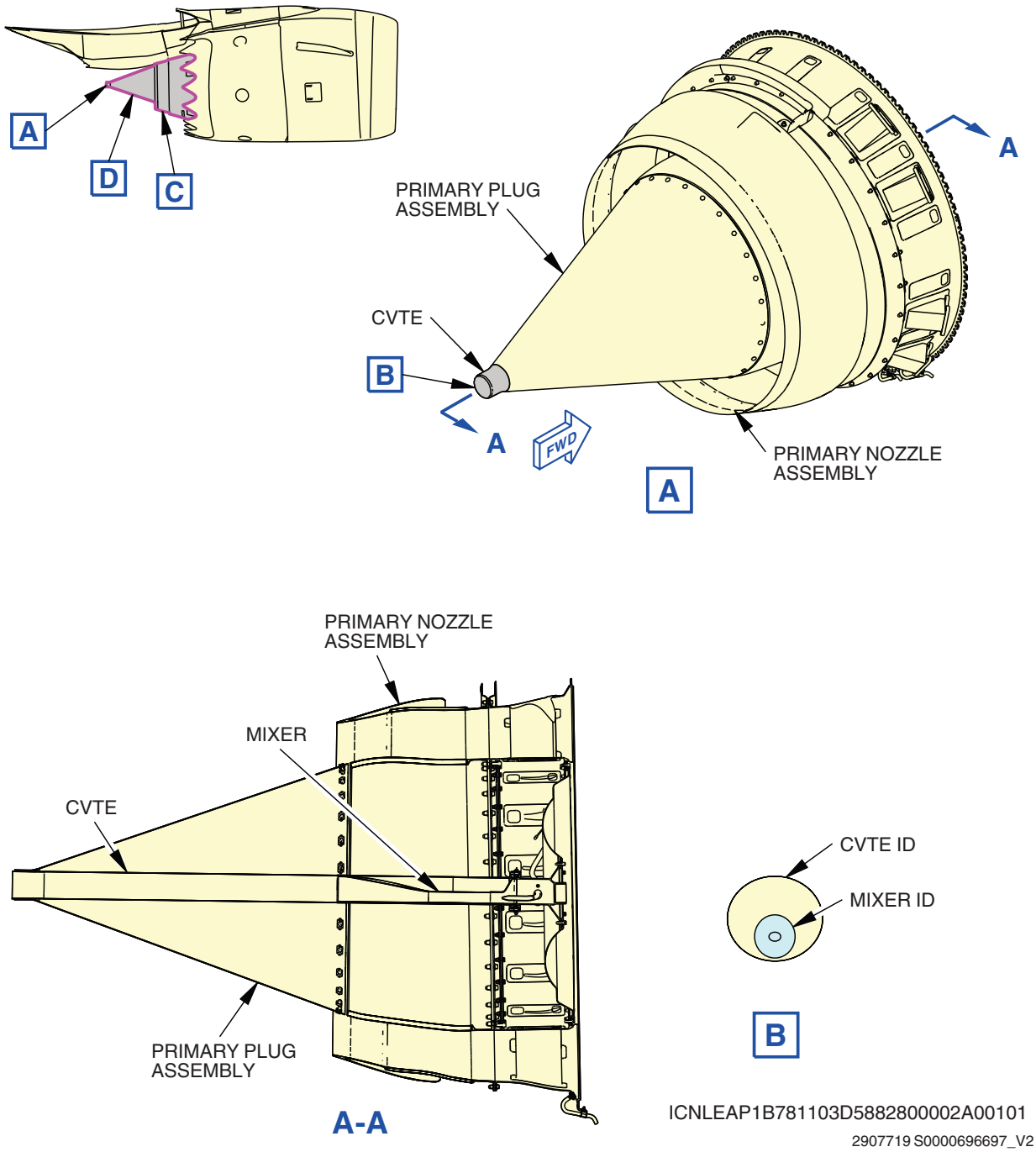
(1) Do these steps to remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

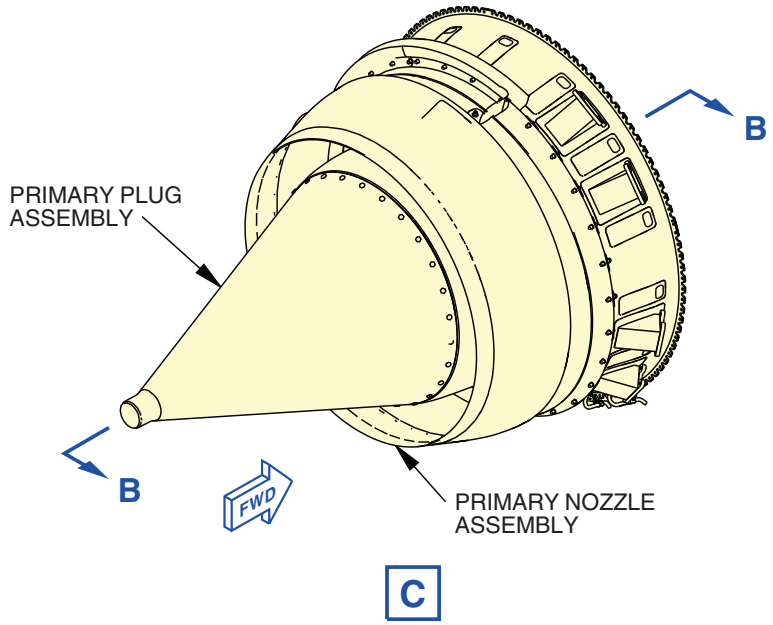
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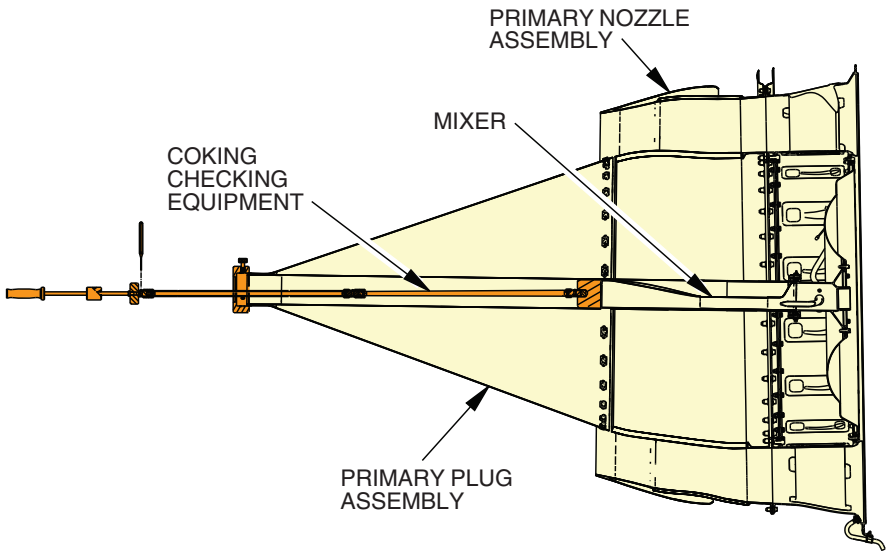
ECCN 9E991 BOEING PROPRIETARY - See title page for details



Center Vent Tube Extension (CVTE) - Inspection
Figure 601/78-11-03-990-802-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

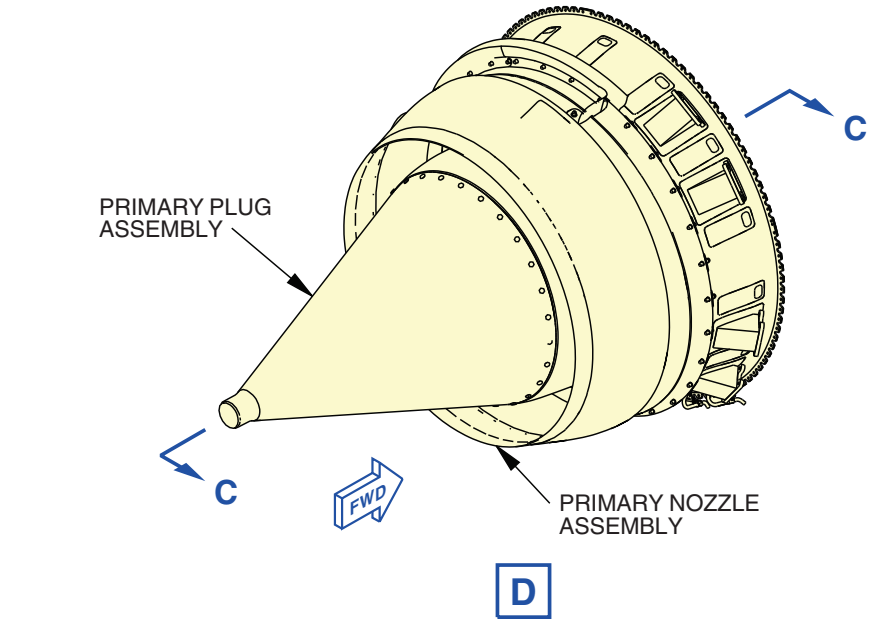


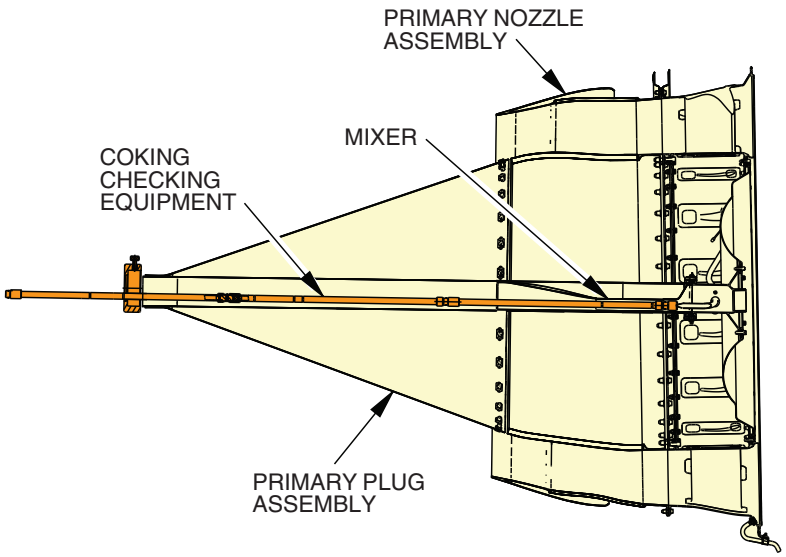


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Center Vent Tube Extension (CVTE) - Inspection
Figure 601/78-11-03-990-802-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





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Center Vent Tube Extension (CVTE) - Inspection
Figure 601/78-11-03-990-802-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

THRUST REVERSER SYSTEM - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks to operate the thrust reverser system:

(1) Open the thrust reverser (Selection).

(2) Close the thrust reverser (Selection).

(3) Open the thrust reverser (Hydraulic Pump Procedure).

(4) Close the thrust reverser (Hydraulic Pump Procedure).

(5) Open the thrust reverser (Sling Procedure).

(6) Close the thrust reverser (Sling Procedure).

(7) Thrust Reverser Operation - Extend (Selection).

(8) Thrust Reverser Operation - Retract (Selection).

(9) Thrust Reverser Operation - Extend (Power Procedure).

(10) Thrust Reverser Operation - Retract (Power Procedure).

(11) Thrust Reverser Operation - Extend (Manual Procedure).

(12) Thrust Reverser Operation - Retract (Manual Procedure).

(13) Thrust Reverser Deactivation for Ground Maintenance.

(14) Thrust Reverser Activation after Ground Maintenance.

TASK 78-31-00-010-801-G00

2. Open the Thrust Reverser (Selection)

A. General

(1) This procedure gives the mechanics the task selection for opening the thrust reverser.

B. Open the Thrust Reverser (Selection)

SUBTASK 78-31-00-010-001-G00

(1) Do one of these tasks to open the applicable thrust reverser:

(a) Do this task: Open the Thrust Reverser (Hydraulic Pump Procedure), TASK 78-31-00-000-801-G00.

(b) Do this task: Open the Thrust Reverser (Sling Procedure), TASK 78-31-00-010-803-G00.

——— END OF TASK ———

TASK 78-31-00-010-802-G00

3. Close the Thrust Reverser (Selection)

A. General

(1) This procedure gives the mechanics the task selection for closing the thrust reverser.

B. Close the Thrust Reverser (Selection)

SUBTASK 78-31-00-010-002-G00

(1) Do one of these tasks to close the applicable thrust reverser:

(a) Do this task: Close the Thrust Reverser (Hydraulic Pump Procedure), TASK 78-31-00-410-801-G00.

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(b) Do this task: Close the Thrust Reverser (Sling Procedure), TASK 78-31-00-410-802-G00.

——— END OF TASK ———

TASK 78-31-00-000-801-G00

4. Open the Thrust Reverser (Hydraulic Pump Procedure)

(Figure 201, Figure 202, Figure 203)

A. General

(1) This task gives the instructions to open the thrust reverser with a hydraulic pump.

(2) This is applicable for the left or right thrust reverser on the applicable engine.

(3) The left and right fan cowl panels must be opened fully to the 61-degree open position.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
54-51-06-000-801	Thrust Reverser Cross Tie Rod Lanyard Removal (P/B 401)
54-51-06-400-801	Thrust Reverser Cross Tie Rod Lanyard Installation (P/B 401)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-2417	Pump - Hand, Cowl Opening Part #: A78019-64 Supplier: 81205 Part #: C78005-68 Supplier: 81205 Part #: FT-HDC-101200D Supplier: 46A73
SPL-14551	Lock - Thrust Reverser Opening Actuator, Fully Extended, LEAP-1B Part #: C78035-1 Supplier: 81205
SPL-14993	Lock - Thrust Reverser Opening Actuator, 15 degree open, LEAP-1B Part #: C78035-14 Supplier: 81205
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00068	Oil - Aircraft Turbine Engine, Synthetic Base	MIL-PRF-23699 Class STD (Standard)
D00071	Oil - Aircraft Turbine Engine, Synthetic Base	MIL-PRF-7808 Grade 3

E. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

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F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2
521AB	Outboard Leading Edge Blowout Door
521BB	Outboard Lower Fixed Leading Edge Access Panel
621AB	Outboard Leading Edge Blowout Door
621BB	Outboard Lower Fixed Leading Edge Access Panel

G. Preparation

SUBTASK 78-31-00-860-241-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 78-31-00-860-143-G00

WARNING RETRACT THE LEADING EDGE FLAPS AND SLATS AND DO THE DEACTIVATION PROCEDURE BEFORE YOU OPEN THE LEFT OR THE RIGHT THRUST REVERSER ON AN ENGINE. IF YOU DO NOT OBEY THIS INSTRUCTION INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 78-31-00-040-009-G00

(3) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 78-31-00-040-010-G00

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(4) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 78-31-00-010-018-G00

CAUTION	IF FAN COWLS ARE INSTALLED, MAKE SURE THAT LEFT AND RIGHT FAN COWLS ARE IN THE FULL OPEN POSITION. MAKE SURE THAT THE SPRING DOOR OPENING-SYSTEM (SDOS) AND HOLD OPEN RODS (HOR) ARE LOCKED IN THEIR POSITION. IF YOU DO NOT, STRUCTURAL DAMAGE TO THE FAN COWL AND THRUST REVERSER CAN OCCUR.

(5) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

(a) On the applicable engine, open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 78-31-00-860-144-G00

(6) Make sure that these access panels are closed:

Number	Name/Location
521AB	Outboard Leading Edge Blowout Door
521BB	Outboard Lower Fixed Leading Edge Access Panel
621AB	Outboard Leading Edge Blowout Door
621BB	Outboard Lower Fixed Leading Edge Access Panel

H. Thrust Reverser Open (Hydraulic Pump Procedure)

SUBTASK 78-31-00-010-003-G00

WARNING	DO NOT OPEN OR CLOSE THE THRUST REVERSER IN HIGH WINDS OR SUDDEN-STRONG-WINDS CONDITIONS. MAKE SURE THAT THE CONSTANT WINDS OR SUDDEN WINDS ARE LESS THAN 40 KNOTS. INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

WARNING	DO NOT GO BETWEEN THE ENGINE AND THE THRUST REVERSER WITHOUT THE ACTUATOR SAFETY LOCK INSTALLED. IF THE THRUST REVERSER SUDDENLY CLOSES, INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

WARNING	DO NOT GO BELOW OR NEAR THE THRUST REVERSER WHILE YOU OPEN OR CLOSE THE THRUST REVERSER. IF THE THRUST REVERSER SUDDENLY CLOSES, INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION	DO NOT EXTEND THE THRUST REVERSER SLEEVES WHILE THE THRUST REVERSER HALVES ARE OPEN. THIS WILL CAUSE DAMAGE TO THE THRUST REVERSER SLEEVES.

(1) Do these steps to open the thrust reverser:

(a) These are the panel identification numbers:

Number	Name/Location
415	Left Thrust Reverser, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Disengage the three latches at the bottom of the aft cowls.

CAUTION

MAKE SURE THAT YOU FULLY DISENGAGE THE LATCHES 8 AND 9. IF YOU DO NOT OBEY, DAMAGE TO THE LANYARD OR THE DEFLECTION LIMITERS CAN OCCUR.

1) Disengage the latch No. 8 and latch No. 9 before the latch No. 7.

(c) Disengage the six latches along the bottom centerline of the thrust reverser.

1) Disengage the latches in sequence from latch No. 6 to latch No. 1.

CAUTION

MAKE SURE THAT THE HAND PUMP OR MANUAL PUMP CONTAINS CLEAN OIL WITHOUT CONTAMINATION. IF YOU DO NOT OBEY, OIL CONTAMINATION CAN CAUSE DAMAGE TO EQUIPMENT.

(d) Remove the dust cap from the inlet fitting on the opening actuator and from the cowl opening hand pump, SPL-2417.

(e) Make sure that the cowl opening hand pump, SPL-2417, has sufficient capacity of oil, D00071, or oil, D00068, per cowl opening hand pump, SPL-2417, placard to be able to fully open the thrust reverser.

NOTE: If the cowl opening hand pump does not have sufficient oil, it may not be possible to fully open thrust reversers, especially if the same hand pump is used to open more than one thrust reverser.

(f) Close the return valve on the hand pump.

(g) Connect the hand pump hose to the inlet fitting on the opening actuator.

WARNING

MAKE SURE THAT THE PUMP HOSE FITTING DOES NOT CONTACT THE THRUST-REVERSER CENTER LOCKING-ACTUATOR. THE CONTACT COULD CAUSE DAMAGE TO THE ACTUATOR OR THE HOSE FITTING. THIS COULD CAUSE THE THRUST REVERSER SUDDENLY CLOSES DUE TO THE FAST LOSS OF FLUID FROM THE OPENING ACTUATOR THROUGH THE BROKEN FITTING. INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION

MAKE SURE THAT THE FILTER ON THE HAND PUMP HOSE DOES NOT INTERFERE WITH THE MOVING THRUST REVERSER. THE FILTER CAN TURN ON THE END OF THE HAND PUMP HOSE AND CAUSE DAMAGE TO THE EQUIPMENT.

(h) Hold the filter on the hand pump hose forward at approximately a 45° angle to prevent contact with the center locking actuator as the thrust reverser opens.

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1) If the engine has MDOS-GSE hose support, route the hand pump hose at the hose support (Figure 203).

NOTE: The MDOS-GSE hose support is near the top of the oil tank on the RH side of the engine, and on the top of the AGB on the LH side of the engine. The MDOS-GSE hose supports are on airplanes beginning with Line Number 7200. The supports can be installed on engines which were delivered on airplanes prior to Line Number 7200 per Boeing SB 737-78-1095.

(i) Pump oil into the opening actuator with the cowl opening hand pump, SPL-2417, to lift the thrust reverser.

(j) Look for these indications that indicates the thrust reverser is in the full open position and the opening actuator is locked:

1) Listen for the click sound of the lock collar.

2) Make sure that the word LOCKED shows on the bottom of the extended piston.

(k) Install the actuator safety lock assembly, SPL-14551, on the extended piston rod.

NOTE: The part number of the actuator safety lock is C78035-1. The part number (C78035-1) is for the set of two actuator safety locks.

1) If it is necessary to open the thrust reverser to a 15-degree open position, install the actuator safety 15 degree lock assembly, SPL-14993, on the extended piston rod.

NOTE: The part number of the actuator safety lock is C78035–14. The part number (C78035–14) is for the set of two actuator safety locks.

(l) Open the return valve on the hand pump to let the weight of the thrust reverser be held by the locked opening actuator.

WARNING

MAKE SURE THAT THE RETURN VALVE ON THE HAND PUMP IS CLOSED WHEN THE THRUST REVERSER IS HELD BY THE LOCKED OPENING ACTUATOR. IF THE RETURN VALVE IS LEFT OPEN, THE ENGINE OIL CAN CONTINUE TO DRAIN OUT OF THE ACTUATOR INTO THE HAND PUMP. DECREASED ENGINE OIL CAN PERMIT THE THRUST REVERSER TO CLOSE QUICKLY DURING SUBSEQUENT OPERATIONS TO CLOSE THE THRUST REVERSER. INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

(m) Close the return valve on the hand pump when the pump pressure is approximately 100 psi (689 kPa) or less. Avoid 0 psi (0 kPa) readings..

WARNING

MAKE SURE THAT YOU RELEASE THE PRESSURE BEFORE YOU DISCONNECT THE HAND PUMP. A SPRAY OF OIL CAN COME FROM THE HOSE. ENGINE OIL IS POISONOUS AND CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

(n) If it is necessary to disconnect the hand pump, install dust caps on the inlet fitting on the opening actuator and the hand pump hose.

PTF ALL; AIRPLANES WITH LANYARD

CAUTION

MAKE SURE THAT THE LANYARD IS NOT DAMAGED OR BROKEN. A BROKEN LANYARD CAN CAUSE A DEFORMATION OR A BENT AFT COWL-DEFLECTION LIMITER STRAP. DAMAGE TO THE ENGINE PARTS CAN OCCUR.

(o) Examine the cross tie rod lanyard for damage.

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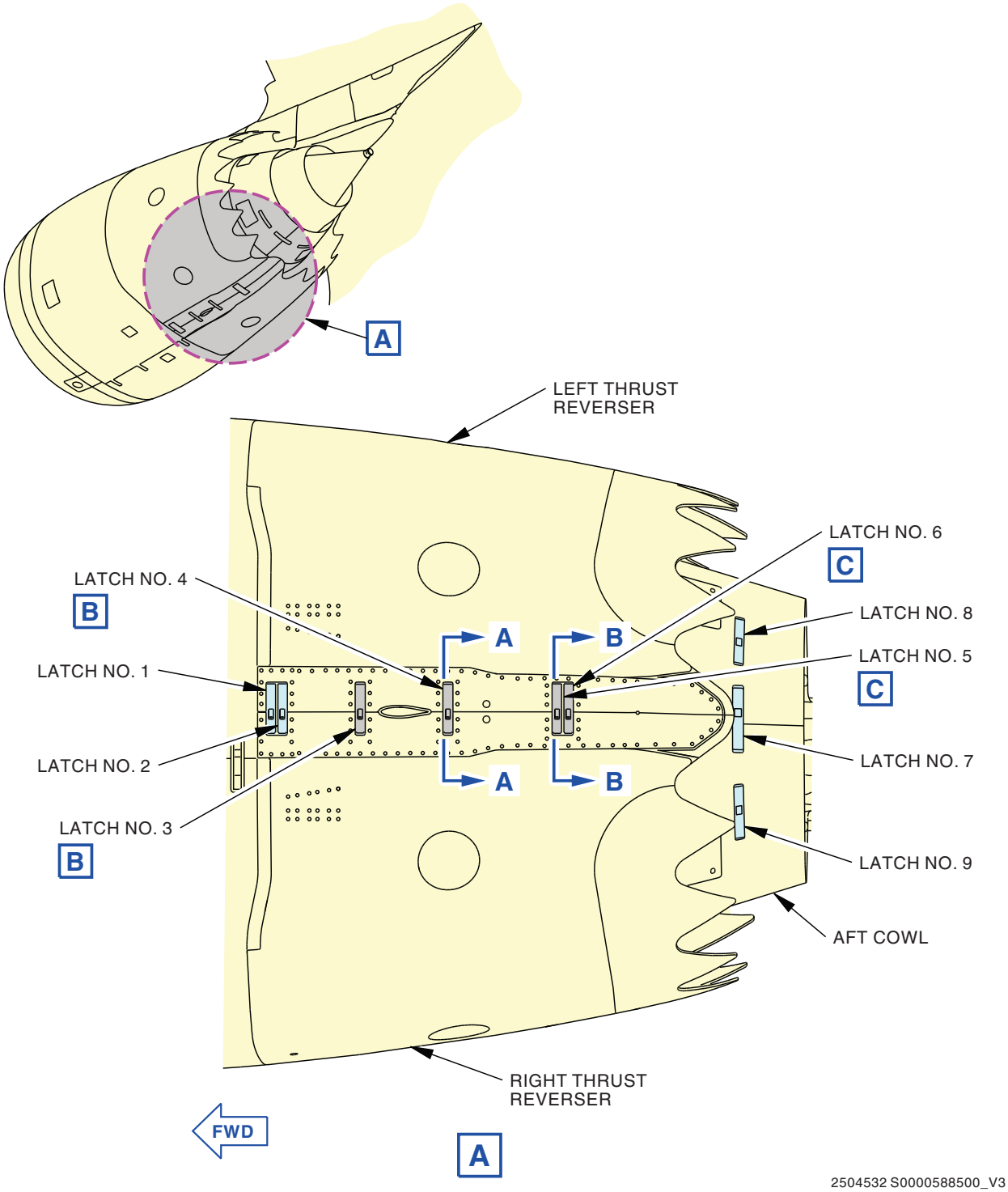
PTF ALL; AIRPLANES WITH LANYARD (Continued)

1) If the lanyard is damaged or broken, replace it (TASK 54-51-06-000-801 and TASK 54-51-06-400-801).

PTF ALL

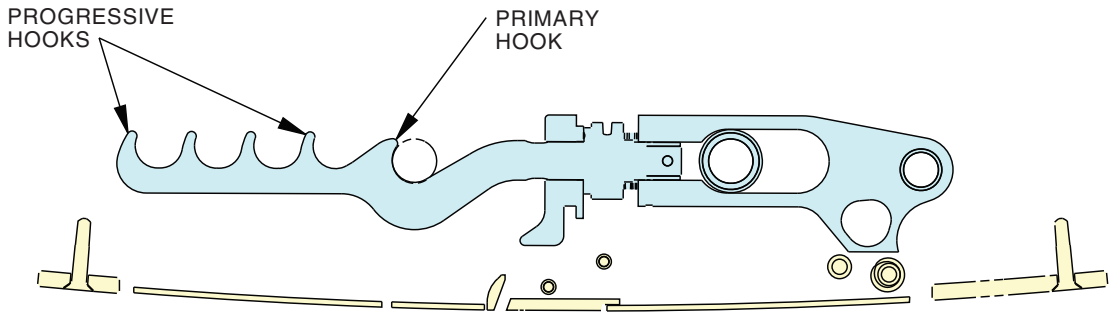
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ECCN 9E991 BOEING PROPRIETARY - See title page for details



Thrust Reverser Latch Release
Figure 201/78-31-00-990-801-G00 (Sheet 1 of 2)

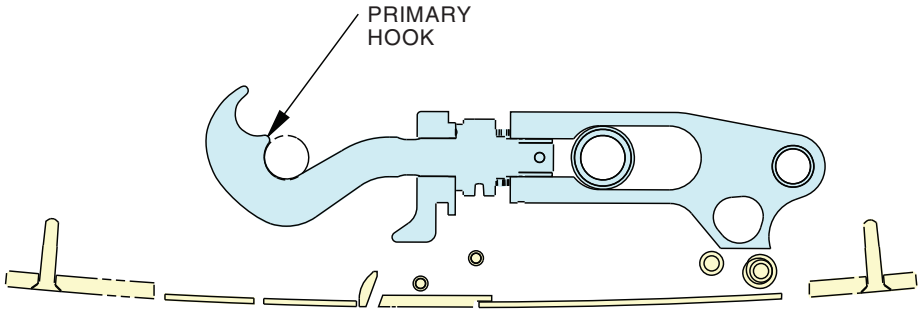
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LATCH NO. 3 & 4

A-A

B



LATCH NO. 5 & 6

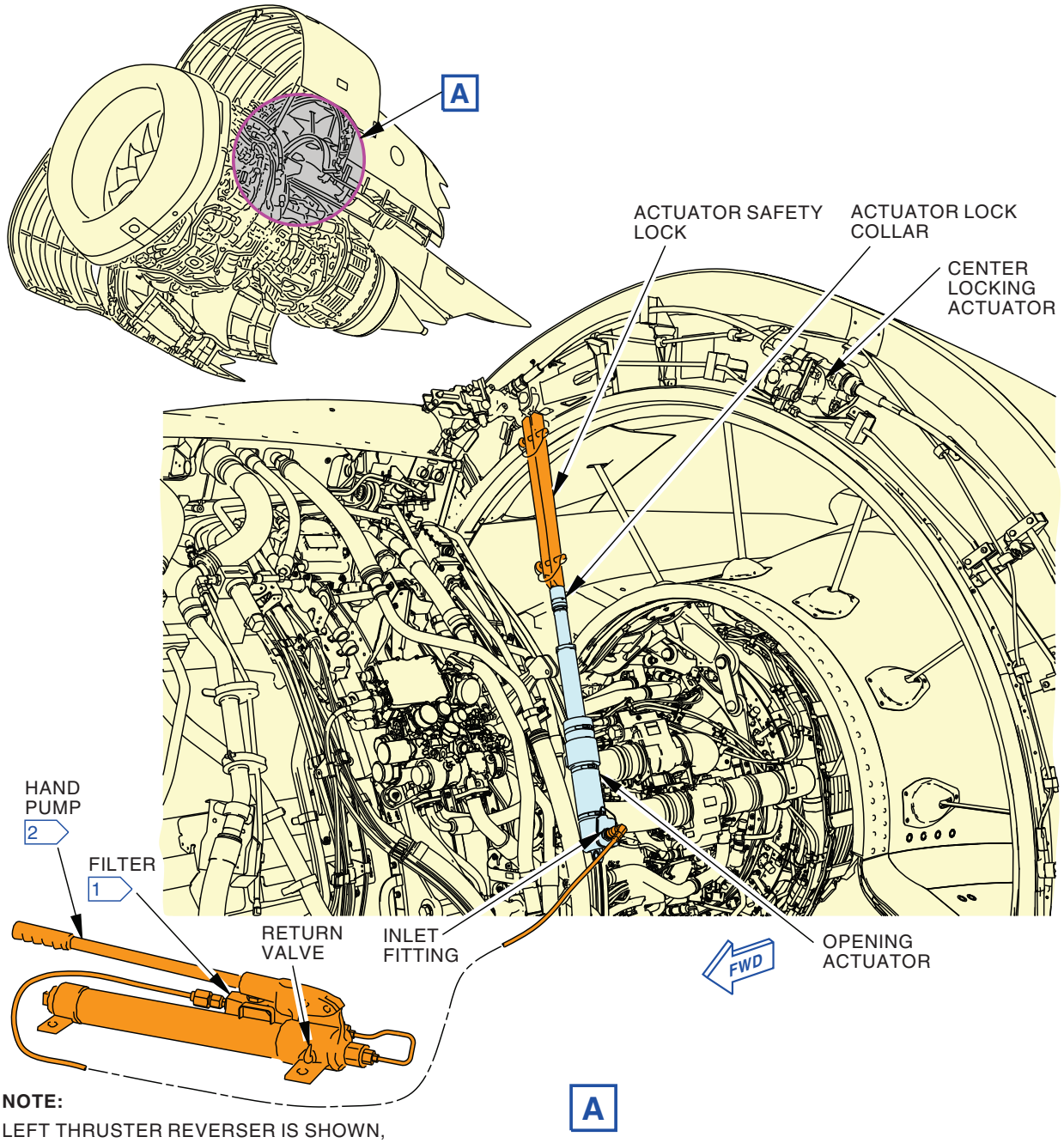
B-B

C

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Thrust Reverser Latch Release
Figure 201/78-31-00-990-801-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

LEFT THRUSTER REVERSER IS SHOWN, RIGHT THRUST REVERSER IS OPPOSITE.

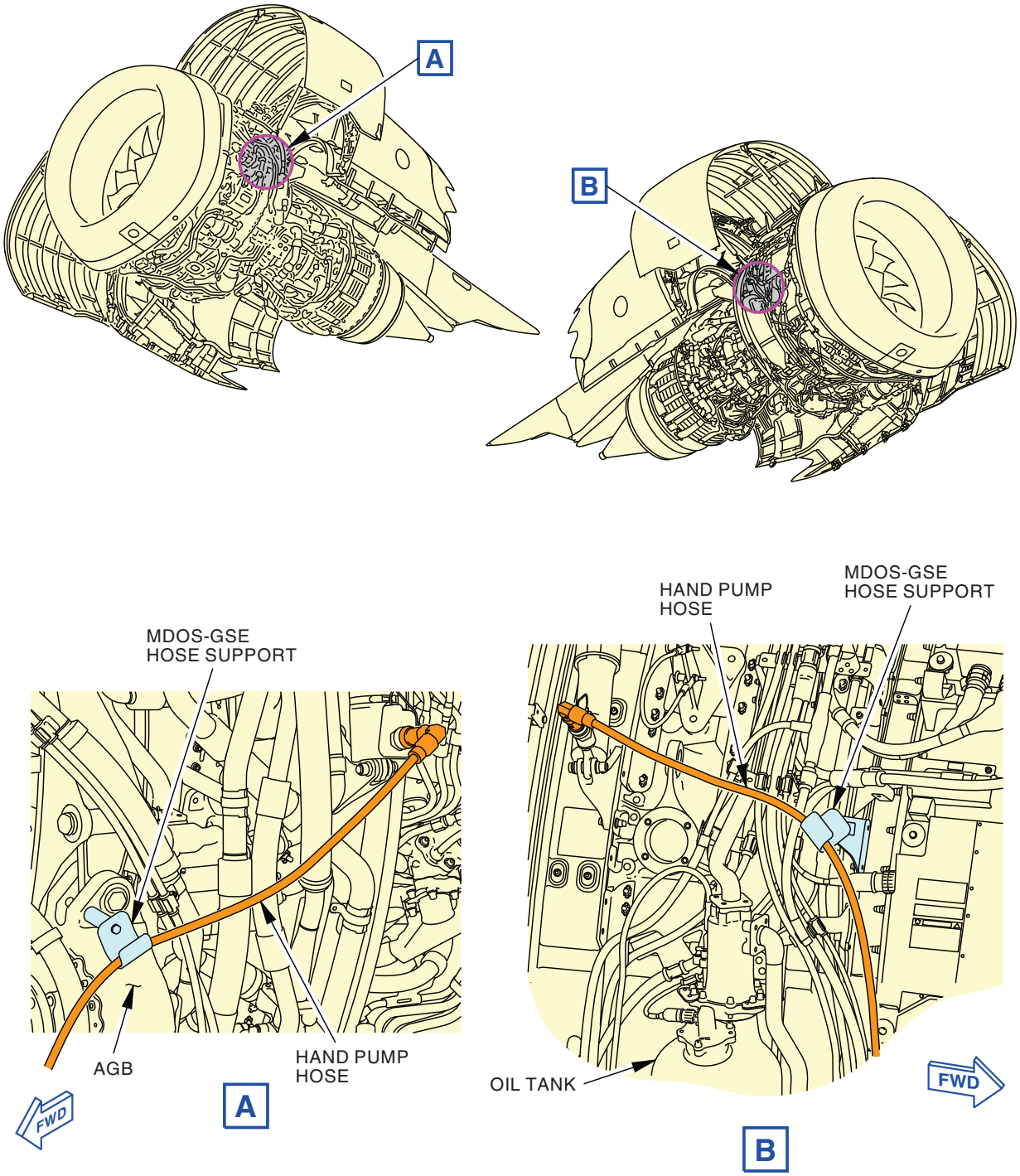
1 HOLD THE FILTER ON THE HAND PUMP HOSE FORWARD AT APPROXIMATELY A 45 DEGREE ANGLE TO PREVENT CONTACT WITH THE CENTER LOCKING ACTUATOR AS THE THRUST REVERSER OPENS OR CLOSES.

2 C78005-60 IS SHOWN.

2355133 S00061516613_V4

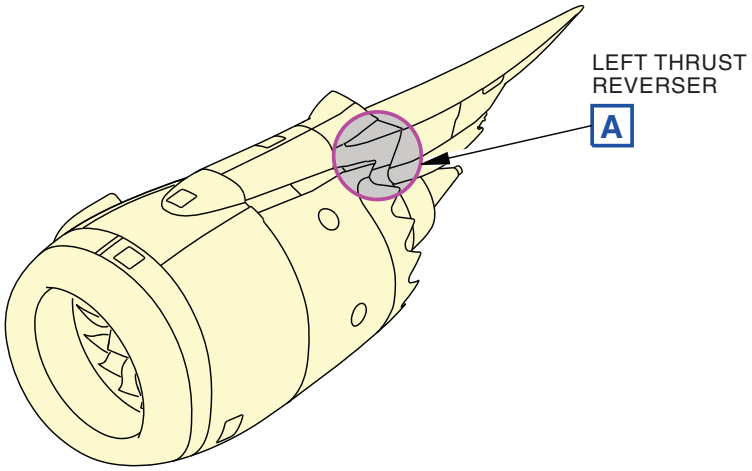
Thrust Reverser Actuator Safety Lock and Hand Pump
Figure 202/78-31-00-990-802-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

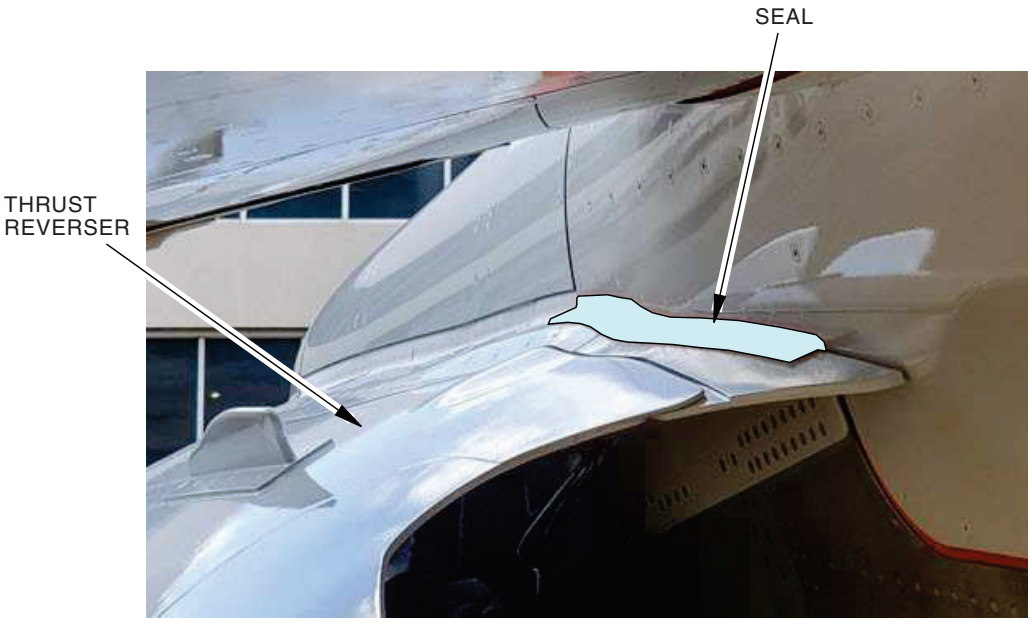


MDOS-GSE Hose Support
Figure 203/78-31-00-990-821-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEFT STRUT
(RIGHT STRUT IS OPPOSITE)



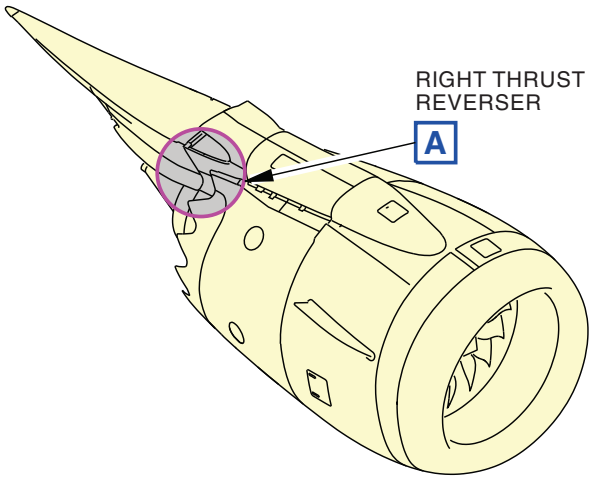
LEFT THRUST REVERSER
(RIGHT THRUST REVERSER IS ALMOST THE SAME)

A

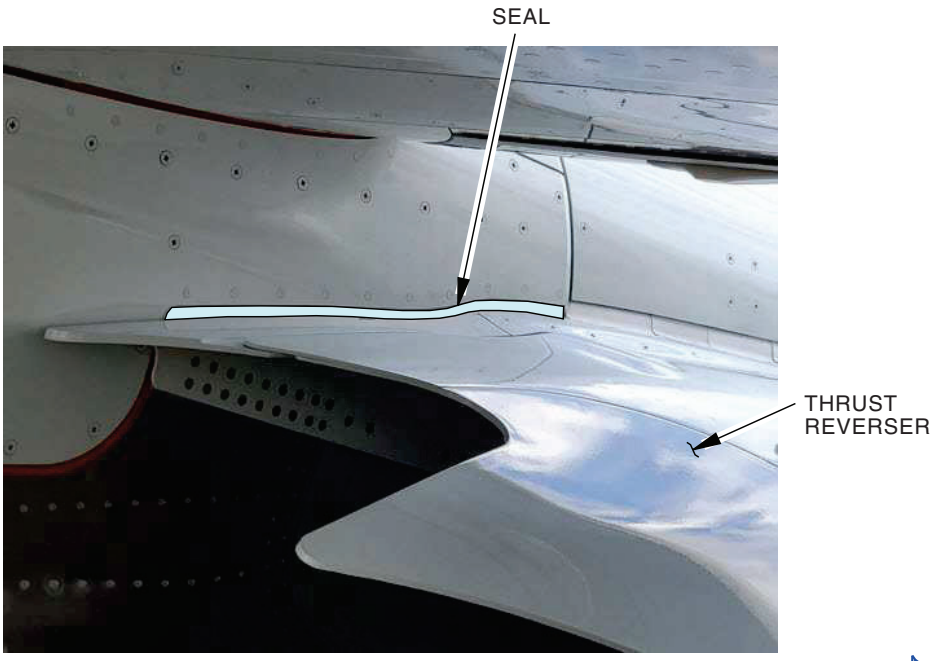
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Thrust Reverser Seal
Figure 204/78-31-00-990-822-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



RIGHT STRUT
(LEFT STRUT IS OPPOSITE)



RIGHT THRUST REVERSER
(LEFT THRUST REVERSER IS ALMOST THE SAME)

A

2998472 S0000774588_V1

Thrust Reverser Seal
Figure 204/78-31-00-990-822-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-31-00-410-801-G00

5. Close the Thrust Reverser (Hydraulic Pump Procedure)

(Figure 201, Figure 202, and Figure 204)

A. General

(1) This task gives the instructions to close the thrust reverser with a hydraulic pump.

(2) This is applicable for the left or right thrust reverser on the applicable engine.

(3) The left and right fan cowl panels must be opened fully to the 61-degree open position.

B. References

Reference	Title
54-51-06-000-801	Thrust Reverser Cross Tie Rod Lanyard Removal (P/B 401)
54-51-06-400-801	Thrust Reverser Cross Tie Rod Lanyard Installation (P/B 401)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-08-870-801-G00	Fill and Bleed Procedure (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-2417	Pump - Hand, Cowl Opening Part #: A78019-64 Supplier: 81205 Part #: C78005-68 Supplier: 81205 Part #: FT-HDC-101200D Supplier: 46A73
SPL-2434	Tool - Latching, Thrust Reverser C-Duct Halves Part #: C78020-21 Supplier: 81205
SPL-14551	Lock - Thrust Reverser Opening Actuator, Fully Extended, LEAP-1B Part #: C78035-1 Supplier: 81205
SPL-14993	Lock - Thrust Reverser Opening Actuator, 15 degree open, LEAP-1B Part #: C78035-14 Supplier: 81205
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00068	Oil - Aircraft Turbine Engine, Synthetic Base	MIL-PRF-23699 Class STD (Standard)
D00071	Oil - Aircraft Turbine Engine, Synthetic Base	MIL-PRF-7808 Grade 3

E. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1

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(Continued)

Number	Name/Location
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Thrust Reverser Closure (Hydraulic Pump Procedure)

SUBTASK 78-31-00-410-001-G00

WARNING DO NOT OPEN OR CLOSE THE THRUST REVERSER IN HIGH WINDS OR SUDDEN-STRONG-WINDS CONDITIONS. MAKE SURE THAT THE CONSTANT WINDS OR SUDDEN WINDS ARE LESS THAN 40 KNOTS. INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

WARNING DO NOT GO BETWEEN THE ENGINE AND THE THRUST REVERSER WITHOUT THE ACTUATOR SAFETY LOCK INSTALLED. IF THE THRUST REVERSER SUDDENLY CLOSES, INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

WARNING DO NOT GO BELOW OR NEAR THE THRUST REVERSER WHILE YOU OPEN OR CLOSE THE THRUST REVERSER. IF THE THRUST REVERSER SUDDENLY CLOSES, INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT THE AREA BETWEEN THE THRUST REVERSER AND THE TURBINE EXHAUST SLEEVE FIRESEALS AND THE ENGINE IS CLEAR OF ALL OBJECTS. THIS WILL PREVENT DAMAGE TO THE EQUIPMENT WHEN YOU CLOSE THE THRUST REVERSER.

CAUTION MAKE SURE THAT THE LATCH RING V-BLADE GOES INTO THE ENGINE V-GROOVE WHEN YOU CLOSE THE THRUST REVERSER. DAMAGE TO THE THRUST REVERSER CAN OCCUR.

(1) Do these steps to close the thrust reverser:

(a) These are the panel identification numbers:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Close the left thrust reverser first.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION MAKE SURE THAT THE HAND PUMP OR MANUAL PUMP CONTAINS CLEAN OIL WITHOUT CONTAMINATION. IF YOU DO NOT OBEY, OIL CONTAMINATION CAN CAUSE DAMAGE TO EQUIPMENT.

(c) Remove the dust caps from the inlet fitting on the opening actuator and from the cowl opening hand pump, SPL-2417.

(d) Make sure that the cowl opening hand pump, SPL-2417, is not full of oil but has sufficient capacity of oil, D00071, or oil, D00068, per cowl opening hand pump, SPL-2417, placard to be able to lower the thrust reverser.

NOTE: The hand pump cannot be completely empty or completely full of oil, D00071, or oil, D00068. Some oil must be in the hand pump to relieve the load on the opening actuator locks. A full hand pump will not permit oil to flow from the opening actuator and thrust reverser will not fully close.

1) If it is possible, use the same hand pump that was used to open the thrust reverser.

NOTE: If the same hand pump was used to open the thrust reverser is also used to close the thrust reverser, the hand pump oil level will be acceptable.

(e) Make sure that the return valve on the hand pump is closed.

(f) Make sure that the cross tie rod lanyard is attached and in a serviceable condition.

1) If the lanyard is damaged or broken, replace it (TASK 54-51-06-000-801 and TASK 54-51-06-400-801).

(g) Connect the cowl opening hand pump, SPL-2417, hose to the inlet fitting on the opening actuator.

CAUTION MAKE SURE THAT THE FILTER ON THE HAND PUMP HOSE DOES NOT INTERFERE WITH THE MOVING THRUST REVERSER. THE FILTER CAN TURN ON THE END OF THE HAND PUMP HOSE AND CAUSE DAMAGE TO THE EQUIPMENT.

(h) Hold the filter on the hand pump hose forward at approximately a 45° angle to prevent contact with the center locking actuator as the thrust reverser closes.

WARNING MAKE SURE YOU USE THE HAND PUMP TO EXTEND THE OPENING ACTUATOR AND LIFT THE WEIGHT OF THE THRUST REVERSER OFF THE ACTUATOR LOCK. IF YOU DO NOT USE THE HAND PUMP, AIR OR A VACUUM COULD GET INTO THE OPENING ACTUATOR. THIS COULD CAUSE THE THRUST REVERSER TO CLOSE QUICKLY WHEN THE ACTUATOR LOCK COLLAR IS DISENGAGED. INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

WARNING DO NOT OPEN THE RETURN VALVE ON THE HAND PUMP UNTIL YOU REMOVE THE SAFETY LOCK AND DISENGAGE THE ACTUATOR LOCK COLLAR. THIS WILL PREVENT INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(i) With the return valve closed on the hand pump, operate the hand pump to remove the load from the opening actuator locks.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(j) Remove the actuator safety lock assembly, SPL-14551, or 15 degree lock assembly, SPL-14993, from the extended piston rod.

NOTE: The part number of the actuator safety lock is C78035-1 or C78035–14. The part number (C78035-1 or C78035–14) is for the set of two actuator safety locks.

(k) Push up on the actuator lock collar to disengage the lock.

1) If the hand pump cannot sufficiently open the thrust reverser to unlock the opening actuator, install the opening actuator safety lock assembly, SPL-14551, on the extended piston rod and service the hand pump with oil, D00071, or oil, D00068, per cowl opening hand pump, SPL-2417, placard.

NOTE: Do not fully fill the hand pump with oil.

(l) Make sure that the v-blade on the thrust reverser and v-groove on the fan case are aligned and fully engaged.

WARNING

DO THE ACTUATOR FILL AND BLEED PROCEDURE IF THE OPENING ACTUATOR RETRACTS TOO QUICKLY FOR MORE THAN THE LIMIT. IF YOU DO NOT OBEY THIS INSTRUCTION, INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(m) Open the return valve on the hand pump and lower the thrust reverser.

NOTE: The opening actuator has a controlled rate at that it should retract. Usually the actuator piston rod will initially move quickly a small amount, less than 0.5 in. (12.7 mm).

1) If the opening actuator retracts quickly for more than 0.5 in. (12.7 mm), do this task: TASK 78-31-08-870-801-G00.

2) If the thrust reverser will not fully close, do these steps:

a) Do this task: Open the Thrust Reverser (Hydraulic Pump Procedure), TASK 78-31-00-000-801-G00.

b) Drain but not fully drain the oil from the cowl opening hand pump, SPL-2417.

c) Return to the step in this task that has instructions to close the thrust reverser.

d) After the thrust reverser is closed, make sure that the cowl opening hand pump, SPL-2417, with oil, D00071, or oil, D00068, are fully serviced per the hand pump placard.

(n) Close the return valve on the hand pump.

(o) Disconnect the hand pump hose from the opening actuator.

(p) Install the dust caps on the inlet fitting on the opening actuator and the hand pump hose.

(q) Repeat steps to close the right thrust reverser.

(r) Do the steps that follow to engage the latches along the bottom centerline of the thrust reverser:

NOTE: The latching tool, SPL-2434, can be used to assist with opening or closing the thrust reverser latches.

1) Engage the first progressive hook on latch No. 4 and close the latch.

2) Engage the first progressive hook on latch No. 3 and close the latch.

NOTE: Latch No. 4 will drop open.

3) Engage the next progressive hook on latch No. 4 and close the latch.

NOTE: Latch No. 3 will drop open.

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4) Alternate between closing latch No. 3 and latch No. 4 on the next progressive hooks until both latches are engaged on the progressive hook nearest the primary hook.

5) Engage the progressive hook on latch No. 5 and close the latch.

6) Engage the primary hook on latch No. 3 and close the latch.

7) Engage the primary hook on latch No. 4 and close the latch.

8) Engage the primary hook on latch No. 5 and close the latch.

9) Engage the primary hook on latch No. 6 and close the latch.

10) Close latch No. 1 and latch No. 2.

11) Close latch No. 7.

12) Close latch No. 8 and latch No. 9.

PTF ALL; AIRPLANES WITH 313A6535-11/-12 SEAL INSTALLED

(s) Make sure that the aft end of the seal at the upper aft end of the thrust reverser is not pinched (Figure 204).

NOTE: The aft end of the seal immediately above the serrated aft edge.

1) If the seal is pinched, use a flexible plastic squeegee to carefully pull it up into its position.

PTF ALL; AIRPLANES WITH 313A6535-15/-16 SEAL INSTALLED

(t) Make sure that the seal sits inside of the thrust reverser (Figure 204).

PTF ALL

H. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-00-860-001-G00

(1) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

(a) On the applicable engine, close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 78-31-00-440-001-G00

(2) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 78-31-00-860-243-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-31-00-010-803-G00

6. Open the Thrust Reverser (Sling Procedure)

(Figure 201 and Figure 202)

A. General

(1) This task gives the instructions to open the left or right thrust reverser on the applicable engine if the opening actuator cannot be used.

(2) It is recommended that the Hand Pump Procedure be used. Use the Sling Procedure only if the opening actuator is not operational or the hand pump is not available.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
54-51-06-000-801	Thrust Reverser Cross Tie Rod Lanyard Removal (P/B 401)
54-51-06-400-801	Thrust Reverser Cross Tie Rod Lanyard Installation (P/B 401)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14551	Lock - Thrust Reverser Opening Actuator, Fully Extended, LEAP-1B Part #: C78035-1 Supplier: 81205
SPL-14554	Sling - Opening, LEAP-1B Thrust Reverser Part #: C78031-1 Supplier: 81205
SPL-14993	Lock - Thrust Reverser Opening Actuator, 15 degree open, LEAP-1B Part #: C78035-14 Supplier: 81205
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
521AB	Outboard Leading Edge Blowout Door
521BB	Outboard Lower Fixed Leading Edge Access Panel
621AB	Outboard Leading Edge Blowout Door
621BB	Outboard Lower Fixed Leading Edge Access Panel

F. Preparation

SUBTASK 78-31-00-860-242-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 78-31-00-040-001-G00

WARNING RETRACT THE LEADING EDGE FLAPS AND SLATS AND DO THE DEACTIVATION PROCEDURE BEFORE YOU OPEN THE LEFT OR THE RIGHT THRUST REVERSER ON AN ENGINE. IF YOU DO NOT OBEY THIS INSTRUCTION INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 78-31-00-040-002-G00

(3) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 78-31-00-040-003-G00

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(4) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

SUBTASK 78-31-00-860-245-G00

CAUTION IF FAN COWLS ARE INSTALLED, MAKE SURE THAT LEFT AND RIGHT FAN COWLS ARE IN THE FULL OPEN POSITION. MAKE SURE THAT THE SPRING DOOR OPENING-SYSTEM (SDOS) AND HOLD OPEN RODS (HOR) ARE LOCKED IN THEIR POSITION. IF YOU DO NOT, STRUCTURAL DAMAGE TO THE FAN COWL AND THRUST REVERSER CAN OCCUR.

(5) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

(a) On the applicable engine, open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 78-31-00-860-003-G00

(6) Make sure that these access panels are closed:

Number	Name/Location
521AB	Outboard Leading Edge Blowout Door
521BB	Outboard Lower Fixed Leading Edge Access Panel
621AB	Outboard Leading Edge Blowout Door
621BB	Outboard Lower Fixed Leading Edge Access Panel

G. Thrust Reverser Open (Sling Procedure)

SUBTASK 78-31-00-010-004-G00

WARNING DO NOT OPEN OR CLOSE THE THRUST REVERSER IN HIGH WINDS OR SUDDEN-STRONG-WINDS CONDITIONS. MAKE SURE THAT THE CONSTANT WINDS OR SUDDEN WINDS ARE LESS THAN 40 KNOTS. INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

WARNING DO NOT GO BETWEEN THE ENGINE AND THE THRUST REVERSER WITHOUT THE ACTUATOR SAFETY LOCK INSTALLED. IF THE THRUST REVERSER SUDDENLY CLOSES, INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

WARNING DO NOT GO BELOW OR NEAR THE THRUST REVERSER WHILE YOU OPEN OR CLOSE THE THRUST REVERSER. IF THE THRUST REVERSER SUDDENLY CLOSES, INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION DO NOT EXTEND THE THRUST REVERSER SLEEVES WHILE THE THRUST REVERSER HALVES ARE OPEN. THIS WILL CAUSE DAMAGE TO THE THRUST REVERSER SLEEVES.

(1) Do these steps to open the thrust reverser:

(a) These are the panel identification numbers:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Disengage the three latches at the bottom of the aft cowls.

1) Disengage the latch No. 8 and latch No. 9 before the latch No. 7.

(c) Disengage the six latches along the bottom centerline of the thrust reverser.

1) Disengage the latches in sequence from latch No. 6 to latch No. 1.

(d) Attach an opening sling, SPL-14554, to the thrust reverser.

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WARNING BE CAREFUL WHEN YOU OPEN OR CLOSE THE THRUST REVERSER, THE THRUST REVERSER IS HEAVY. INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

WARNING WARNING: DO NOT OPEN THE THRUST REVERSER TOO QUICKLY. THIS CAN CAUSE A VACUUM WHICH WILL HAVE AN EFFECT ON THE NORMAL OPERATION (SNUBBING ACTION) OF THE OPENING ACTUATOR. WHEN THE ACTUATOR LOCK IS DISENGAGED FOR THE CLOSE PROCEDURE, THE THRUST REVERSER CAN CLOSE SUDDENLY. INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(e) Slowly lift the thrust reverser with the sling to the fully open position.

NOTE: The thrust reverser weighs approximately 800 lb (363 kg).

(f) Look for these indications that the thrust reverser is in the fully open position and opening actuator is locked:

1) Listen for the click sound of the lock collar.

2) Make sure that the word LOCKED shows on the bottom of the extended piston.

(g) Install the actuator safety lock assembly, SPL-14551, on the extended piston rod.

NOTE: The part number of the actuator safety lock is C78035-1. The part number (C78035-1) is for the set of two actuator safety locks.

1) If it is necessary to open the thrust reverser to the 15-degree open position, install the actuator safety 15 degree lock assembly, SPL-14993, on the extended piston rod.

NOTE: The part number of the actuator safety lock is C78035–14. The part number (C78035–14) is for the set of two actuator safety locks.

PTF ALL; AIRPLANES WITH LANYARD

(h) Examine the cross tie rod lanyard for damage.

1) If the lanyard is damaged or broken, replace it (TASK 54-51-06-000-801 and TASK 54-51-06-400-801).

PTF ALL

——— END OF TASK ———

TASK 78-31-00-410-802-G00

7. Close the Thrust Reverser (Sling Procedure)

(Figure 201, Figure 202, and Figure 204)

A. General

(1) This task gives the instructions to close the left or right thrust reverser on the applicable engine if the opening actuator cannot be used.

(2) It is recommended that the Hand Pump Procedure be used. Use the Sling Procedure if the opening actuator is not operational or hand pump is not available.

B. References

Reference	Title
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Title
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-2434	Tool - Latching, Thrust Reverser C-Duct Halves Part #: C78020-21 Supplier: 81205
SPL-14551	Lock - Thrust Reverser Opening Actuator, Fully Extended, LEAP-1B Part #: C78035-1 Supplier: 81205
SPL-14554	Sling - Opening, LEAP-1B Thrust Reverser Part #: C78031-1 Supplier: 81205
SPL-14993	Lock - Thrust Reverser Opening Actuator, 15 degree open, LEAP-1B Part #: C78035-14 Supplier: 81205
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Thrust Reverser Closure (Sling Procedure)

SUBTASK 78-31-00-410-002-G00

WARNING

MAKE SURE THAT YOU USE A STABLE FORCE (A MINIMUM OF TWO PERSONS AT THE FORWARD END) TO HOLD UP THE THRUST REVERSER BEFORE YOU DISENGAGE THE ACTUATOR LOCKS. THERE COULD BE AIR OR A VACUUM IN THE ACTUATOR WHICH CAN DECREASE THE SNUBBING ACTION OF THE OPENING ACTUATOR. THE THRUST REVERSER CAN CLOSE QUICKLY AND CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING DO THE ACTUATOR FILL AND BLEED PROCEDURE IF THE OPENING ACTUATOR RETRACTS TOO QUICKLY FOR MORE THAN THE LIMIT. IF YOU DO NOT OBEY THIS INSTRUCTION, INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

WARNING DO NOT OPEN OR CLOSE THE THRUST REVERSER IN HIGH WINDS OR SUDDEN-STRONG-WINDS CONDITIONS. MAKE SURE THAT THE CONSTANT WINDS OR SUDDEN WINDS ARE LESS THAN 40 KNOTS. INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

WARNING BE CAREFUL WHEN YOU OPEN OR CLOSE THE THRUST REVERSER, THE THRUST REVERSER IS HEAVY. INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

WARNING DO NOT GO BETWEEN THE ENGINE AND THE THRUST REVERSER WITHOUT THE ACTUATOR SAFETY LOCK INSTALLED. IF THE THRUST REVERSER SUDDENLY CLOSES, INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

WARNING DO NOT GO BELOW OR NEAR THE THRUST REVERSER WHILE YOU OPEN OR CLOSE THE THRUST REVERSER. IF THE THRUST REVERSER SUDDENLY CLOSES, INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT THE LATCH RING V-BLADE GOES INTO THE ENGINE V-GROOVE WHEN YOU CLOSE THE THRUST REVERSER. DAMAGE TO THE THRUST REVERSER CAN OCCUR.

(1) Do these steps to close the thrust reverser:

NOTE: The thrust reverser weighs approximately 800 lb (363 kg).

(a) These are the panel identification numbers:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Lift the thrust reverser a small amount to remove the load from the opening actuator.

(c) Remove the actuator safety lock assembly, SPL-14551, or 15 degree lock assembly, SPL-14993, from the extended piston rod.

(d) Push up on the actuator lock collar to disengage the lock.

(e) Slowly close the thrust reverser.

(f) Make sure that the v-blade on the thrust reverser and v-groove on the fan case are aligned and fully engaged.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(g) Do these steps to engage the latches along the bottom centerline of the thrust reverser:

NOTE: The latching tool, SPL-2434, can be used to assist with opening or closing the thrust reverser latches.

1) Engage the first progressive hook on latch No. 4 and close the latch.
2) Engage the first progressive hook on latch No. 3 and close the latch.

NOTE: Latch No. 4 will drop open.

3) Engage the next progressive hook on latch No. 4 and close the latch.

NOTE: Latch No. 3 will drop open.

4) Alternate between closing latch No. 3 and latch No. 4 on the next progressive hooks until both latches are engaged on the progressive hook nearest the primary hook.
5) Engage the progressive hook on latch No. 5 and close the latch.
6) Engage the primary hook on latch No. 3 and close the latch.
7) Engage the primary hook on latch No. 4 and close the latch.
8) Engage the primary hook on latch No. 5 and close the latch.
9) Engage the primary hook on latch No. 6 and close the latch.
10) Close latch No. 1 and latch No. 2.
11) Close latch No. 7.
12) Close latch No. 8 and latch No. 9.

PTF ALL; AIRPLANES WITH 313A6535-11/-12 SEAL INSTALLED

(h) Make sure that the aft end of the seal at the upper aft end of the thrust reverser is not pinched (Figure 204).

NOTE: The aft end of the seal immediately above the serrated aft edge.

1) If the seal is pinched, use a flexible plastic squeegee to carefully pull it up into its position.

PTF ALL; AIRPLANES WITH 313A6535-15/-16 SEAL INSTALLED

(i) Make sure that the seal sits inside of the thrust reverser (Figure 204).

PTF ALL

SUBTASK 78-31-00-080-009-G00

(2) Remove the opening sling, SPL-14554, from the thrust reverser cowl.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-00-860-137-G00

(1) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

(a) On the applicable engine, close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 78-31-00-440-003-G00

(2) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 78-31-00-440-004-G00

(3) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

SUBTASK 78-31-00-860-244-G00

(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

TASK 78-31-00-980-801-G00

8. Thrust Reverser Operation - Extend (Selection)

A. General

(1) This task gives the mechanic a selection of applicable tasks for thrust reverser extend operation.

B. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

C. Thrust Reverser Operation - Extend (Selection)

SUBTASK 78-31-00-860-013-G00

(1) Do one of these tasks to extend the thrust reverser:

(a) Do this task: Thrust Reverser Operation - Extend (Manual Procedure), TASK 78-31-00-980-803-G00.

(b) Do this task: Thrust Reverser ,Operation - Extend (Power Procedure), TASK 78-31-00-980-805-G00.

——— END OF TASK ———

TASK 78-31-00-980-802-G00

9. Thrust Reverser Operation - Retract (Selection)

A. General

(1) This task gives the mechanic a selection of applicable tasks for thrust reverser retract operation.

B. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

C. Thrust Reverser Operation - Retract (Selection)

SUBTASK 78-31-00-860-014-G00

(1) Do one of these tasks to retract the thrust reverser:

(a) Do this task: Thrust Reverser Operation - Retract (Manual Procedure), TASK 78-31-00-980-804-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Do this task: Thrust Reverser Operation - Retract (Power Procedure), TASK 78-31-00-980-806-G00.

——— END OF TASK ———

TASK 78-31-00-980-803-G00

10. Thrust Reverser Operation - Extend (Manual Procedure)

(Figure 205, Figure 206)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions to manually extend the left or right thrust reverser sleeve.

(2) To engage the manual drive shaft in the sync lock, the lock release pin in the 3/8-inch square opening at the end of the sync lock must be pushed in.

(3) The sync lock is on the lower hydraulic actuator torque box.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-18641	Drive Socket - TRAS, Manual Drive (Fully Depressed) Part #: C78040-10 Supplier: 81205 Opt Part #: C78040-1 Supplier: 88277
STD-858	Tag - DO NOT OPERATE
STD-997	Wrench - Speed, 3/8 Inch Square Drive
STD-6760	Wrench - Standard, 5/8 inch (15.875 mm)
STD-6839	Wrench - Torque, Inspection, 0-150 in-lbs, +/-5% accuracy, Analog or Digital Readout (non-clicker)

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Prepare for the Thrust Reverser Operation

SUBTASK 78-31-00-860-136-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 78-31-00-040-006-G00

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(2) Do the thrust reverser deactivation for ground maintenance (TASK 78-31-00-040-801-G00).

SUBTASK 78-31-00-010-015-G00

(3) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

(a) On the applicable engine, open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 78-31-00-860-247-G00

(4) For Engine 1, make sure that these circuit breakers are open and safety tags are installed:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	5	C00276	ENGINE 1 THRUST REVERSER CONT
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

SUBTASK 78-31-00-860-248-G00

(5) For Engine 2, make sure that these circuit breakers are open and safety tags are installed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK
C	7	C00277	ENGINE 2 THRUST REVERSER CONT

G. Thrust Reverser Operation - Extend (Manual Procedure)

SUBTASK 78-31-00-980-001-G00

WARNING MAKE SURE THAT ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE AREA AFT OF THE APPLICABLE THRUST REVERSER. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

CAUTION STOP THE MANUAL EXTENSION IF THE THRUST REVERSER EXTENDS AT AN ANGLE. THIS IS AN INDICATION THAT THERE ARE SYNC SHAFTS THAT ARE BROKEN OR NOT INSTALLED. IF YOU CONTINUE, DAMAGE TO THE EQUIPMENT CAN OCCUR.

CAUTION DO NOT USE AN EXTENSION BAR OR OTHER TOOL TO MOVE THE UNLOCK HANDLE. IF YOU USE AN EXTENSION BAR OR OTHER TOOLS, DAMAGE TO THE UNLOCK HANDLE OR ACTUATOR CAN OCCUR.

CAUTION MAKE SURE THAT THE THRUST REVERSER SLEEVE DOES NOT HIT THE WING IF THE THRUST REVERSER IS IN THE 45-DEGREE OPEN POSITION. DO NOT EXTEND THE INBOARD SLEEVE. DAMAGE TO THE EQUIPMENT CAN OCCUR.

(1) Do these steps to release the locks on the locking actuators:

(a) Move the manual unlock lever on each locking actuator forward.

(b) Hold the manual unlock lever in the forward position as you push the detent pin in.

(c) Release the manual unlock lever.

SUBTASK 78-31-00-980-002-G00

CAUTION DO NOT HIT THE END OF THE SYNC LOCK OR THE 3/8 INCH DRIVE WRENCH TO PUSH THE LOCK RELEASE PIN. DAMAGE TO THE SYNC LOCK CAN OCCUR.

(2) Unlock the sync lock and engage the manual drive:

(a) If using standard hand tools:

1) Put a standard wrench, STD-6760, or equivalent open end wrench on the hex nut at the end of the sync lock.

2) Put a inspection torque wrench, STD-6839, or equivalent into the 3/8 inch square opening.

NOTE: Use a dial indicator torque wrench to make sure that you do not go over the 50 in-lb (6 N·m) limit.

NOTE: The wrench will go into the opening approximately 0.30 in. (7.62 mm).

3) Apply continuous pressure on the lock release pin with the square drive ratchet.

4) Slowly move the open end wrench a small amount in one or opposite direction until the lock release pin goes in.

(b) If using fully depressed drive socket, SPL-18641:

1) Put a 3/8 inch square drive speed wrench, STD-997, or equivalent into the 3/8 inch square opening in the fully depressed drive socket, SPL-18641.

NOTE: Use a dial indicator torque wrench to make sure that you do not go over the 50 in-lb (6 N·m) limit.

2) Mate the fully depressed drive socket, SPL-18641, with the 5/8 inch hex nut on the end of the sync lock.

3) Apply pressure until the lock release pin goes in.

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a) Move the fully depressed drive socket, SPL-18641, a small amount, in one direction and then the other, to help the lock release pin go in.

NOTE: The 3/8 drive wrench will go into the opening approximately 0.30 in. (7.62 mm).

SUBTASK 78-31-00-860-015-G00

CAUTION

MAKE SURE THAT YOU APPLY CONTINUOUS PRESSURE ON THE SQUARE-DRIVE WRENCH. THIS WILL KEEP THE LOCK RELEASE PIN IN AND THE MANUAL DRIVE FULLY ENGAGED. IF YOU DO NOT, DAMAGE TO THE SYNC LOCK CAN OCCUR.

(3) Turn the inspection torque wrench, STD-6839, with a 6 in. (15 cm) extension to manually extend the thrust reverser sleeve as follows:

NOTE: The sleeve will be in the fully extended position when the override clutch on the manual drive slips.

(a) Turn the inspection torque wrench, STD-6839, no more than 50 in-lb (6 N·m) counter-clockwise to extend the left thrust reverser sleeve.

(b) Turn the inspection torque wrench, STD-6839, no more than 50 in-lb (6 N·m) clockwise to extend the right thrust reverser sleeve.

(c) Make sure that the leading edge flaps are completely retracted.

(d) Monitor the position of the thrust reverser sleeve as you extend it to make sure that it does not touch the wing.

(e) Extend the thrust reverser sleeve to the maximum permitted limit, according to the thrust reverser's open position:

1) If the inboard thrust reverser sleeve is open at the 15-degree position, extend the thrust reverser sleeve no more than 15 in. (381 mm) from the forward edge of the torque box.

2) If the outboard thrust reverser sleeve is open at the 15-degree position, full extension is acceptable.

3) If the inboard thrust reverser sleeve is open at the 37-degree (normal opening) position, do not extend the thrust reverser sleeve.

NOTE: The thrust reverser is too close to the wing leading edge at this position.

4) If the outboard thrust reverser sleeve is open at the 55-degree (normal opening) position, full extension is acceptable.

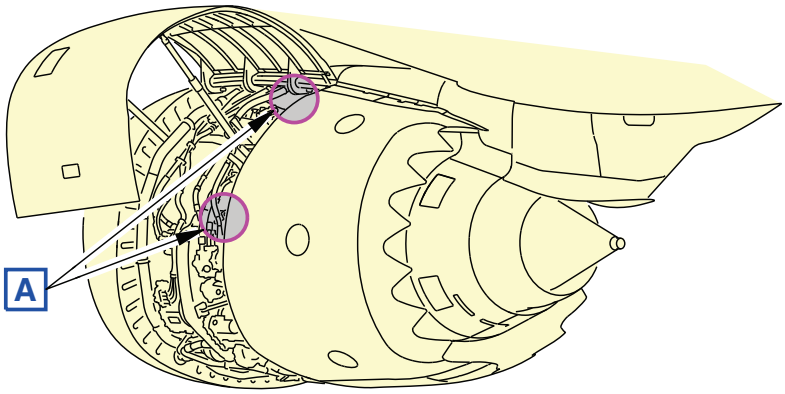
NOTE: Only 15 in. (381 mm) of extension may be needed to get enough access in most cases. Follow the requirements of the specific task you are performing.

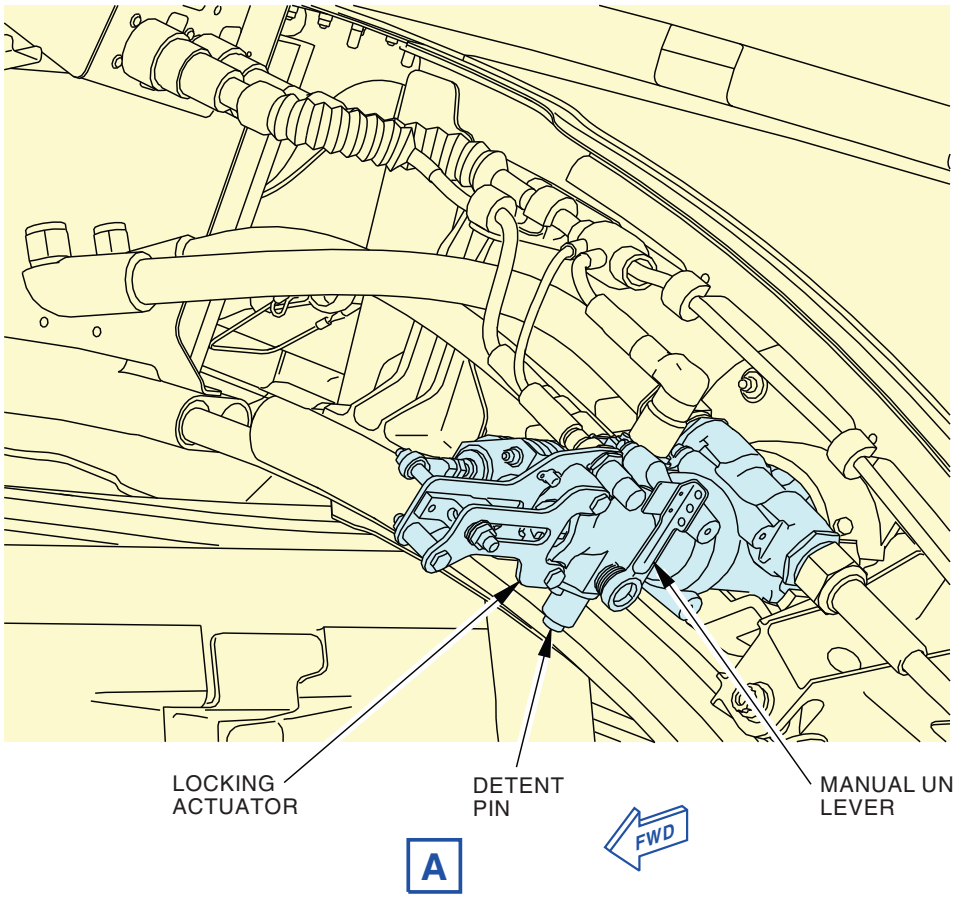
SUBTASK 78-31-00-080-008-G00

(4) Remove the inspection torque wrench, STD-6839, and the extension from the sync lock.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





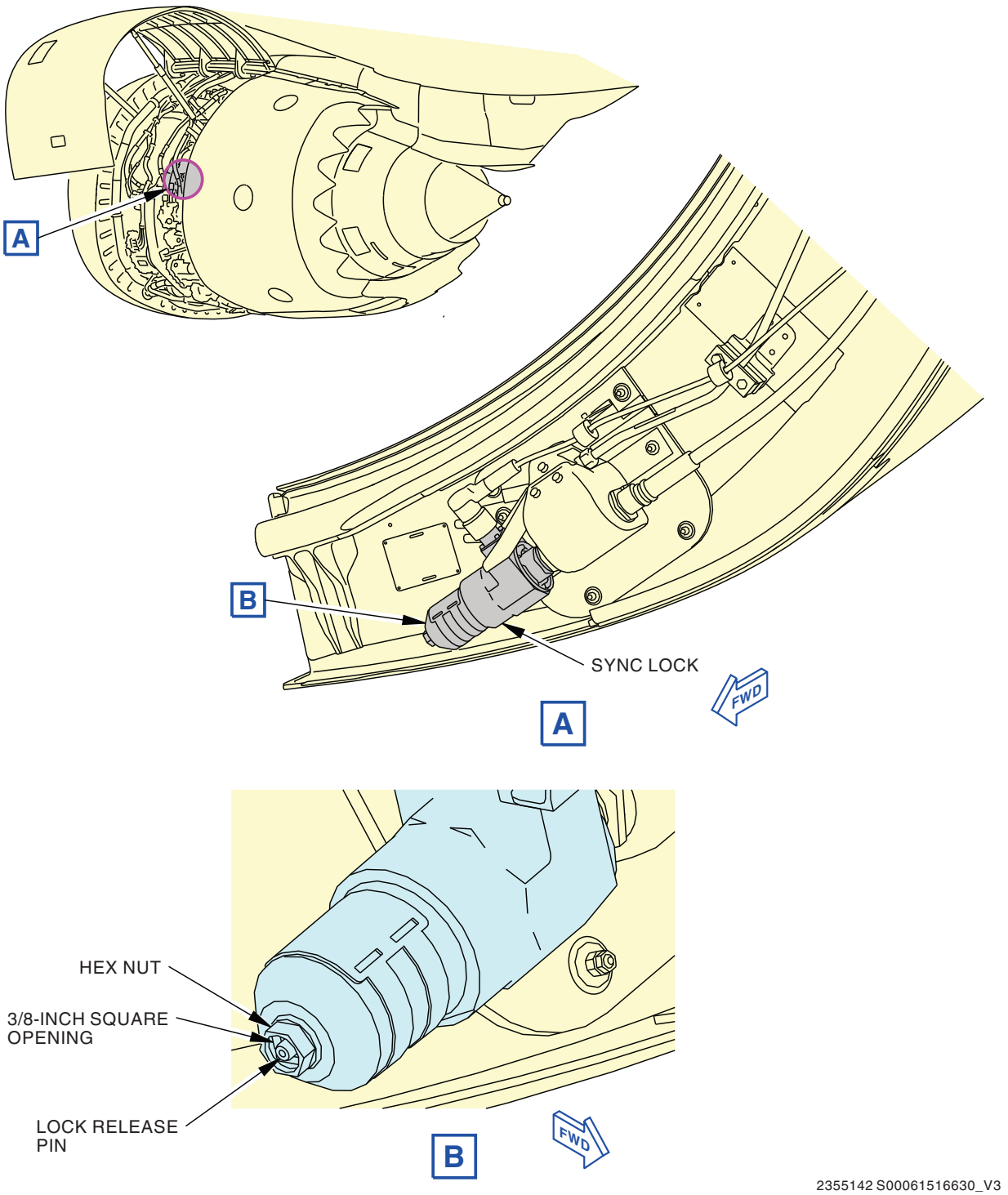
NOTE:

LEFT SIDE IS SHOWN, RIGHT SIDE IS OPPOSITE.

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Manual Unlock of the Locking Actuator
Figure 205/78-31-00-990-808-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Thrust Reverser Sync Lock Manual Drive
Figure 206/78-31-00-990-809-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-31-00-980-804-G00

11. Thrust Reverser Operation - Retract (Manual Procedure)

(Figure 205, Figure 206)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions to manually retract the left or right thrust reverser sleeve on an engine.

(2) To engage the manual drive shaft in the sync lock, the lock release pin in the 3/8-inch square opening at the end of the sync lock must be pushed in.

(3) The sync lock is on the lower hydraulic actuator on the torque box.

(4) Hydraulic power can be necessary to completely retract and lock the thrust reverser.

B. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
71-00-00-700-816-G00	Test No. 38 - Thrust Reverser Cycle Special Functions (P/B 501)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-2418	Lockpin - Equipment, Thrust Reverser Actuator Module Lockout Part #: C78004-1 Supplier: 81205
SPL-18641	Drive Socket - TRAS, Manual Drive (Fully Depressed) Part #: C78040-10 Supplier: 81205 Opt Part #: C78040-1 Supplier: 88277
STD-858	Tag - DO NOT OPERATE
STD-997	Wrench - Speed, 3/8 Inch Square Drive
STD-6760	Wrench - Standard, 5/8 inch (15.875 mm)
STD-6839	Wrench - Torque, Inspection, 0-150 in-lbs, +/-5% accuracy, Analog or Digital Readout (non-clicker)

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

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E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

F. Prepare for the Procedure

SUBTASK 78-31-00-860-016-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 78-31-00-040-011-G00

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(2) Do the thrust reverser deactivation for ground maintenance (TASK 78-31-00-040-801-G00).

SUBTASK 78-31-00-010-016-G00

(3) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

(a) On the applicable engine, open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

G. Thrust Reverser Operation - Retract (Manual Procedure)

SUBTASK 78-31-00-980-003-G00

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE AREA AROUND THE THRUST REVERSERS. THE THRUST REVERSERS EXTEND AND RETRACT QUICKLY. INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION DO NOT APPLY MORE THAN 50 INCH-POUNDS OF TORQUE TO EXTEND OR RETRACT THE THRUST REVERSER. IF YOU DO, DAMAGE TO THE SYNC LOCK CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

DO NOT HIT THE END OF THE SYNC LOCK OR THE 3/8 INCH DRIVE WRENCH TO PUSH THE LOCK RELEASE PIN. DAMAGE TO THE SYNC LOCK CAN OCCUR.

(1) Unlock the sync lock and engage the manual drive as follows:

NOTE: To completely retract and lock the thrust reverser, it can be necessary to use hydraulic power. You can use a wrench to partially retract the thrust reverser; however if the locking actuator does not lock, hydraulic power will be necessary to completely retract and lock the thrust reverser.

(a) If using standard hand tools:

1) Put a standard wrench, STD-6760, or equivalent open end wrench on the hex nut at the end of the sync lock.

2) Put a inspection torque wrench, STD-6839, or equivalent into the 3/8 inch square opening.

NOTE: Use a dial indicator torque wrench to make sure that you do not go over the 50 in-lb (6 N·m) limit.

NOTE: The wrench will go into the opening approximately 0.30 in. (7.62 mm).

3) Apply continuous pressure on the lock release pin with the square drive ratchet.

4) Slowly move the open end wrench a small amount in one or opposite direction until the lock release pin goes in.

(b) If using fully depressed drive socket, SPL-18641:

1) Put a 3/8 inch square drive speed wrench, STD-997, or equivalent into the 3/8 inch square opening in the fully depressed drive socket, SPL-18641.

NOTE: Use a dial indicator torque wrench to make sure that you do not go over the 50 in-lb (6 N·m) limit.

2) Mate the fully depressed drive socket, SPL-18641, with the 5/8 inch hex nut on the end of the sync lock.

3) Apply pressure until the lock release pin goes in.

a) Move the fully depressed drive socket, SPL-18641, a small amount, in one direction and then the other, to help the lock release pin go in.

NOTE: The 3/8 drive wrench will go into the opening approximately 0.30 in. (7.62 mm).

SUBTASK 78-31-00-980-004-G00

CAUTION

MAKE SURE THAT YOU APPLY CONTINUOUS PRESSURE ON THE SQUARE-DRIVE WRENCH. THIS WILL KEEP THE LOCK RELEASE PIN IN AND THE MANUAL DRIVE FULLY ENGAGED. IF YOU DO NOT, DAMAGE TO THE SYNC LOCK CAN OCCUR.

(2) Turn the inspection torque wrench, STD-6839, with a 6 in. (15 cm) extension to manually retract the thrust reverser sleeve as follows:

NOTE: The sleeve will be at the manual track limit when the override clutch on the manual drive slips. If the locking actuator has not locked at this time, hydraulic power will be necessary to completely retract and lock the thrust reverser.

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(a) Turn the inspection torque wrench, STD-6839, no more than 50 in-lb (6 N·m) clockwise to retract the left thrust reverser sleeve.

(b) Turn the inspection torque wrench, STD-6839, no more than 50 in-lb (6 N·m) counter-clockwise to retract the right thrust reverser sleeve.

SUBTASK 78-31-00-080-007-G00

(3) Remove the inspection torque wrench, STD-6839, and the extension from the sync lock.

SUBTASK 78-31-00-860-017-G00

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE AREA AROUND THE THRUST REVERSERS. THE THRUST REVERSERS EXTEND AND RETRACT QUICKLY. INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION DO NOT USE HYDRAULIC POWER TO RETRACT THE THRUST REVERSER IF IT IS OPEN. MONITOR THE POSITION OF THE THRUST REVERSER SLEEVE WHILE YOU MANUALLY RETRACT IT. MAKE SURE THAT THE SLEEVE DOES NOT HIT THE WING. IF IT DOES, DAMAGE CAN OCCUR.

CAUTION MAKE SURE TO EXAMINE THE SYNC LOCK IF ELECTRICAL POWER GOES WHEN THE THRUST REVERSER IS IN OPERATION. DAMAGE TO THE SYNC LOCKS CAN OCCUR.

(4) If it is necessary to use hydraulic power to fully retract the thrust reverser, do these steps:

NOTE: The steps that follow are necessary to unlock the sync locks and subsequently operate the thrust reverser. If the locking actuator is locked, these steps are not necessary, continue to the step for the check of the REVERSER LIMITED light on the engine control panel.

(a) Make sure that the thrust reverser is closed and latched (TASK 78-31-00-010-802-G00).

(b) Pressurize the applicable hydraulic system (TASK 29-11-00-860-801).

NOTE: For Engine 1, pressurize system A. For Engine 2, pressurize system B.

NOTE: Do not remove the hydraulic power prematurely. If you do, it can cause the REVERSER LIMITED and the HYD ISO VALVE SENSOR faults to come on. The hydraulic isolation valve is spring loaded to the closed position. If you remove the hydraulic power, the spring can close the valve. The HYD ISO VALVE SENSOR fault will show if the valve closes while the thrust reverser is extended.

(c) For engine 1, remove the safety tag and close this circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	5	C00276	ENGINE 1 THRUST REVERSER CONT
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

(d) For engine 2, remove the safety tag and close this circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK
C	7	C00277	ENGINE 2 THRUST REVERSER CONT

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(e) Move the reverse thrust lever up and aft to the extended (deployed) position.

NOTE: The reverse thrust lever must be in the extended (deployed) position to unlock the sync locks.

(f) Do this task: Test No. 38 - Thrust Reverser Cycle Special Functions, TASK 71-00-00-700-816-G00.

1) Make sure that the sleeve position match with the reverse thrust lever before you start the thrust reverser special function.

2) Make sure that the thrust reverser is deployed at least 3.0 in. (76.2 mm) before you start the thrust reverser special function with the reverse thrust levers in deployed position.

(g) Remove the lockpin, SPL-2418, from the manual shutoff valve handle on the control valve module.

NOTE: When you remove the lock the lockpin, SPL-2418, the thrust reverser will continue to extend under hydraulic power when the thrust reverser special function is started.

1) Make sure that the manual shutoff valve handle moves counterclockwise to the vertical position.

(h) Move the applicable reverse thrust lever down and forward to retract (stow) the thrust reverser.

(i) Make sure that the applicable REVERSER LIMITED light in the flight compartment is off.

1) If the REVERSER LIMITED light is on, do the applicable fault isolation task in the FIM.

(j) Depressurize the applicable hydraulic system (TASK 29-11-00-860-805).

SUBTASK 78-31-00-200-001-G00

(5) If the locking actuator is locked and hydraulic power is not necessary, do these steps:

(a) Make sure that the applicable REVERSER LIMITED light in the flight compartment is off.

1) If the REVERSER LIMITED light is on, do the applicable fault isolation task in the FIM..

H. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-00-410-007-G00

(1) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

(a) On the applicable engine, close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 78-31-00-440-007-G00

(2) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 78-31-00-860-018-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

TASK 78-31-00-980-805-G00

12. Thrust Reverser ,Operation - Extend (Power Procedure)

A. General

(1) This task gives the instructions to extend the thrust reverser sleeves on the applicable engine with the use of the airplane hydraulic system.

B. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
73-21-00-800-801-G00	EEC Maintenance Power Selection (P/B 201)
78-31-00-700-804-G00	Thrust Reverser Locks Operational Test (P/B 501)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-2418	Lockpin - Equipment, Thrust Reverser Actuator Module Lockout Part #: C78004-1 Supplier: 81205
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

E. Prepare for the Procedure

SUBTASK 78-31-00-210-002-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 78-31-00-860-138-G00

(2) For the applicable engine, open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	1	C04016	ENGINE 1 IGNITION PWR CONT
E	2	C04017	ENGINE 2 IGNITION PWR CONT

SUBTASK 78-31-00-860-151-G00

(3) For the left engine, open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 78-31-00-860-167-G00

(4) For the right engine, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 78-31-00-860-153-G00

(5) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE

SUBTASK 78-31-00-860-154-G00

(6) For engine 1, make sure that these circuit breakers are close.

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	5	C00276	ENGINE 1 THRUST REVERSER CONT
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

SUBTASK 78-31-00-860-155-G00

(7) For engine 2, make sure that these circuit breakers are close.

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	7	C00277	ENGINE 2 THRUST REVERSER CONT

SUBTASK 78-31-00-210-003-G00

(8) Make sure that the applicable thrust reverser is closed (TASK 78-31-00-010-802-G00).

SUBTASK 78-31-00-860-156-G00

(9) Select the left or right Multi-Function Display (MFD).

(a) Push the SYS button on the MFD control panel, P9.

(b) Push the SEL button on SELECTOR knob on the MFD control panel, P9.

SUBTASK 78-31-00-860-149-G00

(10) Select the applicable EEC MAINT POWER to TEST on the MISC SYSTEMS CTRLS page on the MFD (TASK 73-21-00-800-801-G00).

NOTE: The EEC will do a self-test when you set the EEC MAINT POWER to TEST position. Wait for a minimum of 30 seconds to let the EEC complete the test.

SUBTASK 78-31-00-860-157-G00

(11) Set the GND TEST switch on the maintenance BITE panel, P61-4, to the ENABLE position.

SUBTASK 78-31-00-860-158-G00

(12) Set the applicable thrust lever to idle.

SUBTASK 78-31-00-210-005-G00

(13) Make sure that the applicable reverse thrust lever is in the stow position.

SUBTASK 78-31-00-080-005-G00

(14) If it is necessary, remove the lockpin, SPL-2418 from the control valve module (TASK 78-31-00-440-801-G00).

SUBTASK 78-31-00-940-001-G00

(15) Make sure that communication between the person on ground and person at the flight compartment is establish.

SUBTASK 78-31-00-860-159-G00

CAUTION

DO NOT OPERATE THE THRUST REVERSER WITH HYDRAULIC POWER IF ELECTRICAL POWER CAN STOP MOMENTARILY. THE SYNC LOCKS ARE SET TO GO TO THE LOCKED POSITION BY A SPRING WHEN ELECTRICAL POWER IS OFF. IF THE ELECTRICAL POWER STOPS WHEN THE THRUST REVERSER IS IN OPERATION, DAMAGE CAN OCCUR.

(16) Do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

NOTE: For engine 1, pressurize system A. For engine 2, pressurize system B.

F. Thrust Reverser Operation - Extend (Power Procedure)

SUBTASK 78-31-00-860-160-G00

(1) Make these selections on the MFD main function menu:

(a) ONBOARD MAINTENANCE

(b) OTHER FUNCTIONS

(c) SPECIAL FUNCTIONS

(d) 71-80 Left Engine or 71-80 Right Engine

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Thrust Reverser Cycle

(f) CONTINUE.

SUBTASK 78-31-00-860-161-G00

(2) Do the instructions that show on the MFD PRECONDITIONS and then make these selections:

NOTE: Make sure that you verify and follow the inhibited conditions instructions closely. If one of the inhibited conditions is not met, the thrust reverser may not extend.

(a) CONTINUE

(b) START FUNCTION

SUBTASK 78-31-00-860-162-G00

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE AREA AROUND THE THRUST REVERSERS. THE THRUST REVERSERS EXTEND AND RETRACT QUICKLY. INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Extend the applicable thrust reverser as follows:

CAUTION

DO NOT OPERATE THE THRUST REVERSER WITH HYDRAULIC POWER IF ELECTRICAL POWER CAN STOP MOMENTARILY. THE SYNC LOCKS ARE SET TO GO TO THE LOCKED POSITION BY A SPRING WHEN ELECTRICAL POWER IS OFF. IF THE ELECTRICAL POWER STOPS WHEN THE THRUST REVERSER IS IN OPERATION, DAMAGE CAN OCCUR.

(a) Move the applicable reverse thrust lever up and aft to the extend position.

NOTE: If electrical power goes off when the thrust reverser is in operation, the sync lock can shear. Make sure that you do a thrust reverser locks operational test to check for damage to the sync locks (TASK 78-31-00-700-804-G00).

(b) Make sure that the green REV indication shows when the thrust reverser is fully deployed.

SUBTASK 78-31-00-860-187-G00

(4) If it is necessary to retract the thrust reverser to satisfy a step in a separate procedure, do this step:

(a) Wait a minimum of 20 seconds before you move the applicable reverse thrust lever forward and down to the retract (stow) position.

NOTE: This will make sure that the system has sufficient time to detect faults.

SUBTASK 78-31-00-860-163-G00

(5) Select the STOP FUNCTION on the MFD.

SUBTASK 78-31-00-280-001-G00

(6) Do a check of the REVERSER LIMITED light on engine control panel and for TR related status messages.

(a) If the REVERSER LIMITED Light is on, do the applicable IFIM procedure for the thrust reverser system.

(b) If a TR related status message is shown, do the applicable IFIM procedure for the thrust reverser system if it is necessary.

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SUBTASK 78-31-00-860-164-G00

(7) Set the applicable EEC MAINT POWER to NORM on the MISC SYSTEM CTRLS page (TASK 73-21-00-800-801-G00)

SUBTASK 78-31-00-860-165-G00

(8) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 78-31-00-040-012-G00

(9) If it is necessary, do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

NOTE: Deactivation is not necessary if there is no requirement to leave the TR levers in the deployed position.

SUBTASK 78-31-00-860-188-G00

(10) Set the GND TEST switch on the maintenance BITE panel, P61-4, to the NORM position.

SUBTASK 78-31-00-860-168-G00

(11) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE

SUBTASK 78-31-00-860-181-G00

(12) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 78-31-00-860-182-G00

(13) For the applicable engine, remove the safety tags and close these applicable circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	1	C04016	ENGINE 1 IGNITION PWR CONT
E	2	C04017	ENGINE 2 IGNITION PWR CONT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 78-31-00-860-183-G00

(14) For the left engine, remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 78-31-00-860-184-G00

(15) For the right engine, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

——— END OF TASK ———

TASK 78-31-00-980-806-G00

13. Thrust Reverser Operation - Retract (Power Procedure)

A. General

(1) This task gives the instructions to retract the thrust reverser sleeves on the applicable engine with the use of the airplane hydraulic system.

B. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
73-21-00-800-801-G00	EEC Maintenance Power Selection (P/B 201)
78-31-00-700-804-G00	Thrust Reverser Locks Operational Test (P/B 501)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

E. Prepare for the Procedure

SUBTASK 78-31-00-860-029-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 78-31-00-860-169-G00

(2) For the applicable engine, open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	1	C04016	ENGINE 1 IGNITION PWR CONT
E	2	C04017	ENGINE 2 IGNITION PWR CONT

SUBTASK 78-31-00-860-170-G00

(3) For the left engine, open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 78-31-00-860-171-G00

(4) For the right engine, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 78-31-00-860-172-G00

(5) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE

SUBTASK 78-31-00-860-221-G00

(6) For engine 1, make sure that these circuit breakers are close.

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	5	C00276	ENGINE 1 THRUST REVERSER CONT
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

SUBTASK 78-31-00-860-224-G00

(7) For engine 2, make sure that these circuit breakers are close.

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	7	C00277	ENGINE 2 THRUST REVERSER CONT

SUBTASK 78-31-00-860-173-G00

(8) Select the left or right Multi-Function Display (MFD).

(a) Push the SYS button on the MFD control panel, P9.

(b) Push the SEL button on SELECTOR knob on the MFD control panel, P9.

SUBTASK 78-31-00-860-174-G00

(9) Select the applicable EEC MAINT POWER to TEST on the MISC SYSTEMS CTRLS page on the MFD (TASK 73-21-00-800-801-G00).

NOTE: The EEC will do a self-test when you set the EEC MAINT POWER to TEST position. Wait for a minimum of 30 seconds to let the EEC complete the test.

SUBTASK 78-31-00-860-175-G00

(10) Set the GND TEST switch on the maintenance BITE panel, P61-4, to the ENABLE position.

SUBTASK 78-31-00-860-176-G00

(11) Set the applicable thrust lever to idle.

SUBTASK 78-31-00-940-002-G00

(12) Make sure that communication between the person on ground and person at the flight compartment is establish.

SUBTASK 78-31-00-440-008-G00

(13) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 78-31-00-860-031-G00

CAUTION

DO NOT RETRACT THE THRUST REVERSER SLEEVE IF THE THRUST REVERSER IS OPEN. DAMAGE TO THE THRUST REVERSER AND ADJACENT STRUCTURES CAN OCCUR.

(14) Make sure that the thrust reverser is closed and latched (TASK 78-31-00-010-802-G00).

SUBTASK 78-31-00-860-177-G00

(15) Move the applicable reverse thrust lever up and aft to the extend position if necessary.

NOTE: This ensures that the thrust reversers do not move inadvertently to retract.

SUBTASK 78-31-00-860-032-G00

(16) Do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

NOTE: For engine 1, pressurize system A. For engine 2, pressurize system B.

F. Thrust Reverser Operation - Retract (Power Procedure)

SUBTASK 78-31-00-860-222-G00

(1) Make these selections on the MFD main function menu:

(a) ONBOARD MAINTENANCE

(b) OTHER FUNCTIONS

(c) SPECIAL FUNCTIONS

(d) 71-80 Left Engine or 71-80 Right Engine

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Thrust Reverser Cycle

(f) CONTINUE.

SUBTASK 78-31-00-860-223-G00

(2) Do the instructions that show on the MFD PRECONDITIONS and then make these selections:

NOTE: Make sure that you verify and follow the inhibited conditions instructions closely. If one of the inhibited conditions is not met, the thrust reverser may not extend.

(a) CONTINUE

(b) START FUNCTION.

SUBTASK 78-31-00-710-003-G00

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE AREA AROUND THE THRUST REVERSERS. THE THRUST REVERSERS EXTEND AND RETRACT QUICKLY. INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Retract the applicable thrust reverser as follows:

CAUTION

DO NOT OPERATE THE THRUST REVERSER WITH HYDRAULIC POWER IF ELECTRICAL POWER CAN STOP MOMENTARILY. THE SYNC LOCKS ARE SET TO GO TO THE LOCKED POSITION BY A SPRING WHEN ELECTRICAL POWER IS OFF. IF THE ELECTRICAL POWER STOPS WHEN THE THRUST REVERSER IS IN OPERATION, DAMAGE CAN OCCUR.

(a) Move the applicable reverse thrust lever forward and down to the retract (stow) position.

NOTE: If electrical power goes off when the thrust reverser is in operation, the sync lock can shear. Make sure that you do a thrust reverser locks operational test to check for damage to the sync locks (TASK 78-31-00-700-804-G00).

(b) If it is necessary to cycle the thrust reverser, do this step:

1) Wait a minimum of 20 seconds before you move the reverse thrust lever between the extend and retract cycle.

NOTE: This will make sure that the system has sufficient time to detect faults.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-00-860-185-G00

(1) Select the STOP FUNCTION on the MFD.

SUBTASK 78-31-00-210-007-G00

(2) Do a check of the REVERSER LIMITED light on engine control panel and for TR related status messages.

(a) If the REVERSER LIMITED Light is on, do the applicable IFIM procedure for the thrust reverser system.

(b) If a TR related status message is shown, do the applicable IFIM procedure for the thrust reverser system if it is necessary.

SUBTASK 78-31-00-860-186-G00

(3) Set the applicable EEC MAINT POWER to NORM on the MISC SYSTEM CTRLS page (TASK 73-21-00-800-801-G00)

SUBTASK 78-31-00-860-233-G00

(4) Set the GND TEST switch on the maintenance BITE panel, P61-4, to the NORM position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 78-31-00-860-034-G00

(5) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 78-31-00-860-178-G00

(6) For the applicable engine, remove the safety tags and close these applicable circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	1	C04016	ENGINE 1 IGNITION PWR CONT
E	2	C04017	ENGINE 2 IGNITION PWR CONT

SUBTASK 78-31-00-860-179-G00

(7) For the left engine, remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 78-31-00-860-180-G00

(8) For the right engine, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 78-31-00-860-254-G00

(9) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-31-00-040-801-G00

14. Thrust Reverser Deactivation For Ground Maintenance

(Figure 207)

A. General

(1) This task prevents the accidental operation of the thrust reversers on an applicable engine.

(2) To isolate the thrust reverser hydraulic system from the airplane hydraulic system, a lockpin [1] is installed in the manual isolation valve handle [2] and into the control valve module [3].

(3) The two lockpins [1] are in a cotton storage bag and are part of the fly-away kit.

(4) Use this task for ground maintenance only. Do not use this task to deactivate the thrust reversers for a flight dispatch.

B. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
71-00-00-700-816-G00	Test No. 38 - Thrust Reverser Cycle Special Functions (P/B 501)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-2418	Lockpin - Equipment, Thrust Reverser Actuator Module Lockout Part #: C78004-1 Supplier: 81205
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

E. Procedure

SUBTASK 78-31-00-040-007-G00

WARNING USE THIS PROCEDURE FOR GROUND MAINTENANCE ONLY. THE THRUST REVERSER CAN ACCIDENTALLY EXTEND IF YOU USE THIS PROCEDURE TO DO A DEACTIVATION OF THE THRUST REVERSER FOR FLIGHT DISPATCH.

(1) Make sure that the applicable thrust lever is in the idle position.

(a) Attach a DO NOT OPERATE tag, STD-858.

SUBTASK 78-31-00-040-008-G00

(2) Make sure that the applicable reverse thrust lever is forward and down in the retract (stow) position.

(a) Attach a DO NOT OPERATE tag, STD-858.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 78-31-00-860-035-G00

(3) For Engine 1, open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	5	C00276	ENGINE 1 THRUST REVERSER CONT
B	6	C01412	ENGINE 1 THRUST REVERSER INTLK
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

SUBTASK 78-31-00-860-036-G00

(4) For Engine 2, open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK
C	6	C01413	ENGINE 2 THRUST REVERSER INTLK
C	7	C00277	ENGINE 2 THRUST REVERSER CONT

SUBTASK 78-31-00-080-006-G00

(5) Do these steps to install the lockpin, SPL-2418, kit:

NOTE: The lockpins [1] (C78004-2) are contained in lockpin, SPL-2418. The lockpin, SPL-2418, is contained in the fly-away kit (012A8102). The fly-away kit bag is identified with “GROUND SUPPORT EQUIPMENT”.

NOTE: For Engine 1, the control valve module is on the left side of the keel beam [4] in the main gear wheel well.

NOTE: For Engine 2, the control valve module is on the right side of the keel beam [4] in the main gear wheel well.

(a) Turn the manual isolation valve handle [2] clockwise to align the lockpin holes in the manual isolation valve handle [2] and the control valve module [3].

(b) Install the lockpin [1].

1) Make sure that the lockpin [1] is pushed fully in and engages the control valve module [3].

2) Make sure that the “remove before flight” streamer can be seen from the outside of the airplane.

F. Thrust Reverser Deactivation - Tryout

NOTE: This tryout is to make sure that the thrust reverser is in a zero energy state.

SUBTASK 78-31-00-780-001-G00

(1) Do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 78-31-00-860-145-G00

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE APPLICABLE THRUST REVERSER. INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR WHEN THE THRUST REVERSER RETRACTS.

(2) Do this task: Test No. 38 - Thrust Reverser Cycle Special Functions, TASK 71-00-00-700-816-G00

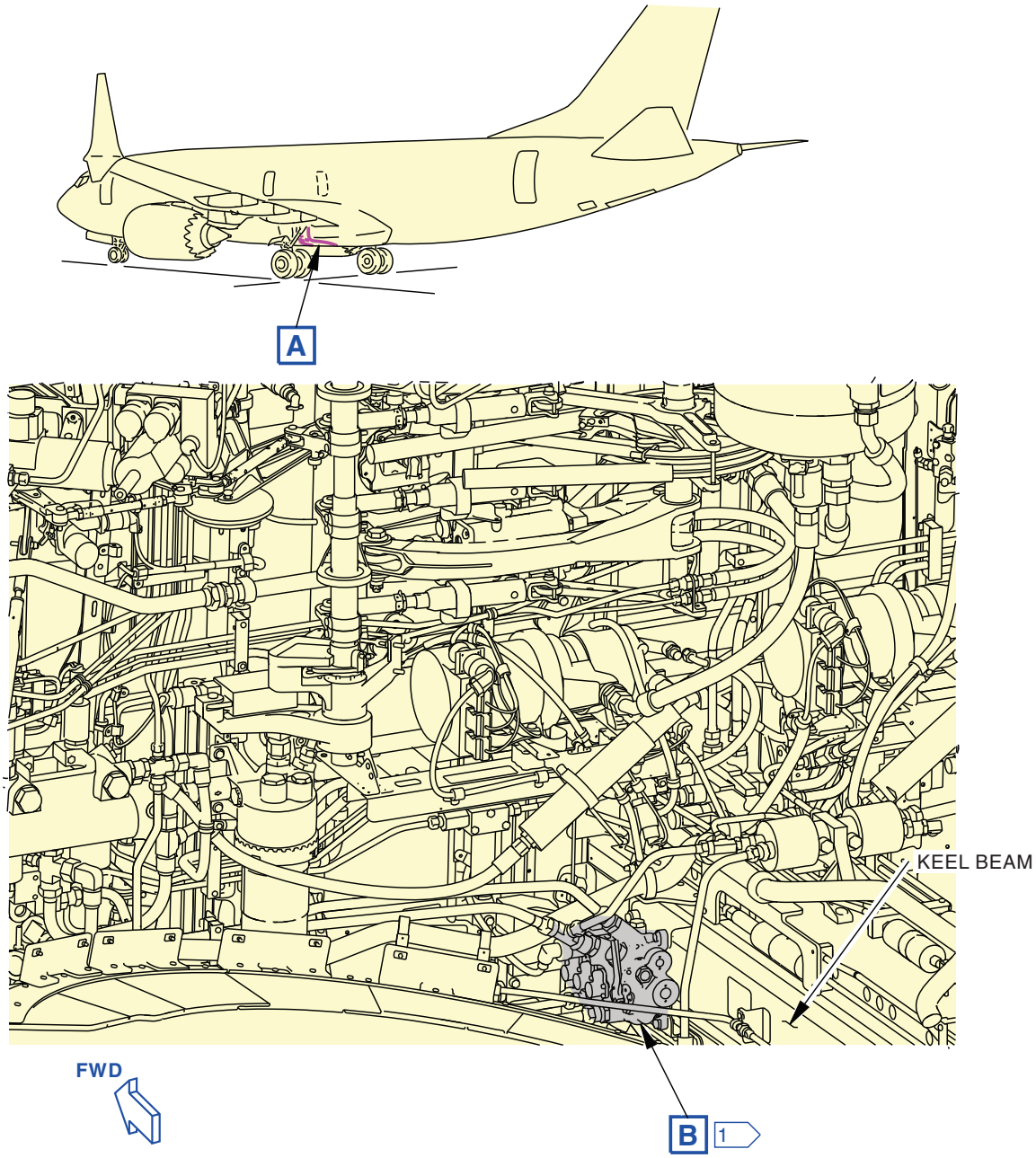
ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the thrust reverser does not move to the extended position when commanded.

NOTE: The ENG X REVERSER status message will be displayed on the MDS, the MAINT light and the REVERSER LIMITED light on the aft P5 overhead panel will illuminate when EEC MAINT POWER function on the MISC SYSTEM CTRLS page is placed in the TEST position.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

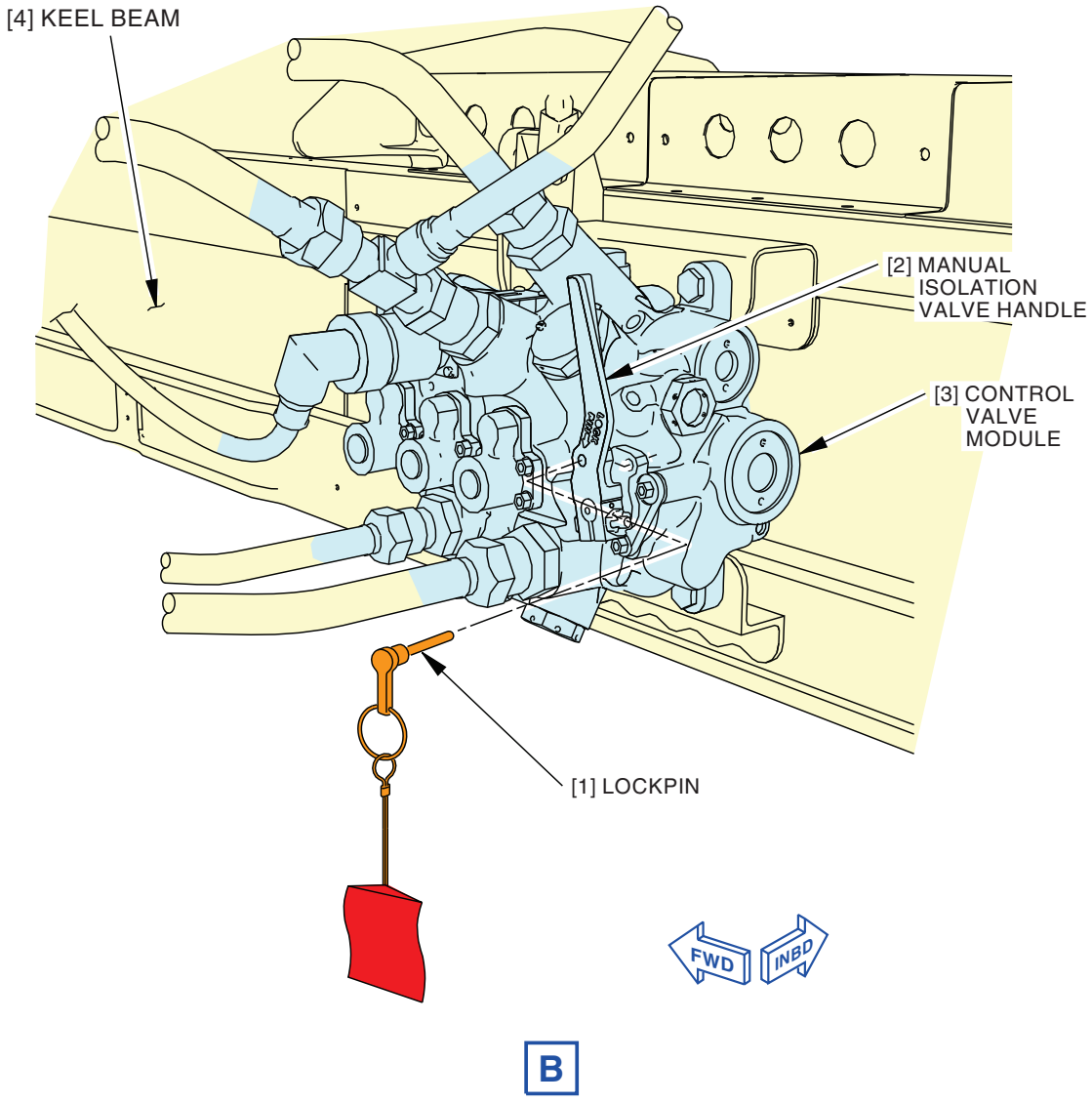


1 ENGINE 1 CONTROL VALVE MODULE IS SHOWN, ENGINE 2 CONTROL VALVE IS ON THE RIGHT SIDE OF THE KEEL BEAM.

2355143 S00061516635_V1

Thrust Reverser Control Valve Module
Figure 207/78-31-00-990-810-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Thrust Reverser Control Valve Module
Figure 207/78-31-00-990-810-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-31-00-440-801-G00

15. Thrust Reverser Activation After Ground Maintenance

(Figure 207)

A. References

Reference	Title
31-65-00-750-801	Fault Interrogation for a Maintenance Message (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-2418	Lockpin - Equipment, Thrust Reverser Actuator Module Lockout Part #: C78004-1 Supplier: 81205

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

D. Procedure

SUBTASK 78-31-00-860-037-G00

(1) For engine 1, make sure that these circuit breakers are open and safety tagged:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	5	C00276	ENGINE 1 THRUST REVERSER CONT
B	6	C01412	ENGINE 1 THRUST REVERSER INTLK
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

SUBTASK 78-31-00-860-038-G00

(2) For engine 2, make sure that these circuit breakers are open and safety tagged:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK
C	6	C01413	ENGINE 2 THRUST REVERSER INTLK
C	7	C00277	ENGINE 2 THRUST REVERSER CONT

SUBTASK 78-31-00-440-009-G00

WARNING

BEFORE YOU REMOVE THE LOCKPIN FROM THE CONTROL VALVE MODULE, MAKE SURE THAT THE POSITION OF THE REVERSE THRUST LEVER AGREES WITH THE POSITION OF THE THRUST REVERSER SLEEVES. IF THE POSITION OF THE REVERSE THRUST LEVER DOES NOT AGREE WITH THE POSITION OF THE THRUST REVERSER SLEEVES, THE THRUST REVERSER SLEEVES COULD EXTEND OR RETRACT WHEN THE HYDRAULIC SYSTEM IS ACTIVATED. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Make sure that the position of the reverse thrust lever agrees with the position of the thrust reverser sleeves.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Remove the lockpin, SPL-2418, kit:

1) Remove the lockpin [1] from the applicable manual isolation valve handle [2] on the control valve module [3].

(b) Make sure that the manual isolation valve handle [2] moves counterclockwise to the vertical position.

(c) Put the lockpin [1] back in the cotton storage bag and store on the airplane.

SUBTASK 78-31-00-860-039-G00

(4) For Engine 1, remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	5	C00276	ENGINE 1 THRUST REVERSER CONT
B	6	C01412	ENGINE 1 THRUST REVERSER INTLK
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

SUBTASK 78-31-00-860-040-G00

(5) For Engine 2, remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK
C	6	C01413	ENGINE 2 THRUST REVERSER INTLK
C	7	C00277	ENGINE 2 THRUST REVERSER CONT

SUBTASK 78-31-00-860-041-G00

(6) Remove the DO-NOT-OPERATE tag from the thrust lever.

SUBTASK 78-31-00-860-042-G00

(7) Remove the DO-NOT-OPERATE tag from the reverse thrust lever.

NOTE: If the EEC is powered when the ENGINE 1 (2) THRUST REVERSER CONT circuit breaker is open, a status message will be set (i.e., ENG 1 REVERSER). Refer to Fault Interrogation for a Maintenance Message, TASK 31-65-00-750-801 to unlatch messages.

——— END OF TASK ———

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THRUST REVERSER SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) Thrust Reverser Operational Test (Selection)

(2) Thrust Reverser System Normal Operation Test

(3) Thrust Reverser Operation Test (Standby Hydraulic System)

(4) Thrust Reverser Locks Operational Test

(5) Thrust Reverser Proximity Sensors Operational Test

(6) Thrust Reverser Annunciator Lights Operational Test

(7) Scheduled Maintenance Task for EEC Faults - Reverser Sensor.

TASK 78-31-00-700-801-G00

2. Thrust Reverser Operational Test (Selection)

A. General

(1) This task is for the selection of the operational test for the thrust reverser system.

B. Procedure

SUBTASK 78-31-00-710-004-G00

(1) Do one of these tasks to do the operational test of the thrust reverser:

(a) Do this task: Thrust Reverser System Normal Operational Test, TASK 78-31-00-700-802-G00.

(b) Do this task: Thrust Reverser Operational Test (Standby Hydraulic System), TASK 78-31-00-700-803-G00.

(c) Do this task: Thrust Reverser Locks Operational Test, TASK 78-31-00-700-804-G00.

(d) Do this task: Thrust Reverser Proximity Sensors Operational Test, TASK 78-31-00-700-805-G00.

——— END OF TASK ———

TASK 78-31-00-700-802-G00

3. Thrust Reverser System Normal Operational Test

(Figure 501)

A. General

(1) This task gives the instructions to do a test of the thrust reverser system for correct operation.

(2) The movements of the thrust lever should be slow, deliberate and smooth whenever possible.

(3) Two persons are necessary for this task, one in the flight deck and one at the engine thrust reversers.

(4) This task is the same for engines 1-2 thrust reversers.

(5) Thrust reverser faults may be inhibited or show “NOT ACTIVE” when the Thrust Reverser Cycle SPECIAL FUNCTION is stopped. Make sure that you verify and follow the operational test instructions closely.

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B. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
31-65-00-750-801	Fault Interrogation for a Maintenance Message (P/B 201)
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-02-700-805	Existing Faults Check (P/B 201)
71-00-00-700-816-G00	Test No. 38 - Thrust Reverser Cycle Special Functions (P/B 501)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-21-00-800-801-G00	EEC Maintenance Power Selection (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205
STD-858	Tag - DO NOT OPERATE
STD-1139	Timer - stopwatch, accurate to within 1 second

D. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

E. Prepare for the Test

SUBTASK 78-31-00-860-043-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 78-31-00-860-044-G00

(2) For the applicable engine, open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT

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(Continued)

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	1	C04016	ENGINE 1 IGNITION PWR CONT
E	2	C04017	ENGINE 2 IGNITION PWR CONT

SUBTASK 78-31-00-860-045-G00

(3) For the left engine, open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 78-31-00-860-046-G00

(4) For the right engine, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 78-31-00-860-047-G00

(5) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE

SUBTASK 78-31-00-480-003-G00

(6) Do these steps to use the universal maintenance device, SPL-11321, to get access to the existing faults function:

(a) Connect the universal maintenance device, SPL-11321 (TASK 46-13-00-480-801).

(b) Use the universal maintenance device, SPL-11321, to get access to the ONS maintenance browser.

NOTE: The maintenance device is required to view ACTIVE faults during Thrust Reverser Cycle SPECIAL FUNCTION.

SUBTASK 78-31-00-860-048-G00

(7) Make sure that the applicable thrust reverser is closed.

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

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SUBTASK 78-31-00-860-049-G00

WARNING DO NOT OPERATE THE ENGINE IN REVERSE THRUST WHILE THE FAN COWL PANELS ARE OPEN. INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(8) Make sure that the applicable fan cowl panels are closed and latched (TASK 71-11-04-410-801-G00).

SUBTASK 78-31-00-860-050-G00

(9) Select the left or right Multi-Function Display (MFD).

(a) Push the SYS button on the MFD control panel, P9.

(b) Push the SEL button on SELECTOR knob on the MFD control panel, P9.

SUBTASK 78-31-00-860-239-G00

(10) Set the applicable EEC MAINT POWER switch on the MISC SYSTEM CTRLS page on the MFD to the TEST position (TASK 73-21-00-800-801-G00).

NOTE: The EEC will do a self-test when you set the EEC MAINT POWER switch to the TEST position. Wait for a minimum of 30 seconds to let the EEC complete the test.

SUBTASK 78-31-00-860-238-G00

(11) Set the GND TEST switch on the maintenance BITE panel, P61-4, to the ENABLE position.

(a) Use the DUAL MODE function on the MFD while the Thrust Reverser Cycle SPECIAL FUNCTION is in operation.

NOTE: The DUAL MODE lets the two screens monitor the EPCS pages while the Thrust Reverser Cycle SPECIAL FUNCTION is in operation.

SUBTASK 78-31-00-860-053-G00

(12) Set the applicable thrust lever to idle.

SUBTASK 78-31-00-860-054-G00

(13) Make sure that the applicable reverse thrust lever is in the stow position.

SUBTASK 78-31-00-860-240-G00

(14) Select MAINT CTRL PGS and then LATCHED MESSAGE ERASE. If the status message shows on the ERASABLE STATUS MESSAGES display, then do this step:

(a) Select ERASE to remove the message from the status display.

SUBTASK 78-31-00-860-056-G00

(15) Do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

F. Thrust Reverser System Normal Operational Test

SUBTASK 78-31-00-710-005-G00

(1) Use the Multi-Function Display (MFD) screen to do the test of the applicable thrust reverser cycle (TASK 71-00-00-700-816-G00):

(a) Make these selections on the MFD main function menu:

1) ONBOARD MAINTENANCE
2) OTHER FUNCTIONS
3) SPECIAL FUNCTIONS
4) 71-80 Left Engine or 71-80 Right Engine
5) Thrust Reverser Cycle

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6) CONTINUE.

(b) Do the instructions that show on the MFD PRECONDITIONS and then make these selections:

NOTE: Make sure that you verify and follow the inhibited conditions instructions closely. If one of the inhibited conditions is not met, the test will be failed.

1) CONTINUE

2) START FUNCTION.

(c) Do these steps to test of the thrust reverser deploy operation:

CAUTION

OBEY THESE PRECAUTIONS. IF YOU IGNORE THESE PRECAUTIONS, DAMAGE TO EQUIPMENT WILL OCCUR.

1) Make sure that communication between person on ground and person at the flight compartment is establish.

2) Use a stopwatch, STD-1139, to record the time period (in seconds).

NOTE: Record the number of seconds for the amber REV indicator on the Center Display Panel (CDP) to appear and then change to a green REV indication as the thrust reverser deploys

WARNING

MAKE SURE THAT PERSONNEL STAY AWAY FROM THE THRUST REVERSER WHEN YOU MOVE THE CONTROLS. THE THRUST REVERSER CAN MOVE VERY QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL.

3) Move the applicable reverse thrust lever up to the deploy position.

NOTE: The two sleeve halves of each thrust reverser are independent of one another. The two sleeves do not have to move in sync.

4) Make sure that the duration time of the amber REV indication is less than three seconds.

5) Make sure that the green REV indication shows when the thrust reverser is fully deployed.

6) Select the EPCS page 1 from the Maintenance Data Pages on the MFD.

7) Make sure that the EPCS page 1 shows T/R SLEEVE L and T/R SLEEVE R from 95.0 to 105.0 percent.

NOTE: Wait for a minimum of 20 seconds to let the maintenance messages to become active and trigger the status messages and flight deck lights.

8) Make sure that the Reverser Limited light on the aft overhead P5 panel is off.

NOTE: The Reverser Limited light will turn on if there are T/R related maintenance messages for ATA chapter 78.

9) If the Reverser Limited light is on, use the maintenance device to view "ACTIVE" faults when Thrust Reverser is Deployed as follows:

a) Do this task: Existing Faults Check, TASK 46-13-02-700-805.

b) If there are thrust reverser related status faults or SMT messages are LATCHED during the check, do this task: (Fault Interrogation for a Maintenance Message, TASK 31-65-00-750-801).

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(d) Do these steps to test the thrust reverser stow operation:

1) Use a stopwatch, STD-1139, to record the duration (in seconds) that the amber REV indication shows.

WARNING MAKE SURE THAT PERSONNEL STAY AWAY FROM THE THRUST REVERSER WHEN YOU MOVE THE CONTROLS. THE THRUST REVERSER CAN MOVE VERY QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL.

2) Move the applicable reverse thrust lever to the stow position.

NOTE: The two sleeve halves of each thrust reverser are independent of one another. The two sleeves do not have to move in sync.

3) Make sure that the time for the thrust reverser to stow is less than five seconds.

4) Make sure that the REV indication does not show on the CDP when the thrust reverser is fully stowed.

5) Select the EPCS page 1 from the Maintenance Data Pages on the MFD.

6) Make sure that the EPCS page 1 shows T/R SLEEVE L and T/R SLEEVE R from -4.0 to +4.0 percent.

NOTE: Wait for a minimum of 20 seconds to let the maintenance messages to become active and trigger the status messages and flight deck lights.

7) Make sure that the Reverser Limited light on the aft overhead P5 panel is off.

NOTE: The Reverser Limited light will turn on if there are T/R related maintenance messages for ATA chapter 78.

8) If the Reverser Limited light is on, use the maintenance device to view "ACTIVE" faults as follows:

a) Do this task: Existing Faults Check, TASK 46-13-02-700-805.

b) If there are thrust reverser related status faults or SMT messages are LATCHED during the check, do this task: (Fault Interrogation for a Maintenance Message, TASK 31-65-00-750-801).

(e) Extend and retract the thrust reverser five more times.

NOTE: You do not have to check the T/R sleeve percentage on the EPCS page.

1) Make sure that the thrust reverser operates smoothly.

2) Wait 10 seconds before you move the reverse thrust lever between the deploy and stow cycle.

(f) After you deployed and stowed the thrust reversers, select the STOP FUNCTION to complete the test.

(g) Make sure that there are no thrust reverser related status or SMT messages.

NOTE: Thrust reverser maintenance messages may show "NOT ACTIVE" when you stop the Thrust Reverser Cycle SPECIAL FUNCTION.

1) If new thrust reverser related status faults or SMT messages are LATCHED during the check, do this task: Thrust Reverser Operational Test (Selection), TASK 78-31-00-700-801-G00.

(h) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

(i) Examine the thrust reverser area for hydraulic fluid leaks.

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Table 501/78-31-00-993-802-G00

Normal Operation Limits	Dispatch Limits to Avoid Delay
8 drops per minute (stopped or in operation)	30 drops per minute (stopped or in operation)

G. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-00-860-057-G00

(1) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 78-31-00-860-058-G00

(2) Set the applicable EEC MAINT POWER switch on the MISC SYSTEM CTRLS page on the MFD to the NORM position (TASK 73-21-00-800-801-G00).

SUBTASK 78-31-00-860-059-G00

(3) Set the GND TEST switch on the maintenance BITE panel, P61-4, to the NORM position.

SUBTASK 78-31-00-860-060-G00

(4) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE

SUBTASK 78-31-00-860-061-G00

(5) For the left engine, remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 78-31-00-860-062-G00

(6) For the right engine, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 78-31-00-860-063-G00

(7) For the applicable engine, remove the safety tags and close these applicable circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	1	C04016	ENGINE 1 IGNITION PWR CONT

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(Continued)

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	2	C04017	ENGINE 2 IGNITION PWR CONT

SUBTASK 78-31-00-860-231-G00

(8) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

(a) On the applicable engine, close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 78-31-00-860-064-G00

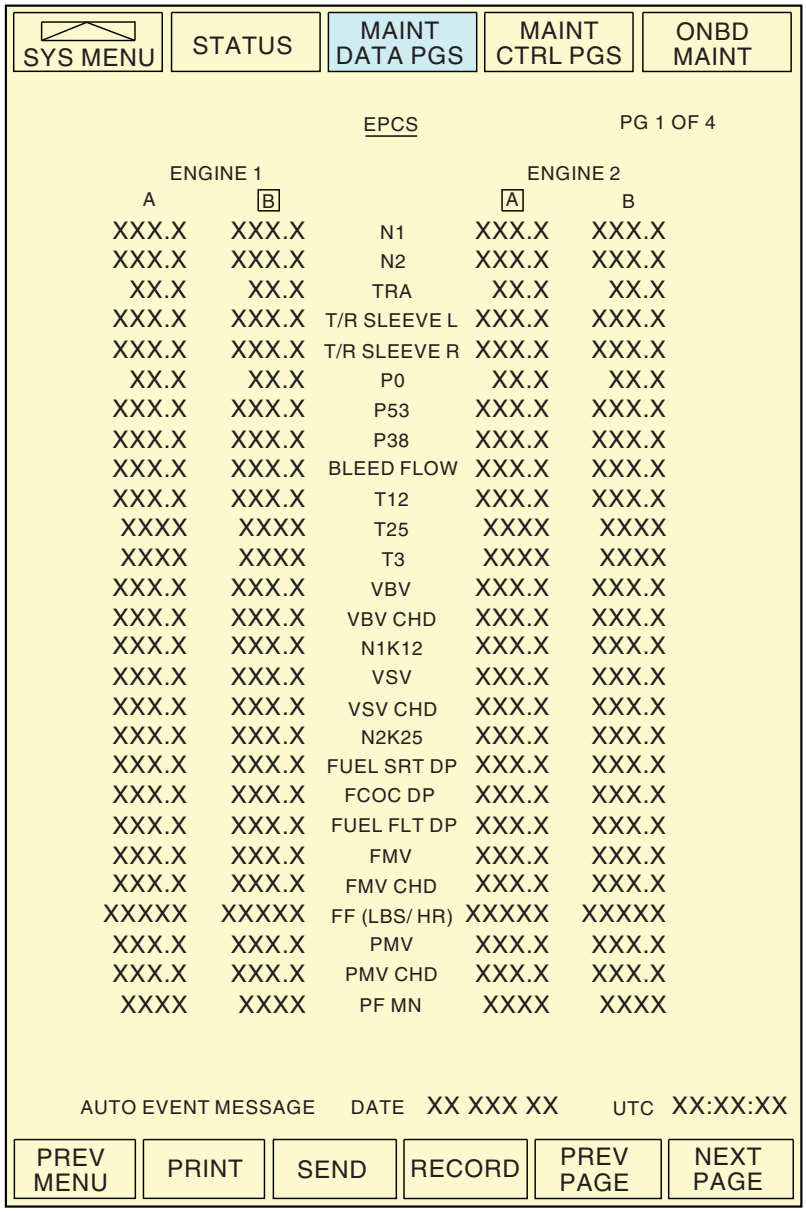
(9) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

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EPCS Page 1 Status Display
Figure 501/78-31-00-990-818-G00

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TASK 78-31-00-700-803-G00

4. Thrust Reverser Operational Test (Standby Hydraulic System)

(Figure 502 and Figure 503)

A. General

(1) This task gives the instructions to do a test of the thrust reverser system for correct operation.

(2) The movements of the thrust lever should be slow, deliberate and smooth whenever possible.

(3) Two persons are necessary to do this task, one in the flight deck and one at the engine thrust reversers.

(4) This task is the same for Engine 1 and Engine 2 thrust reversers.

(5) For Engine 1, when you use the standby hydraulic system, you must connect the return line from hydraulic reservoir A to a hydraulic bench or hydrant, or drain the hydraulic system A reservoir. When the test is completed, fill hydraulic reservoir B. The hydraulic fluid that is used to extend (deploy) the thrust reverser is supplied from the standby hydraulic reservoir and the standby reservoir fills from reservoir B. When the thrust reverser retracts (stows), the hydraulic fluid returns to hydraulic reservoir A.

(6) For Engine 2, when you use the standby hydraulic system, it is not necessary to drain hydraulic reservoir B. The standby hydraulic system fills from reservoir B and returns to reservoir B.

B. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
46-13-02-700-806	Fault History Check (P/B 201)
71-00-00-700-816-G00	Test No. 38 - Thrust Reverser Cycle Special Functions (P/B 501)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-21-00-800-801-G00	EEC Maintenance Power Selection (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE
STD-1139	Timer - stopwatch, accurate to within 1 second

D. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

E. Prepare for the Test

SUBTASK 78-31-00-860-065-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

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(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 78-31-00-860-066-G00

(2) For the applicable engine, open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	1	C04016	ENGINE 1 IGNITION PWR CONT
E	2	C04017	ENGINE 2 IGNITION PWR CONT

SUBTASK 78-31-00-860-067-G00

(3) For the left engine, open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 78-31-00-860-068-G00

(4) For the right engine, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 78-31-00-860-069-G00

(5) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE

SUBTASK 78-31-00-860-070-G00

(6) Make sure that the applicable thrust reverser is closed and latched.

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

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SUBTASK 78-31-00-860-071-G00

WARNING DO NOT OPERATE THE ENGINE IN REVERSE THRUST WHILE THE FAN COWL PANELS ARE OPEN. INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(7) Make sure that the applicable fan cowl panels are closed and latched.

(a) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

SUBTASK 78-31-00-860-072-G00

(8) Select the left or right Multi-Function Display (MFD).

(a) Push the SYS button, on the P9 MFD control panel.

(b) Push the SEL button on SELECTOR knob, on the P9 MFD control panel.

SUBTASK 78-31-00-860-073-G00

(9) Set the applicable EEC MAINT POWER switch on the MISC SYSTEM CTRLS page on the MFD to the TEST position (TASK 73-21-00-800-801-G00).

NOTE: The EEC will do a self-test when you set the EEC MAINT POWER switch to the TEST position. Wait for a minimum of 30 seconds to let the EEC complete the test.

SUBTASK 78-31-00-860-074-G00

(10) Set the GND TEST switch, on the P61-4 maintenance BITE panel, to the ENABLE position.

SUBTASK 78-31-00-860-075-G00

(11) Set the applicable thrust lever to idle.

SUBTASK 78-31-00-860-076-G00

(12) Make sure that the applicable reverse thrust lever is in the stow position.

SUBTASK 78-31-00-860-078-G00

(13) Make sure that the LE STANDBY DRIVE SHUTOFF VALVE is in the No. 1 position.

SUBTASK 78-31-00-860-139-G00

(14) For the Engine 1 thrust reverser, do these steps:

(a) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

(b) For system A reservoir, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

(c) To prevent the leakage of hydraulic fluid when you do the test, connect the system A return line at the system A reservoir to a hydraulic bench or hydrant.

(d) An alternate task to prevent the leakage of hydraulic fluid, is to use the sampling valve to drain the system A reservoir.

NOTE: The reservoir hydraulic level will show on the flight compartment HYDRAULIC gage, on the P9 MFD control panel.

1) If you drain the system A reservoir to the REFILL level, you can operate the thrust reverser for one cycle.

2) If you drain the system A reservoir to the HALF FULL level, you can operate the thrust reverser for three cycles.

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SUBTASK 78-31-00-860-080-G00

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(15) Do these steps to supply hydraulic power for the standby system:

(a) For the Engine 1 thrust reverser, move the FLT CONTROL switch A to the STBY RUD position or the ALTERNATE FLAPS switch to the ARM position.

NOTE: These switches are, on the P5 overhead flight control panel.

(b) For the Engine 2 thrust reverser, move the FLT CONTROL switch B to the STBY RUD position or the ALTERNATE FLAPS switch to the ARM position.

NOTE: These switches are, on the P5 overhead flight control panel.

F. Thrust Reverser Operational Test (Standby Hydraulic System)

SUBTASK 78-31-00-710-006-G00

(1) Use the Multi-Function Display (MFD) screen to do the test of the applicable thrust reverser cycle (TASK 71-00-00-700-816-G00):

(a) Make these selections on the MFD main function menu:

1) ONBOARD MAINTENANCE
2) OTHER FUNCTIONS
3) SPECIAL FUNCTIONS
4) 71-80 Left Engine or 71-80 Right Engine
5) Thrust Reverser Cycle
6) CONTINUE.

(b) Do the instructions that show on the MFD PRECONDITIONS and then make these selections:

NOTE: Make sure that you verify and follow the inhibited conditions instructions closely. If one of the inhibited conditions is not met, the test will be failed.

1) CONTINUE
2) START FUNCTION.

(c) Do these steps to test of the thrust reverser deploy operation:

CAUTION

OBEY THESE PRECAUTIONS. IF YOU IGNORE THESE PRECAUTIONS, DAMAGE TO EQUIPMENT WILL OCCUR.

1) Make sure that communication between person on ground and person at the flight compartment is establish.
2) Use a stopwatch, STD-1139, to record the time period (in seconds) for the amber REV indicator on the Center Display Panel (CDP) to appear and then change to a green REV indication as the thrust reverser deploys.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING MAKE SURE THAT PERSONNEL STAY AWAY FROM THE THRUST REVERSER WHEN YOU MOVE THE CONTROLS. THE THRUST REVERSER CAN MOVE VERY QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL.

3) Move the applicable reverse thrust lever up to the deploy position.

NOTE: The two sleeve halves of each thrust reverser are independent of one another. The two sleeves do not have to move in sync.

4) Make sure that the duration time of the amber REV indication is less than five seconds.

5) Make sure that the green REV indication appears when the thrust reverser is fully deployed.

6) Select the STOP FUNCTION on the MFD.

NOTE: Wait at least 20 seconds to allow faults to trigger before you exit the special function.

NOTE: If you use the maintenance laptop to monitor the faults, you do not have to exit the thrust reverser special function.

7) Select the EPCS page 1 from the Maintenance Data Pages on the MFD.

8) Make sure that the EPCS page 1 shows T/R SLEEVE L and T/R SLEEVE R from 95.0 to 105.0 percent.

9) Make sure that no MDS T/R status messages are shown, except for green REV.

NOTE: Wait for a minimum of 20 seconds to let the maintenance messages to become active and trigger the status messages and flight deck lights.

10) Make sure that the Reverser Limited light, on the P5 aft overhead panel, is off.

NOTE: The Reverser Limited light will turn on if there are T/R related maintenance messages for ATA chapter 78.

11) If there are thrust reverser related MDS messages, do these steps:

a) Erase the thrust reverser related MDS status messages that are currently shown and make sure that they do not appear again.

b) If the MDS status message can not be erased then continue to the steps below to check for the maintenance messages.

c) If not already done, select the STOP FUNCTION on the MFD.

NOTE: If you use the maintenance laptop to monitor the faults, you do not have to exit the thrust reverser special function.

d) Do this task: Fault History Check, TASK 46-13-02-700-806.

(d) Start the T/R cycle special function (TASK 71-00-00-700-816-G00).

(e) Do the steps that follow to do a test of the thrust reverser stow operation:

1) Use a stopwatch, STD-1139, to record the duration (in seconds) that the amber REV indication shows.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING MAKE SURE THAT PERSONNEL STAY AWAY FROM THE THRUST REVERSER WHEN YOU MOVE THE CONTROLS. THE THRUST REVERSER CAN MOVE VERY QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL.

2) Move the applicable reverse thrust lever to the stow position.

NOTE: The two sleeve halves of each thrust reverser are independent of one another. The two sleeves do not have to move in sync.

3) Make sure that the time for the thrust reverser to stow is less than nine seconds.

4) Make sure that the REV indication does not show on the CDP when the thrust reverser is fully stowed.

5) Select the STOP FUNCTION on the MFD.

NOTE: If you use the maintenance laptop to monitor the faults, you do not have to exit the thrust reverser special function.

6) Select the EPCS page 1 from the Maintenance Data Pages on the MFD.

7) Make sure that the EPCS page 1 shows T/R SLEEVE L and T/R SLEEVE R from -4.0 to +4.0 percent.

8) Make sure that no MDS T/R status messages are shown.

NOTE: Wait for a minimum of 20 seconds to let the maintenance messages to become active and trigger the status messages and flight deck lights.

9) Make sure that the Reverser Limited light, on the P5 aft overhead panel, is off.

NOTE: The Reverser Limited light will turn on if there are T/R related maintenance messages for ATA chapter 78.

10) If there are thrust reverser related MDS messages, do these steps:

a) Erase the thrust reverser related MDS status messages that are currently shown and make sure that they do not appear again.

b) If the MDS status message can not be erased then continue to the steps below to check for the maintenance messages.

c) If not already done, select the STOP FUNCTION on the MFD.

NOTE: If you use the maintenance laptop to monitor the faults, you do not have to exit the thrust reverser special function.

d) Do this task: Fault History Check, TASK 46-13-02-700-806.

(f) Start the T/R cycle special function (TASK 71-00-00-700-816-G00).

(g) Move the applicable reverse thrust lever to the deploy position.

(h) Move the applicable reverse thrust lever to the stow position.

(i) Extend and retract the thrust reverser five times.

NOTE: You do not have to check the T/R sleeve percentage on the EPCS page.

1) Make sure that the thrust reverser operates smoothly.

2) Wait 10 seconds before you move the reverse thrust lever between the deploy and stow cycle.

(j) After you deployed and stowed the thrust reversers, select the STOP FUNCTION to complete the test.

(k) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

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(l) Examine the thrust reverser area for hydraulic fluid leaks.

Table 502/78-31-00-993-803-G00

Normal Operation Limits	Dispatch Limits to Avoid Delay
8 drops per minute (stopped or in operation)	30 drops per minute (stopped or in operation)

G. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-00-860-082-G00

(1) Do these steps to remove hydraulic power for the standby system:

(a) For the Engine 1 thrust reverser, move the FLT CONTROL switch A to the OFF position or the ALTERNATE FLAPS switch to the OFF position.

(b) For the Engine 2 thrust reverser, move the FLT CONTROL switch B to the OFF position or the ALTERNATE FLAPS switch to the OFF position.

SUBTASK 78-31-00-860-141-G00

(2) If system A or B return was connected to a hydraulic bench or hydrant, re-connect the return line to the system A or B reservoir.

SUBTASK 78-31-00-860-142-G00

(3) Do a check of the system A and B hydraulic reservoir gages.

(a) To fill the hydraulic reservoirs, do this task: Hydraulic Reservoir Servicing, TASK 12-12-00-610-801.

SUBTASK 78-31-00-860-083-G00

(4) Set the applicable EEC MAINT POWER switch on the MISC SYSTEM CTRLS page on the MFD to the NORM position (TASK 73-21-00-800-801-G00).

SUBTASK 78-31-00-860-084-G00

(5) Set the GND TEST switch, on the P61-4 maintenance BITE panel, to the NORM position.

SUBTASK 78-31-00-860-085-G00

(6) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE

SUBTASK 78-31-00-860-086-G00

(7) For the left engine, remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 78-31-00-860-087-G00

(8) For the right engine, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

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SUBTASK 78-31-00-860-088-G00

(9) For the applicable engine, remove the safety tags and close these applicable circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	1	C04016	ENGINE 1 IGNITION PWR CONT
E	2	C04017	ENGINE 2 IGNITION PWR CONT

SUBTASK 78-31-00-860-232-G00

(10) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

(a) On the applicable engine, close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 78-31-00-860-089-G00

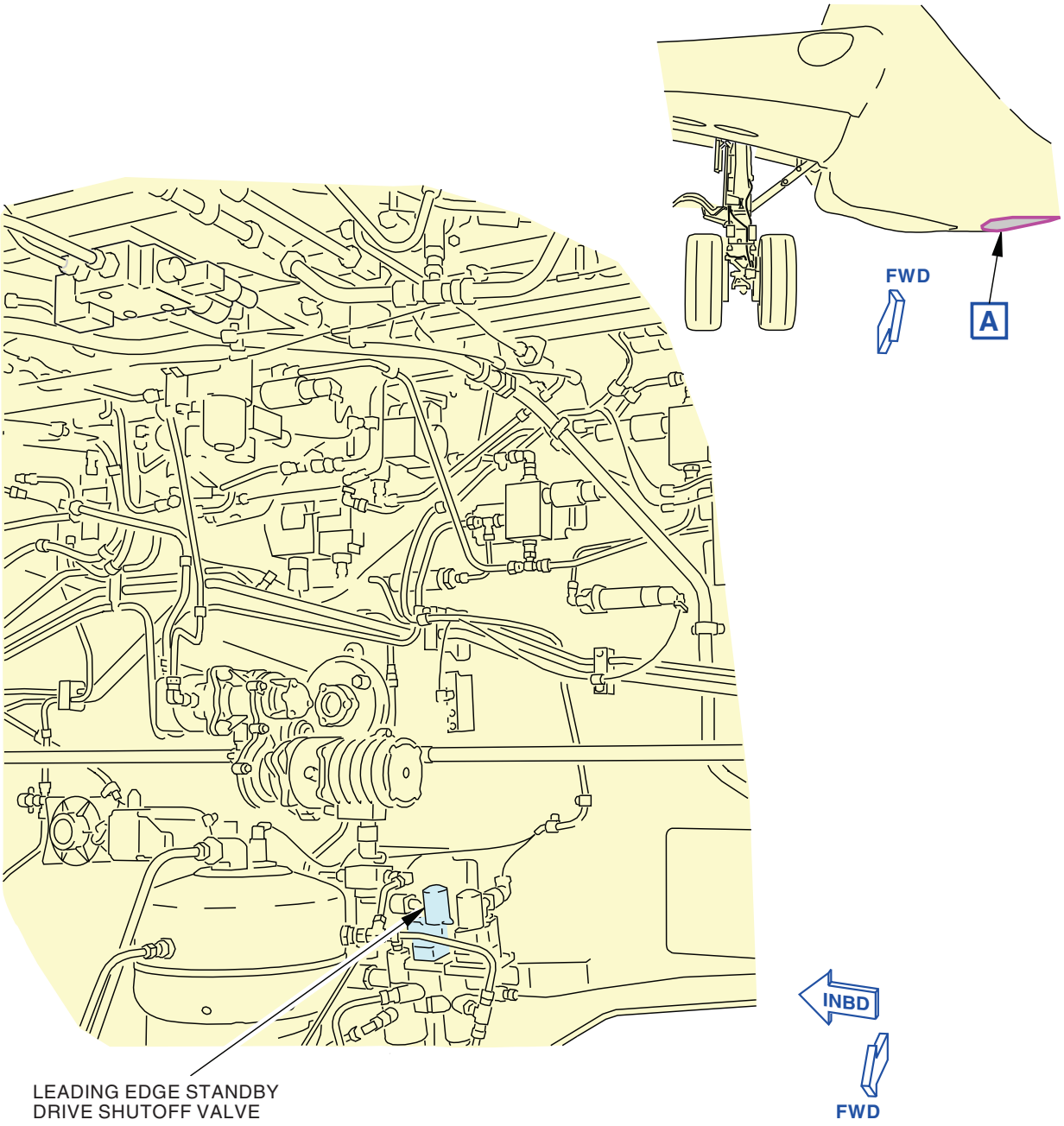
(11) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

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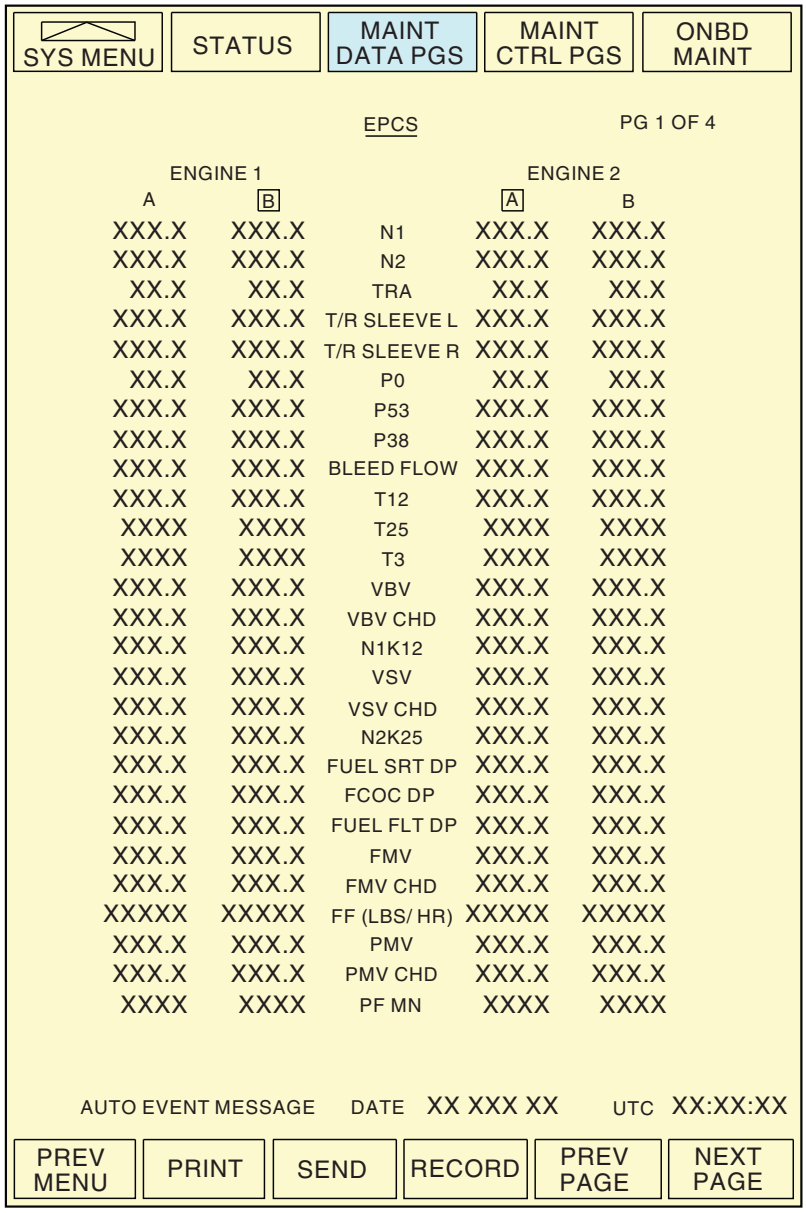
**LEADING EDGE STANDBY DRIVE SHUTOFF VALVE
(MAIN LANDING GEAR WHEEL WELL)**

A

2547816 S0000606262_V1

**Leading Edge Standby Drive Shutoff Valve
Figure 502/78-31-00-990-819-G00**

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2354207 S00061514972_V2

EPCS Page 1 Status Display
Figure 503/78-31-00-990-820-G00

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TASK 78-31-00-700-804-G00

5. Thrust Reverser Locks Operational Test

(Figure 504, Figure 505, and Figure 506)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task makes sure that the thrust reverser locks are serviceable.

(2) There is one actuator lock on the upper actuator of each thrust reverser half.

(3) There is one actuator lock on the center actuator of each thrust reverser half.

(4) To make sure that an actuator lock is serviceable: try to manually extend the thrust reverser sleeve with one actuator lock released and the sync lock released.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-980-802-G00	Thrust Reverser Operation - Retract (Selection) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-2418	Lockpin - Equipment, Thrust Reverser Actuator Module Lockout Part #: C78004-1 Supplier: 81205
STD-653	Platform - Engine and Strut Access
STD-660	Platform - Maintenance, 4 to 8 ft (1.2 to 2.4 m) High
STD-858	Tag - DO NOT OPERATE
STD-6839	Wrench - Torque, Inspection, 0-150 in-lbs, +/-5% accuracy, Analog or Digital Readout (non-clicker)

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1

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(Continued)

Number	Name/Location
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Operational Test

SUBTASK 78-31-00-860-090-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 78-31-00-860-091-G00

(2) Make sure that the applicable forward thrust lever is in the idle position.

SUBTASK 78-31-00-860-092-G00

(3) Make sure that the applicable reverse thrust lever is in the fully forward and down position.

SUBTASK 78-31-00-860-093-G00

(4) Attach DO NOT OPERATE tags, STD-858, to the applicable thrust levers.

SUBTASK 78-31-00-860-094-G00

(5) Open these circuit breakers for the applicable engine and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	5	C00276	ENGINE 1 THRUST REVERSER CONT
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK
C	7	C00277	ENGINE 2 THRUST REVERSER CONT

SUBTASK 78-31-00-860-095-G00

(6) If it is necessary, use a 4 to 8 ft (1.2 to 2.4 m) high maintenance platform, STD-660, or engine and strut access platform, STD-653, to get access to the work area.

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SUBTASK 78-31-00-860-097-G00

WARNING DO ALL OF THE SPECIFIED TASKS IN THE CORRECT SEQUENCE TO PREVENT THRUST REVERSER MOVEMENT. IF YOU DO NOT OBEY THIS INSTRUCTION, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(7) Do these tasks in this sequence to prevent the hydraulic operation of the thrust reverser (Figure 504):

(a) Install the lockpin, SPL-2418, with the red streamer, into the isolation valve handle and the hole on the control valve body.

NOTE: The C78004-2 lockpins are contained in the C78004-1 lockpin kit which is contained in the 012A8102 Fly-Away Kit located in the E&E Compartment.

NOTE: For Engine 1, the control valve module is on the left side of the keel beam in the main gear wheel well.

NOTE: For Engine 2, the control valve module is on the right side of the keel beam in the main gear wheel well.

1) Turn the manual shutoff valve handle clockwise to align the lockpin hole in the manual shutoff valve handle with the hole on the control valve body.

2) Install the lock pin through the hole on the isolation valve handle and the control valve body.

SUBTASK 78-31-00-860-098-G00

CAUTION RETRACT THE LEADING EDGE FLAPS, AND DO THE DEACTIVATION PROCEDURE BEFORE YOU OPEN THE FAN COWL PANELS. IF YOU DO NOT OBEY THESE PRECAUTIONS, THE FAN COWL PANELS WILL HIT THE FLAPS. DAMAGE WILL OCCUR.

(8) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 78-31-00-860-099-G00

WARNING DO THE DEACTIVATION PROCEDURE FOR THE LEADING EDGE FLAPS. THE ACCIDENTAL MOVEMENT OF THE FLAPS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(9) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 78-31-00-860-255-G00

(10) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

(a) On the applicable engine, open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

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G. Thrust Reverser Locks Operational Test

SUBTASK 78-31-00-710-007-G00

WARNING MAKE SURE THAT ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE AREA AFT OF THE APPLICABLE THRUST REVERSER. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION STOP THE MANUAL EXTENSION IF THE THRUST REVERSER EXTENDS AT AN ANGLE. THIS IS AN INDICATION THAT THERE ARE SYNC SHAFTS THAT ARE BROKEN OR NOT INSTALLED. IF YOU CONTINUE, DAMAGE TO THE EQUIPMENT CAN OCCUR.

CAUTION DO NOT USE AN EXTENSION BAR OR OTHER TOOL TO MOVE THE UNLOCK HANDLE. IF YOU USE AN EXTENSION BAR OR OTHER TOOLS, DAMAGE TO THE UNLOCK HANDLE OR ACTUATOR CAN OCCUR.

CAUTION MAKE SURE THAT THE THRUST REVERSER SLEEVE DOES NOT HIT THE WING IF THE THRUST REVERSER IS IN THE 45-DEGREE OPEN POSITION. DO NOT EXTEND THE INBOARD SLEEVE. DAMAGE TO THE EQUIPMENT CAN OCCUR.

(1) Do these steps to release the locks on the applicable thrust reverser upper and center actuators (Figure 505):

(a) Move the manual unlock handle on each thrust reverser actuator forward to the unlock position.

NOTE: The manual unlock handle moves forward approximately 30 degree.

(b) Hold the manual unlock handle in the forward position as you push the detent pin in.

(c) Release the manual unlock handle.

SUBTASK 78-31-00-710-008-G00

CAUTION DO NOT HIT THE END OF THE SYNC LOCK OR THE 3/8 INCH DRIVE WRENCH TO PUSH THE LOCK RELEASE PIN. DAMAGE TO THE SYNC LOCK CAN OCCUR.

(2) Unlock the sync lock and engage the manual drive:

(a) Put an open end wrench on the hex nut at end of the sync lock.

(b) Put a inspection torque wrench, STD-6839, with a 6 in. (15 cm) extension into the 3/8-inch square opening.

NOTE: Use a dial indicator wrench to make sure that you do not go over the 50 in-lb (6 N·m) limit.

NOTE: The wrench will go into the opening approximately 0.30 in. (7.62 mm).

(c) Apply continuous inward pressure on the lock release pin with the inspection torque wrench, STD-6839, and 6 in. (15 cm) extension.

(d) Slowly move the open end wrench a small amount at a time in one or opposite direction until the lock release pin goes in.

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SUBTASK 78-31-00-710-009-G00

CAUTION

MAKE SURE THAT YOU APPLY CONTINUOUS PRESSURE ON THE SQUARE-DRIVE WRENCH. THIS WILL KEEP THE LOCK RELEASE PIN IN AND THE MANUAL DRIVE FULLY ENGAGED. IF YOU DO NOT, DAMAGE TO THE SYNC LOCK CAN OCCUR.

CAUTION

DO NOT APPLY MORE THAN 50 INCH-POUNDS OF TORQUE TO EXTEND OR RETRACT THE THRUST REVERSER. IF YOU DO, DAMAGE TO THE SYNC LOCK CAN OCCUR.

(3) Manually extend (deploy) the thrust reverser sleeve:

(a) Turn the inspection torque wrench, STD-6839, with a 6 in. (15 cm) extension as follows:

NOTE: The sleeve will be in the fully extended position when the override clutch on the manual drive slips.

1) Turn the inspection torque wrench, STD-6839, in the counterclockwise direction to 50 in-lb (6 N·m) or less to extend the left thrust reverser sleeve.

2) Turn the inspection torque wrench, STD-6839, in the clockwise direction to 50 in-lb (6 N·m) or less to extend the right thrust reverser sleeve.

(b) Make sure that the thrust reverser sleeve operates smoothly.

NOTE: It can be necessary to operate the manual drive unit on the other side of the thrust reverser to fully extend the sleeves.

SUBTASK 78-31-00-710-012-G00

CAUTION

MAKE SURE THAT YOU APPLY CONTINUOUS PRESSURE ON THE SQUARE-DRIVE WRENCH. THIS WILL KEEP THE LOCK RELEASE PIN IN AND THE MANUAL DRIVE FULLY ENGAGED. IF YOU DO NOT, DAMAGE TO THE SYNC LOCK CAN OCCUR.

(4) Manually retract (stow) the thrust reverser sleeve:

(a) Turn the inspection torque wrench, STD-6839 with a 6 in. (15 cm) extension as follows:

NOTE: The sleeve will be at the manual track limit when the override clutch on the manual drive slips. If the locking actuator has not locked at this time, hydraulic power will be necessary to completely retract and lock the thrust reverser.

1) Turn the inspection torque wrench, STD-6839 clockwise to not more than 50 in-lb (6 N·m) to retract the left thrust reverser sleeve.

2) Turn the inspection torque wrench, STD-6839 counterclockwise to not more than 50 in-lb (6 N·m) to retract the right thrust reverser sleeve.

(b) Make sure that the thrust reverser sleeve operates smoothly.

(c) Turn the manual drive unit until the thrust reverser sleeve are fully retracted and the manual unlock handle on the upper actuator and center actuator snap to the locked position.

1) Make sure that the manual unlock handle on the upper actuator and center actuator move back to the normal, locked position.

SUBTASK 78-31-00-710-013-G00

(5) Do a check of the lock on the applicable thrust reverser upper actuator:

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(a) Make sure that the manual unlock handle on the applicable thrust reverser upper actuator is in the lock position.

(b) Make sure that the manual unlock handle on the applicable thrust reverser center actuator is in the unlock position.

1) Move the manual unlock handle on the applicable thrust reverser center actuator forward to the unlock position.

NOTE: The manual unlock handle moves forward approximately 30 degree.

a) Hold the manual unlock handle in the forward position as you push the detent pin in.

b) Release the manual unlock handle.

WARNING KEEP ALL PERSONNEL AND EQUIPMENT AWAY FROM THE AREA THAT IS AFT OF THE APPLICABLE THRUST REVERSER. IF YOU IGNORE THIS INSTRUCTION, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR WHEN THE THRUST REVERSER EXTENDS.

CAUTION DO NOT EXTEND THE TRANSLATING SLEEVES IF A THRUST REVERSER HALF IS OPEN. DAMAGE TO THE THRUST REVERSER AND ADJACENT STRUCTURES CAN OCCUR.

(c) Try to extend the thrust reverser sleeve manually.

1) Turn the inspection torque wrench, STD-6839, in the counterclockwise direction to 50 in-lb (6 N·m) or less to extend the left thrust reverser sleeve.

2) Turn the inspection torque wrench, STD-6839, in the clockwise direction to 50 in-lb (6 N·m) or less to extend the right thrust reverser sleeve.

3) Make sure that the thrust reverser sleeve does not extend (deploy).

SUBTASK 78-31-00-710-014-G00

(6) Do a check of the lock on the applicable thrust reverser center actuator:

(a) Make sure that the manual unlock handle on the applicable thrust reverser center actuator is in the lock position.

(b) Make sure that the manual unlock handle on the applicable thrust reverser upper actuator is in the unlock position.

1) Move the manual unlock handle on the applicable thrust reverser upper actuator forward to the unlock position.

NOTE: The manual unlock handle moves forward approximately 30 degree.

a) Hold the manual unlock handle in the forward position as you push the detent pin in.

b) Release the manual unlock handle.

WARNING KEEP ALL PERSONNEL AND EQUIPMENT AWAY FROM THE AREA THAT IS AFT OF THE APPLICABLE THRUST REVERSER. IF YOU IGNORE THIS INSTRUCTION, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR WHEN THE THRUST REVERSER EXTENDS.

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(WARNING PRECEDES)

CAUTION DO NOT EXTEND THE TRANSLATING SLEEVES IF A THRUST REVERSER HALF IS OPEN. DAMAGE TO THE THRUST REVERSER AND ADJACENT STRUCTURES CAN OCCUR.

(c) Try to extend the thrust reverser sleeve manually.

1) Turn the inspection torque wrench, STD-6839, in the counterclockwise direction to 50 in-lb (6 N·m) or less to extend the left thrust reverser sleeve.

2) Turn the inspection torque wrench, STD-6839, in the clockwise direction to 50 in-lb (6 N·m) or less to extend the right thrust reverser sleeve.

3) Make sure that the thrust reverser sleeve does not extend (deploy).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-00-860-101-G00

(1) Do the steps that follow to engage all locks:

NOTE: To completely retract and lock the thrust reverser, it can be necessary to use hydraulic power. You can use a wrench to partially retract the thrust reverser; however if the locking actuator does not lock, hydraulic power will be necessary to completely retract and lock the thrust reverser (TASK 78-31-00-980-802-G00).

(a) Make sure that the manual lock handle on the applicable thrust reverser center actuator is in the lock position.

1) Move the manual unlock handle to the locked position.

(b) Make sure that the manual lock handle on the applicable thrust reverser upper actuator is in the lock position.

1) Move the manual unlock handle to the locked position.

(c) If the upper actuator does not return to the locked position, do these steps:

1) Unlock the center actuator.

2) Insert a inspection torque wrench, STD-6839, with a 6 in. (152 mm) extension into the manual drive unit and turn the manual drive unit in the retract (stow) direction until the upper actuator locks.

3) Lock the center actuator.

(d) Remove the inspection torque wrench, STD-6839, and the 6 in. (152 mm) extension from the manual drive unit in the sync lock.

1) Make sure the lock pin moves out to the usual position.

2) If the sync lock does not return the locked position, it may be necessary to insert a inspection torque wrench, STD-6839, with a 6 in. (152 mm) extension into the manual drive unit and turn the manual drive unit until the sync lock locks.

SUBTASK 78-31-00-860-102-G00

(2) Do the steps that follow to activate the hydraulic isolation valve:

(a) Remove the lock pin from the hole on the isolation valve handle and the control valve body.

(b) Make sure that the manual shutoff handle moves counterclockwise to the vertical position.

(c) Store the lock pin and the red streamer in the T/R Deactivation kit bag in the E&E Compartment.

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SUBTASK 78-31-00-860-104-G00

(3) If it was used, remove the 4 to 8 ft (1.2 to 2.4 m) high maintenance platform, STD-660, or engine and strut access platform, STD-653, from the work area.

SUBTASK 78-31-00-860-256-G00

(4) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

On the applicable engine, close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

SUBTASK 78-31-00-860-106-G00

(5) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

SUBTASK 78-31-00-860-107-G00

(6) Remove the DO NOT OPERATE tags, STD-858, from the thrust levers.

SUBTASK 78-31-00-860-108-G00

(7) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	5	C00276	ENGINE 1 THRUST REVERSER CONT
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK
C	7	C00277	ENGINE 2 THRUST REVERSER CONT

SUBTASK 78-31-00-860-109-G00

(8) Make sure that the reverse thrust levers are in the fully forward and down position.

SUBTASK 78-31-00-860-110-G00

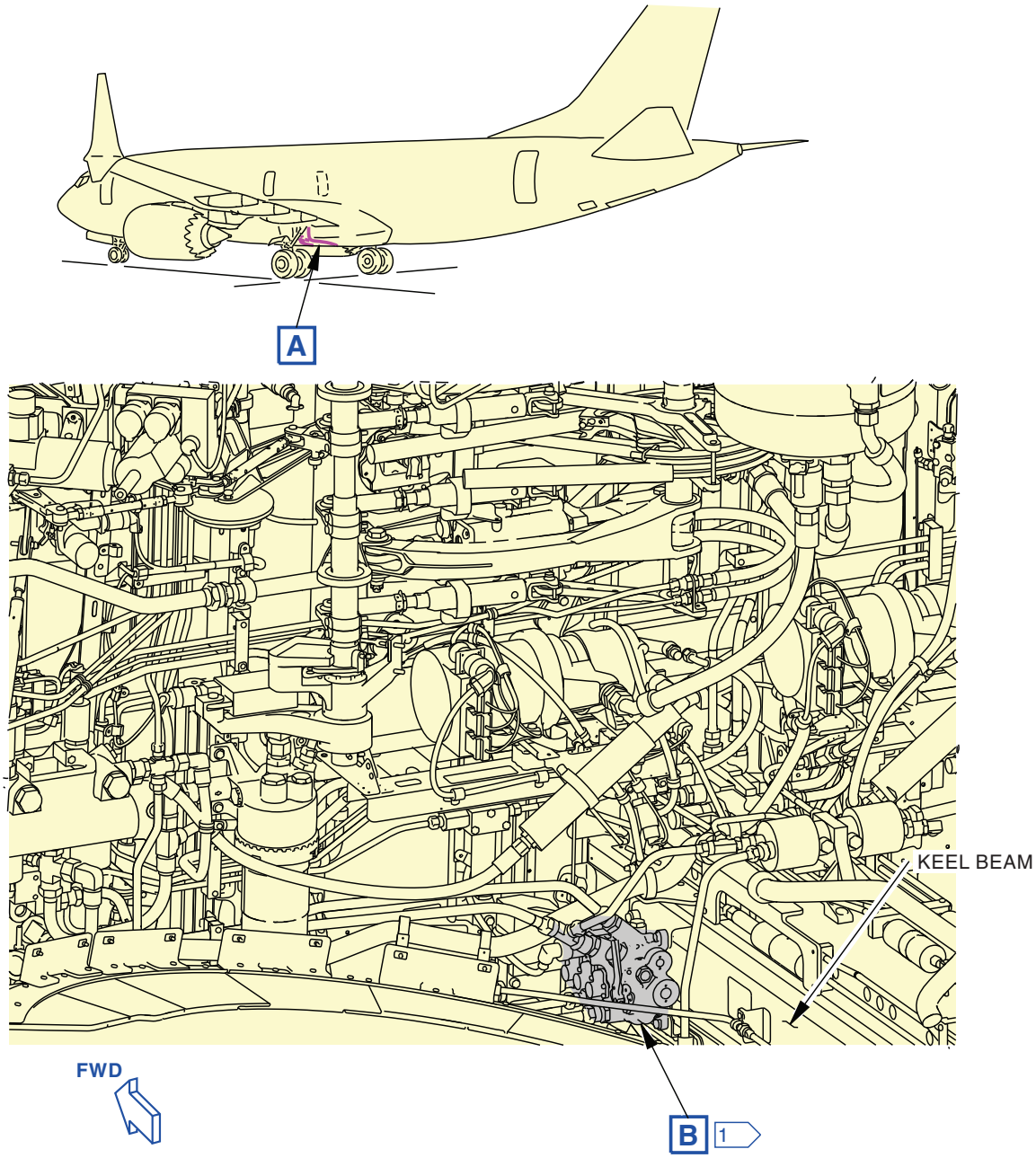
(9) Make sure that the forward thrust levers are in the idle position.

SUBTASK 78-31-00-710-016-G00

(10) Do this task: Thrust Reverser System Normal Operational Test, TASK 78-31-00-700-802-G00.

——— END OF TASK ———

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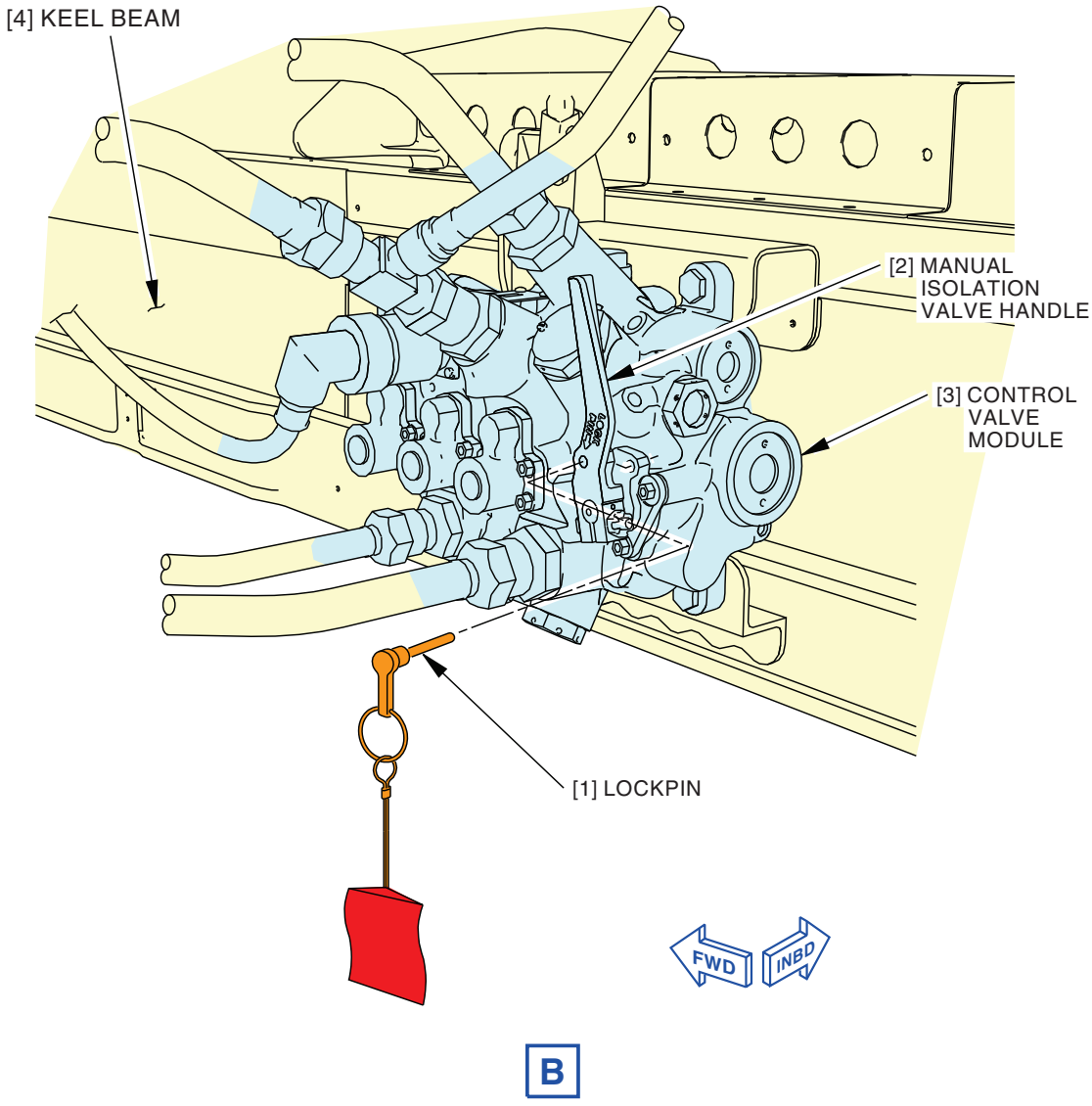


1 ENGINE 1 CONTROL VALVE MODULE IS SHOWN, ENGINE 2 CONTROL VALVE IS ON THE RIGHT SIDE OF THE KEEL BEAM.

2355143 S00061516635_V1

Thrust Reverser Control Valve Module
Figure 504/78-31-00-990-814-G00 (Sheet 1 of 2)

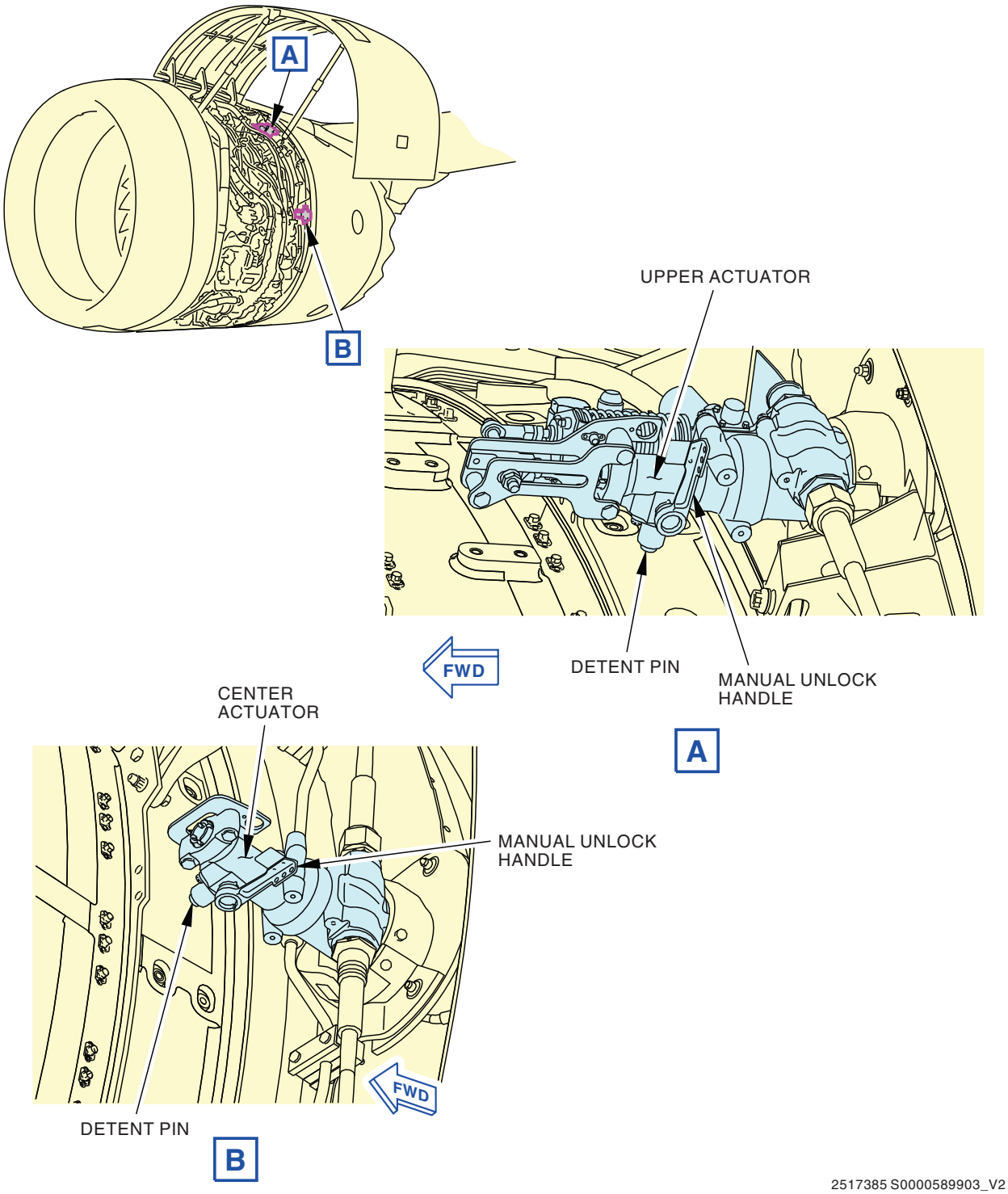
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2355144 S00061516636_V1

Thrust Reverser Control Valve Module
Figure 504/78-31-00-990-814-G00 (Sheet 2 of 2)

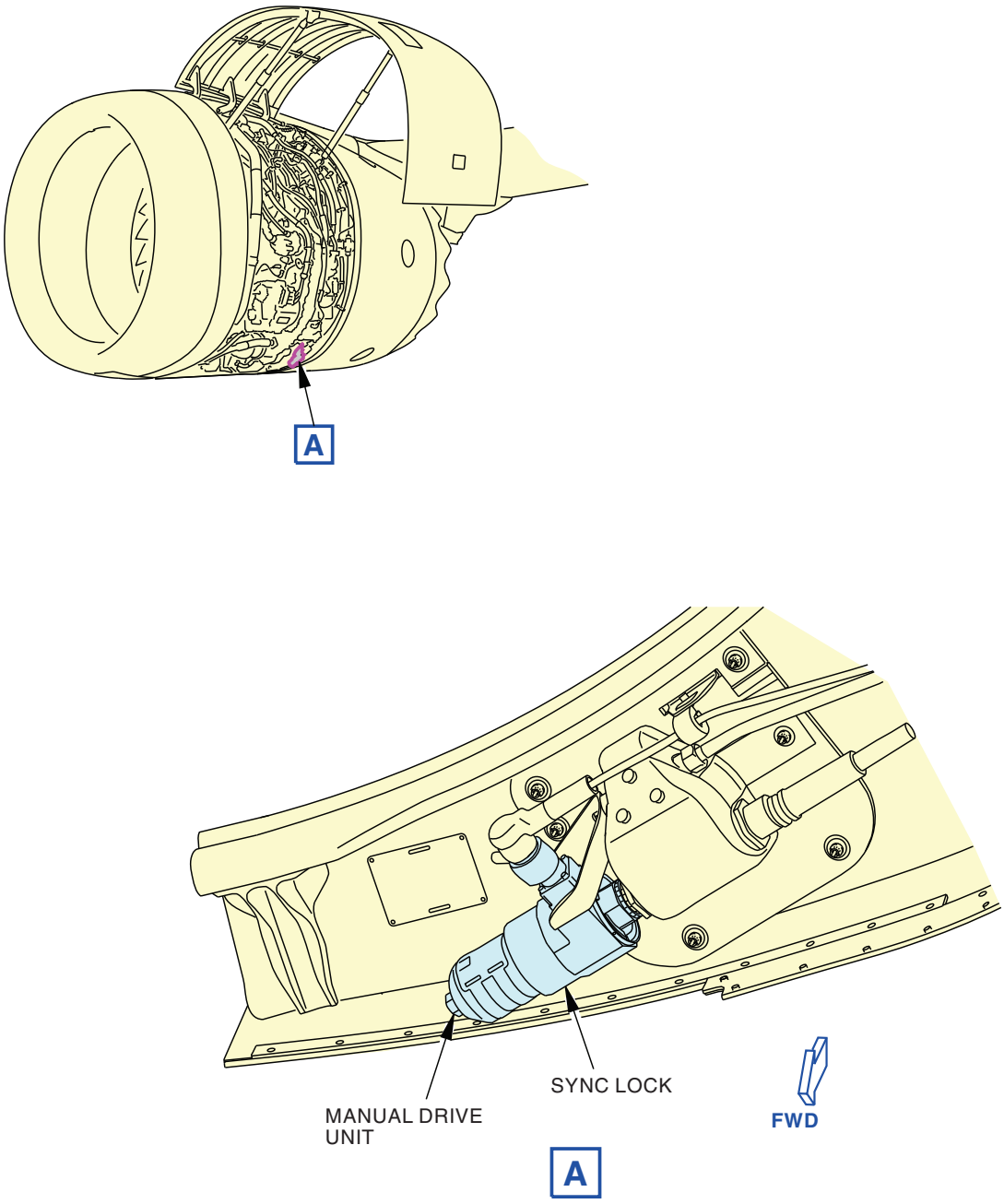
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2517385 S0000589903_V2

Thrust Reverser Hydraulic Actuator Lock
Figure 505/78-31-00-990-815-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2517395 S0000589910_V1

Thrust Reverser Manual Drive Unit
Figure 506/78-31-00-990-816-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-31-00-700-805-G00

6. Thrust Reverser Proximity Sensors Operational Test

(Figure 507)

A. General

(1) This task gives the instructions to do a test of all of the thrust reverser proximity sensors for each engine. The hydraulic isolation valve pressure switch is also tested.

(a) Left sleeve, upper actuator proximity sensor

(b) Left sleeve, center actuator proximity sensor

(c) Aft Strut, Hydraulic Isolation Valve (HIV) pressure switch

(d) Right sleeve, upper actuator proximity sensor

(e) Right sleeve, center actuator proximity sensor.

(2) You must extend and retract the thrust reversers (one engine at a time) while you monitor the MFD.

(3) Make sure that the bit values for the proximity sensors and pressure switch are correct for the applicable thrust reverser position.

B. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
31-65-00-750-801	Fault Interrogation for a Maintenance Message (P/B 201)
71-00-00-700-816-G00	Test No. 38 - Thrust Reverser Cycle Special Functions (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-21-00-800-801-G00	EEC Maintenance Power Selection (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the Operational Test

SUBTASK 78-31-00-860-112-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

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SUBTASK 78-31-00-860-113-G00

(2) For the applicable engine, open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	1	C04016	ENGINE 1 IGNITION PWR CONT
E	2	C04017	ENGINE 2 IGNITION PWR CONT

SUBTASK 78-31-00-860-114-G00

(3) For the left engine, open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 78-31-00-860-115-G00

(4) For the right engine, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 78-31-00-860-116-G00

(5) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE

SUBTASK 78-31-00-860-117-G00

(6) Make sure that the applicable thrust reverser is closed (TASK 78-31-00-010-802-G00).

SUBTASK 78-31-00-860-118-G00

WARNING DO NOT OPERATE THE ENGINE IN REVERSE THRUST WHILE THE FAN COWL PANELS ARE OPEN. INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(7) Make sure that the applicable fan cowl panels are closed and latched (TASK 71-11-04-410-801-G00).

SUBTASK 78-31-00-860-119-G00

(8) Select the left or right Multi-Function Display (MFD).

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(a) Push the SYS button on the MFD control panel, P9.

(b) Push the SEL button on SELECTOR knob on the MFD control panel, P9.

SUBTASK 78-31-00-860-120-G00

(9) Set the applicable EEC MAINT POWER switch on the MISC SYSTEM CTRLS page on the MFD to the TEST position (TASK 73-21-00-800-801-G00).

NOTE: The EEC will do a self-test when you set the EEC MAINT POWER switch to the TEST position. Wait for a minimum of 30 seconds to let the EEC complete the test.

SUBTASK 78-31-00-860-121-G00

(10) Set the GND TEST switch on the maintenance BITE panel, P61-4, to the ENABLE position.

SUBTASK 78-31-00-860-122-G00

(11) Make sure that the forward thrust levers are in the idle position.

SUBTASK 78-31-00-860-123-G00

(12) Make sure that the reverse thrust levers are in the fully forward and down position.

SUBTASK 78-31-00-860-125-G00

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(13) Do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

(a) For engine 1, pressurize hydraulic system A.

(b) For engine 2, pressurize hydraulic system B.

F. Thrust Reverser Proximity Sensors Operational Test

SUBTASK 78-31-00-710-015-G00

(1) Use the Multi-Function Display (MFD) screen to do the test of the applicable thrust reverser cycle (TASK 71-00-00-700-816-G00):

(a) Make these selections on the MFD main function menu:

1) ONBOARD MAINTENANCE
2) OTHER FUNCTIONS
3) SPECIAL FUNCTIONS
4) 71-80 Left Engine or 71-80 Right Engine
5) Thrust Reverser Cycle
6) CONTINUE.

(b) Do the instructions that show on the MFD PRECONDITIONS and then make these selections:

NOTE: Make sure that you verify and follow the inhibited conditions instructions closely. If one of the inhibited conditions is not met, the test will be failed.

1) CONTINUE
2) START FUNCTION.

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WARNING MAKE SURE THAT PERSONNEL STAY AWAY FROM THE THRUST REVERSER WHEN YOU MOVE THE CONTROLS. THE THRUST REVERSER CAN MOVE VERY QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL.

(c) Do the steps that follow to do a test of the proximity sensors:

1) Select the EPCS page 4 from the Maintenance Data Pages on the Multi-Function Display (MFD).

2) Move the applicable reverse thrust lever up to the extend (deploy) position.

a) Compare the EPCS page 4, status 2 display with the Table 503 to make sure that the bit values shown are correct.

3) Move the applicable reverse thrust lever to the retract (stow) position.

a) Compare the EPCS page 4, status 2 display with the Table 503 to make sure that the bit values shown are correct.

4) If you do not get the correct bit values, look at the status message on the MFD for thrust reverser proximity sensor messages (TASK 31-65-00-750-801).

Table 503/78-31-00-993-801-G00 Thrust Reverser Sensor Bit Values

Sensor Location	Bit No.	T/R Extended Bit Values	T/R Retracted Bit Values
Upper Actuator (Left Sleeve)*[1]	14	0 (Target Near)	1 (Target Far)
Center Actuator (Left Sleeve)*[1]	13	0 (Target Near)	1 (Target Far)
Hydraulic Isolation Valve (Aft Strut) and HIV Pressure Switch	4	1 (OPEN)	0 (CLOSE)
Upper Actuator (Right Sleeve)*[2]	16	0 (Target Near)	1 (Target Far)
Center Actuator (Right Sleeve)*[2]	15	0 (Target Near)	1 (Target Far)

*[1] Channel A transmits left sleeve data and HIV pressure switch.

*[2] Channel B transmits right sleeve data.

(d) After you deployed and stowed the thrust reversers, select the STOP FUNCTION to complete the test.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-00-860-126-G00

(1) Put the applicable HYD PUMPS A ENG 1 or HYD PUMPS B ENG 2 switch to the OFF position.

SUBTASK 78-31-00-860-127-G00

(2) Set the applicable EEC MAINT POWER switch on the MISC SYSTEM CTRLS page on the MFD to the NORM position (TASK 73-21-00-800-801-G00).

SUBTASK 78-31-00-860-128-G00

(3) Set the GND TEST switch on the maintenance BITE panel, P61-4, to the NORM position.

SUBTASK 78-31-00-860-129-G00

(4) Make sure that the reverse thrust levers are in the fully forward and down position.

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SUBTASK 78-31-00-860-130-G00

(5) Make sure that the forward thrust levers are in the idle position.

SUBTASK 78-31-00-860-131-G00

(6) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE

SUBTASK 78-31-00-860-132-G00

(7) For the left engine, remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 78-31-00-860-133-G00

(8) For the right engine, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 78-31-00-860-134-G00

(9) For the applicable engine, remove the safety tags and close these applicable circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	1	C04016	ENGINE 1 IGNITION PWR CONT
E	2	C04017	ENGINE 2 IGNITION PWR CONT

SUBTASK 78-31-00-860-135-G00

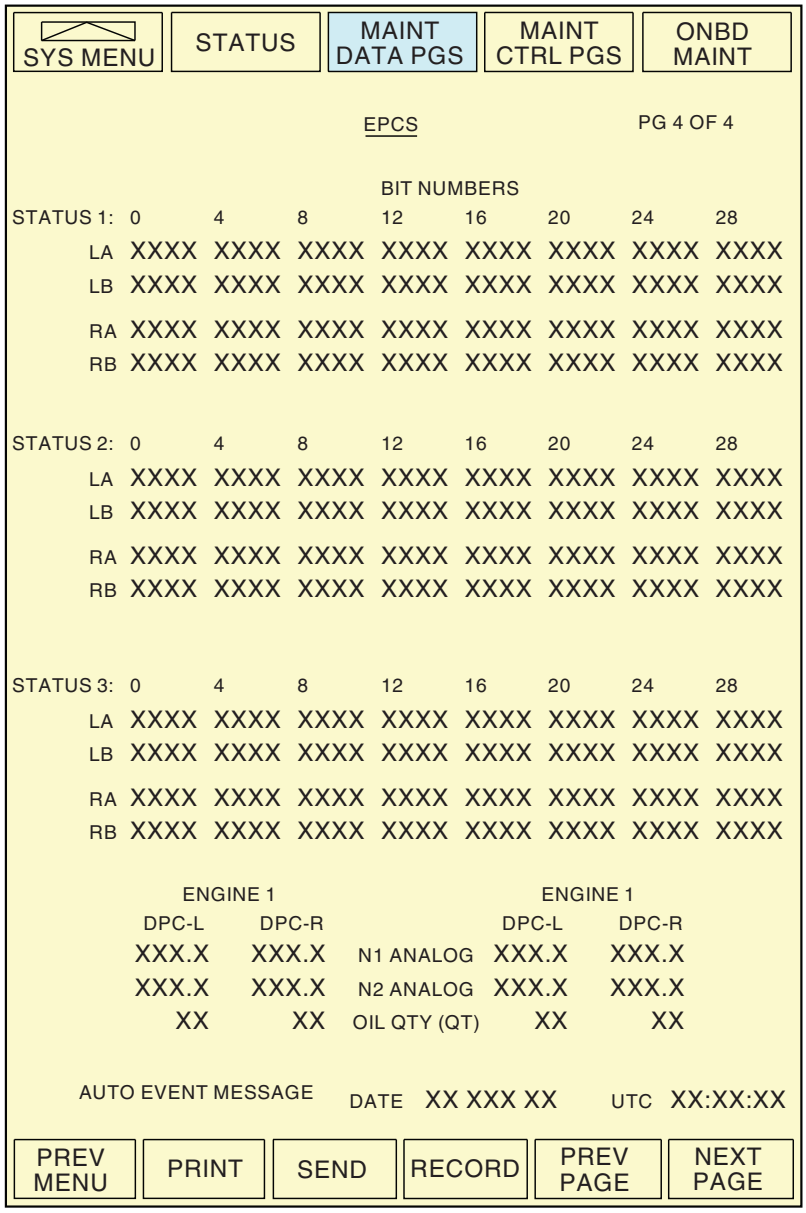
(10) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

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EPCS Page 4 Status Display
Figure 507/78-31-00-990-817-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-31-00-700-806-G00

7. Thrust Reverser Annunciator Lights Operational Test

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions to do a check of the wiring between the Display Processing Computer (DPC) and REVERSER LIMITED, REVERSER COMMAND, and REVERSER AIR/GRD lights, on the P5 pilots aft overhead panel.

B. References

Reference	Title
31-65-00-860-801	Miscellaneous System Controls (MSC) Page Function (P/B 201)
46-13-02-860-802	Special Functions (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Test

SUBTASK 78-31-00-860-189-G00

(1) Set the GND TEST switch to the ENABLE position.

SUBTASK 78-31-00-860-201-G00

(2) Set the ENG 1 (ENG 2) EEC MAINT POWER, on the MAX Display System (MDS) Multi Function Display (MFD), to the TEST mode (TASK 31-65-00-860-801).

NOTE: This test is written for engine 1. The test is the same for engine 2, except as noted by the data in parenthesis. Use the data in parenthesis to do the test for engine 2.

SUBTASK 78-31-00-860-191-G00

(3) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	5	C04023	DISPLAY DPC 1 PRI

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	10	C04022	DISPLAY DPC 1 HOLDUP

E. Thrust Reverser Annunciator Lights Operational Test

SUBTASK 78-31-00-860-192-G00

(1) On the MDS MFD, go to the 70-80 Left (Right) Engine display in the SPECIAL FUNCTIONS (TASK 46-13-02-860-802).

(a) Select the Lamp Check Out to do a test on Channel A.

SUBTASK 78-31-00-860-193-G00

(2) Make sure that the REVERSER AIR/GRD light, REVERSER COMMAND light, and Engine 1 (2) REVERSER LIMITED light, on the P5 panel, come on.

NOTE: If the annunciator lights come on, there is wire continuity between the DPC2 and annunciator lights through Channel A.

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SUBTASK 78-31-00-860-225-G00

(3) Select CONTINUE to do a test on Channel B.

SUBTASK 78-31-00-750-001-G00

(4) Make sure that the REVERSER AIR/GRD light, REVERSER COMMAND light, and Engine 1 (2) REVERSER LIMITED light, on the P5 panel, come on.

NOTE: If the annunciator lights come on, there is wire continuity between the DPC2 and annunciator lights through Channel B.

SUBTASK 78-31-00-860-194-G00

(5) Close the Lamp Check Out function.

SUBTASK 78-31-00-860-249-G00

(6) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	10	C04022	DISPLAY DPC 1 HOLDUP

SUBTASK 78-31-00-860-250-G00

(7) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	5	C04023	DISPLAY DPC 1 PRI

NOTE: If you do not close these circuit breakers in this sequence, the DPC will not operate correctly.

SUBTASK 78-31-00-700-001-G00

(8) Wait a minimum of 45 seconds for DPC 1 to power up.

SUBTASK 78-31-00-860-251-G00

(9) Open these circuit breakers:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	9	C04024	DISPLAY DPC 2 HOLDUP
D	11	C04025	DISPLAY DPC 2 PRI

SUBTASK 78-31-00-860-198-G00

(10) On the MDS MFD, go to the 71-80 Left (Right) Engine display in the SPECIAL FUNCTIONS (TASK 46-13-02-860-802).

(a) Select the Lamp Check Out to do a test on Channel A.

SUBTASK 78-31-00-860-199-G00

(11) Make sure that the REVERSER AIR/GRD light, REVERSER COMMAND light, and Engine 1 (2) REVERSER LIMITED light, on the P5 panel, come on.

NOTE: If the annunciator lights comes on, there is wire continuity between the DPC1 and annunciator lights through Channel A.

SUBTASK 78-31-00-860-226-G00

(12) Select CONTINUE to do a test on Channel B.

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SUBTASK 78-31-00-750-002-G00

(13) Make sure that the REVERSER AIR/GRD light, REVERSER COMMAND light, and Engine 1 (2) REVERSER LIMITED light, on the P5 panel, comes on.

NOTE: If the annunciator lights come on, there is wire continuity between the DPC1 and annunciator lights through Channel B.

(a) Close the Lamp Check Out function.

SUBTASK 78-31-00-860-252-G00

(14) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	9	C04024	DISPLAY DPC 2 HOLDUP

SUBTASK 78-31-00-860-253-G00

(15) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	11	C04025	DISPLAY DPC 2 PRI

NOTE: If you do not close these circuit breakers in this sequence, the DPC will not operate correctly.

SUBTASK 78-31-00-700-002-G00

(16) Wait a minimum of 45 seconds for DPC 2 to power up.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-00-860-246-G00

(1) Set the ENG 1 (ENG2) EEC MAINT POWER, on the MDS MFD, to the NORM mode (TASK 31-65-00-860-801).

SUBTASK 78-31-00-860-202-G00

(2) Set the GND TEST switch to the NORM position.

——— END OF TASK ———

TASK 78-31-00-700-807-G00

8. Scheduled Maintenance Task for EEC Faults - Reverser Sensor

A. General

(1) This task gives the instructions to interrogate the Scheduled Maintenance Task (SMT) message ENG 1/2 REVERSER SENSOR faults.

B. References

Reference	Title
05-00-00-750-801	Scheduled Maintenance Task (SMT) - Repair Time Limits (P/B 201)
73-21-00-800-801-G00	EEC Maintenance Power Selection (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

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D. Prepare for the Fault Interrogation

SUBTASK 78-31-00-860-203-G00

(1) Make sure that the applicable EEC MAINT POWER switch on the MISC SYSTEM CTRLS page on the Multi Function Display (MFD) to the NORM position (TASK 73-21-00-800-801-G00).

SUBTASK 78-31-00-860-204-G00

(2) Select the left or right MFD.

(a) Push the SYS button on the P9 MFD control panel.

(b) Push the SEL button on SELECTOR knob on the P9 MFD control panel.

SUBTASK 78-31-00-860-207-G00

(3) Make sure that the thrust levers are at the idle and down position.

SUBTASK 78-31-00-860-205-G00

(4) Set the GND TEST switch, on the P61-4 maintenance BITE panel, to the ENABLE position.

SUBTASK 78-31-00-860-208-G00

(5) Make sure that there is no other MAX Display System (MDS) maintenance test function is active.

SUBTASK 78-31-00-860-209-G00

(6) Set the applicable ENGINE CONTROL switch to ON.

E. Scheduled Maintenance Task for ENG 1/2 REVERSER SENSOR Faults

SUBTASK 78-31-00-860-210-G00

(1) Use the MFD screen to find if the scheduled maintenance task exists.

(a) Select MAINT CTRL PGS on the MFD.

(b) Select LATCHED MESSAGE ERASE

SUBTASK 78-31-00-810-001-G00

(2) If ENG 1/2 REVERSER SENSOR exists, do one of the steps that follow:

(a) Go to the IFIM and do the applicable procedure(s) to correct the fault.

(b) Continue operation with the fault provided the fault is corrected within remaining deferral period described in this task: Scheduled Maintenance Task (SMT) - Repair Time Limits, TASK 05-00-00-750-801.

NOTE: To help with the future maintenance plan to correct the fault, you may want to use the IFIM to determine and record the maintenance message(s) correlated to the ENG 1/2 REVERSER SENSOR SMT message.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-00-860-229-G00

(1) Set the GND TEST switch, on the P61-4 maintenance BITE panel, to the NORM position.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

THRUST REVERSER - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Thrust Reverser Removal

(2) Thrust Reverser Installation.

TASK 78-31-01-000-801-G00

2. Thrust Reverser Removal

A. General

(1) This task gives the instructions to remove the left or right thrust reverser half.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
71-11-04-000-801-G00	Fan Cowl Panels Removal (P/B 401)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-09-010-801-G00	Krueger Flap Deflector Removal (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-2430	Hoist ? Boom, 800 lbs WLL, 250 lbs Load Positioner Compatible Part #: C20002-267 Supplier: 81205
SPL-14549	Equipment - Sling, Thrust Reverser, LEAP-1B Engine Part #: C78032-75 Supplier: 81205 Opt Part #: C78032-1 Supplier: 81205
SPL-14555	Stand - Storage/Work, LEAP-1B Thrust Reverser Part #: C78036-1 Supplier: 81205
STD-585	Mat - Protective, Closed Cell Neoprene, 30-50 duro, Weather and Oil Resistant, 3/8 Inch Min Thickness, Minimum 42x60 Inches with Warning Streamers
STD-858	Tag - DO NOT OPERATE
STD-1095	Crane - Lift, 2000 lb Capacity, 30 Foot Height
STD-1110	Container - Hydraulic Fluid Resistant, 5 Gallon (19 Liter)
STD-12395	Load Cell Equipment

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D. Consumable Materials

Reference	Description	Specification
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G50625	Strap - Plastic, Adjustable, Self-locking, 14.40 Inches (365.76 mm) Long	BACS38K3

E. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

F. Access Panels

Number	Name/Location
415AL	Left Forward Thrust Reverser Hinge Fairing, Engine 1
415BL	Left Aft Thrust Reverser Hinge Fairing, Engine 1
416AR	Right Forward Thrust Reverser Hinge Fairing, Engine 1
416BR	Right Aft Thrust Reverser Hinge Fairing, Engine 1
425AL	Left Forward Thrust Reverser Hinge Fairing, Engine 2
425BL	Left Aft Thrust Reverser Hinge Fairing, Engine 2
426AR	Right Forward Thrust Reverser Hinge Fairing, Engine 2
426BR	Right Aft Thrust Reverser Hinge Fairing, Engine 2

G. Prepare for the Removal

SUBTASK 78-31-01-040-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 78-31-01-860-001-G00

CAUTION

DO NOT LIFT OR MOVE THE THRUST REVERSER WITH CAUTION: LESS THAN THREE CASCADE SEGMENTS INSTALLED. MAKE SURE THAT THERE IS NO MORE THAN TWO ADJACENT CASCADE SEGMENTS NOT INSTALLED. IF YOU DO NOT OBEY THIS INSTRUCTION, DAMAGE TO THE THRUST REVERSER CAN OCCUR.

(2) Do not lift or move the thrust reverser if there are less than three cascade segments installed.

(a) Make sure that there are no more than two adjacent cascade segments not installed.

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SUBTASK 78-31-01-860-002-G00

(3) For Engine 1, open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	6	C01412	ENGINE 1 THRUST REVERSER INTLK
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

SUBTASK 78-31-01-860-003-G00

(4) For Engine 2, open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK
C	6	C01413	ENGINE 2 THRUST REVERSER INTLK

SUBTASK 78-31-01-860-004-G00

WARNING REVERSER THAT IS NEAR THE LEADING EDGE FLAPS AND SLATS OR BEFORE YOU DO WORK ON THE THRUST REVERSER. IF YOU DO NOT OBEY THIS INSTRUCTION, INJURY TO PERSONS AND DAMAGE TO THE EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT YOU FULLY RETRACT THE LEADING EDGE FLAPS AND SLATS. IF YOU DO NOT FULLY RETRACT THE LEADING EDGE FLAPS AND SLATS, THEN YOU CAN NOT REMOVE THE THRUST REVERSER CORRECTLY.

(5) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 78-31-01-040-002-G00

(6) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 78-31-01-040-003-G00

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(7) Do the thrust reverser deactivation for ground maintenance (TASK 78-31-00-040-801-G00).

SUBTASK 78-31-01-860-005-G00

(8) Deactivate the applicable hydraulic system (TASK 29-11-00-860-805).

(a) For Engine 1, remove power from hydraulic system A.

(b) For Engine 2, remove power from hydraulic system B.

SUBTASK 78-31-01-860-006-G00

(9) Do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 78-31-01-010-001-G00

(10) It is recommended to remove the applicable fan cowl panel (TASK 71-11-04-000-801-G00).

NOTE: If you remove the fan cowl panel, there will be more room to move the thrust reverser during the removal and the fan cowl panel will not get damaged.

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SUBTASK 78-31-01-010-002-G00

(11) If you did not remove the applicable fan cowl panel, do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

(a) Open the fan cowl fully to the 61 degree position.

SUBTASK 78-31-01-030-001-G00

(12) If the replacement thrust reverser does not have the Krueger flap deflectors installed, remove the Krueger flap deflectors from the thrust reverser that you will remove (TASK 78-31-09-010-801-G00).

NOTE: The Krueger flap deflectors are installed only on the inboard translating sleeve.

SUBTASK 78-31-01-010-003-G00

(13) Remove the applicable access panels (Figure 401).

(a) If you are removing the inboard thrust reverser on Engine 1, remove these access panels:

Number	Name/Location
416AR	Right Forward Thrust Reverser Hinge Fairing, Engine 1
416BR	Right Aft Thrust Reverser Hinge Fairing, Engine 1

1) Remove 13 bolts [8] to remove the inboard forward hinge fairing [9].
2) Remove 12 bolts [1] to remove the inboard aft hinge fairing [2].
3) Keep the bolts together in a cloth bag for the installation.

(b) If you are removing the outboard thrust reverser on Engine 1, remove these access panels:

Number	Name/Location
415AL	Left Forward Thrust Reverser Hinge Fairing, Engine 1
415BL	Left Aft Thrust Reverser Hinge Fairing, Engine 1

1) Remove 13 bolts [7] to remove the outboard forward hinge fairing [6].
2) Remove 12 bolts [4] to remove the outboard aft hinge fairing [3].
3) Keep the bolts together in a cloth bag for the installation.

(c) If you are removing the inboard thrust reverser on Engine 2, remove these access panels:

Number	Name/Location
425AL	Left Forward Thrust Reverser Hinge Fairing, Engine 2
425BL	Left Aft Thrust Reverser Hinge Fairing, Engine 2

1) Remove 13 bolts [7] to remove the outboard forward hinge fairing [6].
2) Remove 12 bolts [4] to remove the outboard aft hinge fairing [3].
3) Keep the bolts together in a cloth bag for the installation.

(d) If you are removing the outboard thrust reverser on Engine 2, remove these access panels:

Number	Name/Location
426AR	Right Forward Thrust Reverser Hinge Fairing, Engine 2
426BR	Right Aft Thrust Reverser Hinge Fairing, Engine 2

1) Remove 13 bolts [8] to remove the inboard forward hinge fairing [9].
2) Remove 12 bolts [1] to remove the inboard aft hinge fairing [2].

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3) Keep the bolts together in a cloth bag for the installation.

H. Thrust Reverser Removal

SUBTASK 78-31-01-010-012-G00

(1) Disconnect the aft cowl cross tie rod [113] from the deflection limiting strap [112] on the applicable thrust reverser half (Figure 402).

(a) Open the applicable thrust reverser half (TASK 78-31-00-010-801-G00).

PTF ALL; AIRPLANES WITHOUT LANYARD

(b) Remove the nut [110], bolt [115], bushing [111], and bushing [114] that attach the aft cowl cross tie rod [113] to the deflection limiting strap [112].

PTF ALL; AIRPLANES WITH LANYARD

(c) Remove the nut [110], lanyard [118], washer [117], bolt [115], bushing [111], and bushing [114] that attach the aft cowl cross tie rod [113] to the deflection limiting strap [112].

PTF ALL

(d) Lower the applicable thrust reverser half into the closed position (TASK 78-31-00-010-802-G00).

1) Do not latch the thrust reverser closed.

SUBTASK 78-31-01-020-005-G00

(2) Disconnect the upper end of the thrust reverser opening actuator [75] from the thrust reverser (Figure 403).

(a) Remove the bolt [76], washer [77], bushing [71], barrel nut [74], and retainer [73] that attach the upper end of the opening actuator [75] to the thrust reverser torque box fitting [72].

CAUTION

MAKE SURE THAT YOU SAFETY THE ACTUATOR TO THE ENGINE WITH THE STRAP. IF YOU DO NOT SAFETY THE ACTUATOR, IT CAN FALL AND CAUSE DAMAGE TO THE ENGINE.

(b) Secure the upper end of the opening actuator [75] to the engine with an adjustable plastic strap, G50625.

SUBTASK 78-31-01-020-006-G00

(3) Disconnect the electrical bonding strap [86] from the thrust reverser (Figure 408).

(a) Use a wrench to hold the ground stud [85] to prevent the ground stud [85] from turning when you remove the nut [87].

(b) Remove the nut [87] and washer [91] that attach the bonding strap terminal to the ground stud [85].

(c) Remove the bonding strap [86] from the ground stud [85], and re-install the nut [87] and washer [91] onto the ground stud [85].

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SUBTASK 78-31-01-010-005-G00

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTIONS ARE NOT DIRTY BEFORE YOU CONNECT OR DISCONNECT THEM. SOLID OR LIQUID CONTAMINATION OF THE ELECTRICAL CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

(4) Disconnect the electrical connector [24] and electrical connector [27] from the strut receptacle [25] and strut receptacle [26] (Figure 404).

(a) For the left thrust reverser [5], disconnect the electrical connector [27] and electrical connector [24], D30002 and D30008.

(b) For the right thrust reverser [10], disconnect the electrical connector [27] and electrical connector [24], D30006 and D30010.

(c) Install protective caps on the electrical connectors and strut receptacles.

SUBTASK 78-31-01-020-007-G00

WARNING DO NOT INSTALL PRESSURE CAPS TO THE HYDRAULIC LINES THAT CONNECT TO CONTROL VALVE MODULE PORTS TO DEACTIVATE THE THRUST REVERSER. THE THRUST REVERSER CAN OPERATE WHEN PRESSURIZED. IF YOU DO NOT OBEY, INJURY TO PERSONNEL AND DAMAGE TO THE EQUIPMENT CAN OCCUR.

WARNING MAKE SURE THAT YOU WEAR PROTECTIVE CLOTHES AND GLOVES WHEN YOU DO WORK ON THE HYDRAULIC SYSTEM. HYDRAULIC FLUID CAN CAUSE A LEAK FROM THE OPEN PORTS OF THE COMPONENT OR FROM THE HYDRAULIC LINES. INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION DO NOT LET HYDRAULIC FLUID GET ON EQUIPMENT. IF HYDRAULIC FLUID GETS ON EQUIPMENT, IMMEDIATELY CLEAN IT FROM THE EQUIPMENT. HYDRAULIC FLUID CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE TWO WRENCHES TO LOOSEN THE TUBE COUPLING NUTS. USE ONE WRENCH TO HOLD THE NIPPLE FITTING. USE THE OTHER WRENCH TO LOOSEN THE COUPLING NUT. IF THE NIPPLE FITTING TURNS, DAMAGE CAN OCCUR.

(5) Drain the hydraulic fluid from the hydraulic lines (Figure 404, Figure 405).

NOTE: To decrease hydraulic fluid spray when the coupling nuts are loosened, wrap a cotton wiper, G00034, around the wrench, nut and hydraulic line.

(a) Wrap a cotton wiper, G00034, around the electrical connector [124] on the sync lock [123] receptacle on the lower actuator [122].

NOTE: The cloth will catch the hydraulic fluid and prevent contamination of the electrical connector.

(b) Put a 5 gallon (19 liter) hydraulic fluid resistant container, STD-1110, below the lower actuator [122] to collect hydraulic fluid.

(c) Disconnect the coupling nut on the lower sync shaft and tubing [121] at the upper port of the lower actuator [122].

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(d) Disconnect the coupling nut on the lower hydraulic retract line [120] at the upper port of the lower actuator [122].

(e) Let the hydraulic fluid drain into the container.

(f) Disconnect the extend line flexhose [23] and retract line flexhose [22] from the strut fitting.

(g) Install a hydraulic fitting plug or wrap a cotton wiper, G00034, around the extend line flexhose [23] and retract line flexhose [22] coupling nuts to catch residual hydraulic fluid that will drain from the system.

(h) Re-connect the coupling nut for the lower sync shaft and tubing [121] to the lower actuator [122].

1) Torque the coupling nut to 900 ±45 in-lb (102 ±5 N·m).

2) Loosen the coupling nut.

3) Torque the coupling nut again to 900 ±45 in-lb (102 ±5 N·m).

(i) Re-connect the coupling nut for the lower hydraulic retract line [120] to the lower actuator [122].

1) Torque the coupling nut to 270 ±13 in-lb (30.5 ±1.5 N·m).

2) Loosen the coupling nut.

3) Torque the coupling nut again to 270 ±13 in-lb (30.5 ±1.5 N·m).

(j) Install protective caps on the open ports of the upper locking actuator [21].

SUBTASK 78-31-01-480-001-G00

(6) Install the upper assembly from the thrust reverser sling equipment, SPL-14549, onto the thrust reverser.

(a) Put the forward hinge fitting assembly [51] and aft hinge fitting assembly [54] onto the hinge beam [52] (Figure 406, Figure 408).

1) Put the forward hinge fitting assembly [51] into its position on the hinge beam [52] at the forward attach point marked "GSE HOIST POINT".

2) Install the lockpins [53] to attach the forward hinge fitting assembly [51] onto the hinge beam [52].

3) Install the retention pins [55] in the end of the lockpins [53].

4) Remove the aft lockpins [93] at hinge number 4.

5) Put the aft hinge fitting assembly [54] into its position on the hinge beam [52] at the aft attach point marked "GSE HOIST POINT".

6) Install the lockpins [53] to attach the aft hinge fitting assembly [54] onto the hinge beam [52].

7) Install the retention pins [55] in the end of the lockpins [53].

(b) Put the forward and aft latch fitting assemblies [67] onto the latch beam (View C, Figure 406).

1) Remove the 4 screws from the locations that are marked "GSE" on the latch beam fairing.

2) Put the forward latch fitting assembly [67] into its position on the latch beam.

3) Install the bolts [65] and washers [66] to attach the forward latch fitting assembly [67] onto the latch beam.

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a) It is optional to use nylon or other non-metallic washers [68] between the latch fitting assemblies [67] and the thrust reverser to protect the paint.

4) Put the aft latch fitting assembly [67] into its position on the latch beam.

5) Install the bolts [65] and washers [66] to attach the aft latch fitting assembly [67] onto the latch beam.

a) It is optional to use nylon or other non-metallic washers [68] between the latch fitting assembly [67] and the thrust reverser to protect the paint.

6) Torque the four bolts [65] to 40 ±10 in-lb (4.5 ±1.1 N·m).

SUBTASK 78-31-01-480-002-G00

(7) If you will use an overhead crane to remove the thrust reverser, attach the thrust reverser sling equipment, SPL-14549, to the overhead crane (Figure 406).

(a) Attach the C-beam assembly [36] to the strongback assembly [35] with the collar [42], lockpin [41] and retention pin [43].

NOTE: The C-beam is attached in the correct position for the applicable thrust reverser when the 'FWD' arrow points in the forward direction.

WARNING

DO NOT USE A FORCE MORE THAN 1050 LB (476 KG) TO LIFT THE THRUST REVERSER. IF YOU USE A FORCE MORE THAN 1050 LB (476 KG), YOU CAN BREAK PINS. IF YOU DO NOT OBEY, INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(b) Attach the load cell [38] (load cell, STD-12395) to the triangle lift plate [31] and the master link [39].

1) Make sure that the length of the electrical cable from the load cell [38] to the handheld display has a minimum of 20 ft (6 m).

2) Use the load cell [38] to make sure that the force used when lifting the sling assembly and the thrust reverser is not more than 1050 lb (476 kg).

NOTE: The weight of the sling assembly is approximately 240 lb (109 kg). The weight of the thrust reverser is approximately 800 lb (363 kg).

(c) Connect the master link to the 30 foot height (2000 lb capacity) lift crane, STD-1095.

WARNING

MAKE SURE THAT YOU INSTALL THE HOIST HOOK OPENING WITH THE CLIP POINTED TO THE TRIANGLE LIFT PLATE. MONITOR THE HOOK TURN WHEN YOU OPERATE THE CHAIN HOIST. IF YOU DO NOT OBEY, THE THRUST REVERSER CAN FALL. IF YOU DO NOT FOLLOW INSTRUCTIONS, INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(d) Attach the hand chain hoist [37] to the triangle lift plate [31] and to the C-beam assembly [36].

(e) Attach the web sling [32] to the triangle lift plate [31] and to the strongback assembly [35] using the forward web shackle [33].

(f) Attach the upper assembly [45] to the top fitting on the strongback assembly [35] with the lockpin [46] and the retention pin [47].

NOTE: The upper assembly interfaces with the hinge fittings.

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(g) Attach the lower assembly [44] to the bottom fitting on the strongback assembly [35] with the lockpin [62] and the retention pin [61].

NOTE: The lower assembly interfaces with the latch fittings.

SUBTASK 78-31-01-480-003-G00

(8) If you will use a boom hoist to remove the thrust reverser, attach the boom hoist, SPL-2430, to the thrust reverser sling equipment, SPL-14549 (Figure 407).

(a) Attach the boom hoist [104] to the strongback assembly [35] handling flange with the two lockpins [101] and retention pins [103].

(b) Attach the upper assembly [45] to the top fitting on the strongback assembly [35] with the lockpin [46] and the retention pin [47].

NOTE: The upper assembly interfaces with the hinge fittings.

(c) Attach the lower assembly [44] to the bottom fitting on the strongback assembly [35] with the lockpin [62] and the retention pin [61].

NOTE: The lower assembly interfaces with the latch fittings.

SUBTASK 78-31-01-480-004-G00

WARNING THE TOOL WEIGHS APPROXIMATELY 240 POUNDS (109 KG), MAKE SURE THAT THE TOOL IS CORRECTLY ATTACHED TO THE OVERHEAD CRANE. IF YOU DO NOT OBEY THIS INSTRUCTION, THE TOOL CAN FALL AND INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION DO NOT ATTACH THE HOIST ASSEMBLY TO THE THRUST REVERSER SLEEVE. STRUCTURAL DAMAGE TO THE THRUST REVERSER CAN OCCUR.

(9) Attach the thrust reverser sling equipment, SPL-14549, to the thrust reverser (Figure 406, Figure 407).

(a) If you are using the overhead crane, put a protective mat, STD-585, on the leading edge of the wing to prevent damage to the surface.

NOTE: When the tool is put into position and attached to the thrust reverser the chain may make contact with the leading edge of the wing.

(b) Align the thrust reverser sling equipment, SPL-14549, to the thrust reverser with the overhead crane or the boom hoist.

(c) Position the thrust reverser sling equipment, SPL-14549, so that the flanges on the lower assembly [44] are correctly engaged with the forward latch fitting assembly [67] and aft latch fitting assembly [67].

(d) Attach the lower assembly [44] to the thrust reverser with the lockpins [64] and the retention pins [63].

(e) Position the thrust reverser sling equipment, SPL-14549, so that the clevises on the upper assembly [45] are correctly engaged with the forward hinge fitting assembly [51] and the aft hinge fitting assembly [54].

(f) Attach the upper assembly [45] to the forward hinge fitting assembly [51] and the aft hinge fitting assembly [54] with the lockpins [56] and the retention pins [55].

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SUBTASK 78-31-01-020-008-G00

WARNING THE THRUST REVERSER WEIGHS APPROXIMATELY 800 POUNDS (363 KG), MAKE SURE THAT THE WEIGHT OF THE THRUST REVERSER IS HELD BY THE TOOL. IF YOU DO NOT OBEY THIS INSTRUCTION, THE THRUST REVERSER CAN FALL AND INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

(10) Remove the thrust reverser half (Figure 408).

(a) Rotate the thrust reverser half with the thrust reverser sling equipment, SPL-14549, to get access to and to remove the load from the hinge beam bolts [83] and lockpins [93].

NOTE: At the approximate eight degree open position, the V-band disengages from the V-groove.

(b) Remove the forward lockpin [93] at hinge number 1.

(c) Remove the nuts [81] from the forward and aft hinge bolts [83].

(d) Install the thread protectors [82] on the forward and aft hinge bolt [83].

NOTE: The thread protectors are part of the GSE equipment and are in the storage box with the GSE thrust reverser sling equipment, SPL-14549.

(e) If necessary, adjust the thrust reverser sling equipment, SPL-14549, to remove the load from the bolts [83].

(f) Remove the hinge bolts [83].

WARNING MAKE SURE THAT YOU INSTALL THE HOIST HOOK OPENING WITH THE CLIP POINTED TO THE TRIANGLE LIFT PLATE. MONITOR THE HOOK TURN WHEN YOU OPERATE THE CHAIN HOIST. IF YOU DO NOT OBEY, THE THRUST REVERSER CAN FALL. IF YOU DO NOT FOLLOW INSTRUCTIONS, INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

WARNING DO NOT USE A FORCE MORE THAN 1050 LB (476 KG) TO LIFT THE THRUST REVERSER. IF YOU USE A FORCE MORE THAN 1050 LB (476 KG), YOU CAN BREAK PINS. IF YOU DO NOT OBEY, INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT YOU MONITOR THE THRUST REVERSER POSITION WHEN YOU MOVE IT AWAY FROM THE STRUT AND ENGINE. IF YOU DO NOT OBEY THIS INSTRUCTION, THE THRUST REVERSER CAN HIT THE ADJACENT STRUCTURES AND DAMAGE CAN OCCUR.

(g) Remove the thrust reverser half.

NOTE: The weight of the sling assembly is approximately 240 lb (109 kg). The weight of the thrust reverser is approximately 800 lb (363 kg). Total weight is not more than 1050 lb (476 kg).

1) Make sure that the load cell [38] handheld display electrical cable does not get twisted or tangled up.

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CAUTION

MAKE SURE THAT YOU MONITOR THE POSITION OF THE LATCH HANDLES ON THE LEFT THRUST REVERSER. THE LATCH HANDLES CAN CATCH ON THE SUPPORT EQUIPMENT. IF THE LATCH HANDLES CATCH ON THE SUPPORT EQUIPMENT, DAMAGE TO THE LATCHES CAN OCCUR.

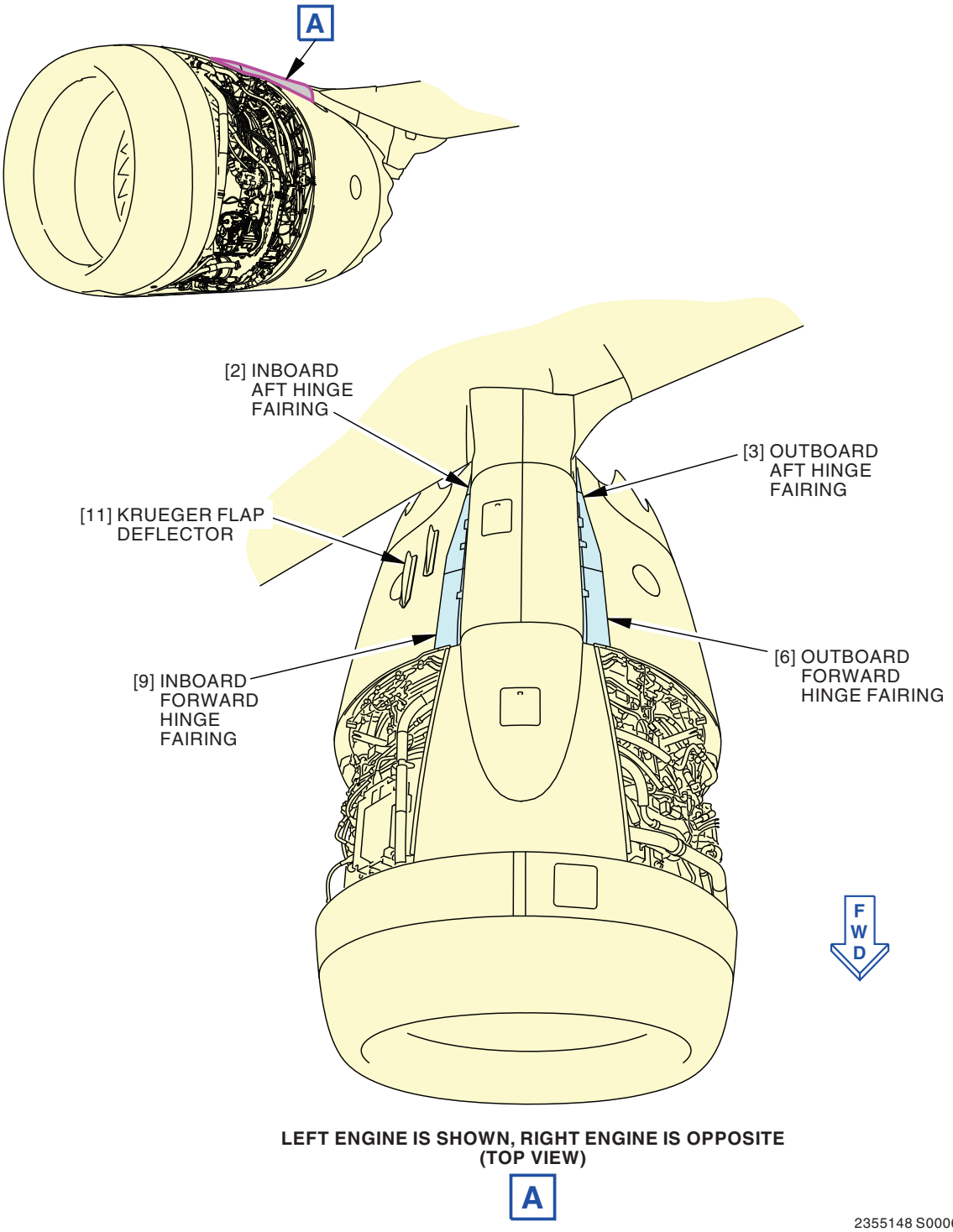
(h) Put the thrust reverser on the TR stand, SPL-14555, or on a suitable pallet and dunnage.

SUBTASK 78-31-01-860-061-G00

(11) If it is necessary, remove the thrust reverser sling equipment, SPL-14549, from the thrust reverser and remove the boom hoist, SPL-2430, from the thrust reverser sling equipment, SPL-14549 (TASK 78-31-01-400-801-G00).

——— END OF TASK ———

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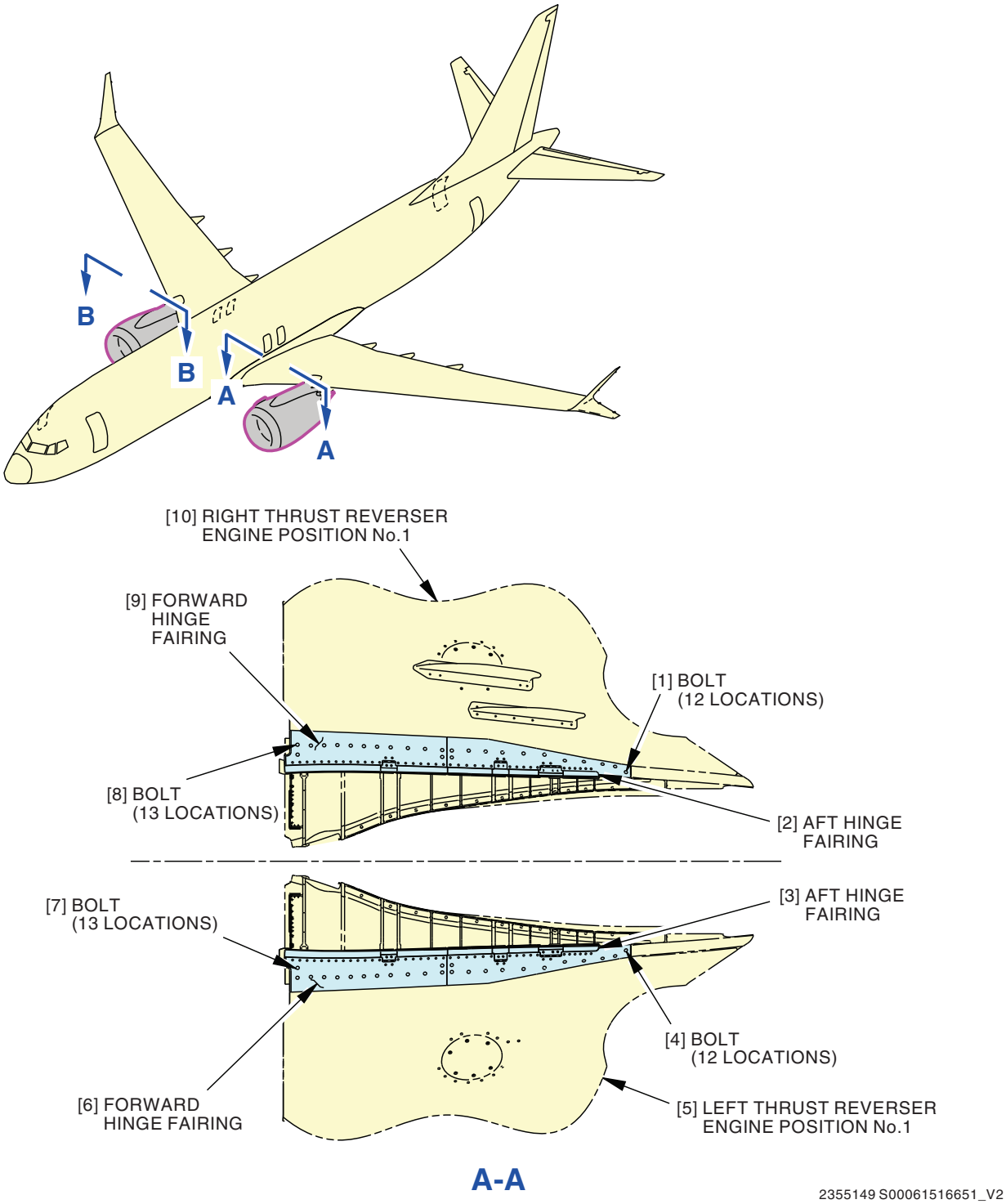


LEFT ENGINE IS SHOWN, RIGHT ENGINE IS OPPOSITE (TOP VIEW)

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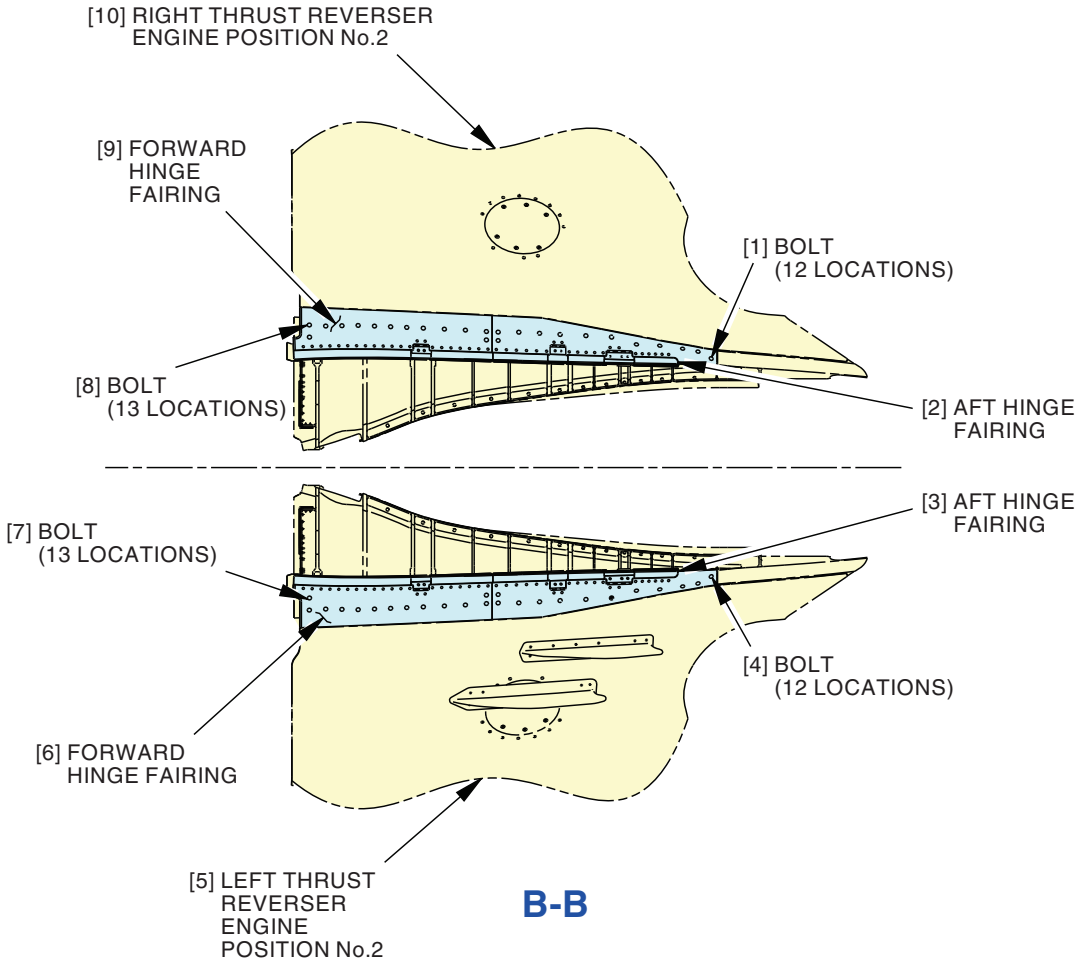
Thrust Reverser Hinge Fairing Installation
Figure 401/78-31-01-990-801-G00 (Sheet 1 of 3)

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Thrust Reverser Hinge Fairing Installation
Figure 401/78-31-01-990-801-G00 (Sheet 2 of 3)

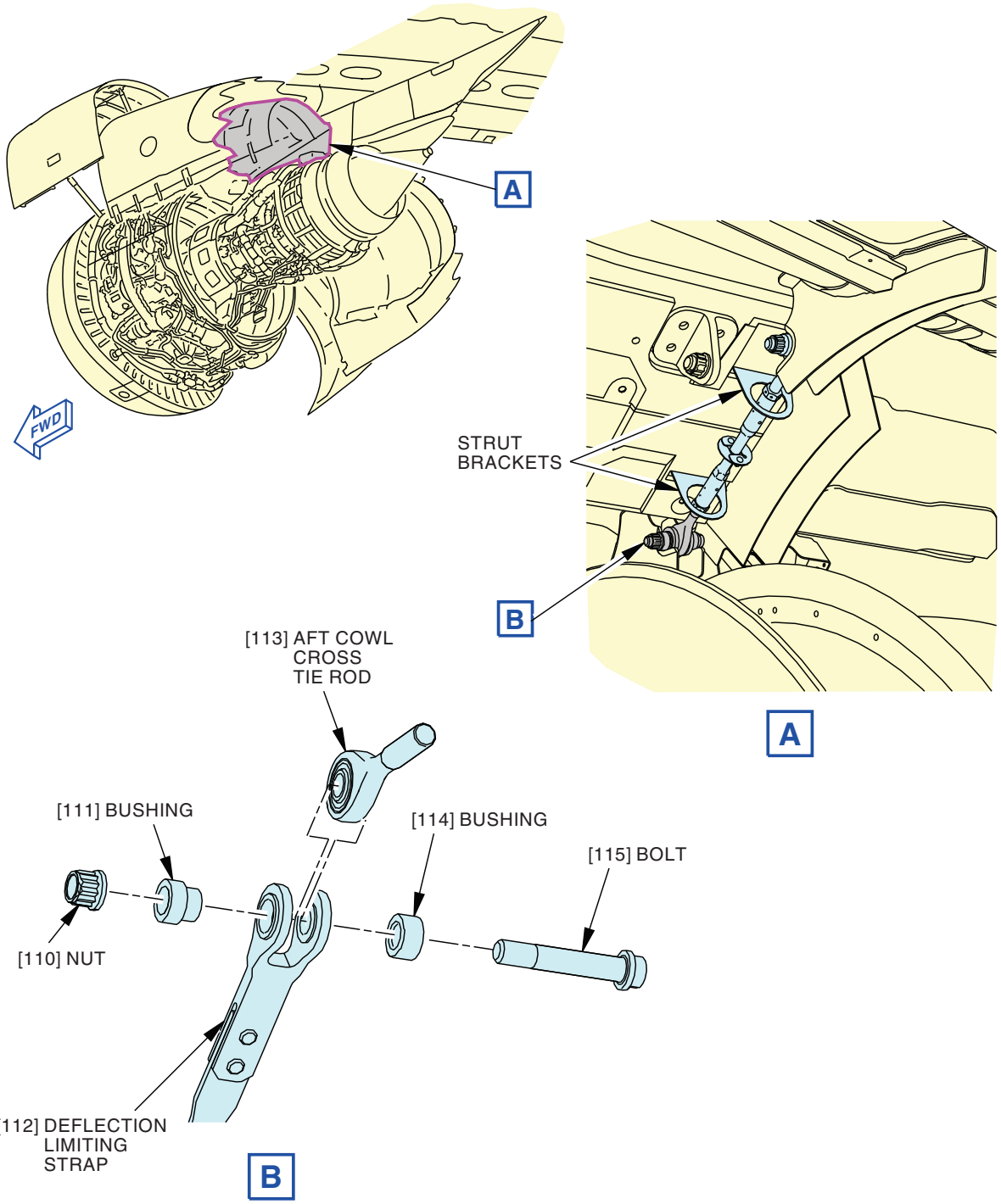
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2355150 S00061516652_V2

Thrust Reverser Hinge Fairing Installation
Figure 401/78-31-01-990-801-G00 (Sheet 3 of 3)

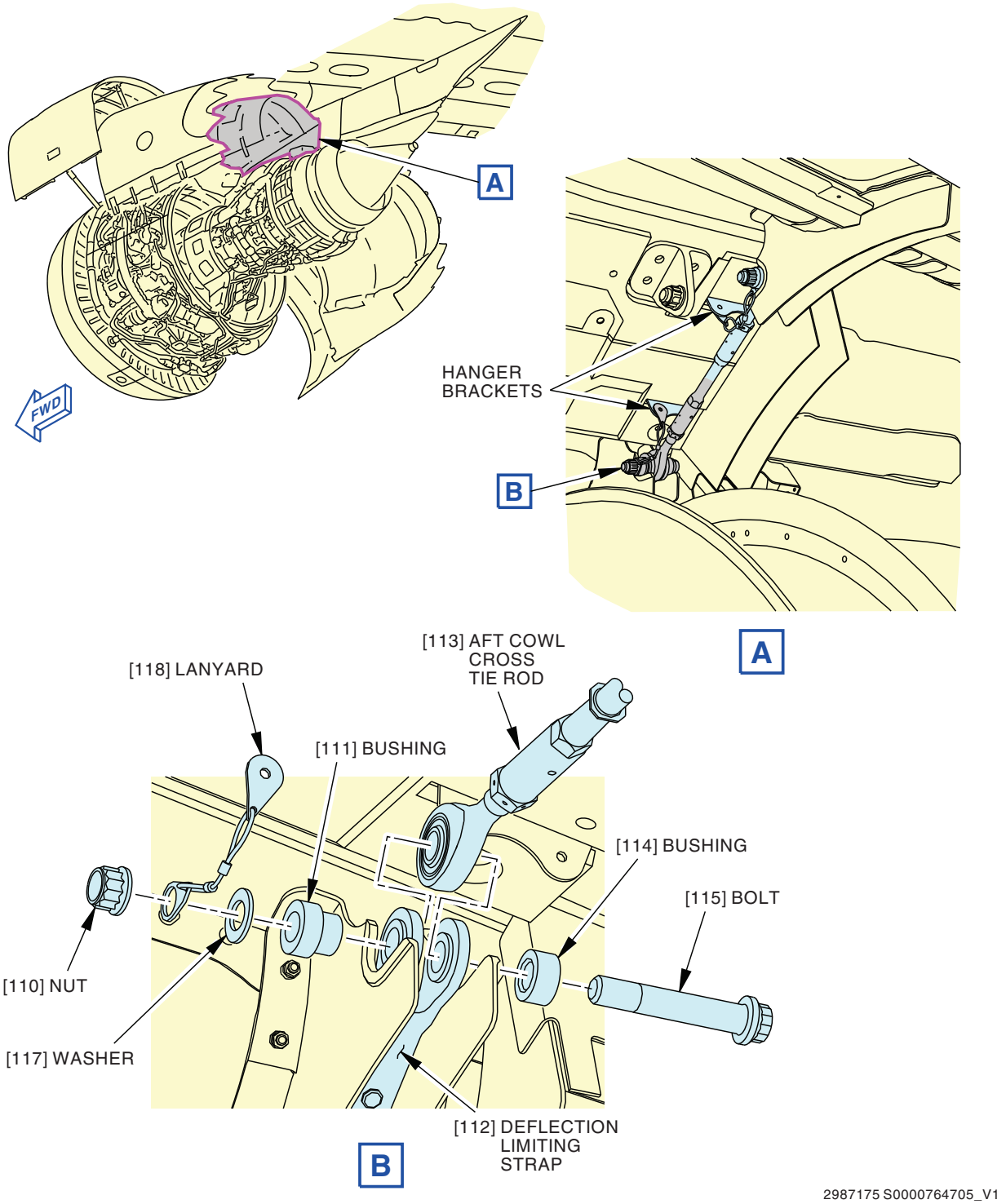
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2471530 S0000578815_V2

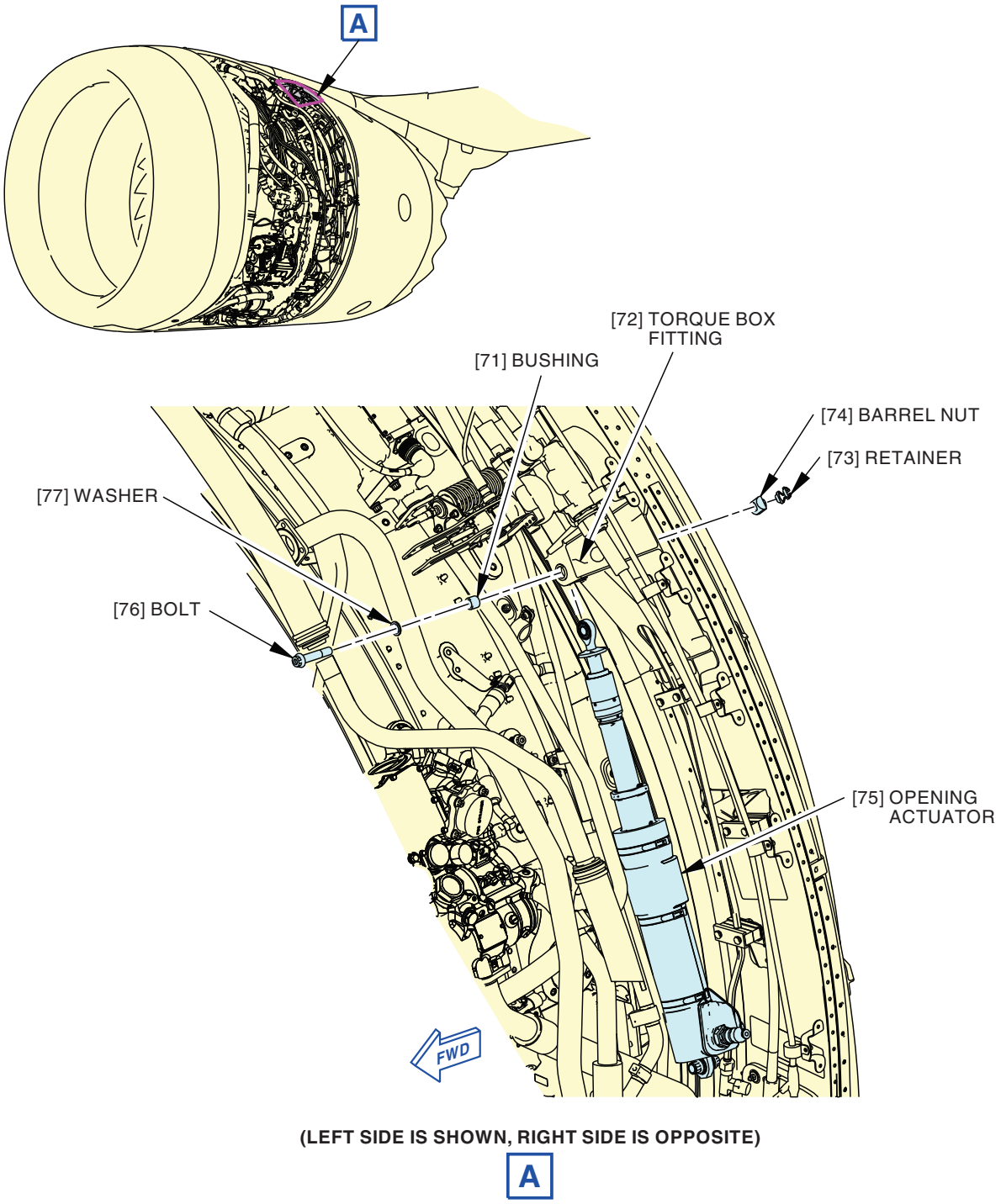
Thrust Reverser Cross Tie Rod Installation
Figure 402/78-31-01-990-814-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Thrust Reverser Cross Tie Rod Installation
Figure 402/78-31-01-990-814-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



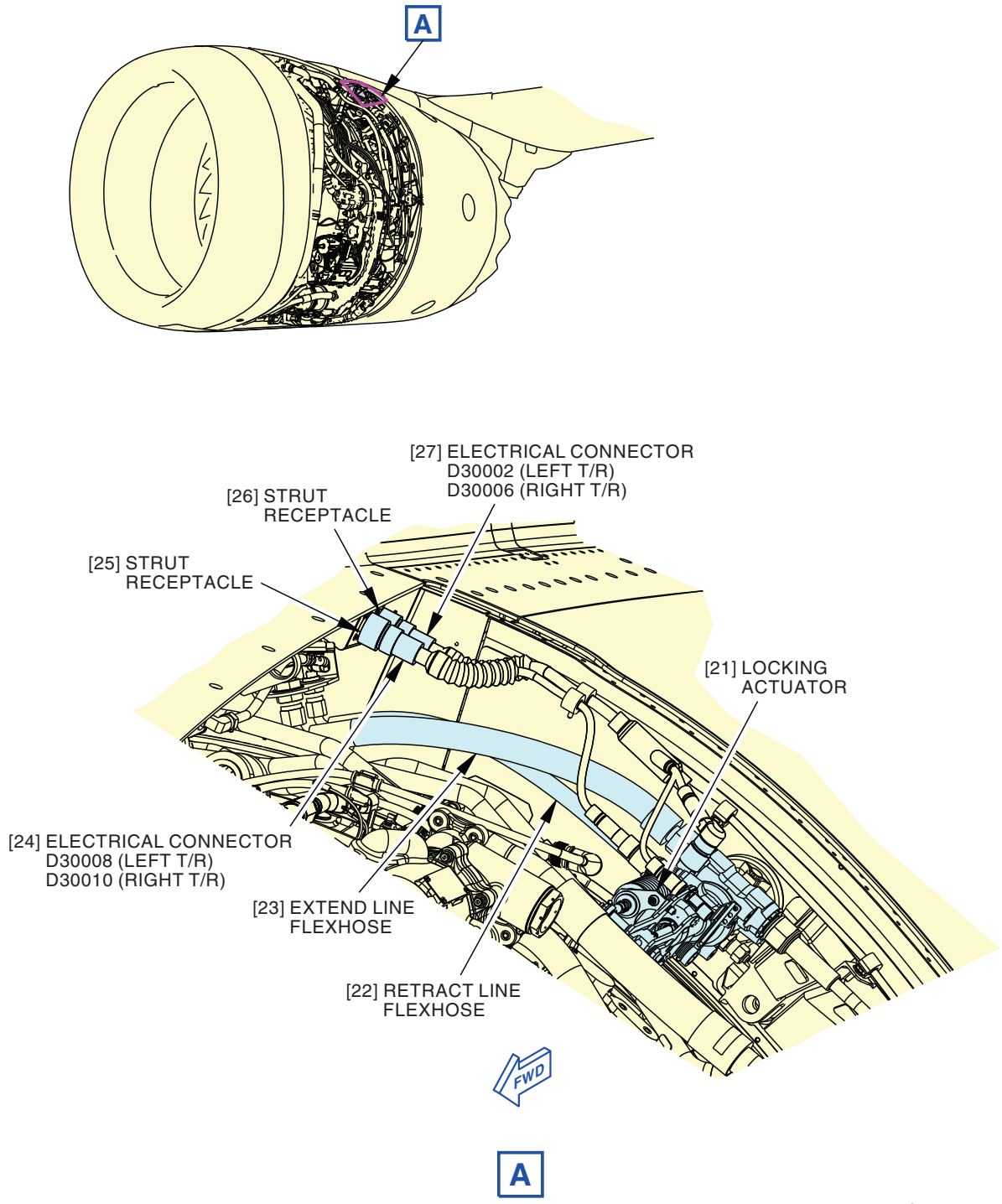
(LEFT SIDE IS SHOWN, RIGHT SIDE IS OPPOSITE)

A

2355160 S00061516662_V4

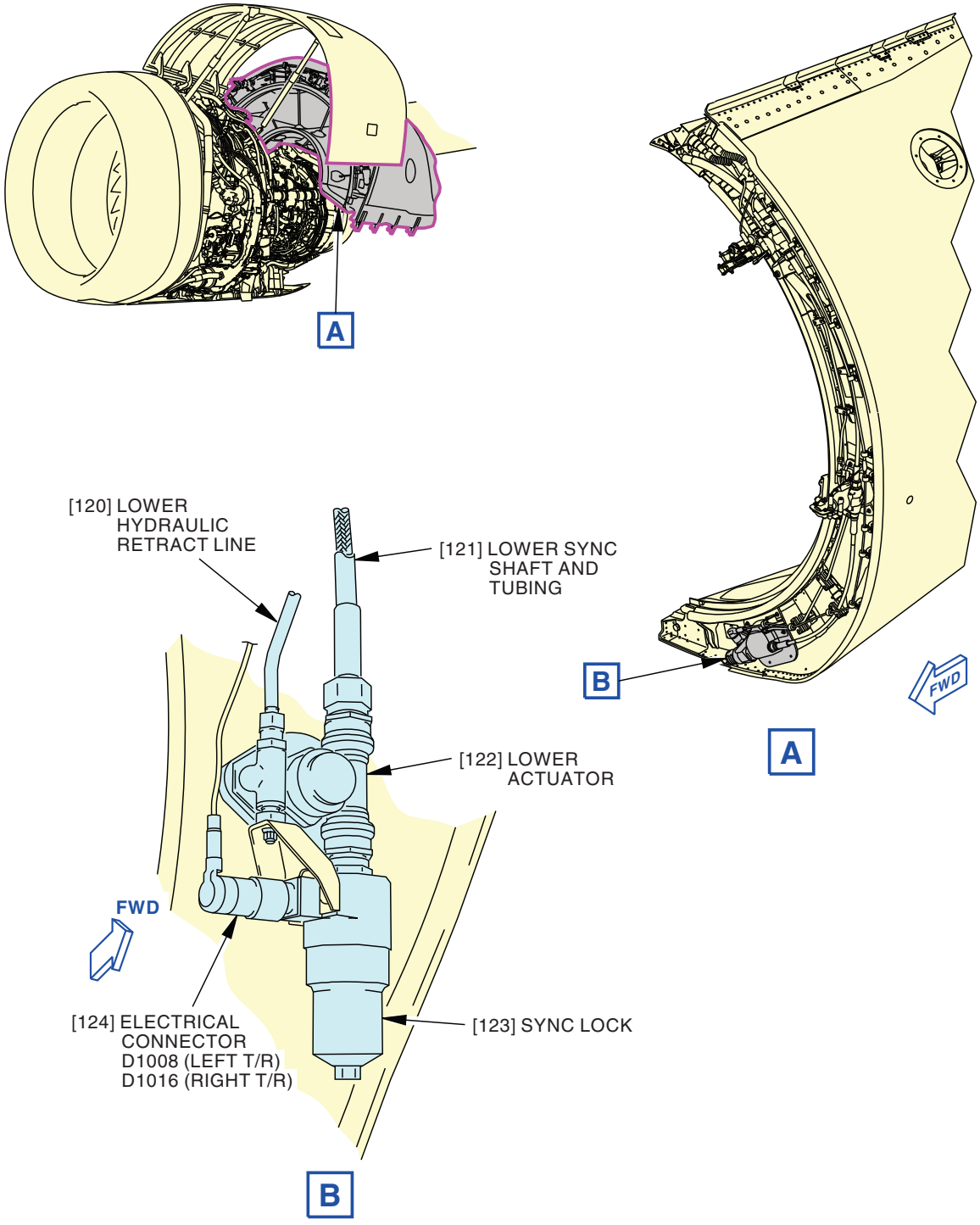
Thrust Reverser Opening Actuator Disconnect
Figure 403/78-31-01-990-815-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Hydraulic and Electric Connections
Figure 404/78-31-01-990-802-G00

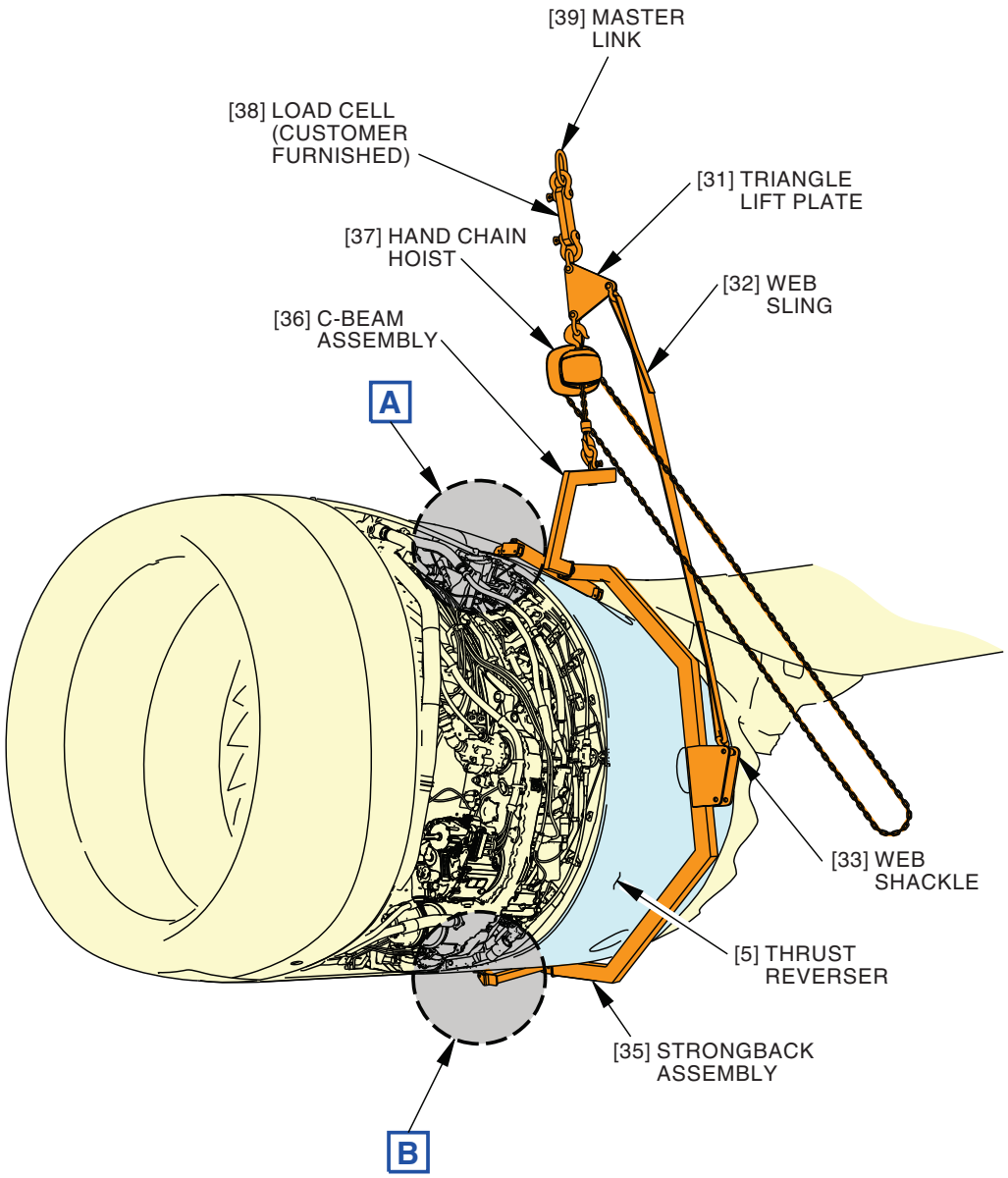
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2480896 S0000582402_V1

Thrust Reverser Lower Hydraulic Actuator
Figure 405/78-31-01-990-816-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

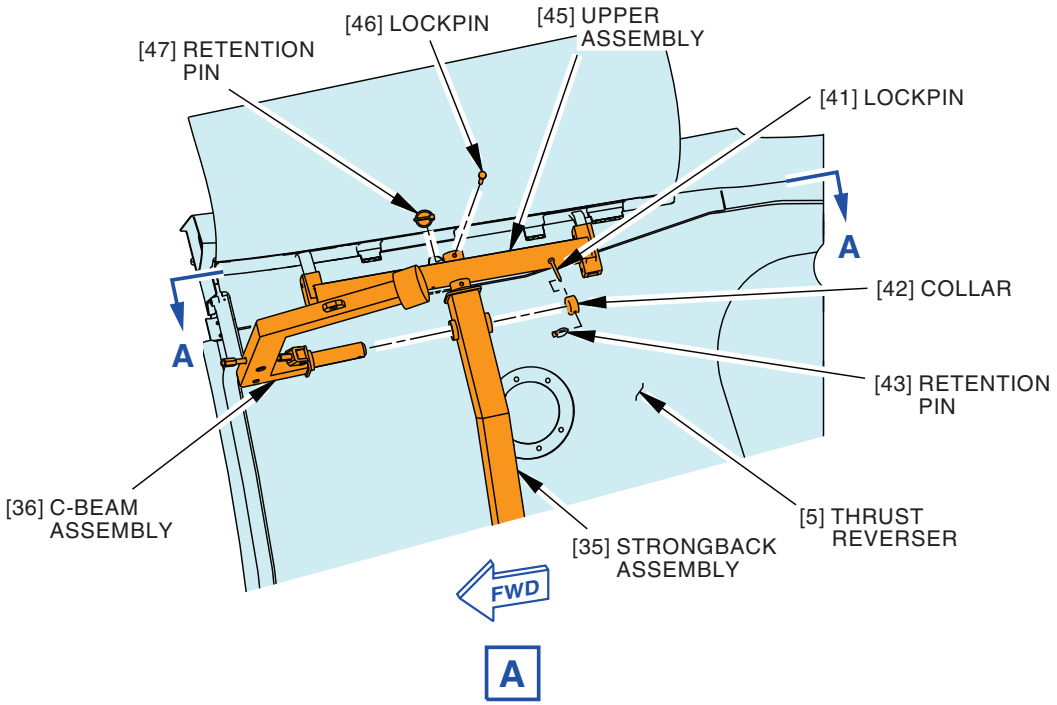


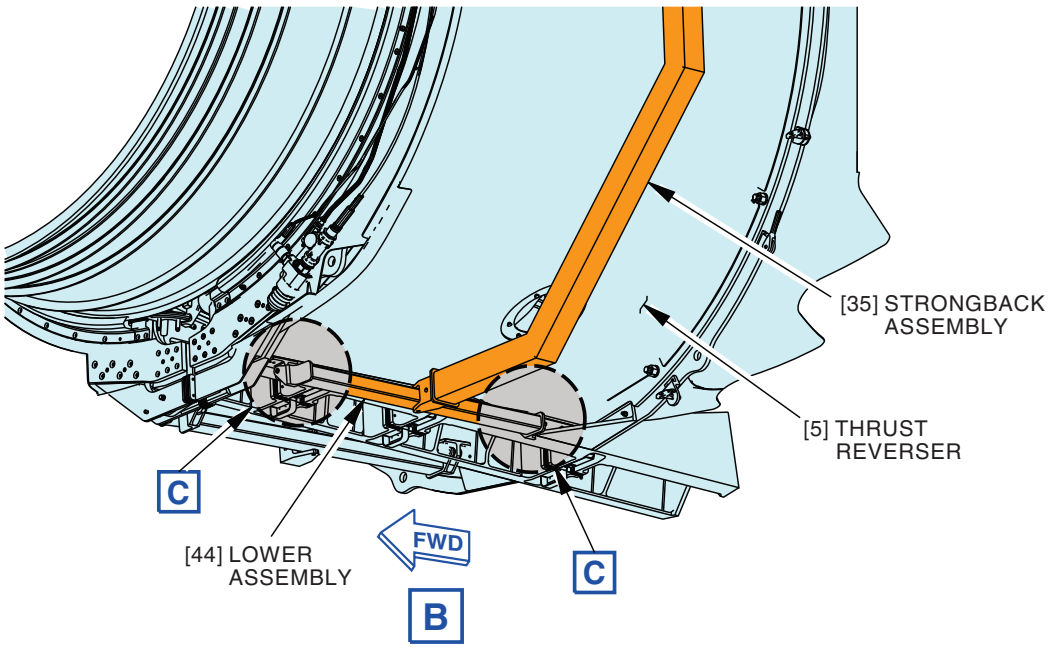
SLING ASSEMBLY WITH OVERHEAD CRANE

2355152 S00061516654_V2

Thrust Reverser (Sling Assembly with Overhead Crane) Installation
Figure 406/78-31-01-990-803-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

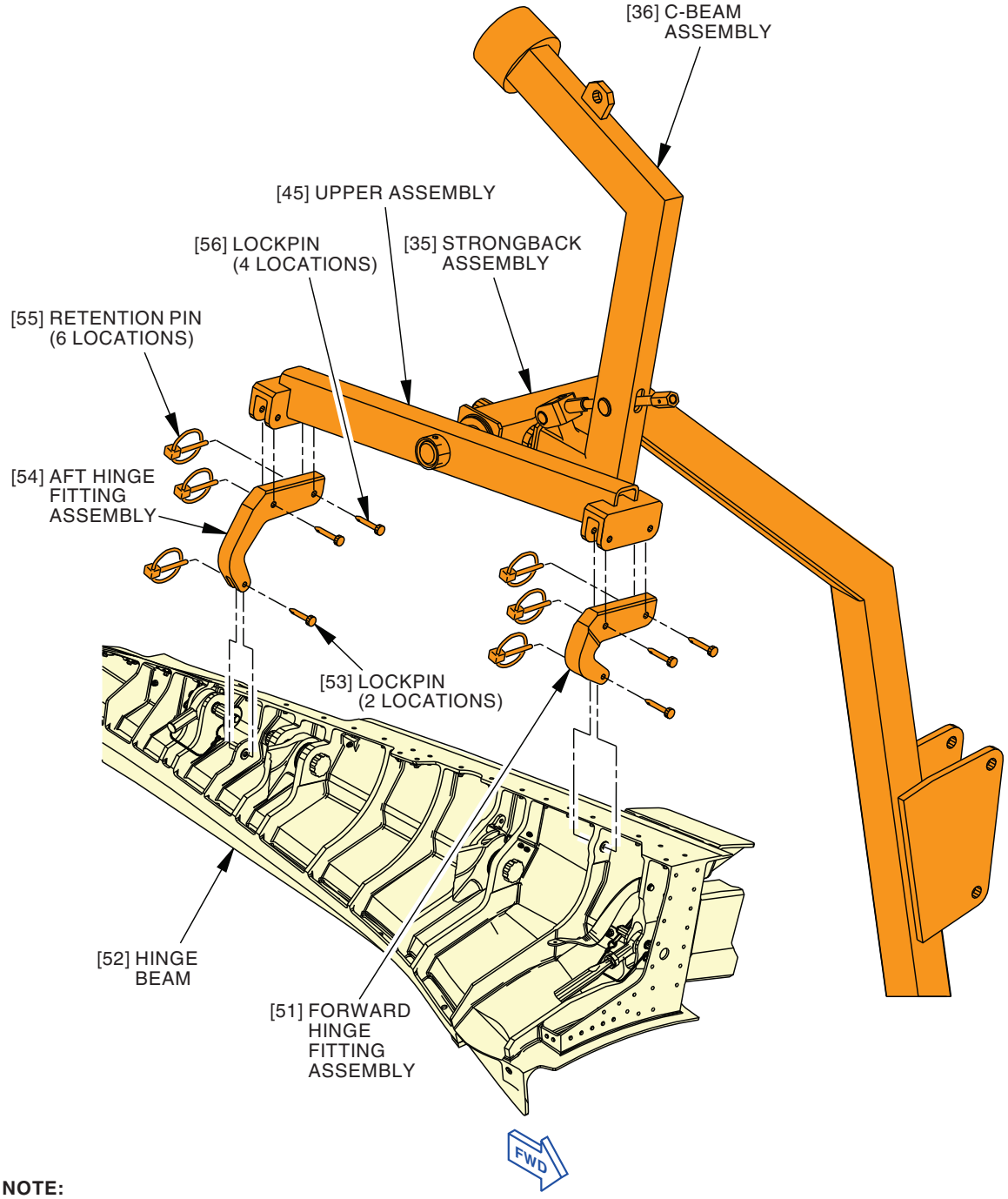




2355153 S00061516655_V1

Thrust Reverser (Sling Assembly with Overhead Crane) Installation
Figure 406/78-31-01-990-803-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



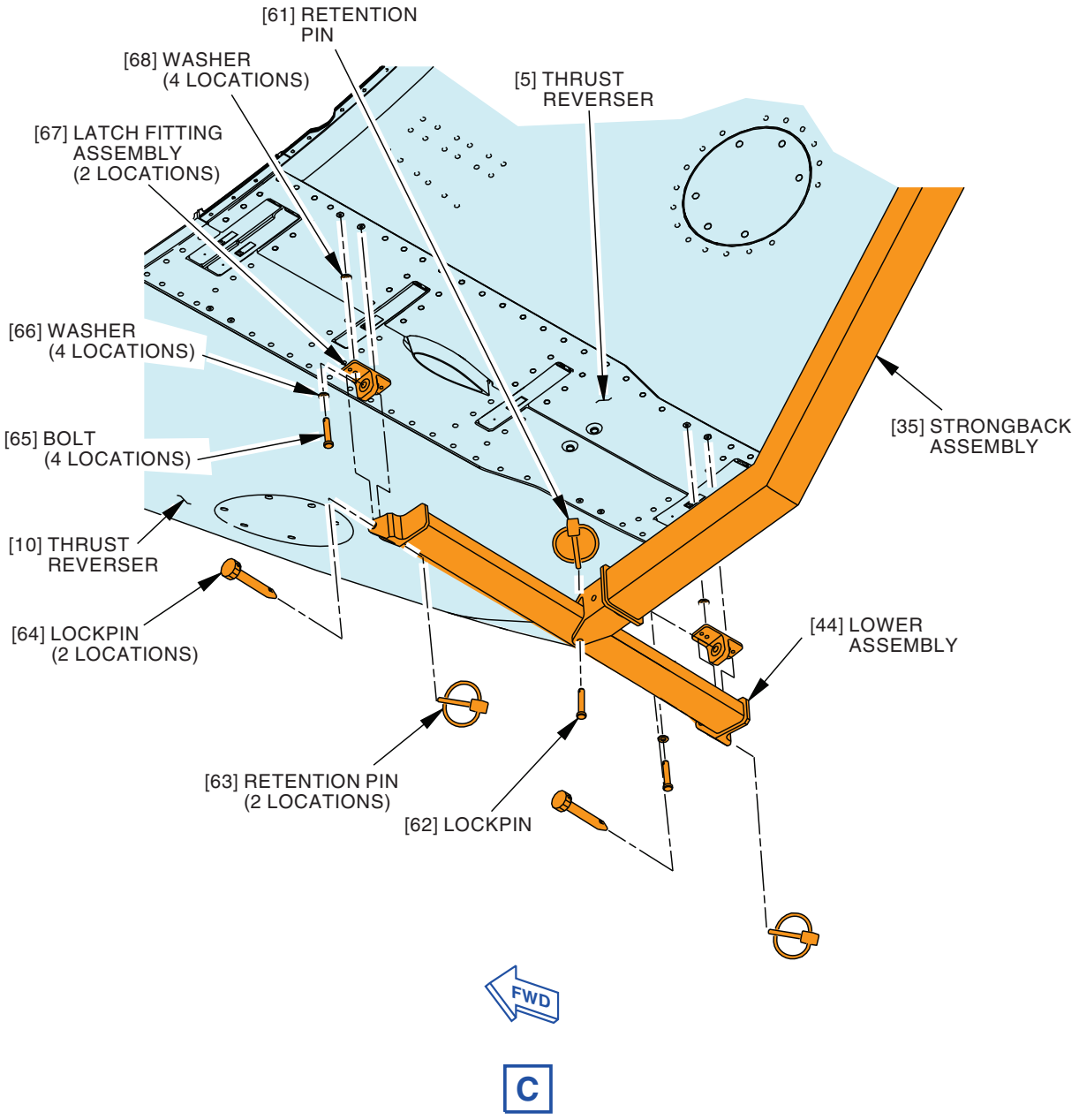
NOTE:

SOME THRUST REVERSER COMPONENTS NOT SHOWN.

2355154 S00061516656_V2

Thrust Reverser (Sling Assembly with Overhead Crane) Installation
Figure 406/78-31-01-990-803-G00 (Sheet 3 of 4)

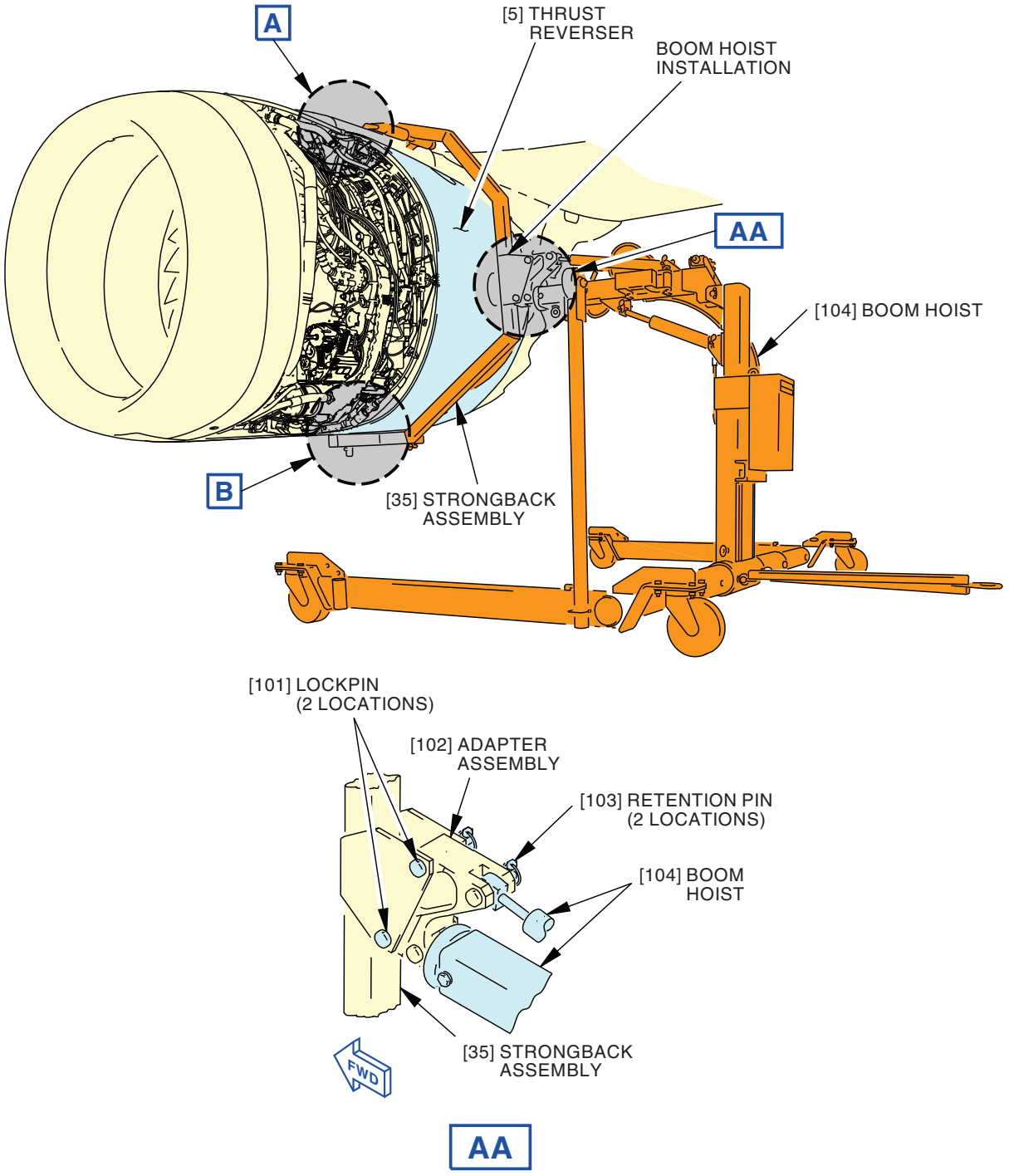
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2355155 S00061516657_V2

Thrust Reverser (Sling Assembly with Overhead Crane) Installation
Figure 406/78-31-01-990-803-G00 (Sheet 4 of 4)

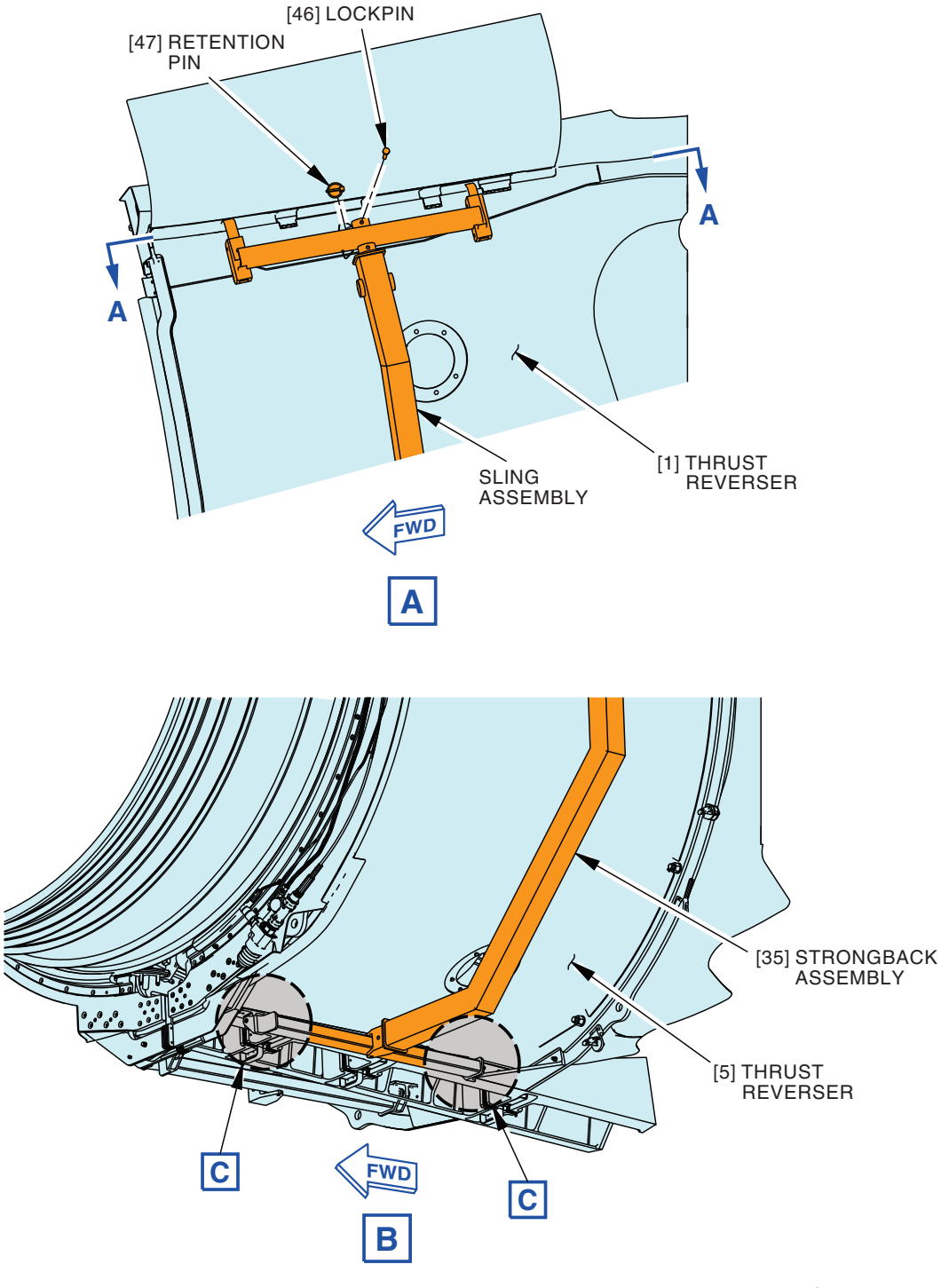
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2355156 S00061516658_V2

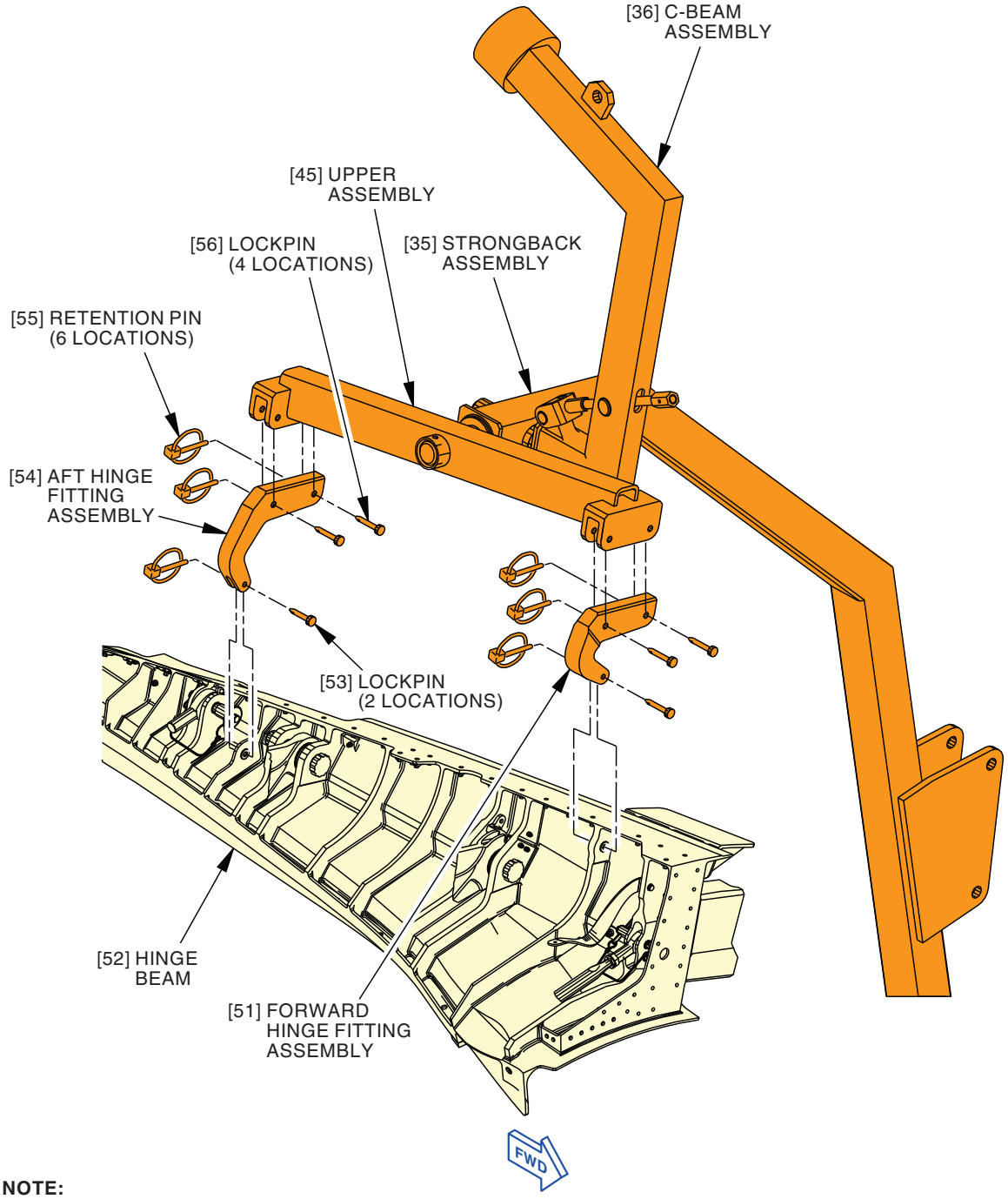
Thrust Reverser (Sling Assembly with Boom Hoist) Installation
Figure 407/78-31-01-990-804-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Thrust Reverser (Sling Assembly with Boom Hoist) Installation
Figure 407/78-31-01-990-804-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



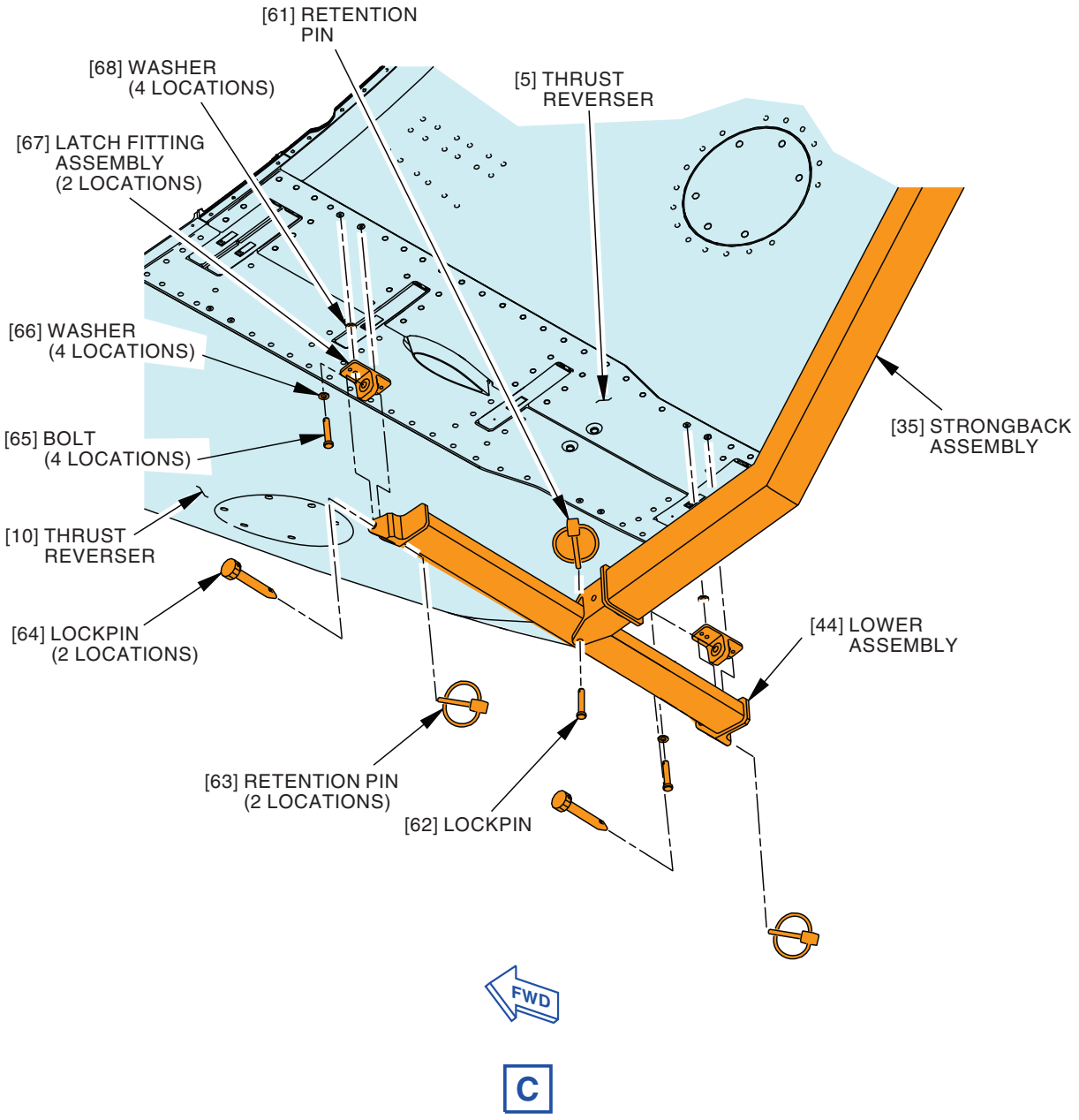
NOTE:

SOME THRUST REVERSER COMPONENTS NOT SHOWN.

2355158 S00061516660_V2

Thrust Reverser (Sling Assembly with Boom Hoist) Installation
Figure 407/78-31-01-990-804-G00 (Sheet 3 of 4)

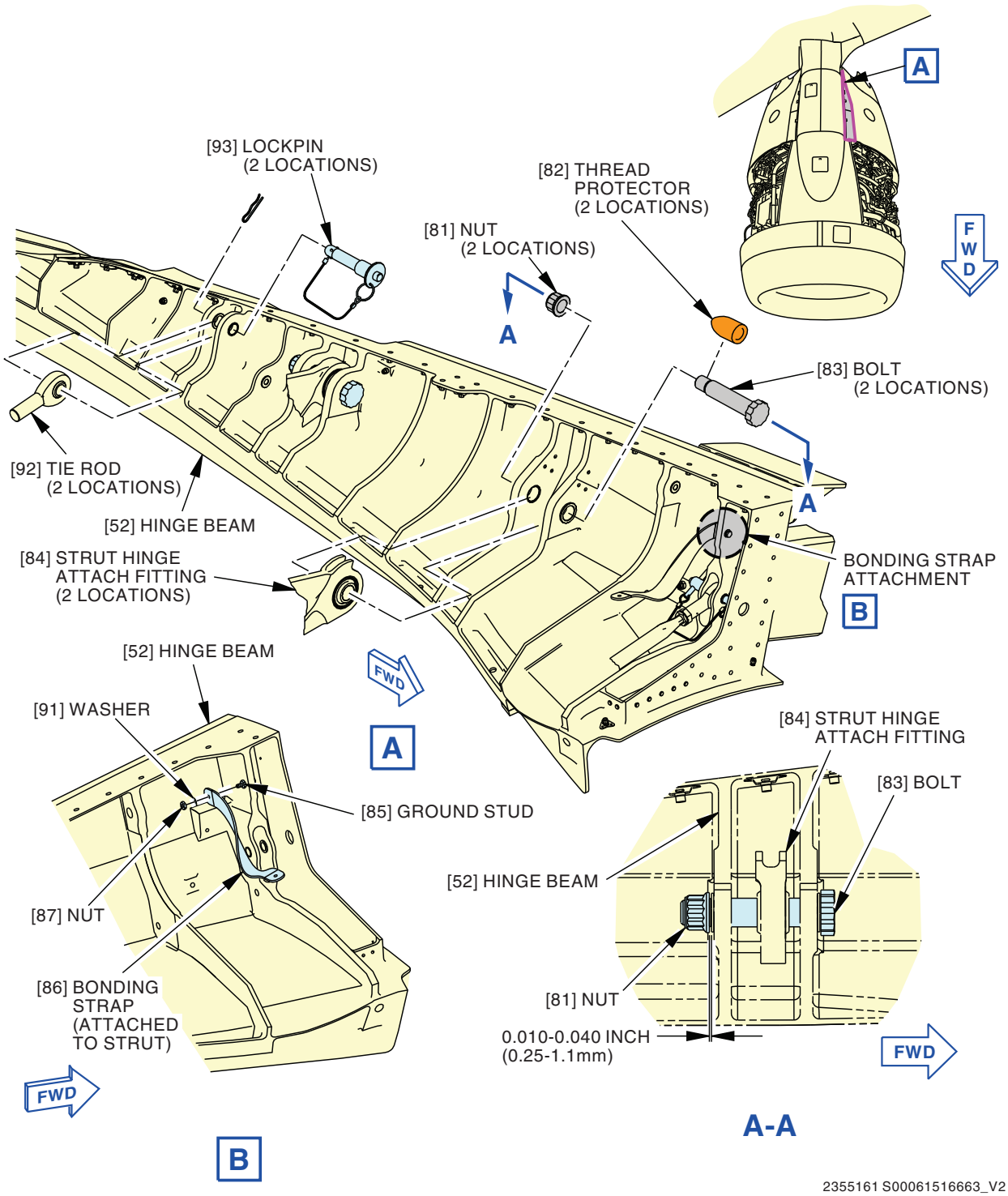
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2355159 S00061516661_V2

Thrust Reverser (Sling Assembly with Boom Hoist) Installation
Figure 407/78-31-01-990-804-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Thrust Reverser Installation
Figure 408/78-31-01-990-806-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-31-01-400-801-G00

3. Thrust Reverser Installation

A. General

(1) This task gives the instructions to install the left or right thrust reverser half.

NOTE: The total acoustic panel blockage for the engine installation must be recalculated when components are changed per SRM SUBJECT 54-00-01.

(2) It is acceptable to have one thrust reverser half with composite inner wall and the other thrust reverser half with titanium inner wall, on the same engine.

(3) You must do the thrust reverser adjustment task after you install the thrust reverser.

NOTE: You must do the thrust reverser adjustment if you re-install the thrust reverser that you removed, or if you install a different thrust reverser.

B. References

Reference	Title
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
29-00-00-790-801	Hydraulic System External Leakage Check (P/B 601)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
54-51-06-000-801	Thrust Reverser Cross Tie Rod Lanyard Removal (P/B 401)
54-51-06-400-801	Thrust Reverser Cross Tie Rod Lanyard Installation (P/B 401)
71-11-04-400-801-G00	Fan Cowl Panels Installation (P/B 401)
78-00-00-040-801-G00	DDG 78-31-01 Thrust Reversers Inoperative - Preparation (P/B 901)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-00-700-801-G00	Thrust Reverser Operational Test (Selection) (P/B 501)
78-31-01-280-801-G00	Thrust Reverser Rigging Check (P/B 601)
78-31-01-820-801-G00	Thrust Reverser Adjustment (P/B 501)
78-31-05-000-801-G00	Cascade Removal (P/B 401)
78-31-05-400-801-G00	Cascade Installation (P/B 401)
78-31-09-420-801-G00	Krueger Flap Deflector Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

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(Continued)

Reference	Description
SPL-2430	Hoist ? Boom, 800 lbs WLL, 250 lbs Load Positioner Compatible Part #: C20002-267 Supplier: 81205
SPL-14549	Equipment - Sling, Thrust Reverser, LEAP-1B Engine Part #: C78032-75 Supplier: 81205 Opt Part #: C78032-1 Supplier: 81205
STD-585	Mat - Protective, Closed Cell Neoprene, 30-50 duro, Weather and Oil Resistant, 3/8 Inch Min Thickness, Minimum 42x60 Inches with Warning Streamers
STD-1095	Crane - Lift, 2000 lb Capacity, 30 Foot Height

D. Consumable Materials

Reference	Description	Specification
A00648	Compound - Retaining - Loctite 242	ASTM D5363 Grp 3 Cl 2 Gd 1 (SUPERSEDES MIL-S-46163)
B50073	Alcohol - Isopropyl	ASTM D 770
D00633	Grease - Aircraft General Purpose	BMS3-33
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G50625	Strap - Plastic, Adjustable, Self-locking, 14.40 Inches (365.76 mm) Long	BACS38K3
G51905	Putty/Lacquer - Tamper-proof, Hydraulic Fluid Resistant, Yellow	BMS8-45 Type III

E. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

F. Access Panels

Number	Name/Location
415AL	Left Forward Thrust Reverser Hinge Fairing, Engine 1
415BL	Left Aft Thrust Reverser Hinge Fairing, Engine 1
416AR	Right Forward Thrust Reverser Hinge Fairing, Engine 1
416BR	Right Aft Thrust Reverser Hinge Fairing, Engine 1
425AL	Left Forward Thrust Reverser Hinge Fairing, Engine 2
425BL	Left Aft Thrust Reverser Hinge Fairing, Engine 2
426AR	Right Forward Thrust Reverser Hinge Fairing, Engine 2
426BR	Right Aft Thrust Reverser Hinge Fairing, Engine 2

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G. Prepare for the Installation

SUBTASK 78-31-01-860-007-G00

(1) Make sure that the thrust reverser you will install is compatible with the engine and airplane.

NOTE: If applicable, refer to the Boeing Service Bulletin for a list of all concurrent requirements.

(a) If an engine will have one thrust reverser half with composite inner wall and one thrust reverser half with titanium inner wall, and if the Fan Air Modulating Valve (FAMV) is inoperative, make sure that the thrust reverser is inoperative (TASK 78-00-00-040-801-G00).

SUBTASK 78-31-01-480-005-G00

WARNING THE TOOL WEIGHS APPROXIMATELY 240 POUNDS (109 KG), MAKE SURE THAT THE TOOL IS CORRECTLY ATTACHED TO THE OVERHEAD CRANE. IF YOU DO NOT OBEY THIS INSTRUCTION, THE TOOL CAN FALL AND INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION DO NOT ATTACH THE HOIST ASSEMBLY TO THE THRUST REVERSER SLEEVE. STRUCTURAL DAMAGE TO THE THRUST REVERSER CAN OCCUR.

(2) Prepare for the installation of the thrust reverser (Figure 401, Figure 408).

(a) If the left thrust reverser half is to be installed, make sure that the latches are taped in the closed position.

NOTE: This will prevent damage to the latches.

(b) If the thrust reverser sling equipment, SPL-14549, is not installed on the thrust reverser, install it (TASK 78-31-01-000-801-G00).

(c) Make sure that the applicable hinge beam fairings are removed from the thrust reverser that you will install (Figure 401).

1) If it is necessary, remove the applicable hinge beam fairings (TASK 78-31-01-000-801-G00).

(d) Install the thread protectors [82] on the two hinge bolts [83].

NOTE: The thread protectors are part of the GSE equipment and are in the storage box with the GSE thrust reverser sling equipment, SPL-14549.

CAUTION DO NOT APPLY GREASE TO OTHER SURFACES. IF YOU APPLY GREASE TO OTHER SURFACES, IT CAN CAUSE DAMAGE.

(e) Apply grease, D00633, to the shanks of the two hinge bolts [83].

(f) Apply grease, D00633, to the forward and aft lockpins [93].

(g) If the inboard thrust reverser is to be installed using the overhead crane, put a protective mat, STD-585, on the leading edge of the wing to prevent damage to the surface.

NOTE: When the tool and the thrust reverser are put into position, the chain on the chain hoist will hit the leading edge of the wing.

(h) If the fan cowl panels are installed, make sure that the panels are fully open and that the fan cowl panel hold open rods are in place and locked.

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1) If it is necessary, open and lock the fan cowl panels in the open position (TASK 78-31-01-000-801-G00).

(i) Make sure that the leading edge flaps and slats devices are retracted and deactivated.

1) If it is necessary, retract and deactivate the leading edge devices (TASK 78-31-01-000-801-G00).

(j) Make sure that the applicable hydraulic system is depressurized and deactivated.

1) If it is necessary, depressurize and deactivate the applicable hydraulic system (TASK 78-31-01-000-801-G00).

(k) Make sure that the applicable circuit breakers are open.

1) If it is necessary, open the applicable circuit breakers (TASK 78-31-01-000-801-G00).

H. Thrust Reverser Installation

SUBTASK 78-31-01-420-002-G00

WARNING THE THRUST REVERSER WEIGHS APPROXIMATELY 800 POUNDS (363 KG), MAKE SURE THAT THE WEIGHT OF THE THRUST REVERSER IS HELD BY THE TOOL. IF YOU DO NOT OBEY THIS INSTRUCTION, THE THRUST REVERSER CAN FALL AND INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT YOU MONITOR THE THRUST REVERSER POSITION WHEN YOU MOVE IT AWAY FROM THE STRUT AND ENGINE. IF YOU DO NOT OBEY THIS INSTRUCTION, THE THRUST REVERSER CAN HIT THE ADJACENT STRUCTURES AND DAMAGE CAN OCCUR.

CAUTION MAKE SURE THAT THE THRUST REVERSER V-BLADE ALIGNS WITH (AND FULLY ENGAGES), THE V-GROOVE ON THE FAN CASE WHEN YOU INSTALL THE THRUST REVERSER. IF YOU DO NOT ALIGN (AND FULLY ENGAGE), THE V-BLADE AND V-GROOVE, DAMAGE TO THE THRUST REVERSER CAN OCCUR.

(1) Install the thrust reverser half (Figure 408).

(a) Position the thrust reverser half so that the thrust reverser hinge beam [52] is near the strut hinge attach fittings [84].

(b) Put the thrust reverser half into the approximately eight degree open position.

NOTE: At the eight degree open position, it will be easier to install the fasteners.

(c) Make sure that the V-blades on the thrust reverser half and the V-grooves on the engine are aligned.

(d) Align the thrust reverser half with the strut hinge attach fittings [84] at hinge numbers 2 and 3.

1) Make sure that the cams located at hinge number 2 are correctly aligned.

(e) Install the forward hinge bolt [83] into hinge number 2 from the forward side.

(f) Remove the thread protector [82] from the forward hinge bolt [83] and install the nut [81] hand tight.

(g) Install the aft hinge bolt [83] into hinge number 3 from the forward side.

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(h) Remove the thread protector [82] from the aft hinge bolt [83] and install the nut [81] hand tight.

(i) Lower the thrust reverser.

1) Make sure that the opening actuator [75] is aligned with the torque box fitting [72].

(j) Remove the thrust reverser sling equipment, SPL-14549, from the thrust reverser (Figure 406, Figure 407).

1) Remove the lockpins [56] and retention pins [55] to disconnect the upper assembly [45] from the forward hinge fitting assembly [51] and the aft hinge fitting assembly [54].

2) Remove the retention pins [63] and lockpins [64] to disconnect the lower assembly [44] from the thrust reverser.

3) Disconnect the thrust reverser sling equipment, SPL-14549, from the thrust reverser.

4) If you used the overhead crane for installation, remove a protective mat, STD-585, from the leading edge of the wing.

5) Remove the forward and aft latch fitting assemblies [67] from the latch beam:

a) Remove the bolts [65] and washers [66] to remove the aft latch fitting assembly [67] from the latch beam.

b) Remove the bolts [65] and washers [66] to remove the forward latch fitting assembly [67] from the latch beam.

c) Install the four screws where the latch fitting assemblies [67] were attached to the thrust reverser, and torque to 50 ±5 in-lb (5.6 ±0.6 N·m).

6) Remove the forward hinge fitting assembly [51] and aft hinge fitting assembly [54] from the hinge beam [52] (Figure 407, Figure 408).

a) Remove the retention pin [55] from the end of the lockpin [53].

b) Remove the lockpin [53] to disconnect the aft hinge fitting assembly [54] from the hinge beam [52].

c) Remove the retention pin [55] from the end of the lockpin [53].

d) Remove the lockpin [53] to disconnect the forward hinge fitting assembly [51] from the hinge beam [52].

(k) Torque the number 2 and 3 hinge bolts [83] to 600 ±100 in-lb (68 ±12 N·m).

1) Apply an approximately 0.125 in. (3.175 mm) wide tamper-proof putty/lacquer, G51905, to extend across the junction of the nut [81] to bolt [83] thread.

(l) Align the ends of the forward and aft tie rods [92] with the hinge number 1 and hinge number 4 locations on the hinge beam (Figure 408).

1) Install the lockpins [93].

a) If adjustment is needed to install the lockpins [93], do these steps:

<1> Break the Loctite seal on the nuts on the ends of the tie rod assembly.

<2> Clean the tie rod assembly of any remaining locking compound.

<3> Adjust the tie rod by adjusting both ends of the tie rod using the same number of turns for each rod end.

<4> Apply Loctite 242 compound, A00648, to the tie rod end assembly.

<5> Torque the nuts to 102.5 ±2.5 in-lb (11.6 ±0.3 N·m).

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b) Install the aft lockpin [93] into hinge no. 4 from the forward side.

c) Install the forward lockpin [93] into hinge no. 1 from the aft side.

d) Make sure that the lockpin [93] ball locks are engaged.

NOTE: When the ball locks are correctly engaged, the lockpin will not pull out of the fitting.

e) Install the retention pins onto the lockpins [93].

(m) If you used the boom hoist to install the thrust reverser, remove the boom hoist, SPL-2430, from the thrust reverser sling equipment, SPL-14549 (Figure 407).

1) Remove the retention pin [61] and the lockpin [62] to disconnect the lower assembly [44] from the bottom fitting on the strongback assembly [35].

2) Remove the retention pin [47] and the lockpin [46] to disconnect the upper assembly [45] from the top fitting on the strongback assembly [35].

3) Remove the two lockpins [101] and the two retention pins [103] to remove the boom hoist [104] from the strongback assembly [35] handling flange.

(n) If you used an overhead crane to install the thrust reverser, remove the thrust reverser sling equipment, SPL-14549, from the overhead crane (Figure 406).

1) Remove the lockpin [62] and the retention pin [61] to disconnect the lower assembly [44] from the bottom fitting on the strongback assembly [35].

2) Remove the lockpin [46] and the retention pin [47] to disconnect the upper assembly [45] from the top fitting on the strongback assembly [35].

3) Disconnect the web sling [32] from the triangle lift plate [31] and from the strongback assembly [35].

4) Remove the hand chain hoist [37] from the triangle lift plate [31] and from the C-beam assembly [36].

5) Disconnect the master link from the 30 foot height (2000 lb capacity) lift crane, STD-1095.

6) Remove the load cell [38] from the triangle lift plate [31] and from the master link [39].

7) Remove the retention pin [43], lockpin [41] and collar [42] to disconnect the C-beam assembly [36] from the strongback assembly [35].

(o) Attach the thrust reverser opening actuator [75] to the thrust reverser (Figure 403).

CAUTION

DO NOT APPLY GREASE TO OTHER SURFACES. IF YOU APPLY GREASE TO OTHER SURFACES, IT CAN CAUSE DAMAGE.

1) Apply a thin film of grease, D00633, onto the shank of the bolt [76].

2) Remove the adjustable plastic strap, G50625, that secures the opening actuator [75] to the engine.

3) Put the opening actuator [75] into its position on the thrust reverser torque box fitting [72].

4) Install the bolt [76], washer [77], bushing [71], barrel nut [74], and retainer [73] that attaches the actuator to the thrust reverser.

5) Torque the bolt [76] to 350 ±10 in-lb (39.5 ±1.2 N·m).

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SUBTASK 78-31-01-410-001-G00

(2) Connect the electrical bonding strap [86] to the hinge beam [52] (Figure 408).

(a) Use alcohol, B50073, or other approved solvent, and a cotton wiper, G00034, to clean the electrical the bonding strap [86], ground stud [85], area around the ground stud, washers [91], and nut [87].

(b) Install the electrical bonding strap [86] onto the ground stud [85].

(c) Install the washer [91] and nut [87].

1) Use a wrench to hold the head of the ground stud [85] fastener so that it will not turn when you torque the nut [87].

2) Torque the nut [87] to 31.5 ±3.5 in-lb (3.6 ±0.4 N·m).

3) Use a intrinsically safe approved bonding meter, COM-1550, to do a check of the electrical resistance between the terminal of the bonding strap [86] and the hinge beam [52] structure.

a) Make sure that the electrical resistance is 0.001 ohm (1 milliohm) or less.

SUBTASK 78-31-01-410-002-G00

WARNING DO NOT INSTALL PRESSURE CAPS TO THE HYDRAULIC LINES THAT CONNECT TO CONTROL VALVE MODULE PORTS TO DEACTIVATE THE THRUST REVERSER. THE THRUST REVERSER CAN OPERATE WHEN PRESSURIZED. IF YOU DO NOT OBEY, INJURY TO PERSONNEL AND DAMAGE TO THE EQUIPMENT CAN OCCUR.

WARNING MAKE SURE THAT YOU WEAR PROTECTIVE CLOTHES AND GLOVES WHEN YOU DO WORK ON THE HYDRAULIC SYSTEM. HYDRAULIC FLUID CAN CAUSE A LEAK FROM THE OPEN PORTS OF THE COMPONENT OR FROM THE HYDRAULIC LINES. INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION USE TWO WRENCHES TO TIGHTEN THE COUPLING NUT. USE ONE TO HOLD THE FITTING, AND THE OTHER TO TURN THE COUPLING NUT. IF YOU DO NOT USE TWO WRENCHES, DAMAGE TO THE EQUIPMENT CAN OCCUR.

(3) Connect the thrust reverser retract line flexhose [22] and extend line flexhose [23] (Figure 404).

(a) If hydraulic fitting plugs were installed on the flexhoses, use a hydraulic resistant container to catch the residual hydraulic fluid that is in the flexhoses.

1) Remove the hydraulic fitting plugs.

(b) If cotton wiper, G00034, is around the flexhoses, remove the cloth.

(c) Connect the extend line flexhose [23] to the strut fitting.

1) Torque the extend line flexhose [23] coupling nut to 900 ±45 in-lb (102 ±5 N·m).

2) Loosen the coupling nut.

3) Torque the coupling nut again to 900 ±45 in-lb (102 ±5 N·m).

(d) Connect the retract line flexhose [22] to the strut fitting.

1) Torque the retract line flexhose [22] coupling nut to 270 ±13 in-lb (30.5 ±1.5 N·m).

2) Loosen the coupling nut.

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3) Torque the coupling nut again to 270 ±13 in-lb (30.5 ±1.5 N·m).

SUBTASK 78-31-01-410-003-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTIONS ARE NOT DIRTY BEFORE YOU CONNECT OR DISCONNECT THEM. SOLID OR LIQUID CONTAMINATION OF THE ELECTRICAL CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

(4) Connect the electrical connector [24] and electrical connector [27] to the strut receptacle [25] and strut receptacle [26] (Figure 404):

(a) For the left thrust reverser [5], connect the electrical connector [27], D30002, and electrical connector [24], D30008, to the strut receptacles.

(b) For the right thrust reverser [10], connect the electrical connector [27], D30006, and electrical connector [24], D30010, to the strut receptacles.

SUBTASK 78-31-01-420-003-G00

(5) Connect the aft cowl cross tie rod [113] to the deflection limiting strap [112] on the applicable thrust reverser half (Figure 402).

(a) Open the applicable thrust reverser half (TASK 78-31-00-010-801-G00).

CAUTION

DO NOT APPLY GREASE TO OTHER SURFACES. IF YOU APPLY GREASE TO OTHER SURFACES, IT CAN CAUSE DAMAGE.

(b) Apply a thin film of grease, D00633, to the shank of the bolt [115].

PTF ALL; AIRPLANES WITHOUT LANYARD

(c) Install the nut [110], bolt [115], bushing [111], and bushing [114] that attach the aft cowl cross tie rod [113] to the deflection limiting strap [112].

PTF ALL; AIRPLANES WITH LANYARD

(d) Install the nut [110], lanyard [118], washer [117], bolt [115], bushing [111], and bushing [114] that attach the aft cowl cross tie rod [113] to the deflection limiting strap [112].

1) If the lanyard [118] is damaged or broken, replace it (TASK 54-51-06-000-801 and TASK 54-51-06-400-801).

PTF ALL

(e) Torque the nut [110] to 320 ±80 in-lb (36 ±9 N·m).

(f) Close the applicable thrust reverser half (TASK 78-31-00-010-802-G00).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-01-860-008-G00

(1) If a new replacement thrust reverser is installed or a thrust reverser from another engine is installed, look at the part number of the cascade segments to make sure that they are installed in the correct position for the left or right engine.

NOTE: The thrust reversers are interchangeable between engine position, but cascades and Krueger deflectors must be correct for the installation location.

(a) If it is necessary, change the cascade segments to the correct positions for the left or right engine location.

1) Do these tasks:

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- Cascade Removal, TASK 78-31-05-000-801-G00.
- Cascade Installation, TASK 78-31-05-400-801-G00.

SUBTASK 78-31-01-410-004-G00

(2) Install the applicable access panels (Figure 401).

(a) If you are installing the inboard thrust reverser on Engine 1, install this access panels:

Number	Name/Location
416AR	Right Forward Thrust Reverser Hinge Fairing, Engine 1
416BR	Right Aft Thrust Reverser Hinge Fairing, Engine 1

1) Apply grease, D00633, to the threads of the bolts before you install the bolts.
2) Install 13 bolts [8] in the forward 13 holes and torque them to 50 ±5 in-lb (5.6 ±0.6 N·m).
3) Install 12 bolts [1] in the aft 12 holes and torque them to 50 ±5 in-lb (5.6 ±0.6 N·m).

(b) If you are installing the outboard thrust reverser on Engine 1, install this access panels:

Number	Name/Location
415AL	Left Forward Thrust Reverser Hinge Fairing, Engine 1
415BL	Left Aft Thrust Reverser Hinge Fairing, Engine 1

1) Apply grease, D00633, to the threads of the bolts before you install the bolts.
2) Install 13 bolts [7] in the forward 13 holes and torque them to 50 ±5 in-lb (5.6 ±0.6 N·m).
3) Install 12 bolts [4] in the aft 12 holes and torque them to 50 ±5 in-lb (5.6 ±0.6 N·m).

(c) If you are installing the inboard thrust reverser on Engine 2, install this access panels:

Number	Name/Location
425AL	Left Forward Thrust Reverser Hinge Fairing, Engine 2
425BL	Left Aft Thrust Reverser Hinge Fairing, Engine 2

1) Apply grease, D00633, to the threads of the bolts before you install the bolts.
2) Install 13 bolts [7] in the forward 13 holes and torque them to 50 ±5 in-lb (5.6 ±0.6 N·m).
3) Install 12 bolts [4] in the aft 12 holes and torque them to 50 ±5 in-lb (5.6 ±0.6 N·m).

(d) If you are installing the outboard thrust reverser on Engine 2, install this access panels:

Number	Name/Location
426AR	Right Forward Thrust Reverser Hinge Fairing, Engine 2
426BR	Right Aft Thrust Reverser Hinge Fairing, Engine 2

1) Apply grease, D00633, to the threads of the bolts before you install the bolts.
2) Install 13 bolts [8] in the forward 13 holes and torque them to 50 ±5 in-lb (5.6 ±0.6 N·m).
3) Install 12 bolts [1] in the aft 12 holes and torque them to 50 ±5 in-lb (5.6 ±0.6 N·m).

(e) Make sure that the lip of the hinge fairing fits under the strut panel.

SUBTASK 78-31-01-430-001-G00

(3) If the inboard thrust reverser was installed, make sure that the Krueger flap deflectors are installed (TASK 78-31-09-420-801-G00).

NOTE: The Krueger flap deflectors are installed on the inboard translating sleeve.

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SUBTASK 78-31-01-820-001-G00

(4) If the thrust reverser installed is a replacement thrust reverser, do the thrust reverser adjustment (TASK 78-31-01-820-801-G00).

SUBTASK 78-31-01-730-001-G00

(5) If the thrust reverser was removed and the same thrust reverser was installed, do the thrust reverser rigging check (TASK 78-31-01-280-801-G00).

SUBTASK 78-31-01-410-006-G00

(6) If you removed the fan cowl, reinstall the fan cowl (TASK 71-11-04-400-801-G00).

SUBTASK 78-31-01-860-010-G00

(7) Activate the applicable hydraulic system (TASK 29-11-00-860-801).

(a) For Engine 1, pressurize system A.

(b) For Engine 2, pressurize system B.

SUBTASK 78-31-01-860-011-G00

(8) For Engine 1, remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	5	C00276	ENGINE 1 THRUST REVERSER CONT
B	6	C01412	ENGINE 1 THRUST REVERSER INTLK
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

For Engine 2, remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK
C	6	C01413	ENGINE 2 THRUST REVERSER INTLK
C	7	C00277	ENGINE 2 THRUST REVERSER CONT

SUBTASK 78-31-01-440-001-G00

(9) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 78-31-01-820-002-G00

(10) Do this task: Thrust Reverser Operational Test (Selection), TASK 78-31-00-700-801-G00.

(a) Move the thrust reverser through a minimum of 4 deploy cycles.

(b) Make sure that the sleeves move smoothly.

(c) Do a check for hydraulic fluid leaks (TASK 29-00-00-790-801).

NOTE: Refer to the Tube Connections, Static Seals, and Other Dynamic Seals sections of the procedure for the leakage limits.

SUBTASK 78-31-01-440-002-G00

(11) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

——— END OF TASK ———

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THRUST REVERSER - ADJUSTMENT/TEST

1. General

A. This procedure has one task:

(1) An adjustment of the thrust reverser.

TASK 78-31-01-820-801-G00

2. Thrust Reverser Adjustment

(Figure 501 or Figure 502, Figure 503, Figure 504, Figure 505)

A. General

(1) This task gives the instructions to rig and adjust the thrust reverser after installation.

(2) This task is required after thrust reverser replacement and if the thrust reverser rig check is not in limits.

B. References

Reference	Title
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-11-820-801-G00	Latches Adjustment (P/B 501)
78-31-12-000-801-G00	Fire Seal Removal (P/B 401)
78-31-12-400-801-G00	Fire Seal Installation (P/B 401)
78-31-24 P/B 401	BLOCKER DOOR AERO SEAL - REMOVAL/INSTALLATION

C. Tools/Equipment

Reference	Description
STD-1097	Caliper - Vernier, 0-6 inch, Readable to 1/1000 inch
STD-1184	Scale - Spring, 0-100 Lbs, Tension

D. Consumable Materials

Reference	Description	Specification
A00160	Sealant - Firewall - Hydraulic Fluid Resistant	BMS5-63
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)
B50073	Alcohol - Isopropyl	ASTM D 770
D00006	Compound - Antiseize, Pure Nickel Special - Never-Seez NSBT-8N/-16N	
D50294	Compound - Antiseize, Pure Nickel Special - Never Seez	BAC5008
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G02020	Clay, Modeling	
G02380	Developer - Inspection - Met-L-Chek D-70	AMS 2644 Form D Type 1, Form E Type 2
G02415	Agent - Parting, Paste Wax (Johnson's Paste Wax or equivalent)	

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(Continued)

Reference	Description	Specification
G50092	Compound - Metrology Casting Material - ReproRubber Quick Setting Putty	
G50146	Developer - Non-acqueous - Dubl-Check D-100	SAE AMS2644
G50347	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

E. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

F. Prepare for the Adjustment

SUBTASK 78-31-01-010-024-G00

(1) Do this task: Open the Thrust Reverser (Selection), TASK 78-31-00-010-801-G00.

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G. Thrust Reverser Adjustment

SUBTASK 78-31-01-210-002-G00

(1) Do a check of the six receiver cups [1] on the upper edge of the inner wall on the left and right thrust reverser (View B and View F, Figure 501 or Figure 502).

NOTE: There are three receiver cups on the left thrust reverser and three on the right thrust reverser.

(a) Remove the applicable receiver cup [1].

(b) Make sure that there is one spacer [21] and no spacers [8] under each receiver cup [1].

1) If there is one or more spacers [8] under each receiver cup [1], do these steps:

a) Remove all spacers [8] and spacer [21].

b) Apply Pure Nickel Special compound, D00006, to the threads of the receiver cup [1].

c) Install the receiver cup [1] with one spacer [21].

d) Tighten the receiver cup [1] to 87.5 ±7.5 in-lb (9.9 ±0.9 N·m).

SUBTASK 78-31-01-820-004-G00

CAUTION

MAKE SURE THAT YOU REMOVE THE WEARPLATE-AND-SHIM COMBINATION. IF YOU DO NOT REMOVE THE WEARPLATE-AND-SHIM COMBINATION, AN INCORRECT CLAY CHECK AND DAMAGE CAN OCCUR TO THE THRUST REVERSER.

(2) Remove the screws [12], washer [33], washers [34] (if applicable), compression pad wear plates [6], and all spacers [11] from the applicable lower TPS retention spacers or inner wall closeout fitting (View E and View G, Figure 501 or Figure 502).

NOTE: There are three compression pads on the left thrust reverser and three on the right thrust reverser. The compression pads are on the lower edge of the inner wall of the thrust reverser.

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PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL (Continued)

(a) Keep the screws [12], washer [33], washers [34] (if applicable), compression pad wear plates [6], and spacers [11] together and record the location from where they were removed for the subsequent installation.

SUBTASK 78-31-01-410-020-G00

(3) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

SUBTASK 78-31-01-820-008-G00

(4) Adjust the thrust reverser centerline and aft cowl side latches.

(a) Measure the closing force 1.0 in. (25.4 mm) from the end of each latch handle.

(b) If it is necessary, adjust the closing force for each latch (TASK 78-31-11-820-801-G00).

SUBTASK 78-31-01-410-018-G00

(5) Do this task: Open the Thrust Reverser (Selection), TASK 78-31-00-010-801-G00.

SUBTASK 78-31-01-820-009-G00

(6) Do the V-blade fit up and latch rigging.

NOTE: The radial positions in the steps below coincide with the engine fan strut locations.

(a) Put a piece of clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, approximately 0.15 in. (3.81 mm) thick into the outer V-groove at the 45°, 90°, 135°, 225°, 270, and 315° radial positions.

NOTE: The top and bottom 10.5 in. (26.7 cm) approximately of the outer V-Blade is reduced in height. Do not apply clay there. If you do, the results will be in error.

(b) Put a piece of clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, a minimum of 0.25 in. (6.35 mm) thick into the inner V-groove at the 30°, 90°, 150°, 210°, 270, and 330° radial positions.

(c) If you will use the clay, G02020, apply a paste wax parting agent, G02415, in the corresponding locations on the outer and inner V-blades of the thrust reverser.

SUBTASK 78-31-01-820-006-G00

(7) Do the upper compression rod and receiver gap verification setup (Figure 501 or Figure 502, Figure 505).

(a) Put a piece of clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, approximately 0.40 in. (10.16 mm) thick in each upper compression pad receiver cups on the left thrust reverser half (3 places).

1) Make sure that no air pocket forms in the clay or Reprorubber compound.

(b) If you are using clay, G02020, apply paste wax parting agent, G02415, to the contact area on the left ends of the upper compression rod assemblies.

SUBTASK 78-31-01-200-001-G00

(8) Do the lower compression pad gap verification setup (Figure 501 or Figure 502).

(a) Put a piece of clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, approximately 0.30 in. (7.62 mm) thick on the applicable lower TPS retention spacers and inner wall closeout fitting on the left thrust reverser half.

(b) If you are using the clay, G02020, apply paste wax parting agent, G02415, on the contact area on the applicable lower TPS retention spacers and inner wall closeout fitting on the right thrust reverser half.

SUBTASK 78-31-01-970-001-G00

(9) Do the aft cowl cross tie rod verification setup (View D, Figure 501 or Figure 502).

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PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL (Continued)

(a) Put a piece of clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, approximately 0.20 in. (5.08 mm) thick on the aft cowl bearing plates on either side of the thrust reverser.

(b) If you are using clay, G02020, apply paste wax parting agent, G02415, to the contact areas on the ends of the aft cowl cross tie rod.

SUBTASK 78-31-01-020-012-G00

(10) Do the fire seal foot print verification setup (Figure 503).

(a) Make sure that the strut, the engine and the thrust reverser fire seals and seal depressors are free of debris.

1) If it is necessary, use a cotton wiper, G00034, and alcohol, B50073, to remove debris.

(b) Apply Met-L-Chek D-70 developer, G02380, or Dubl-Check D-100 developer, G50146, approximately 1.5 in. (38.1 mm) wide along the seal depressors and 12:00 o'clock strut engine seal.

SUBTASK 78-31-01-010-007-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURES TO OPEN AND CLOSE THE THRUST REVERSER. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(11) Close and latch the thrust reverser but do not do the thrust reverser or leading edge activation at this time (TASK 78-31-00-010-802-G00).

(a) If you used ReproRubber Quick Setting Putty compound, G50092, make sure that you leave the thrust reverser closed for the manufacturer's recommended setting time.

SUBTASK 78-31-01-010-025-G00

(12) Do this task: Open the Thrust Reverser (Selection), TASK 78-31-00-010-801-G00.

SUBTASK 78-31-01-970-002-G00

(13) Measure the thickness of the clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, from the aft cowl bearing plates.

(a) Remove the clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, from the aft cowl bearing plates [13].

(b) Measure the thickness of the clay, G02020, or ReproRubber Quick Setting Putty compound, G50092.

NOTE: The clay thickness represents the change required in the length of the cross-tie rod.

(c) Make sure that the thickness of the clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, is a minimum of 0.010 in. (0.254 mm) but not more than 0.030 in. (0.7620 mm).

(d) If it is necessary, adjust the length of the cross-tie rod.

1) Remove the lockwire from the cross-tie rod turnbuckles.

2) Loosen the jam nuts on the cross-tie rod turnbuckles.

3) Adjust the cross-tie rod lengths as required.

NOTE: One full turn of the rod with the rod ends restrained will change the overall length by 0.10 in. (2.54 mm).

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PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL (Continued)

4) Tighten the jam nuts to 80 ±5 in-lb (9 ±1 N·m).

5) Install MS20995NC32 lockwire, G50347, on the jam nuts.

SUBTASK 78-31-01-970-003-G00

(14) Measure the thickness of the clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, on the inner V-groove(Figure 503).

(a) Remove the clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, from the inner V-groove at the six locations.

(b) Measure the thickness of the clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, with a vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

(c) Record the thickness of the clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, from the inner V-groove and the location from where you measured.

(d) Make sure that the thickness at each location is as shown in the table below:

Table 501/78-31-01-993-812-G00 Clay Thickness

EDGE ZONE	SECTION	THICKNESS
[A]	A-A	0.020 in. (0.508 mm) to 0.190 in. (4.826 mm)

SUBTASK 78-31-01-970-004-G00

(15) Measure the thickness of the clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, on the outer V-groove.

(a) Remove the clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, from the outer V-groove at the six locations.

(b) Measure the thickness of each clay, G02020, or ReproRubber Quick Setting Putty compound, G50092.

(c) Record the thickness of each clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, from the outer V-groove and the location from where you measured.

(d) Calculate the average thickness of the three pieces of clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, on the left thrust reverser half.

(e) Calculate the average thickness of the three pieces of clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, on the right thrust reverser half.

(f) Make sure that the average thickness (clearance) for each side is not more than 0.020 in. (0.508 mm).

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PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL (Continued)

SUBTASK 78-31-01-970-005-G00

(16) If it is necessary, adjust the latch beam wear pad shims as required at the wear plate locations (Figure 501 or Figure 502).

NOTE: Remove wear plate and peel shims as required to adjust the V-blade gap. If necessary, use additional wear plate and peel shims to get the required additional thickness. The total wearplate plus shim stack-up at each location must not exceed a maximum thickness of 0.122 in. (3.099 mm). No finish on shim is required during reinstallation. If chatter is present, install the above shims on the T/R that appears to have the most chatter. Install shims of various thickness under the eight lower latch beam pads with limitations. The maximum shim thickness variation between all eight pads is 0.036 in. (0.914 mm). Maximum shim thickness variation between all two pads common to the same set of latches is 0.018 in. (0.457 mm). Shimming shall be done to load latch beam in one continuous curve and not to load latch beam into a wavy form.

(a) Do the above measurement steps again until all clearances meet the specifications.

SUBTASK 78-31-01-160-003-G00

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH, EYES, OR SKIN. DO NOT BREATHE FUMES FROM SOLVENTS. REFER TO PRODUCT MATERIAL SAFETY DATA SHEETS (MSDS AND LOCAL REQUIREMENTS FOR PROPER HANDLING. SOLVENTS ARE HAZARDOUS MATERIALS AND MAY BE FLAMMABLE OR HARMFUL TO THE ENVIRONMENT.

(17) Remove all clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, that remains in the inner and outer V-grooves with solvent, B00062.

SUBTASK 78-31-01-970-006-G00

(18) Adjust the upper receiver cups [1] (Figure 505).

(a) Remove the clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, from the receiver cup [1].

(b) Use a vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097, to measure the thickness of the impression of the compression rod on the clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, on the receiver cup.

(c) Make sure that the thickness of the impression is a minimum of 0.010 in. (0.254 mm) to a maximum of 0.030 in. (0.762 mm).

(d) If the clearances between the compression rods and receiver cups are in the limits, continue to the steps below to do the lower compression pad adjustment.

(e) If the clearances between the compression rods and receiver cups are not in the limits, do these steps:

1) Calculate the amount of spacers [8] that you will need to adjust the clearances.

NOTE: The thickness of a spacer [11] is 0.016 in. (0.406 mm).

2) Install the spacers [8] at each of the three receiver cups [1] on each thrust reverser half.

a) A maximum of five spacers [8] in addition to one spacer [21] are permitted under the receiver cup [1].

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PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL (Continued)

b) A minimum of one spacer [21] must be under the receiver cup [1].

NOTE: If the number or spacers required are more than the maximum permitted, balance the spacers required on the cup on the opposite side.

c) Install the receiver cups [1], spacer [21] and required number of spacers [8] for the left and right thrust reverser halves. The spacers [8] must be installed between the receiver cups [1] and the spacer [21], if used.

d) Tighten the receiver cups [1] to 87.5 ±7.5 in-lb (9.9 ±0.9 N·m).

e) Install the compression rods.

SUBTASK 78-31-01-820-010-G00

(19) Adjust the lower compression pad.

(a) Remove the clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, from the applicable lower TPS retention spacers and inner wall closeout fitting on the left thrust reverser half (Figure 501 or Figure 502).

(b) Use a vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097, to measure the compressed thickness of the clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, at the compression pad wear plates [6] on the three compression pads.

NOTE: The clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, thickness represents the wearplate/shim thickness plus the nominal gap required for an acceptable installation.

(c) Make sure that the compressed thickness at each location is a minimum of 0.010 in. (0.254 mm) to a maximum of 0.030 in. (0.762 mm).

NOTE: If the clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, minus the corresponding combined compression pad wear plate [6] and spacer [11] thickness is within the gap tolerance band, then the existing thickness of the spacer [11] is satisfactory.

(d) Clean the clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, from the applicable lower TPS retention spacers and inner wall closeout fitting with alcohol, B50073.

(e) If the existing spacer [11] thickness is satisfactory, re-install the compression pad wear plate [6], washer [33], washers [34] (if applicable), and spacer [11] that you removed from the compression pads.

1) Make sure that you re-install the compression pad wear plate [6], washer [33], washers [34] (if applicable), and spacer [11] to the same positions from which they were removed.

2) Make sure that there is no gap between the lower compression pad assembly and lower TPS retention spacer #2 [27] or lower TPS retention spacer #1 [28].

a) If it is necessary, use a maximum three washers [34] to accommodate differences in TPS thickness.

3) Tighten the lower TPS retention spacer #2 [27] or lower TPS retention spacer #1 [28] to 40 in-lb (5 N·m) - 50 in-lb (6 N·m).

a) If it is necessary, apply Never Seez, D50294 to the threads of the lower TPS retention spacer #2 [27] or lower TPS retention spacer #1 [28].

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4) Apply Pure Nickel Special compound, D00006, on the threads of the screw [12].

5) Tighten the screw [12] to 25 in-lb (3 N·m) - 35 in-lb (4 N·m).

(f) If the clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, thickness minus the corresponding compression pad wear plate [6] and spacer [11] thickness is outside the gap tolerance band, then add or subtract spacer [11] as required.

(g) Re-install the compression pad wear plate [6], washer [33], washers [34] (if applicable), and spacer [11] with the screws [12].

(h) Increase or decrease the screws [12] grip length as required based on the number of spacers [11] added or removed.

NOTE: The following table shows an example calculation of determining the necessary adjustments.

Table 502/78-31-01-993-803-G00

Example:	
Measured clay or ReproRubber thickness	0.300
Minus the nominal gap	-0.020
Minus the removed wear plane and spacer thickness	-0.210
Change in spacer thickness required	0.070

1) Make sure that there is no gap between the lower compression pad assembly and lower TPS retention spacer #2 [27] or lower TPS retention spacer #1 [28].

a) If it is necessary, use a maximum three washers [34] to accommodate differences in TPS thickness.

2) Tighten the lower TPS retention spacer #2 [27] or lower TPS retention spacer #1 [28] to 40 in-lb (5 N·m) - 50 in-lb (6 N·m).

a) If it is necessary, apply Never Seez, D50294 to the threads of the lower TPS retention spacer #2 [27] or lower TPS retention spacer #1 [28].

3) Apply Pure Nickel Special compound, D00006, on the threads of the screws [12].

4) Tighten the screw [12] to 25 in-lb (3 N·m) - 35 in-lb (4 N·m).

SUBTASK 78-31-01-211-001-G00

(20) Make sure that the fire/aero compression have an acceptable amount of contact with the seal land.

(a) Examine the areas on the seals where you applied Dubl-Check D-100 developer, G50146, or Met-L-Chek D-70 developer, G02380.

(b) For the seal zone definitions and acceptable seal contact widths, refer to Figure 503, cross sections A-A and B-B.

(c) If a section of the fire seal does not have an acceptable width of the developer (footprint), do the following steps:

1) Apply paste wax parting agent, G02415, on the seal depressor surface.

2) Extend paste wax parting agent, G02415, minimum of 3 in. (7.62 cm) beyond the noted discrepant area.

3) Let the parting agent dry.

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4) Apply a continuous and uniform layer of the sealant, A00160, on the fire seal.
   a) Make sure that layer is not more than 0.25 in. (0.64 cm) inches thick.
   b) Make sure that sealant extends to 1 in. (2.54 cm) in each direction past the discrepant area.
5) Close and latch the thrust reverser (TASK 78-31-00-010-802-G00).
6) Let the sealant fully dry.
7) Open the thrust reverser (TASK 78-31-00-010-801-G00).
8) Remove the parting agent.
9) Make sure that the width of the developer is acceptable.
   a) Do the foot print verification setup to make sure that there are no remaining gaps at the discrepant location (Figure 503).

(d) If the seals do not have acceptable seal contact widths and all of the adjustments above are not in the limits, then replace the fire seals (TASK 78-31-12-000-801-G00, TASK 78-31-12-400-801-G00).

H. Prepare for the Aft Cowl Wear Pad Adjustment

SUBTASK 78-31-01-820-030-G00

(1) At the aft cowl on the right thrust reverser, put a piece of clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, 0.10 in. (2.54 mm) thick on the wear plate [31].

NOTE: The aft cowl wear plate is on the lower edge at the aft end of the thrust reverser.

(a) Apply paste wax parting agent, G02415, on the aft cowl on the left thrust reverser that will touch the clay.

SUBTASK 78-31-01-820-031-G00

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURES TO OPEN AND CLOSE THE THRUST REVERSER. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) In the steps that follow, obey all of the WARNINGS and CAUTIONS in the referenced procedures:

(a) Close and latch the thrust reverser but do not do the thrust reverser or leading edge activation at this time (TASK 78-31-00-010-802-G00).

(b) Measure the latch number 7 handle closing force.

1) Make sure that the closing force is 40 lbf (178 N) - 60 lbf (267 N) with a spring scale (0-100 Lbs), STD-1184 no more than 1.0 in. (25.4 mm) from the end of the latch handle to close it.
2) If the closing force is not in the limits, adjust the handle closing force (Latches Adjustment, TASK 78-31-11-820-801-G00).
   a) Make sure the latch remains closed as you do the adjustment for the other latches.
3) If the closing force is in the limits, leave the latch number 7 closed.

(c) Close and measure the latch number 8 and latch number 9 handle closing force.

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1) Make sure that the closing force is 40 lbf (178 N) - 60 lbf (267 N) with a spring scale (0-100 Lbs), STD-1184 no more than 1.0 in. (25.4 mm) from the end of the latch handles to close it.

2) If the closing force is not in the limits, adjust the handle closing force (Latches Adjustment, TASK 78-31-11-820-801-G00).

a) Leave the latch number 8 and latch number 9 closed.

(d) If you used ReproRubber Quick Setting Putty compound, G50092, make sure that you leave the thrust reverser closed for the manufacturer's recommended setting time.

(e) Do this task: Open the Thrust Reverser (Selection), TASK 78-31-00-010-801-G00.

I. Aft Cowl Wear Pad - Adjustment

SUBTASK 78-31-01-820-032-G00

(1) To adjust the clearance between the wear plates [31] at the aft cowl, do the following steps (View H, Figure 501 or Figure 502):

(a) Use a caliper to measure the compressed thickness of the clay at the wear plate [31] on the aft cowl.

1) The limit for clearance between the wear plates [31] is 0.000 +0.100 / -0.000 in. (0.000 +2.540 / -0.000 mm).

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THESE MATERIALS. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE THESE MATERIALS. KEEP THESE MATERIALS AWAY FROM SPARKS, FLAME, AND HEAT. THESE MATERIALS ARE POISONOUS AND FLAMMABLE AND CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(b) Remove the clay from the wear plates [31] and clean off the clay residue with solvent, B00062.

(c) If the clearance between the wear plates [31] at the aft cowl is in the limit, no further action is required.

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(d) If the clearance between the wear plates [31] at the aft cowl is not in the limit, do this step:

1) Re-adjust the aft cowl center latch (number 7) to a higher closing force 40 lbf (178 N) - 75 lbf (334 N) to decrease the clearance between the wear plates [31] at the aft cowl (Latches Adjustment, TASK 78-31-11-820-801-G00).

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J. Thrust Reverser Adjustment

SUBTASK 78-31-01-210-007-G00

(1) Do a check of the six receiver cups [1] on the upper edge of the inner wall on the left and right thrust reverser (View B, Figure 501 or Figure 502).

NOTE: There are three receiver cups on the left thrust reverser and three on the right thrust reverser.

NOTE: Airplanes can have spacers that are 0.8750 in. (22.23 mm) diameter or 2.0000 in. (50.8 mm) in diameter.

(a) Remove the applicable receiver cup [1].

(b) Make sure that there is one spacer [21] under each receiver cup [1].

1) If there is more than one 0.8750 in. (22.23 mm) spacer [21] under each receiver cup [1], do these steps:

a) Remove the receiver cup [1] and all spacers [21].

b) Record the amount of the spacers [21] that are removed and their location for the subsequent installation.

c) Apply Pure Nickel Special compound, D00006, to the threads of the receiver cup [1].

d) Install the receiver cup [1] with one spacer [21].

e) Make sure that the outside diameter of the receiver cup [1] with a 0.8750 in. (22.23 mm) diameter is aligned with the outside diameter of the spacer [21] 0.8750 ±0.0100 in. (22.225 ±0.254 mm).

f) Tighten the receiver cup [1] to 87.5 ±7.5 in-lb (9.9 ±0.9 N·m).

2) If there is more than one 2.0000 in. (50.8 mm) spacer [21] under each receiver cup [1], do these steps:

a) Remove the receiver cup [1] and all but one spacer [21].

NOTE: The spacer against the receiver cup fitting will stay in its position, because there is sealant on the surface of the spacer that is adjacent to the receiver cup fitting.

b) Record the amount of the spacers [21] that are removed and their location for the subsequent installation.

c) Make sure that the spacer [21] against the receiver cup fitting is in its position.

d) Install the receiver cup [1].

e) Tighten the receiver cup [1] to 87.5 ±7.5 in-lb (9.9 ±0.9 N·m).

SUBTASK 78-31-01-820-011-G00

(2) Remove the screws [12], compression pad wear plates [6] and all spacers [11] from the applicable inner wall, (View E, Figure 501 or Figure 502).

NOTE: There are three compression pads on the left thrust reverser and three on the right thrust reverser. The compression pads are on the lower edge of the inner wall of the applicable thrust reverser.

(a) Keep the screws [12], compression pad wear plates [6] and all spacers [11] together and record the location from where they were removed for the subsequent installation.

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SUBTASK 78-31-01-820-012-G00

(3) Remove the two bolts [30], wear plate [31], and spacers [32] from the wear pad on the lower edge of the aft cowl on the left thrust reverser.

(a) Keep the two bolts [30], wear plate [31] and spacers [32] together for the subsequent installation.

K. Prepare for the V-blade to V-groove Clearance and Latch Beam Wear Plate Adjustment

SUBTASK 78-31-01-820-013-G00

(1) Put a piece of clay, G02020, 0.15 in. (3.81 mm) thick into the outer V-groove on the fan frame at the 1:00, 2:00, 4:00, 5:00, 7:00, 8:00, 10:00 and 11:00 o'clock positions (Figure 501 or Figure 502).

NOTE: These positions are the same as the engine fan strut locations.

(a) Apply paste wax parting agent, G02415, on the outer V-blade areas on the thrust reverser where the clay will touch.

SUBTASK 78-31-01-820-014-G00

(2) Put a piece of clay, G02020, not less than 0.25 in. (6.35 mm) thick into the inner V-groove on the engine extension ring at the 1:00, 2:00, 4:00, 5:00, 7:00, 8:00, 10:00 and 11:00 o'clock positions.

NOTE: These positions are the same as the engine fan strut locations.

(a) Apply paste wax parting agent, G02415, on the inner V-blade areas on the thrust reverser where the clay will touch.

SUBTASK 78-31-01-410-022-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURES TO OPEN AND CLOSE THE THRUST REVERSER. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) In the steps that follow, obey all of the WARNINGS and CAUTIONS in the referenced procedures (Figure 501 or Figure 502):

(a) Close and latch the thrust reverser; but, do not do the thrust reverser or leading edge activation at this time (TASK 78-31-00-010-802-G00).

1) Make sure that the tension latches have a closing force of 50 ±10 lbf (222 ±45 N) .

a) If the closing force is not in the limits, do this task: Latches Adjustment, TASK 78-31-11-820-801-G00.

(b) Do this task: Open the Thrust Reverser (Selection), TASK 78-31-00-010-801-G00.

L. V-blade to V-groove Clearance and Latch Beam Wear Plate Clearance Adjustment

SUBTASK 78-31-01-820-015-G00

(1) Do these steps to adjust the clearance between the V-blades and V-grooves (Figure 501 or Figure 502, Figure 504):

(a) For the inner V-groove, use a caliper to measure the compressed thickness of the clay at the six locations.

1) Make sure that you record the thickness of the compressed clay and the location from where you measured.

2) At each radial location, the limit for the inner V-groove to V-blade clearance is as shown in the table below:

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Table 503/78-31-01-993-811-G00 Inner V-groove Clearance

EDGE ZONE	SECTION	CLEARANCE
[A]	A-A	0.0200 in. (0.51 mm) to 0.190 in. (4.83 mm)

(b) For the outer V-groove, use a caliper to measure the compressed thickness of the clay at the eight locations.

1) Make sure that you record the thickness of the compressed clay and the location from where you measured.

2) For each thrust reverser, calculate the average of the four clay thickness measurements.

3) The limit for the outer V-groove to V-blade average clearance is 0.000 in. (0.00 mm) to 0.020 in. (0.508 mm).

WARNING

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(c) Remove the clay from the inner and outer V-groove and clean off the clay residue with solvent, B00062.

(d) If the clearance between the V-grooves and V-blades are in the limits, continue at the steps below to prepare for the compression rod, compression pad, and aft cowl wear pad adjustment.

(e) If the clearance between the V-grooves and V-blades are not in the limits, do these steps:

1) Adjust the wear plate shims [10] at the eight latch beam wear plate locations.

a) Remove the screws [9] that attach the latch beam wear plate [5] and shims [10] to the latch beam.

<1> If it is necessary, remove the screws [9], nut [38], washer [37], and screw [36] that attach the latch beam wear plate [35] and shims [10] to the latch beam.

b) Peel the shims [10] to decrease the clearance or replace the shims [10] to increase the clearance.

<1> Maximum shims [10] thickness between all eight pads is 0.036 in. (0.91 mm).

<2> Maximum shims [10] thickness between any two pads common to the same set of latches is 0.018 in. (0.46 mm).

<3> Shim the latch beam in one continuous curve and not into a wavy form.

NOTE: A continuous curve will make sure that the sleeves operate smoothly.

c) Re-install the latch beam wear plate [5], shims [10], and screws [9].

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<1> If it is necessary, re-install the latch beam wear plate [35], shims [10], screws [9], screw [36], washer [37], and nut [38].

2) Do the clay check again and adjust the shims [10] until the V-groove to V-blade clearance is correct.

a) After the V-groove to V-blade clearance is correct, make sure that when the thrust reverser is closed and latched that the latch beam wear plate [5] and latch beam wear plate [35] shim gap is 0.000 in. (0.00 mm).

M. Prepare for the Compression Rod, Compression Pad and Aft Cowl Wear Pad Adjustment

SUBTASK 78-31-01-820-016-G00

(1) Put a piece of clay, G02020, 0.40 in. (10.16 mm) thick in the three compression rod receiver cups [1] on the left thrust reverser (Figure 505).

NOTE: The three compression receiver cups are on the upper edge of the thrust reverser.

(a) Apply paste wax parting agent, G02415, on the end of the compression rod that will touch the clay.

SUBTASK 78-31-01-820-017-G00

(2) At the three compression pads on the right thrust reverser, put a piece of clay, G02020, 0.30 in. (7.62 mm) thick on the compression pad wear plates [6].

NOTE: The three compression pads are on the lower edge of the thrust reverser.

(a) Apply paste wax parting agent, G02415, on the compression pads on the left thrust reverser that will touch the clay.

SUBTASK 78-31-01-820-018-G00

(3) At the aft cowl on the right thrust reverser, put a piece of clay, G02020, 0.10 in. (2.54 mm) thick on the wear plate [31].

NOTE: The aft cowl wear plate is on the lower edge at the aft end of the thrust reverser.

(a) Apply paste wax parting agent, G02415, on the aft cowl on the left thrust reverser that will touch the clay.

SUBTASK 78-31-01-410-023-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURES TO OPEN AND CLOSE THE THRUST REVERSER. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(4) In the steps that follow, obey all of the WARNINGS and CAUTIONS in the referenced procedures:

(a) Close and latch the thrust reverser but do not do the thrust reverser or leading edge activation at this time (TASK 78-31-00-010-802-G00).

(b) Do this task: Open the Thrust Reverser (Selection), TASK 78-31-00-010-801-G00.

N. Compression Rod - Adjustment

SUBTASK 78-31-01-750-001-G00

(1) To adjust the clearance of the compression rod to the receiver cup [1], do the following steps (Figure 505):

(a) Use a caliper to measure the compressed thickness of the clay at the three receiver cups [1].

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1) Make sure that you record the thickness of the compressed clay and location from where you measured.

2) The limits for the compression rod to the three receiver cups [1] clearance are as follows:

a) For compression rod no. 1 [16] (forward), the limit is 0.010 in. (0.254 mm) to 0.040 in. (1.016 mm).

b) For compression rod no. 2 [15] (middle) and compression rod no. 3 [17] (aft), the limit is 0.020 in. (0.508 mm) to 0.050 in. (1.270 mm).

(b) Remove the clay from the receiver cups [1].

(c) If the clearance between the compression rods and the receiver cups [1] are in the limits, continue at the steps below to do the compression pad adjustment.

(d) If the clearance between the compression rods and receiver cups [1] is not in the limits, do these steps:

1) Remove the receiver cups [1] and spacers [21] from the left and right thrust reversers.

NOTE: If the airplane has 2.000 in. (50.8 mm) diameter spacers, the spacer against the receiver cup fitting will stay in its position, because there is sealant on the surface of the spacer that is adjacent to the receiver cup fitting.

a) If the spacer that is 2.000 in. (50.8 mm) in diameter does not have sealant, clean the spacer and receiver cup surfaces that will touch with solvent, B00062, and fay seal with sealant, A00160.

2) Calculate the amount of spacers [21] that are necessary to adjust the clearance.

NOTE: The thickness of a spacer is 0.016 in. (0.406 mm).

3) Install half of the spacers [21] at the receiver cup [1] on the left thrust reverser and half at thereceiver cup [1] on the right thrust reverser.

a) Make sure that there is at least one spacer [21] under each receiver cup [1] but, no more than seven.

b) Make sure that the outside diameter of the receiver cup [1] with a 0.8750 in. (22.23 mm) diameter is aligned with the outside diameter of the spacer [21] 0.8750 ±0.0100 in. (22.225 ±0.254 mm).

c) Make sure that the spacer with the 2.000 in. (50.8 mm) diameter that is against the receiver cup fitting has sealant.

d) Apply Pure Nickel Special compound, D00006, on the threads of the receiver cup [1].

e) Tighten the receiver cup [1] to 80 in-lb (9.04 N·m) - 95 in-lb (10.73 N·m).

O. Compression Pad Adjustment

SUBTASK 78-31-01-820-019-G00

(1) To adjust the compression pad clearance, do the following steps (Figure 505):

(a) Use a caliper to measure the compressed thickness of the clay at the compression pad wear plates [6] on the three compression pads.

1) Make sure that you record the thickness of the compressed clay and location from where you measured.

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2) The limit for the clearance between the compression pad wear plates [6] on the compression pads is 0.030 +0.020 / -0.000 in. (0.762 +0.508 / -0.000 mm).

3) Calculate the spacer [11] thickness that is necessary.

Table 504/78-31-01-993-804-G00

Example:		
Measured clay thickness		0.300
Minus removed wear plate and spacer thickness		-0.210
Minus the nominal gap		-0.030

Change in spacer thickness	-0.000/+0.020	0.060

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(b) Remove the clay from the compression pad wear plates [6] on the compression pads and clean off the clay residue with solvent, B00062.

(c) If the clearance between the compression pad wear plates [6] on the compression pads is in the limit, do these steps:

1) Re-install the spacers [11], compression pad wear plates [6], and screws [12] that were removed in the previous steps.

a) Apply Pure Nickel Special compound, D00006, on the threads of the screws [12].

b) Tighten the screws [12] to 25.0 in-lb (2.82 N·m) - 35.0 in-lb (3.95 N·m).

(d) If the clearance between the compression pads is not in the limit, do these steps:

1) Remove or add the change (calculated and recorded above) to the spacers [11] that were removed to get the correct clearance.

2) Install the spacers [11], compression pad wear plates [6], and screws [12].

a) Apply Pure Nickel Special compound, D00006, on the threads of the screws [12].

b) Tighten the screws [12] to 25.0 in-lb (2.82 N·m) - 35.0 in-lb (3.95 N·m).

P. Aft Cowl Wear Pad - Adjustment

SUBTASK 78-31-01-820-020-G00

(1) Do these steps to adjust the clearance between the wear plates [31] at the aft cowl (Figure 501 or Figure 502):

(a) Measure the latch number 7 handle closing force.

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1) Make sure that the closing force is 40 lbf (178 N) - 60 lbf (267 N) with a spring scale (0-100 Lbs), STD-1184 no more than 1.0 in. (25.4 mm) from the end of the latch handle to close it.

2) If the closing force is not in the limits, adjust the handle closing force (TASK 78-31-11-820-801-G00).

a) Make sure the latch remains closed as you do the adjustment for the other latches.

3) If the closing force is in the limits, leave the latch number 7 closed.

(b) Close and measure the latch number 8 and latch number 9 handle closing force.

1) Make sure that the closing force is 40 lbf (178 N) - 60 lbf (267 N) with a spring scale (0-100 Lbs), STD-1184 no more than 1.0 in. (25.4 mm) from the end of the latch handle to close it.

2) If the closing force is not in the limits, adjust the handle closing force (TASK 78-31-11-820-801-G00).

a) Leave the latch number 8 and latch number 9 closed.

(c) Use a caliper to measure the compressed thickness of the clay at the wear plate [31] on the aft cowl.

1) The limit for clearance between the wear plates [31] is 0.000 +0.100 / -0.000 in. (0.000 +2.540 / -0.000 mm).

WARNING

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(d) Remove the clay from the wear plates [31] and clean off the clay residue with solvent, B00062.

(e) If the clearance between the wear plates [31] at the aft cowl is in the limit, no further action is required.

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(f) If the clearance between the wear plates [31] at the aft cowl is not in the limit, do this step:

1) Re-adjust the aft cowl center latch (number 7) to a higher closing force 40 lbf (178 N) - 75 lbf (334 N) to decrease the clearance between the wear plates [31] at the aft cowl (TASK 78-31-11-820-801-G00).

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(2) If adjustments were made to the compression rod or compression pad, do a clay rig check again.

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Q. Prepare for the Fireseal and Aero Seal Compression Check

SUBTASK 78-31-01-820-022-G00

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THESE MATERIALS. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE THESE MATERIALS. KEEP THESE MATERIALS AWAY FROM SPARKS, FLAME, AND HEAT. THESE MATERIALS ARE POISONOUS AND FLAMMABLE AND CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(1) Clean the fire seals [4] on the left and right thrust reverser and the areas on the strut, engine and lower edge of the right thrust reverser that the fire seals [4] will compress against with solvent, B00062 (Figure 504).

SUBTASK 78-31-01-820-023-G00

(2) Clean the aero seals [29] on the left and right thrust reverser and clean the aft edge of the outer V-groove on the fan case that the aero seal [29] will compress against with solvent, B00062 (Figure 501 or Figure 502).

SUBTASK 78-31-01-820-028-G00

(3) Apply Met-L-Chek D-70 developer, G02380, or Dubl-Check D-100 developer, G50146, approximately 2.00 in. (50.8 mm) wide on the areas on the strut, engine and lower edge of the right thrust reverser where the fire seals [4] will touch.

SUBTASK 78-31-01-820-024-G00

(4) Apply Met-L-Chek D-70 developer, G02380, approximately 2.00 in. (50.8 mm) wide on the aft edge of the outer V-groove where the aero seal [29] will touch.

SUBTASK 78-31-01-820-025-G00

(5) Make sure that the developer is dry before the thrust reverser is closed.

NOTE: The developer becomes a white powder with a dull finish when it is dry.

SUBTASK 78-31-01-410-024-G00

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURES TO OPEN AND CLOSE THE THRUST REVERSER. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(6) In the steps that follow, obey all of the WARNINGS and CAUTIONS in the referenced procedures:

(a) Close and latch the thrust reverser, but, do not do the thrust reverser or leading edge activation at this time Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

(b) Do this task: Open the Thrust Reverser (Selection), TASK 78-31-00-010-801-G00.

R. Fireseal - Adjustment

SUBTASK 78-31-01-820-026-G00

(1) Measure the width of the developer that is on the fire seals [4], (Figure 504).

(a) Make sure that the minimum width of the developer is 0.35 in. (8.89 mm) along the full length of the fire seals [4].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH TITANIUM INNER WALL (Continued)

(b) If the above limit is not met, use the steps below to measure the developer in each section:

1) Divide the fire seal [4] into three sections as follows:

a) The fire seal [4] on the upper edge of a thrust reverser where it compresses against the strut is one section.

b) The fire seal [4] on the forward edge of a thrust reverser where it compresses against the engine is another section.

c) The fire seal [4] on the lower edge of the left thrust reverser where it compresses against the lower edge of the right thrust reverser is another section.

2) For each section, areas where the developer is 0.20 in. (5.08 mm) to 0.35 in. (8.89 mm) wide is permitted with these condition:

a) The length of each area is not more than 1.50 in. (38.1 mm).

b) The total length of all areas is not more than 3.00 in. (76.2 mm).

3) For each section, areas where the developer is less than 0.20 in. (5.08 mm) wide is permitted with these conditions:

a) The length of each area is not more than 0.25 in. (6.35 mm).

b) The total length of all areas is not more than 1.00 in. (25.4 mm).

4) If a section of the fire seal does not have an acceptable width of the developer (footprint), do the following steps:

a) Apply paste wax parting agent, G02415, on the seal depressor surface.

b) Extend paste wax parting agent, G02415, minimum of 3 in. (7.62 cm) beyond the noted discrepant area.

c) Let the parting agent dry.

d) Apply a continuous and uniform layer of the sealant, A00160, on the fire seal.

<1> Make sure that layer is not more than 0.25 in. (6.35 mm) thick.

<2> Make sure that sealant extends to 1 in. (2.54 cm) in each direction past the discrepant area.

e) Close and latch the thrust reverser (TASK 78-31-00-010-802-G00).

f) Let the sealant fully dry.

g) Open the thrust reverser (TASK 78-31-00-010-801-G00).

h) Remove the parting agent.

i) Make sure that the width of the developer is acceptable.

<1> Do the foot print verification setup to make sure that there are no remaining gaps at the discrepant location.

5) If the fire seal [4] compression is not in the limits and all of the adjustments above are not in the limits, then replace the worn fire seal [4] (TASK 78-31-12-000-801-G00, TASK 78-31-12-400-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH TITANIUM INNER WALL (Continued)

S. Aero Seal - Adjustment

SUBTASK 78-31-01-820-027-G00

(1) Measure the width of the developer that is on the aero seal [29] on the left and right thrust reversers, (Figure 501 or Figure 502).

(a) Make sure that the minimum width of the developer is 0.35 in. (8.89 mm) for the full length of the aero seal [29] on each thrust reverser.

(b) Areas where the developer is 0.20 in. (5.08 mm) to 0.35 in. (8.89 mm) wide are permitted with this condition:

1) The total length of all areas on each thrust reverser is not more than 10.0 in. (254.0 mm).

(c) If the aero seal [29] is not in the limits, replace the aero seal [29], (BLOCKER DOOR AERO SEAL - REMOVAL/INSTALLATION, PAGEBLOCK 78-31-24/401).

SUBTASK 78-31-01-410-025-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURES TO OPEN AND CLOSE THE THRUST REVERSER. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

NOTE: It is not necessary to clean off the developer.

PTF ALL

T. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-01-710-002-G00

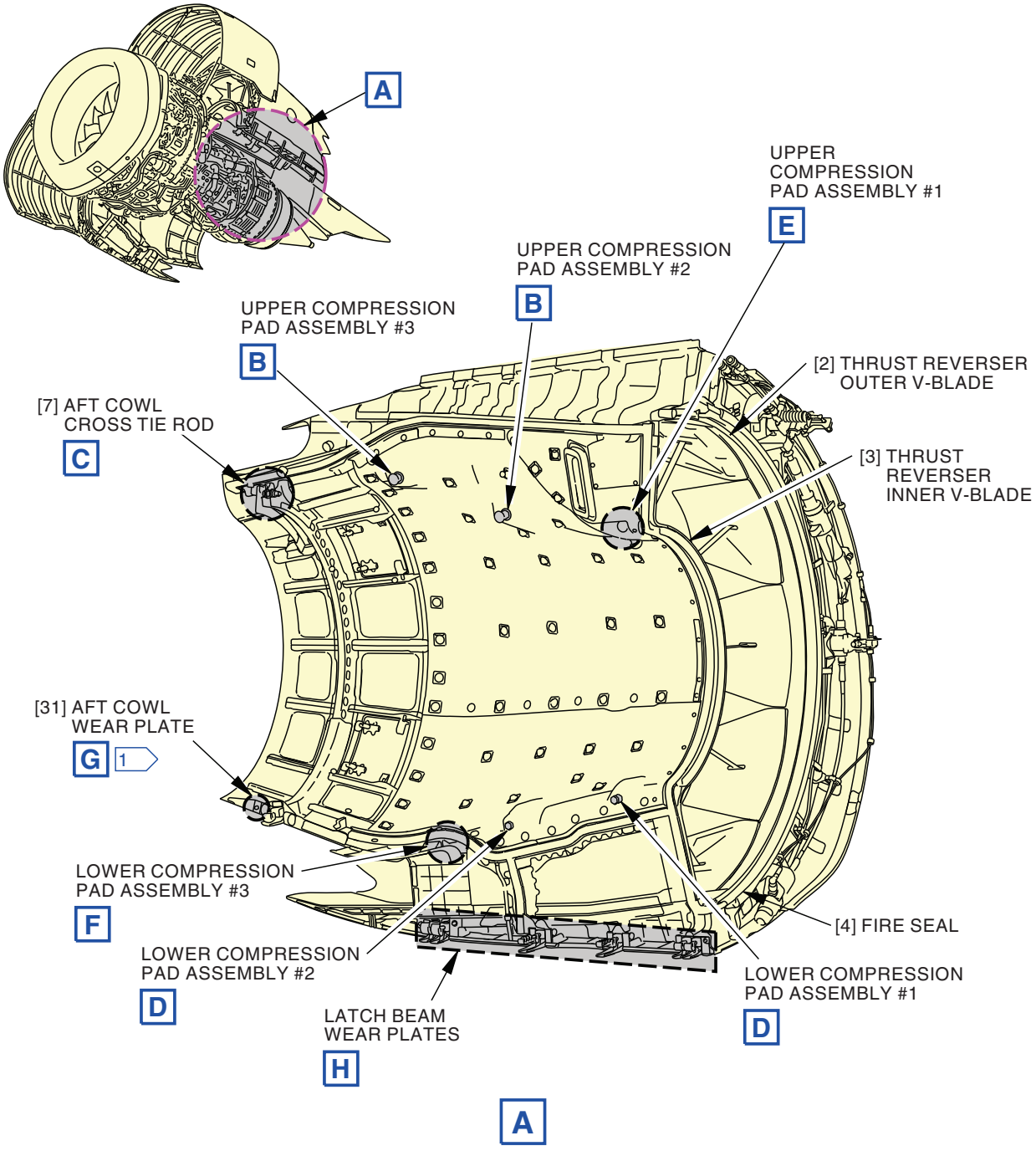
(1) Close the thrust reversers (TASK 78-31-00-010-802-G00).

(a) Make sure that each of the six latches on the thrust reverser latch beam have a closing force of 50.0 ±10.0 lbf (222.4 ±44.5 N).

(b) If it is necessary, adjust the latches (TASK 78-31-11-820-801-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

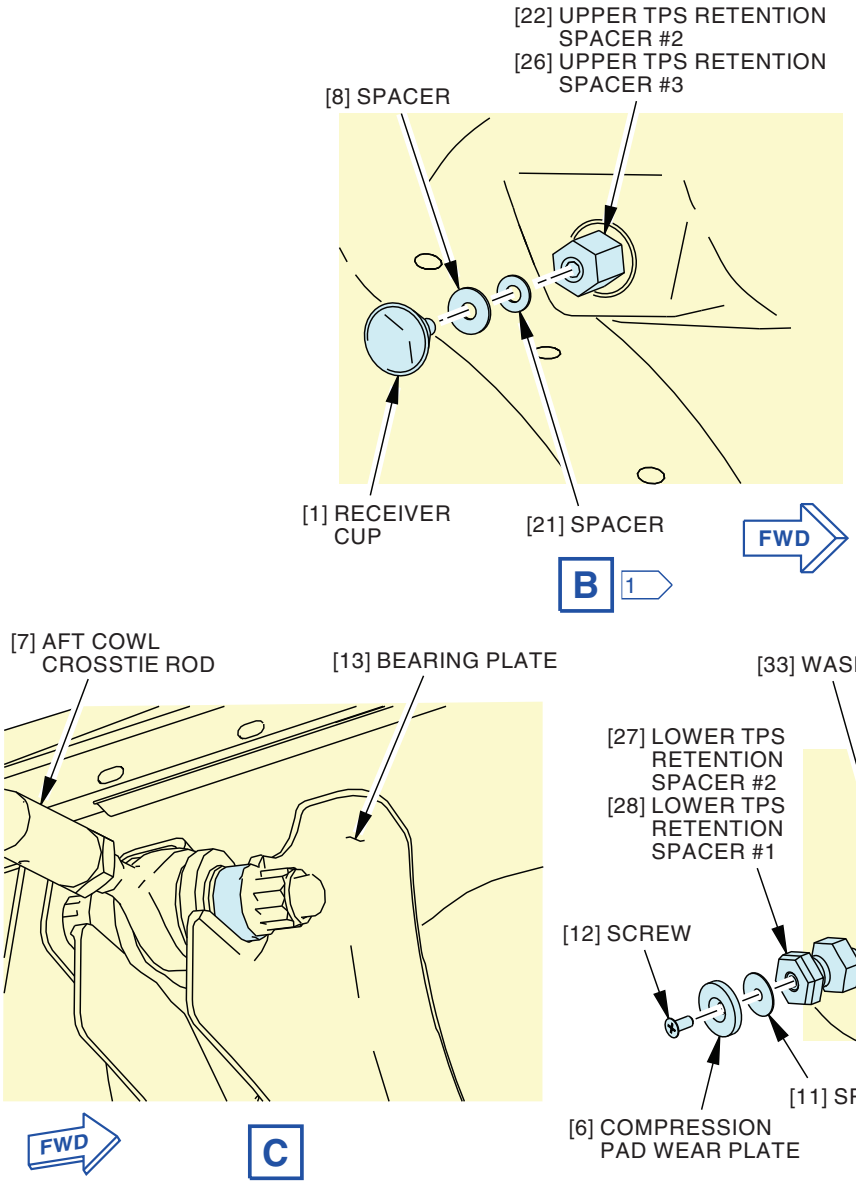


1 CLEARANCE BETWEEN THE AFT COWL AFT-MOST WEAR PLATES AFTER THE T/R IS LATCHED IS 0.000 +0.100/-0.000 INCH (0.000 +2.540/-0.000 mm).

2355162 S00061516667_V5

Thrust Reverser Adjustment
Figure 501/78-31-01-990-807-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 UPPER COMPRESSION PAD ASSEMBLY #2 IS SHOWN. UPPER COMPRESSION PAD #3 IS EQUIVALENT. EXCEPT UPPER TPS RETENTION SPACER #3 IS USED INSTEAD OF UPPER TPS RETENTION SPACER #2.

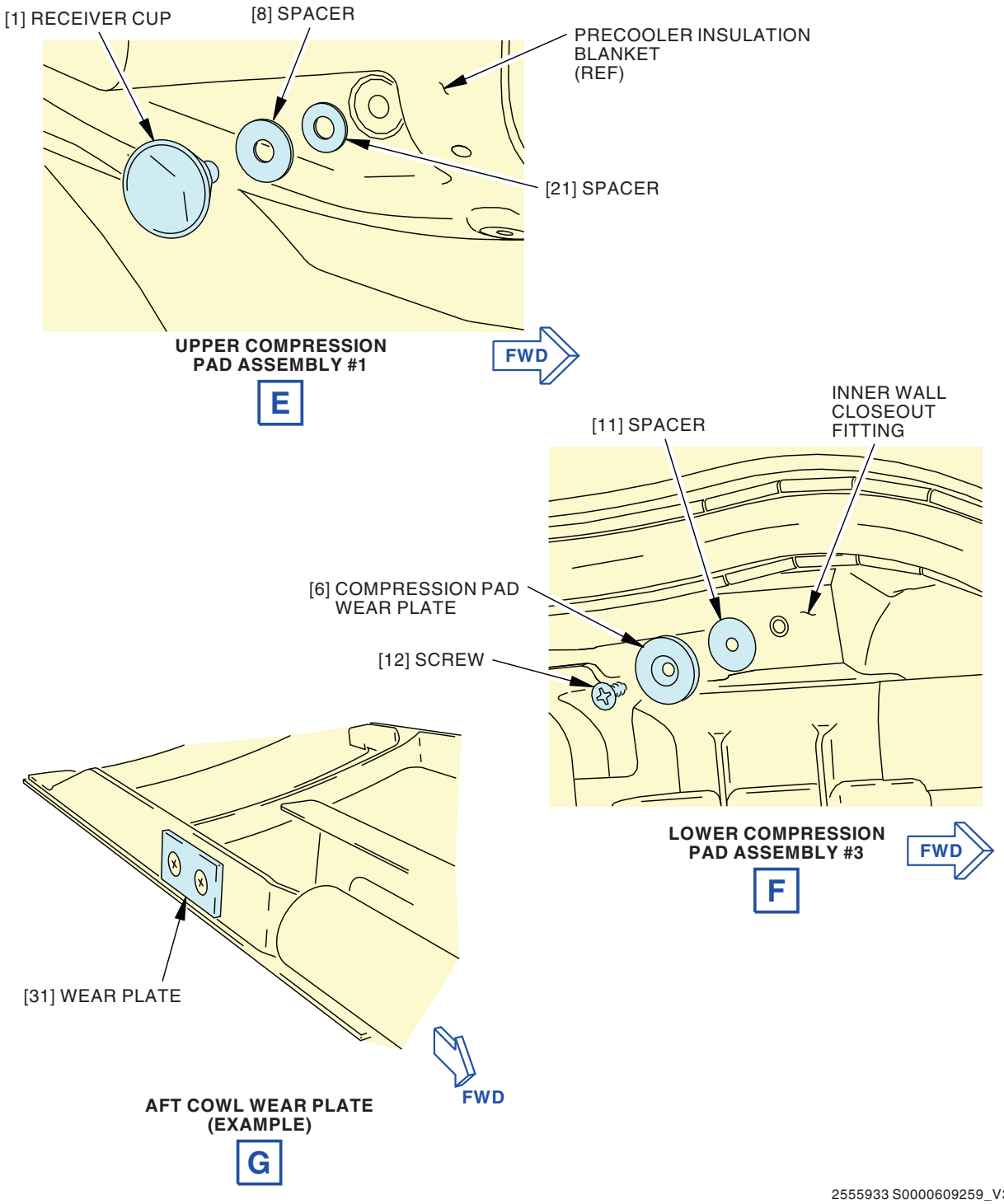
2 LOWER COMPRESSION PAD ASSEMBLY #2 IS SHOWN. LOWER COMPRESSION PAD #1 IS EQUIVALENT. EXCEPT LOWER TPS RETENTION SPACER #1 IS USED INSTEAD OF UPPER TPS RETENTION SPACER #2.

3 USE UP TO THREE WASHERS TO FILL THE GAP BETWEEN THE COMPRESSION PAD ASSEMBLY AND LOWER TPS RETENTION SPACER.

2355163 S00061516668_V5

Thrust Reverser Adjustment
Figure 501/78-31-01-990-807-G00 (Sheet 2 of 4)

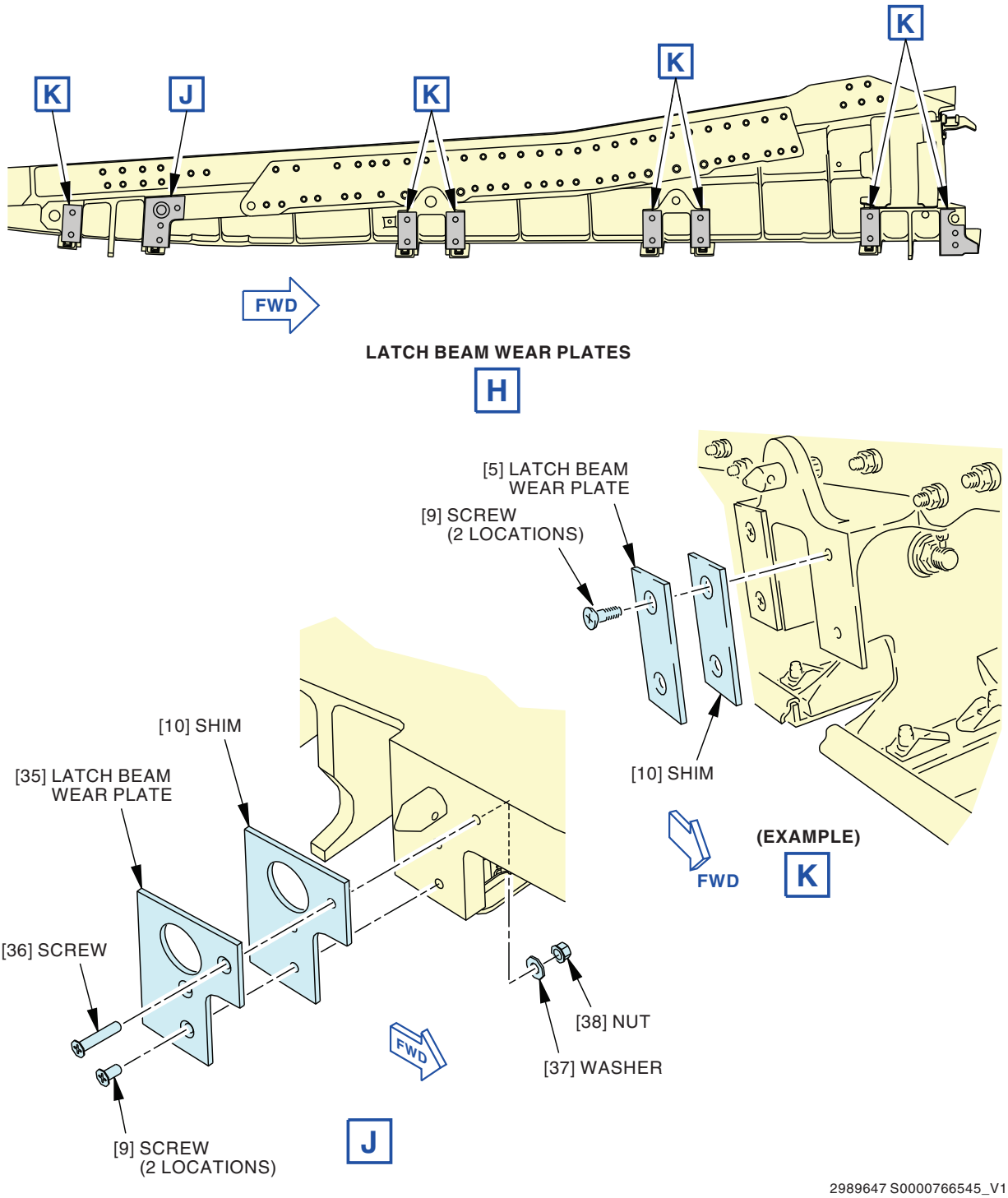
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2555933 S0000609259_V3

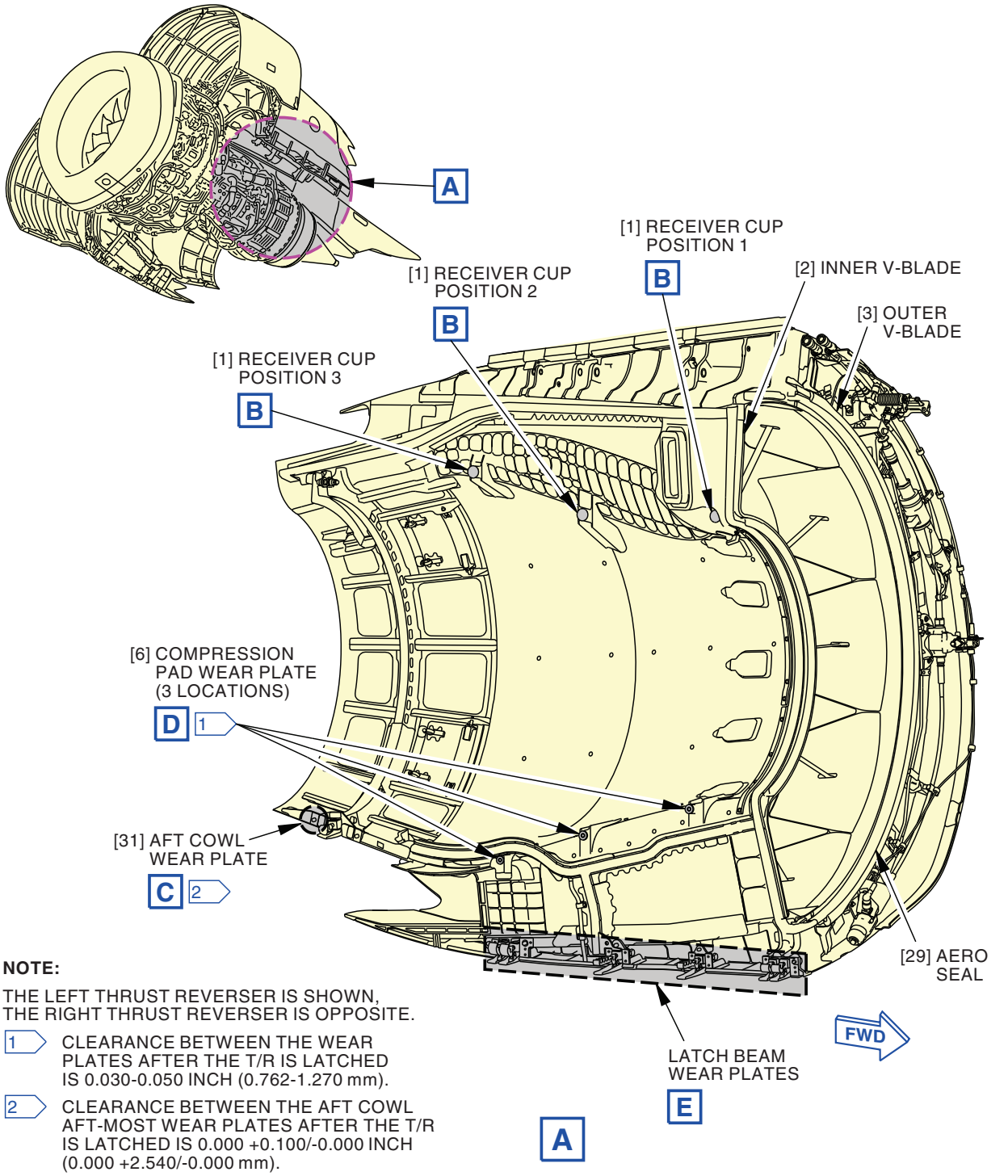
Thrust Reverser Adjustment
Figure 501/78-31-01-990-807-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Thrust Reverser Adjustment
Figure 501/78-31-01-990-807-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

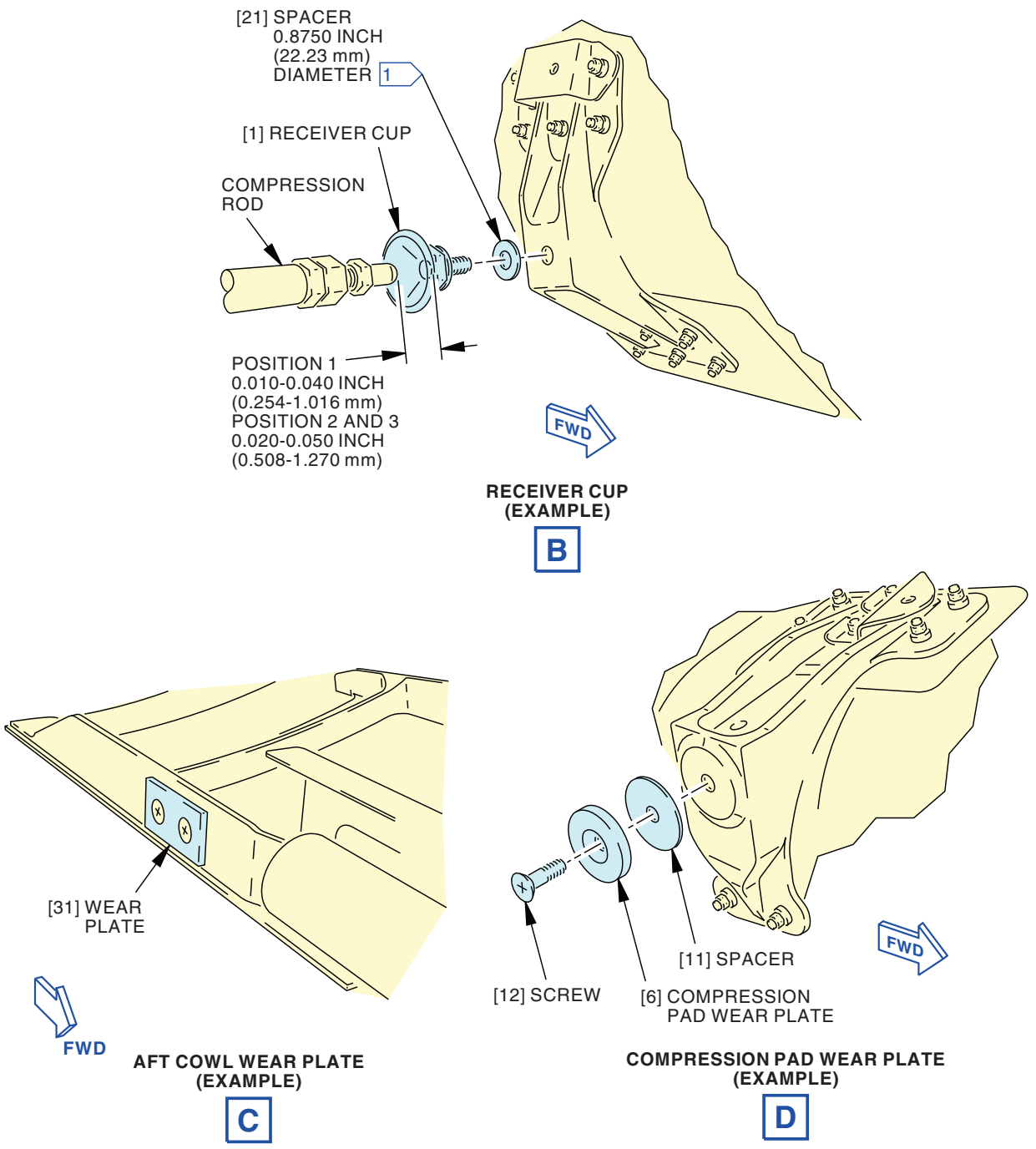
THE LEFT THRUST REVERSER IS SHOWN,
THE RIGHT THRUST REVERSER IS OPPOSITE.

1. CLEARANCE BETWEEN THE WEAR PLATES AFTER THE T/R IS LATCHED IS 0.030-0.050 INCH (0.762-1.270 mm).
2. CLEARANCE BETWEEN THE AFT COWL AFT-MOST WEAR PLATES AFTER THE T/R IS LATCHED IS 0.000 +0.100/-0.000 INCH (0.000 +2.540/-0.000 mm).

2579287 S0000620114_V3

Thrust Reverser Adjustment
Figure 502/78-31-01-990-824-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

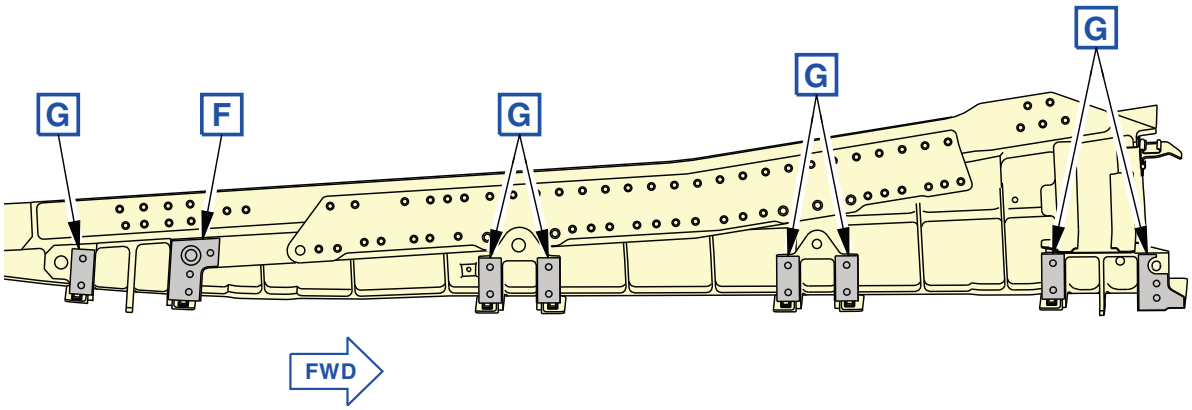


1 SOME AIRPLANES HAVE 2.000 INCH (50.8 mm) DIAMETER SPACERS.

2579288 S0000620115_V4

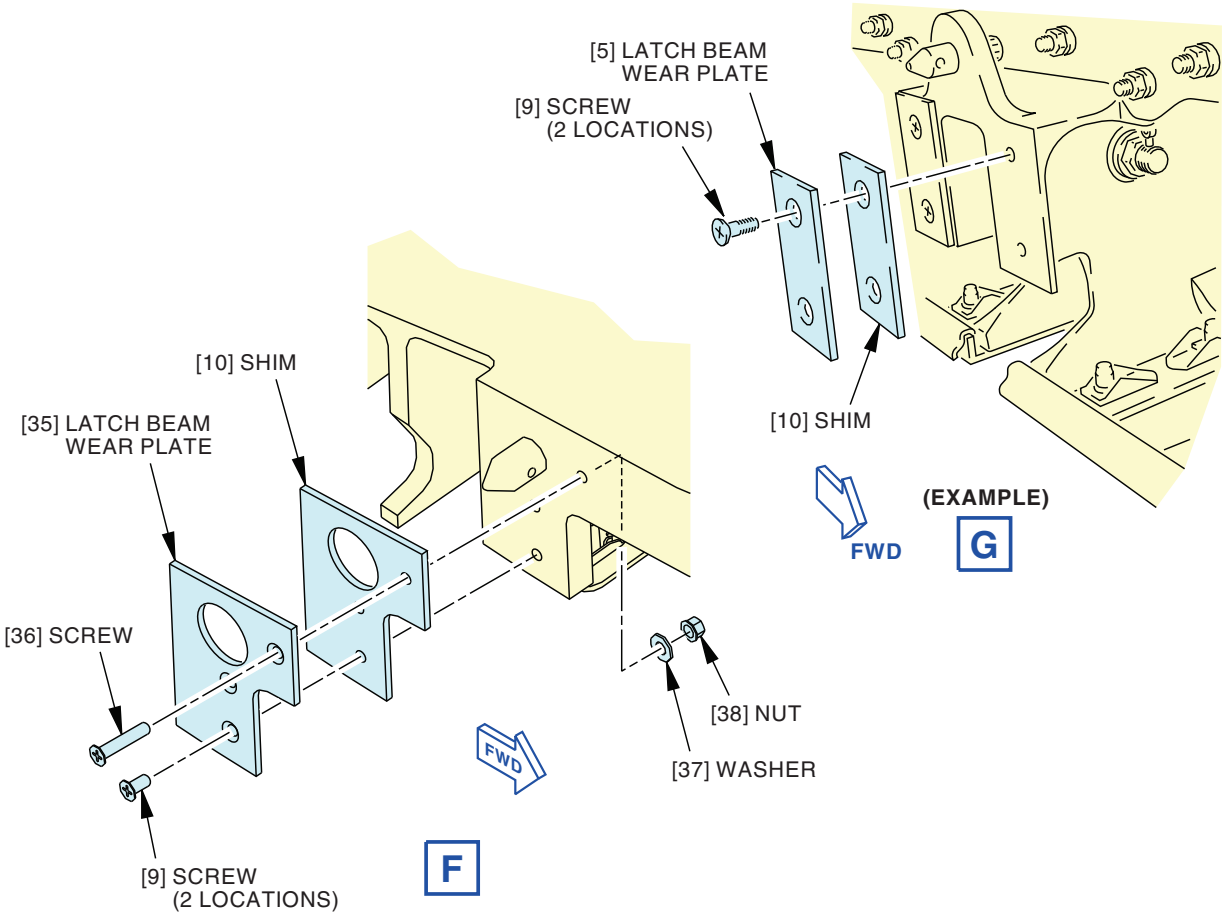
Thrust Reverser Adjustment
Figure 502/78-31-01-990-824-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LATCH BEAM WEAR PLATES

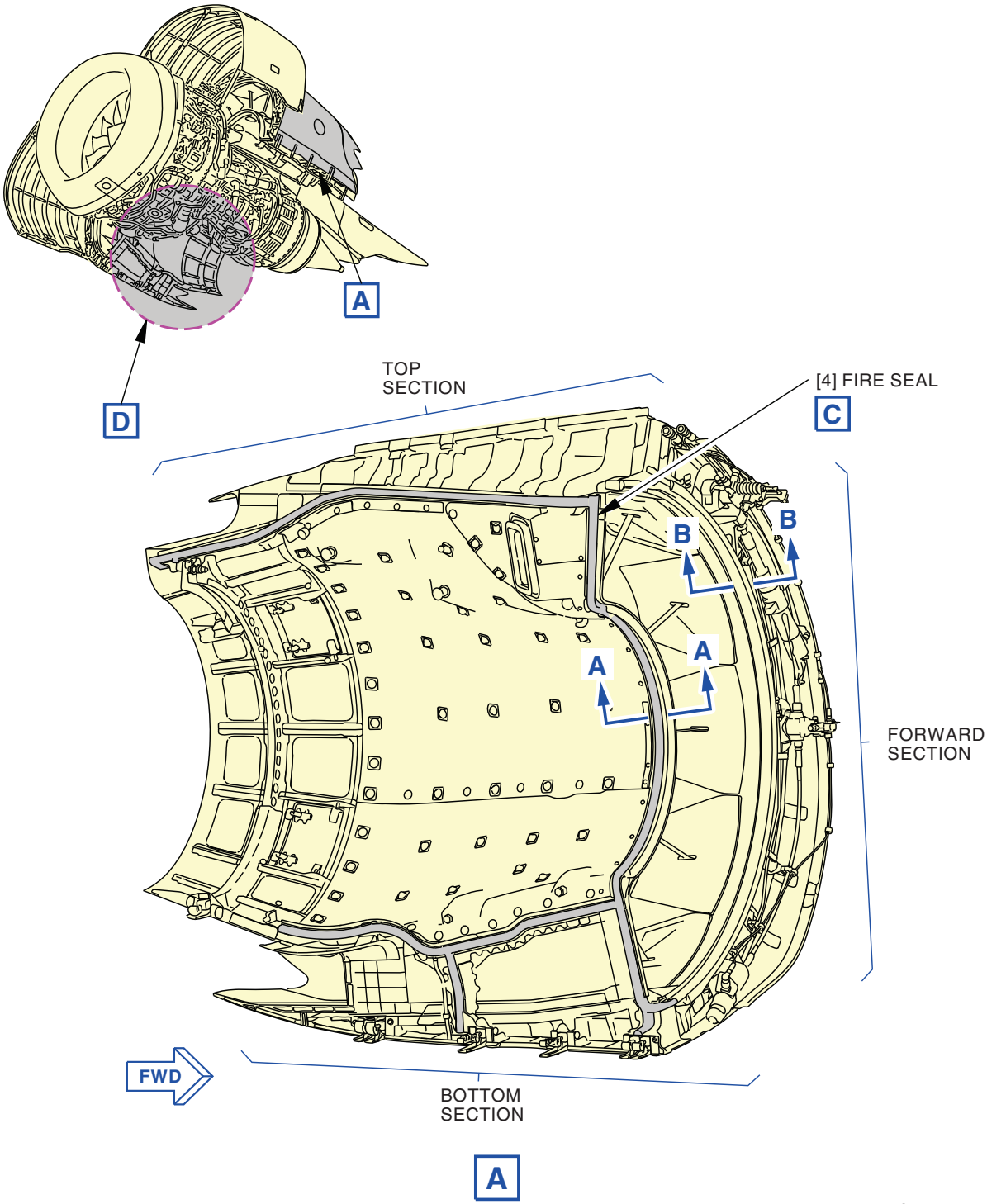
E



2989662 S0000766546_V1

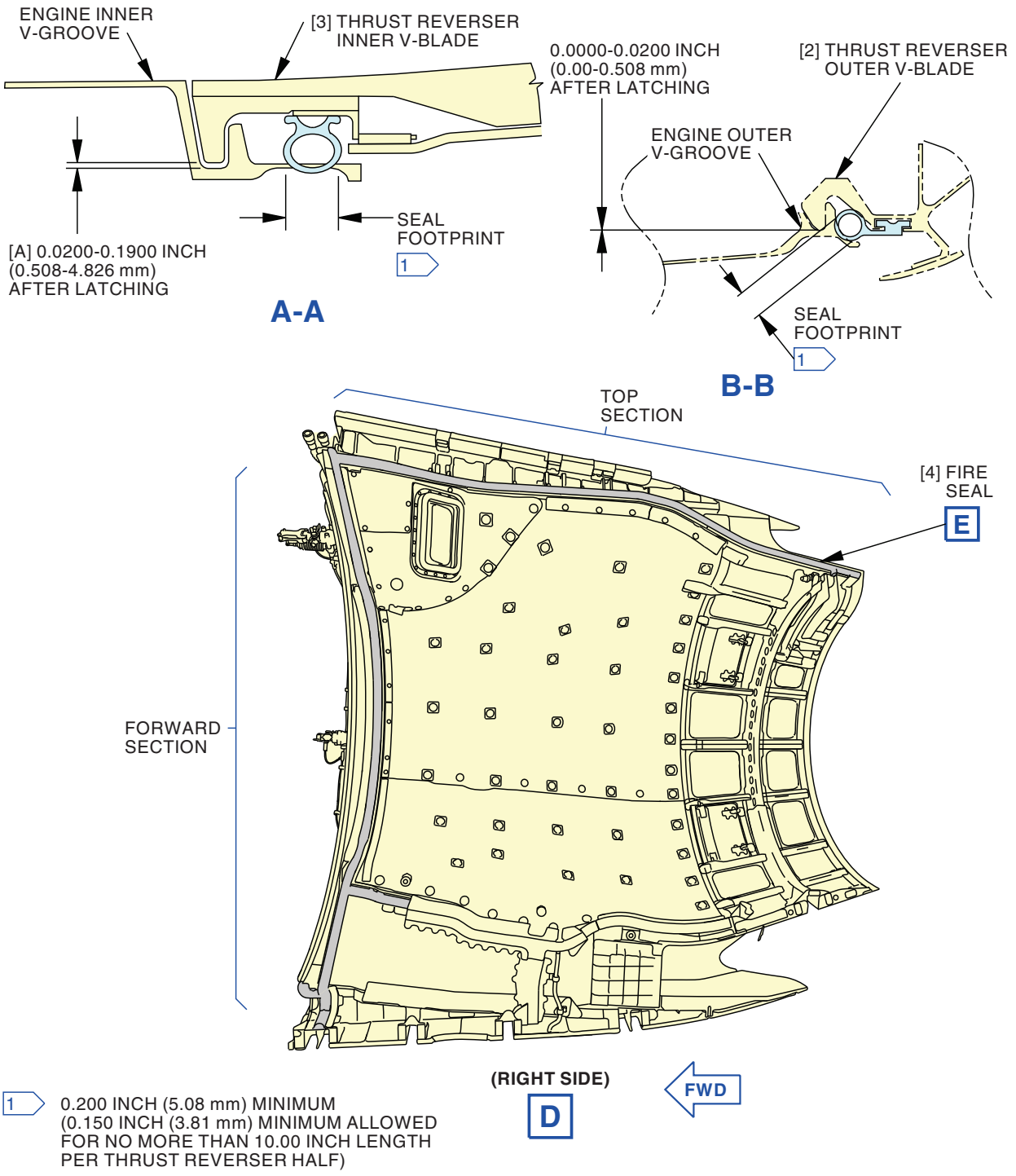
Thrust Reverser Adjustment
Figure 502/78-31-01-990-824-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fireseal Compression Check
Figure 503/78-31-01-990-808-G00 (Sheet 1 of 4)

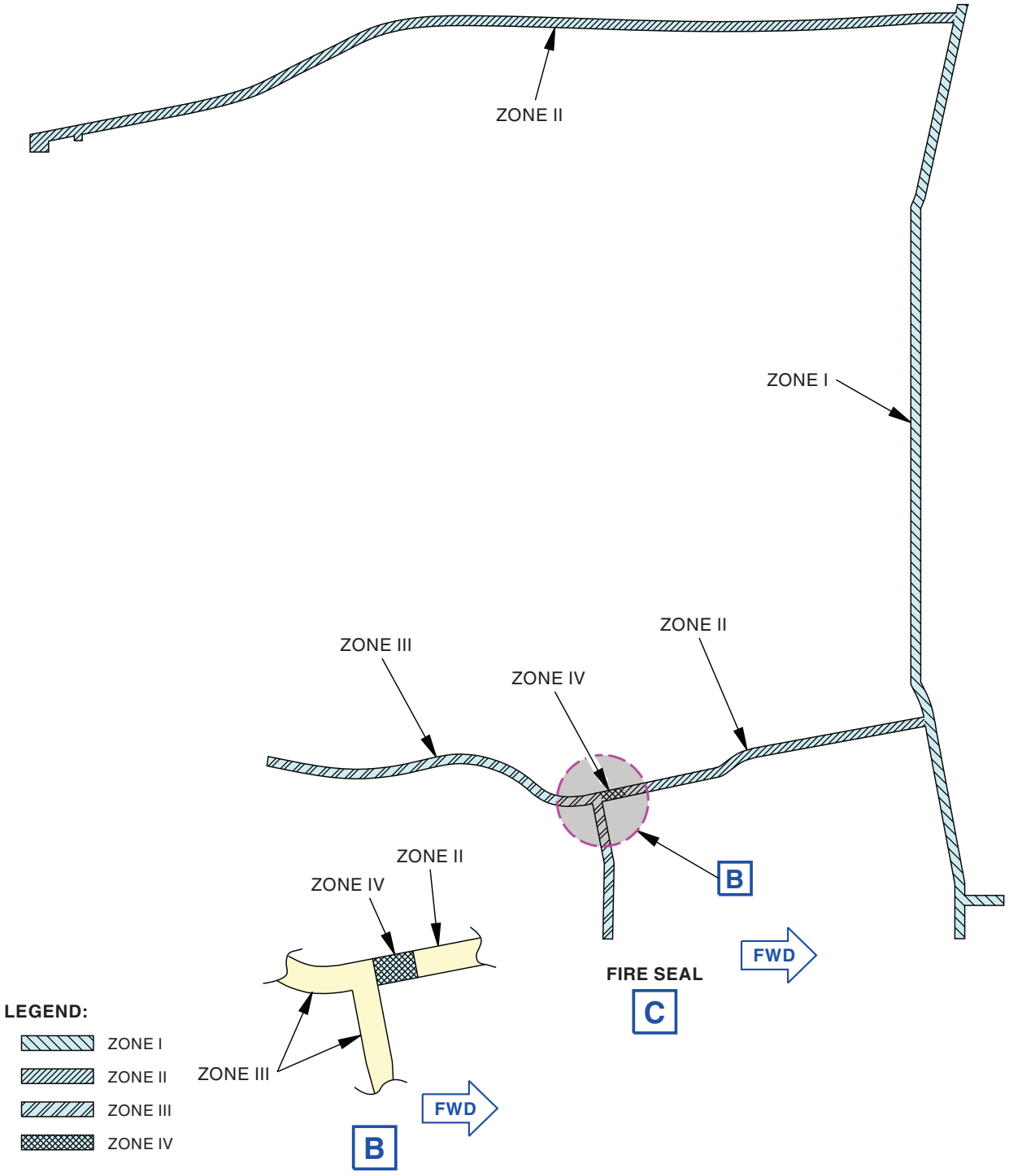
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2355165 S00061516670_V8

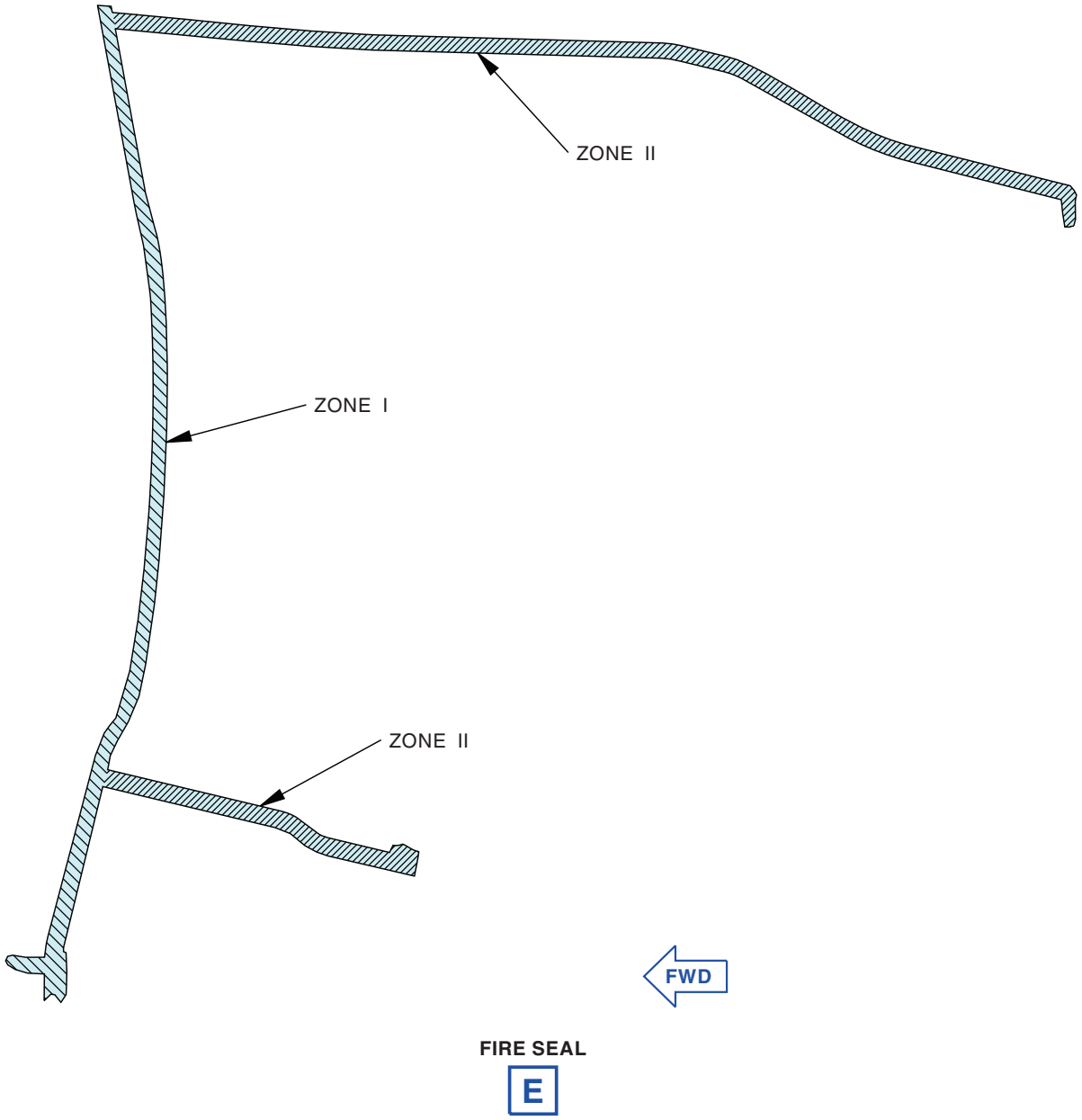
Fireseal Compression Check
Figure 503/78-31-01-990-808-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fireseal Compression Check
Figure 503/78-31-01-990-808-G00 (Sheet 3 of 4)

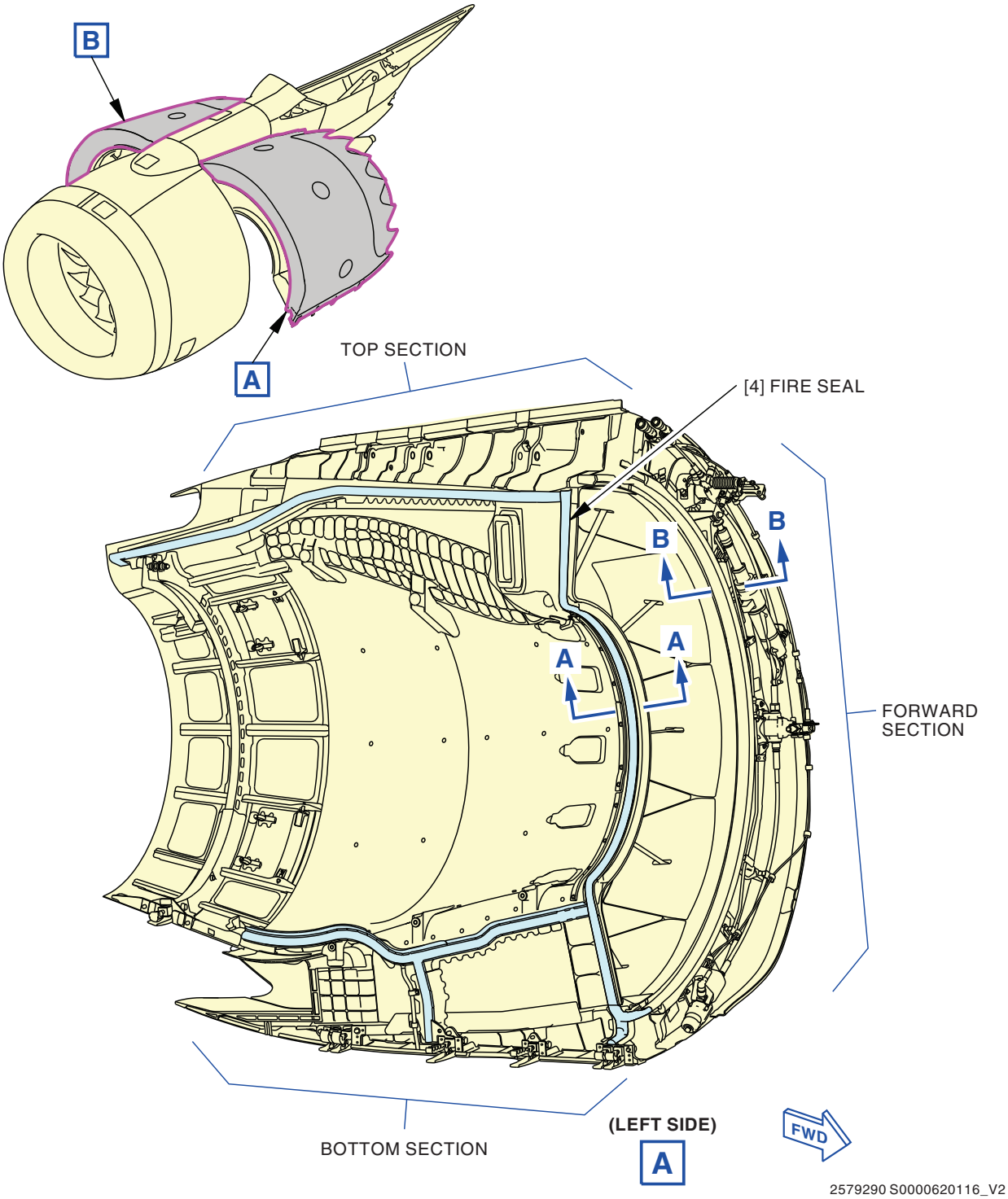
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2501850 S0000588504_V1

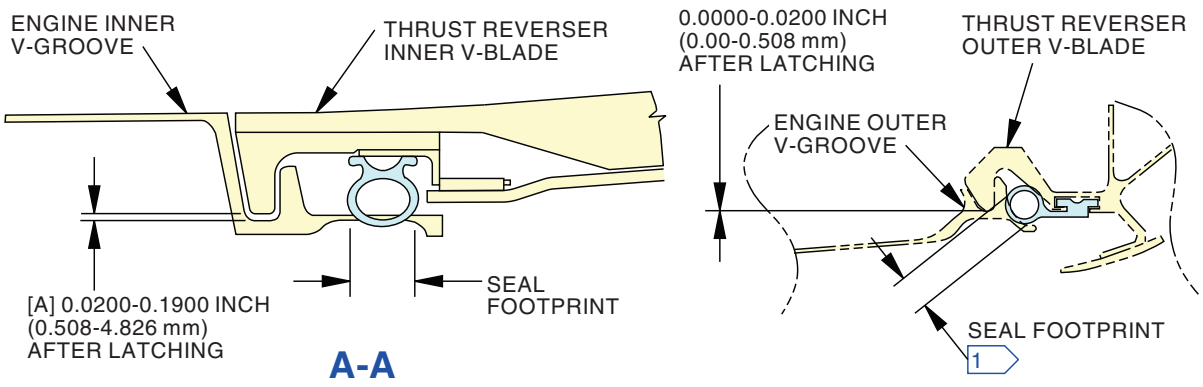
Fireseal Compression Check
Figure 503/78-31-01-990-808-G00 (Sheet 4 of 4)

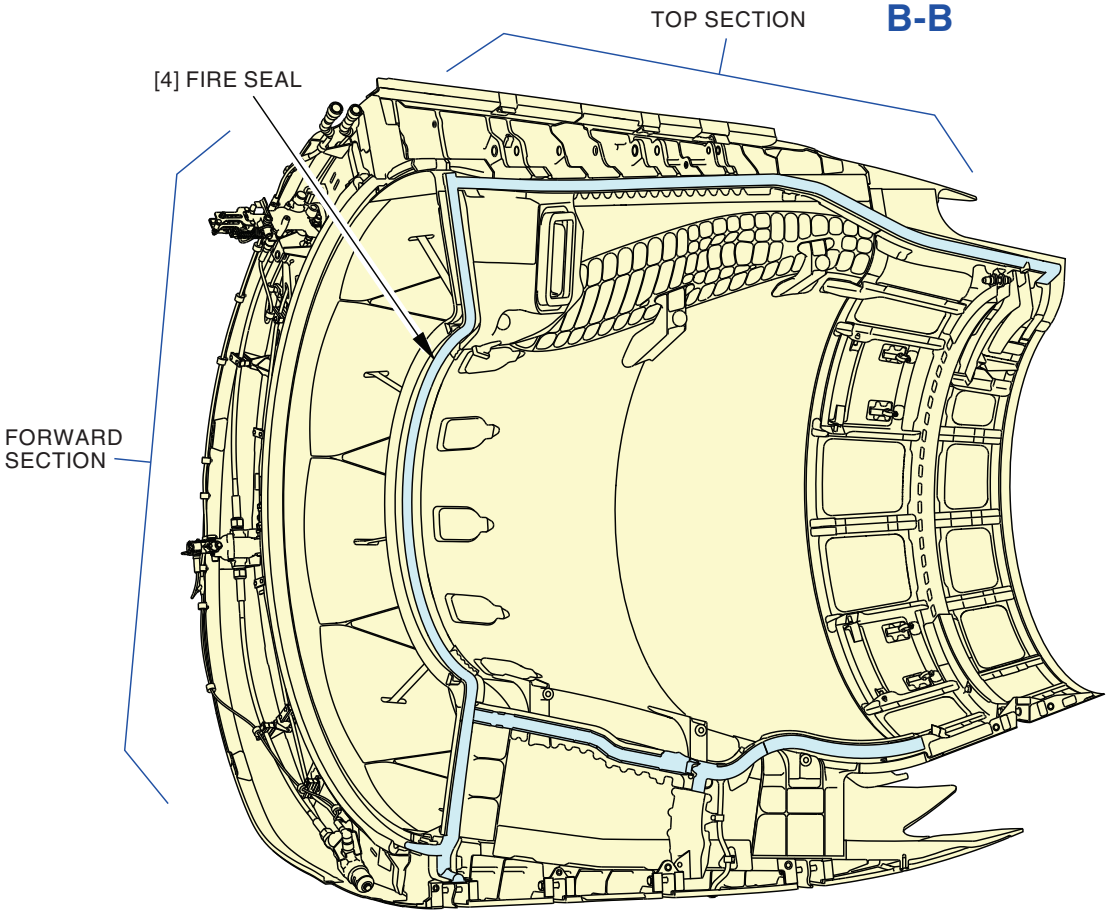
ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fireseal Compression Check
Figure 504/78-31-01-990-825-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





1 0.200 INCH (5.08 mm) MINIMUM
(0.150 INCH (3.81 mm) MINIMUM
ALLOWED FOR NO MORE THAN
10.00 INCH LENGTH PER THRUST
REVERSER HALF)

(RIGHT SIDE)

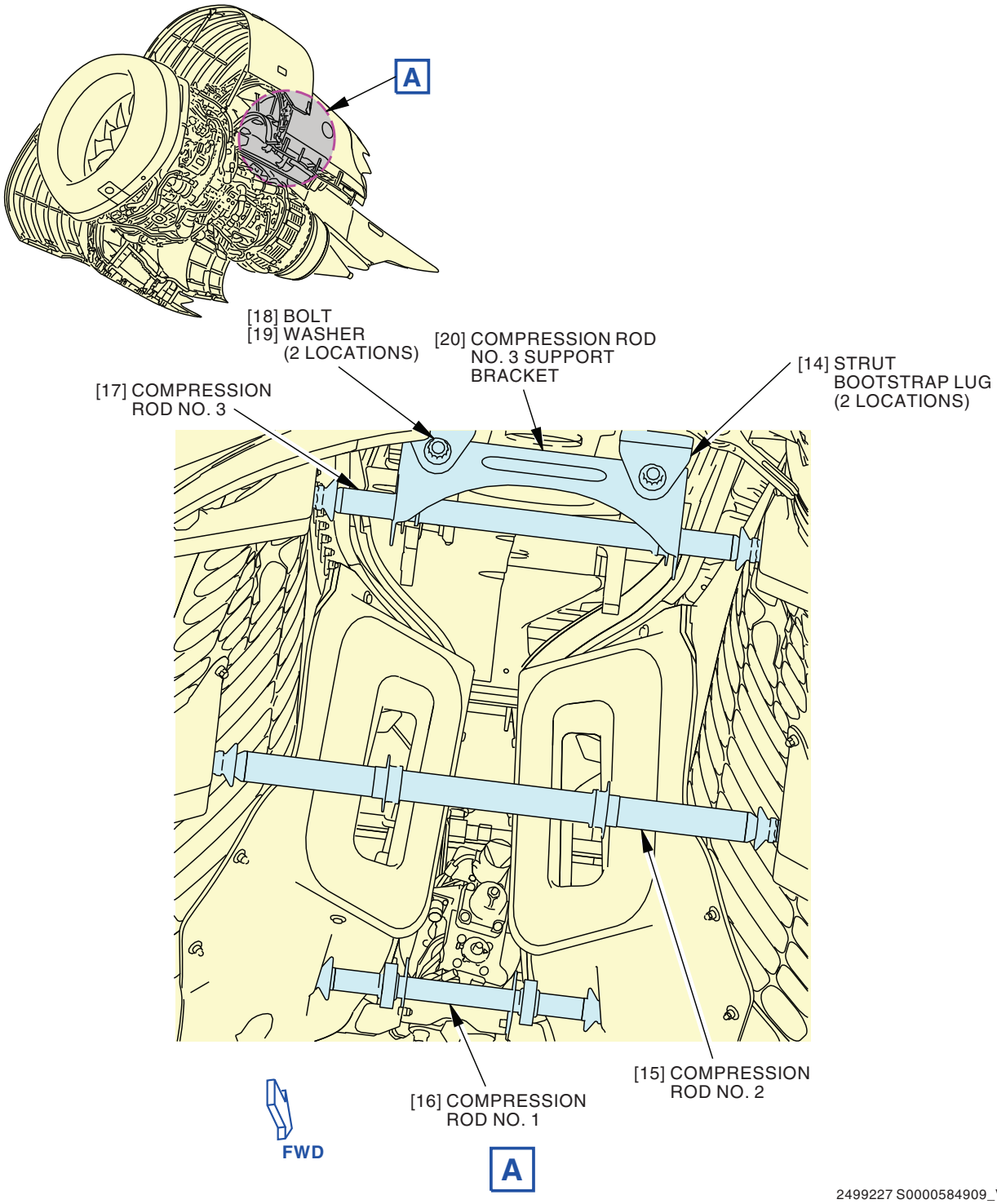
B

FWD

2579291 S0000620118_V2

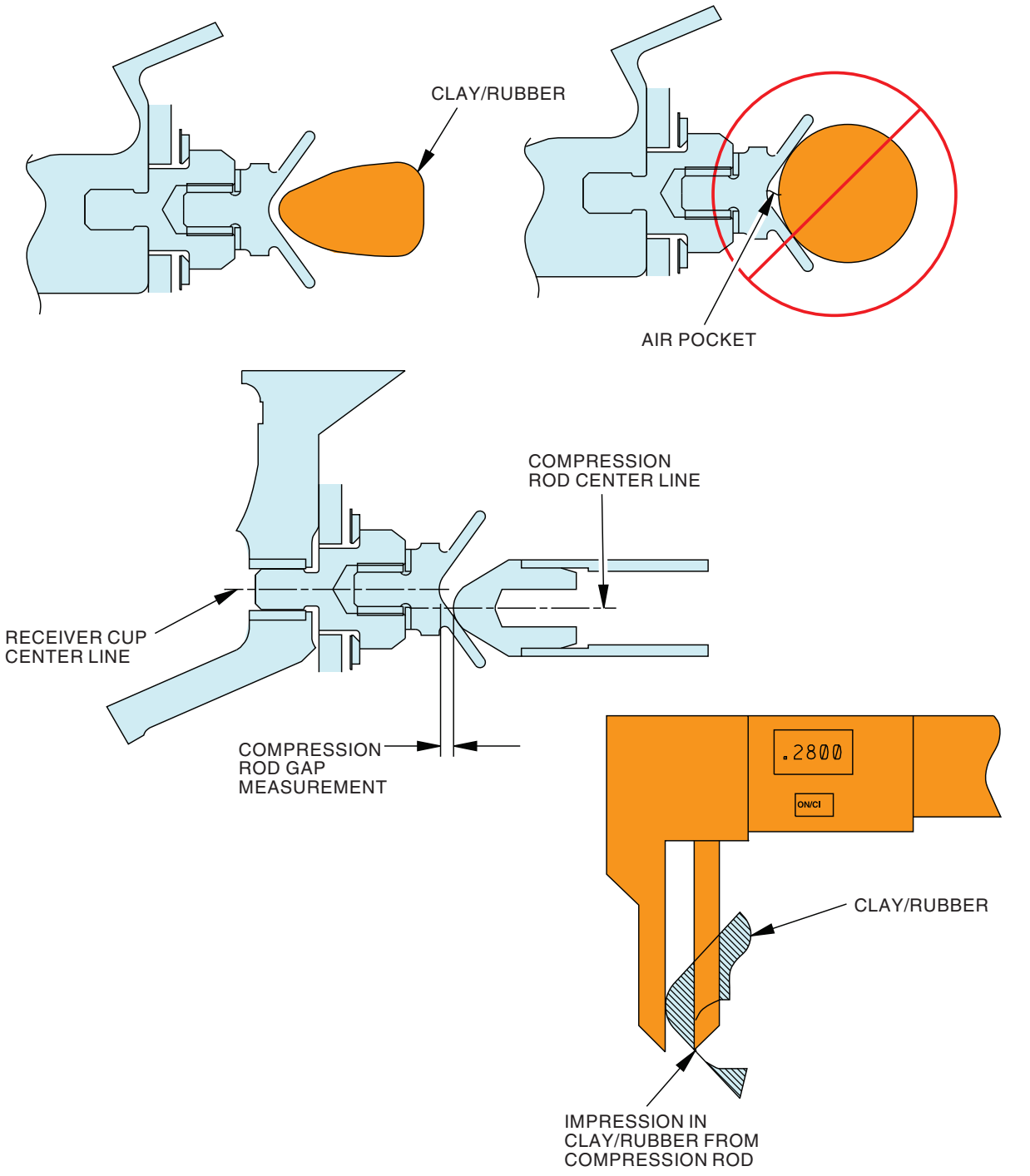
Fireseal Compression Check
Figure 504/78-31-01-990-825-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Compression Rod Gap Check
Figure 505/78-31-01-990-817-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2499719 S0000584912_V2

Compression Rod Gap Check
Figure 505/78-31-01-990-817-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

THRUST REVERSER - INSPECTION/CHECK

1. General

A. This procedure has these tasks:

(1) A visual inspection of the fan duct walls to make sure that they will provide an aerodynamically smooth surface for the passage of fan air or if there was damage from a RTO duct burst.

(2) A visual inspection of the cowl hinges.

(3) A visual inspection of the precooler kiss seal.

(4) A visual inspection of the inner and outer v-blades for damage.

(5) A visual inspection of the outer v-blade seal.

(6) A visual inspection of the outer v-blade.

(7) A visual inspection of the fire seals.

(8) A check of the thrust reverser rigging following an engine change or thrust reverser change.

PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL

(9) A visual inspection of the composite inner wall.

(10) A visual inspection of the composite inner wall inlet and exhaust ports.

PTF ALL

TASK 78-31-01-200-801-G00

2. Thrust Reverser Fan Duct Wall Inspection

(Figure 601, Figure 602)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This is a task to do a visual inspection check of the fan duct inner and outer walls for damage.

B. References

Reference	Title
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)

C. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

D. Prepare for the Inspection

SUBTASK 78-31-01-010-009-G00

WARNING DO ALL OF THE SPECIFIED TASKS IN THE CORRECT SEQUENCE TO OPEN THE THRUST REVERSER. IF YOU DO NOT OBEY THIS INSTRUCTION, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do this task: Open the Thrust Reverser (Selection), TASK 78-31-00-010-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Thrust Reverser Fan Duct Wall Inspection

SUBTASK 78-31-01-210-001-G00

(1) Look through the forward and aft ends of the fan duct to examine the walls for the damage that follows:

(a) Holes, cracks, nicks, gouges, delamination, dents and edge corrosion.

(b) Pitting in the surface layer (small areas where the surface appears chipped away around the perforation of the acoustic panel) (Figure 602).

NOTE: This pitting can occur during the usual manufacturing process of the acoustic panel. This condition has been inspected and approved for in-service use at the time of manufacture.

1) No action is necessary with the following conditions:

a) The surface area adjacent to each pitting location has the original silver finish and does not show the black panel material.

b) There are no signs of edge erosion.

F. Put the Airplane Back to its Usual Condition

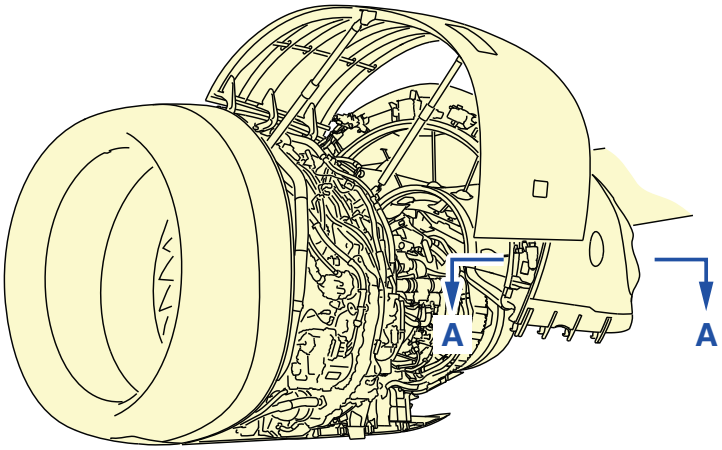
SUBTASK 78-31-01-410-007-G00

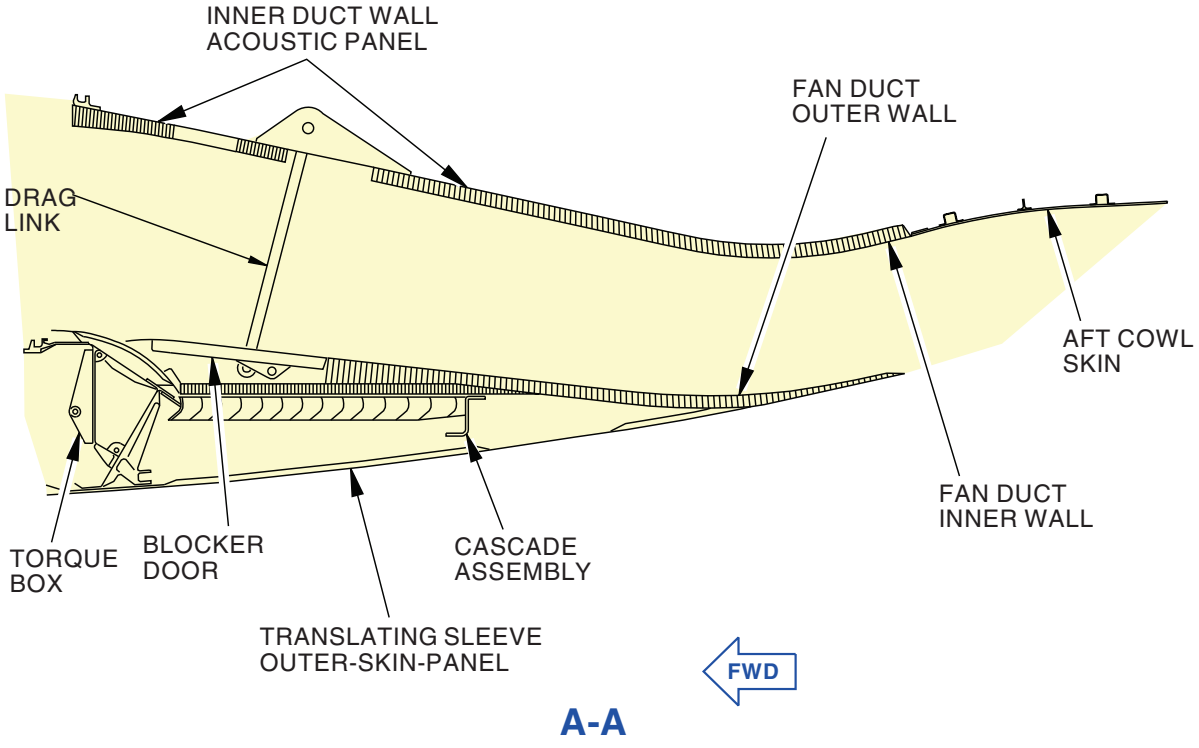
WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

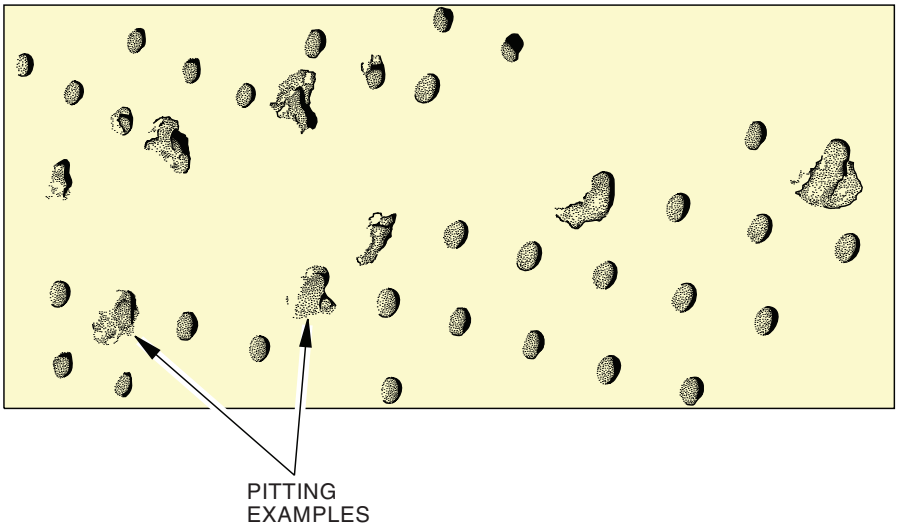




2355166 S00061516673_V1

Thrust Reverser Fan Duct Wall Inspection
Figure 601/78-31-01-990-809-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2355168 S00061516674_V1

Inner and Outer Duct Wall Acoustic Panel Inspection
Figure 602/78-31-01-990-810-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-31-01-211-801-G00

3. Thrust Reverser Cowl Hinge Inspection

(Figure 603)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions for the inspection of the thrust reverser cowl hinge.

B. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

C. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

D. Access Panels

Number	Name/Location
415AL	Left Forward Thrust Reverser Hinge Fairing, Engine 1
415BL	Left Aft Thrust Reverser Hinge Fairing, Engine 1
416AR	Right Forward Thrust Reverser Hinge Fairing, Engine 1
416BR	Right Aft Thrust Reverser Hinge Fairing, Engine 1
425AL	Left Forward Thrust Reverser Hinge Fairing, Engine 2
425BL	Left Aft Thrust Reverser Hinge Fairing, Engine 2
426AR	Right Forward Thrust Reverser Hinge Fairing, Engine 2
426BR	Right Aft Thrust Reverser Hinge Fairing, Engine 2

E. Prepare for the Inspection

SUBTASK 78-31-01-860-020-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 78-31-01-860-021-G00

(2) For Engine 1, open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	5	C00276	ENGINE 1 THRUST REVERSER CONT
B	6	C01412	ENGINE 1 THRUST REVERSER INTLK
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 78-31-01-860-022-G00

(3) For engine 2, open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK
C	6	C01413	ENGINE 2 THRUST REVERSER INTLK
C	7	C00277	ENGINE 2 THRUST REVERSER CONT

SUBTASK 78-31-01-010-017-G00

(4) Remove the applicable access panels:

(a) Open these access panels:

Number	Name/Location
416AR	Right Forward Thrust Reverser Hinge Fairing, Engine 1
416BR	Right Aft Thrust Reverser Hinge Fairing, Engine 1

(b) Open these access panels:

Number	Name/Location
415AL	Left Forward Thrust Reverser Hinge Fairing, Engine 1
415BL	Left Aft Thrust Reverser Hinge Fairing, Engine 1

(c) Open these access panels:

Number	Name/Location
425AL	Left Forward Thrust Reverser Hinge Fairing, Engine 2
425BL	Left Aft Thrust Reverser Hinge Fairing, Engine 2

(d) Open these access panels:

Number	Name/Location
426AR	Right Forward Thrust Reverser Hinge Fairing, Engine 2
426BR	Right Aft Thrust Reverser Hinge Fairing, Engine 2

F. Cowl Hinges Inspection

SUBTASK 78-31-01-210-003-G00

(1) Do a general visual inspection of the thrust reverser cowl hinges for cracks.

(a) Cracks are not permitted.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-01-410-012-G00

(1) Install the applicable access panels:

(a) Close these access panels:

Number	Name/Location
416AR	Right Forward Thrust Reverser Hinge Fairing, Engine 1
416BR	Right Aft Thrust Reverser Hinge Fairing, Engine 1

(b) Close these access panels:

Number	Name/Location
415AL	Left Forward Thrust Reverser Hinge Fairing, Engine 1
415BL	Left Aft Thrust Reverser Hinge Fairing, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Close these access panels:

Number	Name/Location
425AL	Left Forward Thrust Reverser Hinge Fairing, Engine 2
425BL	Left Aft Thrust Reverser Hinge Fairing, Engine 2

(d) Close these access panels:

Number	Name/Location
426AR	Right Forward Thrust Reverser Hinge Fairing, Engine 2
426BR	Right Aft Thrust Reverser Hinge Fairing, Engine 2

SUBTASK 78-31-01-860-024-G00

(2) For Engine 1, remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	5	C00276	ENGINE 1 THRUST REVERSER CONT
B	6	C01412	ENGINE 1 THRUST REVERSER INTLK
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

SUBTASK 78-31-01-860-060-G00

(3) For Engine 2, remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK
C	6	C01413	ENGINE 2 THRUST REVERSER INTLK
C	7	C00277	ENGINE 2 THRUST REVERSER CONT

SUBTASK 78-31-01-860-025-G00

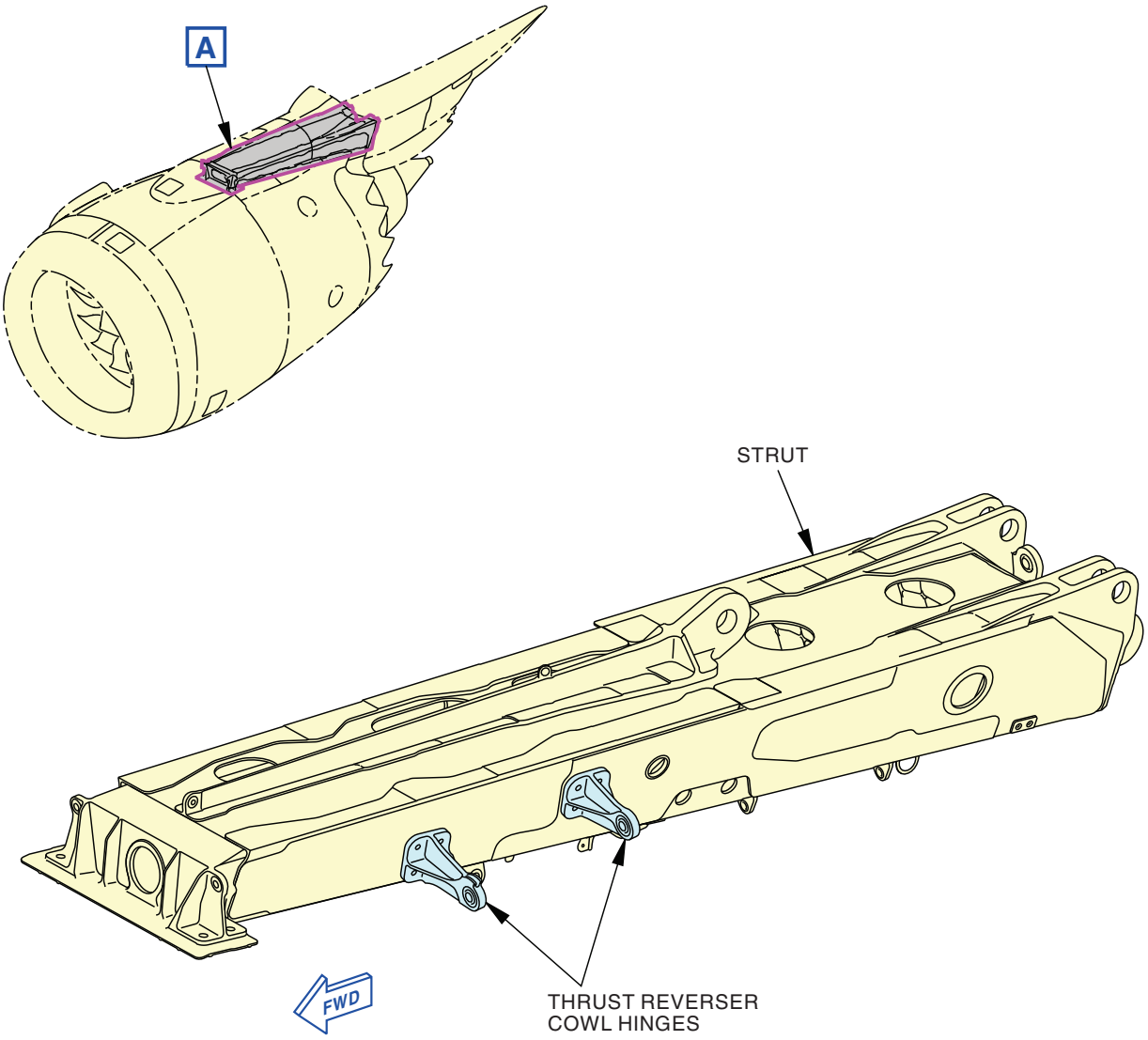
(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



**(LEFT THRUST REVERSER COWL HINGES ARE SHOWN,
RIGHT THRUST REVERSER COWL HINGES ARE OPPOSITE)**

A

2972626 S0000750821_V1

**Thrust Reverser Cowl Hinge Inspection
Figure 603/78-31-01-990-830-G00**

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-31-01-211-802-G00

4. Precooler Kiss Seal Inspection

(Figure 604)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions for the inspection of the precooler kiss seal.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-15-000-801-G00	Precooler Inlet Duct Seal - Removal (P/B 401)
78-31-15-400-801-G00	Precooler Inlet Duct Seal - Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Inspection

SUBTASK 78-31-01-860-026-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 78-31-01-860-027-G00

(2) For Engine 1, open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	5	C00276	ENGINE 1 THRUST REVERSER CONT
B	6	C01412	ENGINE 1 THRUST REVERSER INTLK
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

SUBTASK 78-31-01-860-028-G00

(3) For engine 2, open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK
C	6	C01413	ENGINE 2 THRUST REVERSER INTLK
C	7	C00277	ENGINE 2 THRUST REVERSER CONT

SUBTASK 78-31-01-860-029-G00

(4) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 78-31-01-040-004-G00

(5) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 78-31-01-010-018-G00

(6) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Precooler Kiss Seal Inspection

SUBTASK 78-31-01-210-004-G00

(1) Do a visual inspection of the precooler kiss seal for degradation of the seal.

(a) Degradation of the seal is not permitted.

SUBTASK 78-31-01-900-001-G00

(2) If it is necessary, replace the seal (TASK 78-31-15-000-801-G00 and TASK 78-31-15-400-801-G00).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-01-410-013-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 78-31-01-440-003-G00

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 78-31-01-860-030-G00

(3) For Engine 1, remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	5	C00276	ENGINE 1 THRUST REVERSER CONT
B	6	C01412	ENGINE 1 THRUST REVERSER INTLK
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

For Engine 2, remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK
C	6	C01413	ENGINE 2 THRUST REVERSER INTLK
C	7	C00277	ENGINE 2 THRUST REVERSER CONT

SUBTASK 78-31-01-860-031-G00

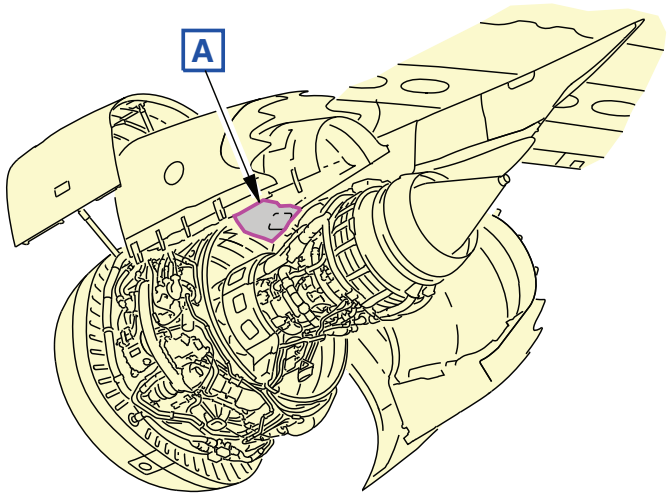
(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

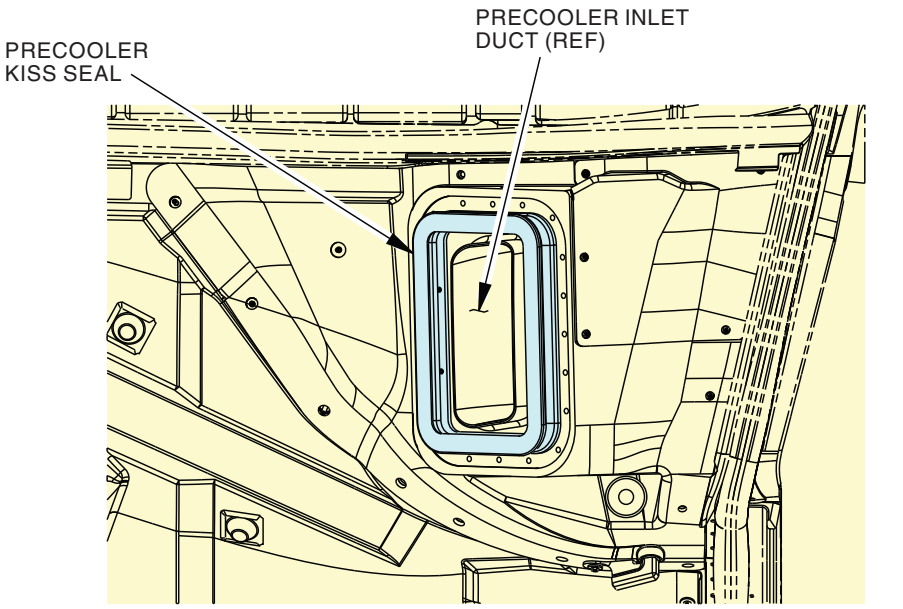
(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





(LEFT THRUST REVERSER IS SHOWN,
RIGHT THRUST REVERSER IS OPPOSITE)

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Precooler Kiss Seal Inspection
Figure 604/78-31-01-990-829-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-31-01-211-803-G00

5. Thrust Reverser Inner and Outer V-Blade Inspection

(Figure 605)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions for the inspection of the inner and outer V-blades on the thrust reverser.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Inspection

SUBTASK 78-31-01-860-032-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 78-31-01-860-033-G00

(2) For Engine 1, open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	5	C00276	ENGINE 1 THRUST REVERSER CONT
B	6	C01412	ENGINE 1 THRUST REVERSER INTLK
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

SUBTASK 78-31-01-860-034-G00

(3) For engine 2, open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK
C	6	C01413	ENGINE 2 THRUST REVERSER INTLK
C	7	C00277	ENGINE 2 THRUST REVERSER CONT

SUBTASK 78-31-01-860-035-G00

(4) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804

SUBTASK 78-31-01-040-005-G00

(5) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801

SUBTASK 78-31-01-010-019-G00

(6) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Thrust Reverser Inner and Outer V-Blade Inspection

SUBTASK 78-31-01-210-005-G00

(1) Do a detailed inspection of the thrust reverser inner and outer V-blade for damage and degradation.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-01-410-014-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 78-31-01-440-004-G00

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 78-31-01-860-036-G00

(3) For Engine 1, remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	5	C00276	ENGINE 1 THRUST REVERSER CONT
B	6	C01412	ENGINE 1 THRUST REVERSER INTLK
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

For Engine 2, remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK
C	6	C01413	ENGINE 2 THRUST REVERSER INTLK
C	7	C00277	ENGINE 2 THRUST REVERSER CONT

SUBTASK 78-31-01-860-037-G00

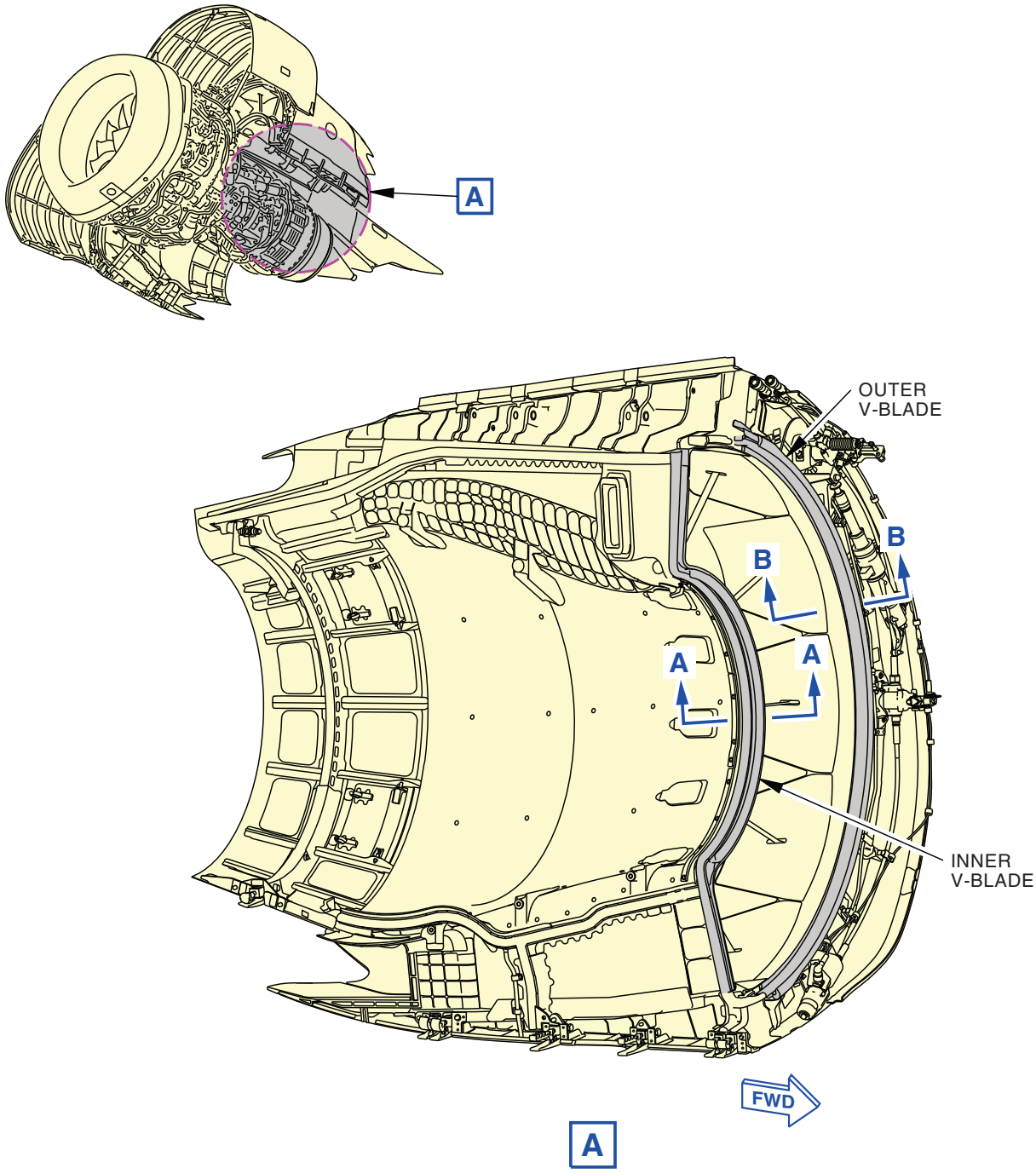
(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



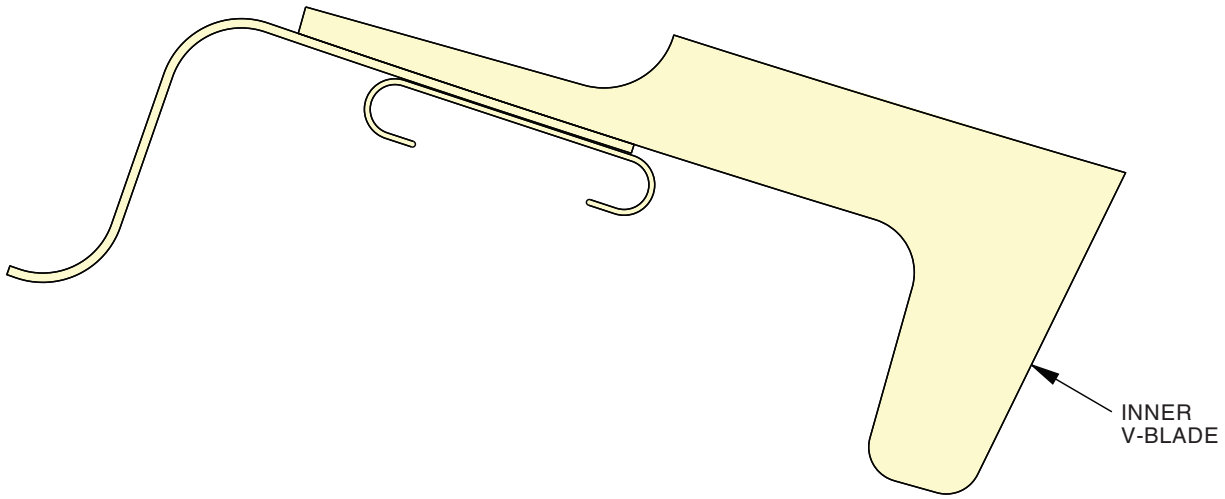
NOTE:

THE LEFT THRUST REVERSER IS SHOWN,
THE RIGHT THRUST REVERSER IS OPPOSITE.

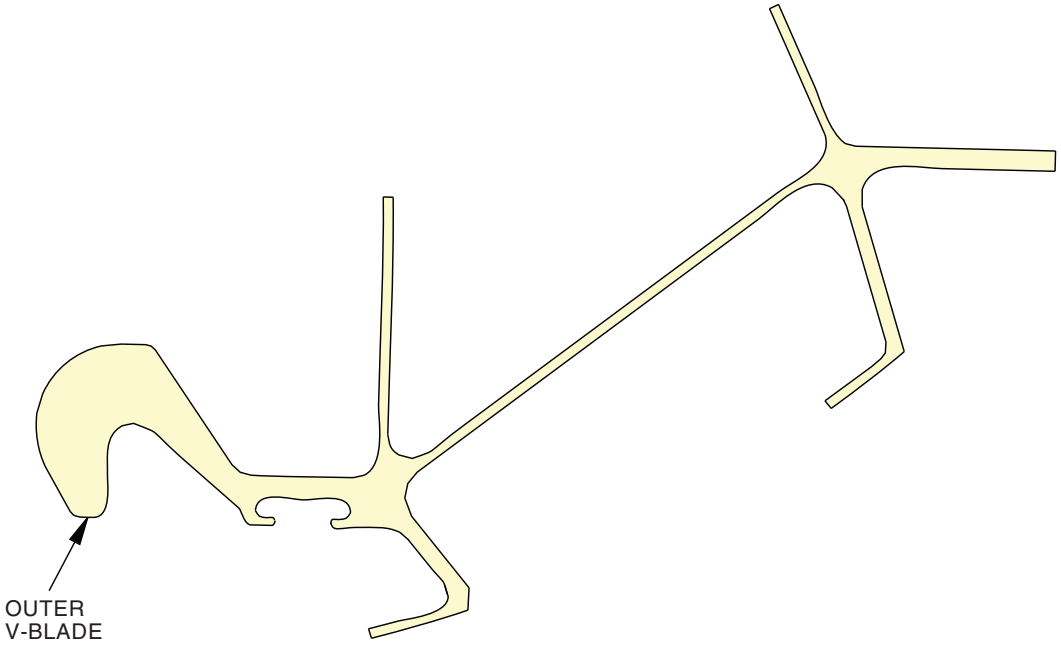
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Thrust Reverser Inner and Outer V-Blade Inspection
Figure 605/78-31-01-990-828-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



A-A



B-B

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Thrust Reverser Inner and Outer V-Blade Inspection
Figure 605/78-31-01-990-828-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-31-01-211-804-G00

6. Thrust Reverser Outer V-Blade Seal Inspection

(Figure 606)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions for the inspection of the outer V-blade seal on the thrust reverser.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Inspection

SUBTASK 78-31-01-860-038-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 78-31-01-860-039-G00

(2) For Engine 1, open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	5	C00276	ENGINE 1 THRUST REVERSER CONT
B	6	C01412	ENGINE 1 THRUST REVERSER INTLK
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

SUBTASK 78-31-01-860-040-G00

(3) For engine 2, open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK
C	6	C01413	ENGINE 2 THRUST REVERSER INTLK
C	7	C00277	ENGINE 2 THRUST REVERSER CONT

SUBTASK 78-31-01-860-041-G00

(4) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 78-31-01-040-006-G00

(5) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 78-31-01-010-020-G00

(6) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

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WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. V-Blade Seal Inspection

SUBTASK 78-31-01-211-002-G00

(1) Do a detailed visual inspection of the thrust reverser outer V-blade seal for fretting and cracking.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-01-410-015-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 78-31-01-440-005-G00

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 78-31-01-860-042-G00

(3) For Engine 1, remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	5	C00276	ENGINE 1 THRUST REVERSER CONT
B	6	C01412	ENGINE 1 THRUST REVERSER INTLK
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

SUBTASK 78-31-01-860-062-G00

(4) For Engine 2, remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK
C	6	C01413	ENGINE 2 THRUST REVERSER INTLK
C	7	C00277	ENGINE 2 THRUST REVERSER CONT

SUBTASK 78-31-01-600-001-G00

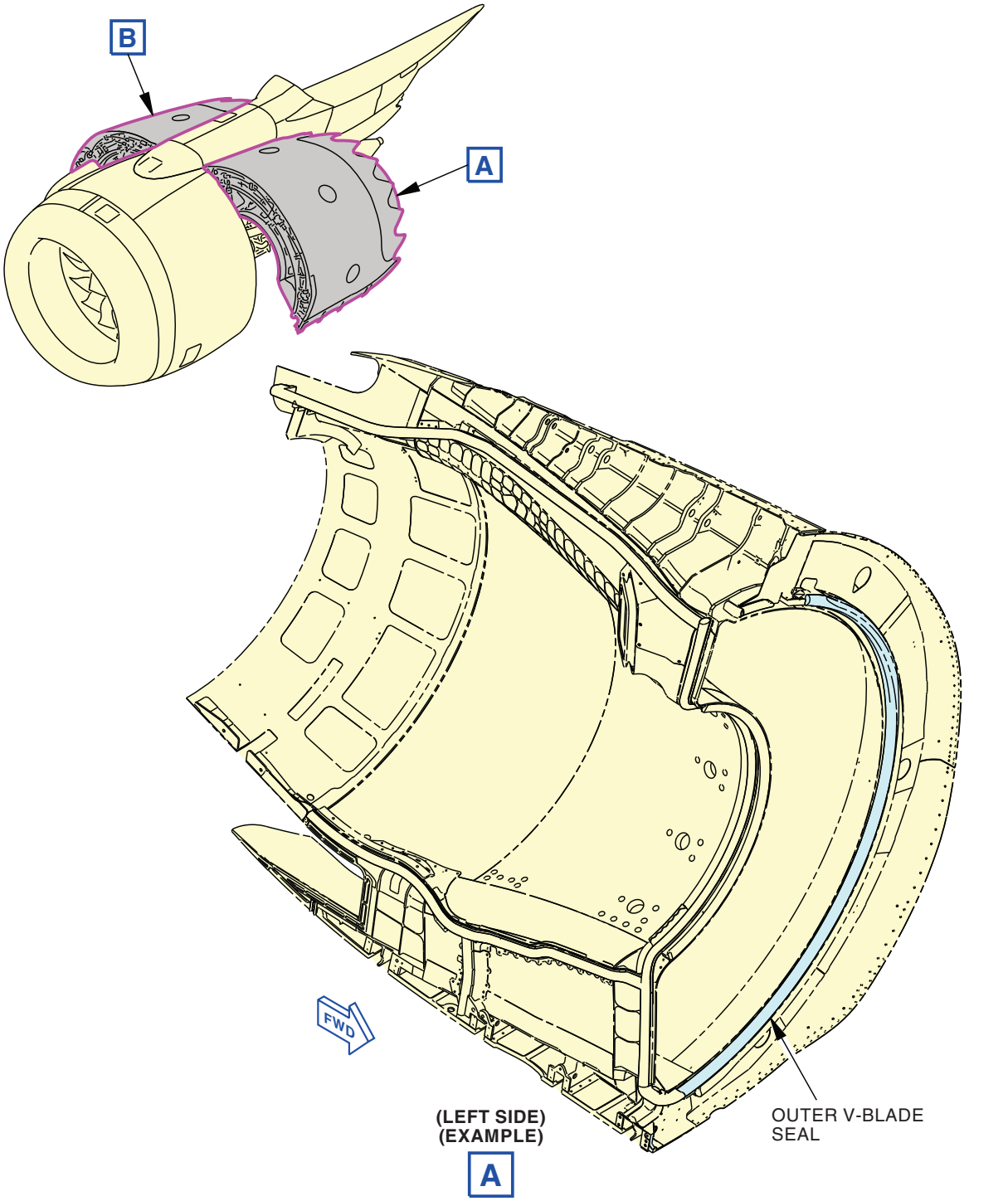
(5) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

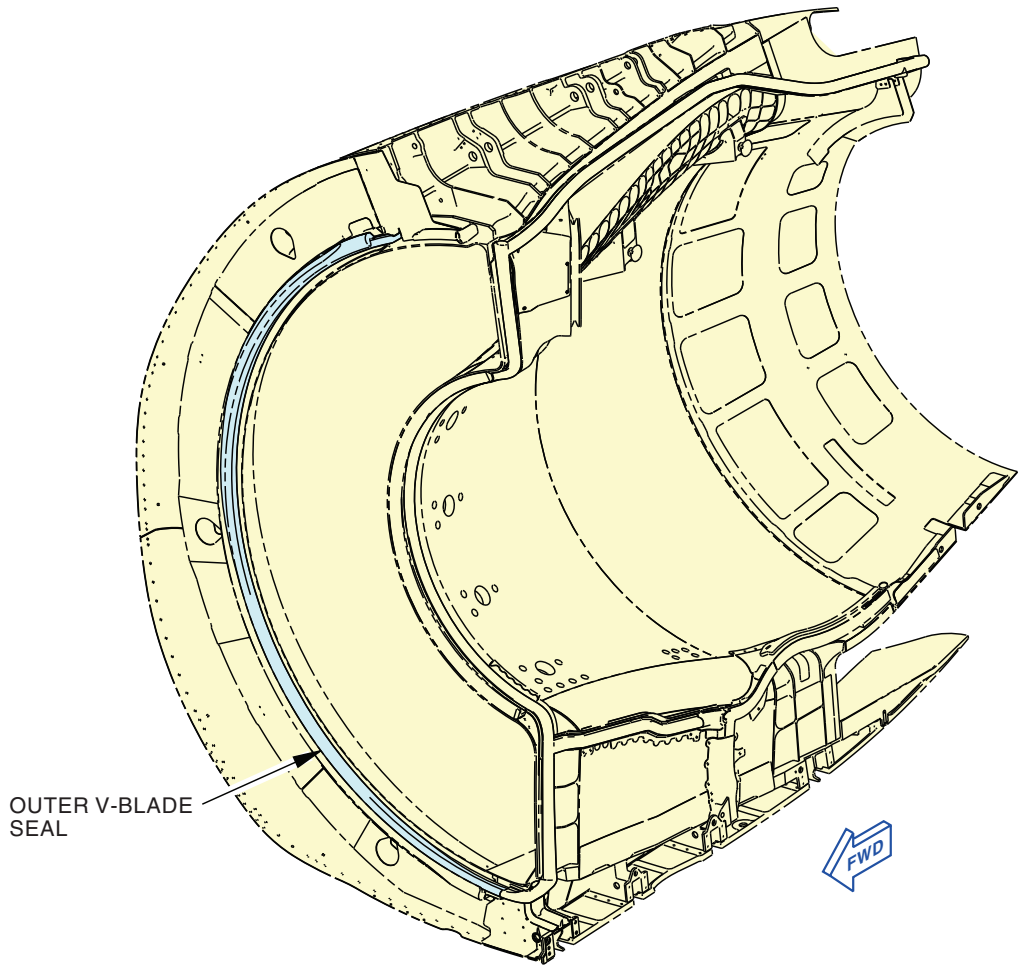
——— END OF TASK ———

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Thrust Reverser Outer V-Blade Seal Inspection
Figure 606/78-31-01-990-831-G00 (Sheet 1 of 2)

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(RIGHT SIDE)
(EXAMPLE)

B

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Thrust Reverser Outer V-Blade Seal Inspection
Figure 606/78-31-01-990-831-G00 (Sheet 2 of 2)

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TASK 78-31-01-211-805-G00

7. Thrust Reverser Outer V-Blade Inspection

(Figure 605)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions for the inspection of the outer V-blade on the thrust reverser.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Inspection

SUBTASK 78-31-01-860-043-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 78-31-01-860-044-G00

(2) For Engine 1, open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	5	C00276	ENGINE 1 THRUST REVERSER CONT
B	6	C01412	ENGINE 1 THRUST REVERSER INTLK
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

SUBTASK 78-31-01-860-045-G00

(3) For engine 2, open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK
C	6	C01413	ENGINE 2 THRUST REVERSER INTLK
C	7	C00277	ENGINE 2 THRUST REVERSER CONT

SUBTASK 78-31-01-860-046-G00

(4) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 78-31-01-040-007-G00

(5) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 78-31-01-010-021-G00

(6) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

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1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Outer V-Blade Inspection

SUBTASK 78-31-01-211-003-G00

(1) Do a detailed visual inspection of the thrust reverser outer V-blade for fretting and cracking.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-01-410-016-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 78-31-01-440-006-G00

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

SUBTASK 78-31-01-860-047-G00

(3) For Engine 1, remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	5	C00276	ENGINE 1 THRUST REVERSER CONT
B	6	C01412	ENGINE 1 THRUST REVERSER INTLK
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

For Engine 2, remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK
C	6	C01413	ENGINE 2 THRUST REVERSER INTLK
C	7	C00277	ENGINE 2 THRUST REVERSER CONT

SUBTASK 78-31-01-860-048-G00

(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

TASK 78-31-01-211-806-G00

8. Thrust Reverser Fire Seal Inspection

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions for the inspection of the fire seal on the thrust reverser.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

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E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Inspection

SUBTASK 78-31-01-860-049-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 78-31-01-860-050-G00

(2) For Engine 1, open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	5	C00276	ENGINE 1 THRUST REVERSER CONT
B	6	C01412	ENGINE 1 THRUST REVERSER INTLK
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

SUBTASK 78-31-01-860-051-G00

(3) For engine 2, open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK
C	6	C01413	ENGINE 2 THRUST REVERSER INTLK
C	7	C00277	ENGINE 2 THRUST REVERSER CONT

SUBTASK 78-31-01-860-052-G00

(4) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804

SUBTASK 78-31-01-040-008-G00

(5) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801

SUBTASK 78-31-01-010-022-G00

(6) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

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WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Thrust Reverser Fire Seal Inspection

SUBTASK 78-31-01-210-006-G00

(1) Do a general visual inspection of the thrust reverser fire seal for degradation.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-01-410-017-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

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1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 78-31-01-440-007-G00

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

SUBTASK 78-31-01-860-053-G00

(3) For Engine 1, remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	5	C00276	ENGINE 1 THRUST REVERSER CONT
B	6	C01412	ENGINE 1 THRUST REVERSER INTLK
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

For Engine 2, remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK
C	6	C01413	ENGINE 2 THRUST REVERSER INTLK
C	7	C00277	ENGINE 2 THRUST REVERSER CONT

SUBTASK 78-31-01-860-054-G00

(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

TASK 78-31-01-280-801-G00

9. Thrust Reverser Rigging Check

(Figure 607)

A. General

(1) This task gives the instructions on how to check the rigging of the thrust reverser after installation of an engine or thrust reverser.

(a) It is strongly recommended that this rig check be accomplished after engine replacement or after removal of one or both thrust reverser halves and re-installation of the same thrust reverser halves.

NOTE: The intent of the rigging check is to prevent excessive wear to the inner V-blade interface if the rigging gaps have changed from the original installation.

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(2) If one or both thrust reverser halves were replaced with different thrust reverser halves, do this task: Thrust Reverser Adjustment, TASK 78-31-01-820-801-G00.

B. References

Reference	Title
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-01-820-801-G00	Thrust Reverser Adjustment (P/B 501)
78-31-11-820-801-G00	Latches Adjustment (P/B 501)

C. Tools/Equipment

Reference	Description
STD-1097	Caliper - Vernier, 0-6 inch, Readable to 1/1000 inch
STD-1184	Scale - Spring, 0-100 Lbs, Tension

D. Consumable Materials

Reference	Description	Specification
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)
G02020	Clay, Modeling	
G02415	Agent - Parting, Paste Wax (Johnson's Paste Wax or equivalent)	
G50092	Compound - Metrology Casting Material - ReproRubber Quick Setting Putty	

E. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

F. Prepare for the Check

SUBTASK 78-31-01-010-026-G00

(1) Do this task: Open the Thrust Reverser (Selection), TASK 78-31-00-010-801-G00.

G. Rigging Check

SUBTASK 78-31-01-860-055-G00

(1) Do the V-blade rigging check.

(a) Put a piece of clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, that is coned shaped (approximately 0.20 in. (5.08 mm) in diameter by 0.15 in. (3.81 mm) thick) into the outer V-groove at the 45°, 90°, 135°, 225°, 270, and 315° radial positions.

NOTE: The top and bottom 10.5 in. (26.7 cm) approximately of the outer V-Blade is reduced in height. Do not apply clay there. If you do, the results will be in error.

(b) Put a piece of clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, that is coned shaped (approximately 0.20 in. (5.08 mm) in diameter by 0.25 in. (6.35 mm) thick) into the inner V-groove at the 30°, 90°, 150°, 210°, 270, and 330° radial positions.

(c) If you will use clay, G02020, apply paste wax parting agent, G02415, in the corresponding locations on the outer and inner V-blades of the thrust reverser.

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SUBTASK 78-31-01-860-056-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN AND CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Close and latch the thrust reverser but do not do the thrust reverser or leading edge activation at this time (TASK 78-31-00-010-802-G00).

(a) If you used ReproRubber Quick Setting Putty compound, G50092, make sure that you leave the thrust reverser closed for the manufacturer's recommended setting time.

SUBTASK 78-31-01-860-057-G00

(3) Do this task: Open the Thrust Reverser (Selection), TASK 78-31-00-010-801-G00.

SUBTASK 78-31-01-220-001-G00

(4) Measure the thickness of clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, on the inner V-groove.

(a) Remove clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, from the inner V-groove at the six locations.

(b) Measure the thickness of clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, with a vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

(c) Record the thickness of clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, from the inner V-groove and location from where it was taken.

(d) Make sure that the thickness at each location is as shown in the table below:

Table 601/78-31-01-993-808-G00 Clay Thicknesses

EDGE ZONE	SECTION	THICKNESS
[A]	B-B	minimum 0.020 in. (0.508 mm) not more 0.190 in. (4.826 mm)

SUBTASK 78-31-01-220-002-G00

(5) Measure the thickness of clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, on the outer V-groove.

(a) Remove the clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, from the outer V-groove at the six locations.

(b) Measure the thickness of each clay, G02020, or ReproRubber Quick Setting Putty compound, G50092.

(c) Record the thickness of each clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, from the outer V-groove and the location from where it was taken.

(d) Calculate the average thickness of the three pieces of clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, on the left thrust reverser half.

(e) Calculate the average thickness of the three pieces of clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, on the right thrust reverser half.

(f) Make sure that the average thickness (clearance) for each side is not more than 0.020 in. (0.508 mm).

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SUBTASK 78-31-01-160-004-G00

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH, EYES, OR SKIN. DO NOT BREATHE FUMES FROM SOLVENTS. REFER TO PRODUCT MATERIAL SAFETY DATA SHEETS (MSDS AND LOCAL REQUIREMENTS FOR PROPER HANDLING. SOLVENTS ARE HAZARDOUS MATERIALS AND MAY BE FLAMMABLE OR HARMFUL TO THE ENVIRONMENT.

(6) Remove all clay, G02020, or ReproRubber Quick Setting Putty compound, G50092, that remains in the inner and outer V-grooves with solvent, B00062.

SUBTASK 78-31-01-820-029-G00

(7) If the thrust reverser rig check is not in the limits, do this task: Thrust Reverser Adjustment, TASK 78-31-01-820-801-G00.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-01-410-021-G00

(1) Close the thrust reversers (TASK 78-31-00-010-802-G00).

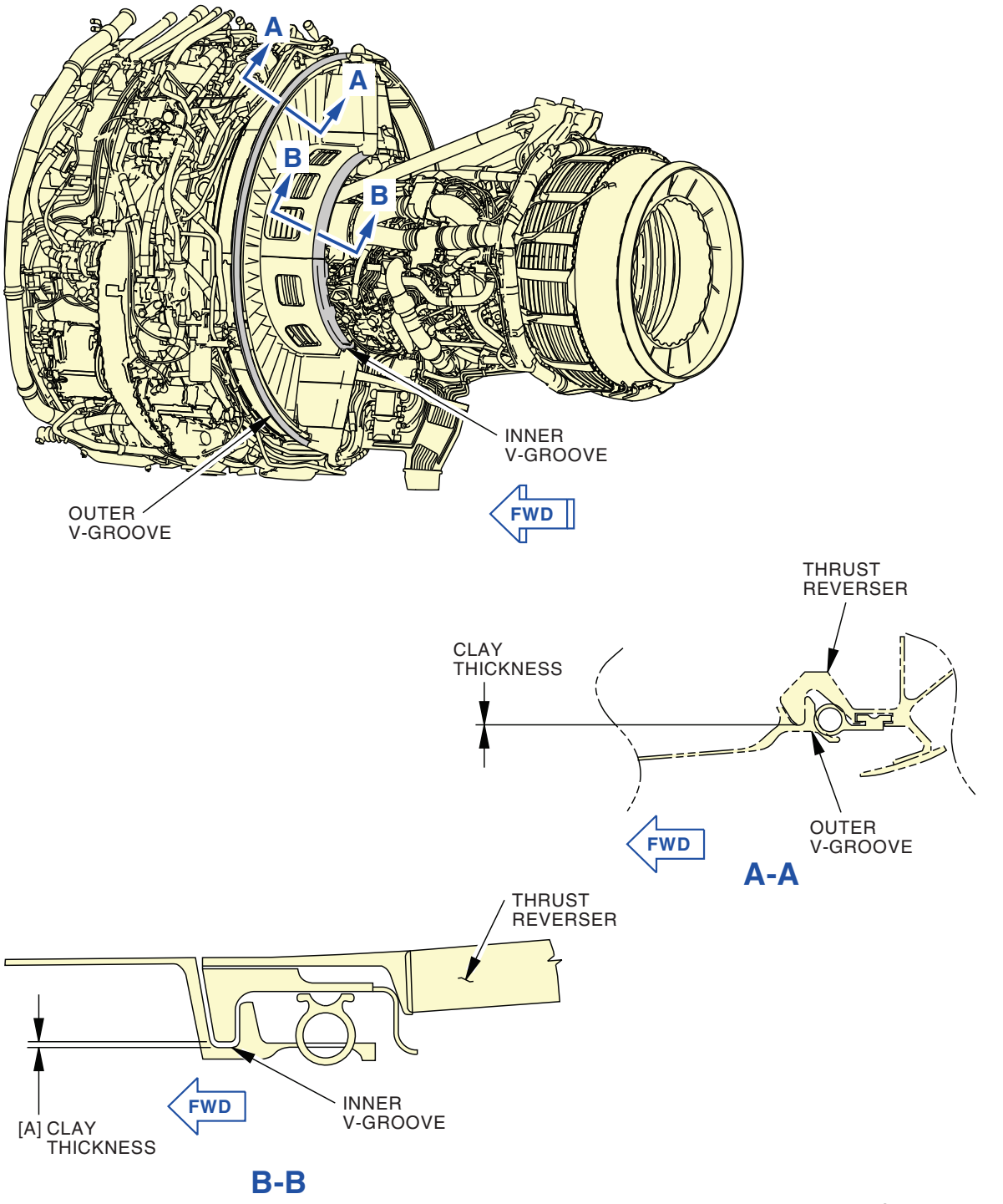
(a) Make sure that latches number 1 through 6, 8, and 9 on the thrust reverser have a closing force of 40 lbf (178 N) - 60 lbf (267 N) using a spring scale (0-100 Lbs), STD-1184.

(b) Make sure that latch number 7 on the thrust reverser aft cowl has a closing force of 40 lbf (178 N) - 75 lbf (334 N) using a spring scale (0-100 Lbs), STD-1184.

(c) If adjustments are necessary, do this task: Latches Adjustment, TASK 78-31-11-820-801-G00.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Thrust Reverser Rigging Check
Figure 607/78-31-01-990-823-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL

TASK 78-31-01-211-807-G00

10. Composite Inner Wall Inspection

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task is for a visual inspection of the composite inner wall of the left and right thrust reversers.

NOTE: See Thrust Reverser Inner Duct Wall Skin /SRM PROCEDURE 54-30-01-0I-3.

(2) The procedures are the same for the left and right thrust reversers.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-01 P/B 401	THRUST REVERSER - REMOVAL/INSTALLATION
78-31-13-000-801-G00	Insulation Blankets - Removal (P/B 401)
78-31-13-400-801-G00	Insulation Blankets - Installation (P/B 401)

C. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

D. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

E. Prepare for the Inspection

SUBTASK 78-31-01-010-027-G00

(1) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL (Continued)

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

SUBTASK 78-31-01-010-029-G00

(2) Remove the insulation blankets, upper thermal protection system (TPS) panel, and lower TPS panel (TASK 78-31-13-000-801-G00).

F. Composite Inner Wall Inspection

SUBTASK 78-31-01-211-004-G00

(1) Inspect the inner wall of the thrust reverser for damage:

NOTE: For allowable damage limits, see Thrust Reverser Composite Inner Duct Wall Skin Panel/SRM PROCEDURE 54-30-01-1A-6.

(a) Cracks

(b) Thermal degradation

(c) Delamination

(d) Disbond.

SUBTASK 78-31-01-300-001-G00

(2) If damage is found, see Thrust Reverser Composite Inner Duct Wall Acoustic Panel/SRM PROCEDURE 54-30-01-2R-6.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL (Continued)

SUBTASK 78-31-01-960-010-G00

(3) If the damage found is beyond repair limits, replace the thrust reverser: THRUST REVERSER - REMOVAL/INSTALLATION, PAGEBLOCK 78-31-01/401.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-01-410-026-G00

(1) Install the lower TPS panel, upper TPS panel, and insulation blankets (TASK 78-31-13-400-801-G00).

SUBTASK 78-31-01-410-028-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

END OF TASK

TASK 78-31-01-210-801-G00

11. Composite Inner Wall Inlet and Exhaust Ports

(Figure 608)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task is for a general visual inspection of the composite inner wall inlet and exhaust ports for the left and right thrust reversers.

(2) The procedures are the same for the left and the right thrust reversers.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL (Continued)

(Continued)

Reference	Title
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

D. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

E. Prepare for the Inspection

SUBTASK 78-31-01-010-028-G00

(1) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL (Continued)

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Composite Inner Wall Inlet and Exhaust Ports Inspection

SUBTASK 78-31-01-210-008-G00

(1) Examine the inlet ports for blockage, FOD, or unwanted material:

(a) Examine the inlet ports on the upstream side of the blocker door drag link anchor fittings (5 locations on each thrust reverser).

(b) Examine the inlet port (1 location) on the inner wall in the fan duct near the NO. 2 compression pad fitting on the lower bifurcation.

SUBTASK 78-31-01-210-009-G00

(2) Examine the exhaust and drain ports for blockage, FOD, or unwanted material.

NOTE: There are six exhaust ports on the vertical flange, forward of the insulation blankets, on each thrust reverser.

NOTE: There are two drain ports (that also operate as exhaust ports) on the bottom of each thrust reverser.

SUBTASK 78-31-01-100-001-G00

(3) Remove all blockage, FOD, or unwanted material from the inlet and exhaust ports.

SUBTASK 78-31-01-211-005-G00

(4) If you find damage, do this task: Composite Inner Wall Inspection, TASK 78-31-01-211-807-G00.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-01-410-027-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL (Continued)

(Continued)

Number	Name/Location
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

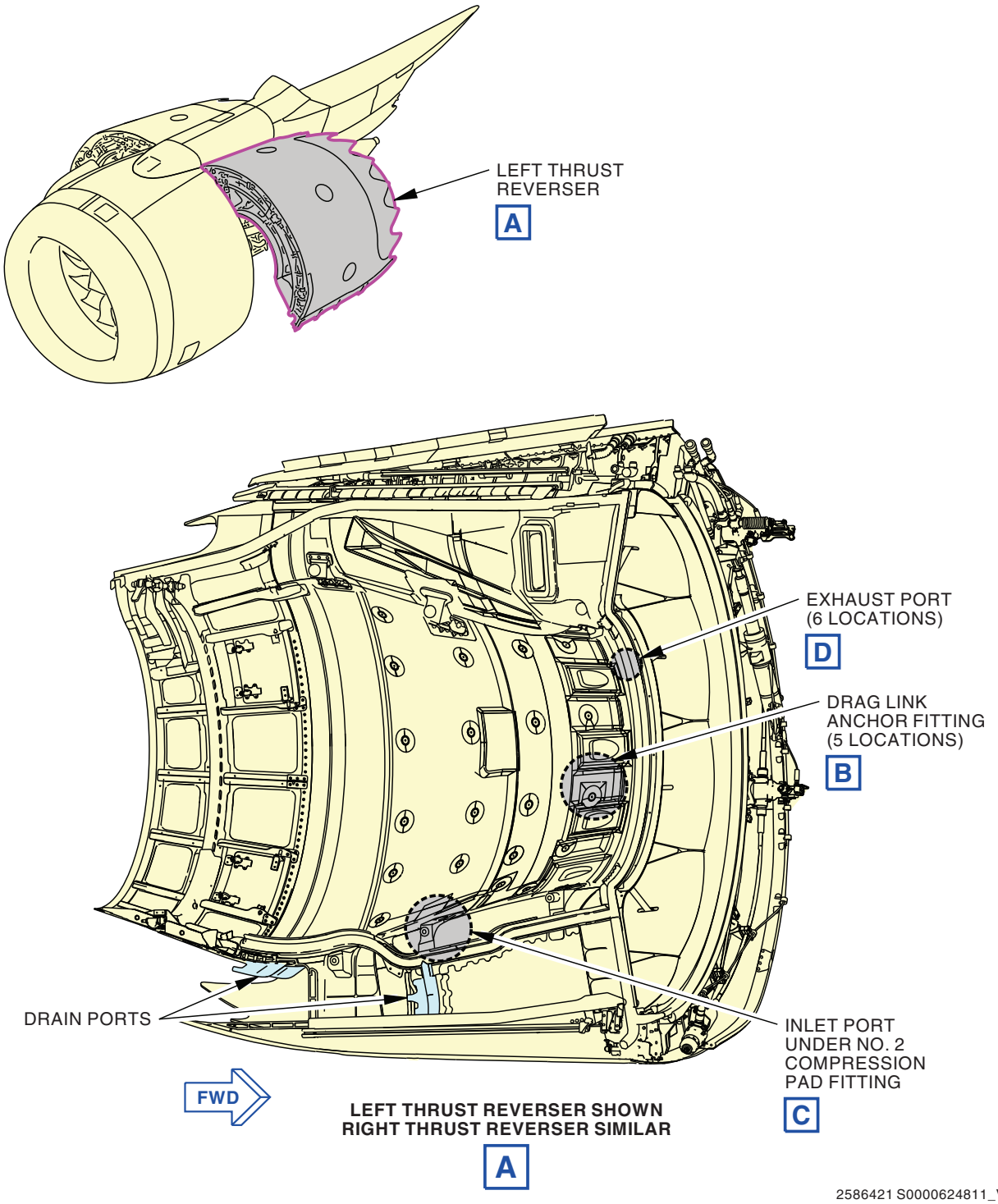
1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

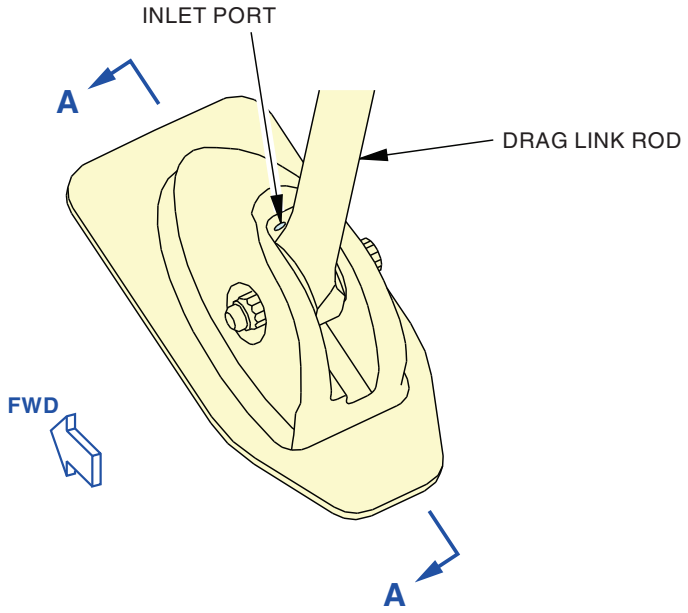
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



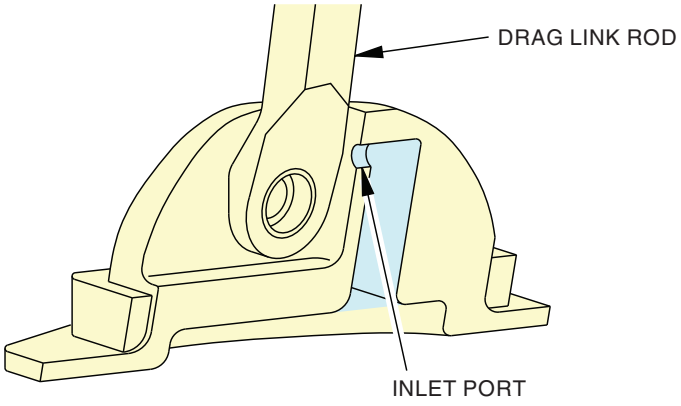
Composite Inner Wall Inlet and Exhaust Ports
Figure 608/78-31-01-990-826-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



DRAG LINK ANCHOR FITTING

B

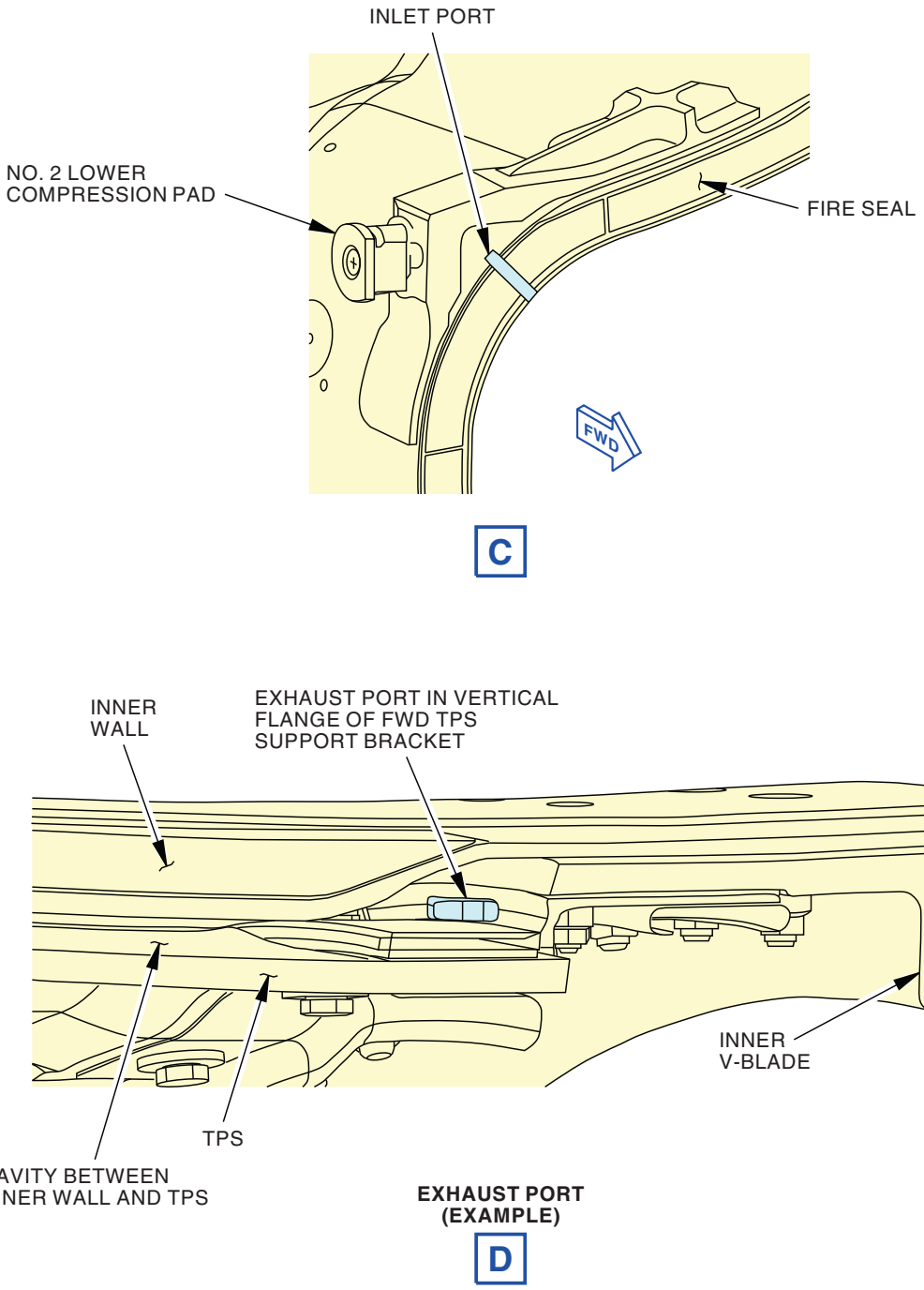


A-A

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Composite Inner Wall Inlet and Exhaust Ports
Figure 608/78-31-01-990-826-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Composite Inner Wall Inlet and Exhaust Ports
Figure 608/78-31-01-990-826-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

THRUST REVERSER- CLEANING/PAINTING

1. General

A. This procedure has these tasks:

(1) Thrust Reverser Witness Mark Installation

(2) Thrust Reverser Acoustic Surfaces Cleaning.

TASK 78-31-01-370-801-G00

2. Thrust Reverser Witness Mark Installation

(Figure 701)

A. General

(1) This task gives the instructions to paint a replacement witness mark on the inside surface of the thrust reverser duct.

B. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
71-00-00-700-816-G00	Test No. 38 - Thrust Reverser Cycle Special Functions (P/B 501)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-128	Brush - Paint

D. Consumable Materials

Reference	Description	Specification
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)
C00033	Coating - Protective Enamel, Flexibility Use	BMS10-60 Type II
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G00251	Abrasive - Mat, Non-Woven, Non-Metallic	A-A-58054
G00270	Tape - Scotch Flatback Masking 250	ASTM D6123 (Supersedes A-A-883)

E. Apply the Witness Mark

SUBTASK 78-31-01-860-012-G00

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Pressurize the applicable hydraulic system (TASK 29-11-00-860-801).

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SUBTASK 78-31-01-860-058-G00

(2) Start the thrust reverser cycle special function (TASK 71-00-00-700-816-G00).

SUBTASK 78-31-01-860-013-G00

(3) Slowly move the engine No. 1 (2) reverse thrust lever up and aft to the extended position.

(a) Make sure that the reverser sleeves on engine No. 1 (2) move to the fully extended position.

SUBTASK 78-31-01-860-014-G00

(4) Move the engine No. 1 (2) reverse thrust lever forward and down to the retracted position.

(a) Make sure that the reverser sleeves on engine No. 1 (2) move to the fully retracted position.

SUBTASK 78-31-01-040-009-G00

(5) Do the thrust reverser deactivation for ground maintenance (TASK 78-31-00-040-801-G00).

SUBTASK 78-31-01-370-001-G00

(6) Prepare the surface to be painted.

(a) Apply a mask around the witness mark area with Scotch Flatback Masking Tape 250, G00270 or equivalent adhesive masking tape.

(b) Roughen and activate the enamel paint on the surface with 400 grit, abrasive mat, G00251 or equivalent abrasive mat.

(c) Use a clean cloth cotton wiper, G00034 that is moist with solvent, B00062 to make the roughened surface clean.

(d) Wipe the surface dry with a dry and clean cotton wiper, G00034 before the solvent becomes dry.

SUBTASK 78-31-01-370-002-G00

(7) Immediately apply 1 coat of black enamel coating, C00033 in the area shown.

(a) Use a paint brush, STD-128 to apply a uniform coat.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-01-370-003-G00

(1) Remove the masking tape.

SUBTASK 78-31-01-440-008-G00

(2) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 78-31-01-860-059-G00

(3) Start the thrust reverser cycle special function (TASK 71-00-00-700-816-G00).

SUBTASK 78-31-01-860-015-G00

(4) Slowly move the engine No. 1 (2) reverse thrust lever up and aft to the extended position.

(a) Make sure that the reverser sleeves on engine No. 1 (2) move to the fully extended position.

SUBTASK 78-31-01-860-016-G00

(5) Move the reverse thrust lever forward and down to the retracted position.

(a) Make sure that the reverser sleeves on engine No. 1 (2) move to the fully retracted position.

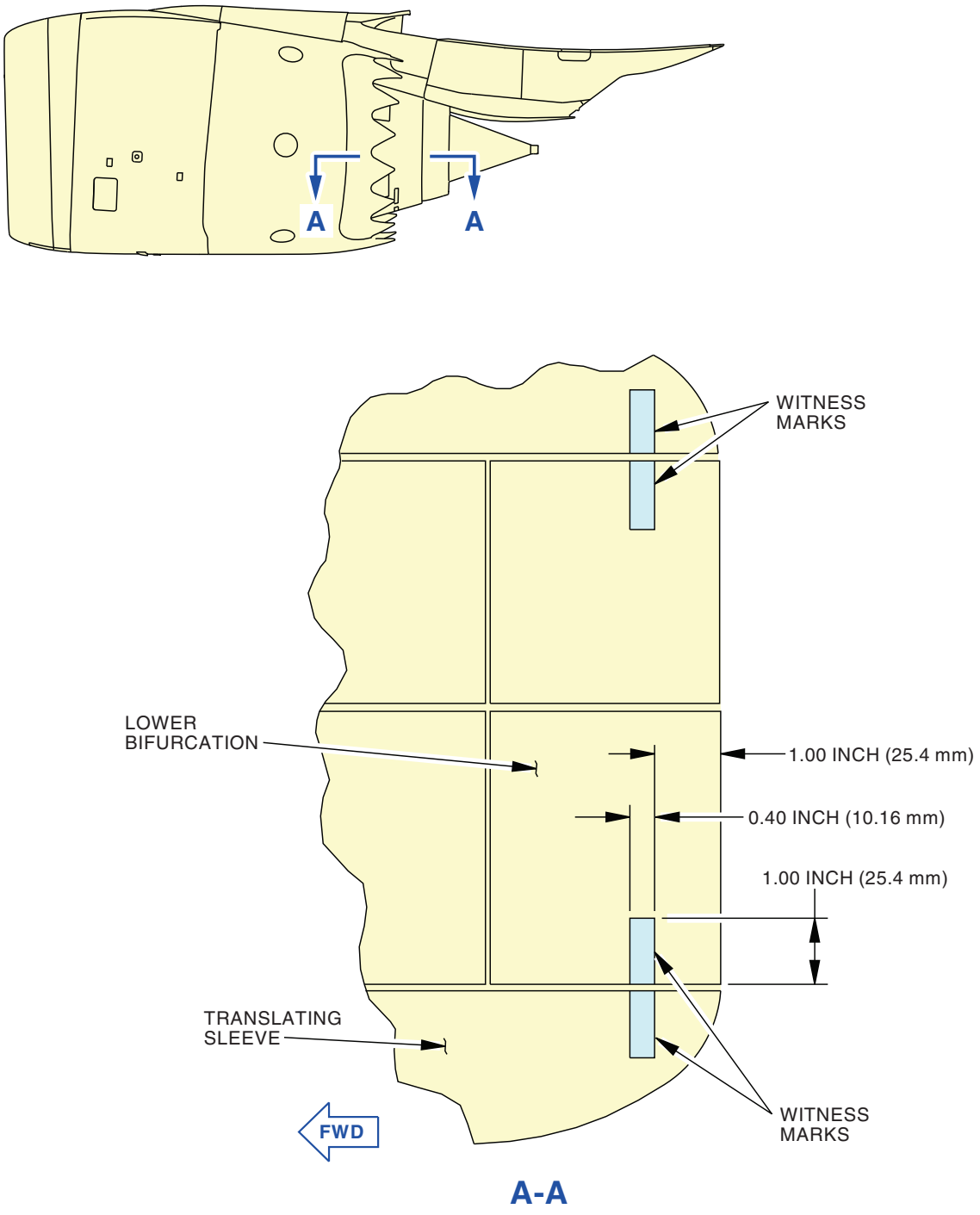
ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 78-31-01-860-017-G00

(6) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Thrust Reverser Witness Mark
Figure 701/78-31-01-990-812-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-31-01-160-801-G00

3. Thrust Reverser Acoustic Surfaces Cleaning

A. General

(1) This task gives the instructions to clean the acoustic panels of the thrust reverser.

(2) Special attention should be given to prevent water and cleaning agents from entering the cells of the acoustic panels. Follow the steps below to minimize the ingestion of cleaning solutions and water into the acoustic panels.

(3) Use only cleaning agents certified for use in cleaning the acoustic surfaces.

(4) If it is necessary, contact your maintenance or Boeing to make sure that the cleaning agent or solvent is certified for use.

B. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

C. Consumable Materials

Reference	Description	Specification
B50135	Solvent - 1,1,1 Trichloroethane	
G50933	Cloth - Cleaning, Low-Lint Cloth (General Use)	A-A-59323 Type II (Supersedes MIL-C-85043)

D. Prepare for the Cleaning

SUBTASK 78-31-01-860-018-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

E. Thrust Reverser Acoustic Surfaces Cleaning

SUBTASK 78-31-01-110-001-G00



CAUTION

DO NOT LET WATER AND CLEANING AGENTS STAY IN THE PANEL. WATER AND CLEANING AGENTS CAN CAUSE CORROSION OF THE ALUMINUM HONEYCOMB CORE.

(1) Wipe the cleaning solution on with a low-lint cleaning cloth, G50933.

NOTE: The cloth must be damp but not saturated with the cleaning solution.

SUBTASK 78-31-01-110-002-G00

(2) Wipe with a low-lint cleaning cloth, G50933 damp with water to remove all residues.

SUBTASK 78-31-01-160-001-G00

(3) Use a clean dry cloth to wipe and remove all moisture.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 78-31-01-950-001-G00

CAUTION

DO NOT USE A SPRAY WITH STRONG PRESSURE ON SKIN AREAS WITH HOLES OR PUT THEM IN SOLVENTS OR WATER. WATER, CLEANING AGENTS OR SOLVENTS CAN CAUSE CORROSION.

(4) If you will pressure wash components around acoustic panels, do the applicable step that follows:

NOTE: This will prevent ingestion of the cleaning solution into acoustic material.

(a) If the thrust reverser contains a composite inner wall, you must cover all the acoustic portions of the nacelle.

(b) If the thrust reverser contains a titanium inner wall, you must cover all the acoustic portions of the nacelle except the titanium inner wall.

SUBTASK 78-31-01-110-005-G00

CAUTION

DO NOT VAPOR-DEGREASE GRAPHITE/ARAMID EPOXY STRUCTURES WITH CHLORINATED AGENTS. DAMAGE TO THE STRUCTURES CAN OCCUR.

CAUTION

DO NOT PUT THE PARTS IN THE SOLVENT OR LET IT SOAK IN THE SOLVENT. DAMAGE TO THE PART CAN OCCUR.

(5) Use 1,1,1-Trichloroethane solvent, B50135 only as a wipe solvent.

NOTE: Trichloroethane solvent is one of the solvents approved to wipe clean composite components.

NOTE: Methylene Chloride, Trichoroethylene and Trichloroethane are chlorinated agents that you cannot use to vapor-degrease graphite/aramid epoxy structures.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-01-860-019-G00

(1) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

THRUST REVERSER - REPAIRS

1. General

A. This procedure has these tasks.

(1) Thrust reverser hinge beam fire wall replacement.

(2) Blocker door support assembly replacement.

PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL

(3) Through stud replacement.

PTF ALL

TASK 78-31-01-960-801-G00

2. Thrust Reverser Hinge Beam Fire Wall Replacement

(Figure 801)

A. General

(1) This task gives the instructions to replace the fire wall plate on the thrust reverser hinge beam.

(2) The fire wall is attached to the aluminum hinge beam of the thrust reverser. The upper flange of the firewall can become worn which would allow a breach of the engine fan compartment fire protection features.

(3) The fire wall helps to contain an engine fan case compartment fire from the components on the fan case mounted gearbox.

B. References

Reference	Title
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-768	Sealant Removal Tool, Non-Metallic Part #: ST982L-9 Supplier: 81205 Part #: ST982LF-9 Supplier: 81205 Part #: ST982LH-A-1 Supplier: 81205
STD-764	Scraper - Non-metallic

D. Consumable Materials

Reference	Description	Specification
A00160	Sealant - Firewall - Hydraulic Fluid Resistant	BMS5-63
C00304	Coating - Teflon Filled, Non Decorative, Sprayable Material	BMS10-86 Type I

E. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Zone	Area
426	Engine 2 - Thrust Reverser, Right

F. Replace the Firewall

SUBTASK 78-31-01-010-010-G00

(1) Open the thrust reverser.

WARNING OBEY THE INSTRUCTIONS IN THIS PROCEDURE WHEN YOU OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

(a) Do this task: Open the Thrust Reverser (Selection), TASK 78-31-00-010-801-G00.

SUBTASK 78-31-01-020-009-G00

(2) Remove the left firewall [1].

(a) Remove the two bolts [3] and nuts [4] that attach the left firewall [1] to the seal retainer [2].

(b) Remove the 19 bolts [6], washers [7] and nuts [8] that attach the left firewall [1] to the hinge beam.

(c) Remove the five bolts [9], washers [10] and nuts [11] that attach the seal retainer [5] and the left firewall [1] to the hinge beam.

(d) Use an non-metallic scraper, STD-764 to pry the fire wall from the hinge beam and remove the firewall.

NOTE: The fire wall was installed with fire wall sealant, between the fire wall and the hinge beam flange.

1) Do not scratch or damage the surface of the metal hinge beam.

SUBTASK 78-31-01-020-010-G00

(3) Remove the right firewall [12].

(a) Remove the two bolts [14] and nuts [15] that attach the right firewall [12] to the seal retainer [13].

(b) Remove the 19 bolts [17], washers [18] and nuts [19] that attach the right firewall [12] to the hinge beam.

(c) Remove the five bolts [20], washers [21] and nuts [22] that attach the seal retainer [16] and the right firewall [12] to the hinge beam.

(d) Use an non-metallic scraper, STD-764 to pry the fire wall from the hinge beam and remove the firewall.

NOTE: The fire wall was installed with fire wall sealant, between the fire wall and the hinge beam flange.

1) Do not scratch or damage the surface of the metal hinge beam.

SUBTASK 78-31-01-160-002-G00

(4) Clean the surface of the hinge beam with sealant removal tool, SPL-768 to remove old firewall sealant.

(a) Do not scratch or damage the surface of the metal hinge beam.

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SUBTASK 78-31-01-350-001-G00

(5) Apply fire wall sealant, A00160 to the forward surface of the hinge beam and to the mating surface of the fire wall.

(a) Prepare the sealant with the manufacturers instructions.

SUBTASK 78-31-01-390-001-G00

(6) Apply fire wall sealant, A00160 to the mating surface of the fire wall and the seal retainer.

(a) Prepare the sealant with the manufacturers instructions.

SUBTASK 78-31-01-420-004-G00

(7) Install the left firewall [1].

(a) Position the fire wall on the forward flange of the hinge beam.

(b) Apply fire wall sealant, A00160 to the fasteners and immediately install the fasteners with wet sealant.

(c) Install the 19 bolts [6], washers [7] and nuts [8] that attach the left firewall [1] to the hinge beam.

(d) Apply a Teflon coating, C00304 to the heads of the fasteners.

(e) Install the five bolts [9], washers [10] and nuts [11] that attach the seal retainer [5] and the left firewall [1] to the hinge beam.

(f) Install the two bolts [3] and nuts [4] that attach the left firewall [1] to the seal retainer [2].

(g) Tighten all the fasteners to 15 in-lb (2 N·m)-25 in-lb (3 N·m).

SUBTASK 78-31-01-420-005-G00

(8) Install the right firewall [12].

(a) Position the fire wall on the forward flange of the hinge beam.

(b) Apply fire wall sealant, A00160 to the fasteners and immediately install the fasteners with wet sealant.

(c) Install the 19 bolts [17], washers [18] and nuts [19] that attach the right firewall [12] to the hinge beam.

(d) Apply a Teflon coating, C00304 to the heads of the fasteners.

(e) Install the five bolts [20], washers [21] and nuts [22] that attach the seal retainer [16] and the right firewall [12] to the hinge beam.

(f) Install the two bolts [14] and nuts [15] that attach the right firewall [12] to the seal retainer [13].

(g) Tighten all the fasteners to 15 in-lb (2 N·m)-25 in-lb (3 N·m).

SUBTASK 78-31-01-410-008-G00

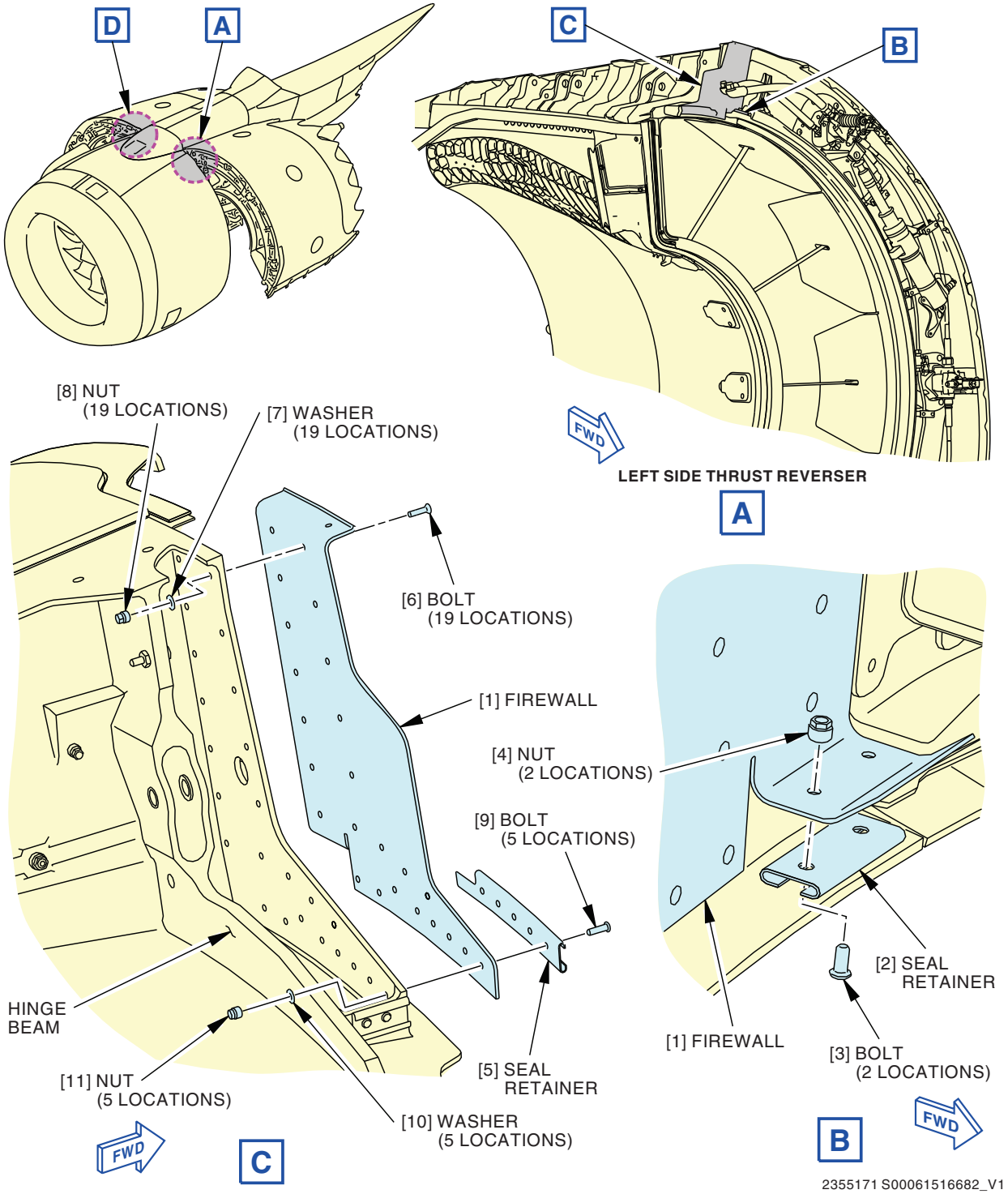
(9) Close the thrust reverser.

WARNING OBEY THE INSTRUCTIONS IN THIS PROCEDURE WHEN YOU CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

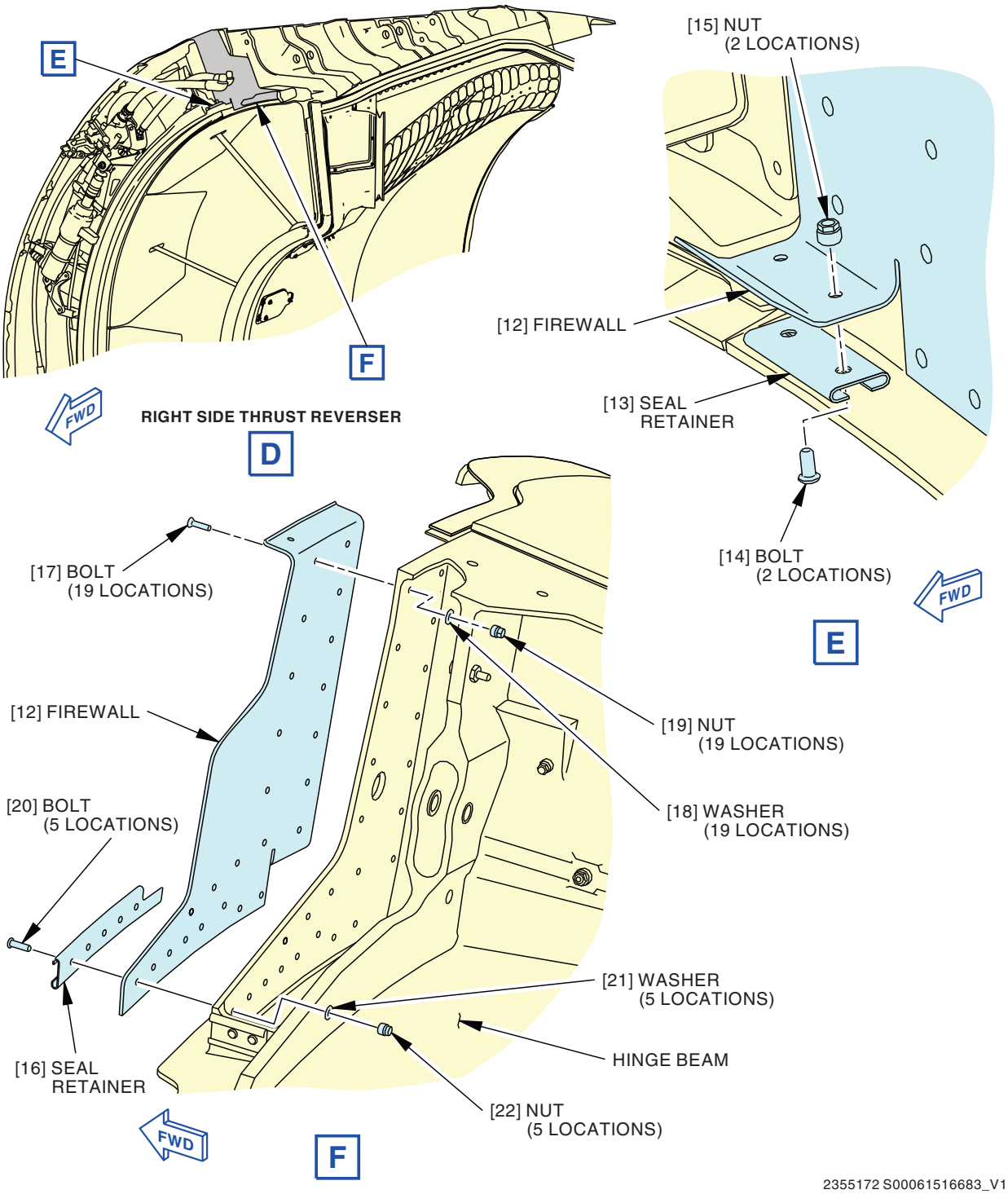
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Hinge Beam Fire Wall
Figure 801/78-31-01-990-813-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Hinge Beam Fire Wall
Figure 801/78-31-01-990-813-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-31-01-960-802-G00

3. Blocker Door Support Assembly Replacement

(Figure 802)

A. General

(1) This task gives the instructions to replace the blocker door bumpers on the support assemblies.

(2) The blocker door support assemblies are the two bumpers that are attached to the translating sleeve below the edge of each blocker door.

(3) For access to the blocker door support assemblies, the thrust reverser translating sleeve must be extended or the blocker door must be removed.

B. References

Reference	Title
78-31-00-980-801-G00	Thrust Reverser Operation - Extend (Selection) (P/B 201)
78-31-00-980-802-G00	Thrust Reverser Operation - Retract (Selection) (P/B 201)
78-31-06-000-801-G00	Blocker Door Removal (P/B 401)
78-31-06-400-801-G00	Blocker Door Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
A00767	Sealant - Fuel Tank	BMS5-45

D. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

E. Prepare for the Repair

SUBTASK 78-31-01-010-011-G00

(1) For access to the support assemblies, do one of these tasks:

(a) Blocker Door Removal, TASK 78-31-06-000-801-G00

(b) Thrust Reverser Operation - Extend (Selection), TASK 78-31-00-980-801-G00

F. Replace the Support Assemblies

SUBTASK 78-31-01-020-011-G00

(1) Remove the support assembly as follow:

(a) Remove the screw [25] that attaches the pad [28], bumper assembly [27], and washer [26] to the support bracket.

(b) Remove the pad [28], bumper assembly [27], and washer [26] from the support bracket.

SUBTASK 78-31-01-420-006-G00

(2) Install a new support assembly as follow:

(a) Put the new pad [28], bumper assembly [27], and washer [26] in the correct position on the support bracket.

1) Install the washers [26] as necessary to make the outside surfaces of the blocker door flush with the surface of the bond assembly within 0.004 in. (0.102 mm).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Make sure the total thickness of the stack of washers [26] is less than 0.158 in. (4.013 mm).

(b) Wet install the screw [25] with sealant, A00767.

1) Tighten the screw [25] to 10 in-lb (1.1 N·m) to 20 in-lb (2.3 N·m) so the pad compresses at least 0.02 in. (0.51 mm) but not more than 0.06 in. (1.52 mm).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-01-410-009-G00

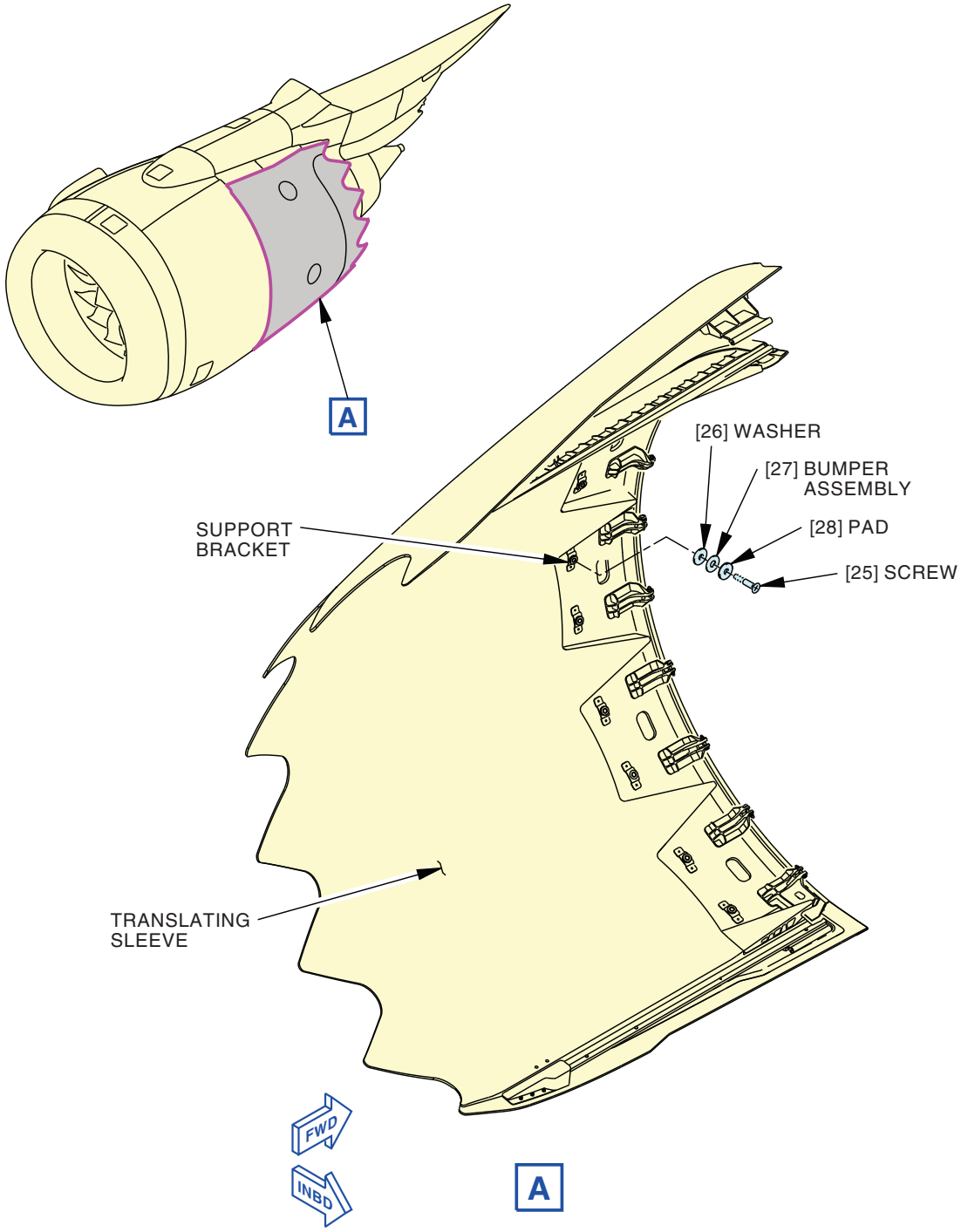
(1) If it is necessary, install the blocker door (TASK 78-31-06-400-801-G00).

SUBTASK 78-31-01-910-001-G00

(2) If it is necessary, retract the thrust reverser (TASK 78-31-00-980-802-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Blocker Door Support Assembly Replacement
Figure 802/78-31-01-990-821-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL

TASK 78-31-01-960-803-G00

4. Through Stud Replacement

Figure 803

A. General

(1) This procedure gives the on-wing instructions to remove and replace damaged or cross-threaded through studs on the thrust reversers.

(a) The through studs are part of the fasteners that hold the thermal insulation blankets to the inner wall of the thrust reverser.

(b) The procedures are the same for the left and right thrust reversers.

B. References

Reference	Title
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)

C. Consumable Materials

Reference	Description	Specification
A00160	Sealant - Firewall - Hydraulic Fluid Resistant	BMS5-63
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)
B00130	Alcohol - Isopropyl	TT-I-735
B00148	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
D00006	Compound - Antiseize, Pure Nickel Special - Never-Seez NSBT-8N/-16N	
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

D. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

E. Prepare for the Replacement

SUBTASK 78-31-01-010-023-G00

WARNING

DO THESE SPECIFIED TASKS IN THE CORRECT SEQUENCE BEFORE YOU OPEN THE THRUST REVERSERS: RETRACT THE LEADING EDGE, DO THE DEACTIVATION PROCEDURES FOR THE LEADING EDGE AND THE THRUST REVERSER (FOR GROUND MAINTENANCE), AND OPEN THE FAN COWL PANELS. IF YOU DO NOT OBEY THE ABOVE SEQUENCE, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do this task: Open the Thrust Reverser (Selection), TASK 78-31-00-010-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL (Continued)

F. Through Stud Replacement Procedure

SUBTASK 78-31-01-960-009-G00

(1) Remove the applicable damaged through stud [1], or through stud [13] or through stud [14] or through stud [15] or through stud [16] or through stud [17] from the applicable inner wall [9] (Preferred Method), as follows:

(a) Unfold the flaps of the fastener cap cover.

(b) Remove the nut [4], washer [6], fastener cap [5], TPS nut [3], and the spring washer [2], if installed, from the applicable through stud.

(c) Do not remove the washer [6] that contacts the inner wall, if installed.

(d) Remove the applicable through stud.

SUBTASK 78-31-01-960-002-G00

(2) Clean the surface of the inner wall.

(a) Wipe or scrub the surface of the inner wall to remove any loose soil, oil, grease, or hydraulic fluid with a clean cotton wiper, G00034.

CAUTION

DO NOT DAMAGE ANY UNDERLYING STRUCTURAL PLIES.

(b) Sand mating surfaces with 180 grit or finer aluminum oxide silicone free sand paper until all sealant and surface gloss is removed.

(c) Wet a clean cotton wiper, G00034 with alcohol, B00130 or solvent, B00062 or solvent, B00148 and wipe the surface of the inner wall.

1) Remove all excess solvent or alcohol with a cloth.

2) Continue to wipe the surface with a new cloth and solvent until the cloth remains clean.

(3) Inspect the inner wall [9] for any damage from the removal procedure.

SUBTASK 78-31-01-960-003-G00

(4) Install the replacement through stud [1] or through stud [18], as follows:

(a) Apply a continuous and uniform layer of sealant, A00160 to the base of the through stud [1] or through stud [18].

(b) Install the through stud [1] or through stud [18], in the inner wall [9] and the washer [6].

(c) Apply a continuous and uniform layer of Pure Nickel Special compound, D00006 to the threads of the through stud [1] or through stud [18].

(d) Install the spring washer [2] and the TPS nut [3].

(e) Tighten the TPS nut [3] 25 in-lb (2.8 N·m) to 35 in-lb (4.0 N·m).

(f) Install the fastener cap [5], washer [6] and the nut [4] on the applicable TPS panel. Orient hinged edge of the fastener cap [5] facing the forward direction.

(g) Tighten the nut [4] 20 in-lb (2.3 N·m) to 22 in-lb (2.5 N·m).

(h) Once final nut torque is met, fold cover over fastener cap [5] and firmly fold flaps to secure cover.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL (Continued)

SUBTASK 78-31-01-960-004-G00

(5) Install the replacement through stud [13] or through stud [19], as follows:

(a) Apply a continuous and uniform layer of sealant, A00160 to the base of the through stud [13] or through stud [19].

(b) Install the through stud [13] or through stud [19], in the inner wall [9] and the washer [6].

(c) Apply a continuous and uniform layer of Pure Nickel Special compound, D00006 to the threads of the through stud [13] or through stud [19].

(d) Install the spring washer [2] and the TPS nut [3].

(e) Tighten the TPS nut [3] 25 in-lb (2.8 N·m) to 35 in-lb (4.0 N·m).

(f) Install the fastener cap [5], washer [6] and the nut [4] on the applicable TPS panel. Orient hinged edge of the fastener cap [5] facing the forward direction.

(g) Tighten the nut [4] 20 in-lb (2.3 N·m) to 22 in-lb (2.5 N·m).

(h) Once final nut torque is met, fold cover over fastener cap [5] and firmly fold flaps to secure cover.

SUBTASK 78-31-01-960-005-G00

(6) Install the replacement through stud [14], as follows:

(a) Apply a continuous and uniform layer of sealant, A00160 to the base of the through stud [14].

(b) Install the through stud [14], in the inner wall [9] and the TPS precooler seal land [11].

(c) Apply a continuous and uniform layer of Pure Nickel Special compound, D00006 to the threads of the through stud [14].

(d) Install the spring washer [2] and the TPS nut [3].

(e) Tighten the TPS nut [3] 25 in-lb (2.8 N·m) to 35 in-lb (4.0 N·m).

(f) Install the fastener cap [5], washer [6] and the nut [4] on the applicable TPS panel. Orient hinged edge of the fastener cap [5] facing the forward direction.

(g) Tighten the nut [4] 20 in-lb (2.3 N·m) to 22 in-lb (2.5 N·m).

(h) Once final nut torque is met, fold cover over fastener cap [5] and firmly fold flaps to secure cover.

SUBTASK 78-31-01-960-006-G00

(7) Install the replacement through stud [15], as follows:

(a) Apply a continuous and uniform layer of sealant, A00160 to the base of the through stud [15].

(b) Install the through stud [15], in the inner wall [9] and the TPS precooler seal land [11].

(c) Apply a continuous and uniform layer of Pure Nickel Special compound, D00006 to the threads of the through stud [15].

(d) Install the TPS nut [3].

(e) Tighten the TPS nut [3] 25 in-lb (2.8 N·m) to 35 in-lb (4.0 N·m).

(f) Install the fastener cap [5], washer [6] and the nut [4] on the applicable TPS panel. Orient hinged edge of the fastener cap [5] facing the forward direction.

(g) Tighten the nut [4] 20 in-lb (2.3 N·m) to 22 in-lb (2.5 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL (Continued)

(h) Once final nut torque is met, fold cover over fastener cap [5] and firmly fold flaps to secure cover.

SUBTASK 78-31-01-960-007-G00

(8) Install the replacement through stud [16], as follows:

(a) Apply a continuous and uniform layer of sealant, A00160 to the base of the through stud [16].

(b) Install the through stud [16], in the inner wall [9], the wear pad [12] and the splice strap [20].

(c) Apply a continuous and uniform layer of Pure Nickel Special compound, D00006 to the threads of the through stud [16].

(d) Install the spring washer [2] and the TPS nut [3].

(e) Tighten the TPS nut [3] 25 in-lb (2.8 N·m) to 35 in-lb (4.0 N·m).

(f) Install the fastener cap [5], washer [6] and the nut [4] on the applicable TPS panel. Orient hinged edge of the fastener cap [5] facing the forward direction.

(g) Tighten the nut [4] 20 in-lb (2.3 N·m) to 22 in-lb (2.5 N·m).

(h) Once final nut torque is met, fold cover over fastener cap [5] and firmly fold flaps to secure cover.

SUBTASK 78-31-01-960-008-G00

(9) Install the replacement through stud [17], as follows:

(a) Apply a continuous and uniform layer of sealant, A00160 to the base of the through stud [17].

(b) Install the through stud [17], in the inner wall [9], the wear pad [12] and the splice strap [20].

(c) Apply a continuous and uniform layer of Pure Nickel Special compound, D00006 to the threads of the through stud [17].

(d) Install the spring washer [2] and the TPS nut [3].

(e) Tighten the TPS nut [3] 25 in-lb (2.8 N·m) to 35 in-lb (4.0 N·m).

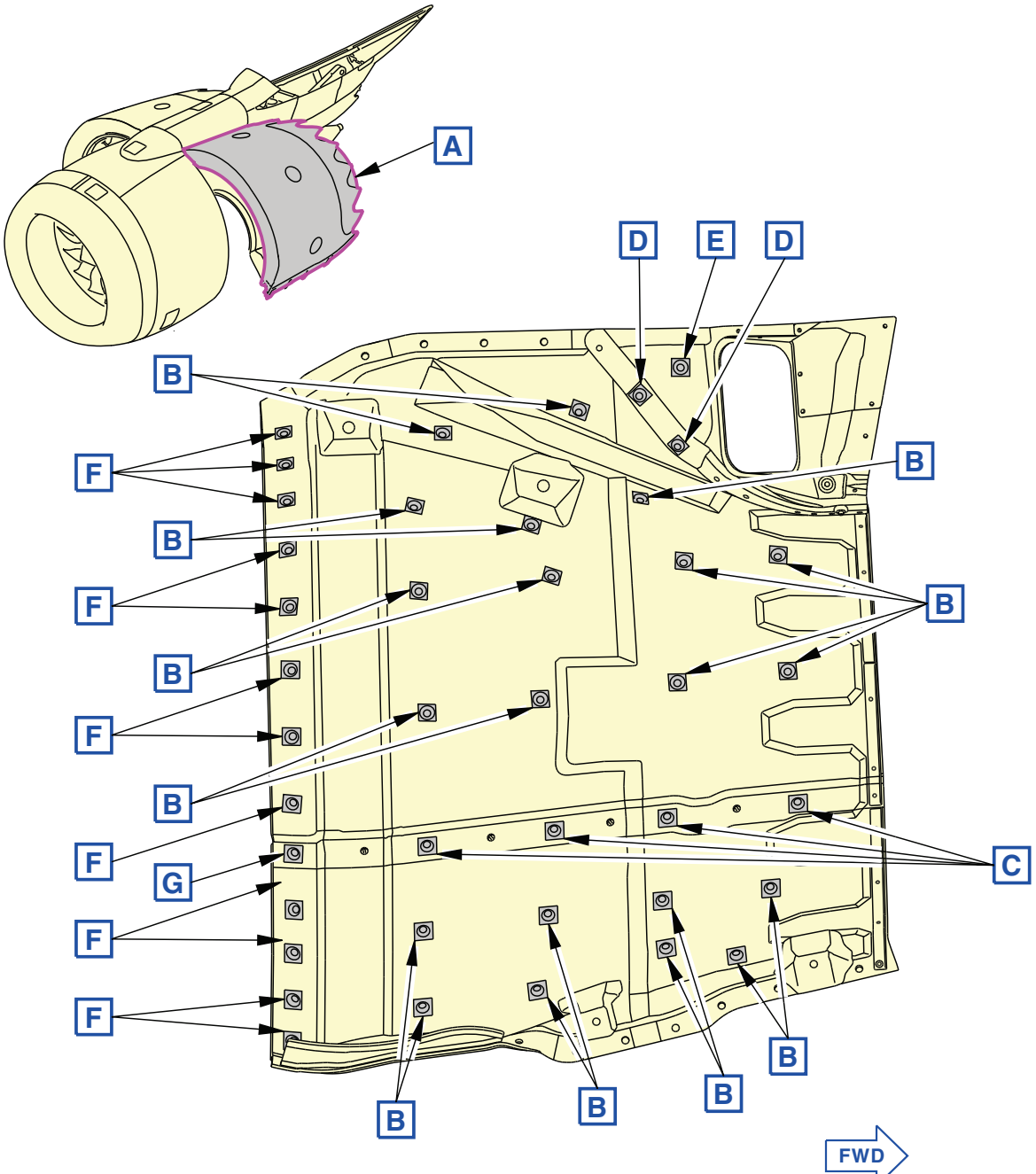
(f) Install the fastener cap [5], washer [6] and the nut [4] on the applicable TPS panel. Orient hinged edge of the fastener cap [5] facing the forward direction.

(g) Tighten the nut [4] 20 in-lb (2.3 N·m) to 22 in-lb (2.5 N·m).

(h) Once final nut torque is met, fold cover over fastener cap [5] and firmly fold flaps to secure cover.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



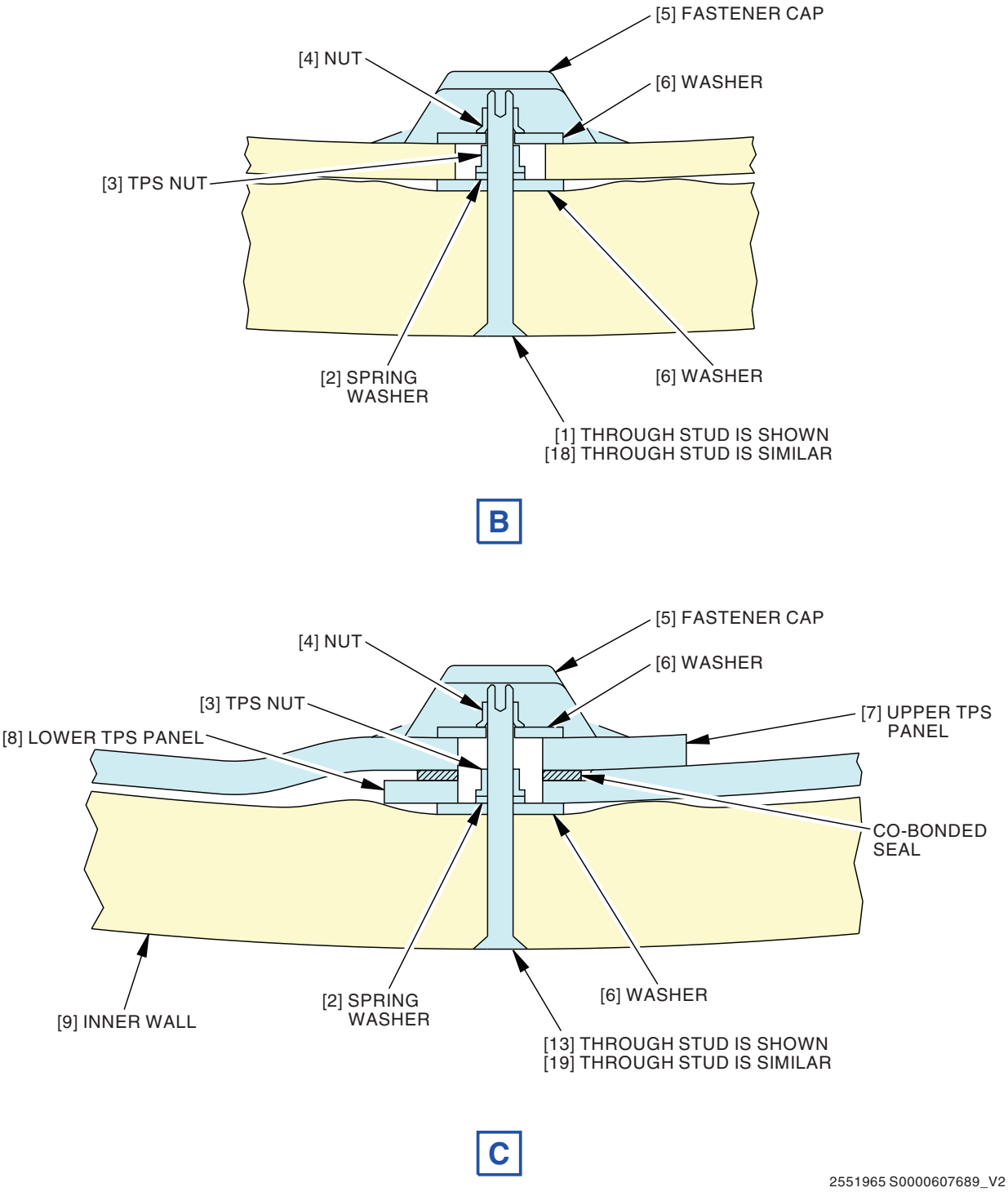
LEFT THRUST REVERSER
(RIGHT THRUST REVERSER IS OPPOSITE)
SIDE VIEW

A

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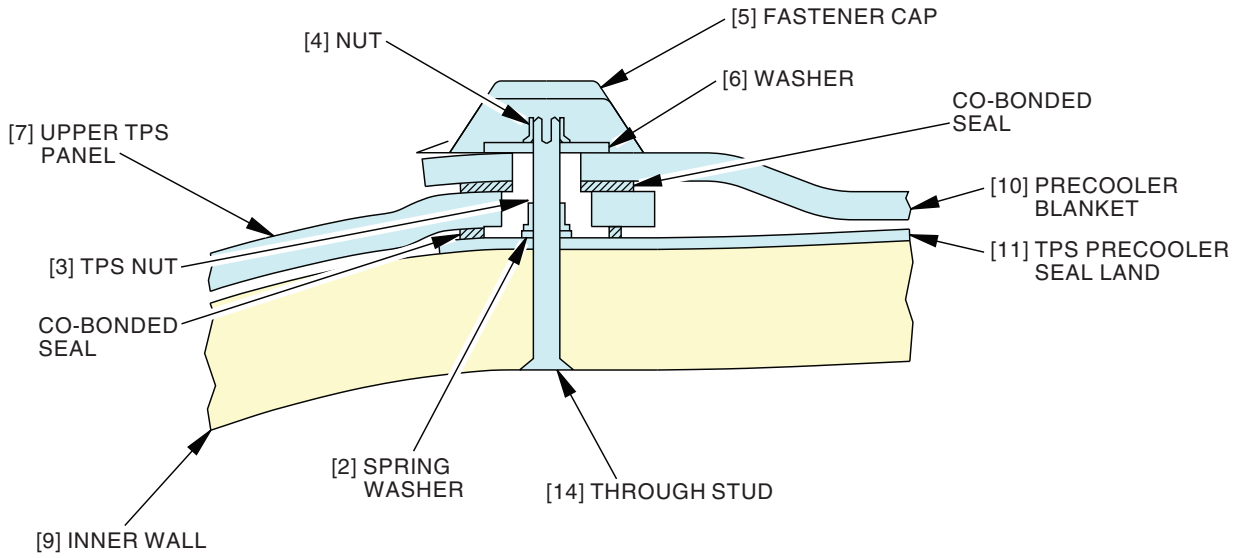
Through Stud Replacement
Figure 803/78-31-01-990-822-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

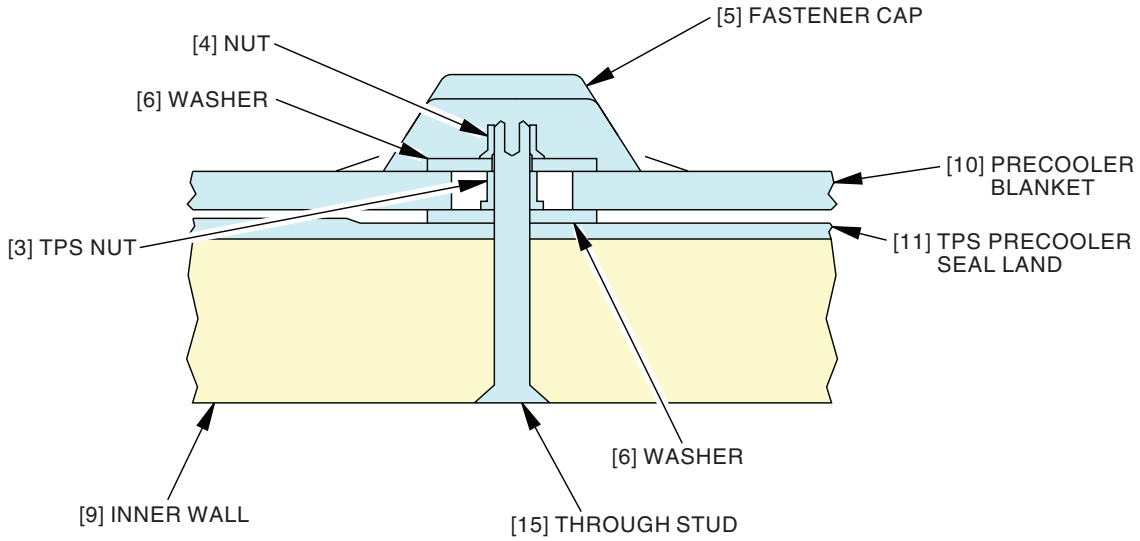


Through Stud Replacement
Figure 803/78-31-01-990-822-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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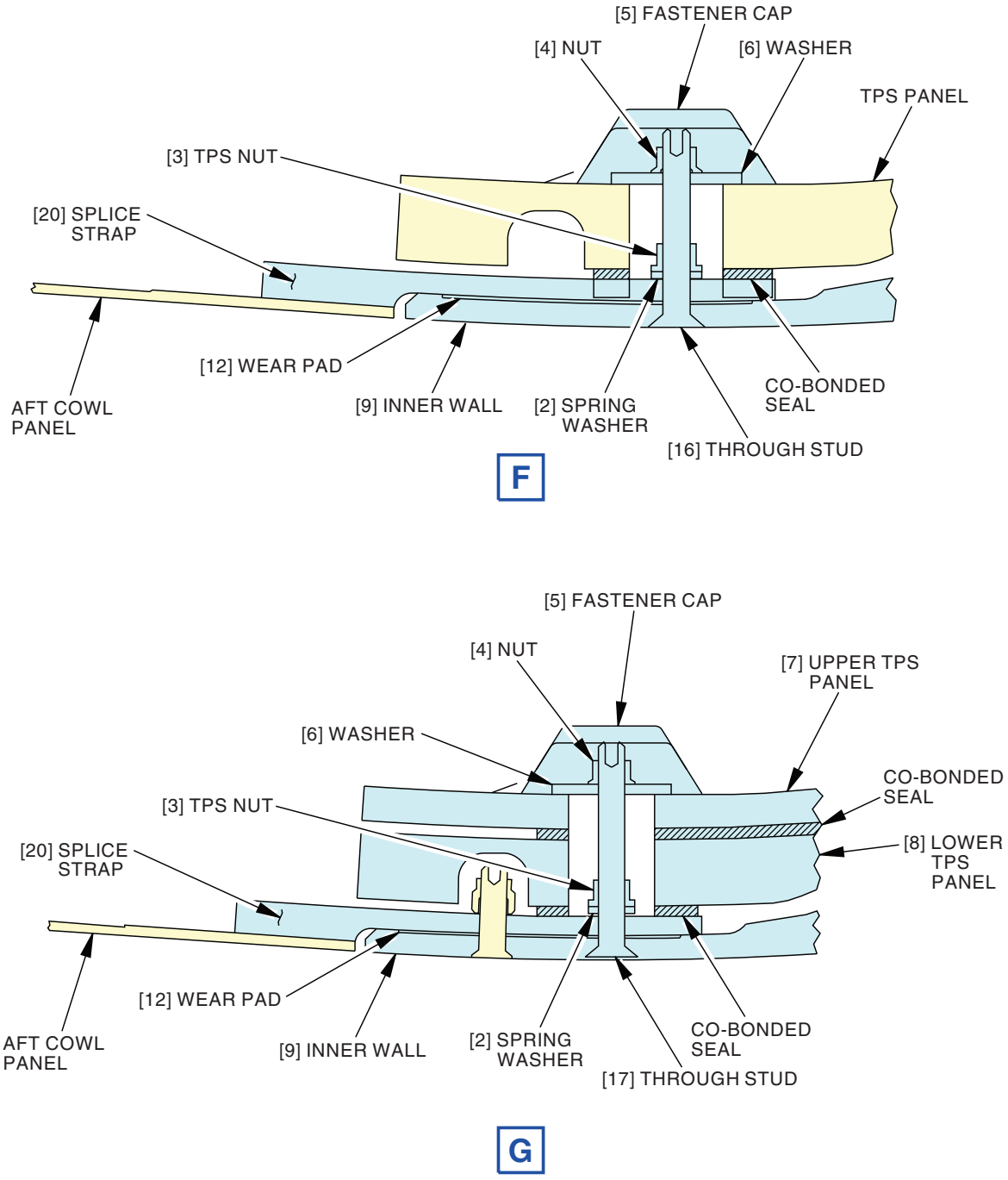


E

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Through Stud Replacement
Figure 803/78-31-01-990-822-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2551967 S0000607692_V1

Through Stud Replacement
Figure 803/78-31-01-990-822-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TRANSLATING SLEEVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the translating sleeve

(2) An installation of the translating sleeve.

TASK 78-31-02-000-801-G00

2. Translating Sleeve Removal

(Figure 401 and Figure 402)

A. General

(1) This task gives the instructions to remove the translating sleeve from the thrust reverser.

B. References

Reference	Title
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-980-803-G00	Thrust Reverser Operation - Extend (Manual Procedure) (P/B 201)
78-31-02-200-802-G00	Main Slider and Auxiliary Slider Inspection (P/B 601)
78-31-07-000-801-G00	Blocker Door Drag Link Removal (P/B 401)
78-31-09-010-801-G00	Krueger Flap Deflector Removal (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-2430	Hoist ? Boom, 800 lbs WLL, 250 lbs Load Positioner Compatible Part #: C20002-267 Supplier: 81205
SPL-14556	Sling - Thrust Reverser Sleeve, LEAP-1B Engine Part #: C78034-1 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G02329	Tape - Aluminum Foil, Pressure Sensitive - Vibration Damping Tape 434	

E. Access Panels

Number	Name/Location
415DL	Left Thrust Reverser Actuator (Upper), Engine 1
415EL	Left Thrust Reverser Actuator (Middle), Engine 1
415FL	Left Thrust Reverser Actuator (Lower), Engine 1
416DR	Right Thrust Reverser Actuator (Upper), Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
416ER	Right Thrust Reverser Actuator (Middle), Engine 1
416FR	Right Thrust Reverser Actuator (Lower), Engine 1
425DL	Left Thrust Reverser Actuator (Upper), Engine 2
425EL	Left Thrust Reverser Actuator (Middle), Engine 2
425FL	Left Thrust Reverser Actuator (Lower), Engine 2
426DR	Right Thrust Reverser Actuator (Upper), Engine 2
426ER	Right Thrust Reverser Actuator (Middle), Engine 2
426FR	Right Thrust Reverser Actuator (Lower), Engine 2

F. Prepare for the Removal

SUBTASK 78-31-02-010-009-G00

(1) If an inboard translating sleeve will be removed, remove the lower Krueger flap deflector to remove the access door for the actuator rod end (TASK 78-31-09-010-801-G00).

SUBTASK 78-31-02-010-016-G00

(2) Remove the access door for the applicable actuator, do these steps:

(a) Remove the six bolts [25] that attach the access doors [26] and access doors [29].

(b) For the upper actuator, remove these access doors [29]:

Number	Name/Location
415DL	Left Thrust Reverser Actuator (Upper), Engine 1
416DR	Right Thrust Reverser Actuator (Upper), Engine 1
425DL	Left Thrust Reverser Actuator (Upper), Engine 2
426DR	Right Thrust Reverser Actuator (Upper), Engine 2

(c) For the middle actuator, remove these access doors [26]:

Number	Name/Location
415EL	Left Thrust Reverser Actuator (Middle), Engine 1
416ER	Right Thrust Reverser Actuator (Middle), Engine 1
425EL	Left Thrust Reverser Actuator (Middle), Engine 2
426ER	Right Thrust Reverser Actuator (Middle), Engine 2

(d) For the lower actuator, remove these access doors [29]:

Number	Name/Location
415FL	Left Thrust Reverser Actuator (Lower), Engine 1
416FR	Right Thrust Reverser Actuator (Lower), Engine 1
425FL	Left Thrust Reverser Actuator (Lower), Engine 2
426FR	Right Thrust Reverser Actuator (Lower), Engine 2

SUBTASK 78-31-02-010-001-G00

WARNING

DO THESE SPECIFIED TASKS IN THE CORRECT SEQUENCE BEFORE YOU OPEN THE THRUST REVERSERS: RETRACT THE LEADING EDGE, DO THE DEACTIVATION PROCEDURES FOR THE LEADING EDGE AND THE THRUST REVERSER (FOR GROUND MAINTENANCE), AND OPEN THE FAN COWL PANELS. IF YOU DO NOT OBEY THE ABOVE SEQUENCE, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Do this task: Open the Thrust Reverser (Selection), TASK 78-31-00-010-801-G00.

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SUBTASK 78-31-02-980-001-G00

(4) Remove the five drag links [1] at the blocker doors from the thrust reverser inner fixed structure (TASK 78-31-07-000-801-G00).

(a) Secure a cotton wiper, G00034, with Vibration Damping Tape 434 tape, G02329, around the ends of the drag links so that they do not cause contact damage to the inner fixed structure.

(b) Use Vibration Damping Tape 434 tape, G02329, to hold the five drag links [1] in position so that they will not contact the inner fixed structure when you remove the translating sleeve.

G. Translating Sleeve Removal

SUBTASK 78-31-02-410-001-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS, BUT DO NOT DO THE THRUST REVERSER OR LEADING EDGE ACTIVATION. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Close the thrust reverser. Do this task:Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

SUBTASK 78-31-02-020-001-G00

(2) Manually extend the thrust reverser translating sleeve [21] until it is approximately 3.0 in. (76.2 mm) from the stowed position (TASK 78-31-00-980-803-G00).

SUBTASK 78-31-02-020-002-G00

CAUTION DO NOT LET THE ROD ENDS OF THE HYDRAULIC ACTUATOR TURN WHEN YOU REMOVE THE BOLTS. IF THE ROD END TURNS, IT CAN CAUSE DAMAGE TO THE HYDRAULIC ACTUATOR AND THRUST REVERSER STRUCTURE.

(3) Disconnect the three thrust reverser actuator rod ends from the applicable translating sleeve (Figure 401).

(a) Remove the nut [28], washer [27], bolt [22], washer [23], and bushing [24] at each of the three actuator rod ends.

(b) Manually slide the translating sleeve [21] aft approximately 2 in. (5 cm) until the actuator rod ends move out of the translating sleeve actuator attach fittings.

(c) Safety the actuator rod ends.

1) Make sure that the actuator rod ends will not turn.

SUBTASK 78-31-02-480-001-G00

WARNING MAKE SURE THAT YOU HOLD THE GSE SLING SATISFACTORILY DURING SLEEVE INSTALLATION. THE SLING IS HEAVY. IF YOU DO NOT OBEY THIS INSTRUCTION, THE SLING CAN FALL AND CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(4) Install the thrust reverser sleeve sling, SPL-14556, on the thrust reverser translating sleeve [21].

NOTE: The sling weights approximately 225.0 lb (102.1 kg).

(a) Install the two adapter fittings [54] onto the upper and middle actuator attach fittings [48] with the thumb screws [57] and thumb nuts [58].

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(b) Put the frame assembly [47] into its position on the two adapter fittings [54].

(c) Attach the frame assembly [47] to the two adapter fittings [54] with the lockpins [55] and retention pins [56].

(d) Align the cross arm assembly [46] with the frame assembly [47] and deactivation pin fitting on the translating sleeve [21].

1) Make sure that the deactivation pin is not installed in the deactivation pin fitting.

(e) Attach the cross arm assembly [46] to the frame assembly [47] with the two lock pins and retaining pins.

(f) Install the hand knob [59] that attach the cross arm assembly [46] to the translating sleeve [21] at the deactivation pin fitting.

SUBTASK 78-31-02-020-003-G00

WARNING MAKE SURE THAT THE GSE SLING CAN HOLD THE WEIGHT OF THE SLEEVE BEFORE YOU REMOVE THE SLEEVE. THE SLEEVE IS HEAVY. IF YOU DO NOT OBEY THIS INSTRUCTION, THE SLEEVE CAN FALL AND CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

CAUTION MAKE SURE THAT YOU MONITOR THE POSITION OF THE SLEEVE AND SLING AS YOU MOVE THE SLEEVE AWAY FROM THE THRUST REVERSER. IF YOU DO NOT OBEY THIS INSTRUCTION, THE SLEEVE CAN HIT THE ADJACENT STRUCTURES AND DAMAGE CAN OCCUR.

(5) Remove the translating sleeve [21].

NOTE: The sleeve weights approximately 196 lb (89 kg).

(a) Manually push the sleeve aft approximately 15 in. (38 cm).

NOTE: If you move the sleeve aft before you attach the crane or boom hoist to the sling, there will be less distance that the crane or boom hoist can move with it attached to the sleeve.

(b) If an overhead crane will be used to remove the translating sleeve, do these steps to attach the frame assembly to the overhead crane.

NOTE: The overhead crane can only be used with thrust reverser sleeve sling, SPL-14556, for the outboard sleeve. For the inboard sleeve, the boom hoist, SPL-2430, must be used with thrust reverser sleeve sling, SPL-14556.

1) Use the shackles [53] and web shackle [42] to attach the hand chain hoist [51], web sling [43], and a 500.0 lb (226.8 kg) capacity load cell [52] to the triangle lift plate [41].

2) Attach the load cell [52] to the overhead crane with the master link [40].

3) Lift the hand chain hoist [51] into position with the overhead crane.

4) Attach the hand chain hoist [51] to the frame assembly [47] with the hook and shackle [50].

5) Attach the web sling to the frame assembly with the web shackle [42].

(c) If the boom hoist [70] (boom hoist, SPL-2430) will be used to remove the translating sleeve [21], do these steps:

1) Attach the adjuster assembly [60] to the frame assembly [47] with the lockpin and retainer pin.

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2) Attach the boom hoist [70] to the adjuster assembly [60] with the lockpin and retainer pin.

(d) Adjust the overhead crane or boom hoist [70] so that the full weight of the translating sleeve [21] is supported.

(e) Manually push the translating sleeve [21] aft and off the tracks.

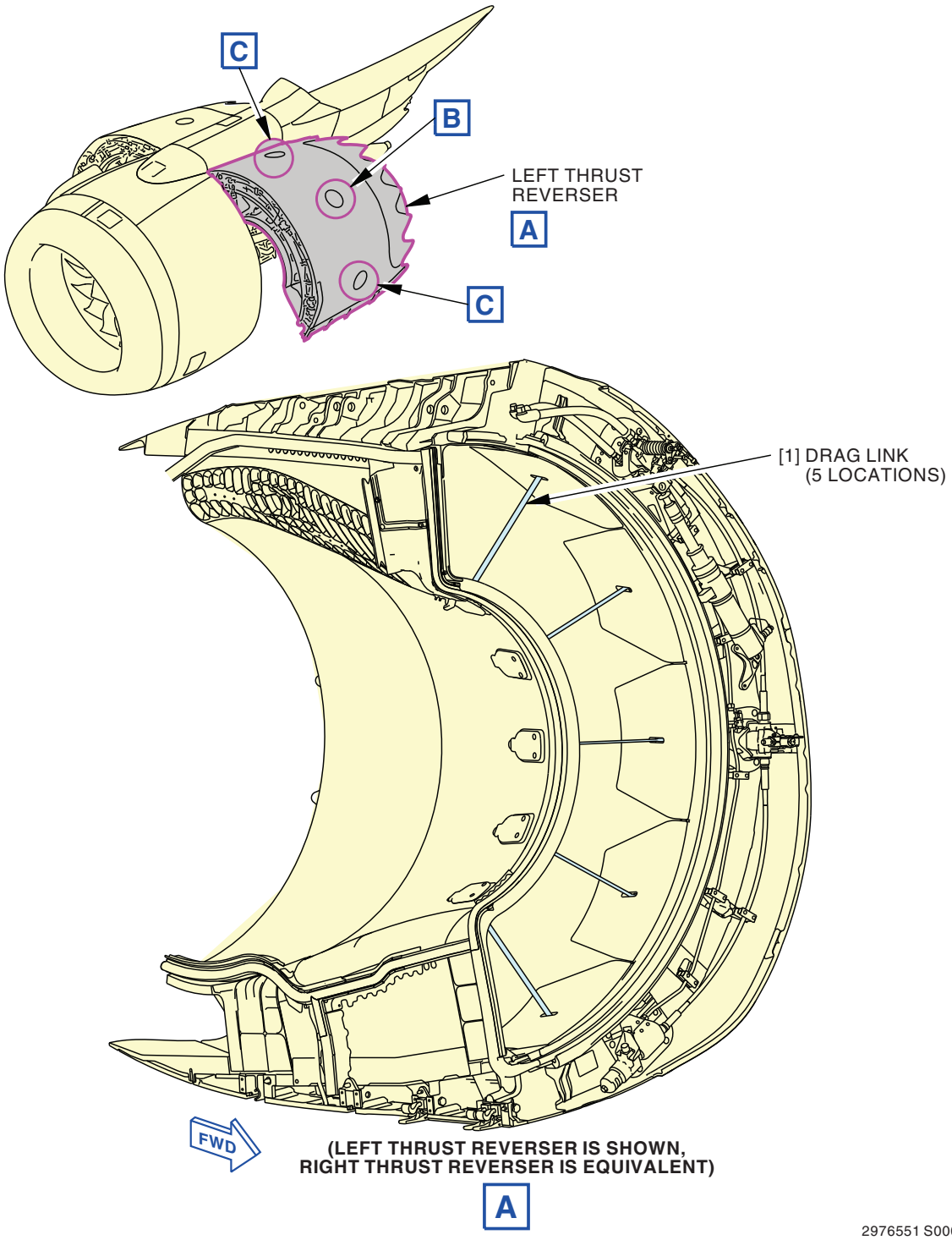
(f) Put the translating sleeve [21] on a suitable pallet and dunnage.

SUBTASK 78-31-02-210-001-G00

(6) Do this task: Main Slider and Auxiliary Slider Inspection, TASK 78-31-02-200-802-G00.

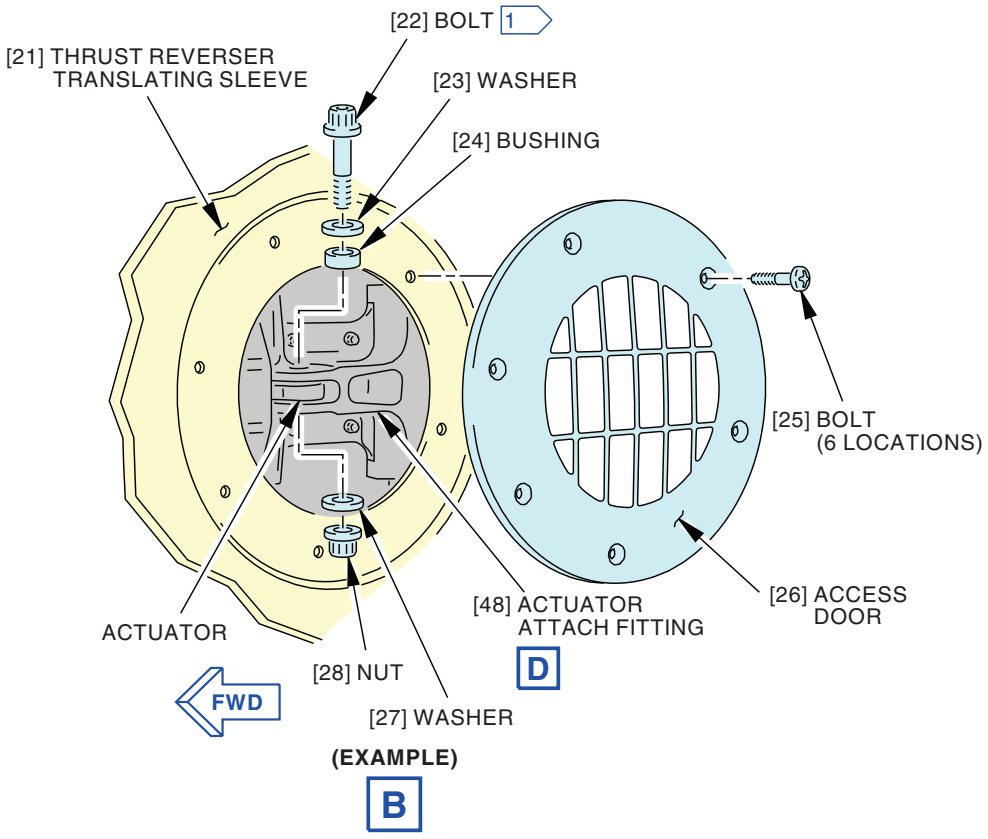
——— END OF TASK ———

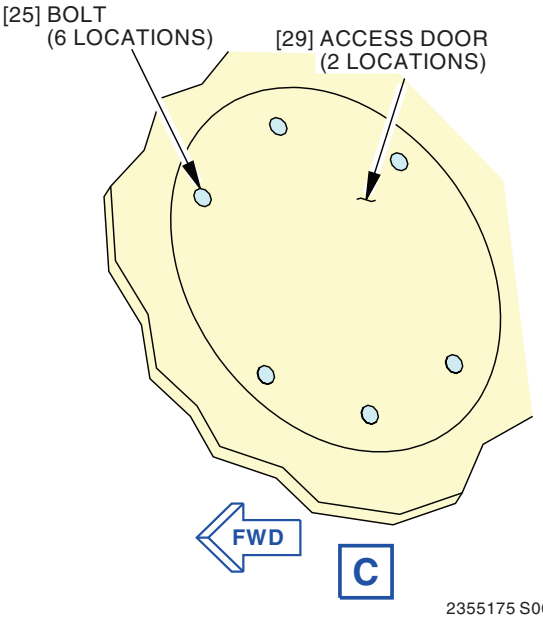
ECCN 9E991 BOEING PROPRIETARY - See title page for details



Translating Sleeve Installation
Figure 401/78-31-02-990-802-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



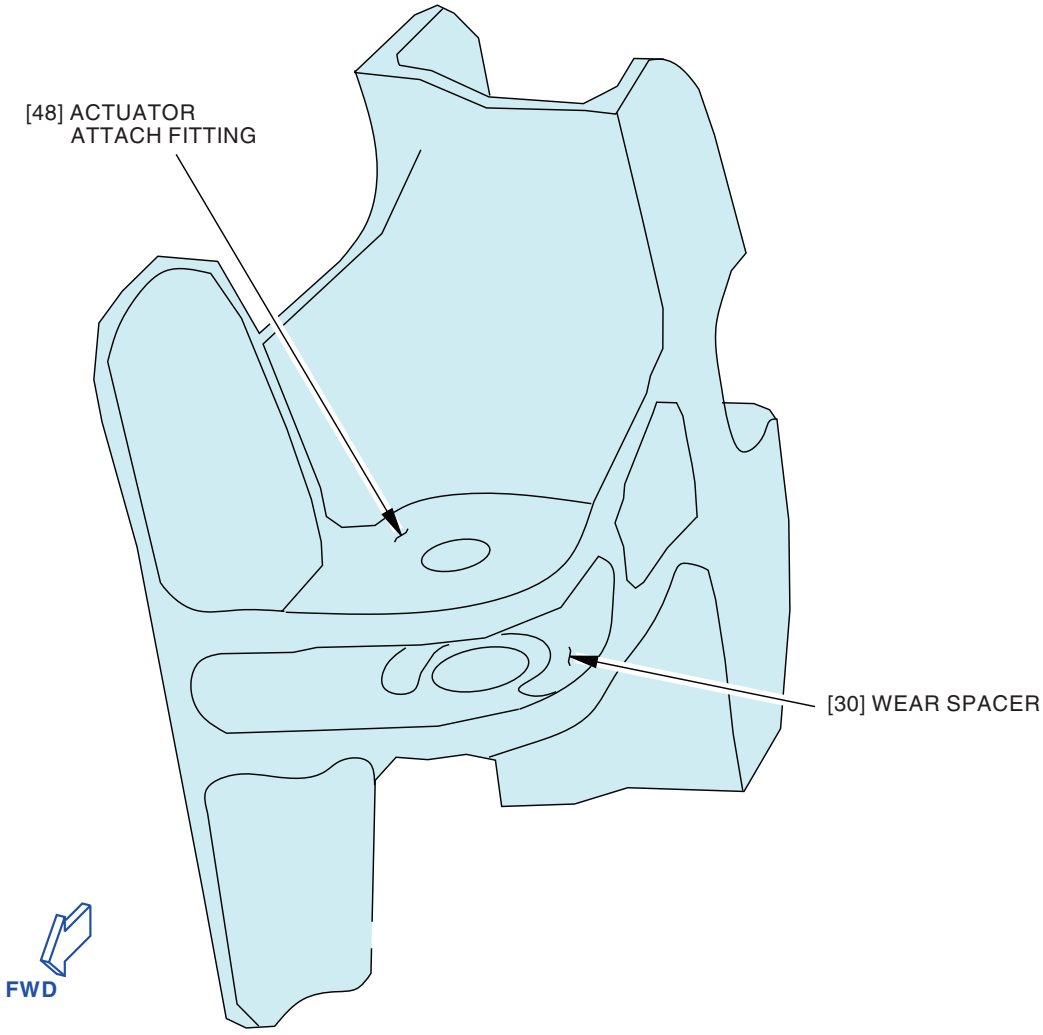


1 INSTALL THE BOLT WITH THE BOLT HEAD ON THE TOP SIDE OF THE FITTING.

2355175 S00061516689_V5

Translating Sleeve Installation
Figure 401/78-31-02-990-802-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



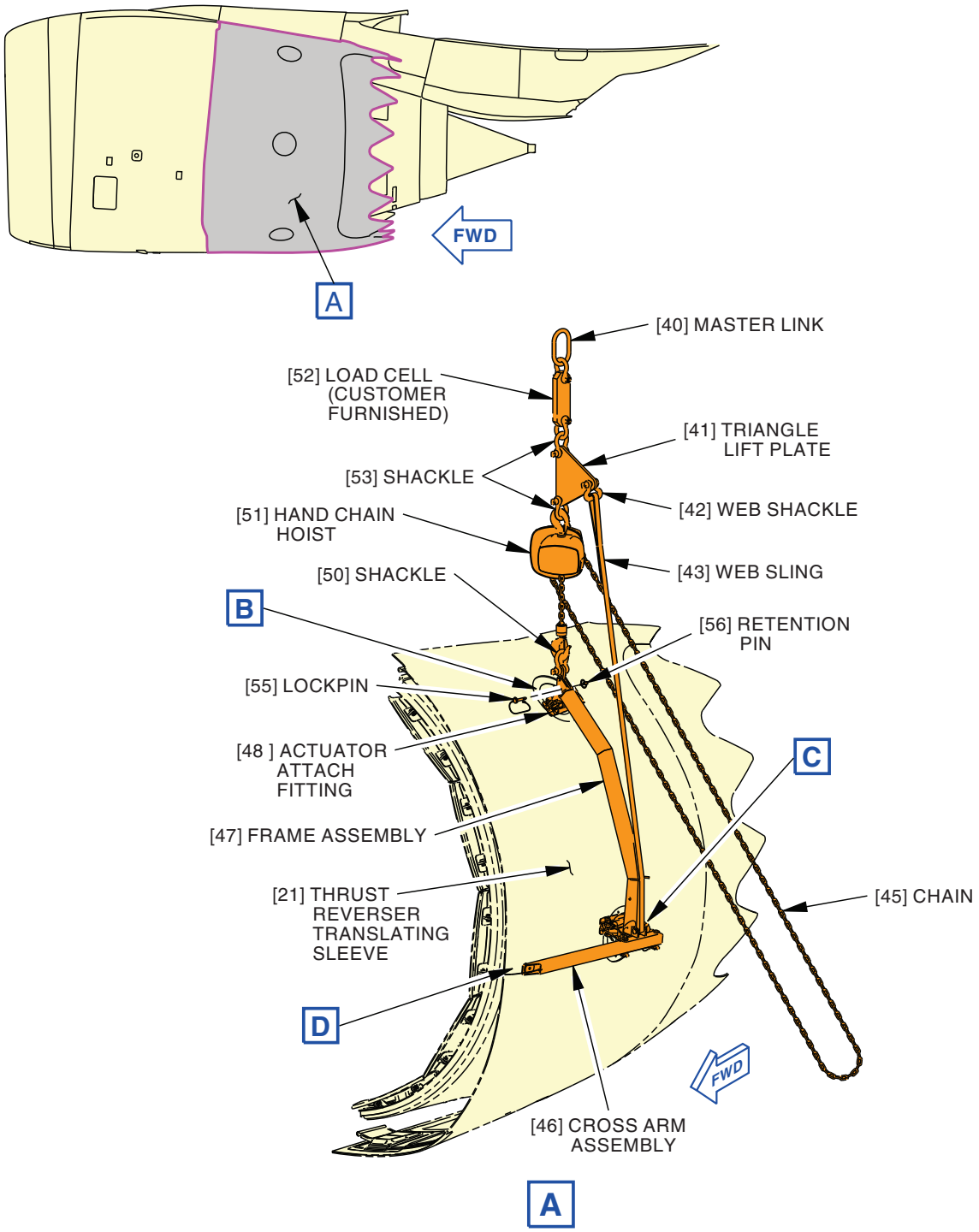
ACTUATOR ATTACH FITTING

D

2355177 S00061516691_V4

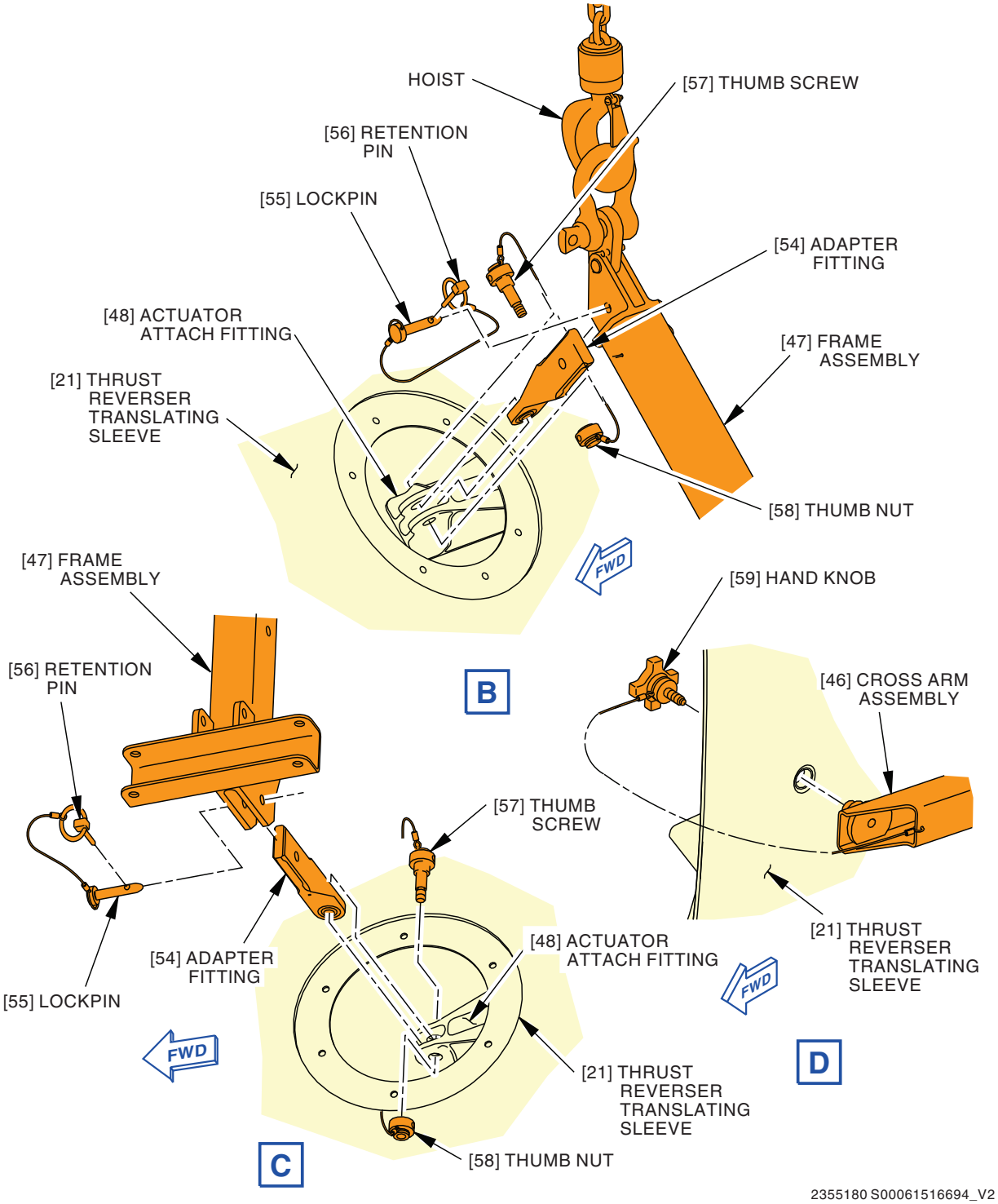
Translating Sleeve Installation
Figure 401/78-31-02-990-802-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



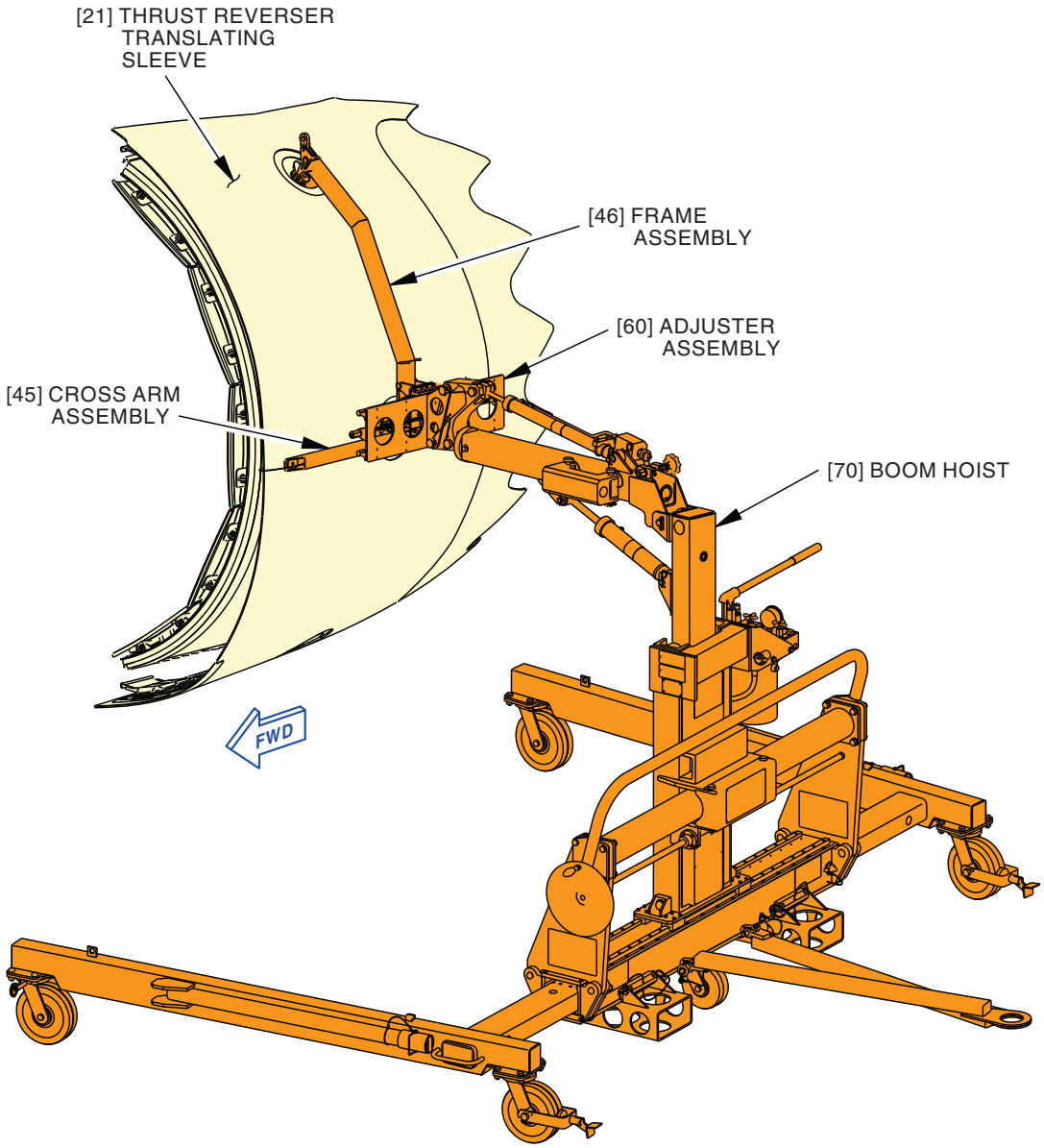
Translating Sleeve Sling Installation
Figure 402/78-31-02-990-803-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Translating Sleeve Sling Installation
Figure 402/78-31-02-990-803-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



HOIST EQUIPMENT WITH THRUST REVERSER BOOM HOIST (C20002)

2355181 S00061516695_V2

Translating Sleeve Sling Installation
Figure 402/78-31-02-990-803-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-31-02-400-801-G00

3. Translating Sleeve Installation

(Figure 401, Figure 402)

A. General

(1) This task gives the instructions to install the translating sleeve on the thrust reverser.

NOTE: The total acoustic panel blockage for the engine installation must be recalculated when components are changed per SRM SUBJECT 54-00-01.

B. References

Reference	Title
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-00-00-040-801-G00	DDG 78-31-01 Thrust Reversers Inoperative - Preparation (P/B 901)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-00-700-801-G00	Thrust Reverser Operational Test (Selection) (P/B 501)
78-31-00-980-803-G00	Thrust Reverser Operation - Extend (Manual Procedure) (P/B 201)
78-31-09-420-801-G00	Krueger Flap Deflector Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-2430	Hoist ? Boom, 800 lbs WLL, 250 lbs Load Positioner Compatible Part #: C20002-267 Supplier: 81205
SPL-14556	Sling - Thrust Reverser Sleeve, LEAP-1B Engine Part #: C78034-1 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
A00803	Sealant - Firewall - Hydraulic Fluid Resistant	BMS5-63 Type I
A50110	Sealant - Fuel Tank	BMS5-45 Class B-2
D50004	Compound - Antiseize	BMS3-28

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Drag link	78-31-07-01-125	PTF ALL
21	Translating sleeve	78-31-02-01-050	PTF ALL

F. Access Panels

Number	Name/Location
415DL	Left Thrust Reverser Actuator (Upper), Engine 1
415EL	Left Thrust Reverser Actuator (Middle), Engine 1
415FL	Left Thrust Reverser Actuator (Lower), Engine 1

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(Continued)

Number	Name/Location
416DR	Right Thrust Reverser Actuator (Upper), Engine 1
416ER	Right Thrust Reverser Actuator (Middle), Engine 1
416FR	Right Thrust Reverser Actuator (Lower), Engine 1
425DL	Left Thrust Reverser Actuator (Upper), Engine 2
425EL	Left Thrust Reverser Actuator (Middle), Engine 2
425FL	Left Thrust Reverser Actuator (Lower), Engine 2
426DR	Right Thrust Reverser Actuator (Upper), Engine 2
426ER	Right Thrust Reverser Actuator (Middle), Engine 2
426FR	Right Thrust Reverser Actuator (Lower), Engine 2

G. Translating Sleeve Installation

SUBTASK 78-31-02-410-010-G00

(1) Make sure that the thrust reversers are closed (Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00).

SUBTASK 78-31-02-210-002-G00

(2) Examine the tracks and track liners for damage and contamination:

(a) Clean all contamination from the tracks.

(b) Repair all damage that you find on the tracks.

SUBTASK 78-31-02-420-001-G00

(3) If it is necessary, do the steps in the removal task to attach the thrust reverser sleeve sling, SPL-14556, to the translating sleeve [21], and overhead crane or boom hoist [70] (TASK 78-31-02-000-801-G00).

SUBTASK 78-31-02-420-002-G00

CAUTION

MAKE SURE THAT THE UPPER AND LOWER SLIDERS ALIGN WITH THE TRACKS BEFORE YOU TRY TO MOVE THE SLEEVE. MAKE SURE THAT THE TOP AND BOTTOM OF THE SLEEVE MOVE TOGETHER WHILE YOU MOVE THE SLEEVE ALONG THE TRACKS. IF YOU DO NOT OBEY THIS INSTRUCTION, DAMAGE TO THE "RULON J" CAN OCCUR.

(4) Install the translating sleeve [21] on the thrust reverser (Figure 402):

(a) Lift the translating sleeve [21] and move it into its position at the rear of the thrust reverser.

(b) Align the sliders on the translating sleeve [21] with the tracks on the hinge and latch beam.

(c) Engage the sliders in the tracks and push the translating sleeve [21] forward until the actuator rod ends align with the actuator attach fittings on the sleeve.

1) Make sure that the top and bottom of the translating sleeve [21] move together.

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SUBTASK 78-31-02-080-001-G00

WARNING BE CAREFUL WHEN YOU DISCONNECT THE SLING ASSEMBLY. THE SLING IS HEAVY. INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

(5) Remove the thrust reverser sleeve sling, SPL-14556, from the translating sleeve [21] as follows:

NOTE: The sling weights approximately 225.0 lb (102.1 kg).

(a) If the overhead crane will be used to install the translating sleeve [21], disconnect the hand chain hoist [51] and web sling [43] from the frame assembly [47].

NOTE: The overhead crane can only be used with thrust reverser sleeve sling, SPL-14556, for the outboard sleeve. For the inboard sleeve, the boom hoist, SPL-2430, must be used with thrust reverser sleeve sling, SPL-14556.

(b) If the boom hoist [70] will be used to install the translating sleeve [21], do these steps:

1) Disconnect the boom hoist [70] (boom hoist, SPL-2430) from the adjuster assembly [60].

2) Remove the adjuster assembly [60] from the frame assembly [47].

(c) Remove the hand knob [59] that attaches the cross arm assembly [46] to the translating sleeve [21] at the deactivation pin fitting.

(d) Remove the cross arm assembly [46] from the frame assembly [47].

(e) Remove the lockpin [49] that attaches the adapter fitting [54] on the middle actuator attach fitting [48] to the frame assembly [47].

(f) Support the frame assembly [47] and remove the lockpin [49] that attaches the upper actuator attach fitting [48] to the frame assembly [47].

(g) Remove the frame assembly [47] from upper and middle adapter fitting [54].

(h) Remove the thumb nuts [58] and thumb screws [57] that attach the upper and middle adapter fittings [54] to the actuator attach fittings [48] on the translating sleeve [21].

(i) Remove the upper and middle adapter fittings [54] from the translating sleeve [21].

SUBTASK 78-31-02-420-003-G00

(6) Connect the three actuator rod ends to the actuator attach fittings [48] on the translating sleeve [21] (Figure 401):

(a) Lubricate the shanks of the three bolts [22] with compound, D50004.

(b) Make sure that each of the actuator attach fittings [48] have a wear spacer [30].

1) If there is no wear spacer [30], use sealant, A00803, to secure a new wear spacer [30] into place.

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CAUTION DO NOT LET THE ROD ENDS OF THE HYDRAULIC ACTUATOR TURN WHEN YOU REMOVE THE BOLTS. IF THE ROD END TURNS, IT CAN CAUSE DAMAGE TO THE HYDRAULIC ACTUATOR AND THRUST REVERSER STRUCTURE.

(c) Pull the actuator rod ends by hand to align them with the clevis on the three actuator attach fittings [48].

NOTE: Do not let the rod end turn when you extend it manually. Make sure that you hold the rod end when you extend the actuator manually. Do not use hydraulic power to extend the rod end.

(d) Install the bolt [22], washer [23], bushing [24], washer [27], and nut [28] to attach each actuator rod end to its actuator attach fitting [48].

1) Tighten the nuts [28] to 530 ±16 in-lb (60 ±2 N·m).

SUBTASK 78-31-02-410-002-G00

(7) Install the access door for the applicable actuator, do these steps:

(a) For the upper actuator, put the access doors [29] in the correct position to align the bolt holes:

Number	Name/Location
415DL	Left Thrust Reverser Actuator (Upper), Engine 1
416DR	Right Thrust Reverser Actuator (Upper), Engine 1
425DL	Left Thrust Reverser Actuator (Upper), Engine 2
426DR	Right Thrust Reverser Actuator (Upper), Engine 2

(b) For the middle actuator, put the access doors [26] in the correct position to align the bolt holes:

Number	Name/Location
415EL	Left Thrust Reverser Actuator (Middle), Engine 1
416ER	Right Thrust Reverser Actuator (Middle), Engine 1
425EL	Left Thrust Reverser Actuator (Middle), Engine 2
426ER	Right Thrust Reverser Actuator (Middle), Engine 2

(c) For the lower actuator access doors [29], do these steps:

1) Make sure that the installation surfaces are clean.

2) Use sealant, A50110, to create a prepack or injection seal between the access doors [29] and installation surface on the thrust reverser.

3) For the lower actuator, put the access doors [29] in the correct position to align the bolt holes:

Number	Name/Location
415FL	Left Thrust Reverser Actuator (Lower), Engine 1
416FR	Right Thrust Reverser Actuator (Lower), Engine 1
425FL	Left Thrust Reverser Actuator (Lower), Engine 2
426FR	Right Thrust Reverser Actuator (Lower), Engine 2

(d) Install the six bolts [25] to attach the access doors [26] and access doors [29].

1) Tighten the bolts [25] to 30.0 in-lb (3.4 N·m) - 50 in-lb (5.6 N·m).

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SUBTASK 78-31-02-010-002-G00

WARNING OBEY THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE PROCEDURE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(8) Open the inboard and outboard thrust reverser(TASK 78-31-00-010-801-G00).

SUBTASK 78-31-02-980-007-G00

(9) Manually extend the inboard and outboard translating sleeve just enough to connect the drag links(TASK 78-31-00-980-803-G00).

(a) Make sure to have enough clearance between the thrust reverser and the wing leading edge.

SUBTASK 78-31-02-420-004-G00

(10) Attach each of the drag links [1] to the inner fixed structure as follows:

(a) Remove the tape and protective cloth from the blocker doors and five drag links [1].

(b) Make sure that the five drag links [1] are installed correctly on the blocker doors.

(c) Align the end of the drag links [1] with the drag link attach fittings on the inner fixed structure.

(d) Make sure that there is no damage or burrs on the bushings [4] before installation.

(e) On each drag link [1] install the bolt, washer, bushing, second washer, and nut.

1) Make sure that the bushing is installed from the bottom of each drag link.

(f) Tighten the nut to 400 ±12 in-lb (45.2 ±1.4 N·m).

SUBTASK 78-31-02-410-003-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS, BUT DO NOT DO THE THRUST REVERSER OR LEADING EDGE ACTIVATION. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(11) Close and latch the thrust reverser, but do not do the thrust reverser or leading edge activation at this time (TASK 78-31-00-010-802-G00).

SUBTASK 78-31-02-980-003-G00

(12) Manually translate the sleeve through an extend and retract cycle (TASK 78-31-00-980-803-G00).

(a) Make sure that the translating sleeve [21] moves smoothly.

SUBTASK 78-31-02-440-001-G00

(13) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 78-31-02-710-001-G00

(14) Do this task: Thrust Reverser Operational Test (Selection), TASK 78-31-00-700-801-G00.

(a) Operate the thrust reverser through an extend and retract cycle four times to make sure that it operates correctly.

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SUBTASK 78-31-02-410-018-G00

(15) If installing the translating sleeve on a new thrust reverser, make sure that the deactivation pin fits the deactivation pin fitting (TASK 78-00-00-040-801-G00).

NOTE: A deactivation pin check is necessary only for translating sleeve assemblies 315A6570-3, -4, -5, -6, -7 and -8. It is not necessary for all other translating sleeve numbers to do a deactivation pin check. If the deactivation pin will not install, contact Boeing for further assistance.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-02-410-004-G00

(1) If the inboard translating sleeve was replaced, install the lower Krueger flap deflector (TASK 78-31-09-420-801-G00).

SUBTASK 78-31-02-410-005-G00

(2) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

SUBTASK 78-31-02-440-002-G00

(3) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

——— END OF TASK ———

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TRANSLATING SLEEVE - INSPECTION/CHECK

1. General

A. This procedure has these tasks:

(1) Translating sleeve inspection

(2) Main slider and auxiliary slider inspection

(3) Main slider and auxiliary slider general visual inspection

(4) Thrust reverser actuator attach fitting visual inspection

(5) Thrust reverser actuator attach fitting detailed inspection.

TASK 78-31-02-200-801-G00

2. Translating Sleeve Inspection

(Figure 601)

A. General

(1) This task gives the instructions on how to examine the outer-skin-panel and the inner-acoustic-panel on the translating sleeve for damage.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Prepare for the Inspection

SUBTASK 78-31-02-860-001-G00

WARNING RETRACT THE LEADING EDGE FLAPS AND SLATS AND DO THE DEACTIVATION PROCEDURE BEFORE YOU OPEN THE LEFT OR THE RIGHT THRUST REVERSER ON AN ENGINE. IF YOU DO NOT OBEY THIS INSTRUCTION INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 78-31-02-040-001-G00

(2) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 78-31-02-040-002-G00

WARNING DO THE DEACTIVATION PROCEDURE TO PREVENT THE OPERATION OF THE THRUST REVERSER. THE ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Translating Sleeve Outer-Skin-Panel Inspection

SUBTASK 78-31-02-210-003-G00

(1) Examine the outer-skin-panel on the translating sleeve for the damage that follows:

(a) Holes, cracks, nicks, gouges, delamination, dents and edge corrosion.

1) Cracks in the paint layer that do not affect the sealant or the structure of the panel are acceptable.

2) If you find damage, refer to the Thrust Reverser Translating Sleeve Outer Skin/SRM PROCEDURE 54-30-01-1A-1 for the permitted limits and repair procedures.

E. Translating Sleeve Inner-Acoustic-Panel Inspection

SUBTASK 78-31-02-210-004-G00

(1) Look through the aft end of the fan duct to examine the aft end of the inner-acoustic-panel on the translating sleeve for the damage that follows:

(a) Holes, cracks, nicks, gouges, delamination, dents and edge corrosion.

1) Cracks in the paint layer that do not affect the sealant or the structure of the panel are acceptable.

2) If you find damage, refer to the Thrust Reverser Translating Sleeve Outer Skin/SRM PROCEDURE 54-30-01-1A-1 for the permitted limits and repair procedures.

(b) Pitting in the surface layer (small areas where the surface appears chipped away around the perforation of the acoustic panel) (Figure 602).

NOTE: This pitting can occur during the usual manufacturing process of the acoustic panel. This condition has been inspected and approved for in-service use at the time of manufacture.

1) No action is necessary with the following conditions:

a) The surface area adjacent to each pitting location has the original silver finish and does not show the black panel material.

b) There are no signs of edge erosion.

SUBTASK 78-31-02-010-003-G00

WARNING DO THESE SPECIFIED TASKS IN THE CORRECT SEQUENCE BEFORE YOU OPEN THE THRUST REVERSERS: RETRACT THE LEADING EDGE, DO THE DEACTIVATION PROCEDURES FOR THE LEADING EDGE AND THE THRUST REVERSER (FOR GROUND MAINTENANCE), AND OPEN THE FAN COWL PANELS. IF YOU DO NOT OBEY THE ABOVE SEQUENCE, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do this task: Open the Thrust Reverser (Selection), TASK 78-31-00-010-801-G00.

SUBTASK 78-31-02-210-005-G00

(3) Look through the forward end of the fan duct to examine the forward end of the inner-acoustic-panel on the translating sleeve for the damage that follows:

(a) Holes, cracks, nicks, gouges, delamination, dents and edge corrosion.

1) Cracks in the paint layer that do not affect the sealant or the structure of the panel are acceptable.

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2) If you find damage, refer to the Thrust Reverser Translating Sleeve Outer Skin/SRM PROCEDURE 54-30-01-1A-1 for the permitted limits and repair procedures.

(b) Pitting in the surface layer (small areas where the surface appears chipped away around the perforation of the acoustic panel) (Figure 602)

NOTE: This pitting can occur during the usual manufacturing process of the acoustic panel. This condition has been inspected and approved for in-service use at the time of manufacture.

1) No action is necessary with the following conditions:

a) The surface area adjacent to each pitting location has the original silver finish and does not show the black panel material.

b) There are no signs of edge erosion.

F. Put the Airplane Back to its Usual Condition

SUBTASK 78-31-02-410-006-G00

WARNING OBEY THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE PROCEDURE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

SUBTASK 78-31-02-440-003-G00

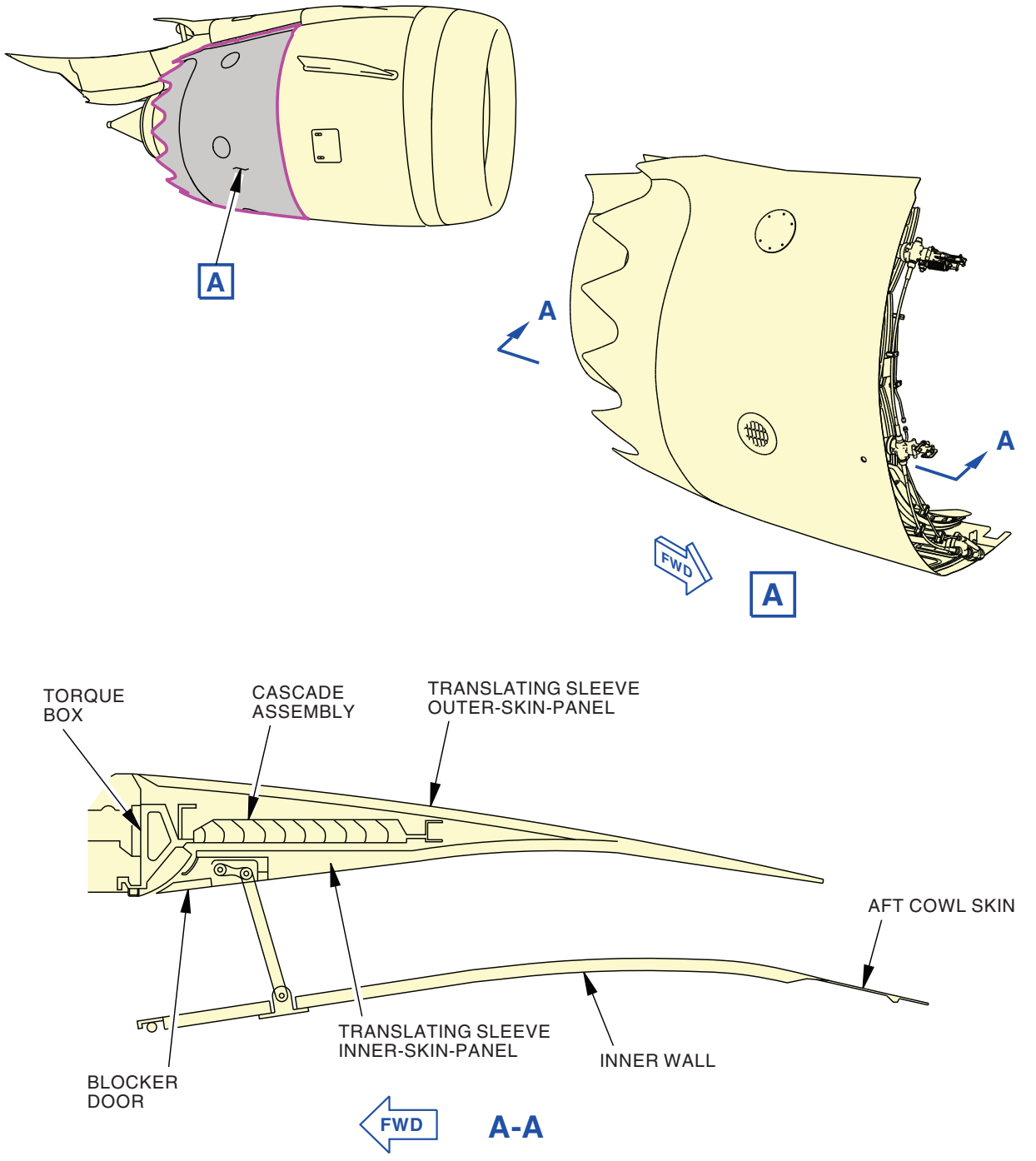
(2) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 78-31-02-440-004-G00

(3) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

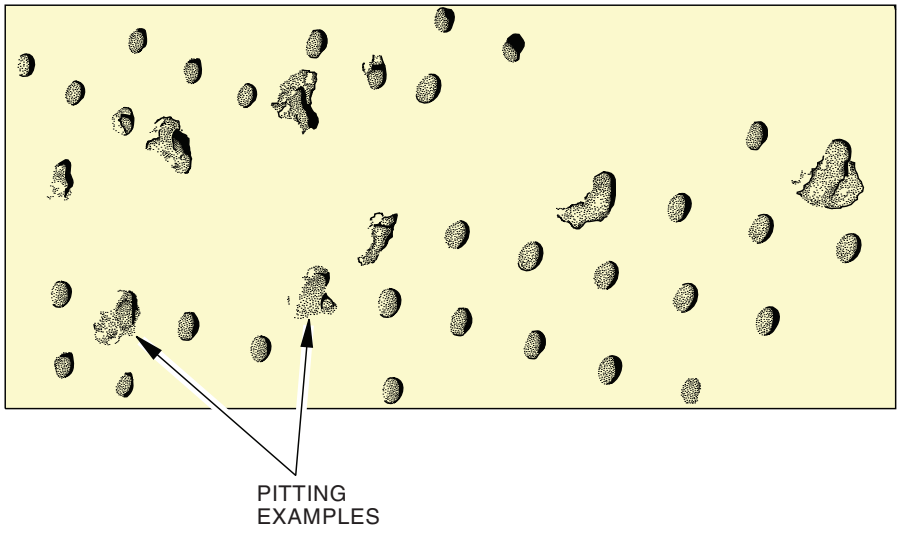
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Translating Sleeve Inspection
Figure 601/78-31-02-990-804-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Translating Sleeve Inner-Acoustic-Panel Inspection
Figure 602/78-31-02-990-805-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-31-02-200-802-G00

3. Main Slider and Auxiliary Slider Inspection

(Figure 603)

A. General

(1) This task gives the instructions on how to examine the Rulon J on the auxiliary slider and on the shoe assembly that is installed on the main slider, when the translating sleeve is removed.

B. References

Reference	Title
78-31-02-000-801-G00	Translating Sleeve Removal (P/B 401)
78-31-02-400-801-G00	Translating Sleeve Installation (P/B 401)
CMM 78-31-24	Thrust Reverser Sleeve Assembly

C. Inspection

SUBTASK 78-31-02-010-004-G00

(1) Do this task: Translating Sleeve Removal, TASK 78-31-02-000-801-G00.

SUBTASK 78-31-02-210-006-G00

(2) Examine the Rulon J on the auxiliary slider and on the shoe assembly that is installed on the main slider for the damage that follows:

(a) Obvious damage, wear through the tape thickness, and disbonding.

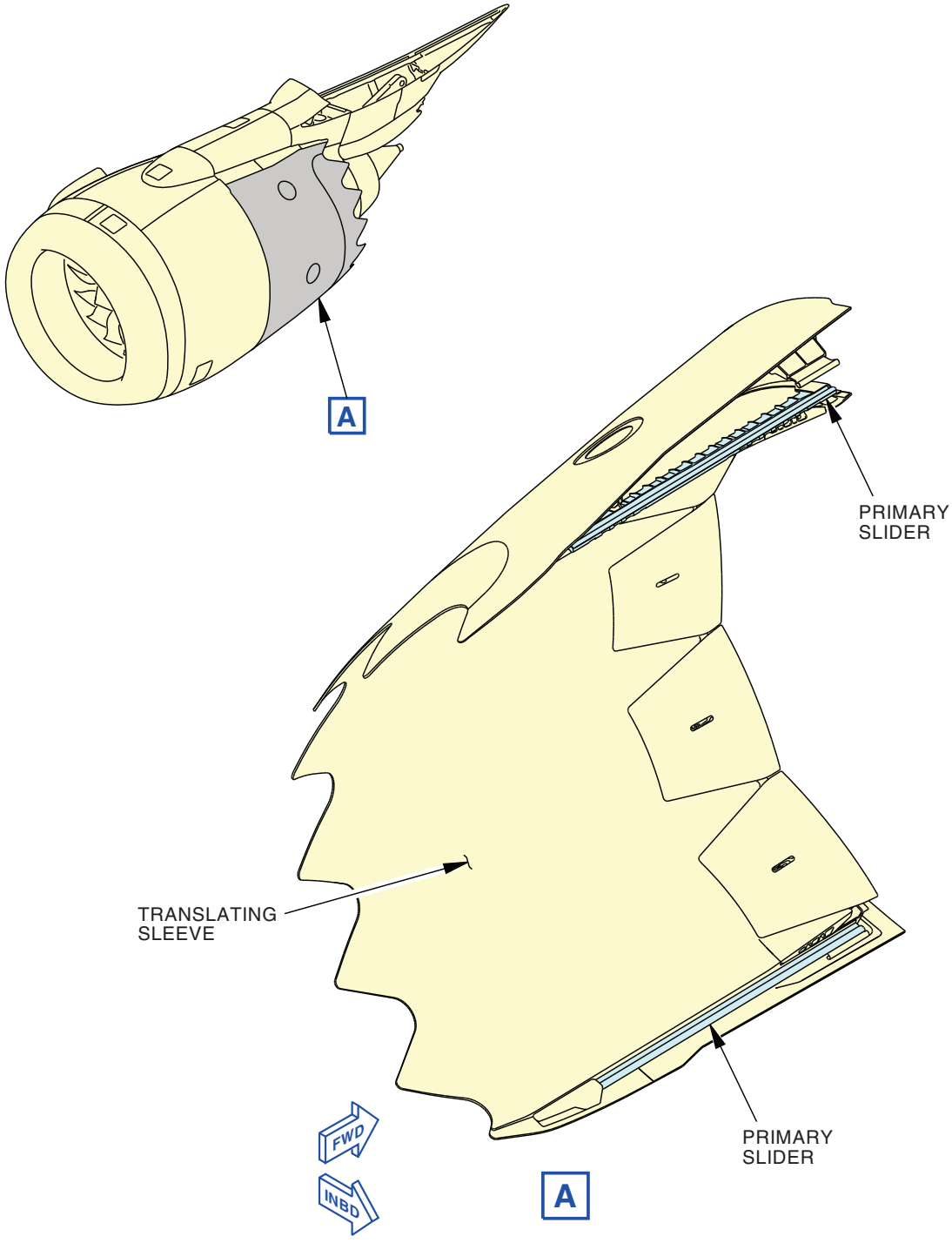
1) If you find damage, refer to the CMM 78-31-24 for the permitted limits and repair procedures.

SUBTASK 78-31-02-010-005-G00

(3) Do this task: Translating Sleeve Installation, TASK 78-31-02-400-801-G00.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Slider and Auxiliary Slider Inspection
Figure 603/78-31-02-990-809-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-31-02-210-801-G00

4. Engine Thrust Reverser Hinge Beam and Latch Beam Primary Track and Translating Sleeve Primary Slider General Visual Inspection for Wear with the Thrust Reverser Deployed

(Figure 604)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task provides instructions to do general visual inspection of engine thrust reverser hinge beam and latch beam primary track and translating sleeve primary slider.

B. References

Reference	Title
78-31-00-980-801-G00	Thrust Reverser Operation - Extend (Selection) (P/B 201)
78-31-00-980-802-G00	Thrust Reverser Operation - Retract (Selection) (P/B 201)

C. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

D. Prepare for the Inspection

SUBTASK 78-31-02-860-006-G00

(1) Do this task: Thrust Reverser Operation - Extend (Selection), TASK 78-31-00-980-801-G00.

E. Inspection

SUBTASK 78-31-02-210-007-G00

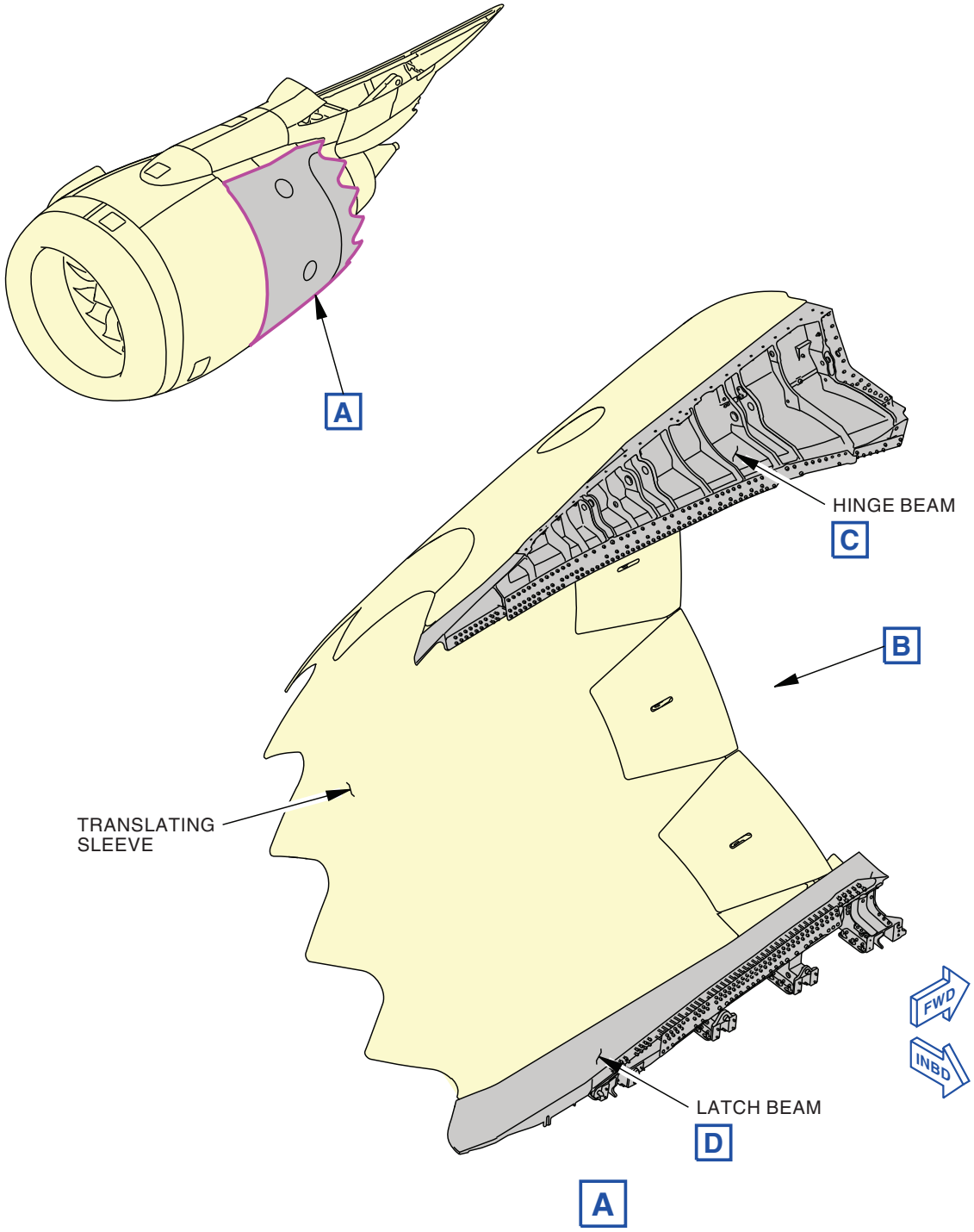
(1) Do general visual inspection of the engine thrust reverser hinge beam and latch beam primary track and translating sleeve primary slider for wear with the thrust reverser deployed.

SUBTASK 78-31-02-860-007-G00

(2) Do this task: Thrust Reverser Operation - Retract (Selection), TASK 78-31-00-980-802-G00.

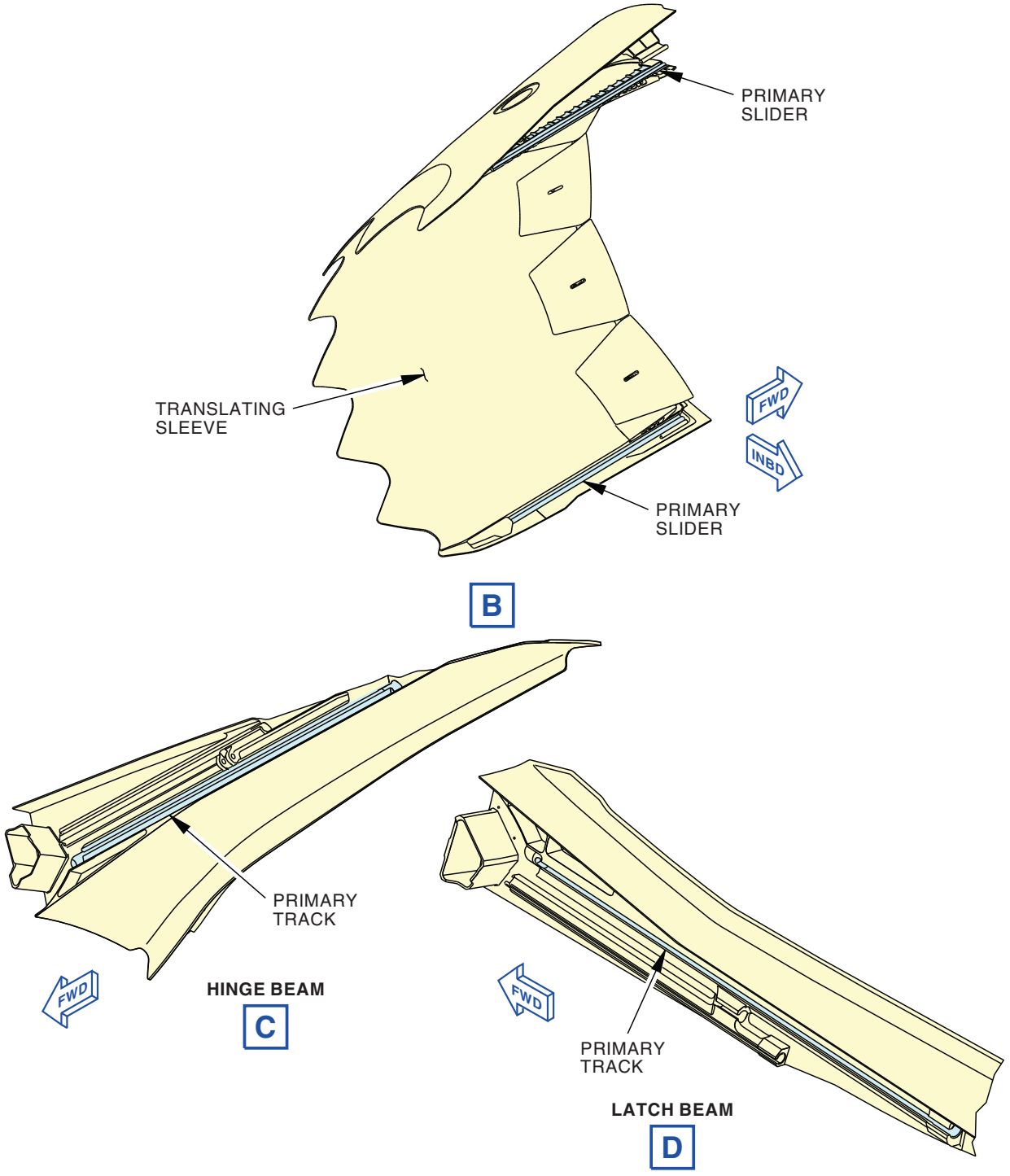
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Engine Thrust Reverser Hinge Beam and Latch Beam Primary Track and Translating Sleeve Primary Slider General Visual Inspection
Figure 604/78-31-02-990-810-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Engine Thrust Reverser Hinge Beam and Latch Beam Primary Track and Translating Sleeve Primary Slider General Visual Inspection
Figure 604/78-31-02-990-810-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-31-02-200-803-G00

5. Thrust Reverser Actuator Attach Fitting Visual Inspection

(Figure 605)

A. General

(1) This task gives the instructions for a visual inspection of the thrust reverser actuator attach fittings.

(a) The thrust reverser actuator attach fittings are installed on the translating sleeve.

(b) There are three actuator attach fittings per thrust reverser half.

B. References

Reference	Title
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-03-610-801-G00	Thrust Reverser Hydraulic Actuator Rod End Inspection (P/B 601)
78-31-09-010-801-G00	Krueger Flap Deflector Removal (P/B 401)
78-31-09-420-801-G00	Krueger Flap Deflector Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
A50110	Sealant - Fuel Tank	BMS5-45 Class B-2

D. Access Panels

Number	Name/Location
415DL	Left Thrust Reverser Actuator (Upper), Engine 1
415EL	Left Thrust Reverser Actuator (Middle), Engine 1
415FL	Left Thrust Reverser Actuator (Lower), Engine 1
416DR	Right Thrust Reverser Actuator (Upper), Engine 1
416ER	Right Thrust Reverser Actuator (Middle), Engine 1
416FR	Right Thrust Reverser Actuator (Lower), Engine 1
425DL	Left Thrust Reverser Actuator (Upper), Engine 2
425EL	Left Thrust Reverser Actuator (Middle), Engine 2
425FL	Left Thrust Reverser Actuator (Lower), Engine 2
426DR	Right Thrust Reverser Actuator (Upper), Engine 2
426ER	Right Thrust Reverser Actuator (Middle), Engine 2
426FR	Right Thrust Reverser Actuator (Lower), Engine 2

E. Prepare for the Inspection

SUBTASK 78-31-02-040-003-G00

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(1) Do the deactivation procedure for the thrust reverser for ground maintenance (TASK 78-31-00-040-801-G00).

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SUBTASK 78-31-02-010-018-G00

(2) To remove the upper actuator access doors [2], do these steps:

Number	Name/Location
416DR	Right Thrust Reverser Actuator (Upper), Engine 1
425DL	Left Thrust Reverser Actuator (Upper), Engine 2

(a) Remove the lower Krueger flap deflector (TASK 78-31-09-010-801-G00).

(b) Remove the six bolts [1].

SUBTASK 78-31-02-010-008-G00

(3) To remove the upper actuator access doors [2], remove the six bolts [1].

Number	Name/Location
415DL	Left Thrust Reverser Actuator (Upper), Engine 1
426DR	Right Thrust Reverser Actuator (Upper), Engine 2

SUBTASK 78-31-02-010-020-G00

(4) To remove the middle actuator access doors [2], remove the six bolts [1].

Number	Name/Location
415EL	Left Thrust Reverser Actuator (Middle), Engine 1
416ER	Right Thrust Reverser Actuator (Middle), Engine 1
425EL	Left Thrust Reverser Actuator (Middle), Engine 2
426ER	Right Thrust Reverser Actuator (Middle), Engine 2

SUBTASK 78-31-02-010-021-G00

(5) To remove the lower actuator access doors [2], remove the six bolts [1].

Number	Name/Location
415FL	Left Thrust Reverser Actuator (Lower), Engine 1
416FR	Right Thrust Reverser Actuator (Lower), Engine 1
425FL	Left Thrust Reverser Actuator (Lower), Engine 2
426FR	Right Thrust Reverser Actuator (Lower), Engine 2

F. Thrust Reverser Actuator Attach Fitting Visual Inspection

SUBTASK 78-31-02-212-001-G00

(1) Examine the thrust reverser actuator attach fitting for these items:

(a) Cracks

1) Not permitted.

(b) Nicks

1) Not permitted.

(c) Dents

1) Not permitted.

(d) Scratches

1) Not permitted.

SUBTASK 78-31-02-869-001-G00

(2) If you find damage on the actuator attach fitting, do these steps:

(a) Do this task: Thrust Reverser Actuator Attach Fitting Detailed Inspection, TASK 78-31-02-200-804-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Do this task: Thrust Reverser Hydraulic Actuator Rod End Inspection, TASK 78-31-03-610-801-G00.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-02-410-009-G00

(1) For the upper actuator, put the access doors [2] in the correct position to align the bolt holes.

Number	Name/Location
415DL	Left Thrust Reverser Actuator (Upper), Engine 1
416DR	Right Thrust Reverser Actuator (Upper), Engine 1
425DL	Left Thrust Reverser Actuator (Upper), Engine 2
426DR	Right Thrust Reverser Actuator (Upper), Engine 2

(a) Install the six bolts [1] onto the access door [2].

1) Tighten the bolts [1] to 30 in-lb (3.4 N·m) - 50 in-lb (5.6 N·m).

SUBTASK 78-31-02-410-021-G00

(2) For the middle actuator, put the access doors [2] in the correct position to align the bolt holes.

Number	Name/Location
415EL	Left Thrust Reverser Actuator (Middle), Engine 1
416ER	Right Thrust Reverser Actuator (Middle), Engine 1
425EL	Left Thrust Reverser Actuator (Middle), Engine 2
426ER	Right Thrust Reverser Actuator (Middle), Engine 2

(a) Install the six bolts [1] onto the access door [2].

1) Tighten the bolts [1] to 30 in-lb (3.4 N·m) - 50 in-lb (5.6 N·m).

SUBTASK 78-31-02-410-022-G00

(3) For the lower actuator access door [2], do the steps that follow:

(a) Make sure that the installation surfaces are clean.

(b) Use sealant, A50110 to create a prepack or injection seal between the access door [2] and the installation surface on the thrust reverser.

(c) Put the access doors [2] in the correct position to align the bolt holes.

Number	Name/Location
415FL	Left Thrust Reverser Actuator (Lower), Engine 1
416FR	Right Thrust Reverser Actuator (Lower), Engine 1
425FL	Left Thrust Reverser Actuator (Lower), Engine 2
426FR	Right Thrust Reverser Actuator (Lower), Engine 2

(d) Install the six bolts [1] onto the access door [2].

1) Tighten the bolts [1] to 30 in-lb (3.4 N·m) - 50 in-lb (5.6 N·m).

SUBTASK 78-31-02-410-019-G00

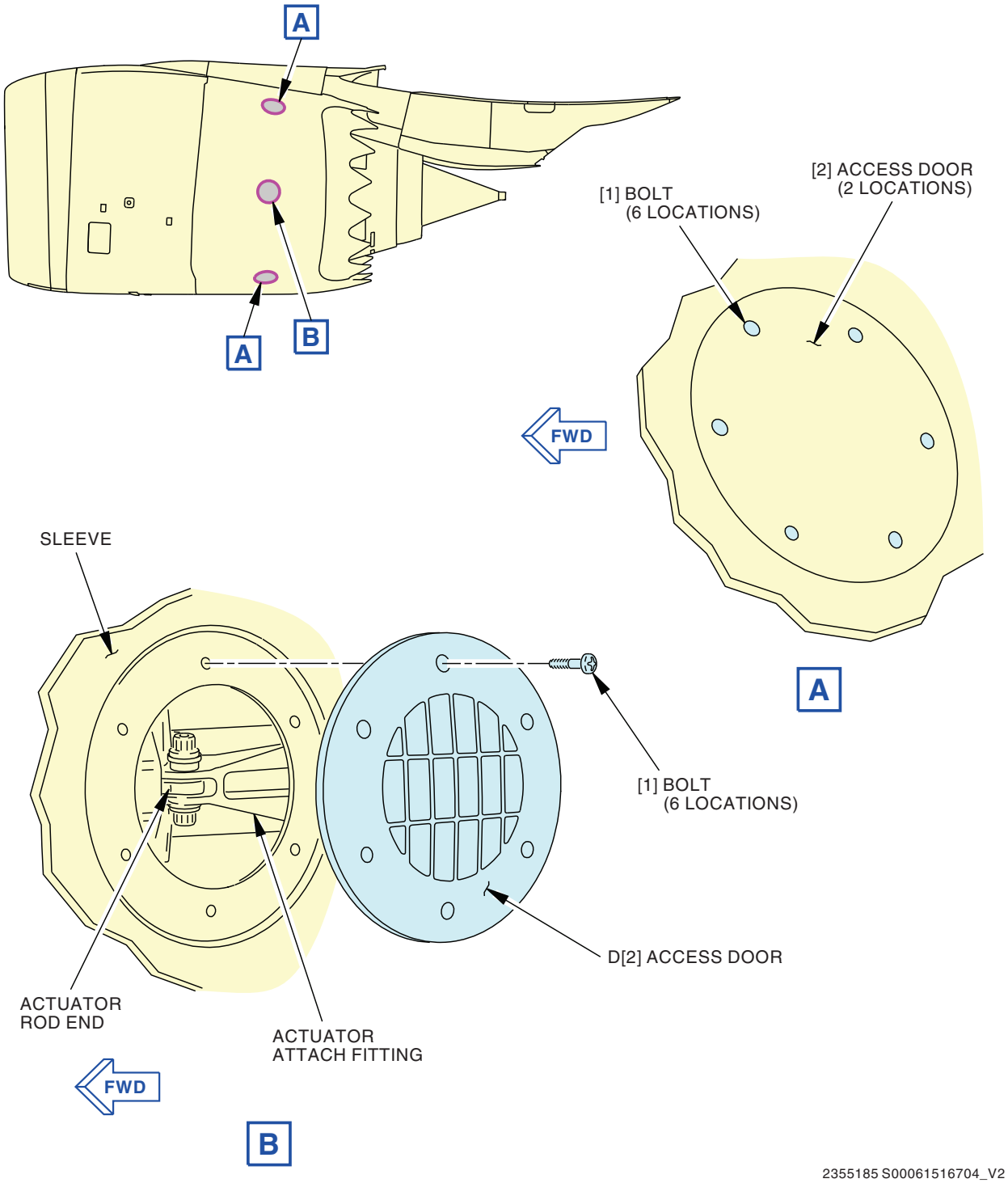
(4) If removed, install the lower Krueger flap deflector (TASK 78-31-09-420-801-G00).

SUBTASK 78-31-02-440-005-G00

(5) Do the activation procedure for the thrust reverser (TASK 78-31-00-440-801-G00).

——— END OF TASK ———

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Thrust Reverser Actuator Attach Fitting Visual Inspection
Figure 605/78-31-02-990-807-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-31-02-200-804-G00

6. Thrust Reverser Actuator Attach Fitting Detailed Inspection

(Figure 606)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions for a detailed inspection of the thrust reverser actuator attach fittings.

(a) The thrust reverser actuator attach fittings are installed on the translating sleeve.

(b) There are three actuator attach fittings per thrust reverser half.

B. References

Reference	Title
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-00-700-802-G00	Thrust Reverser System Normal Operational Test (P/B 501)
78-31-02-000-801-G00	Translating Sleeve Removal (P/B 401)
78-31-02-400-801-G00	Translating Sleeve Installation (P/B 401)
78-31-09-010-801-G00	Krueger Flap Deflector Removal (P/B 401)
78-31-09-420-801-G00	Krueger Flap Deflector Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
A00803	Sealant - Firewall - Hydraulic Fluid Resistant	BMS5-63 Type I
A50110	Sealant - Fuel Tank	BMS5-45 Class B-2
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)
C00259	Coating - Chemical And Solvent Resistant Finish, Corrosion Inhibiting Primer	BMS10-11 Type I
C50294	Alodine - 1000L, 1001 or 1500. Clear Film.	BAC5719, Type II, Class B. MIL-DTL-5541, Class3. F-17.11.
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

D. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

E. Access Panels

Number	Name/Location
415DL	Left Thrust Reverser Actuator (Upper), Engine 1
415EL	Left Thrust Reverser Actuator (Middle), Engine 1

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(Continued)

Number	Name/Location
415FL	Left Thrust Reverser Actuator (Lower), Engine 1
416DR	Right Thrust Reverser Actuator (Upper), Engine 1
416ER	Right Thrust Reverser Actuator (Middle), Engine 1
416FR	Right Thrust Reverser Actuator (Lower), Engine 1
425DL	Left Thrust Reverser Actuator (Upper), Engine 2
425EL	Left Thrust Reverser Actuator (Middle), Engine 2
425FL	Left Thrust Reverser Actuator (Lower), Engine 2
426DR	Right Thrust Reverser Actuator (Upper), Engine 2
426ER	Right Thrust Reverser Actuator (Middle), Engine 2
426FR	Right Thrust Reverser Actuator (Lower), Engine 2

F. Prepare for the Inspection

SUBTASK 78-31-02-040-004-G00

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(1) Do the deactivation procedure for the thrust reverser for ground maintenance (TASK 78-31-00-040-801-G00).

SUBTASK 78-31-02-010-019-G00

(2) To remove the upper actuator access doors [2], do these steps:

Number	Name/Location
416DR	Right Thrust Reverser Actuator (Upper), Engine 1
425DL	Left Thrust Reverser Actuator (Upper), Engine 2

(a) Remove the lower Krueger flap deflector (TASK 78-31-09-010-801-G00).

(b) Remove the six bolts [1].

SUBTASK 78-31-02-860-004-G00

(3) To remove the upper actuator access doors [2], remove the six bolts [1].

Number	Name/Location
415DL	Left Thrust Reverser Actuator (Upper), Engine 1
426DR	Right Thrust Reverser Actuator (Upper), Engine 2

SUBTASK 78-31-02-010-022-G00

(4) To remove the middle actuator access doors [2], remove the six bolts [1].

Number	Name/Location
415EL	Left Thrust Reverser Actuator (Middle), Engine 1
416ER	Right Thrust Reverser Actuator (Middle), Engine 1
425EL	Left Thrust Reverser Actuator (Middle), Engine 2
426ER	Right Thrust Reverser Actuator (Middle), Engine 2

SUBTASK 78-31-02-010-023-G00

(5) To remove the lower actuator access doors [2], remove the six bolts [1].

Number	Name/Location
415FL	Left Thrust Reverser Actuator (Lower), Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
416FR	Right Thrust Reverser Actuator (Lower), Engine 1
425FL	Left Thrust Reverser Actuator (Lower), Engine 2
426FR	Right Thrust Reverser Actuator (Lower), Engine 2

SUBTASK 78-31-02-020-004-G00

CAUTION

DO NOT LET THE ROD ENDS OF THE HYDRAULIC ACTUATOR TURN WHEN YOU REMOVE THE BOLTS. IF THE ROD END TURNS, IT CAN CAUSE DAMAGE TO THE HYDRAULIC ACTUATOR AND THRUST REVERSER STRUCTURE.

(6) Disconnect the actuator rod end from the actuator attach fitting on the translating sleeve (TASK 78-31-02-000-801-G00).

SUBTASK 78-31-02-860-002-G00

(7) Move the translating sleeve away from the actuator rod 2 in. (51 mm) to 3 in. (76 mm) to get access for the inspection.

G. Thrust Reverser Actuator Attach Fitting Detailed Inspection

SUBTASK 78-31-02-211-001-G00

(1) Examine the wear spacer [3] for these conditions:

(a) Tears

1) Not permitted.

(b) Holes

1) Not permitted.

(c) Disbonds

1) Not permitted.

(d) Thickness reduction

1) Thickness less than 0.020 in. (0.508 mm) is not permitted.

(e) If you find a condition that is not permitted, replace the wear spacer [3].

SUBTASK 78-31-02-211-002-G00

(2) Examine the applicable thrust reverser actuator attach fitting for damage:

(a) Do a detailed inspection on the attach fitting for damage caused by the rod end to the lower side of the fitting lug.

1) The lug thickness less than 0.20 in. (5.08 mm) is not permitted.

2) If the minimum lug thickness is more than 0.20 in. (5.08 mm), do the steps that follow to repair the attach fitting:

a) Remove the wear spacer [3] from the attach fitting.

b) Remove the surface damage to a maximum depth of 0.005 in. (0.127 mm) and a minimum radius of 1.0 in. (25.4 mm).

c) Surface finish to 125 micro inch or better the rework areas.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING MAKE SURE TO PREVENT ACETONE CONTACT WITH SKIN. ACETONE IRRITATES SKIN AND IS EXPLOSIVE. USE FUME CONTROL OR USE A RESPIRATOR TO PREVENT INJURY.

d) Clean the attach fitting with a clean cotton wiper, G00034, moist with solvent, B00062.

<1> Wipe the solvent before it becomes dry with another clean cloth wiper.

e) Apply Alodine, C50294, and then apply one coat of primer, C00259, to the attach fitting, except bushing holes.

f) Apply faying surface seal with sealant, A00803, on the etched surface of the wear spacer.

<1> Install the wear spacer [3] to the attach fitting.

g) Remove all tools and items used to do the repair.

SUBTASK 78-31-02-211-003-G00

(3) If the surface damage is more than the allowable limits, replace the actuator attach fitting.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-02-860-003-G00

CAUTION DO NOT TURN THE ACTUATOR ROD END. THE ACTUATOR LENGTH IS SET. IF YOU TURN THE ACTUATOR ROD END, DAMAGE TO THE ACTUATOR OR THE STRUCTURE CAN OCCUR.

(1) Move the translating sleeve to align the actuator rod end with the clevis on the attach fitting.

SUBTASK 78-31-02-420-005-G00

(2) Connect the actuator rod end to the actuator attach fitting on the sleeve (TASK 78-31-02-400-801-G00).

SUBTASK 78-31-02-860-005-G00

(3) For the upper actuator, put the access doors [2] in the correct position to align the bolt holes.

Number	Name/Location
415DL	Left Thrust Reverser Actuator (Upper), Engine 1
416DR	Right Thrust Reverser Actuator (Upper), Engine 1
425DL	Left Thrust Reverser Actuator (Upper), Engine 2
426DR	Right Thrust Reverser Actuator (Upper), Engine 2

(a) Install the six bolts [1] onto the access door [2].

1) Tighten the bolts [1] to 30 in-lb (3.4 N·m) - 50 in-lb (5.6 N·m).

SUBTASK 78-31-02-410-023-G00

(4) For the middle actuator, put the access doors [2] in the correct position to align the bolt holes.

Number	Name/Location
415EL	Left Thrust Reverser Actuator (Middle), Engine 1
416ER	Right Thrust Reverser Actuator (Middle), Engine 1
425EL	Left Thrust Reverser Actuator (Middle), Engine 2
426ER	Right Thrust Reverser Actuator (Middle), Engine 2

(a) Install the six bolts [1] onto the access door [2].

1) Tighten the bolts [1] to 30 in-lb (3.4 N·m) - 50 in-lb (5.6 N·m).

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SUBTASK 78-31-02-410-024-G00

(5) For the lower actuator access door [2], do the steps that follow:

(a) Make sure that the installation surfaces are clean.

(b) Use sealant, A50110, to create a prepack or injection seal between the access door [2] and the installation surface on the thrust reverser.

(c) Put the access doors [2] in the correct position to align the bolt holes.

Number	Name/Location
415FL	Left Thrust Reverser Actuator (Lower), Engine 1
416FR	Right Thrust Reverser Actuator (Lower), Engine 1
425FL	Left Thrust Reverser Actuator (Lower), Engine 2
426FR	Right Thrust Reverser Actuator (Lower), Engine 2

(d) Install the six bolts [1] onto the access door [2].

1) Tighten the bolts [1] to 30 in-lb (3.4 N·m) - 50 in-lb (5.6 N·m).

SUBTASK 78-31-02-410-020-G00

(6) If removed, install the lower Krueger flap deflector (TASK 78-31-09-420-801-G00).

SUBTASK 78-31-02-440-006-G00

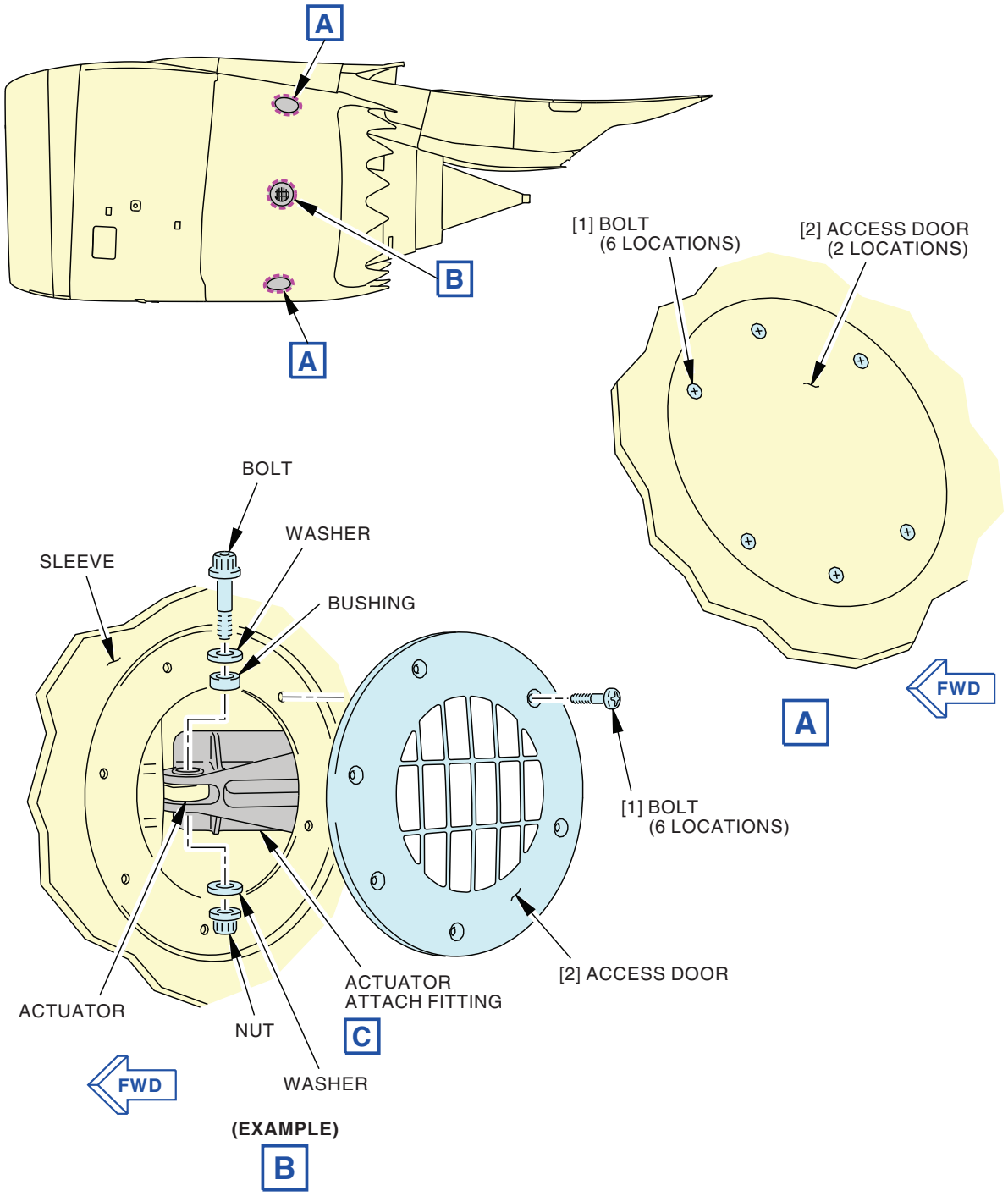
(7) Do the activation procedure for the thrust reverser (TASK 78-31-00-440-801-G00).

SUBTASK 78-31-02-710-002-G00

(8) Move the thrust reverser through the deploy and stow cycles at least 3 times until the sleeves move smoothly (TASK 78-31-00-700-802-G00).

——— END OF TASK ———

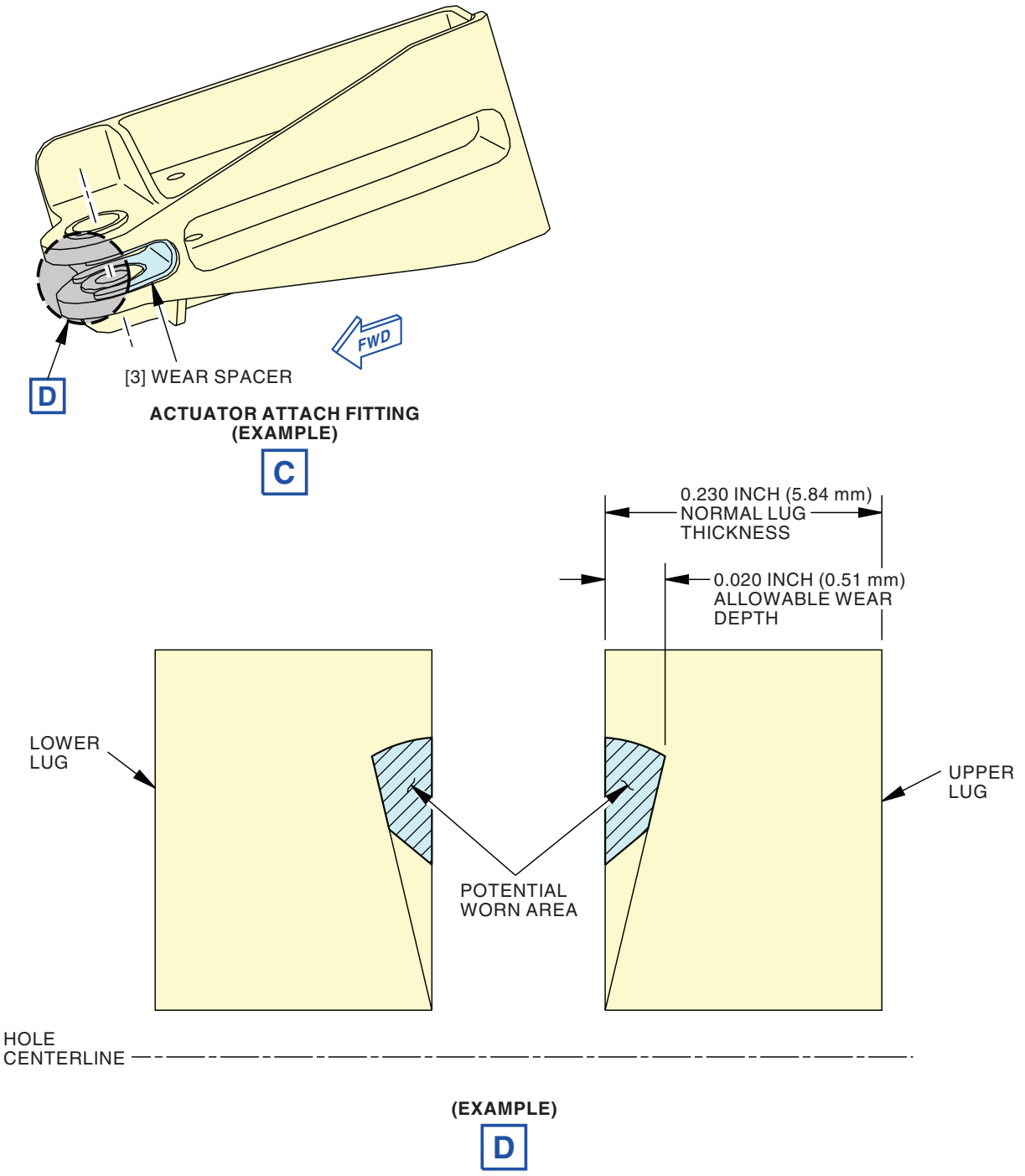
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Thrust Reverser Actuator Attach Fitting Detailed Inspection
Figure 606/78-31-02-990-808-G00 (Sheet 1 of 2)

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Thrust Reverser Actuator Attach Fitting Detailed Inspection
Figure 606/78-31-02-990-808-G00 (Sheet 2 of 2)

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TASK 78-31-02-210-802-G00

7. Engine Thrust Reverser Hinge Beam and Latch Beam Secondary Track and Translating Sleeve Secondary Slider General Visual Inspection for Wear with the Thrust Reverser Deployed

(Figure 607)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task provides instructions to do general visual inspection of engine thrust reverser hinge beam and latch beam secondary track and translating sleeve secondary slider.

B. References

Reference	Title
78-31-00-980-801-G00	Thrust Reverser Operation - Extend (Selection) (P/B 201)
78-31-00-980-802-G00	Thrust Reverser Operation - Retract (Selection) (P/B 201)

C. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

D. Prepare for the Inspection

SUBTASK 78-31-02-860-008-G00

(1) Do this task: Thrust Reverser Operation - Extend (Selection), TASK 78-31-00-980-801-G00.

E. Inspection

SUBTASK 78-31-02-210-010-G00

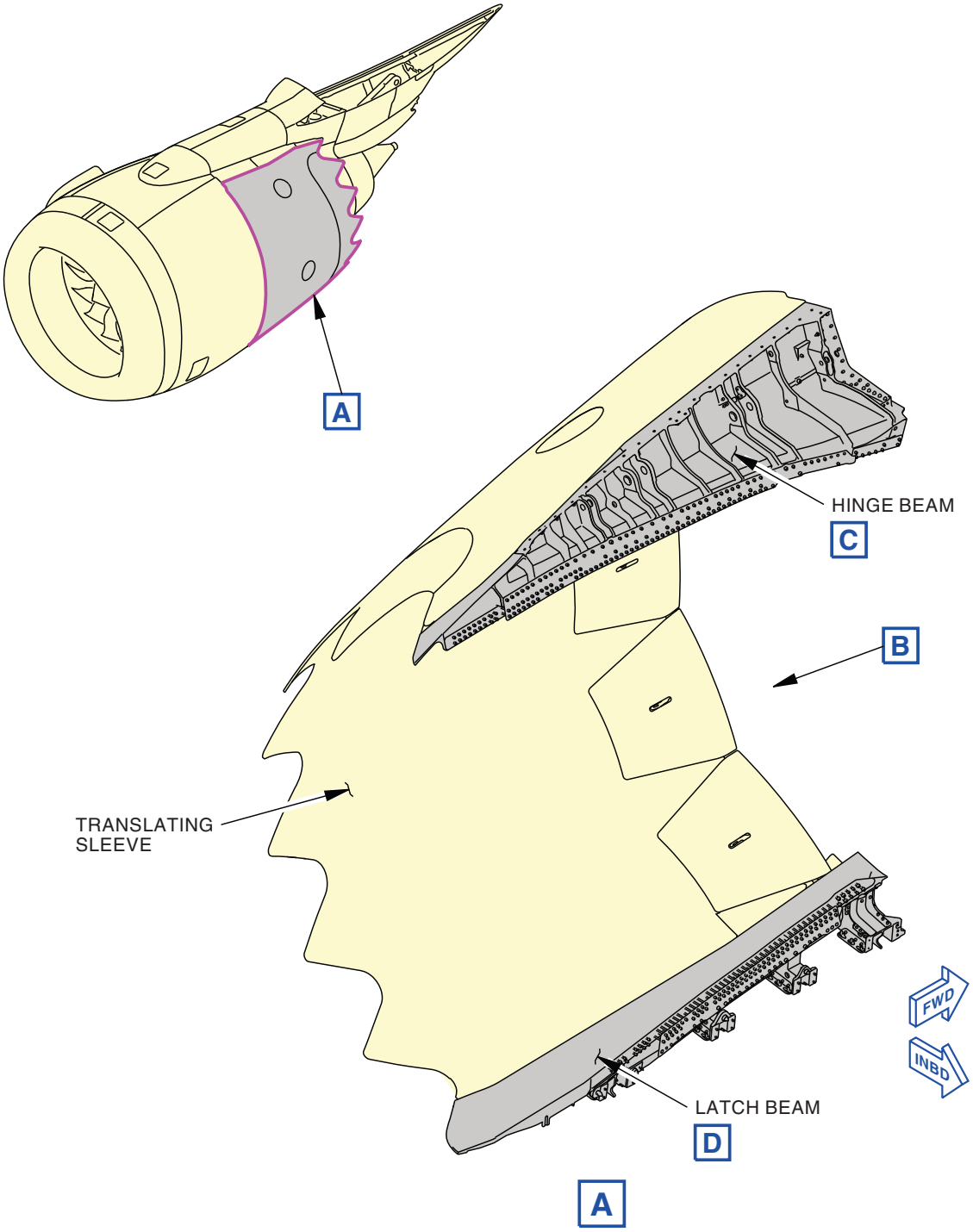
(1) Do general visual inspection of the engine thrust reverser hinge beam and latch beam secondary track and translating sleeve secondary slider for wear with the thrust reverser deployed.

SUBTASK 78-31-02-860-009-G00

(2) Do this task: Thrust Reverser Operation - Retract (Selection), TASK 78-31-00-980-802-G00.

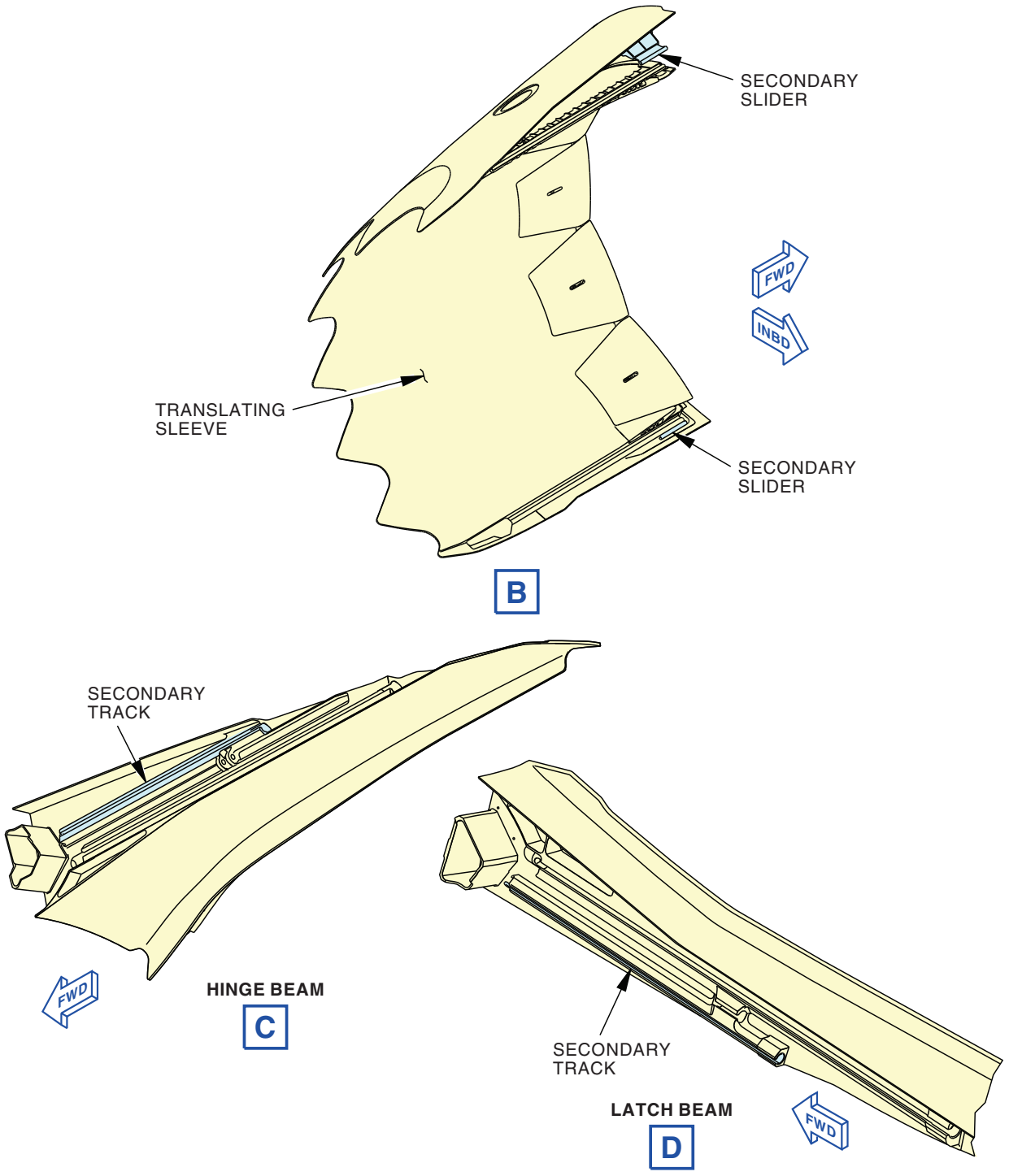
——— END OF TASK ———

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Engine Thrust Reverser Hinge Beam and Latch Beam Secondary Track and Translating Sleeve Secondary Slider General Visual Inspection
Figure 607/78-31-02-990-813-G00 (Sheet 1 of 2)

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Engine Thrust Reverser Hinge Beam and Latch Beam Secondary Track and Translating Sleeve Secondary Slider General Visual Inspection
Figure 607/78-31-02-990-813-G00 (Sheet 2 of 2)

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THRUST REVERSER HYDRAULIC ACTUATORS - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Thrust Reverser Hydraulic Actuator Removal (selection)

(2) Thrust Reverser Hydraulic Actuator Installation (selection)

(3) Upper Locking Hydraulic Actuator Removal

(4) Upper Locking Hydraulic Actuator Installation

(5) Middle Locking Hydraulic Actuator Removal

(6) Middle Locking Hydraulic Actuator Installation

(7) Lower Hydraulic Actuator Removal

(8) Lower Hydraulic Actuator Installation.

TASK 78-31-03-000-801-G00

2. Thrust Reverser Hydraulic Actuator Removal (Selection)

A. Procedure

SUBTASK 78-31-03-020-001-G00

(1) Do the procedure for the applicable thrust reverser hydraulic actuator:

(a) Do this task: Upper Locking Hydraulic Actuator Removal, TASK 78-31-03-000-802-G00.

(b) Do this task: Middle Locking Hydraulic Actuator Removal, TASK 78-31-03-000-803-G00.

(c) Do this task: Lower Hydraulic Actuator Removal, TASK 78-31-03-000-804-G00.

——— END OF TASK ———

TASK 78-31-03-400-801-G00

3. Thrust Reverser Hydraulic Actuator Installation (Selection)

A. Procedure

SUBTASK 78-31-03-420-001-G00

(1) Do the procedure for the applicable thrust reverser hydraulic actuator:

(a) Do this task: Upper Locking Hydraulic Actuator Installation, TASK 78-31-03-400-802-G00.

(b) Do this task: Middle Locking Hydraulic Actuator Installation, TASK 78-31-03-400-803-G00.

(c) Do this task: Lower Hydraulic Actuator Installation, TASK 78-31-03-400-804-G00.

——— END OF TASK ———

TASK 78-31-03-000-802-G00

4. Upper Locking Hydraulic Actuator Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the upper locking hydraulic actuator.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)

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(Continued)

Reference	Title
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
29-21-00-000-802	Standby Hydraulic System Power Removal (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-980-803-G00	Thrust Reverser Operation - Extend (Manual Procedure) (P/B 201)
78-31-00-980-804-G00	Thrust Reverser Operation - Retract (Manual Procedure) (P/B 201)
78-31-09-010-801-G00	Krueger Flap Deflector Removal (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-2439	Tool - Thrust Reverser Actuator, Installation/Removal Part #: C78025-30 Supplier: 81205
SPL-14551	Lock - Thrust Reverser Opening Actuator, Fully Extended, LEAP-1B Part #: C78035-1 Supplier: 81205
STD-1110	Container - Hydraulic Fluid Resistant, 5 Gallon (19 Liter)
STD-7895	Wrench - Allen, 3/16 inch
STD-14286	Allen Wrench - Hex, 1/4 Inch

D. Consumable Materials

Reference	Description	Specification
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

E. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415DL	Left Thrust Reverser Actuator (Upper), Engine 1
416DR	Right Thrust Reverser Actuator (Upper), Engine 1

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(Continued)

Number	Name/Location
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425DL	Left Thrust Reverser Actuator (Upper), Engine 2
426DR	Right Thrust Reverser Actuator (Upper), Engine 2

G. Prepare for the Removal

SUBTASK 78-31-03-040-032-G00

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(1) Do the thrust reverser deactivation for ground maintenance (TASK 78-31-00-040-801-G00).

SUBTASK 78-31-03-860-001-G00

(2) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 78-31-03-040-002-G00

(3) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 78-31-03-040-003-G00

(4) Remove the hydraulic power from the applicable engine as follows (TASK 29-11-00-860-805):

(a) For Engine 1, remove the system A hydraulic power.

(b) For Engine 2, remove the system B hydraulic power.

SUBTASK 78-31-03-040-004-G00

(5) Do this task: Standby Hydraulic System Power Removal, TASK 29-21-00-000-802.

SUBTASK 78-31-03-040-005-G00

(6) Do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 78-31-03-010-017-G00

(7) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

(a) On the applicable engine, open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 78-31-03-010-023-G00

(8) Remove the applicable access doors:

Number	Name/Location
415DL	Left Thrust Reverser Actuator (Upper), Engine 1
426DR	Right Thrust Reverser Actuator (Upper), Engine 2

(a) Remove the bolts [14].

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SUBTASK 78-31-03-010-029-G00

(9) Remove the applicable access doors:

Number	Name/Location
416DR	Right Thrust Reverser Actuator (Upper), Engine 1
425DL	Left Thrust Reverser Actuator (Upper), Engine 2

(a) Remove the lower Krueger flap deflector (TASK 78-31-09-010-801-G00).

(b) Remove the bolts [14].

SUBTASK 78-31-03-980-006-G00

(10) Unlock the upper locking actuator as follows (Figure 401 (Sheet 1), View B):

NOTE: Make sure that the thrust reverser is fully closed and retracted. If it is necessary, retract the thrust reverser hydraulically. If there is a small gap, the thrust reverser is not fully closed and you cannot move the manual unlock lever.

(a) Move and hold forward the manual unlock lever [5].

(b) Push the detent pin in.

(c) Release the manual unlock lever.

SUBTASK 78-31-03-010-002-G00

(11) Expose the gimbal pins and prepare for the use of the actuator removal and installation tool as follows:

NOTE: Removal of the gimbal pins from the aft side of the torque box and to use the actuator removal and installation tool requires extension of the thrust reverser sleeve.

CAUTION DO NOT MANUALLY EXTEND THE INBOARD THRUST REVERSER SLEEVE MORE THAN THE PERMITTED LIMIT. MAKE SURE THAT THE SLEEVE DOES NOT TOUCH THE WING LEADING EDGE WHEN YOU EXTEND THE THRUST REVERSER. DAMAGE CAN OCCUR.

(a) For the inboard thrust reverser sleeve, manually extend the sleeve no more than 15 in. (38 cm) from the forward edge of the torque box (TASK 78-31-00-980-803-G00).

(b) For the outboard thrust reverser sleeve, manually fully extend the thrust reverser sleeve (TASK 78-31-00-980-803-G00).

NOTE: The outboard thrust reverser sleeve will not touch the leading edge of the wing. The tool will be easier to use with the sleeve fully extended.

SUBTASK 78-31-03-020-002-G00

WARNING DO NOT INSTALL PRESSURE CAPS TO THE HYDRAULIC LINES THAT CONNECT TO CONTROL VALVE MODULE PORTS TO DEACTIVATE THE THRUST REVERSER. THE THRUST REVERSER CAN OPERATE WHEN PRESSURIZED. IF YOU DO NOT OBEY, INJURY TO PERSONNEL AND DAMAGE TO THE EQUIPMENT CAN OCCUR.

WARNING MAKE SURE THAT YOU WEAR PROTECTIVE CLOTHES AND GLOVES WHEN YOU DO WORK ON THE HYDRAULIC SYSTEM. HYDRAULIC FLUID CAN CAUSE A LEAK FROM THE OPEN PORTS OF THE COMPONENT OR FROM THE HYDRAULIC LINES. INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

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(WARNING PRECEDES)

CAUTION DO NOT LET HYDRAULIC FLUID GET ON EQUIPMENT. IF HYDRAULIC FLUID GETS ON EQUIPMENT, IMMEDIATELY CLEAN IT FROM THE EQUIPMENT. HYDRAULIC FLUID CAN CAUSE DAMAGE TO EQUIPMENT.

(12) Drain the hydraulic fluid from the hydraulic lines at the lower hydraulic actuator as follows (TASK 78-31-03-000-804-G00):

NOTE: To decrease hydraulic fluid spray when the coupling nuts are loosened, wrap cotton wiper, G00034, around the wrench, nut and hydraulic line.

(a) Wrap cotton wiper, G00034, around the electrical connector on the sync lock receptacle on the lower actuator.

NOTE: The cloth will catch the hydraulic fluid and prevent damage to the electrical connector.

(b) Loosen the two bolts on each of the two clamp blocks that hold the lower hydraulic retract line.

NOTE: To remove the sleeve fitting on the hydraulic retract line from the actuator port, the hydraulic retract line must be carefully flexed. If the clamp blocks are loose, the hydraulic retract line can be flexed easier.

(c) Disconnect the hydraulic retract line at the lower actuator.

1) Carefully remove the sleeve fitting on the hydraulic retract line from the actuator port.

2) Let the hydraulic fluid drain into a 5 gallon (19 liter) hydraulic fluid resistant container, STD-1110.

CAUTION USE TWO WRENCHES TO LOOSEN THE TUBE COUPLING NUTS. USE ONE TO HOLD THE MATING PART, AND THE OTHER TO LOOSEN THE COUPLING NUT. IF YOU DO NOT USE TWO WRENCHES, DAMAGE TO THE TUBES AND MATING PART CAN OCCUR.

(d) Disconnect the coupling nut on the sync shaft tubing at the lower actuator.

1) Let the hydraulic fluid drain into the 5 gallon (19 liter) hydraulic fluid resistant container, STD-1110.

CAUTION USE TWO WRENCHES TO TIGHTEN THE COUPLING NUT. USE ONE TO HOLD THE FITTING, AND THE OTHER TO TURN THE COUPLING NUT. IF YOU DO NOT USE TWO WRENCHES, DAMAGE TO THE EQUIPMENT CAN OCCUR.

(e) Re-connect the coupling nut for the lower sync shaft to the lower actuator.

1) Tighten the coupling nut to 855 in-lb (96.6 N·m) - 945 in-lb (106.8 N·m).

2) Loosen the coupling nut.

3) Tighten the coupling nut again to 855 in-lb (96.6 N·m) - 945 in-lb (106.8 N·m).

CAUTION USE TWO WRENCHES TO TIGHTEN THE COUPLING NUT. USE ONE TO HOLD THE FITTING, AND THE OTHER TO TURN THE COUPLING NUT. IF YOU DO NOT USE TWO WRENCHES, DAMAGE TO THE EQUIPMENT CAN OCCUR.

(f) Re-connect the coupling nut for the lower retract line to the lower actuator.

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1) Tighten the coupling nut to 256 in-lb (28.9 N·m) - 283 in-lb (32.0 N·m).

2) Loosen the coupling nut.

3) Tighten the coupling nut again to 256 in-lb (28.9 N·m) - 283 in-lb (32.0 N·m).

(g) Tighten the two bolts on each of the two clamp blocks that hold the lower hydraulic retract line.

1) Tighten the bolts to 30 in-lb (3.4 N·m) - 35 in-lb (4.0 N·m).

H. Upper Locking Actuator Removal

SUBTASK 78-31-03-020-003-G00

(1) Disconnect the electrical connectors from the Linear Variable Differential Transformer (LVDT) receptacle on the upper locking actuator [1] as follows Figure 401 (Sheet 2):

(a) For the left thrust reverser, disconnect the electrical connectors, D30072 and D30076.

(b) For the right thrust reverser, disconnect the electrical connectors, D30074 and D30078.

SUBTASK 78-31-03-020-004-G00

(2) Remove the sleeve lock proximity sensor [8] from the upper locking actuator [1] ((Figure 401 (Sheet 2)) as follows:

(a) Remove the jamnut [38] from the sleeve lock proximity sensor [8].

NOTE: It is necessary to remove only the jamnut adjacent to the target.

(b) Remove the sleeve lock proximity sensor [8] and key washer [39].

(c) Re-install the key washer [39] and jamnut [38] on the sleeve lock proximity sensor [8] for storage.

(d) Safety the sleeve lock proximity sensor [8] away from the actuator.

SUBTASK 78-31-03-020-006-G00

(3) Remove the two gimbal pins [19] that attach the upper locking actuator [1] to the gimbal ring on the aft side of the torque box as follows (Figure 401 (Sheet 3), View F):

(a) If you use the clamp assembly tool [42] (actuator tool, SPL-2439) to remove the two gimbal pins [19], do these steps (Figure 401 (Sheet 5), View I):

NOTE: Use a allen wrench, STD-7895 to turn the clamp block screws [40] and use a 1/4 inch allen wrench, STD-14286 to turn the adjustment screw [41].

1) Loosen the clamp block screws [40] sufficiently so that the clamp block will fit over the upper locking actuator [1].

2) Make sure that the adjustment screw [41] is retracted.

NOTE: The adjustment screw is on the forward end of the tool.

3) Tilt the aft end of the clamp assembly tool down and over the aft cascade support ring.

4) Lower the forward end of the clamp assembly tool until the clamp block engages the upper locking actuator [1].

5) Tighten the adjustment screw [41] to move the clamp block aft.

a) Move the clamp block until the aft edge touches where the diameter of the upper locking actuator [1] increases.

6) Tighten the clamp block screws [40].

7) Tighten the adjustment screw [41] to compress the fire seal [30] or fire seal [31] against the torque box.

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(b) Remove a bolt [20] and a washer [21] from each of the two gimbal pins [19].

(c) Remove the two gimbal pins [19].

(d) If the clamp assembly tool [42] (actuator tool, SPL-2439) was used, loosen the adjustment screw [41] and clamp block screw [40] to remove the tool.

SUBTASK 78-31-03-010-022-G00

(4) Retract the inboard and outboard thrust reverser sleeve to the stowed position (TASK 78-31-00-980-804-G00).

SUBTASK 78-31-03-020-028-G00

(5) Disconnect the rod end of the upper locking actuator [1] as follows (Figure 401 (Sheet 3), View E):

(a) Remove the nut [17], bolt [11], washer [16], washer [12], and bushing [13] that attach the rod end of the actuator to the attach fitting on the sleeve.

(b) Disconnect the rod end of the upper locking actuator [1] from the actuator attach fitting.

SUBTASK 78-31-03-020-035-G00

(6) Disconnect the hydraulic lines from the upper locking actuator [1] as follows (Figure 401 (Sheet 1), View B and Figure 401 (Sheet 2), View C):

(a) Disconnect the extend line flexhose [7] from the upper extend port.

(b) Disconnect the retract line flexhose [6] from the upper retract port.

(c) Loosen the two bolts on the clamp block that holds the upper hydraulic retract line [2].

NOTE: To remove the sleeve fitting on the hydraulic retract line from the actuator port, the hydraulic retract line must be carefully flexed. With the clamp blocks loose, the hydraulic retract line can be flexed easier.

(d) Disconnect the upper hydraulic retract line [2] from the union [3].

1) Carefully remove the sleeve fitting on the upper hydraulic retract line from the actuator port.

(e) Disconnect the sync shaft tubing at the upper locking actuator.

1) Move the coupling nut and tube adapter back along the tubing to get access to the sync shaft end.

2) Flex the sync shaft to remove it from the internal spline port in the upper locking actuator [1].

NOTE: If it is necessary, remove the supporting block to allow removal of the sync shaft.

CAUTION

DO NOT LET DIRT OR OTHER CONTAMINATION GET ON THE SYNC SHAFT. DAMAGE TO THE SYNC SHAFT CAN OCCUR.

3) Remove the sync shaft.

NOTE: Make sure that dirt or contamination does not get on the sync shaft.

(f) Remove the union [3] and packing [4] from the lower retract port (Figure 401 (Sheet 1), View B).

(g) Keep the union [3] for the subsequent installation.

(h) Discard the packing [4].

(i) Remove the union [9] and packing [10] from the upper retract port (Figure 401 (Sheet 2)).

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(j) Keep the union [9] for the installation.

(k) Discard the packing [10].

(l) Install protective covers on the hydraulic lines and the actuator ports.

SUBTASK 78-31-03-010-004-G00

(7) Do these steps to get the necessary clearance to remove the upper locking actuator [1]:

(a) For the inboard thrust reverser, open the thrust reverser.

1) Make sure the leading edge flaps are completely retracted.

NOTE: Without hydraulics to hold the flaps in the retract position, the weight of the flaps can cause them to extend a small amount.

WARNING

MAKE SURE THAT YOU MONITOR THE POSITION OF THE INBOARD THRUST REVERSER WHILE IT OPENS. MAKE SURE THAT THE INBOARD THRUST REVERSER DOES NOT TOUCH THE WING LEADING EDGE. INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

2) For the removal of the upper locking actuator [1] on the inboard thrust reverser, do this task: Open the Thrust Reverser (Selection), TASK 78-31-00-010-801-G00.

a) Make sure to install the actuator safety lock assembly, SPL-14551.

<1> Make sure to use safety lock PN C78035-3. This is the full open inboard tool.

SUBTASK 78-31-03-010-015-G00

(8) For the removal of the upper locking actuator [1] on the outboard thrust reverser, do this task: Open the Thrust Reverser (Selection), TASK 78-31-00-010-801-G00.

(a) Make sure to install the actuator safety lock assembly, SPL-14551.

1) Make sure to use safety lock PN C78035-2. This is the full open outboard tool.

NOTE: The outboard thrust reverser will not touch the leading edge of the wing.

SUBTASK 78-31-03-010-005-G00

CAUTION

DO NOT LET THE PUSH-ROD HIT THE AFT CASCADE-SUPPORT-RING OR THE TORQUE BOX. DAMAGE TO THE PUSH-ROD CAN OCCUR.

(9) Carefully remove the upper locking actuator [1].

NOTE: You will move the actuator and its push rod through an opening in the cascade support ring and torque box.

SUBTASK 78-31-03-020-007-G00

(10) Remove the fire shield [29] and the fire seal [30] from the upper locking actuator [1] (Figure 401 (Sheet 4), View H).

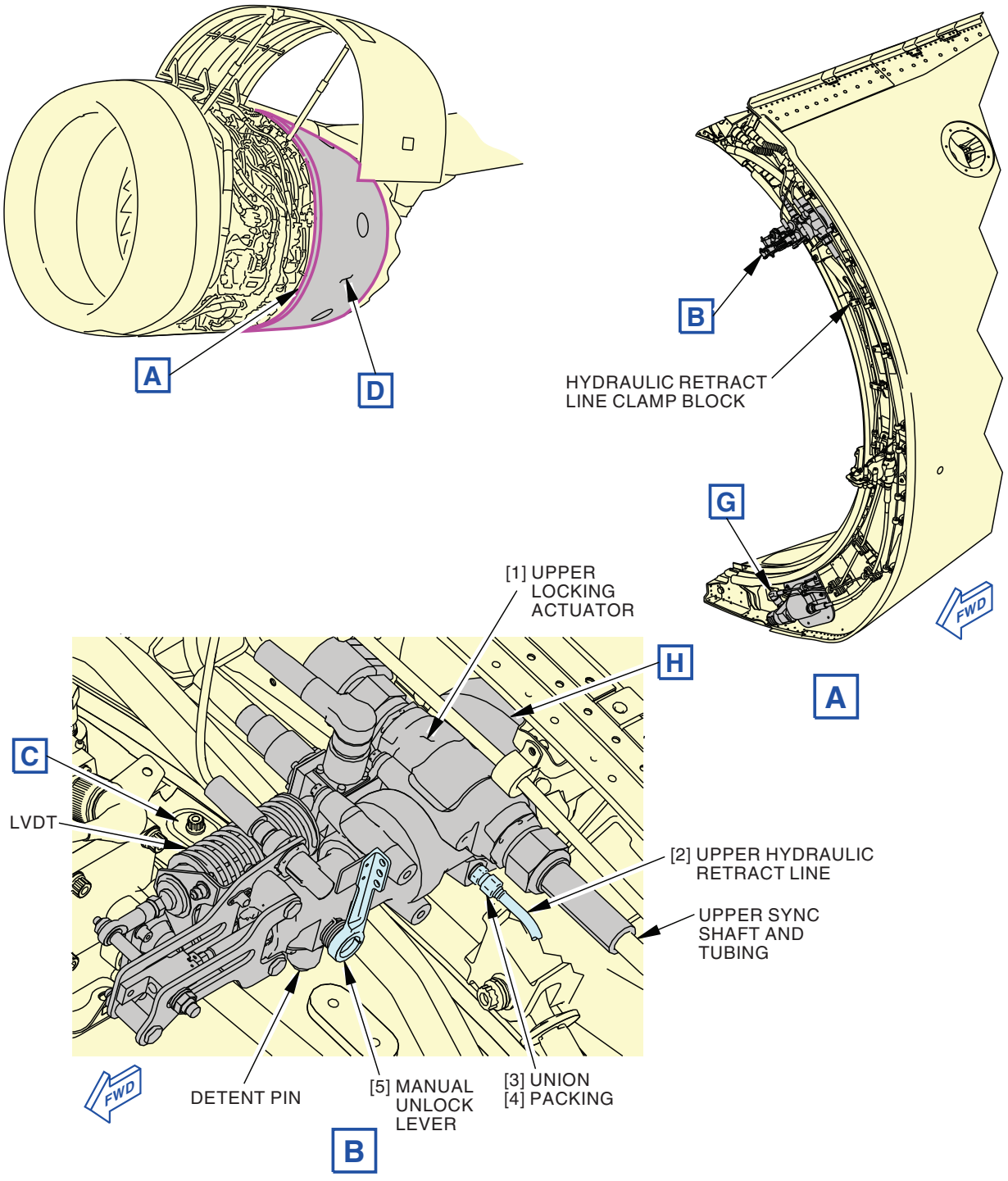
(a) Examine the fire seal [30] for damage.

(b) If you find no damage, keep the fire seal [30] for the installation.

(c) If you find damage, replace the fire seal [30].

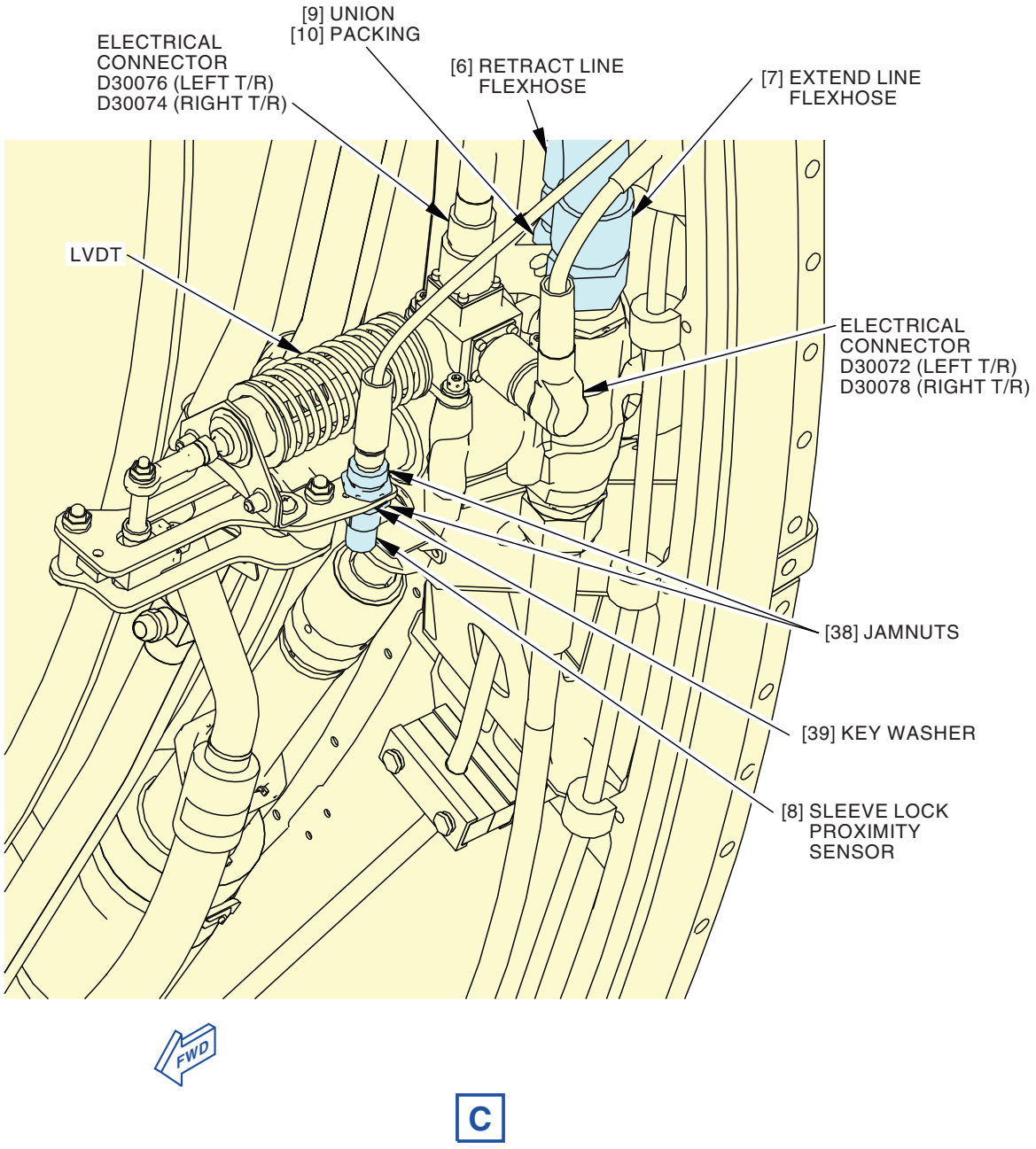
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



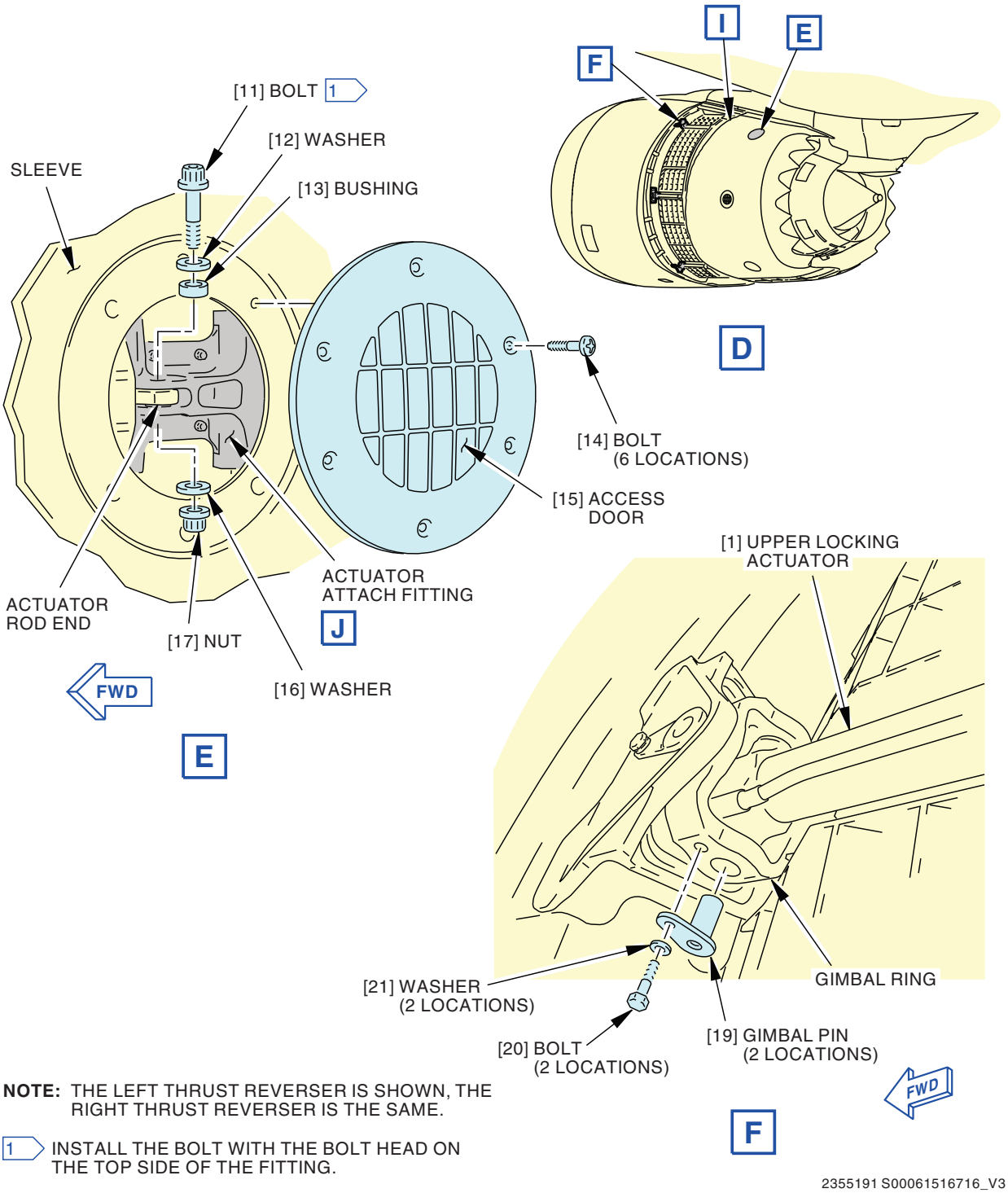
Upper Locking Actuator Installation
Figure 401/78-31-03-990-801-G00 (Sheet 1 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Upper Locking Actuator Installation
Figure 401/78-31-03-990-801-G00 (Sheet 2 of 6)

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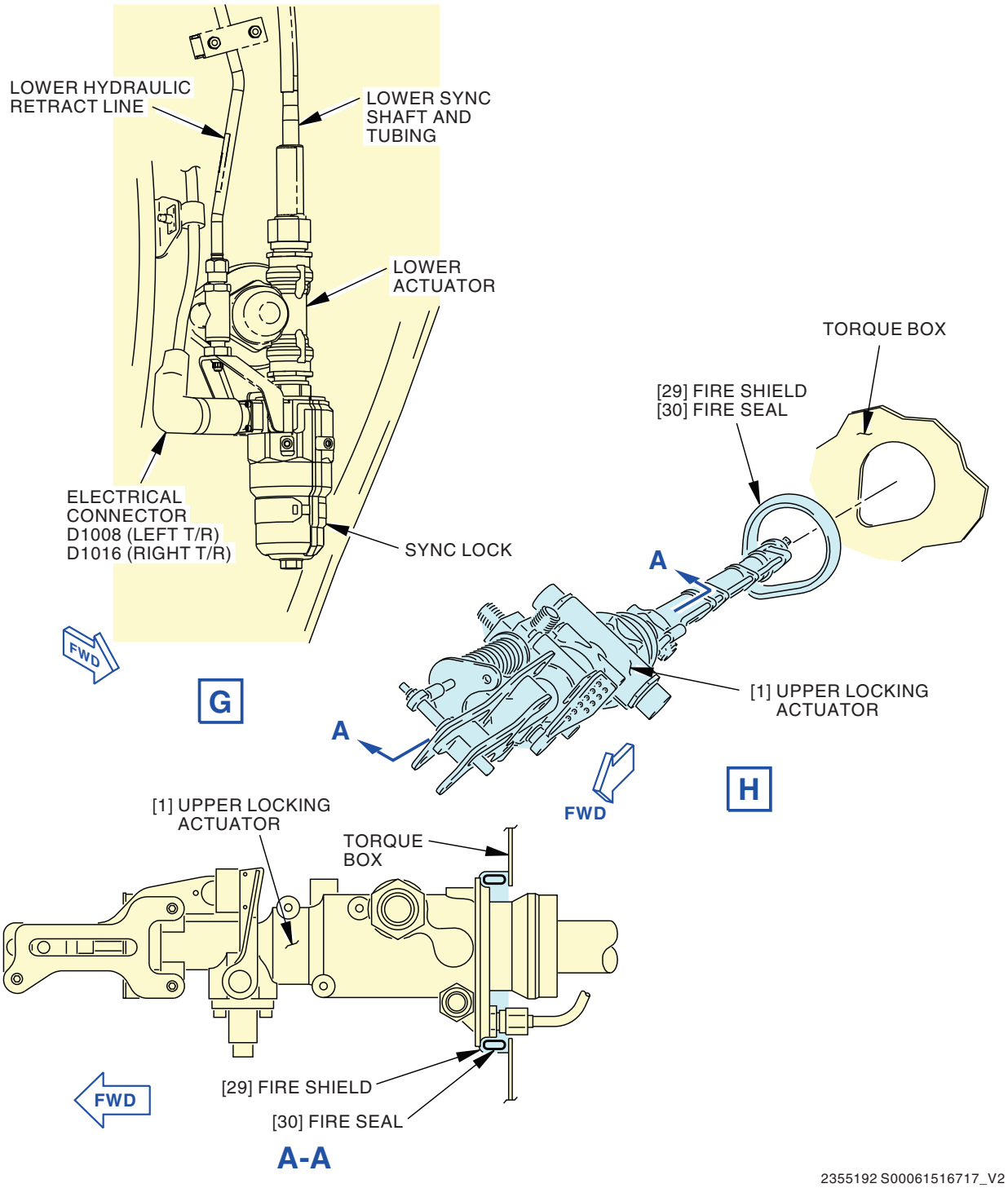
NOTE: THE LEFT THRUST REVERSER IS SHOWN, THE RIGHT THRUST REVERSER IS THE SAME.

1 INSTALL THE BOLT WITH THE BOLT HEAD ON THE TOP SIDE OF THE FITTING.

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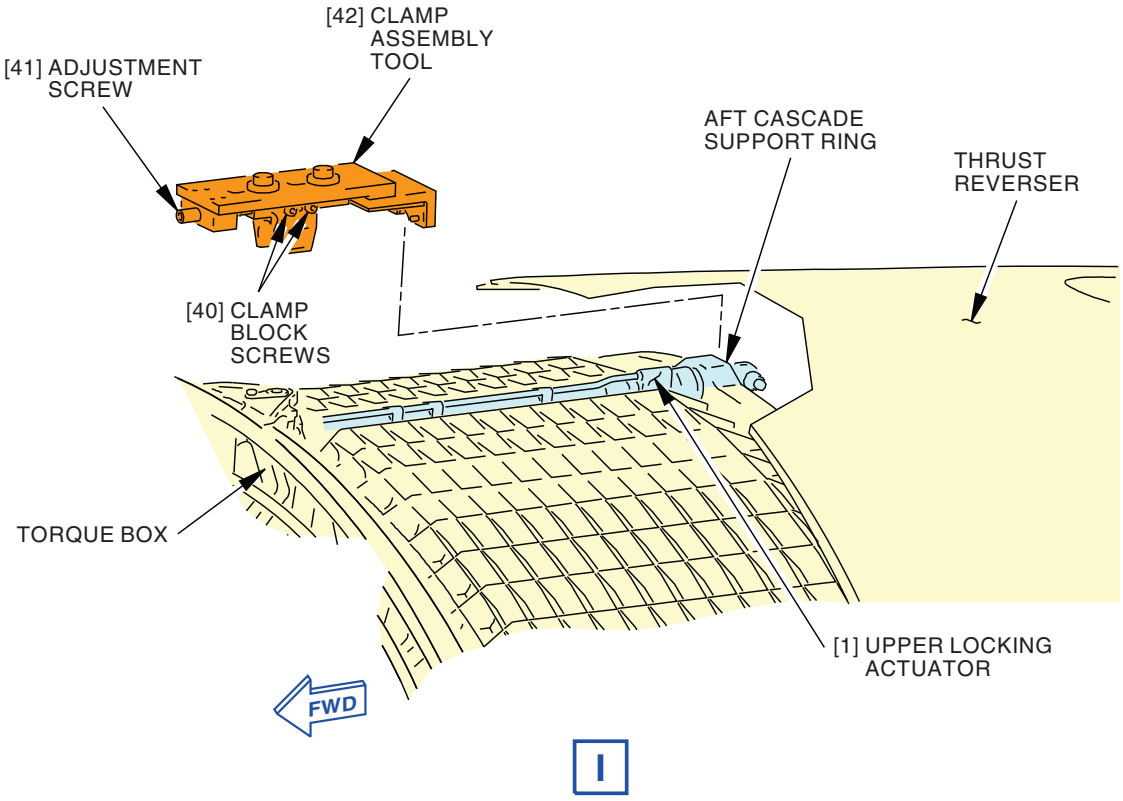
Upper Locking Actuator Installation
Figure 401/78-31-03-990-801-G00 (Sheet 3 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Upper Locking Actuator Installation
Figure 401/78-31-03-990-801-G00 (Sheet 4 of 6)

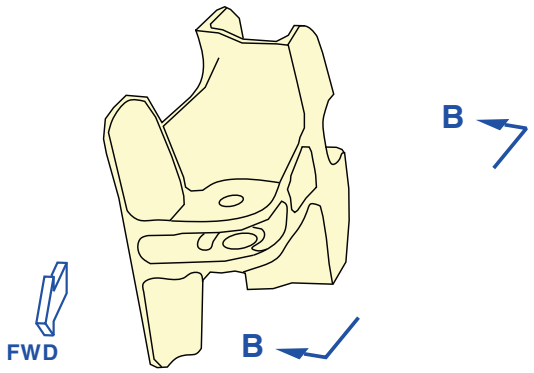
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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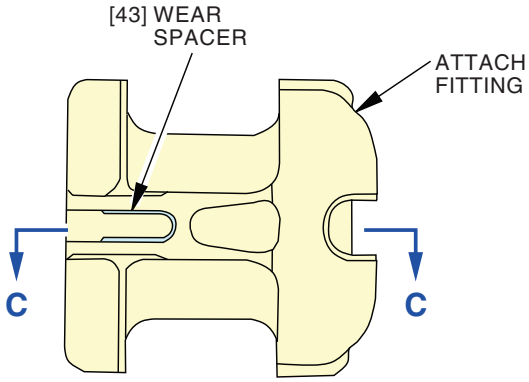
Upper Locking Actuator Installation
Figure 401/78-31-03-990-801-G00 (Sheet 5 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

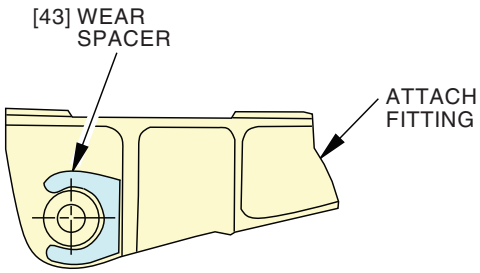


ACTUATOR ATTACH FITTING

J



B-B



C-C

2355194 S00061516719_V2

Upper Locking Actuator Installation
Figure 401/78-31-03-990-801-G00 (Sheet 6 of 6)

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TASK 78-31-03-400-802-G00

5. Upper Locking Hydraulic Actuator Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the upper locking hydraulic actuator.

B. References

Reference	Title
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
29-00-00-790-801	Hydraulic System External Leakage Check (P/B 601)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-00-700-801-G00	Thrust Reverser Operational Test (Selection) (P/B 501)
78-31-00-700-805-G00	Thrust Reverser Proximity Sensors Operational Test (P/B 501)
78-31-00-980-803-G00	Thrust Reverser Operation - Extend (Manual Procedure) (P/B 201)
78-31-00-980-804-G00	Thrust Reverser Operation - Retract (Manual Procedure) (P/B 201)
78-31-04-000-801-G00	Sync Shaft Removal (P/B 401)
78-31-04-200-801-G00	Sync Shaft Inspection (P/B 601)
78-31-04-400-801-G00	Sync Shaft Installation (P/B 401)
78-31-09-420-801-G00	Krueger Flap Deflector Installation (P/B 401)
78-34-03-400-801-G00	Thrust Reverser Sleeve Lock Proximity Sensor Installation (P/B 401)
78-34-03-800-801-G00	Sleeve Lock Proximity Sensor Adjustment and Test (P/B 501)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-2439	Tool - Thrust Reverser Actuator, Installation/Removal Part #: C78025-30 Supplier: 81205
SPL-9002	Lock Equipment - Thrust Reverser Maintenance Part #: B78009-26 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
A00803	Sealant - Firewall - Hydraulic Fluid Resistant	BMS5-63 Type I
D00054	Fluid - Hydraulic, Assembly Lubricant, MCS 352B	
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
D00633	Grease - Aircraft General Purpose	BMS3-33

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(Continued)

Reference	Description	Specification
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
4	Packing	78-31-03-01-150	PTF ALL
10	Packing	78-31-03-01-160	PTF ALL

F. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

G. Access Panels

Number	Name/Location
415DL	Left Thrust Reverser Actuator (Upper), Engine 1
416DR	Right Thrust Reverser Actuator (Upper), Engine 1
425DL	Left Thrust Reverser Actuator (Upper), Engine 2
426DR	Right Thrust Reverser Actuator (Upper), Engine 2

H. Install the Upper Locking Actuator

SUBTASK 78-31-03-420-002-G00

(1) Install the fire seal [30] and fire shield [29] on the upper locking actuator [1] as follows:

(a) Put the fire seal [30] in the fire shield [29].

(b) Put the fire seal [30] and fire shield [29] over the rod end of the upper locking actuator [1].

1) Make sure that the fire seal [30] contacts the forward side of the torque box.

SUBTASK 78-31-03-420-003-G00

CAUTION

DO NOT LET THE PUSH-ROD HIT THE AFT CASCADE-SUPPORT-RING OR THE TORQUE BOX. DAMAGE TO THE PUSH-ROD CAN OCCUR.

CAUTION

DO NOT LET THE ROD END TURN AFTER THE ACTUATOR IS UNLOCKED. IF YOU LET THE ROD END TURN IT WILL AFFECT VERY IMPORTANT INTERNAL MECHANISM ADJUSTMENT.

(2) Install the upper locking actuator [1]:

(a) Unlock the upper locking actuator [1].

1) If applicable, remove the rig shipping pin from the rig pin hole in the proximity sensor bracket and feedback rod end.

2) Move the manual unlock lever [5] forward.

3) Hold it in the forward position as you push the detent pin in.

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4) Release the manual unlock lever [5].

(b) To extend the rod end, pull the rod end out by hand.

NOTE: Do not let the rod end turn when you extend it manually. Make sure that you hold the rod end when you extend the actuator manually. Do not use hydraulic power to extend the rod end.

1) For the inboard and outboard thrust reverser, extend the rod end approximately the same length as the one removed.

NOTE: It is also acceptable to extend the rod 1 in. (3 cm) - 2 in. (5 cm) longer to facilitate installation of the rod end bolt.

(c) Carefully insert the rod end through the opening in the torque box and aft cascade support ring.

SUBTASK 78-31-03-420-027-G00

(3) Align the rod end of the upper locking actuator [1] with the clevis on the attach fitting.

(a) Make sure that the actuator attach fitting has a wear spacer [43].

1) If it is necessary, attach a wear spacer [43] into place with sealant, A00803.

NOTE: Seal around the edge of the fitting and inner sleeve.

(b) Install the bushing [13], washer [16], washer [12], bolt [11], and nut [17] that attach the rod end to the actuator attach fitting.

(c) Tighten the nut [17] to 370 in-lb (41.8 N·m) - 690 in-lb (78.0 N·m).

SUBTASK 78-31-03-410-001-G00

(4) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

NOTE: This will make the attachment of the rod end of the upper locking actuator and the installation of the access door easier.

SUBTASK 78-31-03-010-021-G00

(5) For the inboard thrust reverser sleeve, manually extend the sleeve no more than 15 in. (38 cm) from the forward edge of the torque box (TASK 78-31-00-980-803-G00).

(a) For the outboard thrust reverser sleeve, manually fully extend the thrust reverser sleeve (TASK 78-31-00-980-803-G00).

NOTE: The outboard thrust reverser sleeve will not touch the leading edge of the wing. The tool will be easier to use with the sleeve fully extended.

SUBTASK 78-31-03-420-004-G00

(6) Install the two gimbal pins [19] that attach the upper locking actuator [1] to the gimbal ring on the aft side of the torque box:

(a) Apply grease, D00633, on the shanks of the two gimbal pins [19].

(b) Align the attachment holes in the upper locking actuator [1] and the gimbal ring.

(c) If will you use the clamp assembly tool [42] (actuator tool, SPL-2439) to compress the fire seal [30] or fire seal [31] against the torque box, do these steps:

1) Loosen the clamp block screws [40] sufficiently so that the clamp block will fit over the upper locking actuator [1].

2) Make sure that the adjustment screw [41] is retracted.

NOTE: The adjustment screw is on the forward end of the tool.

3) Tilt the aft end of the clamp assembly tool down and over the aft cascade support ring.

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4) Lower the forward end of the clamp assembly tool until the clamp block engages the upper locking actuator [1].

5) Tighten the adjustment screw [41] to move the clamp block aft.

a) Move the clamp block until the aft edge touches where the diameter of the upper locking actuator [1] increases.

6) Tighten the clamp block screws [40].

7) Tighten the adjustment screw [41] to compress the fire seal [30] or fire seal [31] against the torque box.

8) Turn the housing one direction or the other to align the attachment holes.

(d) Manually compress the fire seal [30] or fire seal [31] against the torque box as follows:

1) Push aft on the forward end of the upper locking actuator [1] to compress the fire seal against the torque box.

2) Turn the housing one direction or the other to align the attachment holes.

(e) Install the two gimbal pins [19] as follows:

1) Install a washer [21] and a bolt [20] in each of the gimbal pins [19].

2) Tighten the bolts to 30 in-lb (3.4 N·m) - 35 in-lb (4.0 N·m).

(f) If you used the clamp assembly tool [42] (actuator tool, SPL-2439), loosen the adjustment screw [41] and the clamp block screws [40] to remove the tool.

SUBTASK 78-31-03-410-002-G00

(7) Install the applicable access doors:

Number	Name/Location
415DL	Left Thrust Reverser Actuator (Upper), Engine 1
426DR	Right Thrust Reverser Actuator (Upper), Engine 2

(a) Install the bolts [14] to attach the access door [15].

1) Tighten the bolts [14] to 30 in-lb (3.4 N·m) - 50 in-lb (5.6 N·m).

SUBTASK 78-31-03-410-012-G00

(8) Install the applicable access doors:

Number	Name/Location
416DR	Right Thrust Reverser Actuator (Upper), Engine 1
425DL	Left Thrust Reverser Actuator (Upper), Engine 2

(a) Install the bolts [14] to attach the access door [15].

1) Tighten the bolts [14] to 30 in-lb (3 N·m) - 50 in-lb (6 N·m).

(b) Install the lower Krueger flap deflector (TASK 78-31-09-420-801-G00).

SUBTASK 78-31-03-420-006-G00

(9) Do these steps if the replacement upper locking actuator [1] does not have the union [3] installed or the union [9]:

(a) Remove the protective covers from the hydraulic actuator ports.

(b) Lubricate a new packing [4] and new packing [10] with hydraulic fluid, D00153, or MCS 352B fluid, D00054.

(c) Install the new packing [4] on the union [3].

(d) Install the new packing [10] on the union [9].

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(e) Lubricate the threads of the union [3] and union [9] with hydraulic fluid, D00153, or MCS 352B fluid, D00054.

(f) Install the union [3] in the lower retract port.

(g) Tighten the union [3] to 256 in-lb (28.9 N·m) - 283 in-lb (32.0 N·m).

(h) Install the union [9] in the upper retract port.

(i) Tighten the union [9] to 256 in-lb (28.9 N·m) - 283 in-lb (32.0 N·m).

SUBTASK 78-31-03-020-008-G00

(10) Connect the hydraulic lines to the upper locking actuator [1]:

(a) Remove the protective covers from the hydraulic lines and the actuator ports.

(b) Connect the upper hydraulic retract line [2] to the union [3].

1) Hand tighten the coupling nut at this time.

2) Tighten the two bolts on the clamp block that hold the upper hydraulic retract line [2] to 30 in-lb (3.4 N·m) - 35 in-lb (4.0 N·m).

3) Tighten the coupling nut on the upper hydraulic retract line [2] as follows:

a) Tighten the coupling nut to 256 in-lb (28.9 N·m) - 283 in-lb (32.0 N·m).

b) Loosen the coupling nut.

c) Tighten the coupling nut again to 256 in-lb (28.9 N·m) - 283 in-lb (32.0 N·m).

(c) Temporarily install the sync shaft and tubing:

NOTE: To manually retract the thrust reverser, the sync shafts must be temporarily installed.

1) Do this task: Sync Shaft Inspection, TASK 78-31-04-200-801-G00.

2) If there is contamination, remove the contamination from the sync shaft with a cotton wiper, G00034.

3) Move the coupling nut and tube adaptor along the tubing.

4) Insert the sync shaft into the tubing.

a) Make sure that the sync shaft engages the internal spline in the middle actuator port.

5) Flex the sync shaft and insert the opposite end into the internal spline in the upper locking actuator port.

6) Tighten the coupling nut to 855 in-lb (96.6 N·m) - 945 in-lb (106.8 N·m).

NOTE: This is a temporary installation and it is not necessary to double-torque the coupling nuts.

(d) Connect the retract line flexhose [6] to the union [9].

1) Tighten the coupling nut to 256 in-lb (28.9 N·m) - 283 in-lb (32.0 N·m).

2) Loosen the coupling nut.

3) Tighten the coupling nut again to 256 in-lb (28.9 N·m) - 283 in-lb (32.0 N·m).

(e) Connect the extend line flexhose [7] to the upper extend port.

1) Tighten the coupling nut to 855 in-lb (96.6 N·m) - 945 in-lb (106.8 N·m).

2) Loosen the coupling nut.

3) Tighten the coupling nut again to 855 in-lb (96.6 N·m) - 945 in-lb (106.8 N·m).

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SUBTASK 78-31-03-020-009-G00

(11) Connect the electrical connectors to the Linear Variable Differential Transformer (LVDT) receptacles on the upper locking actuator [1] as follows:

(a) For the left thrust reverser, connect the electrical connectors, D30072 and D30076.

(b) For the right thrust reverser, connect the electrical connectors, D30074 and D30078.

SUBTASK 78-31-03-420-022-G00

CAUTION

MAKE SURE THAT YOU FOLLOW THE STEPS TO UNLOCK THE UPPER LOCKING ACTUATOR. IF YOU DO NOT OBEY INSTRUCTIONS, THE COMPONENTS WILL NOT CORRECTLY ENGAGE AND DAMAGE CAN OCCUR TO THE SYNC SHAFT.

(12) Unlock the upper locking actuator [1].

NOTE: If sync shaft failure occurs while you unlock the upper locking actuator, Foreign Object Debris (FOD) can be inside the actuator and sync lock.

(a) Move the manual unlock lever [5] forward.

(b) Hold the manual unlock lever [5] in the forward position as you push the detent pin in.

(c) Release the manual unlock lever [5].

NOTE: If the thrust reverser faults occur after the sync shaft replacement, thrust reverser hydraulics can be contaminated with FOD.

SUBTASK 78-31-03-980-002-G00

(13) Do this task: Thrust Reverser Operation - Retract (Manual Procedure), TASK 78-31-00-980-804-G00.

SUBTASK 78-31-03-410-003-G00

(14) Rig the upper sync shaft between the upper locking actuator and the middle actuator as follows:

(a) Do this task: Sync Shaft Removal, TASK 78-31-04-000-801-G00.

(b) Do this task: Sync Shaft Installation, TASK 78-31-04-400-801-G00.

1) Make sure that the thrust reverser is completely retracted when rigging the sync shafts.

NOTE: This task uses the thrust reverser maintenance lock equipment, SPL-9002. There are two -5 wire bundle cable assemblies in the B78009 tool assembly.

SUBTASK 78-31-03-420-023-G00

(15) Install the sleeve lock proximity sensor [8] as follows:

(a) Do this task: Thrust Reverser Sleeve Lock Proximity Sensor Installation, TASK 78-34-03-400-801-G00.

(b) Do this task: Sleeve Lock Proximity Sensor Adjustment and Test, TASK 78-34-03-800-801-G00.

SUBTASK 78-31-03-440-002-G00

(16) If not done, do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 78-31-03-710-001-G00

(17) Do the operational test of the thrust reverser (TASK 78-31-00-700-801-G00).

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SUBTASK 78-31-03-210-003-G00

(18) Examine the thrust reverser area for hydraulic fluid leaks (TASK 29-00-00-790-801).

SUBTASK 78-31-03-710-002-G00

(19) Do this task: Thrust Reverser Proximity Sensors Operational Test, TASK 78-31-00-700-805-G00.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-03-410-004-G00

(1) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

SUBTASK 78-31-03-040-010-G00

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

——— END OF TASK ———

TASK 78-31-03-000-803-G00

6. Middle Locking Hydraulic Actuator Removal

(Figure 402)

A. General

(1) This task gives the instructions to remove the middle locking hydraulic actuator.

(2) The middle locking hydraulic actuator will be referred to as middle actuator.

(3) There are three hydraulic actuators on each torque box of the left and right thrust reversers on an engine.

(4) The upper and middle hydraulic actuators are locking actuators, the lower hydraulic actuator is a non-locking actuator.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
29-21-00-000-802	Standby Hydraulic System Power Removal (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-980-803-G00	Thrust Reverser Operation - Extend (Manual Procedure) (P/B 201)
78-31-00-980-804-G00	Thrust Reverser Operation - Retract (Manual Procedure) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-2439	Tool - Thrust Reverser Actuator, Installation/Removal Part #: C78025-30 Supplier: 81205

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(Continued)

Reference	Description
SPL-14551	Lock - Thrust Reverser Opening Actuator, Fully Extended, LEAP-1B Part #: C78035-1 Supplier: 81205
STD-1110	Container - Hydraulic Fluid Resistant, 5 Gallon (19 Liter)
STD-7895	Wrench - Allen, 3/16 inch
STD-14286	Allen Wrench - Hex, 1/4 Inch

D. Consumable Materials

Reference	Description	Specification
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

E. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415EL	Left Thrust Reverser Actuator (Middle), Engine 1
416ER	Right Thrust Reverser Actuator (Middle), Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425EL	Left Thrust Reverser Actuator (Middle), Engine 2
426ER	Right Thrust Reverser Actuator (Middle), Engine 2

G. Prepare for the Removal

SUBTASK 78-31-03-040-011-G00

(1) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

SUBTASK 78-31-03-860-002-G00

(2) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 78-31-03-040-012-G00

(3) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 78-31-03-040-013-G00

(4) Remove the hydraulic power from the applicable engine as follows (TASK 29-11-00-860-805):

(a) For Engine 1, remove the system A hydraulic power.

(b) For Engine 2, remove the system B hydraulic power.

SUBTASK 78-31-03-040-014-G00

(5) Do this task: Standby Hydraulic System Power Removal, TASK 29-21-00-000-802.

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SUBTASK 78-31-03-040-015-G00

(6) Do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 78-31-03-010-014-G00

(7) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

(a) On the applicable engine, open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 78-31-03-020-036-G00

(8) Remove the applicable access door [64]:

(a) Open these access panels:

Number	Name/Location
415EL	Left Thrust Reverser Actuator (Middle), Engine 1
416ER	Right Thrust Reverser Actuator (Middle), Engine 1
425EL	Left Thrust Reverser Actuator (Middle), Engine 2
426ER	Right Thrust Reverser Actuator (Middle), Engine 2

SUBTASK 78-31-03-980-007-G00

(9) Unlock the middle locking actuator as follows:

NOTE: Make sure that the thrust reverser is fully closed and retracted. If it is necessary, retract the thrust reverser hydraulically. If there is a small gap, the thrust reverser is not fully closed and you cannot move the manual unlock lever.

(a) Move the manual unlock lever forward.

(b) Hold the manual unlock lever in the forward position as you push the detent pin in.

(c) Release the manual unlock lever.

SUBTASK 78-31-03-980-003-G00

(10) Expose the gimbal pins and prepare for the use of the actuator removal and installation tool:

NOTE: The thrust reverser sleeve must be extended to remove the gimbal pins from the aft side of the torque box and to use the actuator removal and installation tool.

CAUTION

DO NOT MANUALLY EXTEND THE INBOARD THRUST REVERSER SLEEVE MORE THAN THE PERMITTED LIMIT. MAKE SURE THAT THE SLEEVE DOES NOT TOUCH THE WING LEADING EDGE WHEN YOU EXTEND THE THRUST REVERSER. DAMAGE CAN OCCUR.

(a) For the inboard thrust reverser sleeve, manually extend the thrust reverser sleeve no more than 15 in. (381 mm) from the forward edge of the torque box (TASK 78-31-00-980-803-G00).

(b) For the outboard thrust reverser sleeve, manually extend the thrust reverser sleeve to the fully extended position (TASK 78-31-00-980-803-G00).

NOTE: The outboard thrust reverser sleeve will not touch the leading edge of the wing. The tool will be easier to use with the sleeve fully extended.

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SUBTASK 78-31-03-020-011-G00

WARNING

DO NOT INSTALL PRESSURE CAPS TO THE HYDRAULIC LINES THAT CONNECT TO CONTROL VALVE MODULE PORTS TO DEACTIVATE THE THRUST REVERSER. THE THRUST REVERSER CAN OPERATE WHEN PRESSURIZED. IF YOU DO NOT OBEY, INJURY TO PERSONNEL AND DAMAGE TO THE EQUIPMENT CAN OCCUR.

WARNING

MAKE SURE THAT YOU WEAR PROTECTIVE CLOTHES AND GLOVES WHEN YOU DO WORK ON THE HYDRAULIC SYSTEM. HYDRAULIC FLUID CAN CAUSE A LEAK FROM THE OPEN PORTS OF THE COMPONENT OR FROM THE HYDRAULIC LINES. INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION

DO NOT LET HYDRAULIC FLUID GET ON EQUIPMENT. IF HYDRAULIC FLUID GETS ON EQUIPMENT, IMMEDIATELY CLEAN IT FROM THE EQUIPMENT. HYDRAULIC FLUID CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE TWO WRENCHES TO LOOSEN THE TUBE COUPLING NUTS. USE ONE TO HOLD THE MATING PART, AND THE OTHER TO LOOSEN THE COUPLING NUT. IF YOU DO NOT USE TWO WRENCHES, DAMAGE TO THE TUBES AND MATING PART CAN OCCUR.

(11) Drain the hydraulic fluid from the hydraulic lines at the lower hydraulic actuator as follows (TASK 78-31-03-000-804-G00):

NOTE: To decrease hydraulic fluid spray when the coupling nuts are loosened, wrap a cotton wiper, G00034, around the wrench, nut and hydraulic line.

(a) Wrap a cotton wiper, G00034, around the electrical connector on the sync lock receptacle on the lower actuator .

NOTE: The cloth will catch the hydraulic fluid and prevent damage to the electrical connector.

(b) Loosen the clamp block screws on the clamp blocks that hold the lower hydraulic retract line.

NOTE: To remove the sleeve fitting on the hydraulic retract line from the actuator port, the hydraulic retract line must be carefully flexed. If the clamp blocks are loose, the hydraulic retract line can be flexed easier.

(c) Disconnect the hydraulic retract line at the lower actuator.

1) Carefully remove the sleeve fitting on the hydraulic retract line from the actuator port.

2) Let the hydraulic fluid drain into the 5 gallon (19 liter) hydraulic fluid resistant container, STD-1110.

CAUTION

USE TWO WRENCHES TO LOOSEN THE TUBE COUPLING NUTS. USE ONE TO HOLD THE MATING PART, AND THE OTHER TO LOOSEN THE COUPLING NUT. IF YOU DO NOT USE TWO WRENCHES, DAMAGE TO THE TUBES AND MATING PART CAN OCCUR.

(d) Disconnect the coupling nut on the sync shaft tubing at the lower actuator.

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1) Let the hydraulic fluid drain into a 5 gallon (19 liter) hydraulic fluid resistant container, STD-1110.

CAUTION USE TWO WRENCHES TO TIGHTEN THE COUPLING NUT. USE ONE TO HOLD THE FITTING, AND THE OTHER TO TURN THE COUPLING NUT. IF YOU DO NOT USE TWO WRENCHES, DAMAGE TO THE EQUIPMENT CAN OCCUR.

(e) Re-connect the coupling nut for the lower sync shaft to the lower actuator.

1) Torque the coupling nut to 855 in-lb (96.6 N·m) - 945 in-lb (106.8 N·m).

2) Loosen the coupling nut.

3) Torque the coupling nut again to 855 in-lb (96.6 N·m) - 945 in-lb (106.8 N·m).

CAUTION USE TWO WRENCHES TO TIGHTEN THE COUPLING NUT. USE ONE TO HOLD THE FITTING, AND THE OTHER TO TURN THE COUPLING NUT. IF YOU DO NOT USE TWO WRENCHES, DAMAGE TO THE EQUIPMENT CAN OCCUR.

(f) Re-connect the coupling nut for the lower retract line to the lower actuator.

1) Torque the coupling nut to 256 in-lb (28.9 N·m) - 283 in-lb (32.0 N·m).

2) Loosen the coupling nut.

3) Torque the coupling nut again to 256 in-lb (28.9 N·m) - 283 in-lb (32.0 N·m).

H. Middle Hydraulic Actuator Removal

SUBTASK 78-31-03-020-012-G00

(1) Remove the sleeve lock proximity sensor [73] from the middle actuator [51] as follows (Figure 402 (Sheet 3), View D):

(a) Remove the jamnut [71] from the sleeve lock proximity sensor [73].

NOTE: It is necessary to remove only the jamnut adjacent to the target.

(b) Remove the sleeve lock proximity sensor [73] and key washer [72].

(c) Re-install the key washer [72] and jamnut [71] on the sleeve lock proximity sensor [73] for storage.

(d) Safety the sleeve lock proximity sensor [73] away from the actuator.

SUBTASK 78-31-03-020-037-G00

(2) Remove the two gimbal pins [68] that attach the middle actuator [51] to the gimbal ring on the aft side of the torque box (Figure 402 (Sheet 2), View H):

(a) If you use the actuator tool, SPL-2439 to remove the gimbal pins [68], do these steps:

NOTE: Use a allen wrench, STD-7895 to turn the clamp block screw [40] and use a 1/4 inch allen wrench, STD-14286 to turn the middle actuator [51].

1) If not already done, loosen the clamp block screws [40] sufficiently so that the clamp block will fit over the middle actuator [51].

2) Make sure that the adjustment screw [41] is retracted.

NOTE: The adjustment screw is on the forward end of the tool.

3) Tilt the aft end of the clamp assembly tool [42] down and over the aft cascade support ring.

4) Lower the forward end of the clamp assembly tool [42] until the clamp block engages the middle actuator [51].

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5) Tighten the adjustment screw [41] to move the clamp block aft.
   a) Move the clamp block until the aft edge touches where the diameter of the middle actuator [51] increases.
6) Tighten the clamp block screws [40].
7) Tighten the adjustment screw [41] to compress the fire seal [80] or fire seal [81] against the torque box.

(b) Remove a bolt [69] and washer [70] from each of the two gimbal pins [68].
1) Remove the two gimbal pins [68].

(c) If you used the clamp assembly tool [42] (actuator tool, SPL-2439), do these steps:
1) Loosen the adjustment screw [41] and two clamp block screws [40].
2) Remove the clamp assembly tool [42].

SUBTASK 78-31-03-010-024-G00

(3) Retract the inboard and outboard thrust reverser sleeve to the stowed position (TASK 78-31-00-980-804-G00).

SUBTASK 78-31-03-020-038-G00

(4) Disconnect the rod end of the middle actuator [51] as follows (Figure 402 (Sheet 2), View G):
(a) Remove the nut [66], the bolt [60], the washer [61], the washer [65], and the bushing [62] that attach the rod end to the actuator attach fitting on the sleeve.
(b) Disconnect the rod end of the middle actuator [51] from the actuator attach fitting.

SUBTASK 78-31-03-020-014-G00

(5) Disconnect the hydraulic lines from the middle actuator [51] (Figure 402 (Sheet 1), View B):
(a) Loosen the two bolts on each of the two clamp blocks that hold the upper retract line.
(b) Disconnect the elbow [50] from the tee fitting [56] and the restrictor [54].
(c) Install protective covers on the fittings.
(d) Disconnect the upper and lower sync shaft tubing at the middle actuator.
1) Move the coupling nuts and tube adapters back along the tubing to get access to the sync shaft ends.
2) Flex the sync shaft to remove it from the internal splined port in the actuator.

CAUTION

DO NOT LET DIRT OR OTHER CONTAMINATION GET ON THE SYNC SHAFT. DAMAGE TO THE SYNC SHAFT CAN OCCUR.

3) Remove the sync shafts from the upper and lower sync shaft tubing.
   a) Make sure that dirt or contamination does not get on the sync shafts.

(e) Remove the restrictor [54] and packing [55] from the upper retract port on the middle actuator (Figure 402 (Sheet 1), View B).
1) Keep the restrictor [54] for the subsequent installation and discard the packing [55].

(f) Remove the bleed plug [52] and packing [53] from the lower retract port on the middle actuator (Figure 402 (Sheet 1), View B).
1) Keep the bleed plug [52] for the subsequent installation and discard the packing [53].

(g) Install protective covers on the hydraulic lines and the actuator ports.

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SUBTASK 78-31-03-010-007-G00

(6) Do these steps to get the necessary clearance to remove the middle actuator [51]:

(a) For the inboard thrust reverser, open the thrust reverser:

1) Make sure that the leading edge flaps are completely retracted.

NOTE: Without hydraulics to hold the flaps in the retract position, the weight of the flaps can cause them to extend a small amount.

WARNING

MAKE SURE THAT YOU MONITOR THE POSITION OF THE INBOARD THRUST REVERSER WHILE IT OPENS. MAKE SURE THAT THE INBOARD THRUST REVERSER DOES NOT TOUCH THE WING LEADING EDGE. INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

2) For the removal of the middle actuator [51] on the inboard thrust reverser, do this task: Open the Thrust Reverser (Selection), TASK 78-31-00-010-801-G00.

a) Install the actuator safety lock assembly, SPL-14551, on the inboard thrust reverser only.

<1> Make sure to use safety lock PN C78035–3. This is the full open inboard tool.

(b) For the removal of the middle actuator [51] on the outboard thrust reverser, do this task: Open the Thrust Reverser (Selection), TASK 78-31-00-010-801-G00.

1) Make sure to use safety lock PN C78035–2. This is the full open outboard tool.

NOTE: The outboard thrust reverser will not touch the leading edge of the wing.

SUBTASK 78-31-03-010-008-G00

CAUTION

DO NOT LET THE PUSH-ROD HIT THE AFT CASCADE-SUPPORT-RING OR THE TORQUE BOX. DAMAGE TO THE PUSH-ROD CAN OCCUR.

(7) Carefully remove the middle actuator [51].

SUBTASK 78-31-03-020-017-G00

(8) Remove the fire shield [79] and the fire seal [80] from the middle actuator [51] (Figure 402 (Sheet 4), View F).

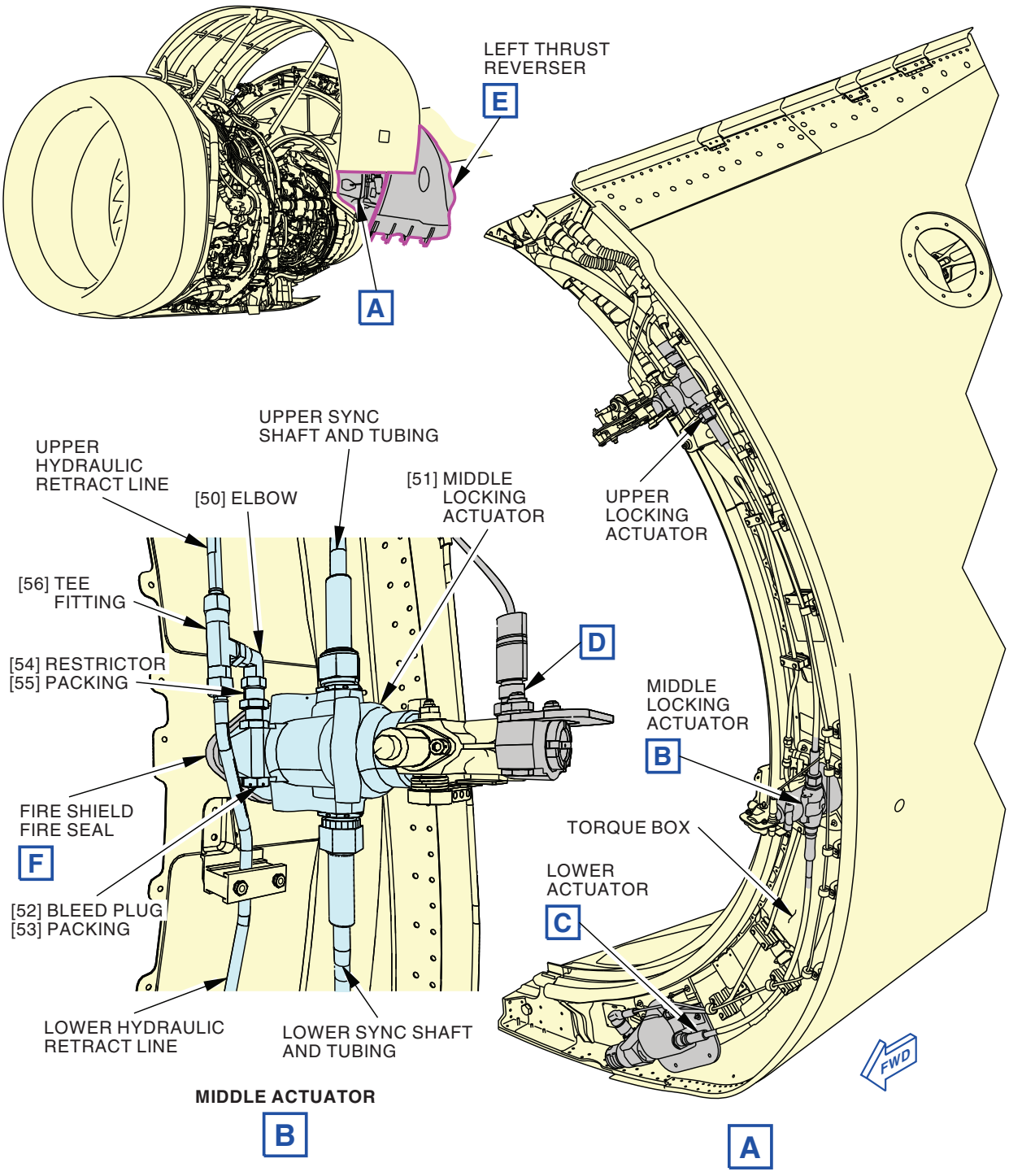
(a) Examine the fire seal [80] for damage.

(b) If you find no damage, keep the fire seal for the subsequent installation.

(c) If you find damage, replace the fire seal [80].

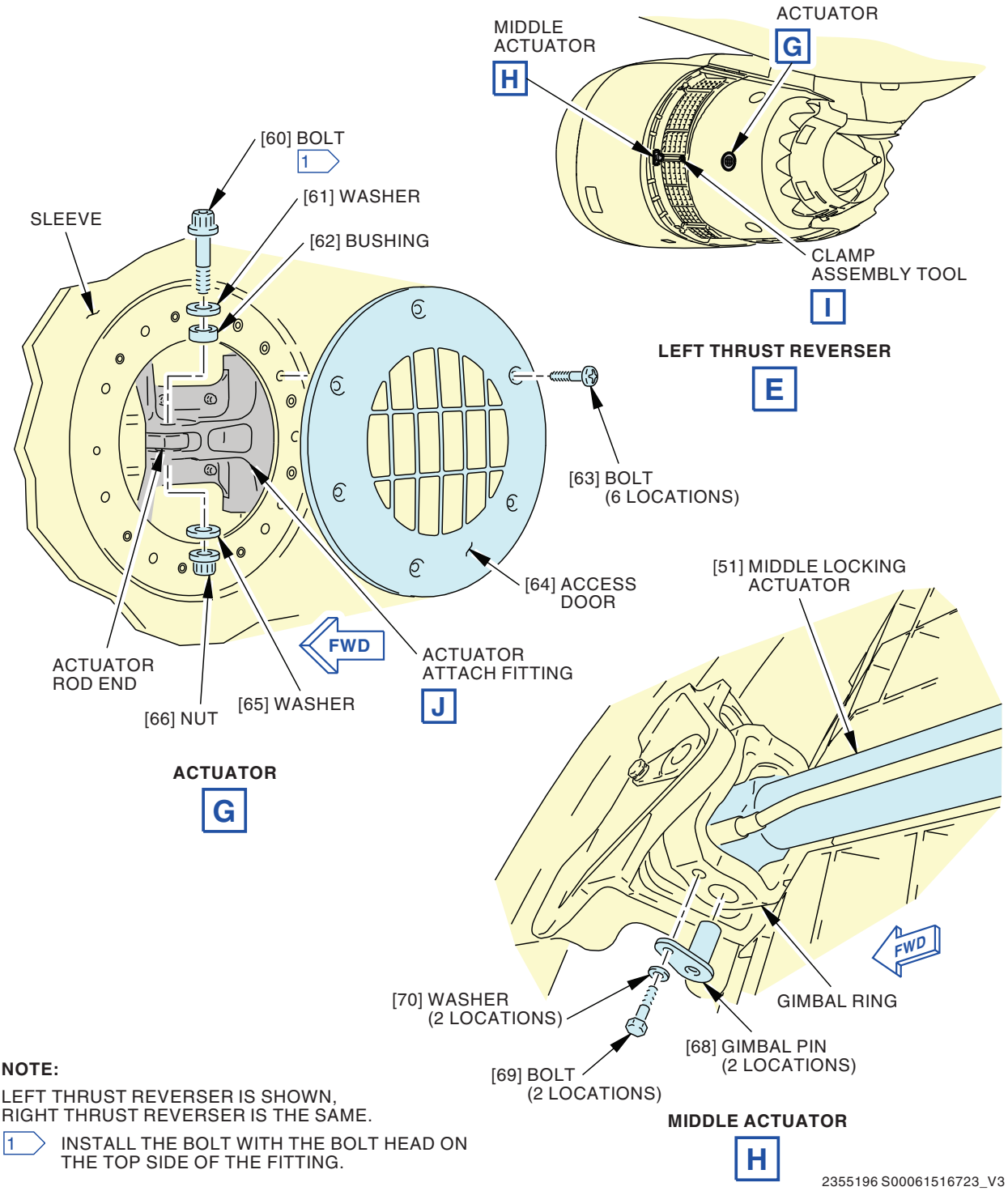
——— END OF TASK ———

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Middle Locking Actuator Installation
Figure 402/78-31-03-990-802-G00 (Sheet 1 of 6)

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NOTE:

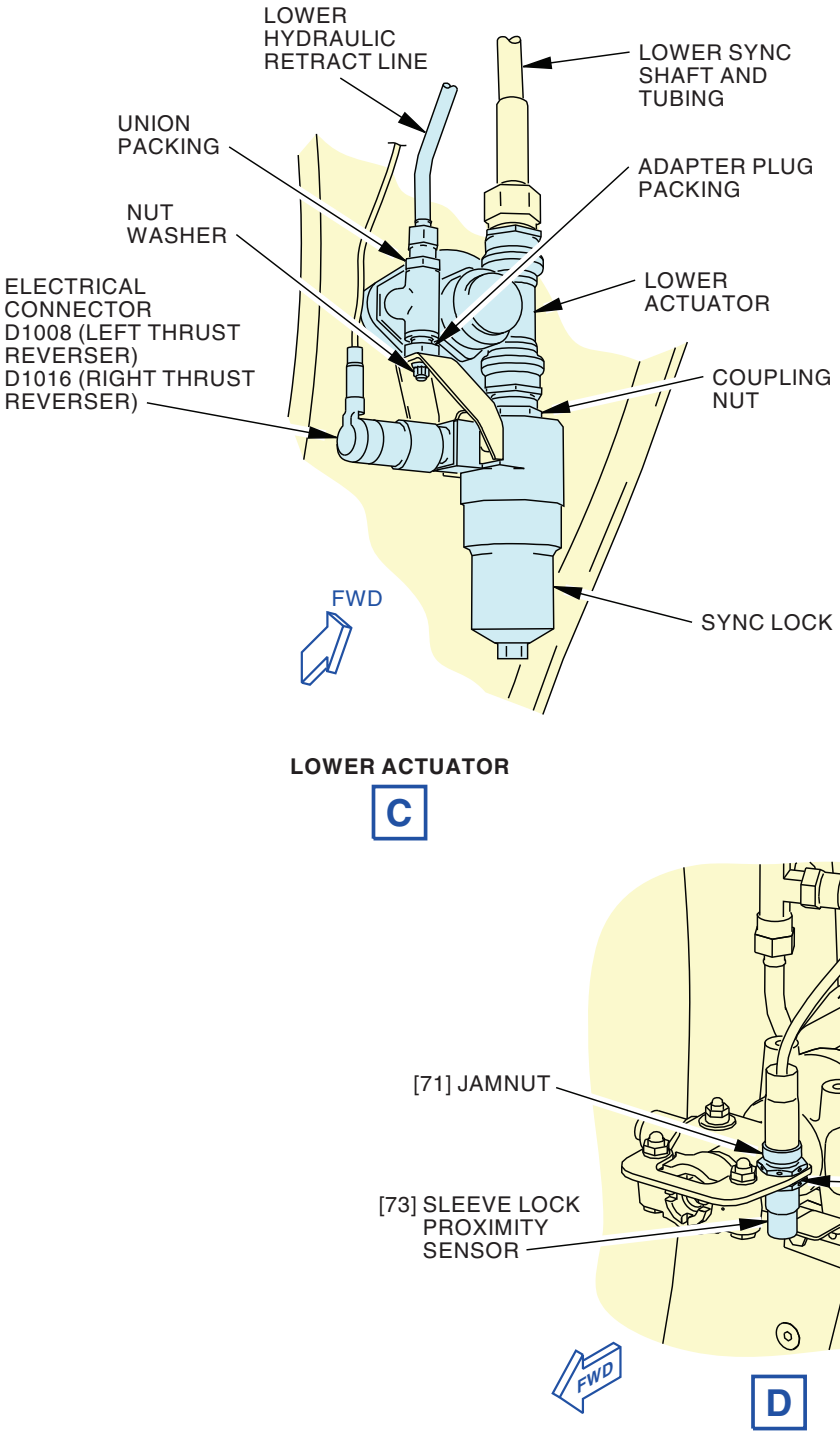
LEFT THRUST REVERSER IS SHOWN,
RIGHT THRUST REVERSER IS THE SAME.

1 INSTALL THE BOLT WITH THE BOLT HEAD ON THE TOP SIDE OF THE FITTING.

2355196 S00061516723_V3

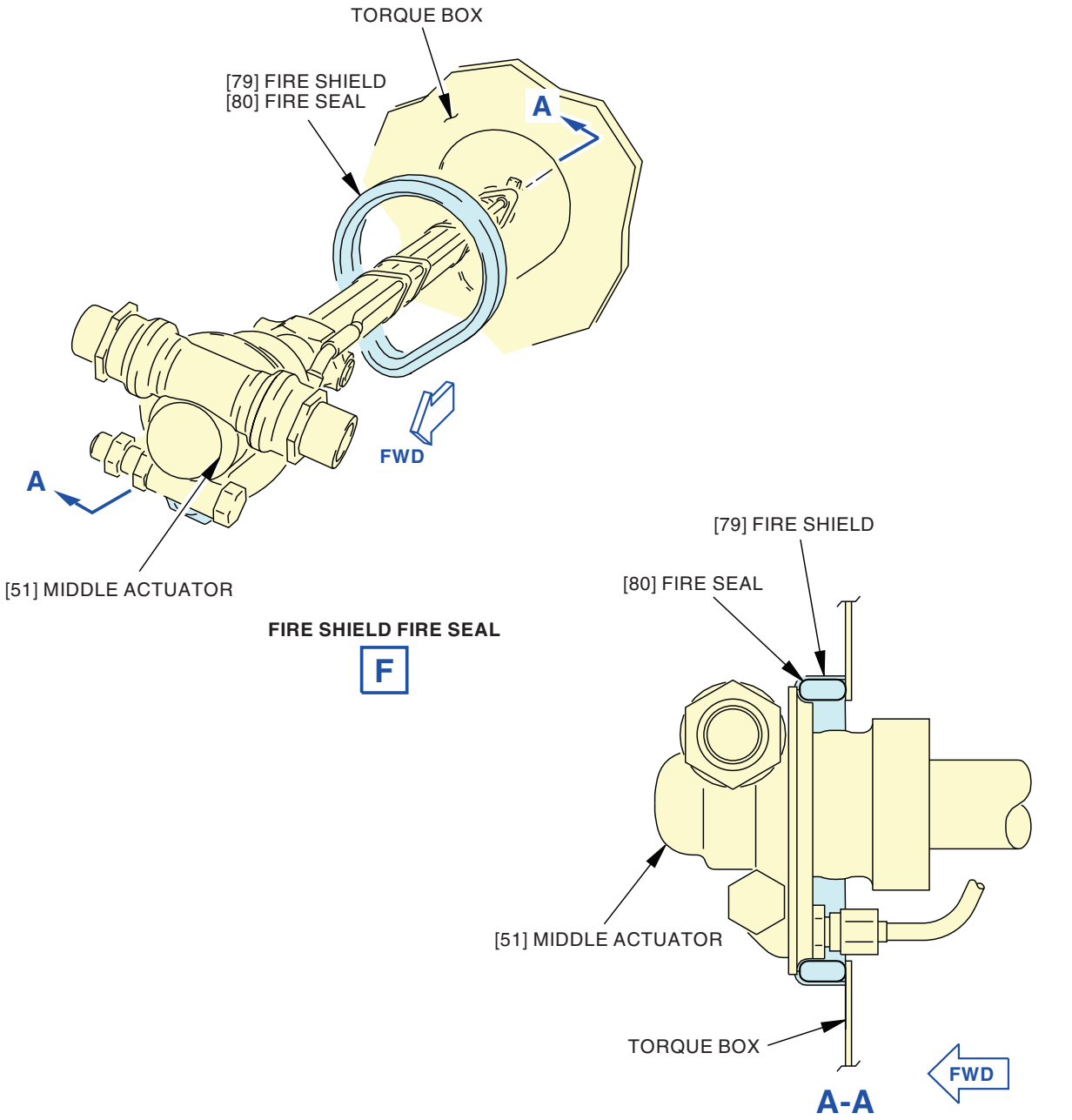
Middle Locking Actuator Installation
Figure 402/78-31-03-990-802-G00 (Sheet 2 of 6)

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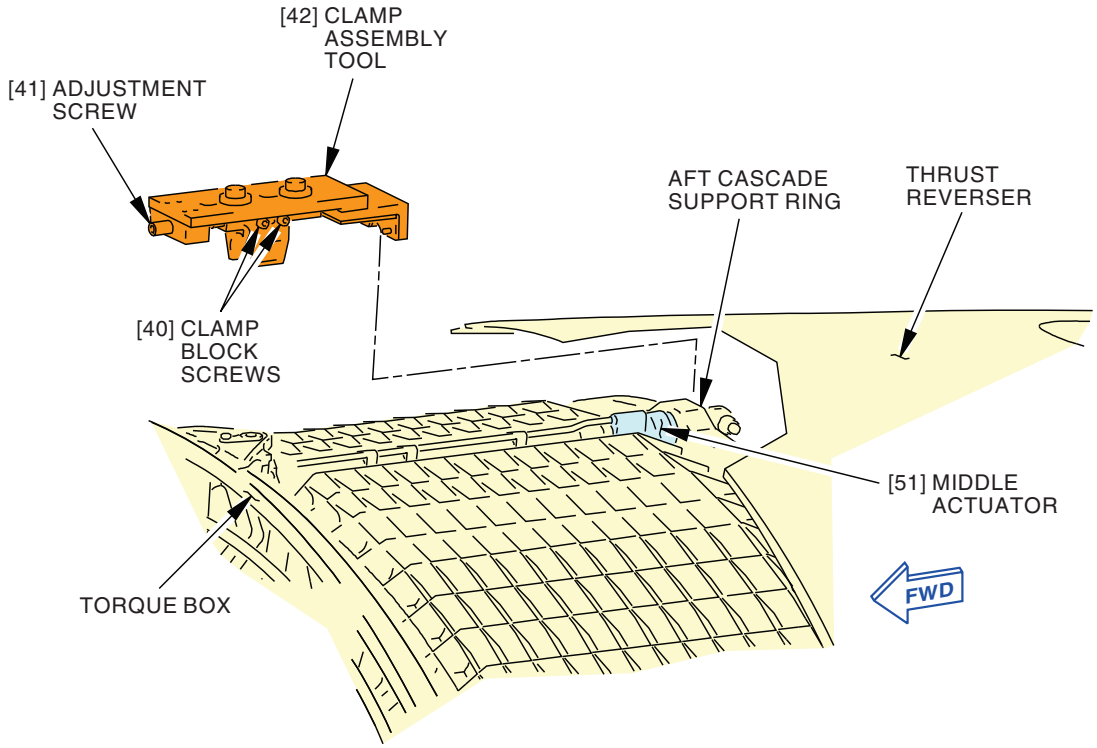
Middle Locking Actuator Installation
Figure 402/78-31-03-990-802-G00 (Sheet 3 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Middle Locking Actuator Installation
Figure 402/78-31-03-990-802-G00 (Sheet 4 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

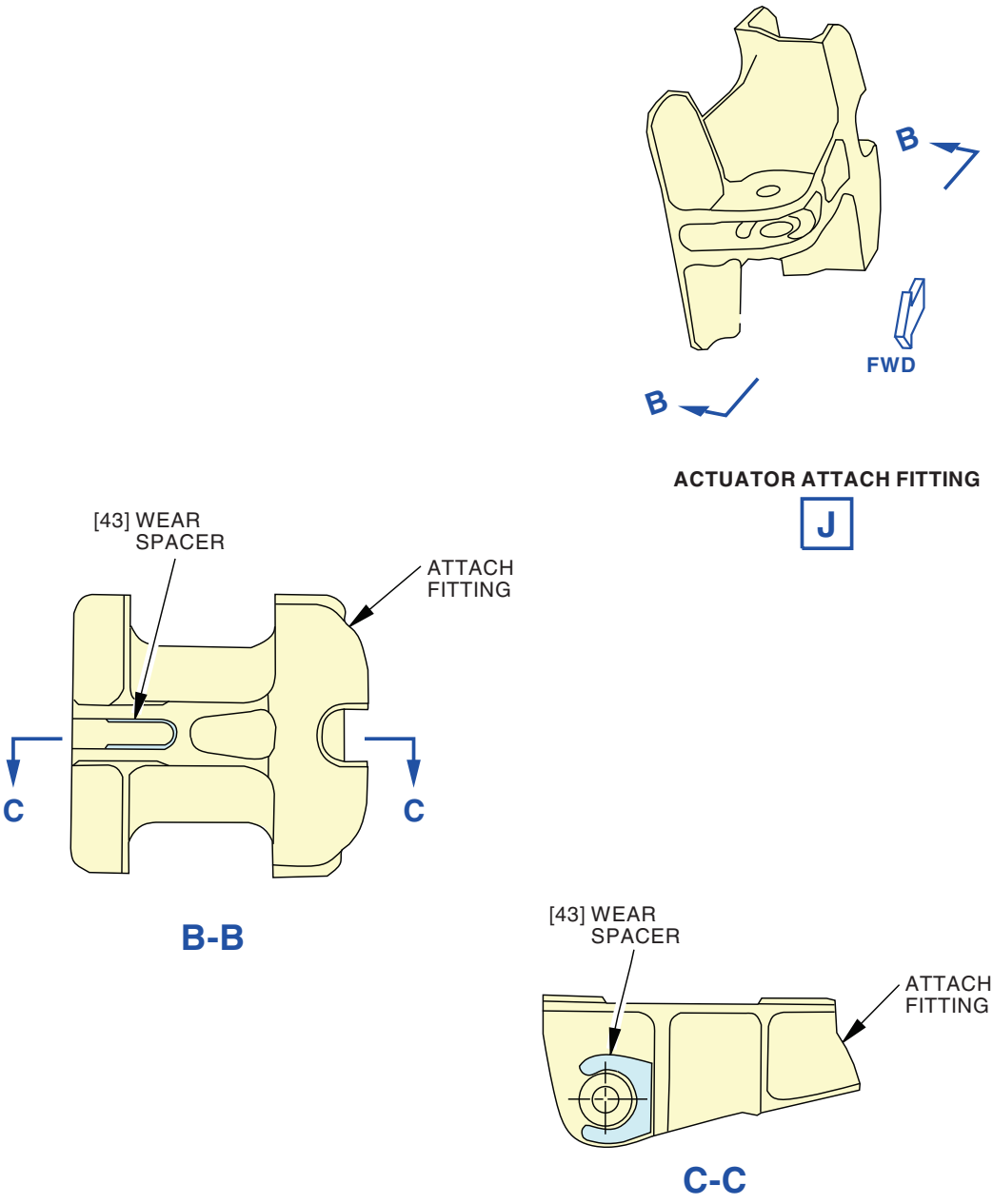


I

2355199 S00061516726_V2

Middle Locking Actuator Installation
Figure 402/78-31-03-990-802-G00 (Sheet 5 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2355200 S00061516727_V2

Middle Locking Actuator Installation
Figure 402/78-31-03-990-802-G00 (Sheet 6 of 6)

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TASK 78-31-03-400-803-G00

7. Middle Locking Hydraulic Actuator Installation

(Figure 402)

A. General

(1) This task gives the instructions to install the middle locking hydraulic actuator.

(2) The middle locking hydraulic actuator will be referred to as middle actuator.

B. References

Reference	Title
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
29-00-00-790-801	Hydraulic System External Leakage Check (P/B 601)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-00-700-801-G00	Thrust Reverser Operational Test (Selection) (P/B 501)
78-31-00-980-803-G00	Thrust Reverser Operation - Extend (Manual Procedure) (P/B 201)
78-31-00-980-804-G00	Thrust Reverser Operation - Retract (Manual Procedure) (P/B 201)
78-31-04-000-801-G00	Sync Shaft Removal (P/B 401)
78-31-04-200-801-G00	Sync Shaft Inspection (P/B 601)
78-31-04-400-801-G00	Sync Shaft Installation (P/B 401)
78-34-03-400-801-G00	Thrust Reverser Sleeve Lock Proximity Sensor Installation (P/B 401)
78-34-03-800-801-G00	Sleeve Lock Proximity Sensor Adjustment and Test (P/B 501)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-2439	Tool - Thrust Reverser Actuator, Installation/Removal Part #: C78025-30 Supplier: 81205
SPL-9002	Lock Equipment - Thrust Reverser Maintenance Part #: B78009-26 Supplier: 81205
SPL-14551	Lock - Thrust Reverser Opening Actuator, Fully Extended, LEAP-1B Part #: C78035-1 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
A00803	Sealant - Firewall - Hydraulic Fluid Resistant	BMS5-63 Type I
D00054	Fluid - Hydraulic, Assembly Lubricant, MCS 352B	
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
D00633	Grease - Aircraft General Purpose	BMS3-33

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(Continued)

Reference	Description	Specification
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
53	Packing	78-31-03-02-185	PTF ALL
55	Packing	78-31-03-02-185	PTF ALL

F. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

G. Access Panels

Number	Name/Location
415EL	Left Thrust Reverser Actuator (Middle), Engine 1
416ER	Right Thrust Reverser Actuator (Middle), Engine 1
425EL	Left Thrust Reverser Actuator (Middle), Engine 2
426ER	Right Thrust Reverser Actuator (Middle), Engine 2

H. Prepare for the Installation

SUBTASK 78-31-03-420-007-G00

(1) Install the fire seal [80] and fire shield [79] on the middle actuator [51] as follows:

(a) Put the fire seal [80] in the fire shield [79].

(b) Put the fire seal [80] and fire shield [79] over the rod end of the middle actuator [51].

1) Make sure that the fire seal [80] will contact the forward side of the torque box.

I. Middle Locking Hydraulic Actuator Installation

SUBTASK 78-31-03-860-006-G00

(1) Unlock the middle actuator [51] as follows:

(a) Move the manual unlock lever forward.

(b) Hold the manual unlock lever in the forward position as you push the detent pin in.

(c) Release the manual unlock lever.

SUBTASK 78-31-03-420-008-G00

(2) Install the middle actuator [51] as follows:

NOTE: Recommended to not turn the actuator rod end. For actuators without a Linear Variable Differential Transformer (LVDT), accidental rotation of the rod end will not cause detrimental effects.

(a) Pull the rod end out of the middle actuator [51] by hand to extend as follows:

NOTE: Do not let the rod end turn when you extend it manually. Make sure that you hold the rod end when you extend the actuator manually. Do not use hydraulic power to extend the rod end.

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1) For the inboard and outboard thrust reverser, extend the rod end approximately the same length as the one removed.

NOTE: It is also acceptable to extend the rod 1 in. (3 cm) - 2 in. (5 cm) longer to facilitate installation of the rod end bolt.

CAUTION

DO NOT LET THE PUSH-ROD HIT THE AFT CASCADE-SUPPORT-RING OR THE TORQUE BOX. DAMAGE TO THE PUSH-ROD CAN OCCUR.

(b) Carefully insert the rod end through the opening in the torque box and aft cascade support ring.

SUBTASK 78-31-03-420-028-G00

(3) Align the rod end of the middle actuator [51] with the clevis on the attach fitting.

(a) Make sure that the actuator attach fitting has a wear spacer [43] (Figure 402 (Sheet 6)).

1) If it is necessary, attach a wear spacer [43] into place with sealant, A00803.

NOTE: Seal around the edge of the fitting and inner sleeve.

(b) Install the bushing [62], washer [61], washer [65], bolt [60] and nut [66] that attach the rod end to the actuator attach fitting.

(c) Tighten the nut [66] to 370 in-lb (41.8 N·m) - 690 in-lb (78.0 N·m).

SUBTASK 78-31-03-410-005-G00

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS, BUT DO NOT DO THE THRUST REVERSER OR LEADING EDGE ACTIVATION. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(4) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

NOTE: This will make the attachment of the rod end of the middle actuator and the installation of the access door easier.

(a) Remove the actuator safety lock assembly, SPL-14551, from the inboard thrust reverser only.

(b) Do not do the thrust reverser or leading edge activation.

(c) Do not close the fan cowl panels at this time.

SUBTASK 78-31-03-010-027-G00

(5) For the inboard thrust reverser sleeve, manually extend the sleeve no more than 15 in. (38 cm) from the forward edge of the torque box (TASK 78-31-00-980-803-G00).

(a) For the outboard thrust reverser sleeve, manually fully extend the thrust reverser sleeve (TASK 78-31-00-980-803-G00).

NOTE: The outboard thrust reverser sleeve will not touch the leading edge of the wing. The tool will be easier to use with the sleeve fully extended.

SUBTASK 78-31-03-420-009-G00

(6) Install the two gimbal pins [68] that attach the middle actuator [51] to the gimbal ring on the aft side of the torque box:

(a) Apply grease, D00633, on the shanks of the two gimbal pins [68].

(b) Align the attachment holes in the middle actuator [51] and the gimbal ring.

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(c) If you use the clamp assembly tool [42] (actuator tool, SPL-2439) to compress the fire seal [80] or fire seal [81] against the torque box, do these steps:

1) Loosen the clamp block screws [40] sufficiently, so that the clamp block will fit over the middle actuator [51].

2) Make sure that the adjustment screw [41] is retracted.

NOTE: The adjustment screw is on the forward end of the tool.

3) Tilt the aft end of the clamp assembly tool [42] down and over the aft cascade support ring.

4) Lower the forward end of the clamp assembly tool [42] until the clamp block engages the middle actuator [51].

5) Tighten the adjustment screw [41] to move the clamp block aft.

NOTE: Move the clamp block until the aft edge touches where the diameter of the middle actuator [51] increases.

6) Tighten the clamp block screw [40].

7) Tighten the adjustment screw [41] to compress the fire seal [80] or fire seal [81] against the torque box.

8) Turn the housing one direction or the other to align the attachment holes.

(d) Manually compress the fire seal [80] or fire seal [81] against the torque box as follows:

1) Push aft on the forward end of the middle actuator [51] to compress the fire seal [80] or fire seal [81] against the torque box.

2) Turn the housing one direction or the other to align the attachment holes.

(e) Install the two gimbal pins [68] as follows:

1) Install a washer [70] and a bolt [69] in each gimbal pin [68].

2) Tighten the bolts to 30 in-lb (3.4 N·m) - 35 in-lb (4.0 N·m).

(f) If you used the clamp assembly tool [42] (actuator tool, SPL-2439), loosen the adjustment screw [41] and clamp block screws [40] to remove the tool.

SUBTASK 78-31-03-410-006-G00

(7) Close these access panels:

Number	Name/Location
415EL	Left Thrust Reverser Actuator (Middle), Engine 1
416ER	Right Thrust Reverser Actuator (Middle), Engine 1
425EL	Left Thrust Reverser Actuator (Middle), Engine 2
426ER	Right Thrust Reverser Actuator (Middle), Engine 2

(a) Install the six bolts [63] to attach the access door [64].

(b) Tighten the bolts [63] to 30 in-lb (3.4 N·m) - 50 in-lb (5.6 N·m).

SUBTASK 78-31-03-020-018-G00

(8) Do these steps if the replacement middle actuator [51] does not have the restrictor [54] or bleed plug [52]:

(a) Remove the protective covers from the hydraulic actuator ports.

(b) Lubricate the new packing [53] and new packing [55] with hydraulic fluid, D00153, or MCS 352B fluid, D00054.

(c) Install the new packing [53] on the bleed plug [52].

(d) Install the new packing [55] on the restrictor [54].

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(e) Lubricate the threads of the bleed plug [52] and restrictor [54] with hydraulic fluid, D00153, or MCS 352B fluid, D00054.

(f) Install the bleed plug [52] in the lower retract port.

(g) Tighten the bleed plug [52] to 256 in-lb (28.9 N·m) - 283 in-lb (32.0 N·m).

(h) Install the restrictor [54] in the upper retract port.

NOTE: The restrictor can be installed in either direction.

(i) Tighten the restrictor [54] to 256 in-lb (28.9 N·m) - 283 in-lb (32.0 N·m).

SUBTASK 78-31-03-020-019-G00

(9) Connect and tighten the hydraulic lines:

(a) Remove the protective covers from the hydraulic fittings.

(b) Connect the elbow [50] to the tee fitting [56] and the restrictor [54].

NOTE: Hand tighten the nuts on the elbow at this time.

(c) Tighten the bolts on the clamp blocks that hold the upper and lower retract lines to 30 in-lb (3.4 N·m) - 35 in-lb (4.0 N·m).

(d) Tighten the nuts on the elbow [50] at the tee fitting [56] and the restrictor [54] as follows:

1) Tighten the nuts to 256 in-lb (28.9 N·m) - 283 in-lb (32.0 N·m).

2) Loosen the nuts.

3) Tighten the nuts again to 256 in-lb (28.9 N·m) - 283 in-lb (32.0 N·m).

(e) Temporarily install the upper and lower sync shaft and tubing:

NOTE: To manually retract the thrust reverser, the sync shafts must be temporarily installed.

1) Do this task: Sync Shaft Inspection, TASK 78-31-04-200-801-G00.

2) If there is contamination, remove the contamination from the sync shaft with a cotton wiper, G00034.

3) Move the coupling nut and tube adaptor along the tubing.

4) Insert the sync shaft into the tubing.

a) Make sure that the sync shaft engages the internal spline in the lower and upper actuator port.

5) Flex the sync shaft and insert the opposite end into the internal spline in the middle actuator port.

6) Tighten the coupling nut to 855 in-lb (96.6 N·m) - 945 in-lb (106.8 N·m).

NOTE: This is a temporary installation and it is not necessary to double-torque the coupling nut.

SUBTASK 78-31-03-420-011-G00

(10) Install the sleeve lock proximity sensor [73] (TASK 78-34-03-400-801-G00).

SUBTASK 78-31-03-700-001-G00

(11) Do this task: Sleeve Lock Proximity Sensor Adjustment and Test, TASK 78-34-03-800-801-G00.

SUBTASK 78-31-03-980-004-G00

(12) Do this task: Thrust Reverser Operation - Retract (Manual Procedure), TASK 78-31-00-980-804-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details